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Fig. 0.1

Chapter

0

Vectors

Introduction of Vector

Physical quantities having magnitude, direction and obeying laws of vector algebra are called vectors.

Example : Displacement, velocity, acceleration, momentum, force, impulse, weight, thrust, torque, angular momentum, angular velocity *etc.*

If a physical quantity has magnitude and direction both, then it does not always imply that it is a vector. For it to be a vector the third condition of obeying laws of vector algebra has to be satisfied.

Example : The physical quantity current has both magnitude and direction but is still a scalar as it disobeys the laws of vector algebra.

Types of Vector

(1) **Equal vectors :** Two vectors $\vec{A}$ and $\vec{B}$ are said to be equal when they have equal magnitudes and same direction.

(2) **Parallel vector :** Two vectors $\vec{A}$ and $\vec{B}$ are said to be parallel when

(i) Both have same direction.

(ii) One vector is scalar (positive) non-zero multiple of another vector.

(3) **Anti-parallel vectors :** Two vectors $\vec{A}$ and $\vec{B}$ are said to be anti-parallel when

(i) Both have opposite direction.

(ii) One vector is scalar non-zero negative multiple of another vector.

(4) **Collinear vectors :** When the vectors under consideration can share the same support or have a common support then the considered vectors are collinear.

(5) **Zero vector** $(\vec{0})$**:** A vector having zero magnitude and arbitrary direction (not known to us) is a zero vector.

(6) **Unit vector :** A vector divided by its magnitude is a unit vector. Unit vector for $\vec{A}$ is $\hat{A}$ (read as A cap or A hat).

Since, $\hat{A} = \dfrac{\vec{A}}{A} \Rightarrow \vec{A} = A\hat{A}$.

Thus, we can say that unit vector gives us the direction.

(7) **Orthogonal unit vectors** $\hat{i}, \hat{j}$ and $\hat{k}$ are called orthogonal unit vectors. These vectors must form a Right Handed Triad (It is a coordinate system such that when we Curl the fingers of right hand from *x* to *y* then we must get the direction of *z* along thumb). The

$$\hat{i} = \frac{\vec{x}}{x}, \hat{j} = \frac{\vec{y}}{y}, \hat{k} = \frac{\vec{z}}{z}$$

$$\therefore \ \vec{x} = x\hat{i}, \ \vec{y} = y\hat{j}, \ \vec{z} = z\hat{k}$$

(8) **Polar vectors :** These have starting point or point of application . Example displacement and force *etc.*

(9) **Axial Vectors :** These represent rotational effects and are always along the axis of rotation in accordance with right hand screw rule. Angular velocity, torque and angular momentum, etc., are example of physical quantities of this type.

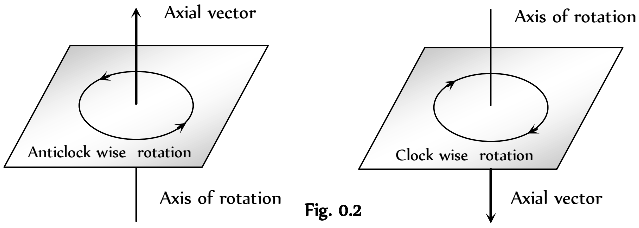


Fig. 0.2

(10) **Coplanar vector** : Three (or more) vectors are called coplanar vector if they lie in the same plane. Two (free) vectors are always coplanar.

Triangle Law of Vector Addition of Two Vectors

If two non zero vectors are represented by the two sides of a triangle taken in same order then the resultant is given by the closing side of triangle in opposite order. *i.e.* $\vec{R} = \vec{A} + \vec{B}$

$$\because \ \overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB}$$

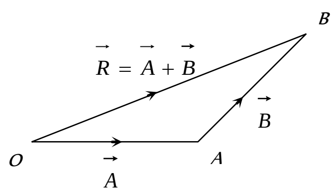


Fig. 0.3

(1) **Magnitude of resultant vector**

In $\Delta\, ABN$, $\cos\theta = \dfrac{AN}{B}$ $\therefore\ AN = B\cos\theta$

$\sin\theta = \dfrac{BN}{B}$ $\therefore\ BN = B\sin\theta$

In ΔOBN, we have $OB^2 = ON^2 + BN^2$

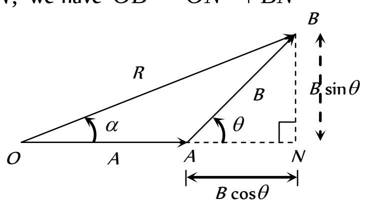


Fig. 0.4

$\Rightarrow\ R^2 = (A + B\cos\theta)^2 + (B\sin\theta)^2$

$\Rightarrow\ R^2 = A^2 + B^2\cos^2\theta + 2AB\cos\theta + B^2\sin^2\theta$

$\Rightarrow\ R^2 = A^2 + B^2(\cos^2\theta + \sin^2\theta) + 2AB\cos\theta$

$\Rightarrow\ R^2 = A^2 + B^2 + 2AB\cos\theta$

$\Rightarrow\ R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

(2) **Direction of resultant vectors :** If θ is angle between $\vec{A}$ and $\vec{B}$, then

$|\vec{A} + \vec{B}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

If $\vec{R}$ makes an angle α with $\vec{A}$, then in ΔOBN,

$\tan\alpha = \dfrac{BN}{ON} = \dfrac{BN}{OA + AN}$

$\tan\alpha = \dfrac{B\sin\theta}{A + B\cos\theta}$

Parallelogram Law of Vector Addition

If two non zero vectors are represented by the two adjacent sides of a parallelogram then the resultant is given by the diagonal of the parallelogram passing through the point of intersection of the two vectors.

(1) **Magnitude**

Since, $R^2 = ON^2 + CN^2$

$\Rightarrow\ R^2 = (OA + AN)^2 + CN^2$

$\Rightarrow\ R^2 = A^2 + B^2 + 2AB\cos\theta$

$\therefore\quad R = |\vec{R}| = |\vec{A} + \vec{B}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

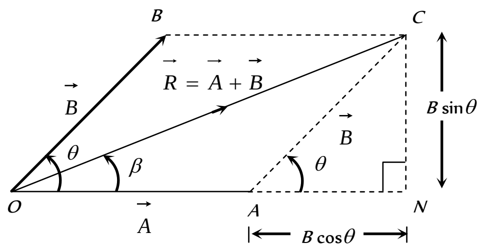


Fig. 0.5

Special cases : $R = A + B$ when $\theta = 0$

$R = A - B$ when $\theta = 180$

$R = \sqrt{A^2 + B^2}$ when $\theta = 90$

(2) **Direction**

$\tan\beta = \dfrac{CN}{ON} = \dfrac{B\sin\theta}{A + B\cos\theta}$

Polygon Law of Vector Addition

If a number of non zero vectors are represented by the $(n-1)$ sides of an n-sided polygon then the resultant is given by the closing side or the n side of the polygon taken in opposite order. So,

$\vec{R} = \vec{A} + \vec{B} + \vec{C} + \vec{D} + \vec{E}$

$\overrightarrow{OA} + \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DE} = \overrightarrow{OE}$

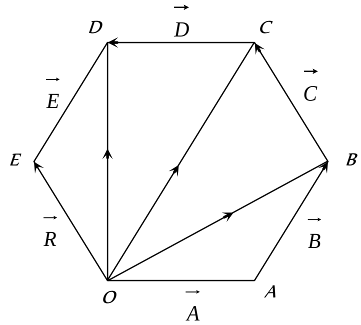


Fig. 0.6

Note : ❑ Resultant of two unequal vectors can not be zero.

❑ Resultant of three co-planar vectors may or may not be zero

❑ Resultant of three non co- planar vectors can not be zero.

Subtraction of vectors

Since, $\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$ and

$|\vec{A} + \vec{B}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

$\Rightarrow\ |\vec{A} - \vec{B}| = \sqrt{A^2 + B^2 + 2AB\cos(180^o - \theta)}$

Since, $\cos(180 - \theta) = -\cos\theta$

$\Rightarrow\ |\vec{A} - \vec{B}| = \sqrt{A^2 + B^2 - 2AB\cos\theta}$

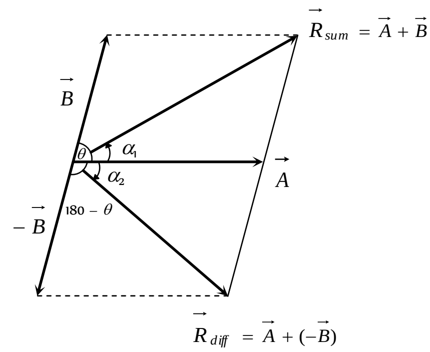


Fig. 0.7

$\tan\alpha_1 = \dfrac{B\sin\theta}{A + B\cos\theta}$

and $\tan\alpha_2 = \dfrac{B\sin(180 - \theta)}{A + B\cos(180 - \theta)}$

But $\sin(180-\theta)=\sin\theta$ and $\cos(180-\theta)=-\cos\theta$

$$\Rightarrow \tan\alpha_2=\frac{B\sin\theta}{A-B\cos\theta}$$

Resolution of Vector Into Components

Consider a vector $\vec{R}$ in X-Y plane as shown in fig. If we draw orthogonal vectors $\vec{R}_x$ and $\vec{R}_y$ along x and y axes respectively, by law of vector addition, $\vec{R}=\vec{R}_x+\vec{R}_y$

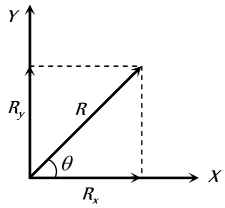


Fig. 0.8

Now as for any vector $\vec{A}=A\hat{n}$ so, $\vec{R}_x=\hat{i}R_x$ and $\vec{R}_y=\hat{j}R_y$

so $\vec{R}=\hat{i}R_x+\hat{j}R_y$ ...(i)

But from figure $R_x=R\cos\theta$ ...(ii)

and $R_y=R\sin\theta$ ...(iii)

Since R and θ are usually known, Equation (ii) and (iii) give the magnitude of the components of $\vec{R}$ along x and y-axes respectively.

Here it is worthy to note once a vector is resolved into its components, the components themselves can be used to specify the vector as

(1) The magnitude of the vector $\vec{R}$ is obtained by squaring and adding equation (ii) and (iii), *i.e.*

$$R=\sqrt{R_x^2+R_y^2}$$

(2) The direction of the vector $\vec{R}$ is obtained by dividing equation (iii) by (ii), *i.e.*

$$\tan\theta=(R_y/R_x) \text{ or } \theta=\tan^{-1}(R_y/R_x)$$

Rectangular Components of 3-D Vector

$$\vec{R}=\vec{R}_x+\vec{R}_y+\vec{R}_z q \quad \text{or} \quad \vec{R}=R_x\hat{i}+R_y\hat{j}+R_z\hat{k}$$

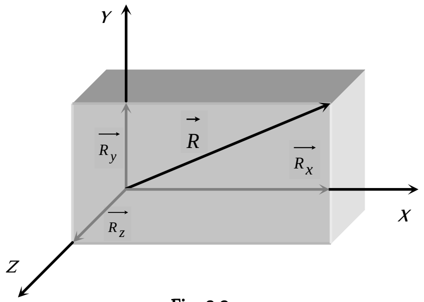


Fig. 0.9

If $\vec{R}$ makes an angle α with x axis, β with y axis and γ with z axis, then

$$\Rightarrow \cos\alpha=\frac{R_x}{R}=\frac{R_x}{\sqrt{R_x^2+R_y^2+R_z^2}}=l$$

$$\Rightarrow \cos\beta=\frac{R_y}{R}=\frac{R_y}{\sqrt{R_x^2+R_y^2+R_z^2}}=m$$

$$\Rightarrow \cos\gamma=\frac{R_z}{R}=\frac{R_z}{\sqrt{R_x^2+R_y^2+R_z^2}}=n$$

Where *l, m, n* are called Direction Cosines of the vector $\vec{R}$ and

$$l^2+m^2+n^2=\cos^2\alpha+\cos^2\beta+\cos^2\gamma=\frac{R_x^2+R_y^2+R_z^2}{R_x^2+R_y^2+R_z^2}=1$$

Note : ❑ When a point P have coordinate (x, y, z) then its position vector $\overrightarrow{OP}=x\hat{i}+y\hat{j}+z\hat{k}$

❑ When a particle moves from point (x, y, z) to (x, y, z) then its displacement vector

$$\vec{r}=(x_2-x_1)\hat{i}+(y_2-y_1)\hat{j}+(z_2-z_1)\hat{k}$$

Scalar Product of Two Vectors

(1) **Definition :** The scalar product (or dot product) of two vectors is defined as the product of the magnitude of two vectors with cosine of angle between them.

Thus if there are two vectors $\vec{A}$ and $\vec{B}$ having angle θ between them, then their scalar product written as $\vec{A}.\vec{B}$ is defined as $\vec{A}.\vec{B}$ $=AB\cos\theta$

(2) **Properties :** (i) It is always a scalar which is positive if angle between the vectors is acute (*i.e.*, < 90°) and negative if angle between them is obtuse (*i.e.* 90°<θ < 180°).

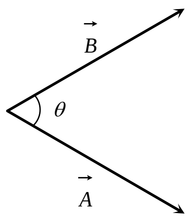


Fig. 0.10

(ii) It is commutative, *i.e.* $\vec{A}.\vec{B}=\vec{B}.\vec{A}$

(iii) It is distributive, *i.e.* $\vec{A}.(\vec{B}+\vec{C})=\vec{A}.\vec{B}+\vec{A}.\vec{C}$

(iv) As by definition $\vec{A}.\vec{B}=AB\cos\theta$

The angle between the vectors $\theta=\cos^{-1}\left[\frac{\vec{A}.\vec{B}}{AB}\right]$

(v) Scalar product of two vectors will be maximum when $\cos\theta=\max=1$, *i.e.* $\theta=0^o$, *i.e.*, vectors are parallel

$$(\vec{A}.\vec{B})_{\max}=AB$$

(vi) Scalar product of two vectors will be minimum when $|\cos\theta|=\min=0$, *i.e.* $\theta=90^o$

$$(\vec{A}.\vec{B})_{\min}=0$$

i.e. if the scalar product of two nonzero vectors vanishes the vectors are orthogonal.

(vii) The scalar product of a vector by itself is termed as self dot product and is given by $(\vec{A})^2=\vec{A}.\vec{A}=AA\cos\theta=A^2$

i.e. $A = \sqrt{\vec{A}.\vec{A}}$

(viii) In case of unit vector $\hat{n}$

$\hat{n}.\hat{n} = 1\times 1\times \cos 0 = 1$ so $\hat{n}.\hat{n} = \hat{i}.\hat{i} = \hat{j}.\hat{j} = \hat{k}.\hat{k} = 1$

(ix) In case of orthogonal unit vectors $\hat{i}, \hat{j}$ and $\hat{k}$, $\hat{i}.\hat{j} = \hat{j}.\hat{k} = \hat{k}.\hat{i} = 1\times 1\cos 90° = 0$

(x) In terms of components

$\vec{A}.\vec{B} = (\hat{i}A_x + \hat{j}A_y + \hat{k}A_z).(\hat{i}B_x + \hat{j}B_y + \hat{k}B_z) = [A_xB_x + A_yB_y + A_ZB_z]$

(3) **Example :** (i) Work W : In physics for constant force work is defined as, $W = Fs\cos\theta$ ...(i)

But by definition of scalar product of two vectors, $\vec{F}.\vec{s} = Fs\cos\theta$ ...(ii)

So from *eq* (i) and (ii) $W = \vec{F}.\vec{s}$ *i.e.* work is the scalar product of force with displacement.

(ii) Power P:

As $W = \vec{F}.\vec{s}$ or $\frac{dW}{dt} = \vec{F}.\frac{d\vec{s}}{dt}$ [As $\vec{F}$ is constant]

or $P = \vec{F}.\vec{v}$ *i.e.*, power is the scalar product of force with velocity. $\left[\text{As} \frac{dW}{dt} = P \text{ and } \frac{d\vec{s}}{dt} = \vec{v}\right]$

(iii) Magnetic Flux ϕ :

Magnetic flux through an area is given by $d\phi = B\,ds\cos\theta$ ...(i)

But by definition of scalar product $\vec{B}.d\vec{s} = Bds\cos\theta$ ...(ii)

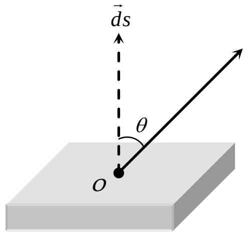


Fig. 0.11

So from *eq* (i) and (ii) we have

$d\phi = \vec{B}.d\vec{s}$ or $\phi = \int \vec{B}.d\vec{s}$

(iv) Potential energy of a dipole U : If an electric dipole of moment $\vec{p}$ is situated in an electric field $\vec{E}$ or a magnetic dipole of moment $\vec{M}$ in a field of induction $\vec{B}$, the potential energy of the dipole is given by :

$U_E = -\vec{p}.\vec{E}$ and $U_B = -\vec{M}.\vec{B}$

Vector Product of Two Vectors

(1) **Definition** : The vector product or cross product of two vectors is defined as a vector having a magnitude equal to the product of the magnitudes of two vectors with the sine of angle between them, and direction perpendicular to the plane containing the two vectors in accordance with right hand screw rule.

$$\vec{C} = \vec{A}\times\vec{B}$$

Thus, if $\vec{A}$ and $\vec{B}$ are two vectors, then their vector product written as $\vec{A}\times\vec{B}$ is a vector $\vec{C}$ defined by

$$\vec{C} = \vec{A}\times\vec{B} = AB\sin\theta\,\hat{n}$$

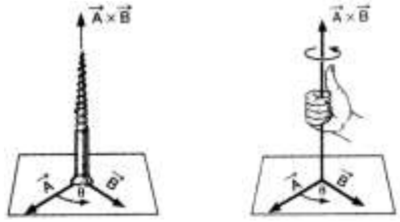


Fig. 0.12

The direction of $\vec{A}\times\vec{B}$, *i.e.* $\vec{C}$ is perpendicular to the plane containing vectors $\vec{A}$ and $\vec{B}$ and in the sense of advance of a right handed screw rotated from $\vec{A}$ (first vector) to $\vec{B}$ (second vector) through the smaller angle between them. Thus, if a right handed screw whose axis is perpendicular to the plane framed by $\vec{A}$ and $\vec{B}$ is rotated from $\vec{A}$ to $\vec{B}$ through the smaller angle between them, then the direction of advancement of the screw gives the direction of $\vec{A}\times\vec{B}$ *i.e.* $\vec{C}$

(2) **Properties**

(i) Vector product of any two vectors is always a vector perpendicular to the plane containing these two vectors, *i.e.*, orthogonal to both the vectors $\vec{A}$ and $\vec{B}$, though the vectors $\vec{A}$ and $\vec{B}$ may or may not be orthogonal.

(ii) Vector product of two vectors is not commutative, *i.e.*, $\vec{A}\times\vec{B} \neq \vec{B}\times\vec{A}$ [but $= -\vec{B}\times\vec{A}$]

Here it is worthy to note that

$|\vec{A}\times\vec{B}| = |\vec{B}\times\vec{A}| = AB\sin\theta$

i.e. in case of vector $\vec{A}\times\vec{B}$ and $\vec{B}\times\vec{A}$ magnitudes are equal but directions are opposite.

(iii) The vector product is distributive when the order of the vectors is strictly maintained, *i.e.*

$$\vec{A}\times(\vec{B}+\vec{C}) = \vec{A}\times\vec{B} + \vec{A}\times\vec{C}$$

(iv) The vector product of two vectors will be maximum when $\sin\theta = \max = 1$, *i.e.*, $\theta = 90^o$

$[\vec{A}\times\vec{B}]_{\max} = AB\,\hat{n}$

i.e. vector product is maximum if the vectors are orthogonal.

(v) The vector product of two non- zero vectors will be minimum when $|\sin\theta| =$ minimum = 0, *i.e.*, $\theta = 0^o$ or 180^o

$[\vec{A}\times\vec{B}]_{\min} = 0$

i.e. if the vector product of two non-zero vectors vanishes, the vectors are collinear.

(vi) The self cross product, *i.e.*, product of a vector by itself vanishes, *i.e.*, is null vector $\vec{A}\times\vec{A} = AA\sin 0^o\,\hat{n} = \vec{0}$

(vii) In case of unit vector $\hat{n}\times\hat{n} = \vec{0}$ so that $\hat{i}\times\hat{i} = \hat{j}\times\hat{j} = \hat{k}\times\hat{k} = \vec{0}$

(viii) In case of orthogonal unit vectors, $\hat{i}, \hat{j}, \hat{k}$ in accordance with right hand screw rule :

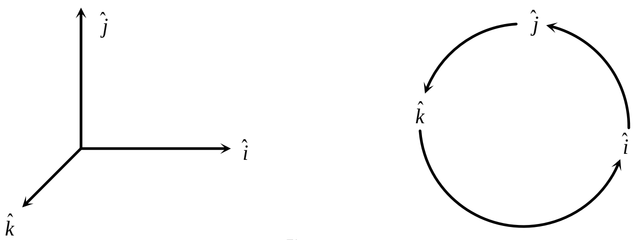


Fig. 0.13

$\hat{i}\times\hat{j}=\hat{k},\;\; \hat{j}\times\hat{k}=\hat{i}$ and $\hat{k}\times\hat{i}=\hat{j}$

And as cross product is not commutative,

$\hat{j}\times\hat{i}=-\hat{k}$, $\hat{k}\times\hat{j}=-\hat{i}$ and $\hat{i}\times\hat{k}=-\hat{j}$

(x) In terms of components

$$\vec{A}\times\vec{B}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ A_x & A_y & A_z\\ B_x & B_y & B_z\end{vmatrix}$$

$$=\hat{i}(A_yB_z-A_zB_y)+\hat{j}(A_zB_x-A_xB_z)+\hat{k}(A_xB_y-A_yB_x)$$

(3) **Example :** Since vector product of two vectors is a vector, vector physical quantities (particularly representing rotational effects) like torque, angular momentum, velocity and force on a moving charge in a magnetic field and can be expressed as the vector product of two vectors. It is well – established in physics that :

(i) Torque $\vec{\tau}=\vec{r}\times\vec{F}$

(ii) Angular momentum $\vec{L}=\vec{r}\times\vec{p}$

(iii) Velocity $\vec{v}=\vec{\omega}\times\vec{r}$

(iv) Force on a charged particle q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ is given by $\vec{F}=q(\vec{v}\times\vec{B})$

(v) Torque on a dipole in a field $\vec{\tau}_E=\vec{p}\times\vec{E}$ and $\vec{\tau}_B=\vec{M}\times\vec{B}$

Lami's Theorem

In any ΔABC with sides $\vec{a},\vec{b},\vec{c}$

$$\frac{\sin\alpha}{a}=\frac{\sin\beta}{b}=\frac{\sin\gamma}{c}$$

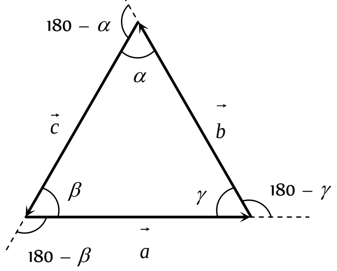


Fig. 0.14

i.e. for any triangle the ratio of the sine of the angle containing the side to the length of the side is a constant.

For a triangle whose three sides are in the same order we establish the Lami's theorem in the following manner. For the triangle shown

$\vec{a}+\vec{b}+\vec{c}=\vec{0}$ [All three sides are taken in order] ...(i)

$\Rightarrow\; \vec{a}+\vec{b}=-\vec{c}$ ...(ii)

Pre-multiplying both sides by $\vec{a}$

$\vec{a}\times(\vec{a}+\vec{b})=-\vec{a}\times\vec{c}\;\Rightarrow\;\vec{0}+\vec{a}\times\vec{b}=-\vec{a}\times\vec{c}$

$\Rightarrow\; \vec{a}\times\vec{b}=\vec{c}\times\vec{a}$ ...(iii)

Pre-multiplying both sides of (ii) by $\vec{b}$

$\vec{b}\times(\vec{a}+\vec{b})=-\vec{b}\times\vec{c}\;\Rightarrow\;\vec{b}\times\vec{a}+\vec{b}\times\vec{b}=-\vec{b}\times\vec{c}$

$\Rightarrow\; -\vec{a}\times\vec{b}=-\vec{b}\times\vec{c}\;\Rightarrow\;\vec{a}\times\vec{b}=\vec{b}\times\vec{c}$ ...(iv)

From (iii) and (iv), we get $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}$

Taking magnitude, we get $|\vec{a}\times\vec{b}|=|\vec{b}\times\vec{c}|=|\vec{c}\times\vec{a}|$

$\Rightarrow\; ab\sin(180-\gamma)=bc\sin(180-\alpha)=ca\sin(180-\beta)$

$\Rightarrow\; ab\sin\gamma=bc\sin\alpha=ca\sin\beta$

Dividing through out by *abc,* we have

$$\Rightarrow\;\frac{\sin\alpha}{a}=\frac{\sin\beta}{b}=\frac{\sin\gamma}{c}$$

Relative Velocity

(1) **Introduction :** When we consider the motion of a particle, we assume a fixed point relative to which the given particle is in motion. For example, if we say that water is flowing or wind is blowing or a person is running with a speed *v*, we mean that these all are relative to the earth (which we have assumed to be fixed).

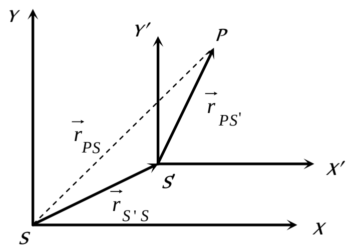


Fig. 0.15

Now to find the velocity of a moving object relative to another moving object, consider a particle *P* whose position relative to frame *S* is $\vec{r}_{PS}$ while relative to S' is $\vec{r}_{PS'}$.

If the position of frames S' relative to *S* at any time is $\vec{r}_{S'S}$ then from figure, $\vec{r}_{PS}=\vec{r}_{PS'}+\vec{r}_{S'S}$

Differentiating this equation with respect to time

$$\frac{d\vec{r}_{PS}}{dt}=\frac{d\vec{r}_{PS'}}{dt}+\frac{d\vec{r}_{S'S}}{dt}$$

or $\vec{v}_{PS}=\vec{v}_{PS'}+\vec{v}_{S'S}$ [as $\vec{v}=d\vec{r}/dt$]

or $\vec{v}_{PS'}=\vec{v}_{PS}-\vec{v}_{S'S}$

(2) **General Formula :** The relative velocity of a particle P_1 moving with velocity $\vec{v}_1$ with respect to another particle P_2 moving with velocity $\vec{v}_2$ is given by, $\vec{v}_{r_{12}} = \vec{v}_1 - \vec{v}_2$

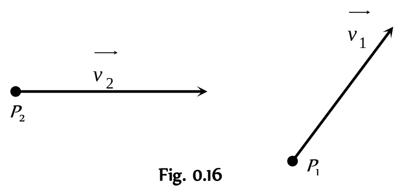


Fig. 0.16

(i) If both the particles are moving in the same direction then :

$\upsilon_{r_{12}} = \upsilon_1 - \upsilon_2$

(ii) If the two particles are moving in the opposite direction, then :

$\upsilon_{r_{12}} = \upsilon_1 + \upsilon_2$

(iii) If the two particles are moving in the mutually perpendicular directions, then:

$$\upsilon_{r_{12}} = \sqrt{\upsilon_1^2 + \upsilon_2^2}$$

(iv) If the angle between $\vec{\upsilon}_1$ and $\vec{\upsilon}_2$ be θ, then $\upsilon_{r_{12}} = \left[\upsilon_1^2 + \upsilon_2^2 - 2\upsilon_1\upsilon_2\cos\theta\right]^{1/2}$.

(3) **Relative velocity of satellite :** If a satellite is moving in equatorial plane with velocity $\vec{v}_s$ and a point on the surface of earth with $\vec{v}_e$ relative to the centre of earth, the velocity of satellite relative to the surface of earth

$$\vec{v}_{se} = \vec{v}_s - \vec{v}_e$$

So if the satellite moves form west to east (in the direction of rotation of earth on its axis) its velocity relative to earth's surface will be $v_{se} = v_s - v_e$

And if the satellite moves from east to west, *i.e.,* opposite to the motion of earth, $v_{se} = v_s - (-v_e) = v_s + v_e$

(4) **Relative velocity of rain :** If rain is falling vertically with a velocity $\vec{v}_R$ and an observer is moving horizontally with speed $\vec{v}_M$ the velocity of rain relative to observer will be $\vec{v}_{RM} = \vec{v}_R - \vec{v}_M$

which by law of vector addition has magnitude

$$v_{RM} = \sqrt{v_R^2 + v_M^2}$$

direction $\theta = \tan^{-1}(v_M / v_R)$ with the vertical as shown in fig.

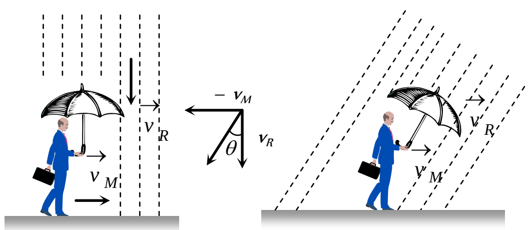


Fig. 0.17

(5) **Relative velocity of swimmer :** If a man can swim relative to water with velocity $\vec{v}$ and water is flowing relative to ground with velocity $\vec{v}_R$ velocity of man relative to ground $\vec{v}_M$ will be given by:

$$\vec{v} = \vec{v}_M - \vec{v}_R, \text{ i.e., } \vec{v}_M = \vec{v} + \vec{v}_R$$

So if the swimming is in the direction of flow of water, $v_M = v + v_R$

And if the swimming is opposite to the flow of water, $v_M = v - v_R$

(6) **Crossing the river :** Suppose, the river is flowing with velocity $\vec{\upsilon}_r$. A man can swim in still water with velocity $\vec{\upsilon}_m$. He is standing on one bank of the river and wants to cross the river, two cases arise.

(i) To cross the river over shortest distance : That is to cross the river straight, the man should swim making angle θ with the upstream as shown.

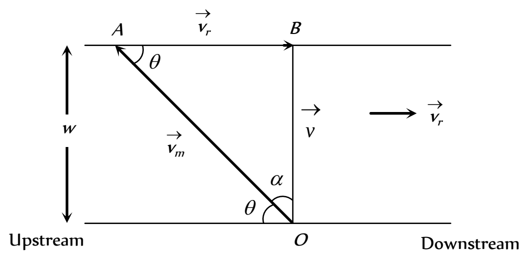


Fig. 0.18

Here OAB is the triangle of vectors, in which $\vec{OA} = \vec{v}_m, \vec{AB} = \vec{\upsilon}_r$.

Their resultant is given by $\vec{OB} = \vec{\upsilon}$. The direction of swimming makes angle θ with upstream. From the triangle OBA, we find,

$$\cos\theta = \frac{\upsilon_r}{\upsilon_m} \quad \text{Also } \sin\alpha = \frac{\upsilon_r}{\upsilon_m}$$

Where α is the angle made by the direction of swimming with the shortest distance (OB) across the river.

Time taken to cross the river : If w be the width of the river, then time taken to cross the river will be given by

$$t_1 = \frac{w}{\upsilon} = \frac{w}{\sqrt{\upsilon_m^2 - \upsilon_r^2}}$$

(ii) To cross the river in shortest possible time : The man should swim perpendicular to the bank.

The time taken to cross the river will be:

$$t_2 = \frac{w}{\upsilon_m}$$

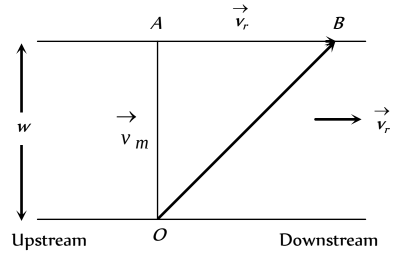


Fig. 0.19

In this case, the man will touch the opposite bank at a distance AB down stream. This distance will be given by:

$$AB = \upsilon_r t_2 = \upsilon_r \frac{w}{\upsilon_m} \quad \text{or} \quad AB = \frac{\upsilon_r}{\upsilon_m} w$$

Tips & Tricks

- All physical quantities having direction are not vectors. For example, the electric current possesses direction but it is a scalar quantity because it can not be added or multiplied according to the rules of vector algebra.
- A vector can have only two rectangular components in plane and only three rectangular components in space.
- A vector can have any number, even infinite components. (minimum 2 components)
- Following quantities are neither vectors nor scalars : Relative density, density, viscosity, frequency, pressure, stress, strain, modulus of elasticity, poisson's ratio, moment of inertia, specific heat, latent heat, spring constant loudness, resistance, conductance, reactance, impedance, permittivity, dielectric constant, permeability, susceptibility, refractive index, focal length, power of lens, Boltzman constant, Stefan's constant, Gas constant, Gravitational constant, Rydberg constant, Planck's constant etc.
- Distance covered is a scalar quantity.
- The displacement is a vector quantity.
- Scalars are added, subtracted or divided algebraically.
- Vectors are added and subtracted geometrically.
- Division of vectors is not allowed as directions cannot be divided.
- Unit vector gives the direction of vector.
- Magnitude of unit vector is 1.
- Unit vector has no unit. For example, velocity of an object is 5 ms^{-1} due East.

i.e. $\vec{v} = 5ms^{-1}$ due east.

$$\hat{v} = \frac{\vec{v}}{|\vec{v}|} = \frac{5ms^{-1}(\text{East})}{5ms^{-1}} = \text{East}$$

So unit vector $\hat{v}$ has no unit as East is not a physical quantity.

- Unit vector has no dimensions.
- $\hat{i}.\hat{i} = \hat{j}.\hat{j} = \hat{k}.\hat{k} = 1$
- $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \vec{0}$
- $\hat{i} \times \hat{j} = \hat{k},\ \hat{j} \times \hat{k} = \hat{i},\ \hat{k} \times \hat{i} = \hat{j}$
- $\hat{i}.\hat{j} = \hat{j}.\hat{k} = \hat{k}.\hat{i} = 0$
- $\vec{A} \times \vec{A} = \vec{0}$. Also $\vec{A} - \vec{A} = \vec{0}$ But $\vec{A} \times \vec{A} \neq \vec{A} - \vec{A}$

Because $\vec{A} \times \vec{A} \perp \vec{A}$ and $\vec{A} - \vec{A}$ is collinear with $\vec{A}$

- Multiplication of a vector with –1 reverses its direction.

If $\vec{A} = \vec{B}$, then $A = B$ and $\hat{A} = \hat{B}$.

If $\vec{A} + \vec{B} = \vec{0}$, then $A = B$ but $\hat{A} = -\hat{B}$.

- Minimum number of collinear vectors whose resultant can be zero is two.
- Minimum number of coplaner vectors whose resultant is zero is three.
- Minimum number of non coplaner vectors whose resultant is zero is four.
- Two vectors are perpendicular to each other if $\vec{A}.\vec{B} = 0$.
- Two vectors are parallel to each other if $\vec{A} \times \vec{B} = 0$.
- Displacement, velocity, linear momentum and force are polar vectors.
- Angular velocity, angular acceleration, torque and angular momentum are axial vectors.
- Division with a vector is not defined because it is not possible to divide with a direction.
- Distance covered is always positive quantity.
- The components of a vectors can have magnitude than that of the vector itself.
- The rectangular components cannot have magnitude greater than that of the vector itself.
- When we multiply a vector with 0 the product becomes a null vector.
- The resultant of two vectors of unequal magnitude can never be a null vector.
- Three vectors not lying in a plane can never add up to give a null vector.
- A quantity having magnitude and direction is not necessarily a vector. For example, time and electric current. These quantities have magnitude and direction but they are scalar. This is because they do not obey the laws of vector addition.
- A physical quantity which has different values in different directions is called a tensor. For example : Moment of inertia has different values in different directions. Hence moment of inertia is a tensor. Other examples of tensor are refractive index, stress, strain, density etc.
- The magnitude of rectangular components of a vector is always less than the magnitude of the vector
- If $\vec{A} = \vec{B}$, then $A_x = B_x$, $A_y = B_y$ and $A_z = B_z$.
- If $\vec{A} + \vec{B} = \vec{C}$. Or if $\vec{A} + \vec{B} + \vec{C} = \vec{0}$, then $\vec{A}$, $\vec{B}$ and $\vec{C}$ lie in one plane.
- If $\vec{A} \times \vec{B} = \vec{C}$, then $\vec{C}$ is perpendicular to $\vec{A}$ as well as $\vec{B}$.
- If $|\vec{A} \times \vec{B}| = |\vec{A} - \vec{B}|$, then angle between $\vec{A}$ and $\vec{B}$ is 90°.
- Resultant of two vectors will be maximum when θ = 0° *i.e.* vectors

are parallel.

$$R_{\max} = \sqrt{P^2 + Q^2 + 2PQ\cos 0^\circ} = |P + Q|$$

✍ Resultant of two vectors will be minimum when $\theta = 180^\circ$ *i.e.* vectors are anti-parallel.

$$R_{\min} = \sqrt{P^2 + Q^2 + 2PQ\cos 180^\circ} = |P - Q|$$

Thus, minimum value of the resultant of two vectors is equal to the difference of their magnitude.

✍ Thus, maximum value of the resultant of two vectors is equal to the sum of their magnitude.

When the magnitudes of two vectors are unequal, then

$$R_{\min} = P - Q \neq 0$$

$$[\because |\vec{P}| \neq |\vec{Q}|]$$

Thus, two vectors $\vec{P}$ and $\vec{Q}$ having different magnitudes can never be combined to give zero resultant. From here, we conclude that the minimum number of vectors of unequal magnitude whose resultant can be zero is three. On the other hand, the minimum number of vectors of equal magnitude whose resultant can be zero is two.

✍ Angle between two vectors $\vec{A}$ and $\vec{B}$ is given by

$$\cos\theta = \frac{\vec{A}.\vec{B}}{|\vec{A}||\vec{B}|}$$

✍ Projection of a vector $\vec{A}$ in the direction of vector $\vec{B}$

$$= \frac{\vec{A}.\vec{B}}{|\vec{B}|}$$

✍ Projection of a vector $\vec{B}$ in the direction of vector $\vec{A}$

$$= \frac{\vec{A}.\vec{B}}{|\vec{A}|}$$

✍ If vectors $\vec{A}, \vec{B}$ and $\vec{C}$ are represented by three sides *ab, bc* and *ca* respectively taken in a order, then

$$\frac{|\vec{A}|}{ab} = \frac{|\vec{B}|}{bc} = \frac{|\vec{C}|}{ca}$$

✍ The vectors $\hat{i} + \hat{j} + \hat{k}$ is equally inclined to the coordinate axes at an angle of 54.74 degrees.

✍ If $\vec{A} \pm \vec{B} = \vec{C}$, then $\vec{A}.\vec{B} \times \vec{C} = 0$.

✍ If $\vec{A}.\vec{B} \times \vec{C} = 0$, then $\vec{A}.\vec{B}$ and $\vec{C}$ are coplanar.

✍ If angle between $\vec{A}$ and $\vec{B}$ is 45°,

then $\vec{A}.\vec{B} = |\vec{A} \times \vec{B}|$

✍ If $\vec{A}_1 + \vec{A}_2 + \vec{A}_3 + ... + \vec{A}_n = \vec{0}$ and $A_1 = A_2 = A_3 = ... = A_n$ then the adjacent vector are inclined to each other at angle $2\pi / n$.

✍ If $\vec{A} + \vec{B} = \vec{C}$ and $A^2 + B^2 = C^2$, then the angle between $\vec{A}$ and $\vec{B}$ is 90°. Also *A, B* and *C* can have the following values.

(i) $A = 3$, $B = 4$, $C = 5$

(ii) $A = 5$, $B = 12$, $C = 13$

(iii) $A = 8$, $B = 15$, $C = 17$.

Ordinary Thinking

Objective Questions

Fundamentals of Vectors

1. The vector projection of a vector $3\hat{i}+4\hat{k}$ on *y*-axis is [RPMT 2004]

(a) 5 (b) 4
(c) 3 (d) Zero

2. Position of a particle in a rectangular-co-ordinate system is (3, 2, 5). Then its position vector will be

(a) $3\hat{i}+5\hat{j}+2\hat{k}$ (b) $3\hat{i}+2\hat{j}+5\hat{k}$
(c) $5\hat{i}+3\hat{j}+2\hat{k}$ (d) None of these

3. If a particle moves from point *P* (2,3,5) to point *Q* (3,4,5). Its displacement vector be

(a) $\hat{i}+\hat{j}+10\hat{k}$ (b) $\hat{i}+\hat{j}+5\hat{k}$
(c) $\hat{i}+\hat{j}$ (d) $2\hat{i}+4\hat{j}+6\hat{k}$

4. A force of 5 *N* acts on a particle along a direction making an angle of 60° with vertical. Its vertical component be

(a) 10 *N* (b) 3 *N*
(c) 4 *N* (d) 2.5 *N*

5. If $A=3\hat{i}+4\hat{j}$ and $B=7\hat{i}+24\hat{j}$, the vector having the same magnitude as *B* and parallel to *A* is

(a) $5\hat{i}+20\hat{j}$ (b) $15\hat{i}+10\hat{j}$
(c) $20\hat{i}+15\hat{j}$ (d) $15\hat{i}+20\hat{j}$

6. Vector $\vec{A}$ makes equal angles with *x*, *y* and *z* axis. Value of its components (in terms of magnitude of $\vec{A}$) will be

(a) $\frac{A}{\sqrt{3}}$ (b) $\frac{A}{\sqrt{2}}$
(c) $\sqrt{3}A$ (d) $\frac{\sqrt{3}}{A}$

7. If $\vec{A}=2\hat{i}+4\hat{j}-5\hat{k}$ the direction of cosines of the vector $\vec{A}$ are

(a) $\frac{2}{\sqrt{45}},\frac{4}{\sqrt{45}}$ and $\frac{-5}{\sqrt{45}}$ (b) $\frac{1}{\sqrt{45}},\frac{2}{\sqrt{45}}$ and $\frac{3}{\sqrt{45}}$
(c) $\frac{4}{\sqrt{45}},0$ and $\frac{4}{\sqrt{45}}$ (d) $\frac{3}{\sqrt{45}},\frac{2}{\sqrt{45}}$ and $\frac{5}{\sqrt{45}}$

8. The vector that must be added to the vector $\hat{i}-3\hat{j}+2\hat{k}$ and $3\hat{i}+6\hat{j}-7\hat{k}$ so that the resultant vector is a unit vector along the *y*-axis is

(a) $4\hat{i}+2\hat{j}+5\hat{k}$ (b) $-4\hat{i}-2\hat{j}+5\hat{k}$
(c) $3\hat{i}+4\hat{j}+5\hat{k}$ (d) Null vector

9. How many minimum number of coplanar vectors having different magnitudes can be added to give zero resultant

(a) 2 (b) 3
(c) 4 (d) 5

10. A hall has the dimensions $10\,m\times12\,m\times14\,m$. A fly starting at one corner ends up at a diametrically opposite corner. What is the magnitude of its displacement

(a) 17 *m* (b) 26 *m*
(c) 36 *m* (d) 20 *m*

11. 100 coplanar forces each equal to 10 N act on a body. Each force makes angle $\pi/50$ with the preceding force. What is the resultant of the forces

(a) 1000 *N* (b) 500 *N*
(c) 250 *N* (d) Zero

12. The magnitude of a given vector with end points (4, – 4, 0) and (– 2, – 2, 0) must be

(a) 6 (b) $5\sqrt{2}$
(c) 4 (d) $2\sqrt{10}$

13. The expression $\left(\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}\right)$ is a

(a) Unit vector (b) Null vector
(c) Vector of magnitude $\sqrt{2}$ (d) Scalar

14. Given vector $\vec{A}=2\hat{i}+3\hat{j}$, the angle between $\vec{A}$ and *y*-axis is [CPMT 1993]

(a) $\tan^{-1}3/2$ (b) $\tan^{-1}2/3$
(c) $\sin^{-1}2/3$ (d) $\cos^{-1}2/3$

15. The unit vector along $\hat{i}+\hat{j}$ is

(a) $\hat{k}$ (b) $\hat{i}+\hat{j}$
(c) $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$ (d) $\frac{\hat{i}+\hat{j}}{2}$

16. A vector is represented by $3\hat{i}+\hat{j}+2\hat{k}$. Its length in *XY* plane is

(a) 2 (b) $\sqrt{14}$
(c) $\sqrt{10}$ (d) $\sqrt{5}$

17. Five equal forces of 10 *N* each are applied at one point and all are lying in one plane. If the angles between them are equal, the resultant force will be [CBSE PMT 1995]

(a) Zero (b) 10 *N*
(c) 20 *N* (d) $10\sqrt{2}N$

18. The angle made by the vector $A=\hat{i}+\hat{j}$ with *x*- axis is [EAMCET (Engg.) 1999]

(a) 90° (b) 45°
(c) 22.5° (d) 30°

19. Any vector in an arbitrary direction can always be replaced by two (or three)

(a) Parallel vectors which have the original vector as their resultant
(b) Mutually perpendicular vectors which have the original vector as their resultant

(c) Arbitrary vectors which have the original vector as their resultant

(d) It is not possible to resolve a vector

20. Angular momentum is [MNR 1986]

(a) A scalar (b) A polar vector

(c) An axial vector (d) None of these

21. Which of the following is a vector

(a) Pressure (b) Surface tension

(c) Moment of inertia (d) None of these

22. If $\vec{P} = \vec{Q}$ then which of the following is NOT correct

(a) $\hat{P} = \hat{Q}$ (b) $|\vec{P}| = |\vec{Q}|$

(c) $P\hat{Q} = Q\hat{P}$ (d) $\vec{P} + \vec{Q} = \hat{P} + \hat{Q}$

23. The position vector of a particle is $\vec{r} = (a\cos\omega t)\hat{i} + (a\sin\omega t)\hat{j}$. The velocity of the particle is [CBSE PMT 1995]

(a) Parallel to the position vector

(b) Perpendicular to the position vector

(c) Directed towards the origin

(d) Directed away from the origin

24. Which of the following is a scalar quantity [AFMC 1998]

(a) Displacement (b) Electric field

(c) Acceleration (d) Work

25. If a unit vector is represented by $0.5\hat{i} + 0.8\hat{j} + c\hat{k}$, then the value of '$c$' is [CBSE PMT 1999; EAMCET 1994]

(a) 1 (b) $\sqrt{0.11}$

(c) $\sqrt{0.01}$ (d) $\sqrt{0.39}$

26. A boy walks uniformally along the sides of a rectangular park of size 400 m× 300 m, starting from one corner to the other corner diagonally opposite. Which of the following statement is incorrect [HP PMT 1999]

(a) He has travelled a distance of 700 m

(b) His displacement is 700 m

(c) His displacement is 500 m

(d) His velocity is not uniform throughout the walk

27. The unit vector parallel to the resultant of the vectors $\vec{A} = 4\hat{i} + 3\hat{j} + 6\hat{k}$ and $\vec{B} = -\hat{i} + 3\hat{j} - 8\hat{k}$ is [EAMCET 2000]

(a) $\frac{1}{7}(3\hat{i} + 6\hat{j} - 2\hat{k})$ (b) $\frac{1}{7}(3\hat{i} + 6\hat{j} + 2\hat{k})$

(c) $\frac{1}{49}(3\hat{i} + 6\hat{j} - 2\hat{k})$ (d) $\frac{1}{49}(3\hat{i} - 6\hat{j} + 2\hat{k})$

28. Surface area is [J&K CET 2002]

(a) Scalar (b) Vector

(c) Neither scalar nor vector (d) Both scalar and vector

29. With respect to a rectangular cartesian coordinate system, three vectors are expressed as

$\vec{a} = 4\hat{i} - \hat{j}$, $\vec{b} = -3\hat{i} + 2\hat{j}$ and $\vec{c} = -\hat{k}$

where $\hat{i}, \hat{j}, \hat{k}$ are unit vectors, along the X, Y and Z-axis respectively. The unit vectors $\hat{r}$ along the direction of sum of these vector is [Kerala CET (Engg.) 2003]

(a) $\hat{r} = \frac{1}{\sqrt{3}}(\hat{i} + \hat{j} - \hat{k})$ (b) $\hat{r} = \frac{1}{\sqrt{2}}(\hat{i} + \hat{j} - \hat{k})$

(c) $\hat{r} = \frac{1}{3}(\hat{i} - \hat{j} + \hat{k})$ (d) $\hat{r} = \frac{1}{\sqrt{2}}(\hat{i} + \hat{j} + \hat{k})$

30. The angle between the two vectors $\vec{A} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ and $\vec{B} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ is [DPMT 2000]

(a) 60° (b) Zero

(c) 90° (d) None of these

31. The position vector of a particle is determined by the expression $\vec{r} = 3t^2\hat{i} + 4t^2\hat{j} + 7\hat{k}$

The distance traversed in first 10 sec is [DPMT 2002]

(a) 500 m (b) 300 m

(c) 150 m (d) 100 m

32. Unit vector parallel to the resultant of vectors $\vec{A} = 4\hat{i} - 3\hat{j}$ and $\vec{B} = 8\hat{i} + 8\hat{j}$ will be [BHU 1995]

(a) $\frac{24\hat{i} + 5\hat{j}}{13}$ (b) $\frac{12\hat{i} + 5\hat{j}}{13}$

(c) $\frac{6\hat{i} + 5\hat{j}}{13}$ (d) None of these

33. The component of vector $A = 2\hat{i} + 3\hat{j}$ along the vector $\hat{i} + \hat{j}$ is [KCET 1997]

(a) $\frac{5}{\sqrt{2}}$ (b) $10\sqrt{2}$

(c) $5\sqrt{2}$ (d) 5

34. The angle between the two vectors $\vec{A} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ and $\vec{B} = 3\hat{i} + 4\hat{j} - 5\hat{k}$ will be [Pb. CET 2001]

(a) 90° (b) 0°

(c) 60° (d) 45°

Addition and Subtraction of Vectors

1. There are two force vectors, one of 5 N and other of 12 N at what angle the two vectors be added to get resultant vector of 17 N, 7 N and 13 N respectively

(a) 0°, 180° and 90° (b) 0°, 90° and 180°

(c) 0°, 90° and 90° (d) 180°, 0° and 90°

2. If $\vec{A} = 4\hat{i} - 3\hat{j}$ and $\vec{B} = 6\hat{i} + 8\hat{j}$ then magnitude and direction of $\vec{A} + \vec{B}$ will be

(a) $5, \tan^{-1}(3/4)$ (b) $5\sqrt{5}, \tan^{-1}(1/2)$

(c) $10, \tan^{-1}(5)$ (d) $25, \tan^{-1}(3/4)$

3. A truck travelling due north at 20 *m/s* turns west and travels at the same speed. The change in its velocity be [UPSEAT 1999]

(a) 40 *m/s N–W* (b) $20\sqrt{2}$ *m/s N–W*

(c) 40 *m/s S–W* (d) $20\sqrt{2}$ *m/s S–W*

4. If the sum of two unit vectors is a unit vector, then magnitude of difference is [CPMT 1995; CBSE PMT 1989]

(a) $\sqrt{2}$ (b) $\sqrt{3}$

(c) $1/\sqrt{2}$ (d) $\sqrt{5}$

5. $\vec{A} = 2\hat{i} + \hat{j}, B = 3\hat{j} - \hat{k}$ and $\vec{C} = 6\hat{i} - 2\hat{k}$.

Value of $\vec{A} - 2\vec{B} + 3\vec{C}$ would be

(a) $20\hat{i} + 5\hat{j} + 4\hat{k}$ (b) $20\hat{i} - 5\hat{j} - 4\hat{k}$

(c) $4\hat{i} + 5\hat{j} + 20\hat{k}$ (d) $5\hat{i} + 4\hat{j} + 10\hat{k}$

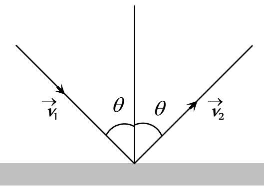
6. An object of *m kg* with speed of *v m/s* strikes a wall at an angle θ and rebounds at the same speed and same angle. The magnitude of the change in momentum of the object will be

(a) $2mv\cos\theta$

(b) $2mv\sin\theta$

(c) 0

(d) $2mv$



7. Two forces, each of magnitude [illegible] ne magnitude *F*. The angle between the two forces is [CBSE PMT 1990]

(a) 45° (b) 120°

(c) 150° (d) 60°

8. For the resultant of the two vectors to be maximum, what must be the angle between them

(a) 0° (b) 60°

(c) 90° (d) 180°

9. A particle is simultaneously acted by two forces equal to 4 *N* and 3 *N*. The net force on the particle is [CPMT 1979]

(a) 7 *N* (b) 5 *N*

(c) 1 *N* (d) Between 1 *N* and 7 *N*

10. Two vectors $\vec{A}$ and $\vec{B}$ lie in a plane, another vector $\vec{C}$ lies outside this plane, then the resultant of these three vectors *i.e.,* $\vec{A} + \vec{B} + \vec{C}$

(a) Can be zero

(b) Cannot be zero

(c) Lies in the plane containing $\vec{A} + \vec{B}$

(d) Lies in the plane containing $\vec{C}$

11. If the resultant of the two forces has a magnitude smaller than the magnitude of larger force, the two forces must be

(a) Different both in magnitude and direction

(b) Mutually perpendicular to one another

(c) Possess extremely small magnitude

(d) Point in opposite directions

12. Forces F_1 and F_2 act on a point mass in two mutually perpendicular directions. The resultant force on the point mass will be [CPMT 1991]

(a) $F_1 + F_2$ (b) $F_1 - F_2$

(c) $\sqrt{F_1^2 + F_2^2}$ (d) $F_1^2 + F_2^2$

13. If $|\vec{A} - \vec{B}| = |\vec{A}| = |\vec{B}|$, the angle between $\vec{A}$ and $\vec{B}$ is

(a) 60° (b) 0°

(c) 120° (d) 90°

14. Let the angle between two nonzero vectors $\vec{A}$ and $\vec{B}$ be 120° and resultant be $\vec{C}$

(a) $\vec{C}$ must be equal to $|\vec{A} - \vec{B}|$

(b) $\vec{C}$ must be less than $|\vec{A} - \vec{B}|$

(c) $\vec{C}$ must be greater than $|\vec{A} - \vec{B}|$

(d) $\vec{C}$ may be equal to $|\vec{A} - \vec{B}|$

15. The magnitude of vector $\vec{A}, \vec{B}$ and $\vec{C}$ are respectively 12, 5 and 13 units and $\vec{A} + \vec{B} = \vec{C}$ then the angle between $\vec{A}$ and $\vec{B}$ is

(a) 0 (b) π

(c) $\pi/2$ (d) $\pi/4$

16. Magnitude of vector which comes on addition of two vectors, $6\hat{i} + 7\hat{j}$ and $3\hat{i} + 4\hat{j}$ is [BHU 2000]

(a) $\sqrt{136}$ (b) $\sqrt{13.2}$

(c) $\sqrt{202}$ (d) $\sqrt{160}$

17. A particle has displacement of 12 *m* towards east and 5 *m* towards north then 6 *m* vertically upward. The sum of these displacements is

(a) 12 (b) 10.04 *m*

(c) 14.31 *m* (d) None of these

18. The three vectors $\vec{A} = 3\hat{i} - 2\hat{j} + \hat{k}$, $\vec{B} = \hat{i} - 3\hat{j} + 5\hat{k}$ and $\vec{C} = 2\hat{i} + \hat{j} - 4\hat{k}$ form

(a) An equilateral triangle (b) Isosceles triangle

(c) A right angled triangle (d) No triangle

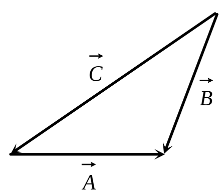
19. For the figure

(a) $\vec{A} + \vec{B} = \vec{C}$

(b) $\vec{B} + \vec{C} = \vec{A}$

(c) $\vec{C} + \vec{A} = \vec{B}$

(d) $\vec{A} + \vec{B} + \vec{C} = 0$ [CPMT 1983]



20. Let $\vec{C} = \vec{A} + \vec{B}$ then

(a) $|\vec{C}|$ is always greater then $|\vec{A}|$

(b) It is possible to have $|\vec{C}| < |\vec{A}|$ and $|\vec{C}| < |\vec{B}|$

(c) *C* is always equal to *A* + *B*

(d) *C* is never equal to *A* + *B*

21. The value of the sum of two vectors $\vec{A}$ and $\vec{B}$ with θ as the angle between them is [BHU 1996]

(a) $\sqrt{A^2 + B^2 + 2AB\cos\theta}$ (b) $\sqrt{A^2 - B^2 + 2AB\cos\theta}$

(c) $\sqrt{A^2+B^2-2AB\sin\theta}$ (d) $\sqrt{A^2+B^2+2AB\sin\theta}$

22. Following sets of three forces act on a body. Whose resultant cannot be zero [CPMT 1985]
(a) 10, 10, 10 (b) 10, 10, 20
(c) 10, 20, 23 (d) 10, 20, 40

23. When three forces of 50 *N*, 30 *N* and 15 *N* act on a body, then the body is
(a) At rest
(b) Moving with a uniform velocity
(c) In equilibrium
(d) Moving with an acceleration

24. The sum of two forces acting at a point is 16 *N*. If the resultant force is 8 *N* and its direction is perpendicular to minimum force then the forces are [CPMT 1997]
(a) 6 *N* and 10 *N* (b) 8 *N* and 8 *N*
(c) 4 *N* and 12 *N* (d) 2 *N* and 14 *N*

25. If vectors *P*, *Q* and *R* have magnitude 5, 12 and 13 units and $\vec{P}+\vec{Q}=\vec{R}$, the angle between *Q* and *R* is [CEET 1998]
(a) $\cos^{-1}\frac{5}{12}$ (b) $\cos^{-1}\frac{5}{13}$
(c) $\cos^{-1}\frac{12}{13}$ (d) $\cos^{-1}\frac{7}{13}$

26. The resultant of two vectors *A* and *B* is perpendicular to the vector *A* and its magnitude is equal to half the magnitude of vector *B*. The angle between *A* and *B* is
(a) 120° (b) 150°
(c) 135° (d) None of these

27. What vector must be added to the two vectors $\hat{i}-2\hat{j}+2\hat{k}$ and $2\hat{i}+\hat{j}-\hat{k}$, so that the resultant may be a unit vector along *x*-axis [BHU 1990]
(a) $2\hat{i}+\hat{j}-\hat{k}$ (b) $-2\hat{i}+\hat{j}-\hat{k}$
(c) $2\hat{i}-\hat{j}+\hat{k}$ (d) $-2\hat{i}-\hat{j}-\hat{k}$

28. What is the angle between $\vec{P}$ and the resultant of $(\vec{P}+\vec{Q})$ and $(\vec{P}-\vec{Q})$
(a) Zero (b) $\tan^{-1}(P/Q)$
(c) $\tan^{-1}(Q/P)$ (d) $\tan^{-1}(P-Q)/(P+Q)$

29. The resultant of $\vec{P}$ and $\vec{Q}$ is perpendicular to $\vec{P}$. What is the angle between $\vec{P}$ and $\vec{Q}$
(a) $\cos^{-1}(P/Q)$ (b) $\cos^{-1}(-P/Q)$
(c) $\sin^{-1}(P/Q)$ (d) $\sin^{-1}(-P/Q)$

30. Maximum and minimum magnitudes of the resultant of two vectors of magnitudes *P* and *Q* are in the ratio $3:1$. Which of the following relations is true
(a) $P=2Q$ (b) $P=Q$
(c) $PQ=1$ (d) None of these

31. The resultant of two vectors $\vec{P}$ and $\vec{Q}$ is $\vec{R}$. If *Q* is doubled, the new resultant is perpendicular to *P*. Then *R* equals
(a) *P* (b) (*P*+*Q*)
(c) *Q* (d) (*P*–*Q*)

32. Two forces, F_1 and F_2 are acting on a body. One force is double that of the other force and the resultant is equal to the greater force. Then the angle between the two forces is
(a) $\cos^{-1}(1/2)$ (b) $\cos^{-1}(-1/2)$
(c) $\cos^{-1}(-1/4)$ (d) $\cos^{-1}(1/4)$

33. Given that $\vec{A}+\vec{B}=\vec{C}$ and that $\vec{C}$ is $\perp$ to $\vec{A}$. Further if $|\vec{A}|=|\vec{C}|$, then what is the angle between $\vec{A}$ and $\vec{B}$
(a) $\frac{\pi}{4}$ *radian* (b) $\frac{\pi}{2}$ *radian*
(c) $\frac{3\pi}{4}$ *radian* (d) π *radian*

34. A body is at rest under the action of three forces, two of which are $\vec{F}_1=4\hat{i}, \vec{F}_2=6\hat{j}$, the third force is [AMU 1996]
(a) $4\hat{i}+6\hat{j}$ (b) $4\hat{i}-6\hat{j}$
(c) $-4\hat{i}+6\hat{j}$ (d) $-4\hat{i}-6\hat{j}$

35. A plane is revolving around the earth with a speed of 100 *km/hr* at a constant height from the surface of earth. The change in the velocity as it travels half circle is [RPET 1998; KCET 2000]
(a) 200 *km/hr* (b) 150 *km/hr*
(c) $100\sqrt{2}\ km/hr$ (d) 0

36. What displacement must be added to the displacement $25\hat{i}-6\hat{j}\ m$ to give a displacement of 7.0 *m* pointing in the *x*-direction
(a) $18\hat{i}-6\hat{j}$ (b) $32\hat{i}-13\hat{j}$
(c) $-18\hat{i}+6\hat{j}$ (d) $-25\hat{i}+13\hat{j}$

37. A body moves due East with velocity 20 *km/hour* and then due North with velocity 15 *km/hour*. The resultant velocity [AFMC 1995]
(a) 5 *km/hour* (b) 15 *km/hour*
(c) 20 *km/hour* (d) 25 *km/hour*

38. The magnitudes of vectors $\vec{A}, \vec{B}$ and $\vec{C}$ are 3, 4 and 5 units respectively. If $\vec{A}+\vec{B}=\vec{C}$, the angle between $\vec{A}$ and $\vec{B}$ is [CBSE PMT 1990]
(a) $\frac{\pi}{2}$ (b) $\cos^{-1}(0.6)$
(c) $\tan^{-1}\left(\frac{7}{5}\right)$ (d) $\frac{\pi}{4}$

39. While travelling from one station to another, a car travels 75 *km* North, 60 *km* North-east and 20 *km* East. The minimum distance between the two stations is [AFMC 1993]
(a) 72 *km* (b) 112 *km*
(c) 132 *km* (d) 155 *km*

40. A scooter going due east at 10 *ms* turns right through an angle of 90°. If the speed of the scooter remains unchanged in taking turn, the change is the velocity of the scooter is

[BHU 1994]

(a) 20.0 ms^{-1} south eastern direction

(b) Zero

(c) 10.0 ms^{-1} in southern direction

(d) 14.14 ms^{-1} in south-west direction

41. A person goes 10 *km* north and 20 *km* east. What will be displacement from initial point [AFMC 1994, 2003]

(a) 22.36 *km* (b) 2 *km*

(c) 5 *km* (d) 20 *km*

42. Two forces $\vec{F}_1 = 5\hat{i} + 10\hat{j} - 20\hat{k}$ and $\vec{F}_2 = 10\hat{i} - 5\hat{j} - 15\hat{k}$ act on a single point. The angle between $\vec{F}_1$ and $\vec{F}_2$ is nearly

[AMU 1995]

(a) 30° (b) 45°

(c) 60° (d) 90°

43. Which pair of the following forces will never give resultant force of 2 *N* [HP PMT 1999]

(a) 2 *N* and 2 *N* (b) 1 *N* and 1 *N*

(c) 1 *N* and 3 *N* (d) 1 *N* and 4 *N*

44. Two forces 3*N* and 2 *N* are at an angle θ such that the resultant is *R*. The first force is now increased to 6*N* and the resultant become 2*R*. The value of θ is [HP PMT 2000]

(a) 30° (b) 60°

(c) 90° (d) 120°

45. Three concurrent forces of the same magnitude are in equilibrium. What is the angle between the forces ? Also name the triangle formed by the forces as sides

[JIPMER 2000]

(a) 60° equilateral triangle

(b) 120° equilateral triangle

(c) 120°, 30°, 30° an isosceles triangle

(d) 120° an obtuse angled triangle

46. If $|\vec{A}+\vec{B}| = |\vec{A}| + |\vec{B}|$, then angle between $\vec{A}$ and $\vec{B}$ will be

[CBSE PMT 2001]

(a) 90° (b) 120°

(c) 0° (d) 60°

47. The maximum and minimum magnitude of the resultant of two given vectors are 17 *units* and 7 *unit* respectively. If these two vectors are at right angles to each other, the magnitude of their resultant is [Kerala CET (Engg.) 2000]

(a) 14 (b) 16

(c) 18 (d) 13

48. The vector sum of two forces is perpendicular to their vector differences. In that case, the forces [CBSE PMT 2003]

(a) Are equal to each other in magnitude

(b) Are not equal to each other in magnitude

(c) Cannot be predicted

(d) Are equal to each other

49. *y* component of velocity is 20 and *x* component of velocity is 10. The direction of motion of the body with the horizontal at this instant is

(a) $\tan^{-1}(2)$ (b) $\tan^{-1}(1/2)$

(c) 45° (d) 0°

50. Two forces of 12 *N* and 8 *N* act upon a body. The resultant force on the body has maximum value of [Manipal 2003]

(a) 4 *N* (b) 0 *N*

(c) 20 *N* (d) 8 *N*

51. Two equal forces (*P* each) act at a point inclined to each other at an angle of 120°. The magnitude of their resultant is

(a) $P/2$ (b) $P/4$

(c) P (d) 2*P*

52. The vectors $5i + 8j$ and $2i + 7j$ are added. The magnitude of the sum of these vector is [BHU 2000]

(a) $\sqrt{274}$ (b) 38

(c) 238 (d) 560

53. Two vectors $\vec{A}$ and $\vec{B}$ are such that $\vec{A}+\vec{B} = \vec{A}-\vec{B}$. Then

[AMU (Med.) 2000]

(a) $\vec{A}.\vec{B} = 0$ (b) $\vec{A}\times\vec{B} = 0$

(c) $\vec{A} = 0$ (d) $\vec{B} = 0$

Multiplication of Vectors

1. If a vector $2\hat{i} + 3\hat{j} + 8\hat{k}$ is perpendicular to the vector $4\hat{j} - 4\hat{i} + \alpha\hat{k}$. Then the value of α is [CBSE PMT 2005]

(a) –1 (b) $\frac{1}{2}$

(c) $-\frac{1}{2}$ (d) 1

2. If two vectors $2\hat{i} + 3\hat{j} - \hat{k}$ and $-4\hat{i} - 6\hat{j} - \lambda\hat{k}$ are parallel to each other then value of λ be

(a) 0 (b) 2

(c) 3 (d) 4

3. A body, acted upon by a force of 50 *N* is displaced through a distance 10 *meter* in a direction making an angle of 60° with the force. The work done by the force be

(a) 200 *J* (b) 100 *J*

(c) 300 (d) 250 *J*

4. A particle moves from position $3\hat{i} + 2\hat{j} - 6\hat{k}$ to $14\hat{i} + 13\hat{j} + 9\hat{k}$ due to a uniform force of $(4\hat{i} + \hat{j} + 3\hat{k})N$. If the displacement in meters then work done will be

[CMEET 1995; Pb. PMT 2002, 03]

(a) 100 *J* (b) 200 *J*

(c) 300 *J* (d) 250 *J*

5. If for two vector $\vec{A}$ and $\vec{B}$, sum $(\vec{A}+\vec{B})$ is perpendicular to the difference $(\vec{A}-\vec{B})$. The ratio of their magnitude is

[Manipal 2003]

(a) 1 (b) 2

(c) 3 (d) None of these

6. The angle between the vectors $\vec{A}$ and $\vec{B}$ is θ. The value of the triple product $\vec{A}.(\vec{B}\times\vec{A})$ is **[CBSE PMT 1991, 2005]**

(a) A^2B (b) Zero

(c) $A^2B\sin\theta$ (d) $A^2B\cos\theta$

7. If $\vec{A}\times\vec{B}=\vec{B}\times\vec{A}$ then the angle between *A* and *B* is **[AIEEE 2004]**

(a) $\pi/2$ (b) $\pi/3$

(c) π (d) $\pi/4$

8. If $\vec{A}=3\hat{i}+\hat{j}+2\hat{k}$ and $\vec{B}=2\hat{i}-2\hat{j}+4\hat{k}$ then value of $|\vec{A}\times\vec{B}|$ will be

(a) $8\sqrt{2}$ (b) $8\sqrt{3}$

(c) $8\sqrt{5}$ (d) $5\sqrt{8}$

9. The torque of the force $\vec{F}=(2\hat{i}-3\hat{j}+4\hat{k})N$ acting at the point $\vec{r}=(3\hat{i}+2\hat{j}+3\hat{k})\,m$ about the origin be **[CBSE PMT 1995]**

(a) $6\hat{i}-6\hat{j}+12\hat{k}$ (b) $17\hat{i}-6\hat{j}-13\hat{k}$

(c) $-6\hat{i}+6\hat{j}-12\hat{k}$ (d) $-17\hat{i}+6\hat{j}+13\hat{k}$

10. If $\vec{A}\times\vec{B}=\vec{C}$, then which of the following statements is wrong

(a) $\vec{C}\perp\vec{A}$ (b) $\vec{C}\perp\vec{B}$

(c) $\vec{C}\perp(\vec{A}+\vec{B})$ (d) $\vec{C}\perp(\vec{A}\times\vec{B})$

11. If a particle of mass *m* is moving with constant velocity *v* parallel to *x*-axis in *x-y* plane as shown in fig. Its angular momentum with respect to origin at any time *t* will be

(a) $mvb\,\hat{k}$ (b) $-mvb\,\hat{k}$

(c) $mvb\,\hat{i}$ (d) $mv\,\hat{i}$

12. Consider two vectors $\vec{F}_1=2\hat{i}+5\hat{k}$ and $\vec{F}_2=3\hat{j}+4\hat{k}$. The magnitude of the scalar product of these vectors is **[MP PMT 1987]**

(a) 20 (b) 23

(c) $5\sqrt{33}$ (d) 26

13. Consider a vector $\vec{F}=4\hat{i}-3\hat{j}$. Another vector that is perpendicular to $\vec{F}$ is

(a) $4\hat{i}+3\hat{j}$ (b) $6\hat{i}$

(c) $7\hat{k}$ (d) $3\hat{i}-4\hat{j}$

14. Two vectors $\vec{A}$ and $\vec{B}$ are at right angles to each other, when

(a) $\vec{A}+\vec{B}=0$ (b) $\vec{A}-\vec{B}=0$

(c) $\vec{A}\times\vec{B}=0$ (d) $\vec{A}.\vec{B}=0$

15. If $|\vec{V}_1+\vec{V}_2|=|\vec{V}_1-\vec{V}_2|$ and V_2 is finite, then **[CPMT 1989]**

(a) V_1 is parallel to V_2

(b) $\vec{V}_1=\vec{V}_2$

(c) V_1 and V_2 are mutually perpendicular

(d) $|\vec{V}_1|=|\vec{V}_2|$

16. A force $\vec{F}=(5\hat{i}+3\hat{j})$ Newton is applied over a particle which displaces it from its origin to the point $\vec{r}=(2\hat{i}-1\hat{j})$ *metres*. The work done on the particle is **[MP PMT 1995]**

(a) $-7\,J$ (b) $+13\,J$

(c) $+7\,J$ (d) $+11\,J$

17. The angle between two vectors $-2\hat{i}+3\hat{j}+\hat{k}$ and $\hat{i}+2\hat{j}-4\hat{k}$ is

(a) 0° (b) 90°

(c) 180° (d) None of the above

18. The angle between the vectors $(\hat{i}+\hat{j})$ and $(\hat{j}+\hat{k})$ is **[EAMCET 1995]**

(a) 30° (b) 45°

(c) 60° (d) 90°

19. A particle moves with a velocity $6\hat{i}-4\hat{j}+3\hat{k}\,m/s$ under the influence of a constant force $\vec{F}=20\hat{i}+15\hat{j}-5\hat{k}\,N$. The instantaneous power applied to the particle is **[CBSE PMT 2000]**

(a) 35 *J/s* (b) 45 *J/s*

(c) 25 *J/s* (d) 195 *J/s*

20. If $\vec{P}.\vec{Q}=PQ$, then angle between $\vec{P}$ and $\vec{Q}$ is **[AIIMS 1999]**

(a) 0° (b) 30°

(c) 45° (d) 60°

21. A force $\vec{F}=5\hat{i}+6\hat{j}+4\hat{k}$ acting on a body, produces a displacement $\vec{S}=6\hat{i}-5\hat{k}$. Work done by the force is **[KCET 1999]**

(a) 10 units (b) 18 units

(c) 11 units (d) 5 units

22. The angle between the two vectors $\vec{A}=5\hat{i}+5\hat{j}$ and $\vec{B}=5\hat{i}-5\hat{j}$ will be **[CPMT 2000]**

(a) Zero (b) 45°

(c) 90° (d) 180°

23. The vector $\vec{P}=a\hat{i}+a\hat{j}+3\hat{k}$ and $\vec{Q}=a\hat{i}-2\hat{j}-\hat{k}$ are perpendicular to each other. The positive value of *a* is **[AFMC 2000; AIIMS 2002]**

(a) 3 (b) 4

(c) 9 (d) 13

24. A body, constrained to move in the *Y*-direction is subjected to a force given by $\vec{F}=(-2\hat{i}+15\hat{j}+6\hat{k})N$. What is the work done by this force in moving the body a distance 10 *m* along the *Y*-axis **[AIIMS 1987]**

(a) 20 *J* (b) 150 *J*

(c) 160 *J* (d) 190 *J*

25. A particle moves in the x-y plane under the action of a force $\vec{F}$ such that the value of its linear momentum $(\vec{P})$ at anytime t is $P_x = 2\cos t, p_y = 2\sin t$. The angle θ between $\vec{F}$ and $\vec{P}$ at a given time t. will be [MNR 1991; UPSEAT 2000]

(a) $\theta = 0°$ (b) $\theta = 30°$
(c) $\theta = 90°$ (d) $\theta = 180°$

26. The area of the parallelogram represented by the vectors $\vec{A} = 2\hat{i} + 3\hat{j}$ and $\vec{B} = \hat{i} + 4\hat{j}$ is

(a) 14 units (b) 7.5 units
(c) 10 units (d) 5 units

27. A vector $\vec{F}_1$ is along the positive X-axis. If its vector product with another vector $\vec{F}_2$ is zero then $\vec{F}_2$ could be [MP PMT 1987]

(a) $4\hat{j}$ (b) $-(\hat{i} + \hat{j})$
(c) $(\hat{j} + \hat{k})$ (d) $(-4\hat{i})$

28. If for two vectors $\vec{A}$ and $\vec{B}$, $\vec{A} \times \vec{B} = 0$, the vectors

(a) Are perpendicular to each other
(b) Are parallel to each other
(c) Act at an angle of 60°
(d) Act at an angle of 30°

29. The angle between vectors $(\vec{A} \times \vec{B})$ and $(\vec{B} \times \vec{A})$ is

(a) Zero (b) π
(c) $\pi/4$ (d) $\pi/2$

30. What is the angle between $(\vec{P} + \vec{Q})$ and $(\vec{P} \times \vec{Q})$

(a) 0 (b) $\frac{\pi}{2}$
(c) $\frac{\pi}{4}$ (d) π

31. The resultant of the two vectors having magnitude 2 and 3 is 1. What is their cross product

(a) 6 (b) 3
(c) 1 (d) 0

32. Let $\vec{A} = \hat{i}A\cos\theta + \hat{j}A\sin\theta$ be any vector. Another vector $\vec{B}$ which is normal to A is [BHU 1997]

(a) $\hat{i}\,B\cos\theta + j\,B\sin\theta$ (b) $\hat{i}\,B\sin\theta + j\,B\cos\theta$
(c) $\hat{i}\,B\sin\theta - j\,B\cos\theta$ (d) $\hat{i}\,B\cos\theta - j\,B\sin\theta$

33. The angle between two vectors given by $6\bar{i} + 6\bar{j} - 3\bar{k}$ and $7\bar{i} + 4\bar{j} + 4\bar{k}$ is [EAMCET (Engg.) 1999]

(a) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (b) $\cos^{-1}\left(\frac{5}{\sqrt{3}}\right)$
(c) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (d) $\sin^{-1}\left(\frac{\sqrt{5}}{3}\right)$

34. A vector $\vec{A}$ points vertically upward and $\vec{B}$ points towards north. The vector product $\vec{A} \times \vec{B}$ is [UPSEAT 2000]

(a) Zero (b) Along west
(c) Along east (d) Vertically downward

35. Angle between the vectors $(\hat{i} + \hat{j})$ and $(\hat{j} - \hat{k})$ is

(a) 90° (b) 0°
(c) 180° (d) 60°

36. The position vectors of points A, B, C and D are $A = 3\hat{i} + 4\hat{j} + 5\hat{k}$, $B = 4\hat{i} + 5\hat{j} + 6\hat{k}$, $C = 7\hat{i} + 9\hat{j} + 3\hat{k}$ and $D = 4\hat{i} + 6\hat{j}$ then the displacement vectors AB and CD are

(a) Perpendicular
(b) Parallel
(c) Antiparallel
(d) Inclined at an angle of 60°

37. If force $(\vec{F}) = 4\hat{i} + 5\hat{j}$ and displacement $(\vec{s}) = 3\hat{i} + 6\hat{k}$ then the work done is [Manipal 1995]

(a) 4×3 (b) 5×6
(c) 6×3 (d) 4×6

38. If $|\vec{A} \times \vec{B}| = |\vec{A}.\vec{B}|$, then angle between $\vec{A}$ and $\vec{B}$ will be [AIIMS 2000; Manipal 2000]

(a) 30° (b) 45°
(c) 60° (d) 90°

39. In an clockwise system [CPMT 1990]

(a) $\hat{j} \times \hat{k} = \hat{i}$ (b) $\hat{i}.\hat{i} = 0$
(c) $\hat{j} \times \hat{j} = 1$ (d) $\hat{k}.\hat{j} = 1$

40. The linear velocity of a rotating body is given by $\vec{v} = \vec{\omega} \times \vec{r}$, where $\vec{\omega}$ is the angular velocity and $\vec{r}$ is the radius vector. The angular velocity of a body is $\vec{\omega} = \hat{i} - 2\hat{j} + 2\hat{k}$ and the radius vector $\vec{r} = 4\hat{j} - 3\hat{k}$, then $|\vec{v}|$ is

(a) $\sqrt{29}$ units (b) $\sqrt{31}$ units
(c) $\sqrt{37}$ units (d) $\sqrt{41}$ units

41. Three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ satisfy the relation $\vec{a}.\vec{b} = 0$ and $\vec{a}.\vec{c} = 0$. The vector $\vec{a}$ is parallel to [AIIMS 1996]

(a) $\vec{b}$ (b) $\vec{c}$
(c) $\vec{b}.\vec{c}$ (d) $\vec{b} \times \vec{c}$

42. The diagonals of a parallelogram are $2\hat{i}$ and $2\hat{j}$. What is the area of the parallelogram

(a) 0.5 units (b) 1 unit
(c) 2 units (d) 4 units

43. What is the unit vector perpendicular to the following vectors $2\hat{i} + 2\hat{j} - \hat{k}$ and $6\hat{i} - 3\hat{j} + 2\hat{k}$

(a) $\dfrac{\hat{i}+10\hat{j}-18\hat{k}}{5\sqrt{17}}$ (b) $\dfrac{\hat{i}-10\hat{j}+18\hat{k}}{5\sqrt{17}}$

(c) $\dfrac{\hat{i}-10\hat{j}-18\hat{k}}{5\sqrt{17}}$ (d) $\dfrac{\hat{i}+10\hat{j}+18\hat{k}}{5\sqrt{17}}$

44. The area of the parallelogram whose sides are represented by the vectors $\hat{j}+3\hat{k}$ and $\hat{i}+2\hat{j}-\hat{k}$ is

(a) $\sqrt{61}$ sq.unit (b) $\sqrt{59}$ sq.unit

(c) $\sqrt{49}$ sq.unit (d) $\sqrt{52}$ sq.unit

45. The position of a particle is given by $\vec{r}=(\vec{i}+2\vec{j}-\vec{k})$ momentum $\vec{P}=(3\vec{i}+4\vec{j}-2\vec{k})$. The angular momentum is perpendicular to [EAMCET (Engg.) 1998]

(a) *x*-axis

(b) *y*-axis

(c) *z*-axis

(d) Line at equal angles to all the three axes

46. Two vector *A* and *B* have equal magnitudes. Then the vector *A* + *B* is perpendicular to

(a) $A\times B$ (b) $A-B$

(c) $3A-3B$ (d) All of these

47. Find the torque of a force $\vec{F}=-3\hat{i}+\hat{j}+5\hat{k}$ acting at the point $\vec{r}=7\hat{i}+3\hat{j}+\hat{k}$

[CPMT 1997; CBSE PMT 1997; CET 1998; DPMT 2004]

(a) $14\hat{i}-38\hat{j}+16\hat{k}$ (b) $4\hat{i}+4\hat{j}+6\hat{k}$

(c) $21\hat{i}+4\hat{j}+4\hat{k}$ (d) $-14\hat{i}+34\hat{j}-16\hat{k}$

48. The value of $(\vec{A}+\vec{B})\times(\vec{A}-\vec{B})$ is

[RPET 1991, 2002; BHU 2002]

(a) 0 (b) A^2-B^2

(c) $\vec{B}\times\vec{A}$ (d) $2(\vec{B}\times\vec{A})$

49. If $\vec{A}$ and $\vec{B}$ are perpendicular vectors and vector $\vec{A}=5\hat{i}+7\hat{j}-3\hat{k}$ and $\vec{B}=2\hat{i}+2\hat{j}-a\hat{k}$. The value of *a* is

[EAMCET 1991]

(a) – 2 (b) 8

(c) – 7 (d) – 8

50. A force vector applied on a mass is represented as $\vec{F}=6\hat{i}-8\hat{j}+10\hat{k}$ and accelerates with $1\ m/s^2$. What will be the mass of the body in kg.

[CMEET 1995]

(a) $10\sqrt{2}$ (b) 20

(c) $2\sqrt{10}$ (d) 10

51. Two adjacent sides of a parallelogram are represented by the two vectors $\hat{i}+2\hat{j}+3\hat{k}$ and $3\hat{i}-2\hat{j}+\hat{k}$. What is the area of parallelogram

[AMU 1997]

(a) 8 (b) $8\sqrt{3}$

(c) $3\sqrt{8}$ (d) 192

52. The position vectors of radius are $2\hat{i}+\hat{j}+\hat{k}$ and $2\hat{i}-3\hat{j}+\hat{k}$ while those of linear momentum are $2\hat{i}+3\hat{j}-\hat{k}$. Then the angular momentum is [BHU 1997]

(a) $2\hat{i}-4\hat{k}$ (b) $4\hat{i}-8\hat{k}$

(c) $2\hat{i}-4\hat{j}+2\hat{k}$ (d) $4\hat{i}-8\hat{k}$

53. What is the value of linear velocity, if $\vec{\omega}=3\hat{i}-4\hat{j}+\hat{k}$ and $\vec{r}=5\hat{i}-6\hat{j}+6\hat{k}$ [CBSE PMT 1999; CPMT 1999, 2001; Pb. PMT 2000; Pb. CET 2000]

(a) $6\hat{i}-2\hat{j}+3\hat{k}$ (b) $6\hat{i}-2\hat{j}+8\hat{k}$

(c) $4\hat{i}-13\hat{j}+6\hat{k}$ (d) $-18\hat{i}-13\hat{j}+2\hat{k}$

54. Dot product of two mutual perpendicular vector is

[Haryana CEET 2002]

(a) 0 (b) 1

(c) ∞ (d) None of these

55. When $\vec{A}.\vec{B}=-|A||B|$, then [Orissa JEE 2003]

(a) $\vec{A}$ and $\vec{B}$ are perpendicular to each other

(b) $\vec{A}$ and $\vec{B}$ act in the same direction

(c) $\vec{A}$ and $\vec{B}$ act in the opposite direction

(d) $\vec{A}$ and $\vec{B}$ can act in any direction

56. If $|\vec{A}\times\vec{B}|=\sqrt{3}\vec{A}.\vec{B}$, then the value of $|\vec{A}+\vec{B}|$ is

[CBSE PMT 2004]

(a) $\left(A^2+B^2+\dfrac{AB}{\sqrt{3}}\right)^{1/2}$ (b) $A+B$

(c) $(A^2+B^2+\sqrt{3}AB)^{1/2}$ (d) $(A^2+B^2+AB)^{1/2}$

57. A force $\vec{F}=3\hat{i}+c\hat{j}+2\hat{k}$ acting on a particle causes a displacement $\vec{S}=-4\hat{i}+2\hat{j}-3\hat{k}$ in its own direction. If the work done is 6*J*, then the value of *c* will be [DPMT 1997]

(a) 12 (b) 6

(c) 1 (d) 0

58. A force $\vec{F}=(5\hat{i}+3\hat{j})\,N$ is applied over a particle which displaces it from its original position to the point $\vec{s}=(2\hat{i}-1\hat{j})\,m$. The work done on the particle is [BHU 2001]

(a) + 11 *J* (b) + 7 *J*

(c) + 13 *J* (d) – 7 *J*

59. If a vector $\vec{A}$ is parallel to another vector $\vec{B}$ then the resultant of the vector $\vec{A} \times \vec{B}$ will be equal to

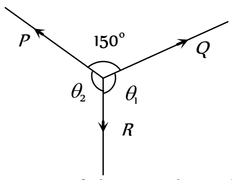
[Pb. CET 1996]

(a) A (b) $\vec{A}$

(c) Zero vector (d) Zero

Lami's Theorem

1. P, Q and R are three coplanar forces acting at a point and are in equilibrium. Given P = 1.9318 *kg wt*, $\sin\theta_1 = 0.9659$, the value of R is (in *kg wt*) [CET 1998]



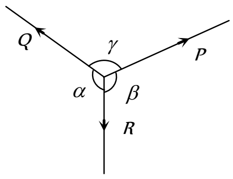
(a) 0.9659

(b) 2

(c) 1

(d) $\frac{1}{2}$

2. A body is in equilibrium under the action of three coplanar forces P, Q and R as shown in the figure. Select the correct statement



(a) $\frac{P}{\sin\alpha} = \frac{Q}{\sin\beta} = \frac{R}{\sin\gamma}$

(b) $\frac{P}{\cos\alpha} = \frac{Q}{\cos\beta} = \frac{R}{\cos\gamma}$

(c) $\frac{P}{\tan\alpha} = \frac{Q}{\tan\beta} = \frac{R}{\tan\gamma}$

(d) $\frac{P}{\sin\beta} = \frac{Q}{\sin\gamma} = \frac{R}{\sin\alpha}$

3. If a body is in equilibrium under a set of non-collinear forces, then the minimum number of forces has to be

[AIIMS 2000]

(a) Four (b) Three

(c) Two (d) Five

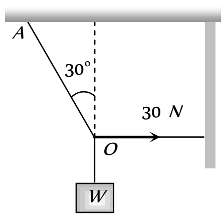
4. How many minimum number of non-zero vectors in different planes can be added to give zero resultant

(a) 2 (b) 3

(c) 4 (d) 5

5. As shown in figure the tension in the horizontal cord is 30 N. The weight W and tension in the string OA in Newton are

[DPMT 1992]



(a) $30\sqrt{3}$, 30

(b) $30\sqrt{3}$, 60

(c) $60\sqrt{3}$, 30

(d) None of these

Relative Velocity

1. Two cars are moving in the same direction with the same speed 30 *km/hr*. They are separated by a distance of 5 *km*, the speed of a car moving in the opposite direction if it meets these two cars at an interval of 4 minutes, will be

(a) 40 *km/hr* (b) 45 *km/hr*

(c) 30 *km/hr* (d) 15 *km/hr*

2. A man standing on a road hold his umbrella at 30° with the vertical to keep the rain away. He throws the umbrella and starts running at 10 *km/hr*. He finds that raindrops are hitting his head vertically, the speed of raindrops with respect to the road will be

(a) 10 *km/hr* (b) 20 *km/hr*

(c) 30 *km/hr* (d) 40 *km/hr*

3. In the above problem, the speed of raindrops *w.r.t.* the moving man, will be

(a) $10/\sqrt{2}\ km/h$ (b) 5 *km/h*

(c) $10\sqrt{3}\ km/h$ (d) $5/\sqrt{3}\ km/h$

4. A boat is moving with a velocity $3i + 4j$ with respect to ground. The water in the river is moving with a velocity $-3i - 4j$ with respect to ground. The relative velocity of the boat with respect to water is

(a) $8j$ (b) $-6i - 8j$

(c) $6i + 8j$ (d) $5\sqrt{2}$

5. A 150 m long train is moving to north at a speed of 10 m/s. A parrot flying towards south with a speed of 5 m/s crosses the train. The time taken by the parrot the cross to train would be: [AFMC 1994]

(a) 30 s (b) 15 s

(c) 8 s (d) 10 s

6. A river is flowing from east to west at a speed of 5 m/min. A man on south bank of river, capable of swimming 10m/min in still water, wants to swim across the river in shortest time. He should swim

(a) Due north

(b) Due north-east

(c) Due north-east with double the speed of river

(d) None of these

7. A person aiming to reach the exactly opposite point on the bank of a stream is swimming with a speed of 0.5 *m/s* at an angle of 120° with the direction of flow of water. The speed of water in the stream is [CBSE PMT 1999]

(a) 1 *m/s* (b) 0.5 *m/s*

(c) 0.25 *m/s* (d) 0.433 *m/s*

8. A moves with 65 *km/h* while B is coming back of A with 80 *km/h*. The relative velocity of B with respect to A is

[AFMC 2000]

(a) 80 *km/h* (b) 60 *km/h*

(c) 15 *km/h* (d) 145 *km/h*

9. A thief is running away on a straight road on a jeep moving with a speed of 9 *m/s*. A police man chases him on a motor cycle moving at a speed of 10 *m/s*. If the instantaneous separation of jeep from the motor cycle is 100 *m*, how long will it take for the policemen to catch the thief

(a) 1 second (b) 19 second

(c) 90 second (d) 100 second

10. A man can swim with velocity v relative to water. He has to cross a river of width d flowing with a velocity u ($u > v$). The distance through which he is carried down stream by the river is x. Which of the following statement is correct

(a) If he crosses the river in minimum time $x = \frac{du}{v}$

(b) x can not be less than $\frac{du}{v}$

(c) For x to be minimum he has to swim in a direction making an angle of $\frac{\pi}{2}+\sin^{-1}\left(\frac{v}{u}\right)$ with the direction of the flow of water

(d) x will be max. if he swims in a direction making an angle of $\frac{\pi}{2}+\sin^{-1}\frac{v}{u}$ with direction of the flow of water

11. A man sitting in a bus travelling in a direction from west to east with a speed of 40 *km/h* observes that the rain-drops are falling vertically down. To the another man standing on ground the rain will appear [HP PMT 1999]

(a) To fall vertically down

(b) To fall at an angle going from west to east

(c) To fall at an angle going from east to west

(d) The information given is insufficient to decide the direction of rain.

12. A boat takes two hours to travel 8 *km* and back in still water. If the velocity of water is 4 *km/h*, the time taken for going upstream 8 *km* and coming back is [EAMCET 1990]

(a) 2*h*

(b) 2*h* 40 *min*

(c) 1*h* 20 *min*

(d) Cannot be estimated with the information given

13. A 120 *m* long train is moving towards west with a speed of 10 *m/s*. A bird flying towards east with a speed of 5 *m/s* crosses the train. The time taken by the bird to cross the train will be

(a) 16 *sec* (b) 12 *sec*

(c) 10 *sec* (d) 8 *sec*

14. A boat crosses a river with a velocity of 8 *km/h*. If the resulting velocity of boat is 10 *km/h* then the velocity of river water is

(a) 4 *km/h* (b) 6 *km/h*

(c) 8 *km/h* (d) 10 *km/h*

CT Critical Thinking

Objective Questions

1. If a vector $\vec{P}$ making angles α, β, and γ respectively with the X, Y and Z axes respectively.

Then $\sin^2\alpha+\sin^2\beta+\sin^2\gamma=$

(a) 0 (b) 1

(c) 2 (d) 3

2. If the resultant of n forces of different magnitudes acting at a point is zero, then the minimum value of n is [SCRA 2000]

(a) 1 (b) 2

(c) 3 (d) 4

3. Can the resultant of 2 vectors be zero [IIIT 2000]

(a) Yes, when the 2 vectors are same in magnitude and direction

(b) No

(c) Yes, when the 2 vectors are same in magnitude but opposite in sense

(d) Yes, when the 2 vectors are same in magnitude making an angle of $\frac{2\pi}{3}$ with each other

4. The sum of the magnitudes of two forces acting at point is 18 and the magnitude of their resultant is 12. If the resultant is at 90° with the force of smaller magnitude, what are the, magnitudes of forces [Roorkee 199

(a) 12, 5 (b) 14, 4

(c) 5, 13 (d) 10, 8

5. A vector $\vec{a}$ is turned without a change in its length through a small angle $d\theta$. The value of $|\Delta\vec{a}|$ and Δa are respectively

(a) $0, a\,d\theta$ (b) $a\,d\theta, 0$

(c) 0, 0 (d) None of these

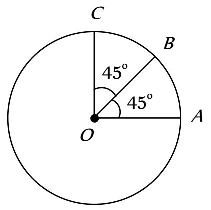
6. Find the resultant of three vectors $\overrightarrow{OA}, \overrightarrow{OB}$ and $\overrightarrow{OC}$ shown in the following figure. Radius of the circle is R.

(a) $2R$

(b) $R(1+\sqrt{2})$

(c) $R\sqrt{2}$ [Manipal 2002]

(d) $R(\sqrt{2}-1)$



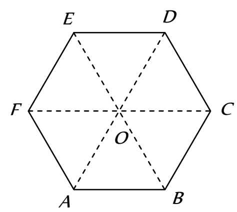
7. Figure shows $ABCDEF$ as a regular hexagon. What is the value of $\overrightarrow{AB}+\overrightarrow{AC}+\overrightarrow{AD}+\overrightarrow{AE}+\overrightarrow{AF}$ [CPMT 2001]

(a) $\overrightarrow{AO}$

(b) $2\overrightarrow{AO}$

(c) $4\overrightarrow{AO}$

(d) $6\overrightarrow{AO}$



8. The length of second's hand in watch is 1 *cm*. The change in velocity of its tip in 15 seconds is [MP PMT 1987]

(a) Zero (b) $\frac{\pi}{30\sqrt{2}}\ cm/\sec$

(c) $\frac{\pi}{30}\ cm/\sec$ (d) $\frac{\pi\sqrt{2}}{30}\ cm/\sec$

9. A particle moves towards east with velocity 5 *m/s*. After 10 seconds its direction changes towards north with same velocity. The average acceleration of the particle is

[CPMT 1997; IIT-JEE 1982]

(a) Zero (b) $\frac{1}{\sqrt{2}} m/s^2\ N-W$

(c) $\frac{1}{\sqrt{2}} m/s^2\ N-E$ (d) $\frac{1}{\sqrt{2}} m/s^2\ S-W$

10. A force $\vec{F} = -K(y\hat{i} + x\hat{j})$ (where K is a positive constant) acts on a particle moving in the x-y plane. Starting from the origin, the particle is taken along the positive x- axis to the point (a, 0) and then parallel to the y-axis to the point (a, a). The total work done by the forces $\vec{F}$ on the particle is

[IIT-JEE 1998]

(a) $-2Ka^2$ (b) $2Ka^2$

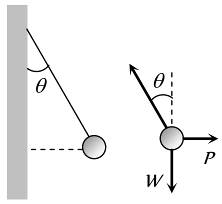
(c) $-Ka^2$ (d) Ka^2

11. The vectors from origin to the points A and B are $\vec{A} = 3\hat{i} - 6\hat{j} + 2\hat{k}$ and $\vec{B} = 2\hat{i} + \hat{j} - 2\hat{k}$ respectively. The area of the triangle OAB be

(a) $\frac{5}{2}\sqrt{17}$ *sq.unit* (b) $\frac{2}{5}\sqrt{17}$ *sq.unit*

(c) $\frac{3}{5}\sqrt{17}$ *sq.unit* (d) $\frac{5}{3}\sqrt{17}$ *sq.unit*

12. A metal sphere is hung by a string fixed to a wall. The sphere is pushed away from the wall by a stick. The forces acting on the sphere are shown in the second diagram. Which of the following statements is wrong



(a) $P = W\tan\theta$

(b) $\vec{T} + \vec{P} + \vec{W} = 0$

(c) $T^2 = P^2 + W^2$

(d) $T = P + W$

13. The speed of a boat is 5 km/h in still water. It crosses a river of width 1 km along the shortest possible path in 15 minutes. The velocity of the river water is

[IIT 1988; CBSE PMT 1998, 2000]

(a) 1 km/h (b) 3 km/h

(c) 4 km/h (d) 5 km/h

14. A man crosses a 320 m wide river perpendicular to the current in 4 minutes. If in still water he can swim with a speed 5/3 times that of the current, then the speed of the current, in m/min is

[Roorkee 1998]

(a) 30 (b) 40

(c) 50 (d) 60.

Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

(a) *If both assertion and reason are true and the reason is the correct explanation of the assertion.*
(b) *If both assertion and reason are true but reason is not the correct explanation of the assertion.*
(c) *If assertion is true but reason is false.*
(d) *If the assertion and reason both are false.*
(e) *If assertion is false but reason is true.*

1. Assertion : $\vec{A} \times \vec{B}$ is perpendicular to both $\vec{A} + \vec{B}$ as well as $\vec{A} - \vec{B}$.
 Reason : $\vec{A} + \vec{B}$ as well as $\vec{A} - \vec{B}$ lie in the plane containing $\vec{A}$ and $\vec{B}$, but $\vec{A} \times \vec{B}$ lies perpendicular to the plane containing $\vec{A}$ and $\vec{B}$.

2. Assertion : Angle between $\hat{i} + \hat{j}$ and $\hat{i}$ is 45°
 Reason : $\hat{i} + \hat{j}$ is equally inclined to both $\hat{i}$ and $\hat{j}$ and the angle between $\hat{i}$ and $\hat{j}$ is 90°

3. Assertion : If θ be the angle between $\vec{A}$ and $\vec{B}$, then $\tan\theta = \frac{\vec{A} \times \vec{B}}{\vec{A}.\vec{B}}$
 Reason : $\vec{A} \times \vec{B}$ is perpendicular to $\vec{A}.\vec{B}$

4. Assertion : If $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$, then angle between $\vec{A}$ and $\vec{B}$ is 90°
 Reason : $\vec{A} + \vec{B} = \vec{B} + \vec{A}$

5. Assertion : Vector product of two vectors is an axial vector
 Reason : If $\vec{v}$ = instantaneous velocity, $\vec{r}$ = radius vector and $\vec{\omega}$ = angular velocity, then $\vec{\omega} = \vec{v} \times \vec{r}$.

6. Assertion : Minimum number of non-equal vectors in a plane required to give zero resultant is three.
 Reason : If $\vec{A} + \vec{B} + \vec{C} = \vec{0}$, then they must lie in one plane

7. Assertion : Relative velocity of A *w.r.t.* B is greater than the velocity of either, when they are moving in opposite directions.
 Reason : Relative velocity of A *w.r.t.* $B = \vec{v}_A - \vec{v}_B$

8. Assertion : Vector addition of two vectors $\vec{A}$ and $\vec{B}$ is commutative.
 Reason : $\vec{A} + \vec{B} = \vec{B} + \vec{A}$

9. Assertion : $\vec{A}.\vec{B} = \vec{B}.\vec{A}$
 Reason : Dot product of two vectors is commutative.

10. Assertion : $\vec{\tau} = \vec{r} \times \vec{F}$ and $\vec{\tau} \neq \vec{F} \times \vec{r}$
 Reason : Cross product of vectors is commutative.

11. Assertion : A negative acceleration of a body is associated with a slowing down of a body.
 Reason : Acceleration is vector quantity.

12. Assertion : A physical quantity cannot be called as a vector if its magnitude is zero.
 Reason : A vector has both, magnitude and direction.

13. Assertion : The sum of two vectors can be zero.
 Reason : The vector cancel each other, when they are equal and opposite.

14. Assertion : Two vectors are said to be like vectors if they have same direction but different magnitude.
 Reason : Vector quantities do not have specific direction.

15. Assertion : The scalar product of two vectors can be zero.
 Reason : If two vectors are perpendicular to each other, their scalar product will be zero.

16. Assertion : Multiplying any vector by an scalar is a meaningful operations.
 Reason : In uniform motion speed remains constant.

17. Assertion : A null vector is a vector whose magnitude is zero and direction is arbitrary.

Reason : A null vector does not exist.

18. Assertion : If dot product and cross product of $\vec{A}$ and $\vec{B}$ are zero, it implies that one of the vector $\vec{A}$ and $\vec{B}$ must be a null vector.

Reason : Null vector is a vector with zero magnitude.

19. Assertion : The cross product of a vector with itself is a null vector.

Reason : The cross-product of two vectors results in a vector quantity.

20. Assertion : The minimum number of non coplanar vectors whose sum can be zero, is four.

Reason : The resultant of two vectors of unequal magnitude can be zero.

21. Assertion : If $\vec{A}.\vec{B} = \vec{B}.\vec{C}$, then $\vec{A}$ may not always be equal to $\vec{C}$

Reason : The dot product of two vectors involves cosine of the angle between the two vectors.

22. Assertion : Vector addition is commutative.

Reason : $(\vec{A}+\vec{B}) \neq (\vec{B}+\vec{A})$.

Answers

Fundamentals of Vectors

1	d	2	b	3	c	4	d	5	d
6	a	7	a	8	b	9	b	10	d
11	d	12	d	13	a	14	b	15	c
16	c	17	a	18	b	19	c	20	c
21	d	22	d	23	b	24	d	25	b
26	b	27	a	28	a	29	a	30	d
31	a	32	b	33	a	34	a		

Addition and Subtraction of Vectors

1	a	2	b	3	d	4	b	5	b
6	a	7	b	8	a	9	d	10	b
11	d	12	c	13	a	14	c	15	c
16	c	17	c	18	c	19	c	20	b
21	a	22	d	23	d	24	a	25	c
26	b	27	b	28	a	29	b	30	a
31	c	32	c	33	c	34	d	35	a
36	c	37	d	38	a	39	c	40	d
41	a	42	b	43	d	44	d	45	a
46	c	47	d	48	a	49	a	50	c
51	c	52	a	53	d				

Multiplication of Vectors

1	c	2	b	3	d	4	a	5	a
6	b	7	c	8	b	9	b	10	d
11	b	12	d	13	c	14	d	15	c
16	c	17	b	18	c	19	b	20	a
21	a	22	c	23	a	24	b	25	c
26	d	27	d	28	b	29	b	30	b
31	d	32	c	33	d	34	b	35	d
36	b	37	a	38	b	39	a	40	a
41	d	42	d	43	c	44	b	45	a
46	a	47	a	48	d	49	d	50	a
51	b	52	b	53	d	54	a	55	c
56	d	57	a	58	b	59	c		

Lami's Theorem

1	c	2	a	3	b	4	c	5	b

Relative Velocity

1	b	2	b	3	c	4	c	5	d
6	a	7	c	8	c	9	d	10	ac
11	b	12	b	13	d	14	b		

Critical Thinking Questions

1	c	2	c	3	c	4	c	5	b
6	b	7	d	8	d	9	b	10	c
11	a	12	d	13	b	14	d		

Assertion and Reason

1	a	2	a	3	d	4	b	5	c
6	b	7	a	8	b	9	a	10	c
11	b	12	e	13	a	14	c	15	a
16	b	17	c	18	b	19	b	20	c
21	a	22	c						

Answers and Solutions

Fundamentals of Vectors

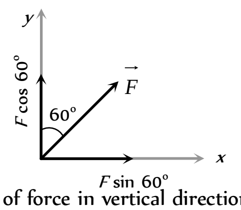
1. (d) As the multiple of $\hat{j}$ in the given vector is zero therefore this vector lies in XZ plane and projection of this vector on y-axis is zero.

2. (b) If a point have coordinate (x, y, z) then its position vector $= x\hat{i} + y\hat{j} + z\hat{k}$.

3. (c) Displacement vector $\vec{r} = \Delta x\hat{i} + \Delta y\hat{j} + \Delta z\hat{k}$

$= (3-2)\hat{i} + (4-3)\hat{j} + (5-5)\hat{k} = \hat{i} + \hat{j}$

4. (d)



The component of force in vertical direction

$= F\cos\theta = F\cos 60° = 5 \times \frac{1}{2} = 2.5\ N$

5. (d) $|B| = \sqrt{7^2 + (24)^2} = \sqrt{625} = 25$

Unit vector in the direction of A will be $\hat{A} = \frac{3\hat{i} + 4\hat{j}}{5}$

So required vector $= 25\left(\frac{3\hat{i} + 4\hat{j}}{5}\right) = 15\hat{i} + 20\hat{j}$

6. (a) Let the components of $\vec{A}$ makes angles α, β and γ with x, y and z axis respectively then $\alpha = \beta = \gamma$

$\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$

$\Rightarrow\ 3\cos^2\alpha=1\ \Rightarrow\ \cos\alpha=\dfrac{1}{\sqrt3}$

$\therefore\ A_x=A_y=A_z=A\cos\alpha=\dfrac{A}{\sqrt3}$

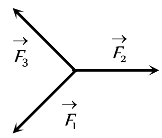
7. (a) $\vec A=2\hat i+4\hat j-5\hat k\ \therefore\ |\vec A|=\sqrt{(2)^2+(4)^2+(-5)^2}=\sqrt{45}$

$\therefore\ \cos\alpha=\dfrac{2}{\sqrt{45}},\quad \cos\beta=\dfrac{4}{\sqrt{45}},\quad \cos\gamma=\dfrac{-5}{\sqrt{45}}$

8. (b) Unit vector along y axis $=\hat j$ so the required vector $=\hat j-[(\hat i-3\hat j+2\hat k)+(3\hat i+6\hat j-7\hat k)]=-4\hat i-2\hat j+5\hat k$

9. (b) $\vec F_3=\vec F_1+\vec F_2$

There should be minimum three coplaner vectors having different magnitude which should be added to give zero resultant

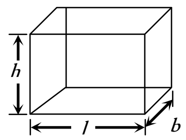


10. (d) Diagonal of the hall = $\sqrt{l^2+b^2+h^2}$

$=\sqrt{10^2+12^2+14^2}$

$=\sqrt{100+144+196}$

$=\sqrt{400}=20m$



11. (d) Total angle = $100\times\dfrac{\pi}{50}=2\pi$

So all the force will pass through one point and all forces will be balanced. *i.e.* their resultant will be zero.

12. (d) $\vec r=\vec r_2-\vec r_1=(-2\hat i-2\hat j+0\hat k)-(4\hat i-4\hat j+0\hat k)$

$\Rightarrow\ \vec r=-6\hat i+2\hat j+0\hat k$

$\therefore\ |\vec r|=\sqrt{(-6)^2+(2)^2+0^2}=\sqrt{36+4}=\sqrt{40}=2\sqrt{10}$

13. (a) $\vec P=\dfrac{1}{\sqrt2}\hat i+\dfrac{1}{\sqrt2}\hat j\ \therefore |\vec P|=\sqrt{\left(\dfrac{1}{\sqrt2}\right)^2+\left(\dfrac{1}{\sqrt2}\right)^2}=1$

$\therefore$ It is a unit vector.

14. (b)

15. (c) $\hat R=\dfrac{\vec R}{|\vec R|}=\dfrac{\hat i+\hat j}{\sqrt{1^2+1^2}}=\dfrac{1}{\sqrt2}\hat i+\dfrac{1}{\sqrt2}\hat j$

16. (c) $\vec R=3\hat i+\hat j+2\hat k$

$\therefore$ Length in XY plane = $\sqrt{R_x^2+R_y^2}=\sqrt{3^2+1^2}=\sqrt{10}$

17. (a) If the angle between all forces which are equal and lying in one plane are equal then resultant force will be zero.

18. (b) $\vec A=\hat i+\hat j\Rightarrow|\vec A|=\sqrt{1^2+1^2}=\sqrt2$

$\cos\alpha=\dfrac{A_x}{|\vec A|}=\dfrac{1}{\sqrt2}=\cos45^\circ\ \therefore\ \alpha=45^\circ$

19. (c)

20. (c)

21. (d) All quantities are tensors.

22. (d) $\vec P+\vec Q=P\hat P+Q\hat Q$

23. (b) $\vec r=(a\cos\omega t)\hat i+(a\sin\omega t)\hat j$

$\vec v=\dfrac{d\vec r}{dt}=-a\omega\sin\omega t\,\hat i+a\omega\cos\omega t\,\hat j$

As $\vec r.\vec v=0$ therefore velocity of the particle is perpendicular to the position vector.

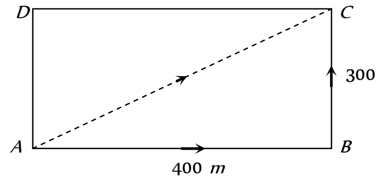
24. (d) Displacement, electrical and acceleration are vector quantities.

25. (b) Magnitude of unit vector = 1

$\Rightarrow\ \sqrt{(0.5)^2+(0.8)^2+c^2}=1$

By solving we get $c=\sqrt{0.11}$

26. (b)



Displacement $\overrightarrow{AC}=\overrightarrow{AB}+\overrightarrow{BC}$

$AC=\sqrt{(AB)^2+(BC)^2}=\sqrt{(400)^2+(300)^2}=500m$

Distance $=AB+BC=400+300=700m$

27. (a) Resultant of vectors $\vec A$ and $\vec B$

$\vec R=\vec A+\vec B=4\hat i+3\hat j+6\hat k-\hat i+3\hat j-8\hat k$

$\vec R=3\hat i+6\hat j-2\hat k$

$\hat R=\dfrac{\vec R}{|\vec R|}=\dfrac{3\hat i+6\hat j-2\hat k}{\sqrt{3^2+6^2+(-2)^2}}=\dfrac{3\hat i+6\hat j-2\hat k}{7}$

28. (a) $\phi=\vec B.\vec A$. In this formula $\vec A$ is a area vector.

29. (a) $\vec r=\vec a+\vec b+\vec c=4\hat i-\hat j-3\hat i+2\hat j-\hat k=\hat i+\hat j-\hat k$

$\hat r=\dfrac{\vec r}{|\vec r|}=\dfrac{\hat i+\hat j-\hat k}{\sqrt{1^2+1^2+(-1)^2}}=\dfrac{\hat i+\hat j-\hat k}{\sqrt3}$

30. (d) $\cos\theta=\dfrac{\vec A.\vec B}{|A||B|}=\dfrac{9+16+25}{\sqrt{9+16+25}\sqrt{9+16+25}}=\dfrac{50}{50}=1$

$\Rightarrow\cos\theta=1\ \therefore\ \theta=\cos^{-1}(1)$

31. (a) $\vec r=3t^2\hat i+4t^2\hat j+7\hat k$

at $t=0$, $\vec r_1=7\hat k$

at $t=10\sec$, $\vec r_2=300\hat i+400\hat j+7\hat k$,

$\overrightarrow{\Delta r}=\vec r_2-\vec r_1=300\hat i+400\hat j$

$|\overrightarrow{\Delta r}|=|\vec r_2-\vec r_1|=\sqrt{(300)^2+(400)^2}=500m$

32. (b) Resultant of vectors $\vec A$ and $\vec B$

$\vec R=\vec A+\vec B=4\hat i-3\hat j+8\hat i+8\hat j=12\hat i+5\hat j$

$\hat R=\dfrac{\vec R}{|\vec R|}=\dfrac{12\hat i+5\hat j}{\sqrt{(12)^2+(5)^2}}=\dfrac{12\hat i+5\hat j}{13}$

33. (a) $\frac{\vec{A}.\vec{B}}{|\hat{i}+\hat{j}|} = \frac{(2\hat{i}+3\hat{j})(\hat{i}+\hat{j})}{\sqrt{2}} = \frac{2+3}{\sqrt{2}} = \frac{5}{\sqrt{2}}$

34. (a) $\cos\theta = \frac{\vec{A}.\vec{B}}{|A||B|} = \frac{(3\hat{i}+4\hat{j}+5\hat{k})(3\hat{i}+4\hat{j}-5\hat{k})}{\sqrt{9+16+25}\sqrt{9+16+25}}$

$= \frac{9+16-25}{50} = 0$

$\Rightarrow \cos\theta = 0$, $\therefore\ \theta = 90°$

Addition and Subtraction of Vectors

1. (a) For 17 *N* both the vector should be parallel *i.e.* angle between them should be zero.
For 7 *N* both the vectors should be antiparallel *i.e.* angle between them should be 180°
For 13 *N* both the vectors should be perpendicular to each other *i.e.* angle between them should be 90°

2. (b) $\vec{A}+\vec{B} = 4\hat{i}-3\hat{j}+6\hat{i}+8\hat{j} = 10\hat{i}+5\hat{j}$

$|\vec{A}+\vec{B}| = \sqrt{(10)^2+(5)^2} = 5\sqrt{5}$

$\tan\theta = \frac{5}{10} = \frac{1}{2} \Rightarrow \theta = \tan^{-1}\left(\frac{1}{2}\right)$

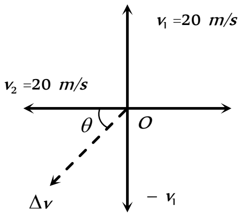
3. (d) From figure

$\vec{v}_1 = 20\hat{j}$ and $\vec{v}_2 = -20\hat{i}$

$\Delta\vec{v} = \vec{v}_2 - \vec{v}_1 = -20(\hat{i}+\hat{j})$

$|\Delta\vec{v}| = 20\sqrt{2}$ and direction

$\theta = \tan^{-1}(1) = 45°$ *i.e. S–W*



4. (b) Let $\hat{n}_1$ and $\hat{n}_2$ are the two unit vectors, then the sum is

$\vec{n}_s = \hat{n}_1 + \hat{n}_2$ or $n_s^2 = n_1^2 + n_2^2 + 2n_1n_2\cos\theta$

$= 1+1+2\cos\theta$

Since it is given that n_s is also a unit vector, therefore

$1 = 1+1+2\cos\theta \Rightarrow \cos\theta = -\frac{1}{2}$ $\therefore \theta = 120°$

Now the difference vector is $\hat{n}_d = \hat{n}_1 - \hat{n}_2$ or

$n_d^2 = n_1^2 + n_2^2 - 2n_1n_2\cos\theta = 1+1-2\cos(120°)$

$\therefore\ n_d^2 = 2-2(-1/2) = 2+1 = 3 \Rightarrow n_d = \sqrt{3}$

5. (b) $\vec{A}-2\vec{B}+3\vec{C} = (2\hat{i}+\hat{j})-2(3\hat{j}-\hat{k})+3(6\hat{i}-2\hat{k})$

$= 2\hat{i}+\hat{j}-6\hat{j}+2\hat{k}+18\hat{i}-6\hat{k} = 20\hat{i}-5\hat{j}-4\hat{k}$

6. (a) $\vec{P}_1 = mv\sin\theta\,\hat{i} - mv\cos\theta\,\hat{j}$

and $\vec{P}_2 = mv\sin\theta\,\hat{i} + mv\cos\theta\,\hat{j}$

So change in momentum

$\overrightarrow{\Delta P} = \vec{P}_2 - \vec{P}_1 = 2mv\cos\theta\,\hat{j}, |\Delta\vec{P}| = 2mv\cos\theta$

7. (b) $R = \sqrt{A^2+B^2+2AB\cos\theta}$

By substituting, $A = F, B = F$ and $R = F$ we get

$\cos\theta = \frac{1}{2}\ \therefore \theta = 120°$

8. (a)

9. (d) If two vectors $\vec{A}$ and $\vec{B}$ are given then the resultant $R_{max} = A+B = 7N$ and $R_{min} = 4-3 = 1N$

i.e. net force on the particle is between 1 *N* and 7 *N*.

10. (b) If $\vec{C}$ lies outside the plane then resultant force can not be zero.

11. (d)

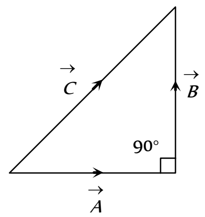
12. (c) $F = \sqrt{F_1^2+F_2^2+2F_1F_2\cos 90°} = \sqrt{F_1^2+F_2^2}$

13. (a)

14. (c)

15. (c) $C = \sqrt{A^2+B^2}$

The angle between *A* and *B* is $\frac{\pi}{2}$



16. (c) $\vec{R} = \vec{A}+\vec{B} = 6\hat{i}+7\hat{j}+3\hat{i}+4\hat{j} = 9\hat{i}+11\hat{j}$

$\therefore |\vec{R}| = \sqrt{9^2+11^2} = \sqrt{81+121} = \sqrt{202}$

17. (c) $R = \sqrt{12^2+5^2+6^2} = \sqrt{144+25+36} = \sqrt{205} = 14.31\ m$

18. (c) $\vec{A} = 3\hat{i}-2\hat{j}+\hat{k}$, $\vec{B} = \hat{i}-3\hat{j}+5\hat{k}$, $\vec{C} = 2\hat{i}-\hat{j}+4\hat{k}$

$|\vec{A}| = \sqrt{3^2+(-2)^2+1^2} = \sqrt{9+4+1} = \sqrt{14}$

$|\vec{B}| = \sqrt{1^2+(-3)^2+5^2} = \sqrt{1+9+25} = \sqrt{35}$

$|\vec{A}| = \sqrt{2^2+1^2+(-4)^2} = \sqrt{4+1+16} = \sqrt{21}$

As $B = \sqrt{A^2+C^2}$ therefore *ABC* will be right angled triangle.

19. (c)

20. (b) $\vec{C}+\vec{A} = \vec{B}$.

The value of *C* lies between $A-B$ and $A+B$

$\therefore |\vec{C}| < |\vec{A}|$ or $|\vec{C}| < |\vec{B}|$

21. (a)

22. (d)

23. (d) Here all the three force will not keep the particle in equilibrium so the net force will not be zero and the particle will move with an acceleration.

24. (a) $A+B = 16$ (given) ...(i)

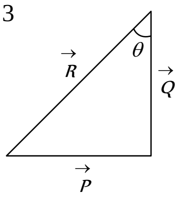
$\tan\alpha = \frac{B\sin\theta}{A+B\cos\theta} = \tan 90°$

$\therefore\ A+B\cos\theta = 0 \Rightarrow \cos\theta = \frac{-A}{B}$ ...(ii)

$8 = \sqrt{A^2 + B^2 + 2AB\cos\theta}$ ...(iii)

By solving eq. (i), (ii) and (iii) we get $A = 6N, \; B = 10N$

25. (c) $|\vec{P}| = 5$, $|\vec{Q}| = 12$ and $|\vec{R}| = 13$



$\cos\theta = \dfrac{Q}{R} = \dfrac{12}{13}$

$\therefore \; \theta = \cos^{-1}\left(\dfrac{12}{13}\right)$

26. (b) $\dfrac{B}{2} = \sqrt{A^2 + B^2 + 2AB\cos\theta}$ ...(i)

$\therefore \; \tan 90° = \dfrac{B\sin\theta}{A + B\cos\theta} \Rightarrow A + B\cos\theta = 0$

$\therefore \; \cos\theta = -\dfrac{A}{B}$

Hence, from (i) $\dfrac{B^2}{4} = A^2 + B^2 - 2A^2 \Rightarrow A = \sqrt{3}\,\dfrac{B}{2}$

$\Rightarrow \; \cos\theta = -\dfrac{A}{B} = -\dfrac{\sqrt{3}}{2} \; \therefore \; \theta = 150°$

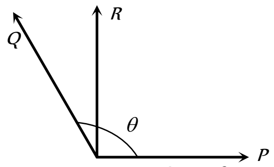
27. (b) $(\hat{i} - 2\hat{j} + 2\hat{k}) + (2\hat{i} + \hat{j} - \hat{k}) + \vec{R} = i$

$\therefore$ Required vector $\vec{R} = -2\hat{i} + \hat{j} - \hat{k}$

28. (a) Resultant $\vec{R} = \vec{P} + \vec{Q} + \vec{P} - \vec{Q} = 2\vec{P}$

The angle between $\vec{P}$ and $2\vec{P}$ is zero.

29. (b)



$\Rightarrow \; \tan 90° = \dfrac{Q\sin\theta}{P + Q\cos\theta} \Rightarrow \; P + Q\cos\theta = 0$

$\cos\theta = \dfrac{-P}{Q} \; \therefore \; \theta = \cos^{-1}\left(\dfrac{-P}{Q}\right)$

30. (a) According to problem $P + Q = 3$ and $P - Q = 1$

By solving we get $P = 2$ and $Q = 1 \; \therefore \; \dfrac{P}{Q} = 2 \Rightarrow \; P = 2Q$

31. (c)

32. (c)

33. (c)

34. (d) $F_1 + F_2 + F_3 = 0 \Rightarrow 4\hat{i} + 6\hat{j} + F_3 = 0$

$\therefore \vec{F}_3 = -4\hat{i} - 6\hat{j}$

35. (a) $\Delta v = 2v\sin\left(\dfrac{\theta}{2}\right) = 2 \times v \times \sin 90°$

$= 2 \times 100 = 200 \; km/hr$

36. (c)

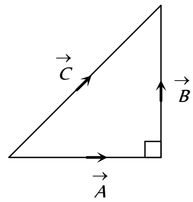
37. (d) Resultant velocity $= \sqrt{20^2 + 15^2}$

$= \sqrt{400 + 225} = \sqrt{625} = 25 \; km/hr$

38. (a) $C = \sqrt{A^2 + B^2}$

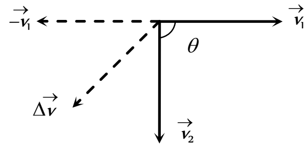
$= \sqrt{3^2 + 4^2} = 5$

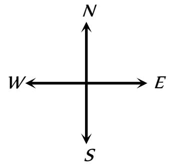
$\therefore$ Angle between $\vec{A}$ and $\vec{B}$ is $\dfrac{\pi}{2}$



39. (c)

40. (d)





If the magnitude of vector remains same, only direction change by θ then

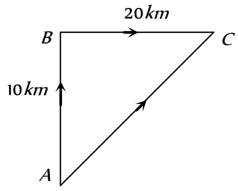
$\overrightarrow{\Delta v} = \vec{v}_2 - \vec{v}_1$, $\overrightarrow{\Delta v} = \vec{v}_2 + (-\vec{v}_1)$

Magnitude of change in vector $|\overrightarrow{\Delta v}| = 2v\sin\left(\dfrac{\theta}{2}\right)$

$|\overrightarrow{\Delta v}| = 2 \times 10 \times \sin\left(\dfrac{90°}{2}\right) = 10\sqrt{2} = 14.14 \; m/s$

Direction is south-west as shown in figure.

41. (a) $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$



$AC = \sqrt{(AB)^2 + (BC)^2}$

$= \sqrt{(10)^2 + (20)^2}$

$= \sqrt{100 + 400} = \sqrt{500} = 22.36 \, km$

42. (b) $\cos\theta = \dfrac{\vec{F}_1 . \vec{F}_2}{|F_1||F_2|}$

$= \dfrac{(5\hat{i} + 10\hat{j} - 20\hat{k}).(10\hat{i} - 5\hat{j} - 15\hat{k})}{\sqrt{25 + 100 + 400}\sqrt{100 + 25 + 225}} = \dfrac{50 - 50 + 300}{\sqrt{525}\sqrt{350}}$

$\Rightarrow \; \cos\theta = \dfrac{1}{\sqrt{2}} \; \therefore \; \theta = 45°$

43. (d) If two vectors A and B are given then Range of their resultant can be written as $(A - B) \le R \le (A + B)$.

i.e. $R_{max} = A + B$ and $R_{min} = A - B$

If $B = 1$ and $A = 4$ then their resultant will lies in between $3N$ and $5N$. It can never be $2N$.

44. (d) $A = 3N$, $B = 2N$ then $R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

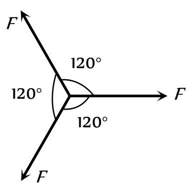
$R = \sqrt{9 + 4 + 12\cos\theta}$ ...(i)

Now $A = 6N$, $B = 2N$ then

$2R = \sqrt{36 + 4 + 24\cos\theta}$ ...(ii)

from (i) and (ii) we get $\cos\theta = -\dfrac{1}{2} \; \therefore \; \theta = 120°$

45. (a) In N forces of equal magnitude works on a single point and their resultant is



zero then angle between any two forces is given $\theta = \frac{360}{N} = \frac{360}{3} = 120°$

If these three vectors are represented by three sides of triangle then they form equilateral triangle

46. (c) Resultant of two vectors $\vec{A}$ and $\vec{B}$ can be given by $\vec{R} = \vec{A} + \vec{B}$

$|\vec{R}| = |\vec{A} + \vec{B}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

If $\theta = 0°$ then $|\vec{R}| = A + B = |\vec{A}| + |\vec{B}|$

47. (d) $R_{\max} = A + B = 17$ when $\theta = 0°$

$R_{\min} = A - B = 7$ when $\theta = 180°$

by solving we get $A = 12$ and $B = 5$

Now when $\theta = 90°$ then $R = \sqrt{A^2 + B^2}$

$\Rightarrow R = \sqrt{(12)^2 + (5)^2} = \sqrt{169} = 13$

48. (a) If two vectors are perpendicular then their dot product must be equal to zero. According to problem

$(\vec{A} + \vec{B}).(\vec{A} - \vec{B}) = 0 \Rightarrow \vec{A}.\vec{A} - \vec{A}.\vec{B} + \vec{B}.\vec{A} - \vec{B}.\vec{B} = 0$

$\Rightarrow A^2 - B^2 = 0 \Rightarrow A^2 = B^2$

$\therefore$ $A = B$ *i.e.* two vectors are equal to each other in magnitude.

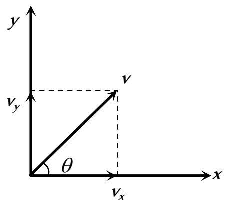
49. (a) $v_y = 20$ and $v_x = 10$

$\therefore$ velocity $\vec{v} = 10\hat{i} + 20\hat{j}$

direction of velocity with *x* axis

$\tan\theta = \frac{v_y}{v_x} = \frac{20}{10} = 2$

$\therefore$ $\theta = \tan^{-1}(2)$



50. (c) $R_{\max} = A + B$ when $\theta = 0°$ $\therefore$ $R_{\max} = 12 + 8 = 20N$

51. (c) $R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

If $A = B = P$ and $\theta = 120°$ then $R = P$

52. (a) Sum of the vectors $\vec{R} = 5\hat{i} + 8\hat{j} + 2\hat{i} + 7\hat{j} = 7\hat{i} + 15\hat{j}$

magnitude of $\vec{R} = |\vec{R}| = \sqrt{49 + 225} = \sqrt{274}$

53. (d)

Multiplication of Vectors

1. (c) Given vectors can be rewritten as $\vec{A} = 2\hat{i} + 3\hat{j} + 8\hat{k}$ and $\vec{B} = -4\hat{i} + 4\hat{j} + \alpha\hat{k}$

Dot product of these vectors should be equal to zero because they are perpendicular.

$\therefore$ $\vec{A}.\vec{B} = -8 + 12 + 8\alpha = 0 \Rightarrow 8\alpha = -4 \Rightarrow \alpha = -1/2$

2. (b) Let $\vec{A} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{B} = -4\hat{i} - 6\hat{j} + \lambda\hat{k}$

$\vec{A}$ and $\vec{B}$ are parallel to each other

$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$ *i.e.* $\frac{2}{-4} = \frac{3}{-6} = \frac{-1}{\lambda} \Rightarrow \lambda = 2$.

3. (d) $W = \vec{F}.\vec{S} = FS\cos\theta$

$= 50 \times 10 \times \cos 60° = 50 \times 10 \times \frac{1}{2} = 250\, J.$

4. (a) $S = \vec{r_2} - \vec{r_1}$

$W = \vec{F}.\vec{S} = (4\hat{i} + \hat{j} + 3\hat{k}).(11\hat{i} + 11\hat{j} + 15\hat{k})$

$= (4 \times 11 + 1 \times 11 + 3 \times 15) = 100\, J.$

5. (a) $(\vec{A} + \vec{B})$ is perpendicular to $(\vec{A} - \vec{B})$. Thus

$(\vec{A} + \vec{B}).(\vec{A} - \vec{B}) = 0$

or $A^2 + \vec{B}.\vec{A} - \vec{A}.\vec{B} - B^2 = 0$

Because of commutative property of dot product $\vec{A}.\vec{B} = \vec{B}.\vec{A}$

$\therefore$ $A^2 - B^2 = 0$ or $A = B$

Thus the ratio of magnitudes $A/B = 1$

6. (b) Let $\vec{A}.(\vec{B} \times \vec{A}) = \vec{A}.\vec{C}$

Here $\vec{C} = \vec{B} \times \vec{A}$ Which is perpendicular to both vector $\vec{A}$ and $\vec{B}$ $\therefore$ $\vec{A}.\vec{C} = 0$

7. (c) We know that $\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})$ because the angle between these two is always 90°.

But if the angle between $\vec{A}$ and $\vec{B}$ is 0 or π. Then $\vec{A} \times \vec{B} = \vec{B} \times \vec{A} = 0$.

8. (b) $\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & 2 \\ 2 & -2 & 4 \end{vmatrix}$

$= (1 \times 4 - 2 \times -2)\hat{i} + (2 \times 2 - 4 \times 3)\hat{j} + (3 \times -2 - 1 \times 2)\hat{k}$

$= 8\hat{i} - 8\hat{j} - 8\hat{k}$

$\therefore$ Magnitude of $\vec{A} \times \vec{B} = |\vec{A} \times \vec{B}| = \sqrt{(8)^2 + (-8)^2 + (-8)^2}$

$= 8\sqrt{3}$

9. (b) $\vec{\tau} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & 3 \\ 2 & -3 & 4 \end{vmatrix}$

$= [(2 \times 4) - (3 \times -3)]\hat{i} + [(2 \times 3) - (3 \times 4)]\hat{j}$

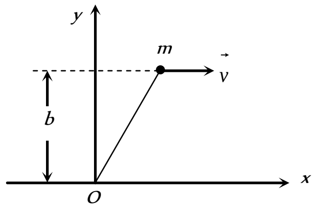
$+ [(3 \times -3) - (2 \times 2)]\hat{k} = 17\hat{i} - 6\hat{j} - 13\hat{k}$

10. (d) From the property of vector product, we notice that $\vec{C}$ must be perpendicular to the plane formed by vector $\vec{A}$ and $\vec{B}$. Thus $\vec{C}$ is perpendicular to both $\vec{A}$ and $\vec{B}$ and $(\vec{A} + \vec{B})$ vector also, must lie in the plane formed by vector $\vec{A}$ and $\vec{B}$. Thus $\vec{C}$ must be perpendicular to $(\vec{A} + \vec{B})$ also but the cross product $(\vec{A}\times\vec{B})$ gives a vector $\vec{C}$ which can not be perpendicular to itself. Thus the last statement is wrong.

11. (b) We know that, Angular momentum

$\vec{L}=\vec{r}\times\vec{p}$ in terms of component becomes

$$\vec{L}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ x & y & z\\ p_x & p_y & p_z\end{vmatrix}$$



As motion is in *x-y* plane (z = 0 and $P_z=0$), so

$\vec{L}=\vec{k}(xp_y-yp_x)$

Here $x = vt$, $y = b$, $p_x=mv$ and $p_y=0$

$\therefore\ \vec{L}=\vec{k}\left[vt\times 0-b\,mv\right]=-mvb\,\hat{k}$

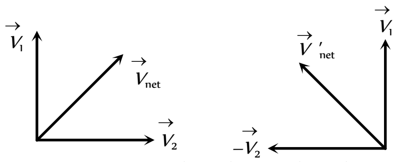
12. (d) $\vec{F}_1.\vec{F}_2=(2\hat{j}+5\hat{k})(3\hat{j}+4\hat{k})$

$=6+20=20+6=26$

13. (c) Force F lie in the *x-y* plane so a vector along *z*-axis will be perpendicular to F.

14. (d) $\vec{A}.\vec{B}=|\vec{A}|.|\vec{B}|.\cos\theta=A.B.\cos 90^\circ=0$

15. (c)



According to problem $|\vec{V}_1+\vec{V}_2|=|\vec{V}_1-\vec{V}_2|$

$\Rightarrow|\vec{V}_{net}|=|\vec{V}'_{net}|$

So V_1 and V_2 will be mutually perpendicular.

16. (c) $W=\vec{F}.\vec{r}=(5\hat{i}+3\hat{j})(2\hat{i}-\hat{j})=10-3=7\,J.$

17. (b) $\cos\theta=\dfrac{\vec{A}.\vec{B}}{|\vec{A}||\vec{B}|}=\dfrac{-2+6-4}{\sqrt{14}\sqrt{21}}=0\ \therefore\ \theta=90^\circ$

18. (c) $(\hat{i}+\hat{j}).(\hat{j}+\hat{k})=0+0+1+0=1$

$\cos\theta=\dfrac{\vec{A}.\vec{B}}{|\vec{A}||\vec{B}|}=\dfrac{1}{\sqrt{2}\times\sqrt{2}}=\dfrac{1}{2}\ \therefore\ \theta=60^\circ$

19. (b) $P=\vec{F}.\vec{v}=20\times 6+15\times(-4)+(-5)\times 3$

$=120-60-15=120-75=45\,J/s$

20. (a) $\cos\theta=\dfrac{\vec{P}.\vec{Q}}{PQ}=1\ \therefore\ \theta=0^\circ$

21. (a) $W=\vec{F}.\vec{s}=(5\hat{i}+6\hat{j}+4\hat{k})(6\hat{i}-5\hat{k})=30-20=10\,J$

22. (c) $\vec{A}.\vec{B}=0\ \therefore\ \theta=90^\circ$

23. (a) $\vec{P}.\vec{Q}=0\ \therefore\ a^2-2a-3=0\Rightarrow a=3$

24. (b) $W=\vec{F}.\vec{r}=(-2\hat{i}+15\hat{j}+6\hat{k})(10\hat{j})=150$

25. (c) $P_x=2\cos t,\ P_y=2\sin t\ \therefore\ \vec{P}=2\cos t\,\hat{i}+2\sin t\,\hat{j}$

$\vec{F}=\dfrac{d\vec{P}}{dt}=-2\sin t\,\hat{i}+2\cos t\,\hat{j}$

$\vec{F}.\vec{P}=0\ \therefore\ \theta=90^\circ$

26. (d) $|\vec{A}\times\vec{B}|=|(2\hat{i}+3\hat{j})\times(\hat{i}+4\hat{j})|=|5\hat{k}|=5\ units$

27. (d)

28. (b) $\vec{A}\times\vec{B}=0\ \therefore\ \sin\theta=0\ \therefore\ \theta=0^\circ$

Two vectors will be parallel to each other.

29. (b) $\vec{A}\times\vec{B}$ and $\vec{B}\times\vec{A}$ are parallel and opposite to each other. So the angle will be π.

30. (b) Vector $(\vec{P}+\vec{Q})$ lies in a plane and vector $(\vec{P}\times\vec{Q})$ is perpendicular to this plane *i.e.* the angle between given vectors is $\dfrac{\pi}{2}$.

31. (d) $\sqrt{2^2+3^2+2\times 2\times 3\times\cos\theta}=1$

By solving we get $\theta=180^\circ\ \therefore\ \vec{A}\times\vec{B}=0$

32. (c) Dot product of two perpendicular vector will be zero.

33. (d) $\cos\theta=\dfrac{\vec{A}\vec{B}}{AB}=\dfrac{42+24-12}{\sqrt{36+36+9}\sqrt{49+16+16}}=\dfrac{56}{9\sqrt{71}}$

$\cos\theta=\dfrac{56}{9\sqrt{71}}\ \therefore\ \sin\theta=\dfrac{\sqrt{5}}{3}$ or $\theta=\sin^{-1}\left(\dfrac{\sqrt{5}}{3}\right)$

34. (b) Direction of vector A is along *z*-axis $\therefore\ \vec{A}=a\hat{k}$

Direction of vector B is towards north $\therefore\ \vec{B}=b\hat{j}$

Now $\vec{A}\times\vec{B}=a\hat{k}\times b\hat{j}=ab(-\hat{j})$

$\therefore$ The direction is $\vec{A}\times\vec{B}$ is along west.

35. (d) $\cos\theta=\dfrac{\vec{A}.\vec{B}}{|\vec{A}||\vec{B}|}=\dfrac{1}{\sqrt{2}\sqrt{2}}=\dfrac{1}{2}\ \therefore\ \theta=60^\circ$

36. (d) $\overrightarrow{AB}=(4\hat{i}+5\hat{j}+6\hat{k})-(3\hat{i}+4\hat{j}+5\hat{k})=\hat{i}+\hat{j}+\hat{k}$

$\overrightarrow{CD}=(4\hat{i}+6\hat{j})-(7\hat{i}+9\hat{j}+3\hat{k})=-3\hat{i}-3\hat{j}-3\hat{k}$

$\overrightarrow{AB}$ and $\overrightarrow{CD}$ are parallel, because its cross-products is 0.

37. (a) $W=\vec{F}\,\vec{S}=(4\hat{i}+5\hat{j})(3\hat{i}+6\hat{j})=12$

38. (b) $|\vec{A}\times\vec{B}|=\vec{A}.\vec{B}\Rightarrow AB\sin\theta=AB\cos\theta\Rightarrow\tan\theta=1$

$\therefore\ \theta=45^\circ$

39. (a)

40. (a) $\vec{v}=\vec{\omega}\times\vec{r}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & -2 & 2\\ 0 & 4 & -3\end{vmatrix}=\hat{i}(6-8)-\hat{j}(-3)+4\hat{k}$

$-2\vec{i}+3\vec{j}+4\vec{k}$

$|\vec{v}|=\sqrt{(-2)^2+(3)^2+4^2}=\sqrt{29}\ unit$

41. (d) $\vec{a}.\vec{b}=0$ *i.e.* $\vec{a}$ and $\vec{b}$ will be perpendicular to each other

$\vec{a}.\vec{c}=0$ *i.e.* $\vec{a}$ and $\vec{c}$ will be perpendicular to each other

$\vec{b}\times\vec{c}$ will be a vector perpendicular to both $\vec{b}$ and $\vec{c}$

So $\vec{a}$ is parallel to $\vec{b}\times\vec{c}$

42. (d) Area $=\left|2\hat{i}\times 2\hat{j}\right| = \left|4\hat{k}\right| = 4\ unit.$

43. (c) $\vec{A}=2\hat{i}+2\hat{j}-\hat{k}$ and $\vec{B}=6\hat{i}-3\hat{j}+2\hat{k}$

$$\vec{C}=\vec{A}\times\vec{B}=\left(2\hat{i}+2\hat{j}-\hat{k}\right)\times\left(6\hat{i}-3\hat{j}+2\hat{k}\right)$$

$$=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 2 & -1\\ 6 & -3 & 2\end{vmatrix}=\hat{i}-10\hat{j}-18\hat{k}$$

Unit vector perpendicular to both $\vec{A}$ and $\vec{B}$

$$=\frac{\hat{i}-10\hat{j}-18\hat{k}}{\sqrt{1^2+10^2+18^2}}=\frac{\hat{i}-10\hat{j}-18\hat{k}}{5\sqrt{17}}$$

44. (b) $\vec{A}=\hat{j}+3\hat{k}$, $\vec{B}=\hat{i}+2\hat{j}-\hat{k}$

$$\vec{C}=\vec{A}\times\vec{B}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 0 & 1 & 3\\ 1 & 2 & -1\end{vmatrix}=-7\hat{i}+3\hat{j}-\hat{k}$$

Hence area = $|\vec{C}|=\sqrt{49+9+1}=\sqrt{59}\ squnit$

45. (a) $\vec{L}=\vec{r}\times\vec{p}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 2 & -1\\ 3 & 4 & -2\end{vmatrix}=-\hat{j}-2\hat{k}$

i.e. the angular momentum is perpendicular to *x*-axis.

46. (a) $\vec{A}\times\vec{B}$ is a vector perpendicular to plane $\vec{A}+\vec{B}$ and hence perpendicular to $\vec{A}+\vec{B}$.

47. (a) $\vec{\tau}=\vec{r}\times\vec{F}=(7\hat{i}+3\hat{j}+\hat{k})(-3\hat{i}+\hat{j}+5\hat{k})$

$$\vec{\tau}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 7 & 3 & 1\\ -3 & 1 & 5\end{vmatrix}=14\hat{i}-38\hat{j}+16\hat{k}$$

48. (d) $(\vec{A}+\vec{B})\times(\vec{A}-\vec{B}) = \vec{A}\times\vec{A}-\vec{A}\times\vec{B}+\vec{B}\times\vec{A}-\vec{B}\times\vec{B}$

$$=0-\vec{A}\times\vec{B}+\vec{B}\times\vec{A}-0=\vec{B}\times\vec{A}+\vec{B}\times\vec{A}=2(\vec{B}\times\vec{A})$$

49. (d) For perpendicular vector $\vec{A}.\vec{B}=0$

$\Rightarrow (5\hat{i}+7\hat{j}-3\hat{k}).(2\hat{i}+2\hat{j}-a\hat{k})=0$

$\Rightarrow 10+14+3a=0 \Rightarrow a=-8$

50. (a) $\text{Mass}=\dfrac{\text{Force}}{\text{Acceleration}}=\dfrac{|\vec{F}|}{a}$

$$=\frac{\sqrt{36+64+100}}{1}=10\sqrt{2}\ kg$$

51. (a) Area of parallelogram $=\vec{A}\times\vec{B}$

$$=(\hat{i}+2\hat{j}+3\hat{k})\times(3\hat{i}-2\hat{j}+\hat{k})$$

$$=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 2 & 3\\ 3 & -2 & 1\end{vmatrix}=(8)\hat{i}+(8)\hat{j}-(8)\hat{k}$$

Magnitude $=\sqrt{64+64+64}=8\sqrt{3}$

52. (b) Radius vector $\vec{r}=\vec{r_2}-\vec{r_1}=(2\hat{i}-3\hat{j}+\hat{k})-(2\hat{i}+\hat{j}+\hat{k})$

$\therefore\ \vec{r}=-4\hat{j}$

Linear momentum $\vec{p}=2\hat{i}+3\hat{j}-\hat{k}$

$$\vec{L}=\vec{r}\times\vec{p}=(-4\hat{j})\times(2\hat{i}+3\hat{j}-\hat{k})$$

$$=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 0 & -4 & 0\\ 2 & 3 & -1\end{vmatrix}=4\hat{i}-8\hat{k}$$

53. (d) $\vec{v}=\vec{\omega}\times\vec{r}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 3 & -4 & 1\\ 5 & -6 & 6\end{vmatrix}=-18\hat{i}-13\hat{j}+2\hat{k}$

54. (a)

55. (c) $\vec{A}.\vec{B}=AB\cos\theta$

In the problem $\vec{A}.\vec{B}=-AB$ i.e. $\cos\theta=-1\ \therefore\ \theta=180°$

i.e. $\vec{A}$ and $\vec{B}$ acts in the opposite direction.

56. (d) $|\vec{A}\times\vec{B}|=\sqrt{3}(\vec{A}.\vec{B})$

$AB\sin\theta=\sqrt{3}AB\cos\theta \Rightarrow \tan\theta=\sqrt{3}\ \therefore\ \theta=60°$

Now $|\vec{R}|=|\vec{A}+\vec{B}|=\sqrt{A^2+B^2+2AB\cos\theta}$

$$=\sqrt{A^2+B^2+2AB\left(\frac{1}{2}\right)}=(A^2+B^2+AB)^{1/2}$$

57. (a) $W=\vec{F}.\vec{s}=(3\hat{i}+c\hat{j}+2\hat{k}).(-4\hat{i}+2\hat{j}-3\hat{k})=-12+2c-6$

Work done $=6J$ (given)

$\therefore\ -12+2c-6=6 \Rightarrow c=12$

58. (b) $W=\vec{F}.\vec{s}=(5\hat{i}+3\hat{j}).(2\hat{i}-\hat{j})=10-3=7\ J$

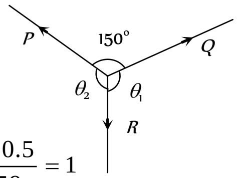
59. (c) $\vec{A}\times\vec{B}=AB\sin\theta\,\hat{n}$

for parallel vectors $\theta=0°$ or $180°$, $\sin\theta=0$

$\therefore \ \vec{A} \times \vec{B} = \hat{0}$

Lami's Theorem

1. (c) $\dfrac{P}{\sin\theta_1} = \dfrac{Q}{\sin\theta_2} = \dfrac{R}{\sin 150°}$



$\Rightarrow \dfrac{1.93}{\sin\theta_1} = \dfrac{R}{\sin 150°}$

$\Rightarrow R = \dfrac{1.93 \times \sin 150°}{\sin\theta_1} = \dfrac{1.93 \times 0.5}{0.9659} = 1$

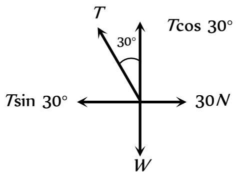
2. (a) According to Lami's theorem

$\dfrac{P}{\sin\alpha} = \dfrac{Q}{\sin\beta} = \dfrac{R}{\sin\gamma}$

3. (b)

4. (c)

5. (b)



From the figure $T\sin 30^o = 30$ ...(i)

$T\cos 30^o = W$ ...(ii)

By solving equation (i) and (ii) we get

$W = 30\sqrt{3}N$ and $T = 60N$

Relative Velocity

1. (b) The two car (say A and B) are moving with same velocity, the relative velocity of one (say B) with respect to the other $A, \vec{v}_{BA} = \vec{v}_B - \vec{v}_A = v - v = 0$

So the relative separation between them (= 5 km) always remains the same.

Now if the velocity of car (say C) moving in opposite direction to A and B, is $\vec{v}_C$ relative to ground then the velocity of car C relative to A and B will be $\vec{v}_{rel.} = \vec{v}_C - \vec{v}$

But as $\vec{v}$ is opposite to v_c

So $v_{rel} = v_c - (-30) = (v_C + 30) km/hr.$

So, the time taken by it to cross the cars A and B

$t = \dfrac{d}{v_{rel}} \Rightarrow \dfrac{4}{60} = \dfrac{5}{v_C + 30}$

$\Rightarrow v_C = 45\,km/hr.$

2. (b) When the man is at rest $w.r.t.$ the ground, the rain comes to him at an angle 30° with the vertical. This is the direction of the velocity of raindrops with respect to the ground.

Here $\vec{v}_{rg}$ = velocity of rain with respect to the ground

$\vec{v}_{mg}$ = velocity of the man with respect to the ground.

and $\vec{v}_{rm}$ = velocity of the rain with respect to the man,

We have $\vec{v}_{rg} = \vec{v}_{rm} + \vec{v}_{mg}$ ......(i)

Taking horizontal components equation (i) gives

$v_{rg}\sin 30° = v_{mg} = 10\,km/hr$

or $v_{rg} = \dfrac{10}{\sin 30°} = 20\,km/hr$

3. (c) Taking vertical components equation (i) gives

$v_{rg}\cos 30° = v_{rm} = 20\dfrac{\sqrt{3}}{2} = 10\sqrt{3}\,km/hr$

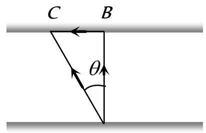
4. (c) Relative velocity = $(3\hat{i} + 4\hat{j}) - (-3\hat{i} - 4\hat{j}) = 6\hat{i} + 8\hat{j}$

5. (d) Relative velocity of parrot with respect to train

$= 5 - (-10) = 5 + 10 = 15\,m/\sec$

Time taken by the parrot $= \dfrac{d}{v_{rel.}} = \dfrac{150}{15} = 10\sec.$

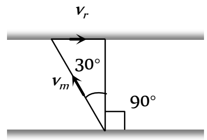
6. (a)



For shortest time, swimmer should swim along AB, so he will reach at point C due to the velocity of river.

i.e. he should swim due north.

7. (c)



$\sin 30^o = \dfrac{v_r}{v_m} = \dfrac{1}{2} \Rightarrow v_r = \dfrac{v_m}{2} = \dfrac{0.5}{2} = 0.25\,m/s$

8. (c) $\vec{v}_B + \vec{v}_A = \vec{v}_B + \vec{v}_A = 80 + 65 = 145\,km/hr$

9. (d) Relative speed of police with respect to thief

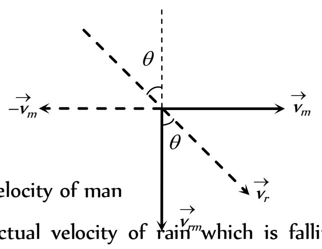
$= 10 - 9 = 1\,m/s$

Instantaneous separation = 100 m

Time = $\dfrac{\text{distance}}{\text{veclotiy}} = \dfrac{100}{1} = 100\sec.$

10. (a,c)

11. (b) A man is sitting in a bus and travelling from west to east, and the rain drops are appears falling vertically down.



v_m = velocity of man

v_r = Actual velocity of rain which is falling at an angle θ with vertical

v_{rm} = velocity of rain $w.r.t.$ to moving man

If the another man observe the rain then he will find that actually rain falling with velocity v_r at an angle going from west to east.

12. (b) Boat covers distance of 16km in a still water in 2 hours.

i.e. $v_B = \frac{16}{2} = 8\,km/hr$

Now velocity of water $\Rightarrow v_w = 4km/hr$.

Time taken for going upstream

$t_1 = \frac{8}{v_B - v_w} = \frac{8}{8-4} = 2hr$

(As water current oppose the motion of boat)

Time taken for going down stream

$t_2 = \frac{8}{v_B + v_w} = \frac{8}{8+4} = \frac{8}{12}hr$

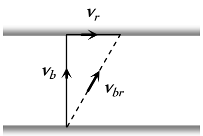
(As water current helps the motion of boat)

$\therefore$ Total time $= t_1 + t_2 = \left(2 + \frac{8}{12}\right)hr$ or $2hr\,40min$

13. (d) Relative velocity $= 10 + 5 = 15\,m/s$.

Time taken by the bird to cross the train $= \frac{120}{15} = 8$ sec

14. (b) $\overrightarrow{v_{br}} = \overrightarrow{v_b} + \overrightarrow{v_r}$



$\Rightarrow v_{br} = \sqrt{v_b^2 + v_r^2}$

$\Rightarrow 10 = \sqrt{8^2 + v_r^2}$

$\Rightarrow v_r = 6km/hr$.

Critical Thinking Questions

1. (c) $\sin^2\alpha + \sin^2\beta + \sin\gamma$

$= 1 - \cos^2\alpha + 1 - \cos^2\beta + 1 - \cos^2\gamma$

$= 3 - (\cos^2\alpha + \cos^2\beta + \cos^2\gamma) = 3 - 1 = 2$

2. (c) If vectors are of equal magnitude then two vectors can give zero resultant, if they works in opposite direction. But if the vectors are of different magnitudes then minimum three vectors are required to give zero resultant.

3. (c)

4. (c) Let P be the smaller force and Q be the greater force then according to problem –

$P + Q = 18$ ......(i)

$R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta} = 12$ ....(ii)

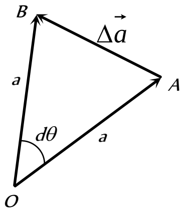
$\tan\phi = \frac{Q\sin\theta}{P + Q\cos\theta} = \tan 90 = \infty$

$\therefore\ P + Q\cos\theta = 0$ ....(iii)

By solving (i), (ii) and (iii) we will get $P = 5$, and $Q = 13$

5. (b) From the figure $|\overrightarrow{OA}| = a$ and $|\overrightarrow{OB}| = a$

Also from triangle rule $\overrightarrow{OB} - \overrightarrow{OA} = \overrightarrow{AB} = \Delta\vec{a}$



$\Rightarrow |\Delta\vec{a}| = AB$

Using angle $= \frac{\text{arc}}{\text{radius}}$

$\Rightarrow\ AB = a\,.\,d\theta$

So $|\Delta\vec{a}| = a\,d\theta$

Δa means change in magnitude of vector *i.e.* $|\overrightarrow{OB}| - |\overrightarrow{OA}|$

$\Rightarrow\ a - a = 0$

So $\Delta a = 0$

6. (b) $R_{net} = R + \sqrt{R^2 + R^2} = R + \sqrt{2}R = R(\sqrt{2} + 1)$

7. (d)

8. (d) $\Delta v = 2v\sin\left(\frac{90°}{2}\right) = 2v\sin 45° = 2v \times \frac{1}{\sqrt{2}} = \sqrt{2}v$

$= \sqrt{2} \times r\omega = \sqrt{2} \times 1 \times \frac{2\pi}{60} = \frac{\sqrt{2}\pi}{`30}\ cm/s$

9. (b) $\Delta v = 2v\sin\left(\frac{\theta}{2}\right) = 2 \times 5 \times \sin 45° = \frac{10}{\sqrt{2}}$

$\therefore\ a = \frac{\Delta v}{\Delta t} = \frac{10/\sqrt{2}}{10} = \frac{1}{\sqrt{2}}\ m/s^2$

10. (c) For motion of the particle from (0, 0) to (a, 0)

$\vec{F} = -K(0\hat{i} + a\hat{j}) \Rightarrow \vec{F} = -Ka\hat{j}$

Displacement $\vec{r} = (a\hat{i} + 0\hat{j}) - (0\hat{i} + 0\hat{j}) = a\hat{i}$

So work done from (0, 0) to (a, 0) is given by

$W = \vec{F}.\vec{r} = -Ka\hat{j}.a\hat{i} = 0$

For motion (a, 0) to (a, a)

$\vec{F} = -K(a\hat{i} + a\hat{j})$ and displacement

$\vec{r} = (a\hat{i} + a\hat{j}) - (a\hat{i} + 0\hat{j}) = a\hat{j}$

So work done from (a, 0) to (a, a) $W = \vec{F}.\vec{r}$

$= -K(a\hat{i} + a\hat{j}).a\hat{j} = -Ka^2$

So total work done $= -Ka^2$

11. (a) Given $\overrightarrow{OA} = \vec{a} = 3\hat{i} - 6\hat{j} + 2\hat{k}$ and $\overrightarrow{OB} = \vec{b} = 2\hat{i} + \hat{j} - 2\hat{k}$

$\therefore\ (\vec{a} \times \vec{b}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -6 & 2 \\ 2 & 1 & -2 \end{vmatrix}$

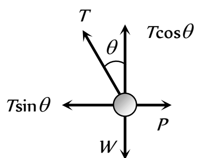
$= (12 - 2)\hat{i} + (4 + 6)\hat{j} + (3 + 12)\hat{k}$

$= 10\hat{i} + 10\hat{j} + 15\hat{k} \Rightarrow |\vec{a} \times \vec{b}| = \sqrt{10^2 + 10^2 + 15^2}$

$= \sqrt{425} = 5\sqrt{17}$

Area of $\Delta OAB = \frac{1}{2}|\vec{a} \times \vec{b}| = \frac{5\sqrt{17}}{2}$ *sq.unit.*

12. (d)



As the metal sphere is in equilibrium under the effect of three forces therefore $\vec{T} + \vec{P} + \vec{W} = 0$

From the figure $T\cos\theta = W$ ...(i)

$T\sin\theta = P$ ...(ii)

From equation (i) and (ii) we get $P = W\tan\theta$

and $T^2 = P^2 + W^2$

13. (b)

14. (d)

Assertion and Reason

1 (a) Cross product of two vectors is perpendicular to the plane containing both the vectors.

2 (a) $\cos\theta = \dfrac{(\hat{i}+\hat{j}).(\hat{i})}{|\hat{i}+\hat{j}||\hat{i}|} = \dfrac{1}{\sqrt{2}}$. Hence $\theta = 45^\circ$.

3 (d) $\dfrac{\vec{A}\times\vec{B}}{\vec{A}.\vec{B}} = \dfrac{AB\sin\theta\,\hat{n}}{AB\cos\theta} = \tan\theta\,\hat{n}$

where $\hat{n}$ is unit vector perpendicular to both $\vec{A}$ and $\vec{B}$.

However $\dfrac{|\vec{A}\times\vec{B}|}{\vec{A}.\vec{B}} = \tan\theta$

4 (b) $|\vec{A}+\vec{B}| = |\vec{A}-\vec{B}|$

$\Rightarrow A^2+B^2+2AB\cos\theta = A^2+B^2+2AB\cos\theta$

Hence $\cos\theta = 0$ which gives $\theta = 90^\circ$

Also vector addition is commutative.

Hence $\vec{A}+\vec{B} = \vec{B}+\vec{A}$.

5 (c) $\vec{v} = \vec{\omega}\times\vec{r}$

The expression $\vec{\omega} = \vec{v}\times\vec{r}$ is wrong.

6 (b) For giving a zero resultant, it should be possible to represent the given vectors along the sides of a closed polygon and minimum number of sides of a polygon is three.

7 (a) Since velocities are in opposite direction, therefore $v_{AB} = |\vec{v}_A - \vec{v}_B| = v_A + v_B$.

Which is greater than v_A or v_B

8 (b) Vector addition of two vectors is commutative *i.e.* $\vec{A}+\vec{B} = \vec{B}+\vec{A}$.

9 (a)

10 (c) Cross-product of two vectors is anticommutative.

i.e. $\vec{A}\times\vec{B} = -\vec{B}\times\vec{A}$

11 (b)

12 (e) If a vector quantity has zero magnitude then it is called a null vector. That quantity may have some direction even if its magnitude is zero.

13 (a) Let $\vec{P}$ and $\vec{Q}$ are two vectors in opposite direction, then their sum $\vec{P}+(-\vec{Q}) = \vec{P}-\vec{Q}$

If $\vec{P} = \vec{Q}$ then sum equal to zero.

14 (c) If two vectors are in opposite direction, then they cannot be like vectors.

15 (a) If θ be the angle between two vectors $\vec{A}$ and $\vec{B}$, then their scalar product, $\vec{A}.\vec{B} = AB\cos\theta$

If $\theta = 90^\circ$ then $\vec{A}.\vec{B} = 0$

i.e. if $\vec{A}$ and $\vec{B}$ are perpendicular to each other then their scalar product will be zero.

16 (b) We can multiply any vector by any scalar.

For example, in equation $\vec{F} = m\vec{a}$ mass is a scalar quantity, but acceleration is a vector quantity.

17 (c) If two vectors equal in magnitude are in opposite direction, then their sum will be a null vector.

A null vector has direction which is intermediate (or depends on direction of initial vectors) even its magnitude is zero.

18 (b) $\vec{A}.\vec{B} = |\vec{A}||\vec{B}|\cos\theta = 0$

$\vec{A}\times\vec{B} = |\vec{A}||\vec{B}|\sin\theta = 0$

If $\vec{A}$ and $\vec{B}$ are not null vectors then it follows that $\sin\theta$ and $\cos\theta$ both should be zero simultaneously. But it cannot be possible so it is essential that one of the vector must be null vector.

19 (b)

20 (c) The resultant of two vectors of unequal magnitude given by $R = \sqrt{A^2+B^2+2AB\cos\theta}$ cannot be zero for any value of θ.

21 (a) $\vec{A}.\vec{B} = \vec{B}.\vec{C} \Rightarrow AB\cos\theta_1 = BC\cos\theta_2$

$\therefore$ $A = C$, only when $\theta_1 = \theta_2$

So when angle between $\vec{A}$ and $\vec{B}$ is equal to angle between $\vec{B}$ and $\vec{C}$ only then $\vec{A}$ equal to $\vec{C}$

22 (c) Since vector addition is commutative, therefore $\vec{A}+\vec{B} = \vec{B}+\vec{A}$.

Vectors

SET Self Evaluation Test - 0

1. $0.4i + 0.8j + ck$ represents a unit vector when c is

(a) – 0.2 (b) $\sqrt{0.2}$

(c) $\sqrt{0.8}$ (d) 0

2. The angles which a vector $\hat{i} + \hat{j} + \sqrt{2}\,\hat{k}$ makes with X, Y and Z axes respectively are

(a) 60°, 60°, 60° (b) 45°, 45°, 45°

(c) 60°, 60°, 45° (d) 45°, 45°, 60°

3. The value of a unit vector in the direction of vector $A = 5\hat{i} - 12\hat{j}$, is

(a) $\hat{i}$ (b) $\hat{j}$

(c) $(\hat{i} + \hat{j})/13$ (d) $(5\hat{i} - 12\hat{j})/13$

4. Which of the following is independent of the choice of co-ordinate system

(a) $\vec{P} + \vec{Q} + \vec{R}$ (b) $(P_x + Q_x + R_x)\hat{i}$

(c) $P_x\hat{i} + Q_y\hat{j} + R_z\hat{k}$ (d) None of these

5. A car travels 6 km towards north at an angle of 45° to the east and then travels distance of 4 km towards north at an angle of 135° to the east. How far is the point from the starting point. What angle does the straight line joining its initial and final position makes with the east

(a) $\sqrt{50}\,km$ and $\tan^{-1}(5)$

(b) 10 km and $\tan^{-1}(\sqrt{5})$

(c) $\sqrt{52}\,km$ and $\tan^{-1}(5)$

(d) $\sqrt{52}\,km$ and $\tan^{-1}(\sqrt{5})$

6. Given that $\vec{A} + \vec{B} + \vec{C} = 0$ out of three vectors two are equal in magnitude and the magnitude of third vector is $\sqrt{2}$ times that of either of the two having equal magnitude. Then the angles between vectors are given by

(a) 30°, 60°, 90° (b) 45°, 45°, 90°

(c) 45°, 60°, 90° (d) 90°, 135°, 135°

7. Two forces $F_1 = 1\,N$ and $F_2 = 2\,N$ act along the lines $x = 0$ and $y = 0$ respectively. Then the resultant of forces would be

(a) $\hat{i} + 2\hat{j}$ (b) $\hat{i} + \hat{j}$

(c) $3\hat{i} + 2\hat{j}$ (d) $2\hat{i} + \hat{j}$

8. At what angle must the two forces $(x + y)$ and $(x - y)$ act so that the resultant may be $\sqrt{(x^2 + y^2)}$

(a) $\cos^{-1}\left(-\frac{x^2 + y^2}{2(x^2 - y^2)}\right)$ (b) $\cos^{-1}\left(-\frac{2(x^2 - y^2)}{x^2 + y^2}\right)$

(c) $\cos^{-1}\left(-\frac{x^2 + y^2}{x^2 - y^2}\right)$ (d) $\cos^{-1}\left(-\frac{x^2 - y^2}{x^2 + y^2}\right)$

9. Following forces start acting on a particle at rest at the origin of the co-ordinate system simultaneously

$\vec{F}_1 = -4\hat{i} - 5\hat{j} + 5\hat{k}$, $\vec{F}_2 = 5\hat{i} + 8\hat{j} + 6\hat{k}$, $\vec{F}_3 = -3\hat{i} + 4\hat{j} - 7\hat{k}$ and $\vec{F}_4 = 2\hat{i} - 3\hat{j} - 2\hat{k}$ then the particle will move

(a) In $x - y$ plane (b) In $y - z$ plane

(c) In $x - z$ plane (d) Along x-axis

10. The resultant of $\vec{A} + \vec{B}$ is $\vec{R}_1$. On reversing the vector $\vec{B}$, the resultant becomes $\vec{R}_2$. What is the value of $R_1^2 + R_2^2$

(a) $A^2 + B^2$ (b) $A^2 - B^2$

(c) $2(A^2 + B^2)$ (d) $2(A^2 - B^2)$

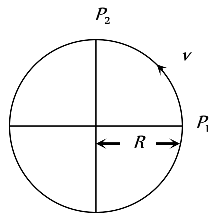
11. Figure below shows a body of mass M moving with the uniform speed on a circular path of radius, R. What is the change in acceleration in going from P_1 to P_2

(a) Zero

(b) $v^2/2R$

(c) $2v^2/R$

(d) $\frac{v^2}{R} \times \sqrt{2}$



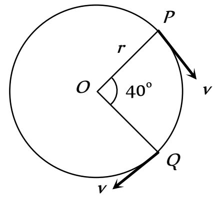
12. A particle is moving on a circular path of radius r with uniform velocity v. The change in velocity when the particle moves from P to Q is $(\angle POQ = 40°)$

(a) $2v\cos 40°$

(b) $2v\sin 40°$

(c) $2v\sin 20°$

(d) $2v\cos 20°$



13. $\vec{A} = 2\hat{i} + 4\hat{j} + 4\hat{k}$ and $\vec{B} = 4\hat{i} + 2\hat{j} - 4\hat{k}$ are two vectors. The angle between them will be

(a) 0° (b) 45°

(c) 60° (d) 90°

14. If $\vec{A} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{B} = -\hat{i} + 3\hat{j} + 4\hat{k}$ then projection of $\vec{A}$ on $\vec{B}$ will be

(a) $\frac{3}{\sqrt{13}}$ (b) $\frac{3}{\sqrt{26}}$

(c) $\sqrt{\dfrac{3}{26}}$ (d) $\sqrt{\dfrac{3}{13}}$

15. In above example a unit vector perpendicular to both $\vec{A}$ and $\vec{B}$ will be

(a) $+\dfrac{1}{\sqrt{3}}(\hat{i}-\hat{j}-\hat{k})$ (b) $-\dfrac{1}{\sqrt{3}}(\hat{i}-\hat{j}-\hat{k})$

(c) Both (a) and (b) (d) None of these

16. Two constant forces $F_1 = 2\hat{i} - 3\hat{j} + 3\hat{k}$ (*N*) and $F_2 = \hat{i} + \hat{j} - 2\hat{k}$ (*N*) act on a body and displace it from the position $r_1 = \hat{i} + 2\hat{j} - 2\hat{k}$ (m) to the position $r_2 = 7\hat{i} + 10\hat{j} + 5\hat{k}$ (m). What is the work done

(a) 9 *J* (b) 41 *J*

(c) − 3 *J* (d) None of these

17. For any two vectors $\vec{A}$ and $\vec{B}$, if $\vec{A}.\vec{B} = |\vec{A} \times \vec{B}|$, the magnitude of $\vec{C} = \vec{A} + \vec{B}$ is equal to

(a) $\sqrt{A^2 + B^2}$ (b) $A + B$

(c) $\sqrt{A^2 + B^2 + \dfrac{AB}{\sqrt{2}}}$ (d) $\sqrt{A^2 + B^2 + \sqrt{2} \times AB}$

18. Which of the following is the unit vector perpendicular to $\vec{A}$ and $\vec{B}$

(a) $\dfrac{\hat{A} \times \hat{B}}{AB\sin\theta}$ (b) $\dfrac{\hat{A} \times \hat{B}}{AB\cos\theta}$

(c) $\dfrac{\vec{A} \times \vec{B}}{AB\sin\theta}$ (d) $\dfrac{\vec{A} \times \vec{B}}{AB\cos\theta}$

19. Two vectors $P = 2\hat{i} + b\hat{j} + 2\hat{k}$ and $Q = \hat{i} + \hat{j} + \hat{k}$ will be parallel if

(a) $b = 0$ (b) $b = 1$

(c) $b = 2$ (d) $b = -4$

20. Which of the following is not true ? If $\vec{A} = 3\hat{i} + 4\hat{j}$ and $\vec{B} = 6\hat{i} + 8\hat{j}$ where *A* and *B* are the magnitudes of $\vec{A}$ and $\vec{B}$

(a) $\vec{A} \times \vec{B} = 0$ (b) $\dfrac{A}{B} = \dfrac{1}{2}$

(c) $\vec{A}.\vec{B} = 48$ (d) A = 5

21. The area of the triangle formed by $2\hat{i} + \hat{j} - \hat{k}$ and $\hat{i} + \hat{j} + \hat{k}$ is

(a) 3 sq.unit

(b) $2\sqrt{3}$ sq. unit

(c) $2\sqrt{14}$ sq. unit

(d) $\dfrac{\sqrt{14}}{2}$ sq. unit

22. Two trains along the same straight rails moving with constant speed 60 *km/hr* and 30 *km/hr* respectively towards each other. If at time $t = 0$, the distance between them is 90 *km,* the time when they collide is

(a) 1 *hr* (b) 2 *hr*

(c) 3 *hr* (d) 4 *hr*

23. A steam boat goes across a lake and comes back (a) On a quite day when the water is still and (b) On a rough day when there is uniform air current so as to help the journey onward and to impede the journey back. If the speed of the launch on both days was same, in which case it will complete the journey in lesser time

(a) Case (a)

(b) Case (b)

(c) Same in both

(d) Nothing can be predicted

24. To a person, going eastward in a car with a velocity of 25 *km/hr*, a train appears to move towards north with a velocity of $25\sqrt{3}$ *km/hr*. The actual velocity of the train will be

(a) 25 *km/hr* (b) 50 *km/hr*

(c) 5 *km/hr* (d) $5\sqrt{3}$ *km/hr*

25. A swimmer can swim in still water with speed υ and the river is flowing with velocity *v*/2. To cross the river in shortest distance, he should swim making angle θ with the upstream. What is the ratio of the time taken to swim across the shortest time to that is swimming across over shortest distance

(a) cos θ (b) sin θ

(c) tan θ (d) cot θ

26. A bus is moving with a velocity 10 m/s on a straight road. A scooterist wishes to overtake the bus in 100 *s*. If the bus is at a distance of 1 *km* from the scooterist, with what velocity should the scooterist chase the bus

(a) 50 *m/s* (b) 40 *m/s*

(c) 30 *m/s* (d) 20 *m/s*

AS Answers and Solutions

(SET -0)

1. (b) $\sqrt{(0.4)^2+(0.8)^2+c^2}=1$

$\Rightarrow 0.16+0.64+c^2=1 \Rightarrow c=\sqrt{0.2}$

2. (c) $\vec{R}=\hat{i}+\hat{j}+\sqrt{2}\hat{k}$

Comparing the given vector with $R=R_x\hat{i}+R_y\hat{j}+R_z\hat{k}$

$R_x=1,\ R_y=1,\ R_z=\sqrt{2}$ and $|\vec{R}|=\sqrt{R_x^2+R_y^2+R_z^2}=2$

$\cos\alpha=\frac{R_x}{R}=\frac{1}{2}\Rightarrow\alpha=60°$

$\cos\beta=\frac{R_y}{R}=\frac{1}{2}\Rightarrow\beta=60°$

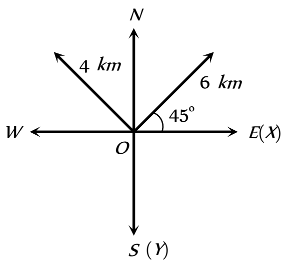
$\cos\gamma=\frac{R_z}{R}=\frac{1}{\sqrt{2}}\Rightarrow\gamma=45°$

3. (d) $\vec{A}=5\hat{i}+12\hat{j}$, $|\vec{A}|=\sqrt{5^2+(-12)^2}=\sqrt{25+144}=13$

Unit vector $\hat{A}=\frac{\vec{A}}{|\vec{A}|}=\frac{5\hat{i}-12\hat{j}}{13}$

4. (a)

5. (c)



Net movement along x-direction $S_x=(6-4)\cos 45°\ \hat{i}$

$=2\times\frac{1}{\sqrt{2}}=\sqrt{2}\ km$

Net movement along y-direction $S_y=(6+4)\sin 45°\ \hat{j}$

$=10\times\frac{1}{\sqrt{2}}=5\sqrt{2}\ km$

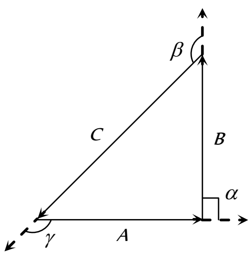
Net movement from starting point

$|\vec{S}|=\sqrt{S_x{}^2+S_y{}^2}=\sqrt{(\sqrt{2})^2+(5\sqrt{2})^2}=\sqrt{52}\ km$

Angle which makes with the east direction

$\tan\theta=\frac{\text{Y-component}}{\text{X-component}}=\frac{5\sqrt{2}}{\sqrt{2}}\ \therefore\ \theta=\tan^{-1}(5)$

6. (d)



From polygon law, three vectors having summation zero should form a closed polygon. (Triangle) since the two vectors are having same magnitude and the third vector is $\sqrt{2}$ times that of either of two having equal magnitude. *i.e.* the triangle should be right angled triangle

Angle between A and B, $\alpha=90°$

Angle between B and C, $\beta=135°$

Angle between A and C, $\gamma=135°$

7. (d) $x=0$ means y-axis $\Rightarrow \vec{F}_1=\hat{j}$

$y=0$ means x-axis $\Rightarrow \vec{F}_2=2\hat{i}$

so resultant $\vec{F}=\vec{F}_1+\vec{F}_2=2\hat{i}+\hat{j}$

8. (a) $R^2=A^2+B^2+2AB\cos\theta$

Substituting, $A=(x+y)$, $B=(x-y)$ and $R=\sqrt{(x^2+y^2)}$

we get $\theta=\cos^{-1}\left(-\frac{(x^2+y^2)}{2(x^2-y^2)}\right)$

9. (b) $F_1+\vec{F}_2+\vec{F}_3+\vec{F}_4$

$=(-4\hat{i}+5\hat{i}-3\hat{i}+2\hat{i})+(-5\hat{j}+8\hat{j}+4\hat{j}-3\hat{j})$

$+(5\hat{k}+6\hat{k}-7\hat{k}-2\hat{k})=4\hat{j}+2\hat{k}$

$\therefore$ the particle will move in $y-z$ plane.

10. (c) $\vec{R}_1=\vec{A}+\vec{B}$, $\vec{R}_2=\vec{A}-\vec{B}$

$R_1^2+R_2^2=\left(\sqrt{A^2+B^2}\right)^2+\left(\sqrt{A^2+B^2}\right)^2=2(A^2+B^2)$

11. (d) $\Delta a=2a\sin\left(\frac{\theta}{2}\right)=2a\times\sin 45°=\sqrt{2}a=\sqrt{2}\frac{v^2}{R}$

12. (b) $\Delta v=2v\sin\left(\frac{\theta}{2}\right)=2v\sin 20°$

13. (c) $\cos\theta=\frac{\vec{A}.\vec{B}}{|\vec{A}|.|\vec{B}|}=\frac{a_1b_1+a_2b_2+a_3b_3}{|\vec{A}|.|\vec{B}|}$

$=\frac{2\times4+4\times2-4\times4}{|\vec{A}|.|\vec{B}|}=0$

$\therefore\theta=\cos^{-1}(0°)\ \Rightarrow\ \theta=90°$

14. (b) $|\vec{A}|=\sqrt{2^2+3^2+(-1)^2}=\sqrt{4+9+1}=\sqrt{14}$

$|\vec{B}|=\sqrt{(-1)^2+3^2+4^2}=\sqrt{1+9+16}=\sqrt{26}$

$\vec{A}.\vec{B}=2(-1)+3\times3+(-1)(4)=3$

The projection of $\vec{A}$ on $\vec{B}$ $=\frac{\vec{A}.\vec{B}}{|\vec{B}|}=\frac{3}{\sqrt{26}}$

15. (c) $\hat{n}=\frac{\vec{A}\times\vec{B}}{|\vec{A}\times\vec{B}|}=\frac{8\hat{i}-8\hat{j}-8\hat{k}}{8\sqrt{3}}=\frac{1}{\sqrt{3}}(\hat{i}-\hat{j}-\hat{k})$

There are two unit vectors perpendicular to both $\vec{A}$ and $\vec{B}$ they are $\hat{n}=\pm\frac{1}{\sqrt{3}}(\hat{i}-\hat{j}-\hat{k})$

16. (a) $W = \bar{F}(\bar{r}_2 - \bar{r}_1)$

$= (3\hat{i} - 2\hat{j} + \hat{k})(6\hat{i} + 8\hat{j} + 7\hat{k}) = 18 - 16 + 7 = 9\,J$

17. (d) $AB\cos\theta = AB\sin\theta \Rightarrow \tan\theta = 1 \therefore \theta = 45°$

$\therefore |\bar{C}| = \sqrt{A^2 + B^2 + 2AB\cos 45°} = \sqrt{A^2 + B^2 + \sqrt{2}AB}$

18. (c) Vector perpendicular to A and B, $\vec{A} \times \vec{B} = AB\sin\theta\,\hat{n}$

$\therefore$ Unit vector perpendicular to A and B

$$\hat{n} = \frac{\vec{A} \times \vec{B}}{|\vec{A}| \times |\vec{B}|\,\sin\theta}$$

19. (c) P and Q will be parallel if $\frac{2}{1} = \frac{b}{1} = \frac{2}{1}$ $\therefore b = 2$

20. (b) $|\vec{A}| = 5$, $|\vec{B}| = 10 \Rightarrow \frac{A}{B} = \frac{1}{2}$

21. (d) $\vec{A} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{B} = \hat{i} + \hat{j} + \hat{k}$

Area of the triangle $= \frac{1}{2}\left(\vec{A} \times \vec{B}\right)$

$$= \frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 1 & 1 & 1 \end{vmatrix} = \frac{1}{2}\left|2\hat{i} - 3\hat{j} + \hat{k}\right| = \frac{1}{2}\sqrt{4 + 9 + 1}$$

$$= \frac{\sqrt{14}}{2}\,sq.unit$$

22. (a) The relative velocity $v_{rel.} = 60 - (-30) = 90\,km/hr.$

Distance between the train $s_{rel.} = 90\,km$,

$\therefore$ Time when they collide $= \frac{s_{rel.}}{v_{rel.}} = \frac{90}{90} = 1\,hr.$

23. (b) If the breadth of the lake is l and velocity of boat is v_b. Time in going and coming back on a quite day

$$t_Q = \frac{l}{v_b} + \frac{l}{v_b} = \frac{2l}{v_b} \qquad .....(i)$$

Now if v_a is the velocity of air- current then time taken in going across the lake,

$$t_1 = \frac{l}{v_b + v_a} \qquad \text{[As current helps the motion]}$$

and time taken in coming back $t_2 = \frac{l}{v_b - v_a}$

[As current opposes the motion]

So $t_R = t_1 + t_2 = \frac{2l}{v_b[1 - (v_a/v_b)^2]}$ .....(ii)

From equation (i) and (ii)

$$\frac{t_R}{t_Q} = \frac{1}{[1 - (v_a/v_b)^2]} > 1 \ \left[\text{as } 1 - \frac{v_a^2}{v_b^2} < 1\right] \quad i.e.\ t_R > t_Q$$

i.e. time taken to complete the journey on quite day is lesser than that on rough day.

24. (a) $v_T = \sqrt{v_{TC}^2 + v_C^2} = \sqrt{(25\sqrt{3})^2 + (25)^2}$

$= \sqrt{1875 + 625} = \sqrt{2500} = 25\,km/hr$

25. (b)

26. (d) Let the velocity of the scooterist $= v$

Relative velocity of scooterist with respect to bus $= (v - 10)$

$\Rightarrow S = (v - 10) \times 100 \Rightarrow 1000 = (v - 10) \times 100$

$\therefore v = 10 + 10 = 20\,m/s$

***

Chapter
1
Units, Dimensions and Measurement

Physical Quantity

A quantity which can be measured and by which various physical happenings can be explained and expressed in the form of laws is called a physical quantity. For example length, mass, time, force *etc.*

On the other hand various happenings in life *e.g.*, happiness, sorrow *etc.* are not physical quantities because these can not be measured.

Measurement is necessary to determine magnitude of a physical quantity, to compare two similar physical quantities and to prove physical laws or equations.

A physical quantity is represented completely by its magnitude and unit. For example, 10 *metre* means a length which is ten times the unit of length. Here 10 represents the numerical value of the given quantity and *metre* represents the unit of quantity under consideration. Thus in expressing a physical quantity we choose a unit and then find that how many times that unit is contained in the given physical quantity, *i.e.*

Physical quantity (Q) = Magnitude × Unit $= n \times u$

Where, n represents the numerical value and u represents the unit. Thus while expressing definite amount of physical quantity, it is clear that as the unit(u) changes, the magnitude(n) will also change but product 'nu' will remain same.

i.e. $n\,u$ = constant, or $n_1 u_1 = n_2 u_2 = \text{constant}$; $\therefore \quad n \propto \frac{1}{u}$

i.e. magnitude of a physical quantity and units are inversely proportional to each other .Larger the unit, smaller will be the magnitude.

(1) **Ratio (numerical value only) :** When a physical quantity is the ratio of two similar quantities, it has no unit.

e.g. Relative density = Density of object/Density of water at $4^\circ C$

Refractive index = Velocity of light in air/Velocity of light in medium

Strain = Change in dimension/Original dimension

(2) **Scalar (magnitude only) :** These quantities do not have any direction *e.g.* Length, time, work, energy *etc.*

Magnitude of a physical quantity can be negative. In that case negative sign indicates that the numerical value of the quantity under consideration is negative. It does not specify the direction.

Scalar quantities can be added or subtracted with the help of ordinary laws of addition or subtraction.

(3) **Vector (magnitude and direction) :** These quantities have magnitude and direction both and can be added or subtracted with the help of laws of vector algebra *e.g.* displacement, velocity, acceleration, force *etc.*

Fundamental and Derived Quantities

(1) **Fundamental quantities :** Out of large number of physical quantities which exist in nature, there are only few quantities which are independent of all other quantities and do not require the help of any other physical quantity for their definition, therefore these are called absolute quantities. These quantities are also called fundamental or basic quantities, as all other quantities are based upon and can be expressed in terms of these quantities.

(2) **Derived quantities :** All other physical quantities can be derived by suitable multiplication or division of different powers of fundamental quantities. These are therefore called derived quantities.

If length is defined as a fundamental quantity then area and volume are derived from length and are expressed in term of length with power 2 and 3 over the term of length.

Note : ❑ In mechanics, Length, Mass and Time are arbitrarily chosen as fundamental quantities. However this set of fundamental quantities is not a unique choice. In fact any three quantities in mechanics can be termed as fundamental as all other quantities in mechanics can be expressed in terms of these. *e.g.* if speed and time are taken as fundamental quantities, length will become a derived quantity because then length will be expressed as Speed × Time. and if force and acceleration are taken as fundamental quantities, then mass will be defined as Force / acceleration and will be termed as a derived quantity.

Fundamental and Derived Units

Normally each physical quantity requires a unit or standard for its specification so it appears that there must be as many units as there are physical quantities. However, it is not so. It has been found that if in *mechanics* we choose arbitrarily units of any *three* physical quantities we can express the units of all other physical quantities in mechanics in terms of these. Arbitrarily the physical quantities *mass*, *length* and *time* are chosen for this purpose. *So any unit of mass, length and time in mechanics is called a* **fundamental, absolute or base unit.** *Other units which can be expressed in terms of fundamental units, are called derived units.* For example light year or *km* is a fundamental unit as it is a unit of length while s^{-1}, m^2 or kg/m are derived units as these are derived from units of time, mass and length.

System of units : A complete set of units, both fundamental and derived for all kinds of physical quantities is called system of units. The common systems are given below

(1) **CGS system :** This system is also called Gaussian system of units. In this length, mass and time have been chosen as the fundamental quantities and corresponding fundamental units are centimetre (*cm*), gram (*g*) and second (*s*) respectively.

(2) **MKS system :** This system is also called Giorgi system. In this system also length, mass and time have been taken as fundamental quantities, and the corresponding fundamental units are *metre*, kilogram and second.

(3) **FPS system :** In this system foot, pound and second are used respectively for measurements of length, mass and time. In this system force is a derived quantity with unit poundal.

(4) **S. I. system :** It is known as International system of units, and is extended system of units applied to whole physics. There are seven fundamental quantities in this system. These quantities and their units are given in the following table

Table 1.1 : Unit and symbol of quantities

Quantity	Unit	Symbol
Length	metre	*m*
Mass	kilogram	*kg*
Time	second	*s*
Electric Current	ampere	*A*
Temperature	Kelvin	*K*
Amount of Substance	mole	*mol*
Luminous Intensity	candela	*cd*

Besides the above seven fundamental units two supplementary units are also defined –

Radian (*rad*) for plane angle and Steradian (*sr*) for solid angle.

Note : ❑ Apart from fundamental and derived units we also use practical units very frequently. These may be fundamental or derived units *e.g.*, light year is a practical unit (fundamental) of distance while horse power is a practical unit (derived) of power.

❑ Practical units may or may not belong to a system but can be expressed in any system of units

e.g., 1 mile = 1.6 *km* = 1.6×10^3 *m*.

S.I. Prefixes

In physics we deal from very small (*micro*) to very large (*macro*) magnitudes, as one side we talk about the atom while on the other side of universe, *e.g.*, the mass of an electron is 9.1×10^{-31} *kg* while that of the sun is 2×10^{30} *kg*. To express such large or small magnitudes we use the following prefixes :

Table 1.2 : Prefixes and symbol

Power of 10	Prefix	Symbol
10^{18}	exa	*E*
10^{15}	peta	*P*
10^{12}	tera	*T*
10^{9}	giga	*G*
10^{6}	mega	*M*
10^{3}	kilo	*k*
10^{2}	hecto	*h*
10^{1}	deca	*da*
10^{-1}	deci	*d*
10^{-2}	centi	*c*
10^{-3}	milli	*m*
10^{-6}	micro	μ
10^{-9}	nano	*n*
10^{-12}	pico	*p*
10^{-15}	femto	*f*
10^{-18}	atto	*a*

Standards of Length, Mass and Time

(1) **Length :** Standard metre is defined in terms of wavelength of light and is called atomic standard of length.

The metre is the distance containing 1650763.73 wavelength in vacuum of the radiation corresponding to orange red light emitted by an atom of krypton-86.

Now a days metre is defined as length of the path travelled by light in vacuum in 1/299,7792, 45 part of a second.

(2) **Mass :** The mass of a cylinder made of platinum-iridium alloy kept at International Bureau of Weights and Measures is defined as 1 *kg*.

On atomic scale, 1 *kilogram* is equivalent to the mass of 5.0188×10^{25} atoms of ${}_6C^{12}$ (an isotope of carbon).

(3) **Time :** 1 *second* is defined as the time interval of 9192631770 vibrations of radiation in *Cs*-133 atom. This radiation corresponds to the transition between two hyperfine level of the ground state of *Cs*-133.

Practical Units

(1) **Length**

(i) 1 fermi = 1 *fm* = 10^{-15} *m*

(ii) 1 *X*-ray unit = 1*XU* = 10^{-13} *m*

(iii) 1 angstrom = 1Å = 10^{-10} *m* = 10^{-8} *cm* = 10^{-7} *mm* = 0.1 μmm

(iv) 1 micron = μm = 10^{-6} *m*

(v) 1 astronomical unit = 1 *A.U.* = 1.49×10^{11} *m*

$\approx 1.5 \times 10^{11}$ *m* $\approx 10^{8}$ *km*

(vi) 1 Light year = 1 *ly* = 9.46×10^{15} *m*

(vii) 1 *Parsec* = 1*pc* = 3.26 light year

(2) **Mass**

(i) Chandra Shekhar unit : 1 *CSU* = 1.4 times the mass of sun = 2.8×10^{30} *kg*

(ii) Metric tonne : 1 Metric tonne = 1000 *kg*

(iii) Quintal : 1 Quintal = 100 *kg*

(iv) Atomic mass unit (*amu*) : *amu* = 1.67×10^{-27} *kg*

Mass of proton or neutron is of the order of 1 *amu*

(3) **Time**

(i) Year : It is the time taken by the Earth to complete 1 revolution around the Sun in its orbit.

(ii) Lunar month : It is the time taken by the Moon to complete 1 revolution around the Earth in its orbit.

1 *L.M.* = 27.3 days

(iii) Solar day : It is the time taken by Earth to complete one rotation about its axis with respect to Sun. Since this time varies from day to day, average solar day is calculated by taking average of the duration of all the days in a year and this is called Average Solar day.

1 Solar year = 365.25 average solar day

or average solar day $= \frac{1}{365.25}$ the part of solar year

(iv) Sedrial day : It is the time taken by earth to complete one rotation about its axis with respect to a distant star.

1 Solar year = 366.25 Sedrial day

= 365.25 average solar day

Thus 1 Sedrial day is less than 1 solar day.

(v) Shake : It is an obsolete and practical unit of time.

1 Shake = 10^{-8} *sec*

Dimensions

When a derived quantity is expressed in terms of fundamental quantities, it is written as a product of different powers of the fundamental quantities. The powers to which fundamental quantities must be raised in order to express the given physical quantity are called its dimensions.

To make it more clear, consider the physical quantity force

Force = mass × acceleration

$$= \frac{\text{mass} \times \text{velocity}}{\text{time}}$$

$$= \frac{\text{mass} \times \text{length/time}}{\text{time}}$$

= mass × length × (time)$^{-2}$ ... (i)

Thus, the dimensions of force are 1 in mass, 1 in length and – 2 in time.

Here the physical quantity that is expressed in terms of the basic quantities is enclosed in square brackets to indicate that the equation is among the dimensions and not among the magnitudes.

Thus equation (i) can be written as [force] = $[MLT^{-2}]$.

Such an expression for a physical quantity in terms of the fundamental quantities is called the dimensional equation. If we consider only the R.H.S. of the equation, the expression is termed as dimensional formula.

Thus, dimensional formula for force is, $[MLT^{-2}]$.

Quantities Having same Dimensions

Dimension	Quantity
$[M^0L^0T^{-1}]$	Frequency, angular frequency, angular velocity, velocity gradient and decay constant
$[M^1L^2T^{-2}]$	Work, internal energy, potential energy, kinetic energy, torque, moment of force
$[M^1L^{-1}T^{-2}]$	Pressure, stress, Young's modulus, bulk modulus, modulus of rigidity, energy density
$[M^1L^1T^{-1}]$	Momentum, impulse
$[M^0L^1T^{-2}]$	Acceleration due to gravity, gravitational field intensity
$[M^1L^1T^{-2}]$	Thrust, force, weight, energy gradient
$[M^1L^2T^{-1}]$	Angular momentum and Planck's constant
$[M^1L^0T^{-2}]$	Surface tension, Surface energy (energy per unit area)
$[M^0L^0T^0]$	Strain, refractive index, relative density, angle, solid angle, distance gradient, relative permittivity (dielectric constant), relative permeability etc.
$[M^0L^2T^{-2}]$	Latent heat and gravitational potential
$[ML^2T^{-2}\theta^{-1}]$	Thermal capacity, gas constant, Boltzmann constant and entropy
$[M^0L^0T^1]$	$\sqrt{l/g}, \sqrt{m/k}, \sqrt{R/g}$, where l = length g = acceleration due to gravity, m = mass, k = spring constant, R = Radius of earth
$[M^0L^0T^1]$	L/R, $\sqrt{LC}$, RC where L = inductance, R = resistance, C = capacitance
$[ML^2T^{-2}]$	$I^2Rt, \frac{V^2}{R}t, VIt, qV, LI^2, \frac{q^2}{C}, CV^2$ where I = current, t = time, q = charge, L = inductance, C = capacitance, R = resistance

Important Dimensions of Complete Physics

Heat

Quantity	Unit	Dimension
Temperature (T)	*Kelvin*	$[M^0L^0T^0\theta^1]$
Heat (Q)	*Joule*	$[ML^2T^{-2}]$
Specific Heat (c)	*Joule/kg-K*	$[M^0L^2T^{-2}\theta^{-1}]$
Thermal capacity	*Joule/K*	$[ML^2T^{-2}\theta^{-1}]$
Latent heat (L)	*Joule/kg*	$[M^0L^2T^{-2}]$
Gas constant (R)	*Joule/mol-K*	$[ML^2T^{-2}\theta^{-1}]$
Boltzmann constant (k)	*Joule/K*	$[ML^2T^{-2}\theta^{-1}]$
Coefficient of thermal conductivity (K)	*Joule/m-s-K*	$[MLT^{-3}\theta^{-1}]$
Stefan's constant (σ)	*Watt/m^2-K^4*	$[ML^0T^{-3}\theta^{-4}]$
Wien's constant (b)	*Metre-K*	$[M^0LT^0\theta]$
Planck's constant (h)	*Joule-s*	$[ML^2T^{-1}]$
Coefficient of Linear Expansion (α)	*Kelvin^{-1}*	$[M^0L^0T^0\theta^{-1}]$
Mechanical equivalent of Heat (J)	*Joule/Calorie*	$[M^0L^0T^0]$
Vander wall's constant (a)	*Newton-m^4*	$[ML^5T^{-2}]$
Vander wall's constant (b)	*m^3*	$[M^0L^3T^0]$

Electricity

Quantity	Unit	Dimension
Electric charge (q)	*Coulomb*	$[M^0L^0T^1A^1]$
Electric current (I)	*Ampere*	$[M^0L^0T^0A^1]$
Capacitance (C)	*Coulomb/volt or Farad*	$[M^{-1}L^{-2}T^4A^2]$
Electric potential (V)	*Joule/coulomb*	$[ML^2T^{-3}A^{-1}]$
Permittivity of free space (ε_0)	$\frac{Coulomb^2}{Newton\text{-}metre^2}$	$[M^{-1}L^{-3}T^4A^2]$
Dielectric constant (K)	Unitless	$[M^0L^0T^0]$
Resistance (R)	*Volt/Ampere* or *ohm*	$[ML^2T^{-3}A^{-2}]$

Quantity	Unit	Dimension
Resistivity or Specific resistance (ρ)	*Ohm-metre*	$[ML^3T^{-3}A^{-2}]$
Coefficient of Self-induction (L)	$\frac{volt-second}{ampere}$ or *henry* or *ohm-second*	$[ML^2T^{-2}A^{-2}]$
Magnetic flux (ϕ)	*Volt-second* or *weber*	$[ML^2T^{-2}A^{-1}]$
Magnetic induction (B)	$\frac{newton}{ampere-metre}$ $\frac{Joule}{ampere-metre^2}$ $\frac{volt-\text{second}}{metre^2}$ or *Tesla*	$[ML^0T^{-2}A^{-1}]$
Magnetic Intensity (H)	*Ampere/metre*	$[M^0L^{-1}T^0A^1]$
Magnetic Dipole Moment (M)	*Ampere-metre*	$[M^0L^2T^0A^1]$
Permeability of Free Space (μ)	$\frac{Newton}{ampere^2}$ or $\frac{Joule}{ampere^2-metre}$ or $\frac{Volt-\text{second}}{ampere-metre}$ or $\frac{Ohm-\text{sec}ond}{metre}$ or $\frac{henry}{metre}$	$[MLT^{-2}A^{-2}]$
Surface charge density (σ)	$Coulomb\,metre^{-2}$	$[M^0L^{-2}TA^1]$
Electric dipole moment (p)	$Coulomb-metre$	$[M^0LTA^1]$
Conductance (G) ($1/R$)	ohm^{-1}	$[M^{-1}L^{-2}T^3A^2]$
Conductivity (σ) ($1/\rho$)	$ohm^{-1}metre^{-1}$	$[M^{-1}L^{-3}T^3A^2]$
Current density (J)	*Ampere/m*^{2}	$M^0L^{-2}T^0A^1$
Intensity of electric field (E)	*Volt/metre, Newton/coulomb*	$ML^1T^{-3}A^{-1}$
Rydberg constant (R)	m^{-1}	$M^0L^{-1}T^0$

Application of Dimensional Analysis

(1) **To find the unit of a physical quantity in a given system of units** : To write the definition or formula for the physical quantity we find its dimensions. Now in the dimensional formula replacing *M*, *L* and *T* by the fundamental units of the required system we get the unit of physical quantity. However, sometimes to this unit we further assign a specific name,

e.g., Work = Force × Displacement

So $[W] = [MLT^{-2}] \times [L] = [ML^2T^{-2}]$

So its unit in C.G.S. system will be *g cm²/s²* which is called *erg* while in M.K.S. system will be *kg-m²/s²* which is called *joule*.

(2) **To find dimensions of physical constant or coefficients** : As dimensions of a physical quantity are unique, we write any formula or equation incorporating the given constant and then by substituting the dimensional formulae of all other quantities, we can find the dimensions of the required constant or coefficient.

(i) Gravitational constant : According to Newton's law of gravitation $F = G\frac{m_1m_2}{r^2}$ or $G = \frac{Fr^2}{m_1m_2}$

Substituting the dimensions of all physical quantities

$$[G] = \frac{[MLT^{-2}][L^2]}{[M][M]} = [M^{-1}L^3T^{-2}]$$

(ii) Plank constant : According to Planck $E = h\nu$ or $h = \frac{E}{\nu}$

Substituting the dimensions of all physical quantities

$$[h] = \frac{[ML^2T^{-2}]}{[T^{-1}]} = [ML^2T^{-1}]$$

(iii) Coefficient of viscosity : According to Poiseuille's formula $\frac{dV}{dt} = \frac{\pi pr^4}{8\eta l}$ or $\eta = \frac{\pi pr^4}{8l(dV/dt)}$

Substituting the dimensions of all physical quantities

$$[\eta] = \frac{[ML^{-1}T^{-2}][L^4]}{[L][L^3/T]} = [ML^{-1}T^{-1}]$$

(3) **To convert a physical quantity from one system to the other** : The measure of a physical quantity is nu = constant

If a physical quantity X has dimensional formula $[M^aL^bT^c]$ and if (derived) units of that physical quantity in two systems are $[M_1^aL_1^bT_1^c]$ and $[M_2^aL_2^bT_2^c]$ respectively and n_1 and n_2 be the numerical values in the two systems respectively, then $n_1[u_1] = n_2[u_2]$

$$\Rightarrow n_1[M_1^aL_1^bT_1^c] = n_2[M_2^aL_2^bT_2^c]$$

$$\Rightarrow n_2 = n_1\left[\frac{M_1}{M_2}\right]^a\left[\frac{L_1}{L_2}\right]^b\left[\frac{T_1}{T_2}\right]^c$$

where M_1, L_1 and T_1 are fundamental units of mass, length and time in the first (known) system and M_2, L_2 and T_2 are fundamental units of mass, length and time in the second (unknown) system. Thus knowing the values of fundamental units in two systems and numerical value in one system, the numerical value in other system may be evaluated.

Example : (i) conversion of *Newton* into *Dyne*.

The Newton is the S.I. unit of force and has dimensional formula $[MLT^{-2}]$.

So 1 *N* = 1 *kg-m/sec²*

By using $n_2 = n_1\left[\frac{M_1}{M_2}\right]^a\left[\frac{L_1}{L_2}\right]^b\left[\frac{T_1}{T_2}\right]^c$

$$= 1\left[\frac{kg}{gm}\right]^1\left[\frac{m}{cm}\right]^1\left[\frac{sec}{sec}\right]^{-2}$$

$$= 1\left[\frac{10^3\,gm}{gm}\right]^1\left[\frac{10^2\,cm}{cm}\right]^1\left[\frac{sec}{sec}\right]^{-2} = 10^5$$

∴ 1 *N* = 10^5 *Dyne*

(ii) Conversion of gravitational constant (G) from C.G.S. to M.K.S. system

The value of G in C.G.S. system is 6.67×10^{-8} C.G.S. units while its dimensional formula is $[M^{-1}L^{3}T^{-2}]$

So $G = 6.67 \times 10^{-8}$ $cm^3/g\ s^2$

By using $n_2 = n_1 \left[\frac{M_1}{M_2}\right]^a \left[\frac{L_1}{L_2}\right]^b \left[\frac{T_1}{T_2}\right]^c$

$$= 6.67 \times 10^{-8} \left[\frac{gm}{kg}\right]^{-1} \left[\frac{cm}{m}\right]^{3} \left[\frac{sec}{sec}\right]^{-2}$$

$$= 6.67 \times 10^{-8} \left[\frac{gm}{10^3 gm}\right]^{-1} \left[\frac{cm}{10^2 cm}\right]^{3} \left[\frac{sec}{sec}\right]^{-2}$$

$$= 6.67 \times 10^{-11}$$

$\therefore$ $G = 6.67 \times 10^{-11}$ M.K.S. units

(4) **To check the dimensional correctness of a given physical relation :** This is based on the '*principle of homogeneity*'. According to this principle the dimensions of each term on both sides of an equation must be the same.

If $X = A \pm (BC)^2 \pm \sqrt{DEF}$,

then according to principle of homogeneity

$[X] = [A] = [(BC)^2] = [\sqrt{DEF}]$

If the dimensions of each term on both sides are same, the equation is dimensionally correct, otherwise not. A dimensionally correct equation may or may not be physically correct.

Example : (i) $F = mv^2/r^2$

By substituting dimension of the physical quantities in the above relation, $[MLT^{-2}] = [M][LT^{-1}]^2/[L]^2$

i.e. $[MLT^{-2}] = [MT^{-2}]$

As in the above equation dimensions of both sides are not same; this formula is not correct dimensionally, so can never be physically.

(ii) $s = ut - (1/2)at^2$

By substituting dimension of the physical quantities in the above relation

$[L] = [LT^{-1}][T] - [LT^{-2}][T^2]$

i.e. $[L] = [L] - [L]$

As in the above equation dimensions of each term on both sides are same, so this equation is dimensionally correct. However, from equations of motion we know that $s = ut + (1/2)at^2$

(5) **As a research tool to derive new relations :** If one knows the dependency of a physical quantity on other quantities and if the dependency is of the product type, then using the method of dimensional analysis, relation between the quantities can be derived.

Example : (i) Time period of a simple pendulum.

Let time period of a simple pendulum is a function of mass of the bob (m), effective length (l), acceleration due to gravity (g) then assuming the function to be product of power function of m, l and g

i.e., $T = Km^x l^y g^z$; where K = dimensionless constant

If the above relation is dimensionally correct then by substituting the dimensions of quantities –

$[T] = [M]^x [L]^y [LT^{-2}]^z$ or $[M^0L^0T^1] = [M^xL^{y+z}T^{-2z}]$

Equating the exponents of similar quantities $x = 0$, $y = 1/2$ and $z = -1/2$

So the required physical relation becomes $T = K\sqrt{\frac{l}{g}}$

The value of dimensionless constant is found (2π) through experiments so $T = 2\pi\sqrt{\frac{l}{g}}$

(ii) Stoke's law : When a small sphere moves at low speed through a fluid, the viscous force F, opposes the motion, is found experimentally to depend on the radius r, the velocity of the sphere v and the viscosity η of the fluid.

So $F = f(\eta, r, v)$

If the function is product of power functions of η, r and v, $F = K\eta^x r^y v^z$; where K is dimensionless constant.

If the above relation is dimensionally correct

$[MLT^{-2}] = [ML^{-1}T^{-1}]^x [L]^y [LT^{-1}]^z$

or $[MLT^{-2}] = [M^x L^{-x+y+z} T^{-x-z}]$

Equating the exponents of similar quantities

$x = 1$; $-x + y + z = 1$ and $-x - z = -2$

Solving these for x, y and z, we get $x = y = z = 1$

So equation (i) becomes $F = K\eta rv$

On experimental grounds, $K = 6\pi$; so $F = 6\pi\eta rv$

This is the famous Stoke's law.

Limitations of Dimensional Analysis

Although dimensional analysis is very useful it cannot lead us too far as,

(1) If dimensions are given, physical quantity may not be unique as many physical quantities have same dimensions. For example if the dimensional formula of a physical quantity is $[ML^2T^{-2}]$ it may be work or energy or torque.

(2) Numerical constant having no dimensions $[K]$ such as (1/2), 1 or 2π *etc.* cannot be deduced by the methods of dimensions.

(3) The method of dimensions can not be used to derive relations other than product of power functions. For example,

$$s = ut + (1/2)at^2 \text{ or } y = a\sin\omega t$$

cannot be derived by using this theory (try if you can). However, the dimensional correctness of these can be checked.

(4) The method of dimensions cannot be applied to derive formula if in mechanics a physical quantity depends on more than 3 physical quantities as then there will be less number (= 3) of equations than the unknowns (>3). However still we can check correctness of the given equation dimensionally. For example $T = 2\pi\sqrt{I/mgl}$ can not be derived by theory of dimensions but its dimensional correctness can be checked.

(5) Even if a physical quantity depends on 3 physical quantities, out of which two have same dimensions, the formula cannot be derived by theory of dimensions, *e.g.*, formula for the frequency of a tuning fork $f = (d/L^2)v$ cannot be derived by theory of dimensions but can be checked.

Significant Figures

Significant figures in the measured value of a physical quantity tell the number of digits in which we have confidence. Larger the number of significant figures obtained in a measurement, greater is the accuracy of the measurement. The reverse is also true.

The following rules are observed in counting the number of significant figures in a given measured quantity.

(1) All non-zero digits are significant.

Example : 42.3 has three significant figures.

243.4 has four significant figures.

24.123 has five significant figures.

(2) A zero becomes significant figure if it appears between two non-zero digits.

Example : 5.03 has three significant figures.

5.604 has four significant figures.

4.004 has four significant figures.

(3) Leading zeros or the zeros placed to the left of the number are never significant.

Example : 0.543 has three significant figures.

0.045 has two significant figures.

0.006 has one significant figure.

(4) Trailing zeros or the zeros placed to the right of the number are significant.

Example : 4.330 has four significant figures.

433.00 has five significant figures.

343.000 has six significant figures.

(5) In exponential notation, the numerical portion gives the number of significant figures.

Example : 1.32×10^{-2} has three significant figures.

1.32×10^{4} has three significant figures.

Rounding Off

While rounding off measurements, we use the following rules by convention:

(1) If the digit to be dropped is less than 5, then the preceding digit is left unchanged.

Example : $x = 7.82$ is rounded off to 7.8,

again $x = 3.94$ is rounded off to 3.9.

(2) If the digit to be dropped is more than 5, then the preceding digit is raised by one.

Example : x = 6.87 is rounded off to 6.9,

again x = 12.78 is rounded off to 12.8.

(3) If the digit to be dropped is 5 followed by digits other than zero, then the preceding digit is raised by one.

Example : x = 16.351 is rounded off to 16.4,

again x = 6.758 is rounded off to 6.8.

(4) If digit to be dropped is 5 or 5 followed by zeros, then preceding digit is left unchanged, if it is even.

Example : x = 3.250 becomes 3.2 on rounding off,

again x = 12.650 becomes 12.6 on rounding off.

(5) If digit to be dropped is 5 or 5 followed by zeros, then the preceding digit is raised by one, if it is odd.

Example : x = 3.750 is rounded off to 3.8,

again x = 16.150 is rounded off to 16.2.

Significant Figures in Calculation

In most of the experiments, the observations of various measurements are to be combined mathematically, *i.e.*, added, subtracted, multiplied or divided to achieve the final result. Since, all the observations in measurements do not have the same precision, it is natural that the final result cannot be more precise than the least precise measurement. The following two rules should be followed to obtain the proper number of significant figures in any calculation.

(1) The result of an addition or subtraction in the number having different precisions should be reported to the same number of decimal places as present in the number having the least number of decimal places. The rule is illustrated by the following examples :

(i)
$$\begin{array}{rl} 33.3 & \leftarrow \text{(has only one decimal place)} \\ 3.11 & \\ +\ 0.313 & \\ \hline 36.723 & \leftarrow \text{(answer should be reported to one decimal place)} \\ \hline \end{array}$$

Answer = 36.7

(ii)
$$\begin{array}{rl} 3.1421 & \\ 0.241 & \\ +\ 0.09 & \leftarrow \text{(has 2 decimal places)} \\ \hline 3.4731 & \leftarrow \text{(answer should be reported to 2 decimal places)} \\ \hline \end{array}$$

Answer = 3.47

(iii)
$$\begin{array}{rl} 62.831 & \leftarrow \text{(has 3 decimal places)} \\ -\ 24.5492 & \\ \hline 38.2818 & \leftarrow \text{(answer should be reported to 3 decimal places after rounding off)} \\ \hline \end{array}$$

Answer = 38.282

(2) The answer to a multiplication or division is rounded off to the same number of significant figures as possessed by the least precise term used in the calculation. The rule is illustrated by the following examples :

(i)
$$\begin{array}{rl} 142.06 & \\ \times\ 0.23 & \leftarrow \text{(two significant figures)} \\ \hline 32.6738 & \leftarrow \text{(answer should have two significant figures)} \\ \hline \end{array}$$

Answer = 33

(ii)
$$\begin{array}{rl} 51.028 & \\ \times\ 1.31 & \leftarrow \text{(three significant figures)} \\ \hline 66.84668 & \\ \hline \end{array}$$

Answer = 66.8

(iii) $\dfrac{0.90}{4.26} = 0.2112676$

Answer = 0.21

Order of Magnitude

In scientific notation the numbers are expressed as, Number $= M \times 10^{x}$. Where M is a number lies between 1 and 10 and x is integer. Order of magnitude of quantity is the power of 10 required to represent the quantity. For determining this power, the value of the quantity has to be rounded off. While rounding off, we ignore the last digit which is less than 5. If the last digit is 5 or more than five, the preceding digit is increased by one. For example,

(1) Speed of light in vacuum

$= 3 \times 10^8 ms^{-1} \approx 10^8 m/s$ (ignoring 3 < 5)

(2) Mass of electron $= 9.1 \times 10^{-31} kg \approx 10^{-30} kg$ (as 9.1 > 5).

Errors of Measurement

The measuring process is essentially a process of comparison. Inspite of our best efforts, the measured value of a quantity is always somewhat different from its actual value, or true value. This difference in the true value and measured value of a quantity is called error of measurement.

(1) **Absolute error :** Absolute error in the measurement of a physical quantity is the magnitude of the difference between the true value and the measured value of the quantity.

Let a physical quantity be measured n times. Let the measured value be $a_1, a_2, a_3, \ldots\ldots a_n$. The arithmetic mean of these value is $a_m = \dfrac{a_1 + a_2 + \ldots\ldots + a_n}{n}$

Usually, a_m is taken as the true value of the quantity, if the same is unknown otherwise.

By definition, absolute errors in the measured values of the quantity are

$$\Delta a_1 = a_m - a_1$$

$$\Delta a_2 = a_m - a_2$$

....

$$\Delta a_n = a_m - a_n$$

The absolute errors may be positive in certain cases and negative in certain other cases.

(2) **Mean absolute error :** It is the arithmetic mean of the magnitudes of absolute errors in all the measurements of the quantity. It is represented by $\overline{\Delta a}$. Thus

$$\overline{\Delta a} = \frac{|\Delta a_1| + |\Delta a_2| + \ldots\ldots |\Delta a_n|}{n}$$

Hence the final result of measurement may be written as $a = a_m \pm \overline{\Delta a}$

This implies that any measurement of the quantity is likely to lie between $(a_m + \overline{\Delta a})$ and $(a_m - \overline{\Delta a})$.

(3) **Relative error or Fractional error :** The relative error or fractional error of measurement is defined as the ratio of mean absolute error to the mean value of the quantity measured. Thus

Relative error or Fractional error $= \dfrac{\text{Mean absolute error}}{\text{Mean value}} = \dfrac{\overline{\Delta a}}{a_m}$

(4) **Percentage error :** When the relative/fractional error is expressed in percentage, we call it percentage error. Thus

Percentage error $= \dfrac{\overline{\Delta a}}{a_m} \times 100\%$

Propagation of Errors

(1) **Error in sum of the quantities :** Suppose $x = a + b$

Let Δa = absolute error in measurement of a

Δb = absolute error in measurement of b

Δx = absolute error in calculation of x

i.e. sum of a and b.

The maximum absolute error in x is $\Delta x = \pm(\Delta a + \Delta b)$

Percentage error in the value of $x = \dfrac{(\Delta a + \Delta b)}{a + b} \times 100\%$

(2) **Error in difference of the quantities :** Suppose $x = a - b$

Let Δa = absolute error in measurement of a,

Δb = absolute error in measurement of b

Δx = absolute error in calculation of x *i.e.* difference of a and b.

The maximum absolute error in x is $\Delta x = \pm(\Delta a + \Delta b)$

Percentage error in the value of $x = \dfrac{(\Delta a + \Delta b)}{a - b} \times 100\%$

(3) **Error in product of quantities :**

Suppose $x = a \times b$

Let Δa = absolute error in measurement of a,

Δb = absolute error in measurement of b

Δx = absolute error in calculation of x *i.e.* product of a and b.

The maximum fractional error in x is $\dfrac{\Delta x}{x} = \pm\left(\dfrac{\Delta a}{a} + \dfrac{\Delta b}{b}\right)$

Percentage error in the value of x

= (% error in value of a) + (% error in value of b)

(4) **Error in division of quantities :** Suppose $x = \dfrac{a}{b}$

Let Δa = absolute error in measurement of a,

Δb = absolute error in measurement of b

Δx = absolute error in calculation of x *i.e.* division of a and b.

The maximum fractional error in x is $\dfrac{\Delta x}{x} = \pm\left(\dfrac{\Delta a}{a} + \dfrac{\Delta b}{b}\right)$

Percentage error in the value of x

= (% error in value of a) + (% error in value of b)

(5) **Error in quantity raised to some power :** Suppose $x = \dfrac{a^n}{b^m}$

Let Δa = absolute error in measurement of a,

Δb = absolute error in measurement of b

Δx = absolute error in calculation of x

The maximum fractional error in x is $\dfrac{\Delta x}{x} = \pm\left(n\dfrac{\Delta a}{a} + m\dfrac{\Delta b}{b}\right)$

Percentage error in the value of x

= n (% error in value of a) + m (%error in value of b)

Tips & Tricks

✍ The standard of Weight and Measures Act was passed in India in 1976. It recommended the use of SI in all fields of science, technology, trade and industry.

✍ The dimensions of many physical quantities, especially those in heat, thermodynamics, electricity and magnetism in terms of mass, length and time alone become irrational. Therefore, SI is adopted which uses 7 basic units.

✍ The dimensions of a physical quantity are the powers to which basic units (not fundamental units alone) should be raised to represent the derived unit of that physical quantity.

✍ The dimensional formula is very helpful in writing the unit of a physical quantity in terms of the basic units.

✍ The dimensions of a physical quantity do not depend on the system of units.

✍ A physical quantity that does not have any unit must be dimensionless.

✍ The pure numbers are dimensionless.

✍ Generally, the symbols of those basic units, whose dimension (power) in the dimensional formula is zero, are omitted from the dimensional formula.

✍ It is wrong to say that the dimensions of force are MLT^{-2}. On the other hand we should say that the dimensional formula for force is MLT^{-2} and that the dimensions of force are 1 in mass, 1 in length and –2 in time.

✍ Physical quantities defined as the ratio of two similar quantities are dimensionless.

✍ The physical relation involving logarithm, exponential, trigonometric ratios, numerical factors etc. cannot be derived by the method of dimensional analysis.

✍ Physical relations involving addition or subtraction sign cannot be derived by the method of dimensional analysis.

✍ If units or dimensions of two physical quantities are same, these need not represent the same physical characteristics. For example torque and work have the same units and dimensions but their physical characteristics are different.

✍ The standard units must not change with space and time. That is why atomic standard of length and time have been defined. Attempts are being made to define the atomic standard for mass as well.

✍ The unit of time, the second, was initially defined in terms of the rotation of the earth around the sun as well as that about its own axis. This time standard is subjected to variation with time. Therefore, the atomic standard of time has been defined.

✍ Any repetitive phenomenon, such as an oscillating pendulum, spinning of earth about its axis, etc can be used to measure time.

✍ The product of numerical value of the physical quantity (n) and its unit (U) remains constant.

That is : nU = constant or $n_1U_1 = n_2U_2$.

✍ The product of numerical value (n) and unit (U) of a physical quantity is called magnitude of the physical quantity.

Thus : Magnitude = nU

✍ Poiseuille (unit of viscosity) = pascal (unit of pressure) × second. That is : $Pl : Pa\text{-}s$.

✍ The unit of power of lens (dioptre) gives the ability of the lens to converge or diverge the rays refracted through it.

✍ The order of magnitude of a quantity means its value (in suitable power of 10) nearest to the actual value of the quantity.

✍ Angle is exceptional physical quantity, which though is a ratio of two similar physical quantities (angle = arc / radius) but still requires a unit (degrees or radians) to specify it along with its numerical value.

✍ Solid angle subtended at a point inside the closed surface is 4π steradian.

✍ A measurement of a physical quantity is said to be accurate if the systematic error in its measurement is relatively very low. On the other hand, the measurement of a physical quantity is said to be precise if the random error is small.

✍ A measurement is most accurate if its observed value is very close to the true value.

✍ Errors are always additive in nature.

✍ For greater accuracy, the quantity with higher power should have least error.

✍ The absolute error in each measurement is equal to the least count of the measuring instrument.

✍ Percentage error = relative error × 100.

✍ The unit and dimensions of the absolute error are same as that of quantity itself.

✍ Absolute error is not dimensionless quantity.

✍ Relative error is dimensionless quantity.

✍ Least Count = $\dfrac{\text{value of 1 part on main scale}(s)}{\text{Number of parts on vernier scale}(n)}$

✍ Least count of vernier callipers

$$= \left\{\begin{matrix}\text{value of 1 part of} \\ \text{main scale}(s)\end{matrix}\right\} - \left\{\begin{matrix}\text{value of 1 part of} \\ \text{vernier scale}(v)\end{matrix}\right\}$$

$\Rightarrow$ Least count of vernier calliper = 1 MSD – 1 VSD

where MSD = Main Scale Division

VSD = Vernier Scale Division

✍ Least count of screw guaze = $\dfrac{\text{Pitch}(p)}{\text{No. of parts on circular scale}(n)}$

✍ Smaller the least count, higher is the accuracy of measurement.

✍ Larger the number of significant figures after the decimal in a measurement, higher is the accuracy of measurement.

✍ Significant figures do not change if we measure a physical quantity in different units.

✍ Significant figures retained after mathematical operation (like addition, subtraction, multiplication and division) should be equal to the minimum significant figures involved in any physical quantity in the given operation.

✍ Significant figures are the number of digits upto which we are sure about their accuracy.

✍ If a number is without a decimal and ends in one or more zeros, then all the zeros at the end of the number may not be significant. To make the number of significant figures clear, it is suggested that the number may be written in exponential form. For example 20300 may be expressed as 203.00×10^{2}, to suggest that all the zeros at the end of 20300 are significant.

✍ 1 $inch$ = 2.54 cm

1 $foot$ = 12 $inches$ = 30.48 cm = 0.3048 m

1 $mile$ = 5280 ft = 1.609 km

✍ 1 $yard$ = 0.9144 m

- 1 $slug = 14.59\ kg$
- 1 $barn = 10^{-28}\ m^2$
- 1 $liter = 10^3\ cm^3 = 10^{-3}\ m^3$
- 1 $km/h = \frac{5}{18}\ m/s$

  1 $m/s = 3.6\ km/h$
- 1 $g/cm^3 = 1000\ kg/m^3$
- 1 $atm. = 76\ cm$ of $Hg = 1.013 \times 10^5\ N/m^2$

  1 $N/m^2 = Pa$ ($Pascal$)
- When we add or subtract two measured quantities, the absolute error in the final result is equal to the sum of the absolute errors in the measured quantities.
- When we multiply or divide two measured quantities, the relative error in the final result is equal to the sum of the relative errors in the measured quantities.

Ordinary Thinking

Objective Questions

Units

1. Light year is a unit of [MP PMT 1989; CPMT 1991; AFMC 1991,2005]
   (a) Time (b) Mass
   (c) Distance (d) Energy

2. The magnitude of any physical quantity
   (a) Depends on the method of measurement
   (b) Does not depend on the method of measurement
   (c) Is more in SI system than in CGS system
   (d) Directly proportional to the fundamental units of mass, length and time

3. Which of the following is not equal to *watt* [SCRA 1991; CPMT 1990]
   (a) *Joule/second* (b) *Ampere* × *volt*
   (c) $(Ampere)^2 \times ohm$ (d) *Ampere/volt*

4. Newton–*second* is the unit of [CPMT 1984, 85; MP PMT 1984]
   (a) Velocity (b) Angular momentum
   (c) Momentum (d) Energy

5. Which of the following is not represented in correct unit [NCERT 1984; MNR 1995]
   (a) $\frac{\text{Stress}}{\text{Strain}} = N/m^2$ (b) Surface tension $= N/m$
   (c) Energy $= kg\text{-}m/\text{sec}$ (d) Pressure $= N/m^2$

6. One *second* is equal to [MNR 1986]
   (a) 1650763.73 time periods of Kr clock
   (b) 652189.63 time periods of Kr clock
   (c) 1650763.73 time periods of Cs clock
   (d) 9192631770 time periods of Cs clock

7. One nanometre is equal to [SCRA 1986; MNR 1986]
   (a) $10^9 mm$ (b) $10^{-6}\ cm$
   (c) $10^{-7} cm$ (d) $10^{-9} cm$

8. A *micron* is related to centimetre as
   (a) $1\,micron = 10^{-8} cm$ (b) $1\,micron = 10^{-6} cm$
   (c) $1\,micron = 10^{-5} cm$ (d) $1\,micron = 10^{-4}\ cm$

9. The unit of power is [CPMT 1985]
   (a) *Joule*
   (b) *Joule* per *second* only
   (c) *Joule* per *second* and *watt* both
   (d) Only *watt*

10. A suitable unit for gravitational constant is [MNR 1988]
   (a) $kg\text{-}m\,\text{sec}^{-1}$ (b) $N\ m^{-1}\ \text{sec}$
   (c) $N\ m^2\ kg^{-2}$ (d) $kg\,m\,\text{sec}^{-1}$

11. SI unit of pressure is [EAMCET 1980; DPMT 1984; CBSE PMT 1988; NCERT 1976; AFMC 1991; USSR MEE 1991]
   (a) *Pascal* (b) $Dynes/cm^2$
   (c) cm of Hg (d) *Atmosphere*

12. The unit of angular acceleration in the SI system is [SCRA 1980; EAMCET 1981]
   (a) $N\,kg^{-1}$ (b) $m\,s^{-2}$
   (c) $rad\,s^{-2}$ (d) $m\,kg^{-1}K$

13. The unit of Stefan's constant σ is [AFMC 1986; MP PET 1992; MP PMT 1992; CBSE PMT 2002]
   (a) $W\,m^{-2}\,K^{-1}$ (b) $W\,m^2\,K^{-4}$
   (c) $W\,m^{-2}\,K^{-4}$ (d) $W\,m^{-2}\,K^4$

14. Which of the following is not a unit of energy [AIIMS 1985]
   (a) $W\text{-}s$ (b) $kg\text{-}m/\text{sec}$
   (c) $N\text{-}m$ (d) *Joule*

15. In $S = a + bt + ct^2$. S is measured in metres and t in *seconds*. The unit of c is [MP PMT 1993]
   (a) None (b) m
   (c) ms^{-1} (d) ms^{-2}

16. *Joule-second* is the unit of [CPMT 1990; CBSE PMT 1993; BVP 2003]
   (a) Work (b) Momentum
   (c) Pressure (d) Angular momentum

17. Unit of energy in SI system is [CPMT 1971; NCERT 1976]
   (a) *Erg* (b) *Calorie*
   (c) *Joule* (d) Electron *volt*

18. A cube has numerically equal volume and surface area. The volume of such a cube is [CPMT 1971, 74]
   (a) 216 *units* (b) 1000 *units*
   (c) 2000 *units* (d) 3000 *units*

19. Wavelength of ray of light is $0.00006\,m$. It is equal to [CPMT 1977]
   (a) 6 *microns* (b) 60 *microns*
   (c) 600 *microns* (d) 0.6 *microns*

20. Electron *volt* is a unit of [MP PMT 1993]
   (a) Charge (b) Potential difference
   (c) Momentum (d) Energy

21. Temperature can be expressed as a derived quantity in terms of any of the following [MP PET 1993; UPSEAT 2001]
   (a) Length and mass
   (b) Mass and time
   (c) Length, mass and time
   (d) None of these

22. Unit of power is [NCERT 1972; CPMT 1971; DCE 1999]

(a) *Kilowatt* (b) *Kilowatt-hour*
(c) *Dyne* (d) *Joule*

23. Density of wood is $0.5 gm/cc$ in the CGS system of units. The corresponding value in MKS units is [CPMT 1983; NCERT 1973; JIPMER 1993]
(a) 500 (b) 5
(c) 0.5 (d) 5000

24. Unit of energy is [NCERT 1974; CPMT 1975]
(a) $J/\sec$ (b) $Watt-day$
(c) $Kilowatt$ (d) $gm\text{-}cm/\sec^2$

25. Which is the correct unit for measuring nuclear radii
(a) *Micron* (b) *Millimetre*
(c) *Angstrom* (d) *Fermi*

26. One Mach number is equal to
(a) Velocity of light
(b) Velocity of sound $(332\ m/\sec)$
(c) $1\ km/\sec$
(d) $1 m/\sec$

27. The unit for nuclear dose given to a patient is
(a) *Fermi* (b) Rutherford
(c) Curie (d) Roentgen

28. *Volt/metre* is the unit of [AFMC 1991; CPMT 1984]
(a) Potential (b) Work
(c) Force (d) Electric intensity

29. $Newton/metre^2$ is the unit of [CPMT 1985; ISM Dhanbad 1994; AFMC 1995]
(a) Energy (b) Momentum
(c) Force (d) Pressure

30. The unit of surface tension in SI system is [MP PMT 1984; AFMC 1986; CPMT 1985, 87; CBSE PMT 1993; KCET 1999; DCE 2000, 01]
(a) $Dyne/cm^2$ (b) $Newton/m$
(c) $Dyne/cm$ (d) $Newton/m^2$

31. The unit of reduction factor of tangent galvanometer is [CPMT 1987; AFMC 2004]
(a) *Ampere* (b) *Gauss*
(c) *Radian* (d) None of these

32. The unit of self inductance of a coil is [MP PMT 1983, 92; SCRA 1986; CBSE PMT 1993; CPMT 1984, 85, 87]
(a) *Farad* (b) *Henry*
(c) *Weber* (d) *Tesla*

33. *Henry/ohm* can be expressed in [CPMT 1987]
(a) *Second* (b) *Coulomb*
(c) *Mho* (d) *Metre*

34. The SI unit of momentum is [SCRA 1986, 89; CPMT 1987]
(a) $\frac{kg}{m}$ (b) $\frac{kg.m}{\sec}$
(c) $\frac{kg.m^2}{\sec}$ (d) $kg \times Newton$

35. The velocity of a particle depends upon as $v = a + bt + ct^2$; if the velocity is in $m/\sec$, the unit of a will be [CPMT 1990]
(a) $m/\sec$ (b) $m/\sec^2$
(c) $m^2/\sec$ (b) $m/\sec^3$

36. One million electron *volt* $(1\ MeV)$ is equal to [JIPMER 1993, 97]
(a) $10^5 eV$ (b) $10^6 eV$
(c) $10^4 eV$ (d) $10^7 eV$

37. $Erg - m^{-1}$ can be the unit of measure for [DCE 1993]
(a) Force (b) Momentum
(c) Power (d) Acceleration

38. The unit of potential energy is [AFMC 1991]
(a) $g(cm/\sec^2)$ (b) $g(cm/\sec)^2$
(c) $g(cm^2/\sec)$ (d) $g(cm/\sec)$

39. Which of the following represents a *volt* [CPMT 1990; AFMC 1991]
(a) *Joule/second* (b) *Watt/Ampere*
(c) *Watt/Coulomb* (d) *Coulomb/Joule*

40. $Kilowatt-hour$ is a unit of [NCERT 1975; AFMC 1991]
(a) Electrical charge (b) Energy
(c) Power (d) Force

41. What is the SI unit of permeability [CBSE PMT 1993]
(a) *Henry* per *metre*
(b) Tesla *metre* per *ampere*
(c) *Weber* per *ampere metre*
(d) All the above units are correct

42. In which of the following systems of unit, $Weber$ is the unit of magnetic flux [SCRA 1991; CBSE PMT 1993; DPMT 2005]
(a) CGS (b) MKS
(c) SI (d) None of these

43. Tesla is a unit for measuring [CBSE PMT 1993]
(a) Magnetic moment (b) Magnetic induction
(c) Magnetic intensity (d) Magnetic pole strength

44. If the unit of length and force be increased four times, then the unit of energy is [Kerala PMT 2005]
(a) Increased 4 times (b) Increased 8 times
(c) Increased 16 times (d) Decreased 16 times

45. Oersted is a unit of [SCRA 1989]
(a) Dip (b) Magnetic intensity
(c) Magnetic moment (d) Pole strength

46. $Ampere - hour$ is a unit of [SCRA 1980, 89; ISM Dhanbad 1994]
(a) Quantity of electricity
(b) Strength of electric current
(c) Power
(d) Energy

47. The unit of specific resistance is [SCRA 1989; MP PET 1984; CPMT 1975]
(a) Ohm/cm^2 (b) Ohm/cm
(c) $Ohm\text{–}cm$ (d) $(Ohm\text{–}cm)^{-1}$

48. The binding energy of a nucleon in a nucleus is of the order of a few [SCRA 1979]

(a) eV (b) $Ergs$
(c) MeV (d) $Volts$

49. $Parsec$ is a unit of [SCRA 1986; BVP 2003; AIIMS 2005]
(a) Distance (b) Velocity
(c) Time (d) Angle

50. If u_1 and u_2 are the units selected in two systems of measurement and n_1 and n_2 their numerical values, then [SCRA 1986]
(a) $n_1u_1 = n_2u_2$ (b) $n_1u_1 + n_2u_2 = 0$
(c) $n_1n_2 = u_1u_2$ (d) $(n_1 + u_1) = (n_2 + u_2)$

51. $1\,eV$ is [SCRA 1986]
(a) Same as one *joule* (b) $1.6\times10^{-19}\,J$
(c) $1V$ (d) $1.6\times10^{-19}\,C$

52. $1kWh =$ [AFMC 1986; SCRA 1986, 91]
(a) $1000W$ (b) $36\times10^5\,J$
(c) $1000J$ (d) $3600\,J$

53. Universal time is based on [SCRA 1989]
(a) Rotation of the earth on its axis
(b) Earth's orbital motion around the earth
(c) Vibrations of cesium atom
(d) Oscillations of quartz crystal

54. The nuclear cross-section is measured in barn, it is equal to
(a) $10^{-20}\,m^2$ (b) $10^{-30}\,m^2$
(c) $10^{-28}\,m^2$ (d) $10^{-14}\,m^2$

55. Unit of moment of inertia in MKS system [MP PMT 1984]
(a) $kg\times cm^2$ (b) kg/cm^2
(c) $kg\times m^2$ (d) $Joule\times m$

56. Unit of stress is [MP PMT 1984]
(a) N/m (b) $N\text{–}m$
(c) N/m^2 (d) $N\text{–}m^2$

57. Unit of Stefan's constant is [MP PMT 1989]
(a) $J\,s^{-1}$ (b) $J\,m^{-2}s^{-1}K^{-4}$
(c) $J\,m^{-2}$ (d) $J\,s$

58. Unit of magnetic moment is [MP PET 1989]
(a) $Ampere\text{–}metre^2$ (b) $Ampere\text{–}metre$
(c) $Weber\text{–}metre^2$ (c) $Weber/metre$

59. Curie is a unit of [CBSE PMT 1992; CPMT 1992]
(a) Energy of γ-rays (b) Half life
(c) Radioactivity (d) Intensity of γ-rays

60. Hertz is the unit for [MNR 1983; SCRA 1983; RPMT 1999]
(a) Frequency (b) Force
(c) Electric charge (d) Magnetic flux

61. One pico *Farad* is equal to
(a) $10^{-24}\,F$ (b) $10^{-18}\,F$
(c) $10^{-12}\,F$ (d) $10^{-6}\,F$

62. In SI, *Henry* is the unit of [MP PET 1984; CBSE PMT 1993; DPMT 1984]
(a) Self inductance (b) Mutual inductance
(c) (a) and (b) both (d) None of the above

63. The unit of $e.m.f.$ is [CPMT 1986; AFMC 1986]
(a) *Joule* (b) *Joule-Coulomb*
(c) *Volt–Coulomb* (d) *Joule/Coulomb*

64. Which of the following is not the unit of time [CPMT 1991; NCERT 1990; DPMT 1987; AFMC 1996]
(a) Micro *second* (b) Leap year
(c) Lunar months (d) Parallactic *second*
(e) Solar day

65. Unit of self inductance is [MP PET 1982]
(a) $\dfrac{Newton\text{ - }second}{Coulomb\times Ampere}$ (b) $\dfrac{Joule/Coulomb\times Second}{Ampere}$
(c) $\dfrac{Volt\times metre}{Coulomb}$ (d) $\dfrac{Newton\times metre}{Ampere}$

66. To determine the Young's modulus of a wire, the formula is $Y = \dfrac{F}{A}\times\dfrac{L}{\Delta L}$; where L = length, A = area of cross-section of the wire, ΔL = change in length of the wire when stretched with a force F. The conversion factor to change it from CGS to MKS system is
(a) 1 (b) 10
(c) 0.1 (d) 0.01

67. Young's modulus of a material has the same units as [MP PMT 1994]
(a) Pressure (b) Strain
(c) Compressibility (d) Force

68. One yard in SI units is equal [MP PMT 1995]
(a) 1.9144 $metre$ (b) 0.9144 $metre$
(c) 0.09144 $kilometre$ (d) 1.0936 $kilometre$

69. Which of the following is smallest unit [AFMC 1996]
(a) *Millimetre* (b) *Angstrom*
(c) *Fermi* (d) *Metre*

70. Which one of the following pairs of quantities and their units is a proper match
(a) Electric field – $Coulomb\,/\,m$
(b) Magnetic flux – $Weber$
(c) Power – *Farad*
(d) Capacitance – *Henry*

71. The units of modulus of rigidity are [MP PMT 1997]
(a) $N\text{–}m$ (b) N/m
(c) $N\text{–}m^2$ (d) N/m^2

72. The unit of absolute permittivity is [CMEET Bihar 1995]
(a) Fm (*Farad*-meter) (b) Fm^{-1} (*Farad*/meter)
(c) Fm^{-2} (*Farad*/ $metre^2$) (d) F (*Farad*)
(e) None of these

73. Match List-I with List-II and select the correct answer using the codes given below the lists [SCRA 1994]

List-I	List-II
I. *Joule*	A. *Henry* × *Amp/sec*
II. *Watt*	B. *Farad* × *Volt*
III. *Volt*	C. *Coulomb* × *Volt*

IV. *Coulomb* D. *Oersted* × *cm*

E. *Amp* × *Gauss*

F. Amp^2 × *Ohm*

Codes:

(a) $I-A, II-F, III-E, IV-D$

(b) $I-C, II-F, III-A, IV-B$

(c) $I-C, II-F, III-A, IV-E$

(d) $I-B, II-F, III-A, IV-C$

74. Which relation is wrong [RPMT 1997]

(a) 1 *Calorie* = 4.18 *Joules*

(b) $1Å = 10^{-10}\, m$

(c) $1\ MeV = 1.6 \times 10^{-13}\ Joules$

(d) $1\ Newton = 10^{-5}\ Dynes$

75. If $x = at + bt^2$, where x is the distance travelled by the body in kilometres while t is the time in seconds, then the units of b are

(a) km/s (b) $km\text{–}s$

(c) km/s^2 (d) $km\text{–}s^2$

76. The equation $\left(P + \frac{a}{V^2}\right)(V-b)$ constant. The units of a are

(a) $Dyne \times cm^5$ (b) $Dyne \times cm^4$

(c) $Dyne/cm^3$ (d) $Dyne/cm^2$

77. Which of the following quantity is expressed as force per unit area

(a) Work (b) Pressure

(c) Volume (d) Area

78. Match List-I with List-II and select the correct answer by using the codes given below the lists [NDA 1995]

List-I	List-II
(a) Distance between earth and stars	1. Microns
(b) Inter-atomic distance in a solid	2. Angstroms
(c) Size of the nucleus	3. Light years
(d) Wavelength of infrared laser	4. *Fermi*
	5. Kilometres

Codes

	a	b	c	d		a	b	c	d
(a)	5	4	2	1	(b)	3	2	4	1
(c)	5	2	4	3	(d)	3	4	1	2

79. Unit of impulse is [CPMT 1997]

(a) Newton (b) $kg\text{–}m$

(c) $kg\text{–}m/s$ (d) *Joule*

80. Which is not a unit of electric field [UPSEAT 1999]

(a) NC^{-1} (b) Vm^{-1}

(c) JC^{-1} (d) $JC^{-1}m^{-1}$

81. The correct value of $0^o C$ on the Kelvin scale is [UPSEAT 2000]

(a) $273.15K$ (b) $272.85K$

(c) $273K$ (d) $273.2K$

82. 'Torr' is the unit of [RPMT 1999, 2000]

(a) Pressure (b) Volume

(c) Density (d) Flux

83. Which of the following is a derived unit [BHU 2000]

(a) Unit of mass (b) Unit of length

(c) Unit of time (d) Unit of volume

84. *Dyne/cm* is not a unit of [RPET 2000]

(a) Pressure (b) Stress

(c) Strain (d) Young's modulus

85. The units of angular momentum are [MP PMT 2000]

(a) $kg\text{–}m^2/s^2$ (b) $Joule\text{–}s$

(c) $Joule/s$ (d) $kg\text{–}m\text{–}s^2$

86. Which of the following is not the unit of energy [MP PET 2000]

(a) *Calorie* (b) *Joule*

(c) Electron *volt* (d) *Watt*

[CBSE PMT 1993]

87. Which of the following is not a unit of time [UPSEAT 2001]

(a) *Leap year* (b) *Micro second*

(c) *Lunar month* (d) *Light year*

88. The S.I. unit of gravitational potential is [AFMC 2001]

(a) J [MNR 1995; AFMC 1995] (b) $J\text{-}kg^{-1}$

(c) $J\text{-}kg$ (d) $J\text{-}kg^{-2}$

89. Which one of the following is not a unit of young's modulus [KCET 2005]

(a) Nm [AFMC 1995] (b) Nm^{-2}

(c) $Dyne\ cm^{-2}$ (d) Mega Pascal

90. In C.G.S. system the magnitutde of the force is 100 *dynes*. In another system where the fundamental physical quantities are kilogram, *metre* and minute, the magnitude of the force is

(a) 0.036 (b) 0.36

(c) 3.6 (d) 36

91. The unit of L/R is (where L = inductance and R = resistance)

(a) sec (b) $\sec^{-1}$

(c) *Volt* (d) *Ampere*

92. Which is different from others by units [Orissa JEE 2002]

(a) Phase difference (b) Mechanical equivalent

(c) Loudness of sound (d) Poisson's ratio

93. Length cannot be measured by [AIIMS 2002]

(a) *Fermi* (b) Debye

(c) *Micron* (d) *Light year*

94. The value of Planck's constant is [CBSE PMT 2002]

(a) $6.63 \times 10^{-34}\ J\text{-sec}$ (b) $6.63 \times 10^{34}\ J/\text{sec}$

(c) $6.63 \times 10^{-34}\ kg\text{-}m^2$ (d) $6.63 \times 10^{34}\ kg/\text{sec}$

95. A physical quantity is measured and its value is found to be nu where $n =$ numerical value and $u =$ unit. Then which of the following relations is true [RPET 2003]

(a) $n \propto u^2$ (b) $n \propto u$

(c) $n \propto \sqrt{u}$ (d) $n \propto \frac{1}{u}$

96. *Faraday* is the unit of [AFMC 2003]

(a) Charge
(b) emf
(c) Mass
(d) Energy

97. *Candela* is the unit of [UPSEAT 1999; CPMT 2003]
(a) Electric intensity
(b) Luminous intensity
(c) Sound intensity
(d) None of these

98. The unit of reactance is [MP PET 2003]
(a) *Ohm*
(b) *Volt*
(c) *Mho*
(d) *Newton*

99. The unit of Planck's constant is [RPMT 1999; MP PET 2003; Pb. PMT 2004]
(a) *Joule*
(b) *Joule*/s
(c) *Joule*/*m*
(d) *Joule*-s

100. Number of base SI units is [MP PET 2003]
(a) 4
(b) 7
(c) 3
(d) 5

101. SI unit of permittivity is [KCET 2004]
(a) $C^2m^2N^{-1}$
(b) $C^{-1}m^2N^{-2}$
(c) $C^2m^2N^2$
(d) $C^2m^{-2}N^{-1}$

102. Which does not has the same unit as others [Orissa PMT 2004]
(a) *Watt-sec*
(b) *Kilowatt-hour*
(c) *eV*
(d) *J-sec*

103. Unit of surface tension is [Orissa PMT 2004]
(a) Nm^{-1}
(b) Nm^{-2}
(c) N^2m^{-1}
(d) Nm^{-3}

104. Which of the following system of units is not based on units of mass, length and time alone [Kerala PMT 2004]
(a) SI
(b) MKS
(c) FPS
(d) CGS

105. The unit of the coefficient of viscosity in S.I. system is [J & K CET 2004]
(a) $m/kg\text{-}s$
(b) $m\text{-}s/kg^2$
(c) $kg/m\text{-}s^2$
(d) $kg/m\text{-}s$

106. The unit of Young's modulus is [Pb. PET 2001]
(a) Nm^2
(b) Nm^{-2}
(c) Nm
(d) Nm^{-1}

107. One femtometer is equivalent to [DCE 2004]
(a) $10^{15}\ m$
(b) $10^{-15}\ m$
(c) $10^{-12}\ m$
(d) $10^{12}\ m$

108. How many wavelength of Kr^{86} are there in one *metre* [MNR 1985; UPSEAT 2000; Pb. PET 2004]
(a) 1553164.13
(b) 1650763.73
(c) 652189.63
(d) 2348123.73

109. Which of the following pairs is wrong [AFMC 2003]
(a) Pressure-Baromter
(b) Relative density-Pyrometer
(c) Temperature-Thermometer
(d) Earthquake-Seismograph

Dimensions

1. Select the pair whose dimensions are same
(a) Pressure and stress
(b) Stress and strain
(c) Pressure and force
(d) Power and force

2. Dimensional formula $ML^{-1}T^{-2}$ does not represent the physical quantity [Manipal MEE 1995]
(a) Young's modulus of elasticity
(b) Stress
(c) Strain
(d) Pressure

3. Dimensional formula ML^2T^{-3} represents [EAMCET 1981; MP PMT 1996, 2001]
(a) Force
(b) Power
(c) Energy
(d) Work

4. The dimensions of *calorie* are [CPMT 1985]
(a) ML^2T^{-2}
(b) MLT^{-2}
(c) ML^2T^{-1}
(d) ML^2T^{-3}

5. Whose dimensions is ML^2T^{-1} [CPMT 1989]
(a) Torque
(b) Angular momentum
(c) Power
(d) Work

6. If L and R are respectively the inductance and resistance, then the dimensions of $\frac{L}{R}$ will be [CPMT 1986; CBSE PMT 1988; Roorkee 1995; MP PET/PMT 1998; DCE 2002]
(a) $M^0L^0T^{-1}$
(b) M^0LT^0
(c) M^0L^0T
(d) Cannot be represented in terms of M, L and T

7. Which pair has the same dimensions [EAMCET 1982; CPMT 1984, 85; Pb. PET 2002; MP PET 1985]
(a) Work and power
(b) Density and relative density
(c) Momentum and impulse
(d) Stress and strain

8. If C and R represent capacitance and resistance respectively, then the dimensions of RC are [CPMT 1981, 85; CBSE PMT 1992, 95; Pb. PMT 1999]
(a) $M^0L^0T^2$
(b) M^0L^0T
(c) ML^{-1}
(d) None of the above

9. Dimensions of one or more pairs are same. Identify the pairs
(a) Torque and work
(b) Angular momentum and work
(c) Energy and Young's modulus
(d) Light year and wavelength

10. Dimensional formula for latent heat is [MNR 1987; CPMT 1978, 86; IIT 1983, 89; RPET 2002]
(a) $M^0L^2T^{-2}$
(b) MLT^{-2}

(c) ML^2T^{-2} (d) ML^2T^{-1}

11. Dimensional formula for volume elasticity is

[MP PMT 1991, 2002; CPMT 1991; MNR 1986]

(a) $M^1L^{-2}T^{-2}$ (b) $M^1L^{-3}T^{-2}$

(c) $M^1L^2T^{-2}$ (d) $M^1L^{-1}T^{-2}$

12. The dimensions of universal gravitational constant are

[MP PMT 1984, 87, 97, 2000; CBSE PMT 1988, 92; 2004 MP PET 1984, 96, 99; MNR 1992; DPMT 1984; CPMT 1978, 84, 89, 90, 92, 96; AFMC 1999; NCERT 1975; DPET 1993; AIIMS 2000; RPET 2001; Pb. PMT 2002, 03; UPSEAT 1999; BCECE 2003, 05;]

(a) $M^{-2}L^2T^{-2}$ (b) $M^{-1}L^3T^{-2}$

(c) $ML^{-1}T^{-2}$ (d) ML^2T^{-2}

13. The dimensional formula of angular velocity is

[JIPMER 1993; AFMC 1996; AIIMS 1998]

(a) $M^0L^0T^{-1}$ (b) MLT^{-1}

(c) $M^0L^0T^1$ (d) ML^0T^{-2}

14. The dimensions of power are

[CPMT 1974, 75; SCRA 1989]

(a) $M^1L^2T^{-3}$ (b) $M^2L^1T^{-2}$

(c) $M^1L^2T^{-1}$ (d) $M^1L^1T^{-2}$

15. The dimensions of couple are [CPMT 1972; JIPMER 1993]

(a) ML^2T^{-2} (b) MLT^{-2}

(c) $ML^{-1}T^{-3}$ (d) $ML^{-2}T^{-2}$

16. Dimensional formula for angular momentum is

[CBSE PMT 1988, 92; EAMCET 1995; DPMT 1987; CMC Vellore 1982; CPMT 1973, 82, 86; MP PMT 1987; BHU 1995; IIT 1983; Pb. PET 2000]

(a) ML^2T^{-2} (b) ML^2T^{-1}

(c) MLT^{-1} (d) $M^0L^2T^{-2}$

17. The dimensional formula for impulse is

[EAMCET 1981; CBSE PMT 1991; CPMT 1978; AFMC 1998; BCECE 2003]

(a) MLT^{-2} (b) MLT^{-1}

(c) ML^2T^{-1} (d) M^2LT^{-1}

18. The dimensional formula for the modulus of rigidity is

[MNR 1984; IIT 1982; MP PET 2000]

(a) ML^2T^{-2} (b) $ML^{-1}T^{-3}$

(c) $ML^{-2}T^{-2}$ (d) $ML^{-1}T^{-2}$

19. The dimensional formula for $r.m.s.$ (root mean square) velocity is

(a) M^0LT^{-1} (b) $M^0L^0T^{-2}$

(c) $M^0L^0T^{-1}$ (d) MLT^{-3}

20. The dimensional formula for Planck's constant (h) is

[DPMT 1987; MP PMT 1983, 96; IIT 1985; MP PET 1995; AFMC 2003; RPMT 1999; Kerala PMT 2002]

(a) $ML^{-2}T^{-3}$ (b) ML^2T^{-2}

(c) ML^2T^{-1} (d) $ML^{-2}T^{-2}$

21. Out of the following, the only pair that does not have identical dimensions is [MP PET/PMT 1998; BHU 1997]

(a) Angular momentum and Planck's constant
(b) Moment of inertia and moment of a force
(c) Work and torque
(d) Impulse and momentum

22. The dimensional formula for impulse is same as the dimensional formula for

[CPMT 1982, 83; CBSE PMT 1993; UPSEAT 2001]

(a) Momentum
(b) Force
(c) Rate of change of momentum
(d) Torque

23. Which of the following is dimensionally correct

(a) Pressure = Energy per unit area
(b) Pressure = Energy per unit volume
(c) Pressure = Force per unit volume
(d) Pressure = Momentum per unit volume per unit time

24. Planck's constant has the dimensions (unit) of

[CPMT 1983, 84, 85, 90, 91; AIIMS 1985; MP PMT 1987; EAMCET 1990; RPMT 1999; CBSE PMT 2001; MP PET 2002; KCET 2004]

(a) Energy (b) Linear momentum

(c) Work (d) Angular momentum

25. The equation of state of some gases can be expressed as $\left(P+\frac{a}{V^2}\right)(V-b)=RT$. Here P is the pressure, V is the volume, T is the absolute temperature and a, b, R are constants. The dimensions of $'a'$ are

[CBSE PMT 1991, 96; NCERT 1984; MP PET 1992; CPMT 1974, 79, 87, 97; MP PMT 1992, 94; MNR 1995; AFMC 1995]

(a) ML^5T^{-2} (b) $ML^{-1}T^{-2}$

(c) $M^0L^3T^0$ (d) $M^0L^6T^0$

26. If V denotes the potential difference across the plates of a capacitor of capacitance C, the dimensions of CV^2 are

[CPMT 1982]

(a) Not expressible in MLT (b) MLT^{-2}

(c) M^2LT^{-1} (d) ML^2T^{-2}

27. If L denotes the inductance of an inductor through which a current i is flowing, the dimensions of Li^2 are

[CPMT 1982, 85, 87]

(a) ML^2T^{-2} (b) Not expressible in MLT

(c) MLT^{-2} (d) $M^2L^2T^{-2}$

28. Of the following quantities, which one has dimensions different from the remaining three

[AIIMS 1987; CBSE PMT 1993]

(a) Energy per unit volume
(b) Force per unit area
(c) Product of voltage and charge per unit volume
(d) Angular momentum per unit mass

29. A spherical body of mass m and radius r is allowed to fall in a medium of viscosity η. The time in which the velocity of the body

increases from zero to 0.63 times the terminal velocity (v) is called time constant (τ). Dimensionally τ can be represented by

(a) $\frac{mr^2}{6\pi\eta}$ (b) $\sqrt{\left(\frac{6\pi mr\eta}{g^2}\right)}$

(c) $\frac{m}{6\pi\eta rv}$ (d) None of the above

30. The frequency of vibration f of a mass m suspended from a spring of spring constant K is given by a relation of this type $f = C m^x K^y$; where C is a dimensionless quantity. The value of x and y are [CBSE PMT 1990]

(a) $x = \frac{1}{2}, y = \frac{1}{2}$ (b) $x = -\frac{1}{2}, y = -\frac{1}{2}$

(c) $x = \frac{1}{2}, y = -\frac{1}{2}$ (d) $x = -\frac{1}{2}, y = \frac{1}{2}$

31. The quantities A and B are related by the relation, $m = A/B$, where m is the linear density and A is the force. The dimensions of B are of

(a) Pressure (b) Work

(c) Latent heat (d) None of the above

32. The velocity of water waves v may depend upon their wavelength λ, the density of water ρ and the acceleration due to gravity g. The method of dimensions gives the relation between these quantities as

[NCERT 1979; CET 1992; MP PET 2001; UPSEAT 2000]

(a) $v^2 \propto \lambda g^{-1} \rho^{-1}$ (b) $v^2 \propto g\lambda\rho$

(c) $v^2 \propto g\lambda$ (d) $v^2 \propto g^{-1}\lambda^{-3}$

33. The dimensions of *Farad* are [MP PET 1993]

(a) $M^{-1}L^{-2}T^2Q^2$ (b) $M^{-1}L^{-2}TQ$

(c) $M^{-1}L^{-2}T^{-2}Q$ (d) $M^{-1}L^{-2}TQ^2$

34. The dimensions of resistivity in terms of M, L, T and Q where Q stands for the dimensions of charge, is

[MP PET 1993]

(a) $ML^3T^{-1}Q^{-2}$ (b) $ML^3T^{-2}Q^{-1}$

(c) $ML^2T^{-1}Q^{-1}$ (d) $MLT^{-1}Q^{-1}$

35. The equation of a wave is given by

$$Y = A\sin\omega\left(\frac{x}{v} - k\right)$$

where ω is the angular velocity and v is the linear velocity. The dimension of k is [MP PMT 1993]

(a) LT (b) T

(c) T^{-1} (d) T^2

36. The dimensions of coefficient of thermal conductivity is

[MP PMT 1993]

(a) $ML^2T^{-2}K^{-1}$ (b) $MLT^{-3}K^{-1}$

(c) $MLT^{-2}K^{-1}$ (d) $MLT^{-3}K$

37. Dimensional formula of stress is

(a) M^0LT^{-2} [AIIMS 1987] (b) $M^0L^{-1}T^{-2}$

(c) $ML^{-1}T^{-2}$ (d) ML^2T^{-2}

38. Dimensional formula of velocity of sound is

(a) M^0LT^{-2} (b) LT^0

(c) M^0LT^{-1} (d) $M^0L^{-1}T^{-1}$

39. Dimensional formula of capacitance is

[CPMT 1978; MP PMT 1979; IIT 1983]

(a) $M^{-1}L^{-2}T^4A^2$ (b) $ML^2T^4A^{-2}$

(c) $MLT^{-4}A^2$ (d) $M^{-1}L^{-2}T^{-4}A^{-2}$

40. MLT^{-1} represents the dimensional formula of

[CPMT 1975]

(a) Power (b) Momentum

(c) Force (d) Couple

41. Dimensional formula of heat energy is

[CPMT 1976, 81, 86, 91]

(a) ML^2T^{-2} (b) MLT^{-1}

(c) $M^0L^0T^{-2}$ (d) None of these

42. If C and L denote capacitance and inductance respectively, then the dimensions of LC are

[CPMT 1981; MP PET 1997]

(a) $M^0L^0T^0$ (b) $M^0L^0T^2$

(c) $M^2L^0T^2$ (d) MLT^2

43. Which of the following quantities has the same dimensions as that of energy [AFMC 1991; CPMT 1976; DPMT 2001]

(a) Power (b) Force

(c) Momentum (d) Work

44. The dimensions of "time constant" $\frac{L}{R}$ during growth and decay of current in all inductive circuit is same as that of

[MP PET 1993; EAMCET 1994]

(a) Constant (b) Resistance

(c) Current (d) Time

45. The period of a body under SHM *i.e.* presented by $T = P^aD^bS^c$; where P is pressure, D is density and S is surface tension. The value of a, b and c are [CPMT 1981]

(a) $-\frac{3}{2}, \frac{1}{2}, 1$ (b) $-1, -2, 3$

(c) $\frac{1}{2}, -\frac{3}{2}, -\frac{1}{2}$ (d) $1, 2, \frac{1}{3}$

46. Which of the following pairs of physical quantities has the same dimensions [CPMT 1978; NCERT 1987]

(a) Work and power (b) Momentum and energy

(c) Force and power (d) Work and energy

47. The velocity of a freely falling body changes as g^ph^q where g is acceleration due to gravity and h is the height. The values of p and q are [NCERT 1983; EAMCET 1994]

(a) $1, \frac{1}{2}$ (b) $\frac{1}{2}, \frac{1}{2}$

(c) $\frac{1}{2}, 1$ (d) $1, 1$

48. Which one of the following does not have the same dimensions

(a) Work and energy
(b) Angle and strain
(c) Relative density and refractive index
(d) Planck constant and energy

49. Dimensions of frequency are [CPMT 1988]

(a) $M^0L^{-1}T^0$ (b) $M^0L^0T^{-1}$
(c) M^0L^0T (d) MT^{-2}

50. Which one has the dimensions different from the remaining three

(a) Power (b) Work
(c) Torque (d) Energy

51. A small steel ball of radius r is allowed to fall under gravity through a column of a viscous liquid of coefficient of viscosity η. After some time the velocity of the ball attains a constant value known as terminal velocity v_T. The terminal velocity depends on (i) the mass of the ball m, (ii) η, (iii) r and (iv) acceleration due to gravity g. Which of the following relations is dimensionally correct

[CPMT 1992; CBSE PMT 1992; NCERT 1983; MP PMT 2001]

(a) $v_T \propto \frac{mg}{\eta r}$ (b) $v_T \propto \frac{\eta r}{mg}$
(c) $v_T \propto \eta rmg$ (d) $v_T \propto \frac{mgr}{\eta}$

52. The quantity $X = \frac{\varepsilon_0 LV}{t}$: ε_0 is the permittivity of free space, L is length, V is potential difference and t is time. The dimensions of X are same as that of [IIT 2001]

(a) Resistance (b) Charge
(c) Voltage (d) Current

53. μ_0 and ε_0 denote the permeability and permittivity of free space, the dimensions of $\mu_0\varepsilon_0$ are

(a) LT^{-1} (b) $L^{-2}T^2$
(c) $M^{-1}L^{-3}Q^2T^2$ (d) $M^{-1}L^{-3}I^2T^2$

54. The expression $[ML^2T^{-2}]$ represents [JIPMER 1993, 97]

(a) Pressure (b) Kinetic energy
(c) Momentum (d) Power

55. The dimensions of physical quantity X in the equation Force $= \frac{X}{\text{Density}}$ is given by [DCE 1993]

(a) $M^1L^4T^{-2}$ (b) $M^2L^{-2}T^{-1}$
(c) $M^2L^{-2}T^{-2}$ (d) $M^1L^{-2}T^{-1}$

56. The dimensions of CV^2 matches with the dimensions of

[DCE 1993]

(a) L^2I (b) L^2I^2
(c) LI^2 (d) $\frac{1}{LI}$

57. The M[CPMT 1985]artians use force (F), acceleration (A) and time (T) as their fundamental physical quantities. The dimensions of length on Martians system are [DCE 1993]

(a) FT^2 (b) $F^{-1}T^2$
(c) $F^{-1}A^2T^{-1}$ (d) AT^2

58. The dimension of $\frac{1}{\sqrt{\varepsilon_0\mu_0}}$ is that of [SCRA 1986]

(a) Velocity (b) Time
(c) Ca[CBSE PMT 1988]pacitance (d) Distance

59. An athletic coach told his team that muscle times speed equals power. What dimensions does he view for muscle

(a) MLT^{-2} (b) ML^2T^{-2}
(c) MLT^2 (d) L

60. The foundations of dimensional analysis were laid down by

(a) Gallileo (b) Newton
(c) Fourier (d) Joule

61. The dimensional formula of wave number is

(a) $M^0L^0T^{-1}$ (b) $M^0L^{-1}T^0$
(c) $M^{-1}L^{-1}T^0$ (d) $M^0L^0T^0$

62. The dimensions of stress are equal to [MP PET 1991, 2003]

(a) Force (b) Pressure
(c) Work (d) $\frac{1}{\text{Pressure}}$

63. The dimensions of pressure are

[CPMT 1977; MP PMT 1994]

(a) MLT^{-2} (b) $ML^{-2}T^2$
(c) $ML^{-1}T^{-2}$ (d) MLT^2

64. Dimensions of permeability are

[CBSE PMT 1991; AIIMS 2003]

(a) $A^{-2}M^1L^1T^{-2}$ (b) MLT^{-2}
(c) ML^0T^{-1} (d) $A^{-1}MLT^2$

65. Dimensional formula of magnetic flux is

[DCE 1993; IIT 1982; CBSE PMT 1989, 99; DPMT 2001; Kerala PMT 2005]

(a) $ML^2T^{-2}A^{-1}$ (b) $ML^0T^{-2}A^{-2}$
(c) $M^0L^{-2}T^{-2}A^{-3}$ (d) $ML^2T^{-2}A^3$

66. If P represents radiation pressure, c represents speed of light and Q represents radiation energy striking a unit area per *second*, then non-zero integers x, y and z such that $P^xQ^yc^z$ is dimensionless, are

[AFMC 1991; CBSE PMT 1992; CPMT 1981, 92; MP PMT 1992]

(a) $x = 1,\ y = 1,\ z = -1$
(b) $x = 1, y = -1, z = 1$
(c) $x = -1, y = 1, z = 1$
(d) $x = 1, y = 1, z = 1$

67. Inductance L can be dimensionally represented as

[CBSE PMT 1989, 92; IIT 1983; CPMT 1992; DPMT 1999; KCET 2004; J&K CET 2005]

(a) $ML^2T^{-2}A^{-2}$ (b) $ML^2T^{-4}A^{-3}$

(c) $ML^{-2}T^{-2}A^{-2}$ (d) $ML^2T^4A^3$

68. Dimensions of strain are [MP PET 1984; SCRA 1986]

(a) MLT^{-1} (b) ML^2T^{-1}

(c) MLT^{-2} (d) $M^0L^0T^0$

69. Dimensions of time in power are [EAMCET 1982]

(a) T^{-1} (b) T^{-2}

(c) T^{-3} (d) T^0

70. Dimensions of kinetic energy are

[Bihar PET 1983; DPET 1993; AFMC 1991]

(a) ML^2T^{-2} (b) M^2LT^{-1}

(c) ML^2T^{-1} (d) ML^3T^{-1}

71. Dimensional formula for torque is

[DPMT 1984; IIT 1983; CBSE PMT 1990; MNR 1988; AIIMS 2002; BHU 1995, 2001; RPMT 1999; RPET 2003; DCE 1999, 2000; DCE 2004]

(a) L^2MT^{-2} (b) $L^{-1}MT^{-2}$

(c) L^2MT^{-3} (d) LMT^{-2}

72. Dimensions of coefficient of viscosity are

[AIIMS 1993; CPMT 1992; Bihar PET 1984; MP PMT 1987, 89, 91; AFMC 1986; CBSE PMT 1992; KCET 1994; DCE 1999; AIEEE 2004; DPMT 2004]

(a) ML^2T^{-2} (b) ML^2T^{-1}

(c) $ML^{-1}T^{-1}$ (d) MLT

73. The dimension of quantity (L/RCV) is [Roorkee 1994]

(a) $[A]$ (b) $[A^2]$

(c) $[A^{-1}]$ (d) None of these

74. The dimension of the ratio of angular to linear momentum is

(a) $M^0L^1T^0$ (b) $M^1L^1T^{-1}$

(c) $M^1L^2T^{-1}$ (d) $M^{-1}L^{-1}T^{-1}$

75. The pair having the same dimensions is

[MP PET 1994; CPMT 1996]

(a) Angular momentum, work

(b) Work, torque

(c) Potential energy, linear momentum

(d) Kinetic energy, velocity

76. The dimensions of surface tension are

[MP PMT 1994, 99; UPSEAT 1999]

(a) $ML^{-1}T^{-2}$ (b) MLT^{-2}

(c) $ML^{-1}T^{-1}$ (d) MT^{-2}

77. In the following list, the only pair which have different dimensions, is [Manipal MEE 1995]

(a) Linear momentum and moment of a force

(b) Planck's constant and angular momentum

(c) Pressure and modulus of elasticity

(d) Torque and potential energy

78. If R and L represent respectively resistance and self inductance, which of the following combinations has the dimensions of frequency

[MP PMT 1996, 2000, 02; MP PET 1999]

(a) $\frac{R}{L}$ (b) $\frac{L}{R}$

(c) $\sqrt{\frac{R}{L}}$ (d) $\sqrt{\frac{L}{R}}$

79. If velocity v, acceleration A and force F are chosen as fundamental quantities, then the dimensional formula of angular momentum in terms of v, A and F would be

(a) $FA^{-1}v$ (b) Fv^3A^{-2}

(c) Fv^2A^{-1} (d) $F^2v^2A^{-1}$

80. The dimensions of permittivity ε_0 are

[MP PET 1997; AIIMS-2004; DCE-2003]

(a) $A^2T^2M^{-1}L^{-3}$ (b) $A^2T^4M^{-1}L^{-3}$

(c) $A^{-2}T^{-4}ML^3$ (d) $A^2T^{-4}M^{-1}L^{-3}$

81. Dimensions of the following three quantities are the same

[MP PET 1997]

(a) Work, energy, force

(b) Velocity, momentum, impulse

(c) Potential energy, kinetic energy, momentum

(d) Pressure, stress, coefficient of elasticity

82. The dimensions of Planck's constant and angular momentum are respectively [CPMT 1999; BCECE 2004]

(a) ML^2T^{-1} and MLT^{-1} (b) ML^2T^{-1} and ML^2T^{-1}

(c) MLT^{-1} and ML^2T^{-1} (d) MLT^{-1} and ML^2T^{-2}

83. Let $[\varepsilon_0]$ denotes the dimensional formula of the permittivity of the vacuum and $[\mu_0]$ that of the permeability of the vacuum. If $M = \text{mass}$, $L = \text{length}$, $T = \text{Time}$ and $I = \text{electric current}$, then [IIT 1998]

(a) $[\varepsilon_0] = M^{-1}L^{-3}T^2I$ (b) $[\varepsilon_0] = M^{-1}L^{-3}T^4I^2$

(c) $[\mu_0] = MLT^{-2}I^{-2}$ (d) $[\mu_0] = ML^2T^{-1}I$

[MNR 1994]

84. Dimensions of CR are those of

[EAMCET (Engg.) 1995; AIIMS 1999]

(a) Frequency (b) Energy

(c) Time period (d) Current

85. The physical quantity that has no dimensions

[EAMCET (Engg.) 1995]

(a) Angular Velocity (b) Linear momentum

(c) Angular momentum (d) Strain

86. $ML^{-1}T^{-2}$ represents

[EAMCET (Med.) 1995; Pb. PMT 2001]

(a) Stress

(b) Young's Modulus

(c) Pressure

(d) All the above three quantities

87. Dimensions of magnetic field intensity is

[RPMT 1997; EAMCET (Med.) 2000; MP PET 2003]

(a) $[M^0L^{-1}T^0A^1]$ (b) $[MLT^{-1}A^{-1}]$

(c) $[ML^0T^{-2}A^{-1}]$ (d) $[MLT^{-2}A]$

88. The force F on a sphere of radius 'a' moving in a medium with velocity 'v' is given by $F = 6\pi\eta av$. The dimensions of η are [CBSE PMT 1997; DPMT 2000]

(a) $ML^{-1}T^{-1}$ (b) MT^{-1}

(c) MLT^{-2} (d) ML^{-3}

89. Which physical quantities have the same dimension [CPMT 1997]

(a) Couple of force and work

(b) Force and power

(c) Latent heat and specific heat

(d) Work and power

90. Two quantities A and B have different dimensions. Which mathematical operation given below is physically meaningful [CPMT 1997]

(a) A/B (b) $A+B$

(c) $A-B$ (d) None

91. Given that v is speed, r is the radius and g is the acceleration due to gravity. Which of the following is dimensionless [CET 1998]

(a) v^2/rg (b) v^2r/g

(c) v^2g/r (d) v^2rg

92. The physical quantity which has the dimensional formula M^1T^{-3} is [CET 1998]

(a) Surface tension (b) Solar constant

(c) Density (d) Compressibility

93. A force F is given by $F = at + bt^2$, where t is time. What are the dimensions of a and b [AFMC 2001; BHU 1998, 2005]

(a) MLT^{-3} and ML^2T^{-4} (b) MLT^{-3} and MLT^{-4}

(c) MLT^{-1} and MLT^0 (d) MLT^{-4} and MLT^1

94. The dimensions of inter atomic force constant are [UPSEAT 1999]

(a) MT^{-2} (b) MLT^{-1}

(c) MLT^{-2} (d) $ML^{-1}T^{-1}$

95. If the speed of light (c), acceleration due to gravity (g) and pressure (p) are taken as the fundamental quantities, then the dimension of gravitational constant is [AMU (Med.) 1999]

(a) $c^2g^0p^{-2}$ (b) $c^0g^2p^{-1}$

(c) cg^3p^{-2} (d) $c^{-1}g^0p^{-1}$

96. If the time period (T) of vibration of a liquid drop depends on surface tension (S), radius (r) of the drop and density (ρ) of the liquid, then the expression of T is [AMU (Med.) 2000]

(a) $T = k\sqrt{\rho r^3/S}$ (b) $T = k\sqrt{\rho^{1/2}r^3/S}$

(c) $T = k\sqrt{\rho r^3/S^{1/2}}$ (d) None of these

97. $ML^3T^{-1}Q^{-2}$ is dimension of [RPET 2000]

(a) Resistivity (b) Conductivity

(c) Resistance (d) None of these

98. Dimension of electric current is [CBSE PMT 2000]

(a) $[M^0L^0T^{-1}Q]$ (b) $[ML^2T^{-1}Q]$

(c) $[M^2LT^{-1}Q]$ (d) $[M^2L^2T^{-1}Q]$

99. The fundamental physical quantities that have same dimensions in the dimensional formulae of torque and angular momentum are

(a) Mass, time (b) Time, length

(c) Mass, length (d) Time, mole

100. If pressure P, velocity V and time T are taken as fundamental physical quantities, the dimensional formula of force is

(a) PV^2T^2 (b) $P^{-1}V^2T^{-2}$

(c) PVT^2 (d) $P^{-1}VT^2$

101. The physical quantity which has dimensional formula as that of $\frac{\text{Energy}}{\text{Mass} \times \text{Length}}$ is [EAMCET (Eng.) 2000]

(a) Force (b) Power

(c) Pressure (d) Acceleration

102. If energy (E), velocity (v) and force (F) be taken as fundamental quantity, then what are the dimensions of mass [AMU 2000]

(a) Ev^2 (b) Ev^{-2}

(c) Fv^{-1} (d) Fv^{-2}

103. Dimensions of luminous flux are [UPSEAT 2001]

(a) ML^2T^{-2} (b) ML^2T^{-3}

(c) ML^2T^{-1} (d) MLT^{-2}

104. A physcial quantity x depends on quantities y and z as follows: $x = Ay + B\tan Cz$, where A, B and C are constants. Which of the following do not have the same dimensions

(a) x and B (b) C and z^{-1}

(c) y and B/A (d) x and A

105. Which of the following pair does not have similar dimensions

(a) Stress and pressure

(b) Angle and strain

(c) Tension and surface tension

(d) Planck's constant and angular momentum

106. Out of the following which pair of quantities do not have same dimensions [RPET 2001]

(a) Planck's constant and angular momentum

(b) Work and energy

(c) Pressure and Young's modulus

(d) Torque & moment of inertia

107. Identify the pair which has different dimensions [KCET 2001]

(a) Planck's constant and angular momentum

(b) Impulse and linear momentum

(c) Angular momentum and frequency

(d) Pressure and Young's modulus

108. The dimensional formula $M^0L^2T^{-2}$ stands for [KCET 2001]

(a) Torque

(b) Angular momentum

(c) Latent heat

(d) Coefficient of thermal conductivity

109. Which of the following represents the dimensions of *Farad* [AMU (Med.) 2002]

(a) $M^{-1}L^{-2}T^4A^2$ (b) $ML^2T^2A^{-2}$

(c) $ML^2T^2A^{-1}$ (d) $MT^{-2}A^{-1}$

110. If L, C and R denote the inductance, capacitance and resistance respectively, the dimensional formula for C^2LR is [UPSEAT 2002]

(a) $[ML^{-2}T^{-1}I^0]$ (b) $[M^0L^0T^3I^0]$

(c) $[M^{-1}L^{-2}T^6I^2]$ (d) $[M^0L^0T^2I^0]$

111. If the velocity of light (c), gravitational constant (G) and Planck's constant (h) are chosen as fundamental units, then the dimensions of mass in new system is [UPSEAT 2002]

(a) $c^{1/2}G^{1/2}h^{1/2}$ (b) $c^{1/2}G^{1/2}h^{-1/2}$

(c) $c^{1/2}G^{-1/2}h^{1/2}$ (d) $c^{-1/2}G^{1/2}h^{1/2}$

112. Dimensions of charge are [DPMT 2002]
(a) $M^0L^0T^{-1}A^{-1}$ (b) $MLTA^{-1}$
(c) $T^{-1}A$ (d) TA

113. According to Newton, the viscous force acting between liquid layers of area A and velocity gradient $\Delta v/\Delta z$ is given by $F = -\eta A\frac{\Delta v}{\Delta z}$ where η is constant called coefficient of viscosity. The dimension of η are [JIPMER 2001, 02]
(a) $[ML^2T^{-2}]$ (b) $[ML^{-1}T^{-1}]$
(c) $[ML^{-2}T^{-2}]$ (d) $[M^0L^0T^0]$

114. Identify the pair whose dimensions are equal [AIEEE 2002]
(a) Torque and work (b) Stress and energy
(c) Force and stress (d) Force and work

115. The dimensions of pressure is equal to [AIEEE 2002]
(a) Force per unit volume
(b) Energy per unit volume
(c) Force
(d) Energy

116. Which of the two have same dimensions [AIEEE 2002]
(a) Force and strain
(b) Force and stress
(c) Angular velocity and frequency
(d) Energy and strain

117. An object is moving through the liquid. The viscous damping force acting on it is proportional to the velocity. Then dimension of constant of proportionality is [Orissa JEE 2002]
(a) $ML^{-1}T^{-1}$ (b) MLT^{-1}
(c) M^0LT^{-1} (d) ML^0T^{-1}

118. The dimensions of emf in MKS is [CPMT 2002]
(a) $ML^{-1}T^{-2}Q^{-2}$ (b) $ML^2T^{-2}Q^{-2}$
(c) $MLT^{-2}Q^{-1}$ (d) $ML^2T^{-2}Q^{-1}$

119. Which of the following quantities is dimensionless [MP PET 2002]
(a) Gravitational constant (b) Planck's constant
(c) Power of a convex lens (d) None

120. The dimensional formula for Boltzmann's constant is [MP PET 2002; Pb. PET 2001]
(a) $[ML^2T^{-2}\theta^{-1}]$ (b) $[ML^2T^{-2}]$
(c) $[ML^0T^{-2}\theta^{-1}]$ (d) $[ML^{-2}T^{-1}\theta^{-1}]$

121. The dimensions of K in the equation $W = \frac{1}{2}Kx^2$ is [Orissa JEE 2003]
(a) $M^1L^0T^{-2}$ (b) $M^0L^1T^{-1}$
(c) $M^1L^1T^{-2}$ (d) $M^1L^0T^{-1}$

122. The physical quantities not having same dimensions are [AIEEE 2003]
(a) Speed and $(\mu_0\varepsilon_0)^{-1/2}$
(b) Torque and work
(c) Momentum and Planck's constant
(d) Stress and Young's modules

123. Dimension of R is [AFMC 2003; AIIMS 2005]
(a) ML^2T^{-1} (b) $ML^2T^{-3}A^{-2}$
(c) $ML^{-1}T^{-2}$ (d) None of these

124. The dimensional formula of relative density is [CPMT 2003]
(a) ML^{-3} (b) LT^{-1}
(c) MLT^{-2} (d) Dimensionless

125. The dimensional formula for young's modulus is [BHU 2003; CPMT 2004]
(a) $ML^{-1}T^{-2}$ (b) M^0LT^{-2}
(c) MLT^{-2} (d) ML^2T^{-2}

126. Frequency is the function of density (ρ), length (a) and surface tension (T). Then its value is [BHU 2003]
(a) $k\rho^{1/2}a^{3/2}/\sqrt{T}$ (b) $k\rho^{3/2}a^{3/2}/\sqrt{T}$
(c) $k\rho^{1/2}a^{3/2}/T^{3/4}$ (d) $k\rho^{1/2}a^{1/2}/T^{3/2}$

127. The dimensions of electric potential are [UPSEAT 2003]
(a) $[ML^2T^{-2}Q^{-1}]$ (b) $[MLT^{-2}Q^{-1}]$
(c) $[ML^2T^{-1}Q]$ (d) $[ML^2T^{-2}Q]$

128. Dimensions of potential energy are [MP PET 2003]
(a) MLT^{-1} (b) ML^2T^{-2}
(c) $ML^{-1}T^{-2}$ (d) $ML^{-1}T^{-1}$

129. The dimension of $\frac{R}{L}$ are [MP PET 2003]
(a) T^2 (b) T
(c) T^{-1} (d) T^{-2}

130. The dimensions of shear modulus are [MP PMT 2004]
(a) MLT^{-1} (b) ML^2T^{-2}
(c) $ML^{-1}T^{-2}$ (d) MLT^{-2}

131. Pressure gradient has the same dimension as that of [AFMC 2004]
(a) Velocity gradient (b) Potential gradient
(c) Energy gradient (d) None of these

132. If force (F), length (L) and time (T) are assumed to be fundamental units, then the dimensional formula of the mass will be
(a) $FL^{-1}T^2$ (b) $FL^{-1}T^{-2}$
(c) $FL^{-1}T^{-1}$ (d) FL^2T^2

133. The dimensions of universal gas constant is [Pb. PET 2003]
(a) $[ML^2T^{-2}\theta^{-1}]$ (b) $[M^2LT^{-2}\theta]$
(c) $[ML^3T^{-1}\theta^{-1}]$ (d) None of these

134. In the relation $y = a\cos(\omega t - kx)$, the dimensional formula for k is
(a) $[M^0L^{-1}T^{-1}]$ (b) $[M^0LT^{-1}]$
(c) $[M^0L^{-1}T^0]$ (d) $[M^0LT]$

135. Position of a body with acceleration 'a' is given by $x = Ka^mt^n$, here t is time. Find dimension of m and n. [Orissa JEE 2005]
(a) $m = 1$, $n = 1$ (b) $m = 1, n = 2$
(c) $m = 2, n = 1$ (d) $m = 2, n = 2$

136. "Pascal-Second" has dimension of [AFMC 2005]
(a) Force (b) Energy
(c) Pressure (d) Coefficient of viscosity

137. In a system of units if force (F), acceleration (A) and time (T) are taken as fundamental units then the dimensional formula of energy is [BHU 2005]
(a) FA^2T (b) FAT^2

(c) F^2AT (d) FAT

138. Out of the following pair, which one does not have identical dimensions **[AIEEE 2005]**

(a) Moment of inertia and moment of force

(b) Work and torque

(c) Angular momentum and Planck's constant

(d) Impulse and momentum

139. The ratio of the dimension of Planck's constant and that of moment of inertia is the dimension of **[CBSE PMT 2005]**

(a) Frequency (b) Velocity

(c) Angular momentum (d) Time

140. Which of the following group have different dimension **[IIT JEE 2005]**

(a) Potential difference, EMF, voltage

(b) Pressure, stress, young's modulus

(c) Heat, energy, work-done

(d) Dipole moment, electric flux, electric field

141. Out of following four dimensional quantities, which one quantity is to be called a dimensional constant **[KCET 2005]**

(a) Acceleration due to gravity

(b) Surface tension of water

(c) Weight of a standard kilogram mass

(d) The velocity of light in vacuum

142. Density of a liquid in CGS system is 0.625 g/cm^3. What is its magnitude in SI system **[J&K CET 2005]**

(a) 0.625 (b) 0.0625

(c) 0.00625 (d) 625

Errors of Measurement

1. The period of oscillation of a simple pendulum is given by $T = 2\pi\sqrt{\frac{l}{g}}$ where l is about 100 *cm* and is known to have 1*mm* accuracy. The period is about 2*s*. The time of 100 oscillations is measured by a stop watch of least count 0.1 s. The percentage error in g is

(a) 0.1% (b) 1%

(c) 0.2% (d) 0.8%

2. The percentage errors in the measurement of mass and speed are 2% and 3% respectively. How much will be the maximum error in the estimation of the kinetic energy obtained by measuring mass and speed

(a) 11% (b) 8%

(c) 5% (d) 1%

3. The random error in the arithmetic mean of 100 observations is x; then random error in the arithmetic mean of 400 observations would be

(a) $4x$ (b) $\frac{1}{4}x$

(c) $2x$ (d) $\frac{1}{2}x$

4. What is the number of significant figures in 0.310×10^{3}

(a) 2 (b) 3

(c) 4 (d) 6

5. Error in the measurement of radius of a sphere is 1%. The error in the calculated value of its volume is **[AFMC 2005]**

(a) 1% (b) 3%

(c) 5% (d) 7%

6. The mean time period of *second*s pendulum is 2.00*s* and mean absolute error in the time period is 0.05*s*. To express maximum estimate of error, the time period should be written as

(a) (2.00 ± 0.01) *s* (b) $(2.00 + 0.025)$ *s*

(c) (2.00 ± 0.05) *s* (d) (2.00 ± 0.10) *s*

7. A body travels uniformly a distance of (13.8 ± 0.2) *m* in a time (4.0 ± 0.3) s. The velocity of the body within error limits is

(a) (3.45 ± 0.2) *ms* (b) (3.45 ± 0.3) *ms*

(c) (3.45 ± 0.4) *ms* (d) (3.45 ± 0.5) *ms*

8. The percentage error in the above problem is

(a) 7% (b) 5.95%

(c) 8.95% (d) 9.85%

9. The unit of percentage error is

(a) Same as that of physical quantity

(b) Different from that of physical quantity

(c) Percentage error is unit less

(d) Errors have got their own units which are different from that of physical quantity measured

10. The decimal equivalent of 1/20 upto three significant figures is

(a) 0.0500 (b) 0.05000

(c) 0.0050 (d) 5.0×10^{-2}

11. Accuracy of measurement is determined by

(a) Absolute error (b) Percentage error

(c) Both (d) None of these

12. The radius of a sphere is (5.3 ± 0.1) *cm*. The percentage error in its volume is

(a) $\frac{0.1}{5.3}\times100$ (b) $3\times\frac{0.1}{5.3}\times100$

(c) $\frac{0.1\times100}{3.53}$ (d) $3+\frac{0.1}{5.3}\times100$

13. A thin copper wire of length *l metre* increases in length by 2% when heated through 10°C. What is the percentage increase in area when a square copper sheet of length *l metre* is heated through $10^\circ C$

(a) 4% (b) 8%

(c) 16% (d) None of the above

14. In the context of accuracy of measurement and significant figures in expressing results of experiment, which of the following is/are correct

(1) Out of the two measurements 50.14 *cm* and 0.00025 *ampere*, the first one has greater accuracy

(2) If one travels 478 *km* by rail and 397 *m*. by road, the total distance travelled is 478 *km*.

(a) Only (1) is correct (b) Only (2) is correct

(c) Both are correct (d) None of them is correct.

15. A physical parameter a can be determined by measuring the parameters b, c, d and e using the relation $a = b^\alpha c^\beta / d^\gamma e^\delta$. If the maximum errors in the measurement of b, c, d and e are b_1 %, c_1 %, d_1 % and e_1 %, then the maximum error in the value of a determined by the experiment is

(a) $(b_1 + c_1 + d_1 + e_1)$%

(b) $(b_1 + c_1 - d_1 - e_1)$%

(c) $(\alpha b_1 + \beta c_1 - \gamma d_1 - \delta e_1)$%

(d) $(\alpha b_1 + \beta c_1 + \gamma d_1 + \delta e_1)\%$

16. The relative density of material of a body is found by weighing it first in air and then in water. If the weight in air is (5.00 ± 0.05) *Newton* and weight in water is (4.00 ± 0.05) *Newton*. Then the relative density along with the maximum permissible percentage error is

(a) $5.0 \pm 11\%$ (b) $5.0 \pm 1\%$

(c) $5.0 \pm 6\%$ (d) $1.25 \pm 5\%$

17. The resistance $R = \frac{V}{i}$ where $V = 100 \pm 5$ *volts* and $i = 10 \pm 0.2$ *amperes*. What is the total error in *R*

(a) 5% (b) 7%

(c) 5.2% (d) $\frac{5}{2}\%$

18. The period of oscillation of a simple pendulum in the experiment is recorded as 2.63 s, 2.56 s, 2.42 s, 2.71 s and 2.80 s respectively. The average absolute error is

(a) 0.1 s (b) 0.11 s

(c) 0.01 s (d) 1.0 s

19. The length of a cylinder is measured with a meter rod having least count 0.1 *cm*. Its diameter is measured with vernier calipers having least count 0.01 *cm*. Given that length is 5.0 *cm*. and radius is 2.0 *cm*. The percentage error in the calculated value of the volume will be

(a) 1% (b) 2%

(c) 3% (d) 4%

20. In an experiment, the following observation's were recorded : L = 2.820 *m*, M = 3.00 *kg*, l = 0.087 *cm*, Diameter D = 0.041 *cm*

Taking g = 9.81 m/s^2 using the formula , $Y = \frac{4MgL}{\pi D^2 l}$, the maximum permissible error in *Y* is

(a) 7.96% (b) 4.56%

(c) 6.50% (d) 8.42%

21. According to *Joule*'s law of heating, heat produced $H = I^2 Rt$, where *I* is current, *R* is resistance and *t* is time. If the errors in the measurement of *I*, *R* and *t* are 3%, 4% and 6% respectively then error in the measurement of *H* is

(a) $\pm 17\%$ (b) $\pm 16\%$

(c) $\pm 19\%$ (d) $\pm 25\%$

22. If there is a positive error of 50% in the measurement of velocity of a body, then the error in the measurement of kinetic energy is

(a) 25% (b) 50%

(c) 100% (d) 125%

23. A physical quantity *P* is given by $P = \frac{A^3 B^{\frac{1}{2}}}{C^{-4} D^{\frac{3}{2}}}$. The quantity which brings in the maximum percentage error in *P* is

(a) *A* (b) *B*

(c) *C* (d) *D*

24. If $L = 2.331\,cm$, $B = 2.1\,cm$, then $L + B =$ [DCE 2003]

(a) 4.431 *cm* (b) 4.43 *cm*

(c) 4.4 *cm* (d) 4 *cm*

25. The number of significant figures in all the given numbers 25.12, 2009, 4.156 and 1.217×10^{-4} is [Pb. PET 2003]

(a) 1 (b) 2

(c) 3 (d) 4

26. If the length of rod *A* is 3.25 ± 0.01 *cm* and that of *B* is 4.19 ± 0.01 *cm* then the rod *B* is longer than rod *A* by [J&K CET 2005]

(a) 0.94 ± 0.00 *cm* (b) 0.94 ± 0.01 *cm*

(c) 0.94 ± 0.02 *cm* (d) 0.94 ± 0.005 *cm*

27. A physical quantity is given by $X = M^a L^b T^c$. The percentage error in measurement of M, L and T are α, β and γ respectively. Then maximum percentage error in the quantity *X* is

(a) $a\alpha + b\beta + c\gamma$ (b) $a\alpha + b\beta - c\gamma$

(c) $\frac{a}{\alpha} + \frac{b}{\beta} + \frac{c}{\gamma}$ (d) None of these

28. A physical quantity *A* is related to four observable a, b, c and d as follows, $A = \frac{a^2 b^3}{c\sqrt{d}}$, the percentage errors of measurement in a, b, c and d are 1%,3%,2% and 2% respectively. What is the percentage error in the quantity *A* [Kerala PET 2005]

(a) 12% (b) 7%

(c) 5% (d) 14%

Critical Thinking

Objective Questions

1. If the acceleration due to gravity is $10\,ms^{-2}$ and the units of length and time are changed in kilometer and hour respectively, the numerical value of the acceleration is [Kerala PET 2002]

(a) 360000 (b) 72,000

(c) 36,000 (d) 129600

2. If L, C and R represent inductance, capacitance and resistance respectively, then which of the following does not represent dimensions of frequency [IIT 1984]

(a) $\frac{1}{RC}$ (b) $\frac{R}{L}$

(c) $\frac{1}{\sqrt{LC}}$ (d) $\frac{C}{L}$

3. Number of particles is given by $n = -D\frac{n_2 - n_1}{x_2 - x_1}$ crossing a unit area perpendicular to *X*-axis in unit time, where n_1 and n_2 are number of particles per unit volume for the value of x meant to x_2 and x_1. Find dimensions of D called as diffusion constant

(a) $M^0 LT^2$ (b) $M^0 L^2 T^{-4}$

(c) $M^0 LT^{-3}$ (d) $M^0 L^2 T^{-1}$

4. With the usual notations, the following equation $S_t = u + \frac{1}{2}a(2t - 1)$ is

(a) Only numerically correct
(b) Only dimensionally correct
(c) Both numerically and dimensionally correct
(d) Neither numerically nor dimensionally correct

5. If the dimensions of length are expressed as $G^x c^y h^z$; where G, c and h are the universal gravitational constant, speed of light and Planck's constant respectively, then [IIT 1992]

(a) $x = \frac{1}{2}, y = \frac{1}{2}$ (b) $x = \frac{1}{2}, z = \frac{1}{2}$

(c) $y = \frac{1}{2}, z = \frac{3}{2}$ (d) $y = -\frac{3}{2}, z = \frac{1}{2}$

6. A highly rigid cubical block A of small mass M and side L is fixed rigidly onto another cubical block B of the same dimensions and of low modulus of rigidity η such that the lower face of A completely covers the upper face of B. The lower face of B is rigidly held on a horizontal surface. A small force F is applied perpendicular to one of the side faces of A. After the force is withdrawn block A executes small oscillations. The time period of which is given by [IIT 1992]

(a) $2\pi\sqrt{\frac{M\eta}{L}}$ (b) $2\pi\sqrt{\frac{L}{M\eta}}$

(c) $2\pi\sqrt{\frac{ML}{\eta}}$ (d) $2\pi\sqrt{\frac{M}{\eta L}}$

7. The pair(s) of physical quantities that have the same dimensions, is (are) [IIT 1995]

(a) Reynolds number and coefficient of friction
(b) Latent heat and gravitational potential
(c) Curie and frequency of a light wave
(d) Planck's constant and torque

8. The speed of light (c), gravitational constant (G) and Planck's constant (h) are taken as the fundamental units in a system. The dimension of time in this new system should be

(a) $G^{1/2}h^{1/2}c^{-5/2}$ (b) $G^{-1/2}h^{1/2}c^{1/2}$

(c) $G^{1/2}h^{1/2}c^{-3/2}$ (d) $G^{1/2}h^{1/2}c^{1/2}$

9. If the constant of gravitation (G), Planck's constant (h) and the velocity of light (c) be chosen as fundamental units. The dimension of the radius of gyration is [AMU (Eng.) 1999]

(a) $h^{1/2}c^{-3/2}G^{1/2}$ (b) $h^{1/2}c^{3/2}G^{1/2}$

(c) $h^{1/2}c^{-3/2}G^{-1/2}$ (d) $h^{-1/2}c^{-3/2}G^{1/2}$

10. $X = 3YZ^2$ find dimension of Y in (MKSA) system, if X and Z are the dimension of capacity and magnetic field respectively

(a) $M^{-3}L^{-2}T^{-4}A^{-1}$ (b) ML^{-2}

(c) $M^{-3}L^{-2}T^4A^4$ (d) $M^{-3}L^{-2}T^8A^4$

11. In the relation $P = \frac{\alpha}{\beta}e^{-\frac{\alpha Z}{k\theta}}$ P is pressure, Z is the distance, k is Boltzmann constant and θ is the temperature. The dimensional formula of β will be [IIT (Screening) 2004]

(a) $[M^0L^2T^0]$ (b) $[M^1L^2T^1]$

(c) $[M^1L^0T^{-1}]$ (d) $[M^0L^2T^{-1}]$

12. The frequency of vibration of string is given by $\nu = \frac{p}{2l}\left[\frac{F}{m}\right]^{1/2}$. Here p is number of segments in the string and l is the length. The dimensional formula for m will be [BHU 2004]

(a) $[M^0LT^{-1}]$ (b) $[ML^0T^{-1}]$

(c) $[ML^{-1}T^0]$ (d) $[M^0L^0T^0]$

13.

Column I	Column II
(i) Curie	(A) MLT^{-2}
(ii) Light year	(B) M
(iii) Dielectric strength	(C) Dimensionless
(iv) Atomic weight	(D) T
(v) Decibel	(E) ML^2T^{-2}
	(F) MT^{-3}
	(G) T^{-1}
	(H) L
	(I) $MLT^{-3}I^{-1}$
	(J) LT^{-1}

Choose the correct match [IIT 1992]

(a) (i) G, (ii) H, (iii) C, (iv) B, (v) C
(b) (i) D, (ii) H, (iii) I, (iv) B, (v) G
(c) (i) G, (ii) H, (iii) I, (iv) B, (v) G
(d) None of the above

14. A wire has a mass $0.3 \pm 0.003\, g$, radius $0.5 \pm 0.005\, mm$ and length $6 \pm 0.06\, cm$. The maximum percentage error in the measurement of its density is [IIT (Screening) 2004]

(a) 1 (b) 2

(c) 3 (d) 4

15. If 97.52 is divided by 2.54, the correct result in terms of significant figures is

(a) 38.4 (b) 38.3937

(c) 38.394 (d) 38.39

AR Assertion & Reason

For AIIMS Aspirants

Choose any one of the following four responses :

(a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
(b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
(c) If assertion is true but reason is false.
(d) If the assertion and reason both are false.
(e) If assertion is false but reason is true.

1. Assertion : '*Light year*' and 'Wavelength' both measure distance. [MP PMT 2003]

Reason : Both have dimensions of time.

2. Assertion : Light year and year, both measure time.

Reason : Because light year is the time that light takes to reach the earth from the sun.

3. Assertion : Force cannot be added to pressure.

Reason : Because their dimensions are different.

4. Assertion : Linear mass density has the dimensions of $[ML^{-1}T^0]$.

Reason : Because density is always mass per unit volume.

5. Assertion : Rate of flow of a liquid represents velocity of flow.

Reason : The dimensions of rate of flow are $[M^0L^3T^{-1}]$.

6. Assertion : Units of Rydberg constant R are m^{-1}

Reason : It follows from Bohr's formula $\bar{\nu} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$, where the symbols have their usual meaning.

7. Assertion : Parallex method cannot be used for measuring distances of stars more than 100 light years away.
Reason : Because parallex angle reduces so much that it cannot be measured accurately.

8. Assertion : Number of significant figures in 0.005 is one and that in 0.500 is three.
Reason : This is because zeros are not significant.

9. Assertion : Out of three measurements $l = 0.7$ *m*; $l = 0.70$ *m* and $l = 0.700$ *m*, the last one is most accurate.
Reason : In every measurement, only the last significant digit is not accurately known.

10. Assertion : Mass, length and time are fundamental physical quantities.
Reason : They are independent of each other.

11. Assertion : Density is a derived physical quantity.
Reason : Density cannot be derived from the fundamental physical quantities.

12. Assertion : Now a days a standard *metre* is defined as in terms of the wavelength of light.
Reason : Light has no relation with length.

13. Assertion : Radar is used to detect an aeroplane in the sky
Reason : Radar works on the principle of reflection of waves.

14. Assertion : Surface tension and surface energy have the same dimensions.
Reason : Because both have the same S.I. unit

15. Assertion : In $y = A\sin(\omega t - kx)$, $(\omega t - kx)$ is dimensionless.
Reason : Because dimension of $\omega = [M^0L^0T]$.

16. Assertion : Radian is the unit of distance.
Reason : One radian is the angle subtended at the centre of a circle by an arc equal in length to the radius of the circle.

17. Assertion : A.U. is much bigger than Å.
Reason : A.U. stands for astronomical unit and Å stands from *Angstrom*.

18. Assertion : When we change the unit of measurement of a quantity, its numerical value changes.
Reason : Smaller the unit of measurement smaller is its numerical value.

19. Assertion : Dimensional constants are the quantities whose value are constant.
Reason : Dimensional constants are dimensionless.

20. Assertion : The time period of a pendulum is given by the formula, $T = 2\pi\sqrt{g/l}$.
Reason : According to the principle of homogeneity of dimensions, only that formula is correct in which the dimensions of L.H.S. is equal to dimensions of R.H.S.

21. Assertion : In the relation $f = \frac{1}{2l}\sqrt{\frac{T}{m}}$, where symbols have standard meaning, *m* represent linear mass density.
Reason : The frequency has the dimensions of inverse of time.

22. Assertion : The graph between *P* and *Q* is straight line, when *P/Q* is constant.
Reason : The straight line graph means that *P* proportional to *Q* or *P* is equal to constant multiplied by *Q*.

23. Assertion : Avogadro number is the number of atoms in one gram mole.
Reason : Avogadro number is a dimensionless constant.

24. Assertion : *L*/*R* and *CR* both have same dimensions.
Reason : *L/R* and *CR* both have dimension of time.

25. Assertion : The quantity $(1/\sqrt{\mu_0\varepsilon_0})$ is dimensionally equal to velocity and numerically equal to velocity of light.
Reason : μ_0 is permeability of free space and ε_0 is the permittivity of free space.

Answers

Units

1	c	2	b	3	d	4	c	5	c
6	d	7	c	8	d	9	c	10	c
11	a	12	c	13	c	14	b	15	d
16	d	17	c	18	a	19	b	20	d
21	d	22	a	23	a	24	b	25	d
26	b	27	d	28	d	29	d	30	b
31	a	32	b	33	a	34	b	35	a
36	b	37	a	38	b	39	b	40	b
41	d	42	c	43	c, b	44	c	45	b
46	a	47	c	48	c	49	a	50	a
51	b	52	b	53	c	54	c	55	c
56	c	57	b	58	a	59	c	60	a
61	c	62	c	63	d	64	d	65	b
66	c	67	a	68	b	69	c	70	b
71	d	72	b	73	b	74	d	75	c
76	b	77	b	78	b	79	c	80	c
81	a	82	a	83	d	84	c	85	b
86	d	87	d	88	b	89	a	90	c
91	a	92	d	93	b	94	a	95	d
96	a	97	b	98	a	99	d	100	b
101	d	102	d	103	a	104	a	105	d
106	b	107	b	108	b	109	b		

Dimensions

1	a	2	c	3	b	4	a	5	b
6	c	7	c	8	b	9	ad	10	a
11	d	12	b	13	a	14	a	15	a
16	b	17	b	18	d	19	a	20	c
21	b	22	a	23	b	24	d	25	a
26	d	27	a	28	d	29	d	30	d
31	c	32	c	33	a	34	a	35	b
36	b	37	c	38	c	39	a	40	b
41	a	42	b	43	d	44	d	45	a

46	d	47	b	48	d	49	b	50	a
51	a	52	d	53	b	54	b	55	c
56	c	57	d	58	a	59	a	60	c
61	b	62	b	63	c	64	a	65	a
66	b	67	a	68	d	69	c	70	a
71	a	72	c	73	c	74	a	75	b
76	d	77	a	78	a	79	b	80	b
81	d	82	b	83	bc	84	c	85	d
86	d	87	c	88	a	89	a	90	a
91	a	92	b	93	b	94	a	95	b
96	a	97	a	98	a	99	c	100	a
101	d	102	b	103	b	104	d	105	c
106	d	107	c	108	c	109	a	110	b
111	c	112	d	113	b	114	a	115	b
116	c	117	d	118	d	119	d	120	a
121	a	122	c	123	b	124	d	125	a
126	a	127	a	128	b	129	c	130	c
131	d	132	a	133	a	134	c	135	b
136	d	137	b	138	a	139	a	140	d
141	d	142	d						

Errors of Measurement

1	c	2	b	3	b	4	b	5	b
6	c	7	b	8	c	9	c	10	a
11	b	12	b	13	a	14	c	15	d
16	a	17	b	18	b	19	c	20	c
21	b	22	d	23	c	24	c	25	d
26	c	27	a	28	d				

Critical Thinking Questions

1	d	2	d	3	d	4	c	5	bd
6	d	7	abc	8	a	9	a	10	d
11	a	12	c	13	a	14	d	15	a

Assertion and Reason

1	c	2	d	3	a	4	c	5	d
6	a	7	a	8	c	9	b	10	a
11	c	12	c	13	a	14	c	15	c
16	e	17	b	18	c	19	c	20	e
21	b	22	a	23	c	24	a	25	b

Answers and Solutions

Units

1. (c) Light year is a distance which light travels in one year.

2. (b) Because magnitude is absolute.

3. (d) *Watt=Joule/second* = *Ampere×volt* = *Ampere×Ohm*

4. (c) Impulse = change in momentum = $F \times t$

So the unit of momentum will be equal to *Newton-sec.*

5. (c) Unit of energy will be $kg\text{-}m^2/\sec^2$

6. (d) It is by standard definition.

7. (c) $1\,nm = 10^{-9}\,m = 10^{-7}\,cm$

8. (d) $1\,micron = 10^{-6}\,m = 10^{-4}\,cm$

9. (c) *Watt* = *Joule/sec.*

10. (c) $F = \dfrac{Gm_1m_2}{d^2}$; $\therefore$ $G = \dfrac{Fd^2}{m_1m_2} = Nm^2/kg^2$

11. (a)

12. (c) Angular acceleration $= \dfrac{\text{Angular velocity}}{\text{Time}} = \dfrac{rad}{\sec^2}$

13. (c) Stefan's law is $E = \sigma(T^4) \Rightarrow \sigma = \dfrac{E}{T^4}$

where, $E = \dfrac{\text{Energy}}{\text{Area} \times \text{Time}} = \dfrac{\text{Watt}}{m^2}$

$\sigma = \dfrac{\text{Watt-}m^{-2}}{K^4} = Watt\text{–}m^{-2}K^{-4}$

14. (b) *Kg-m/sec* is the unit of linear momentum

15. (d) ct^2 must have dimensions of L

$\Rightarrow$ c must have dimensions of L/T^2 *i.e.* LT^{-2}.

16. (d) $\tau = \dfrac{dL}{dt} \Rightarrow dL = \tau \times dt = r \times F \times dt$

i.e. the unit of angular momentum is *joule-second.*

17. (c)

18. (a) Volume of cube = a^3

Surface area of cube = $6a^2$

according to problem $a = 6a \Rightarrow a = 6$

$\therefore V = a^3 = 216\,units.$

19. (b) $6 \times 10^{-5} = 60 \times 10^{-6} = 60\,microns$

20. (d)

21. (d) Because temperature is a fundamental quantity.

22. (a)

23. (a) 1 C.G.S unit of density = 1000 M.K.S. unit of density

$\Rightarrow 0.5\,gm/cc = 500\,kg/m^3$

24. (b)

25. (d)

26. (b) Mach number $= \dfrac{\text{Velocity of object}}{\text{Velocity of sound}}$.

27. (d)

28. (d) $E = -\dfrac{dV}{dx}$

29. (d)

30. (b) Surface tension = $\dfrac{\text{Force}}{\text{Length}}$ = $Newtons/metre$

31. (a)

32. (b) $L = \dfrac{\phi}{I} = \dfrac{Wb}{A} = Henry$.

33. (a) $\dfrac{L}{R}$ is a time constant of *L-R* circuit so *Henry/ohm* can be expressed as *second.*

34. (b) $mv = kg\left(\dfrac{m}{\sec}\right)$

35. (a) Quantities of similar dimensions can be added or subtracted so unit of *a* will be same as that of velocity.

36. (b) $1\ MeV = 10^6\ eV$

37. (a) Energy (E) = $F \times d \Rightarrow F = \dfrac{E}{d}$ so *Erg/metre* can be the unit of force.

38. (b) Potential energy $= mgh = g\left(\dfrac{cm}{\sec^2}\right)cm = g\left(\dfrac{cm}{\sec}\right)^2$

39. (b) $\dfrac{watt}{ampere} = volt$

40. (b)

41. (d)

42. (c)

43. (b,c)

44. (c) Energy = force × distance, so if both are increased by 4 times then energy will increase by 16 times.

45. (b) 1 Oerstead = 1 Gauss = $10^{-4}\,Tesla$

46. (a) Charge = current × time

47. (c) $R = \rho\dfrac{L}{A} \Rightarrow \rho = \dfrac{RA}{L} = ohm \times cm$

48. (c)

49. (a) Astronomical unit of distance.

50. (a) Physical quantity (p) = Numerical value (n) × Unit (u)

If physical quantity remains constant then $n \propto 1/u$ $\therefore$ $n_1u_1 = n_2u_2$.

51. (b) $1\,eV = 1.6 \times 10^{-19}\,coulomb \times 1\,volt = 1.6 \times 10^{-19}\,J$.

52. (b) $1\,kWh = 1 \times 10^3 \times 3600\,W \times \sec = 36 \times 10^5\,J$

53. (c) According to the definition.

54. (c)

55. (c) As $I = MR^2 = kg - m^2$

56. (c) $\text{Stress} = \dfrac{\text{Force}}{\text{Area}} = \dfrac{N}{m^2}$

57. (b) $\dfrac{Q}{t} = \sigma AT^4 \Rightarrow \sigma = Jm^{-2}s^{-1}K^{-4}$

58. (a) M = Pole strength × length

$= amp - metre \times metre = amp - metre^2$

59. (c) *Curie* = *disintegration/second*

60. (a)

61. (c) Pico prefix used for 10^{-12}

62. (c)

63. (d) Unit of $e.m.f.$ = *volt* = *joule/coulomb*

64. (d)

65. (b)

66. (c) $Y = \frac{F}{A}.\frac{L}{\Delta L} = \frac{dyne}{cm^2} = \frac{10^{-5}N}{10^{-4}m^2} = 0.1 N/m^2$

67. (a) $Y = \frac{\text{Stress}}{\text{Strain}} = \frac{\text{Force/Area}}{\text{Dimensionless}} \Rightarrow Y \equiv \text{Pressure}$.

68. (b) $1\ yard = 36\ inches = 36 \times 2.54\ cm = 0.9144 m$.

69. (c) $1\ \text{fermi} = 10^{-15} metre$

70. (b)

71. (d)

72. (b)

73. (b)

74. (d) 1 *Newton* = 10^5 *Dyne*

75. (c) $[x] = [bt^2] \Rightarrow [b] = [x/t^2] = km/s^2$

76. (b) Units of *a* and PV^2 are same and equal to $dyne \times cm$.

77. (b)

78. (b)

79. (c) Impulse = Force × time $= (kg\text{-}m/s^2) \times s = kg\text{-}m/s$

80. (c)

81. (a) $K = C + 273.15$

82. (a)

83. (d)

84. (c)

85. (b)

86. (d) *Watt* is a unit of power

87. (d) $1\,\text{lightyear} = 9.46 \times 10^{15}\ meter$

88. (b) $V = \frac{W}{m}$ so, SI unit = $\frac{Joule}{kg}$

89. (a)

90. (c) $n_2 = n_1 \left(\frac{M_1}{M_2}\right)^1 \left(\frac{L_1}{L_2}\right)^1 \left(\frac{T}{T_2}\right)^{-2}$

$= 100 \left(\frac{gm}{kg}\right)^1 \left(\frac{cm}{m}\right)^1 \left(\frac{\sec}{min}\right)^{-2}$

$= 100 \left(\frac{gm}{10^3 gm}\right)^1 \left(\frac{cm}{10^2 cm}\right)^1 \left(\frac{\sec}{60 \sec}\right)^{-2}$

$n_2 = \frac{3600}{10^3} = 3.6$

91. (a) $[L/R]$ is a time constant so its unit is *Second*.

92. (d) Poission ratio is a unitless quantity.

93. (b)

94. (a)

95. (d) $P = nu \therefore n \propto \frac{1}{u}$

96. (a) 1 *Faraday* = 96500 *coulomb*.

97. (b)

98. (a)

99. (d)

100. (b)

101. (d) $F = \frac{1}{4\pi \in} \frac{q_1 q_2}{r^2} \Rightarrow \in = \frac{1}{4\pi} \frac{q_1 q_2}{Fr^2} = C^2 m^{-2} N^{-1}$

102. (d) *Joule-sec* is the unit of angular momentum where as other units are of energy.

103. (a) $T = \frac{F}{l} = Nm^{-1}$

104. (a) Because in S.I. system there are seven fundamental quantities.

105. (d) $[\eta] = ML^{-1}T^{-1}$ so its unit will be *kg/m-sec.*

106. (b)

107. (b)

108. (b) According to the definition.

109. (b) Pyrometer is used for measurement of temperature.

Dimensions

1. (a) Pressure $= \frac{\text{Force}}{\text{Area}} = ML^{-1}T^{-2}$

Stress $= \frac{\text{Restoring force}}{\text{Area}} = ML^{-1}T^{-2}$

2. (c) Strain $= \frac{\Delta L}{L} \Rightarrow$ dimensionless quantity

3. (b) Power $= \frac{\text{Work}}{\text{Time}} = \frac{ML^2T^{-2}}{T} = ML^2T^{-3}$

4. (a) *Calorie* is the unit of heat *i.e.*, energy.

So dimensions of energy $= ML^2T^{-2}$

5. (b) Angular momentum = $mvr = MLT^{-1} \times L = ML^2T^{-1}$

6. (c) $\frac{L}{R}$ = Time constant

7. (c) Impulse = change in momentum so dimensions of both quantities will be same and equal to MLT^{-1}

8. (b) $RC = T$

$\because\ [R] = [ML^2T^{-3}I^{-2}]$ and $[C] = [M^{-1}L^{-2}T^4I^2]$

9. (a,d) [Torque] = [work] = $[ML^2T^{-2}]$

[Light year] = [Wavelength] = $[L]$

10. (a) $Q = mL \Rightarrow L = \frac{Q}{m}$ (Heat is a form of energy)

$= \frac{ML^2T^{-2}}{M} = [M^0L^2T^{-2}]$

11. (d) Volume elasticity = $\frac{\text{Force/Area}}{\text{Volume strain}}$

Strain is dimensionless, so

$= \frac{\text{Force}}{\text{Area}} = \frac{MLT^{-2}}{L^2} = [ML^{-1}T^{-2}]$

12. (b) $F = \frac{Gm_1m_2}{d^2} \Rightarrow G = \frac{Fd^2}{m_1m_2}$

$\therefore\ [G] = \frac{[MLT^{-2}][L^2]}{[M^2]} = [M^{-1}L^3T^{-2}]$

13. (a) Angular velocity = $\frac{\theta}{t}$, $[\omega]=\frac{[M^0L^0T^0]}{[T]}=[T^{-1}]$

14. (a) Power = $\frac{\text{Work done}}{\text{Time}}=\left[\frac{ML^2T^{-2}}{T}\right]=[ML^2T^{-3}]$

15. (a) Couple = Force × Arm length = $[MLT^{-2}][L]=[ML^2T^{-2}]$

16. (b) Angular momentum = mvr
$=[MLT^{-1}][L]=[ML^2T^{-1}]$

17. (b) Impulse = Force × Time = $[MLT^{-2}][T]=[MLT^{-1}]$

18. (d) Modulus of rigidity = $\frac{\text{Shear stress}}{\text{Shear strain}}=[ML^{-1}T^{-2}]$

19. (a)

20. (c) $E=hv\Rightarrow[ML^2T^{-2}]=[h][T^{-1}]\Rightarrow[h]=[ML^2T^{-1}]$

21. (b) Moment of inertia $=mr^2=[M][L^2]$
Moment of Force = Force × Perpendicular distance
= $[MLT^{-2}][L]=[ML^2T^{-2}]$

22. (a) Momentum = $mv=[MLT^{-1}]$
Impulse = Force × Time = $[MLT^{-2}]\times[T]=[MLT^{-1}]$

23. (b) Pressure = $\frac{\text{Force}}{\text{Area}}=\frac{\text{Energy}}{\text{Volume}}=ML^{-1}T^{-2}$

24. (d) $[h]=[\text{Angular momentum}]=[ML^2T^{-1}]$

25. (a) By principle of dimensional homogenity $\left[\frac{a}{V^2}\right]=[P]$
$\therefore\ [a]=[P][V^2]=[ML^{-1}T^{-2}]\times[L^6]=[ML^5T^{-2}]$

26. (d) $\frac{1}{2}CV^2$ = Stored energy in a capacitor = $[ML^2T^{-2}]$

27. (a) $\frac{1}{2}Li^2$ = Stored energy in an inductor = $[ML^2T^{-2}]$

28. (d) Energy per unit volume = $\frac{[ML^2T^{-2}]}{[L^3]}=[ML^{-1}T^{-2}]$
Force per unit area = $\frac{[MLT^{-2}]}{[L^2]}=[ML^{-1}T^{-2}]$
Product of voltage and charge per unit volume
$=\frac{V\times Q}{\text{Volume}}=\frac{VIt}{\text{Volume}}=\frac{\text{Power}\times\text{Time}}{\text{Volume}}$
$\Rightarrow\ \frac{[ML^2T^{-3}][T]}{[L^3]}=[ML^{-1}T^{-2}]$
Angular momentum per unit mass = $\frac{[ML^2T^{-1}]}{[M]}=[L^2T^{-1}]$
So angular momentum per unit mass has different dimension.

29. (d) Time constant $\tau=[T]$ and Viscosity $\eta=[ML^{-1}T^{-1}]$
For options (a), (b) and (c) dimensions are not matching with time constant.

30. (d) By putting the dimensions of each quantity both the sides we get $[T^{-1}]=[M]^x[MT^{-2}]^y$
Now comparing the dimensions of quantities in both sides we get $x+y=0$ and $2y=1\ \therefore\ x=-\frac{1}{2}, y=\frac{1}{2}$

31. (c) m = linear density = mass per unit length = $\left[\frac{M}{L}\right]$
A= force $=[MLT^{-2}]\ \therefore\ [B]=\frac{[A]}{[m]}=\frac{[MLT^{-2}]}{[ML^{-1}]}=[L^2T^{-2}]$
This is same dimension as that of latent heat.

32. (c) Let $v^x=kg^y\lambda^z\rho^\delta$. Now by substituting the dimensions of each quantities and equating the powers of M, L and T we get $\delta=0$ and $x=2, y=1, z=1$.

33. (a) *Farad* is the unit of capacitance and $C=\frac{Q}{V}=\frac{[Q]}{[ML^2T^{-2}Q^{-1}]}=M^{-1}L^{-2}T^2Q^2$

34. (a) $\rho=\frac{RA}{l}$ *i.e.* dimension of resistivity is $[ML^3T^{-1}Q^{-2}]$

35. (b) From the principle of homogenity $\left(\frac{x}{v}\right)$ has dimensions of T.

36. (b) $\frac{dQ}{dt}=-KA\left(\frac{d\theta}{dx}\right)$
$\Rightarrow\ [K]=\frac{[ML^2T^{-2}]}{[T]}\times\frac{[L]}{[L^2][K]}=MLT^{-3}K^{-1}$

37. (c) Stress = $\frac{\text{Force}}{\text{Area}}=\frac{[MLT^{-2}]}{[L^2]}=[ML^{-1}T^{-2}]$

38. (c)

39. (a) $[C]=\left(\frac{Q}{V}\right)=\left(\frac{Q^2}{W}\right)=\left[\frac{A^2T^2}{ML^2T^{-2}}\right]=[M^{-1}L^{-2}T^4A^2]$

40. (b) Momentum = mv = $[MLT^{-1}]$

41. (a) $Q=[ML^2T^{-2}]$ (All energies have same dimension)

42. (b) $f=\frac{1}{2\pi\sqrt{LC}}\Rightarrow LC=\frac{1}{f^2}=[M^0L^0T^2]$

43. (d) Energy = Work done [Dimensionally]

44. (d) $\frac{L}{R}$ = Time constant.

45. (a) By substituting the dimension of each quantity we get $T=[ML^{-1}T^{-2}]^a[L^{-3}M]^b[MT^{-2}]^c$
By solving we get a = – 3/2, b = 1/2 and c = 1

46. (d)

47. (b) $v\propto g^ph^q$ (given)
By substituting the dimension of each quantity and comparing the powers in both sides we get $[LT^{-1}]=[LT^{-2}]^p[L]^q$
$\Rightarrow\ p+q=1,\ -2p=-1, \therefore\ p=\frac{1}{2}, q=\frac{1}{2}$

48. (d) [Planck constant] = $[ML^2T^{-1}]$ and
[Energy] = $[ML^2T^{-2}]$

49. (b) Frequency $=\frac{1}{T}=[M^0L^0T^{-1}]$

50. (a) Power = $\frac{\text{Energy}}{\text{Time}}$

51. (a) By substituting dimension of each quantity in R.H.S. of option (a) we get $\left[\frac{mg}{\eta r}\right] = \left[\frac{M \times LT^{-2}}{ML^{-1}T^{-1} \times L}\right] = [LT^{-1}]$.

This option gives the dimension of velocity.

52. (d) $[\varepsilon_0 L]$ = [C] $\therefore$ $X = \frac{\varepsilon_0 LV}{t} = \frac{C \times V}{t} = \frac{Q}{t}$ = current

53. (b) $C = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \Rightarrow \mu_0 \varepsilon_0 = \left(\frac{1}{C^2}\right)$ (where C = velocity of light)

$\therefore$ $[\mu_0 \varepsilon_0] = L^{-2}T^2$

54. (b)

55. (c) $[X]$ = $[F] \times [\rho]$ = $[MLT^{-2}] \times \left[\frac{M}{L^3}\right] = [M^2 L^{-2} T^{-2}]$

56. (c) Both are the formula of energy. $\left(E = \frac{1}{2}CV^2 = \frac{1}{2}LI^2\right)$

57. (d) Acceleration = $\frac{\text{distance}}{\text{time}^2} \Rightarrow A = LT^{-2} \Rightarrow L = AT^2$

58. (a) $\frac{1}{\sqrt{\varepsilon_0 \mu_0}} = C =$ velocity of light

59. (a) According to problem muscle × speed = power

$\therefore$ muscle = $\frac{\text{power}}{\text{speed}} = \frac{ML^2T^{-3}}{LT^{-1}} = MLT^{-2}$

60. (c)

61. (b) Wave number = $\frac{1}{\lambda}$ $\therefore$ dimension is $[M^0 L^{-1} T^0]$

62. (b) [Pressure] =[stress] = $[ML^{-1}T^{-2}]$

63. (c)

64. (a) $F = \frac{\mu_0}{4\pi} \frac{2I_1 I_2 l}{r}$ $\Rightarrow$ $\mu_0 = [F][A]^{-2} = [MLT^{-2}A^{-2}]$

65. (a) $\phi = BA = \frac{F}{I \times L} A = \frac{[MLT^{-2}]\,[L^2]}{[A]\,[L]} = [ML^2T^{-2}A^{-1}]$

66. (b) By substituting the dimension of given quantities $[ML^{-1}T^{-2}]^x [MT^{-3}]^y [LT^{-1}]^z = [MLT]^0$

By comparing the power of M, L, T in both sides $x + y = 0$

$-x + z = 0$ .....(ii)

$-2x - 3y - z = 0$ ...(iii)

The only values of x, y, z satisfying (i), (ii) and (iii) corresponds to (b).

67. (a) $E = \frac{1}{2}Li^2$ hence $L = [ML^2T^{-2}A^{-2}]$

68. (d) Strain is dimensionless.

69. (c) Dimensions of power is $[ML^2T^{-3}]$

70. (a) Kinetic energy = $\frac{1}{2}mv^2 = M[LT^{-1}]^2 = [ML^2T^{-2}]$

71. (a) Torque = force × distance = $[ML^2T^{-2}]$

72. (c) $F = -\eta . A \frac{dv}{dx} \Rightarrow [\eta] = [ML^{-1}T^{-1}]$

73. (c) $\frac{L}{RCV} = \left[\frac{L}{R}\right] \frac{1}{CV} = \frac{T}{Q} = [A^{-1}]$

74. (a) $\frac{\text{Angular momentum}}{\text{Linear momentum}} = \frac{mvr}{mv} = r = [M^0 L^1 T^0]$

75. (b) Dimension of work and torque = $[ML^2T^{-2}]$

76. (d) Surface tension = $\frac{\text{Force}}{\text{Length}} = \frac{[MLT^{-2}]}{L} = [MT^{-2}]$

77. (a) Linear momentum = Mass × Velocity = $[MLT^{-1}]$

Moment of a force = Force × Distance = $[ML^2T^{-2}]$

78. (a) $\frac{R}{L} = \frac{V/I}{V \times T/I} = \frac{1}{T}$ = Frequency

79. (b) $L \propto v^x A^y F^z \Rightarrow L = kv^x A^y F^z$

Putting the dimensions in the above relation

$[ML^2T^{-1}] = k[LT^{-1}]^x [LT^{-2}]^y [MLT^{-2}]^z$

$\Rightarrow [ML^2T^{-1}] = k[M^z L^{x+y+z} T^{-x-2y-2z}]$

Comparing the powers of M, L and T

$z = 1$ ...(i)

$x + y + z = 2$ ...(ii)

$-x - 2y - 2z = -1$ ...(iii)

On solving (i), (ii) and (iii) $x = 3, y = -2, z = 1$

So dimension of L in terms of v, A and f

$[L] = [Fv^3A^{-2}]$

80. (b) $F = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r^2}$

$\Rightarrow \varepsilon_0 = \frac{|q_1||q_2|}{[F][r^2]} = \frac{[A^2T^2]}{[MLT^{-2}][L^2]} = [A^2T^4M^{-1}L^{-3}]$

81. (d) [Pressure] = [Stress] = [coefficient of elasticity] = $[ML^{-1}T^{-2}]$

82. (b)

83. (b, c)

84. (c) Capacity × Resistance = $\frac{\text{Charge}}{\text{Potential}} \times \frac{\text{Volt}}{\text{amp}}$

= $\frac{\text{amp} \times \text{second} \times \text{Volt}}{\text{Volt} \times \text{amp}}$ = Second .....(i)

85. (d) Strain has no dimensions.

86. (d)

87. (c) $B = \frac{F}{IL} = \frac{[MLT^{-2}]}{[A][L]} = [MT^{-2}A^{-1}]$

88. (a) $\eta = \frac{F}{av} = \frac{[MLT^{-2}]}{[L][LT^{-1}]} = [ML^{-1}T^{-1}]$

89. (a) Couple of force = $|\vec{r} \times \vec{F}| = [ML^2T^{-2}]$

Work = $[\vec{F}.\vec{d}] = [ML^2T^{-2}]$

90. (a) Quantities having different dimensions can only be divided or multiplied but they cannot be added or subtracted.

91. (a) Angle of banking : $\tan\theta = \frac{v^2}{rg}$. *i.e.* $\frac{v^2}{rg}$ is dimensionless.

92. (b) Solar constant is energy received per unit area per unit time *i.e.*

$$\frac{[ML^2T^{-2}]}{[L^2][T]}=[M^1T^{-3}]$$

93. (b) From the principle of dimensional homogenity $[a]=\left[\frac{F}{t}\right]=[MLT^{-3}]$ and $[b]=\left[\frac{F}{t^2}\right]=[MLT^{-4}]$

94. (a) $K=Y\times r_0$ = $[ML^{-1}T^{-2}]\times[L]$ = $[MT^{-2}]$

Y = Young's modulus and r_0 = Interatomic distance

95. (b) Let $[G]\propto c^x g^y p^z$

by substituting the following dimensions :

$[G]=[M^{-1}L^3T^{-2}],[c]=[LT^{-1}],[g]=[LT^{-2}]$

$[p]=[ML^{-1}T^{-2}]$

and by comparing the powers of both sides

we can get $x=0,y=2,z=-1$

$\therefore\ [G]\propto c^0g^2p^{-1}$

96. (a) Let $T\propto S^x r^y \rho^z$

by substituting the dimension of $[T]=[T]$

$[S]=[MT^{-2}],[r]=[L],[\rho]=[ML^{-3}]$

and by comparing the power of both the sides

$x=-1/2,y=3/2,z=1/2$

so $T\propto\sqrt{\rho r^3/S}\Rightarrow T=k\sqrt{\frac{\rho r^3}{S}}$

97. (a) Resistivity $[\rho]=\frac{[R].[A]}{[l]}$ where $[R]=[ML^2T^{-1}Q^{-2}]$

$\therefore\ [\rho]=[ML^3T^{-1}Q^{-2}]$

98. (a) $I=\frac{Q}{t}=\frac{[Q]}{[T]}=[M^0L^0T^{-1}Q]$

99. (c) Torque = $[ML^2T^{-2}]$, Angular momentum = $[ML^2T^{-1}]$ So mass and length have the same dimensions

100. (a) Let $F\propto P^xV^yT^z$

by substituting the following dimensions :

$[P]=[ML^{-1}T^{-2}]$ $[V]=[LT^{-1}],[T]=[T]$

and comparing the dimension of both sides

$x=1,y=2,z=2$, so $F=PV^2T^2$

101. (d) $\frac{\text{Energy}}{\text{mass}\times\text{length}}=\frac{[ML^2T^{-2}]}{[M][L]}=[LT^{-2}]$

102. (b) Let $m\propto E^xv^yF^z$

By substituting the following dimensions :

$[E]=[ML^2T^{-2}],[v]=[LT^{-1}],[F]=[MLT^{-2}]$

and by equating the both sides

$x=1,y=-2,z=0$. So $[m]=[Ev^{-2}]$

103. (b)

104. (d) $x=Ay+B\tan Cz$

From the dimensional homogenity

$[x]=[Ay]=[B]\Rightarrow\left[\frac{x}{A}\right]=[y]=\left[\frac{B}{A}\right]$

$[Cz]=[M^0L^0T^0]=$ Dimension less

x and B ; C and Z^{-1}; y and $\frac{B}{A}$ have the same dimension but x and A have the different dimensions.

105. (c) Tension = $[MLT^{-2}]$, Surface Tension = $[MT^{-2}]$

106. (d) Torque = $[ML^2T^{-2}]$, Moment of inertia = $[ML^2]$

107. (c) Angular momentum = $[ML^2T^{-1}]$, Frequency = $[T^{-1}]$

108. (c) Latent Heat $L=\frac{Q}{m}=\frac{\text{Energy}}{\text{mass}}=\frac{[ML^2T^{-2}]}{[M]}=[L^2T^{-2}]$

109. (a) $C=\frac{Q}{V}=\frac{[AT]}{[ML^2T^{-3}A^{-1}]}=[M^{-1}L^{-2}T^4A^2]$

110. (b) $C^2LR=[C^2L^2]\times\left[\frac{R}{L}\right]=[T^4]\times\left[\frac{1}{T}\right]=[T^3]$

As $\left[\frac{L}{R}\right]=T$ and $\sqrt{LC}=T$

111. (c) Let $m\propto C^xG^yh^z$

By substituting the following dimensions :

$[C]=LT^{-1};[G]=[M^{-1}L^3T^{-2}]$ and $[h]=[ML^2T^{-1}]$

Now comparing both sides we will get

$x=1/2;y=-1/2,z=+1/2$

So $m\propto c^{1/2}G^{-1/2}h^{1/2}$

112. (d) Charge = Current × Time $=[AT]$

113. (b) $F=-\eta A\frac{\Delta v}{\Delta z}\Rightarrow[\eta]=[ML^{-1}T^{-1}]$

As $F=[MLT^{-2}],\ A=[L^2],\frac{\Delta v}{\Delta z}=[T^{-1}]$

114. (a)

115. (b) $\frac{\text{Energy}}{\text{Volume}}=\frac{ML^2T^{-2}}{L^3}=[ML^{-1}T^{-2}]$ = Pressure

116. (c) $\omega=\frac{d\theta}{dt}=[T^{-1}]$ and frequency $[n]=[T^{-1}]$

117. (d) $F\propto v\Rightarrow F=kv\Rightarrow[k]=\left[\frac{F}{v}\right]=\left[\frac{MLT^{-2}}{LT^{-1}}\right]=[MT^{-1}]$

118. (d) $e=L\frac{di}{dt}\Rightarrow[e]=[ML^2T^{-2}A^{-2}]\left[\frac{A}{T}\right]$

$[e]=\left[\frac{ML^2T^{-2}}{AT}\right]=[ML^2T^{-2}Q^{-1}]$

119. (d) $[G]=[M^{-1}L^3T^{-2}];[h]=[ML^2T^{-1}]$

Power = $\frac{1}{\text{focal length}}=[L^{-1}]$

All quantities have dimensions

120. (a) $k=\left[\frac{R}{N}\right]=[ML^2T^{-2}\theta^{-1}]$

121. (a) $W=\frac{1}{2}kx^2\Rightarrow[k]=\frac{[W]}{[x^2]}=\left[\frac{ML^2T^{-2}}{L^2}\right]=[MT^{-2}]$

122. (c) Momentum $[MLT^{-1}]$, Planck's constant $[ML^2T^{-1}]$

123. (b) $R = \frac{V}{I} = \left[\frac{ML^2T^{-3}A^{-1}}{A}\right] = [ML^2T^{-3}A^{-2}]$

124. (d) Relative density = $\frac{\text{Density of substance}}{\text{density of water}} = [M^0L^0T^0]$

125. (a)

126. (a) Let $n = k\rho^a a^b T^c$ where $[\rho] = [ML^{-3}], [a] = [L]$ and $[T] = [MT^{-2}]$

Comparing both sides, we get

$a = \frac{1}{2}, b = \frac{3}{2}$ and $c = \frac{-1}{2}$ $\therefore$ $\eta = \frac{k\rho^{1/2}a^{3/2}}{\sqrt{T}}$

127. (a) $V = \frac{W}{Q} = [ML^2T^{-2}Q^{-1}]$

128. (b)

129. (c) L/R is a time constant so $(R/L) = T^{-1}$

130. (c) Shear modulus = $\frac{\text{Shearing stress}}{\text{Shearing strain}} = \frac{F}{A\theta} = [ML^{-1}T^{-2}]$

131. (d) Velocity gradient $= \frac{v}{x} = \frac{[LT^{-1}]}{[L]} = [T^{-1}]$

Potential gradient $= \frac{V}{x} = \frac{[ML^2T^{-3}A^{-1}]}{[L]} = [MLT^{-3}A^{-1}]$

Energy gradient $= \frac{E}{x} = \frac{[ML^2T^2]}{[L]} = [MLT^{-2}]$

and pressure gradient $= \frac{P}{x} = \frac{[ML^{-1}T^{-2}]}{[L]} = [ML^{-2}T^{-2}]$

132. (a) Let $m = KF^aL^bT^c$

Substituting the dimension of

$[F] = [MLT^{-2}], \ [C] = [L] \ and \ [T] = [T]$

and comparing both sides, we get $m = FL^{-1}T^{-2}$

133. (a) $\because R = \frac{PV}{T} = \left[\frac{ML^{-1}T^{-2} \times L^3}{\theta}\right] = [ML^2T^{-2}\theta^{-1}]$

134. (c) $[Kx]$ = Dimension of ωt = (dimensionless) hence

$K = \frac{1}{X} = \frac{1}{L} = [L^{-1}]$ $\therefore$ $[K] = [L^{-1}]$

135. (b) As $x = Ka^m \times t^n$

$[M^0LT^0] = [LT^{-2}]^m[T]^n = [L^mT^{-2m+n}]$

$\therefore$ $m = 1$ and $-2m + n = 0 \Rightarrow n = 2$.

136. (d) $NSm^{-2} = Nm^{-2} \times S =$ Pascal-second.

137. (b) $E = KF^aA^bT^c$

$[ML^2T^{-2}] = [MLT^{-2}]^a[LT^{-2}]^b[T]^c$

$[ML^2T^{-2}] = [M^aL^{a+b}T^{-2a-2b+c}]$

$\therefore$ $a = 1$, $a + b = 2 \Rightarrow b = 1$

and $-2a - 2b + c = -2 \Rightarrow c = 2$

$\therefore$ $E = KFAT^2$.

138. (a)

139. (a) $\frac{h}{I} = \left[\frac{ML^2T^{-1}}{ML^2}\right] = [T^{-1}]$

140. (d)

141. (d)

142. (d) CGS SI

$N_1U_1 = N_2U_2$

$N_1[M_1L_1^{-3}] = N_2[M_2L_2^{-3}]$

$\therefore$ $N_2 = N_1\left[\frac{M_1}{M_2}\right] \times \left[\frac{L_1}{L_2}\right]^{-3} = 0.625\left[\frac{1g}{1kg}\right] \times \left[\frac{1cm}{1m}\right]^{-3}$

$= 0.625 \times 10^{-3} \times 10^6 = 625$

Errors of Measurement

1. (c) $T = 2\pi\sqrt{l/g} \Rightarrow T^2 = 4\pi^2 l/g \Rightarrow g = \frac{4\pi^2 l}{T^2}$

Here % error in l = $\frac{1mm}{100cm} \times 100 = \frac{0.1}{100} \times 100 = 0.1\%$

and % error in T = $\frac{0.1}{2 \times 100} \times 100 = 0.05\%$

$\therefore$ % error in g = % error in l + 2(% error in T)

$= 0.1 + 2 \times 0.05$ = 0.2 %

2. (b) $\therefore E = \frac{1}{2}mv^2$

$\therefore$ % Error in K.E.

= % error in mass + 2 × % error in velocity

= 2 + 2 × 3 = 8 %

3. (b)

4. (b) Number of significant figures are 3, because 10^3 is decimal multiplier.

5. (b) $\because V = \frac{4}{3}\pi r^3$

$\therefore$ % error is volume $= 3 \times \%$ error in radius

$= 3 \times 1$ = 3%

6. (c) Mean time period T = 2.00 *sec*

& Mean absolute error $= \Delta T$ = 0.05 *sec.*

To express maximum estimate of error, the time period should be written as (2.00 ± 0.05) *sec*

7. (b) Here, $S = (13.8 \pm 0.2)\, m$

and $t = (4.0 \pm 0.3)$ *sec*

Expressing it in percentage error, we have,

$S = 13.8 \pm \frac{0.2}{13.8} \times 100\% = 13.8 \pm 1.4\%$

and $t = 4.0 \pm \frac{0.3}{4} \times 100\% = 4 \pm 7.5\%$

$\because V = \frac{s}{t} = \frac{13.8 \pm 1.4}{4 \pm 7.5} = (3.45 \pm 0.3)\, m/s.$

8. (c) % error in velocity = %error in L + %error in t

$= \frac{0.2}{13.8} \times 100 + \frac{0.3}{4} \times 100$

= 1.44 + 7.5 = 8.94 %

9. (c)

10. (a) $\frac{1}{20} = 0.05$

$\therefore$ Decimal equivalent upto 3 significant figures is 0.0500

11. (b)

12. (b) $\because V = \frac{4}{3}\pi r^3$

$\therefore$ % error in volume

$= 3 \times \%$ error in radius.

$= \frac{3 \times 0.1}{5.3} \times 100$

13. (a) Since percentage increase in length = 2 %

Hence, percentage increase in area of square sheet

$= 2 \times 2\%$ = 4%

14. (c) Since for 50.14 *cm*, significant number = 4 and for 0.00025, significant numbers = 2

15. (d) $a = b^{\alpha} c^{\beta} / d^{\gamma} e^{\delta}$

So maximum error in *a* is given by

$$\left(\frac{\Delta a}{a} \times 100\right)_{\max} = \alpha . \frac{\Delta b}{b} \times 100 + \beta . \frac{\Delta c}{c} \times 100 + \gamma . \frac{\Delta d}{d} \times 100 + \delta . \frac{\Delta e}{e} \times 100$$

$= (\alpha b_1 + \beta c_1 + \gamma d_1 + \delta e_1)\%$

16. (a) Weight in air $= (5.00 \pm 0.05)\, N$

Weight in water $= (4.00 \pm 0.05)\, N$

Loss of weight in water $= (1.00 \pm 0.1)\, N$

Now relative density $= \frac{\text{weight in air}}{\text{weight loss in water}}$

i.e. $R\,.\,D = \frac{5.00 \pm 0.05}{1.00 \pm 0.1}$

Now relative density with max permissible error

$= \frac{5.00}{1.00} \pm \left(\frac{0.05}{5.00} + \frac{0.1}{1.00}\right) \times 100 = 5.0 \pm (1 + 10)\%$

$= 5.0 \pm 11\%$

17. (b) $\therefore \left(\frac{\Delta R}{R} \times 100\right)_{\max} = \frac{\Delta V}{V} \times 100 + \frac{\Delta I}{I} \times 100$

$= \frac{5}{100} \times 100 + \frac{0.2}{10} \times 100 = (5 + 2)\%$ = 7%

18. (b) Average value $= \frac{2.63 + 2.56 + 2.42 + 2.71 + 2.80}{5}$

$= 2.62 \text{ sec}$

Now $|\Delta T_1| = 2.63 - 2.62 = 0.01$

$|\Delta T_2| = 2.62 - 2.56 = 0.06$

$|\Delta T_3| = 2.62 - 2.42 = 0.20$

$|\Delta T_4| = 2.71 - 2.62 = 0.09$

$|\Delta T_5| = 2.80 - 2.62 = 0.18$

Mean absolute error

$$\Delta T = \frac{|\Delta T_1| + |\Delta T_2| + |\Delta T_3| + |\Delta T_4| + |\Delta T_5|}{5}$$

$= \frac{0.54}{5} = 0.108 = 0.11 sec$

19. (c) Volume of cylinder $V = \pi r^2 l$

Percentage error in volume

$\frac{\Delta V}{V} \times 100 = \frac{2\Delta r}{r} \times 100 + \frac{\Delta l}{l} \times 100$

$= \left(2 \times \frac{0.01}{2.0} \times 100 + \frac{0.1}{5.0} \times 100\right) = (1 + 2)\% = 3\%$

20. (c) $Y = \frac{4MgL}{\pi D^2 l}$ so maximum permissible error in Y

$= \frac{\Delta Y}{Y} \times 100 = \left(\frac{\Delta M}{M} + \frac{\Delta g}{g} + \frac{\Delta L}{L} + \frac{2\Delta D}{D} + \frac{\Delta l}{l}\right) \times 100$

$= \left(\frac{1}{300} + \frac{1}{981} + \frac{1}{2820} + 2 \times \frac{1}{41} + \frac{1}{87}\right) \times 100$

$= 0.065 \times 100 = 6.5\%$

21. (b) $H = I^2 R t$

$\therefore \frac{\Delta H}{H} \times 100 = \left(\frac{2\Delta I}{I} + \frac{\Delta R}{R} + \frac{\Delta t}{t}\right) \times 100$

$= (2 \times 3 + 4 + 6)\% = 16\%$

22. (d) Kinetic energy $E = \frac{1}{2} m v^2$

$\therefore \frac{\Delta E}{E} \times 100 = \frac{v'^2 - v^2}{v^2} \times 100$

$= [(1.5)^2 - 1] \times 100$

$\therefore \frac{\Delta E}{E} \times 100 = 125\%$

23. (c) Quantity C has maximum power. So it brings maximum error in P.

24. (c) Given, $L = 2.331\, cm$

$= 2.33$ (correct upto two decimal places)

and $B = 2.1\, cm = 2.10\, cm$

$\therefore L + B = 2.33 + 2.10 = 4.43\, cm = 4.4\, cm$

Since minimum significant figure is 2.

25. (d) The number of significant figures in all of the given number is 4.

26. (c)

27. (a) Percentage error in $X = a\alpha + b\beta + c\gamma$

28. (d) Percentage error in A

$= \left(2 \times 1 + 3 \times 3 + 1 \times 2 + \frac{1}{2} \times 2\right)\% = 14\%$

Critical Thinking Questions

1. (d) $n_2 = n_1 \left[\frac{L_1}{L_2}\right]^1 \left[\frac{T_1}{T_2}\right]^{-2} = 10 \left[\frac{meter}{km}\right]^1 \left[\frac{\sec}{hr}\right]^{-2}$

$n_2 = 10 \left[\frac{m}{10^3 m}\right]^1 \left[\frac{\sec}{3600 \sec}\right]^{-2} = 129600$

2. (d) $f = \frac{1}{2\pi\sqrt{LC}}$ $\therefore$ $\left(\frac{C}{L}\right)$ does not represent the dimension of frequency

3. (d) $[n]$ = Number of particles crossing a unit area in unit time = $[L^{-2}T^{-1}]$

$[n_2] = [n_1]$ = number of particles per unit volume = $[L^{-3}]$

$[x_2] = [x_1]$ = positions

$\therefore\ D = \frac{[n][x_2 - x_1]}{[n_2 - n_1]} = \frac{[L^{-2}T^{-1}] \times [L]}{[L^{-3}]} = [L^2T^{-1}]$

4. (c) We can derive this equation from equations of motion so it is numerically correct.

S_t = distance travelled in t second = $\frac{\text{Distance}}{\text{time}} = [LT^{-1}]$

u = velocity = $[LT^{-1}]$ and $\frac{1}{2}a(2t-1) = [LT^{-1}]$

As dimensions of each term in the given equation are same, hence equation is dimensionally correct also.

5. (b, d) Length $\propto Gch$

$L = [M^{-1}L^3T^{-2}]^x\ [LT^{-1}]^y [ML^2T^{-1}]^z$

By comparing the power of M, L and T in both sides we get $-x + z = 0$, $3x + y + 2z = 1$ and $-2x - y - z = 0$

By solving above three equations we get

$x = \frac{1}{2}, y = -\frac{3}{2}, z = \frac{1}{2}$

6. (d) By substituting the dimensions of mass [M], length [L] and coefficient of rigidity $[ML^{-1}T^{-2}]$ we get $T = 2\pi\sqrt{\frac{M}{\eta L}}$ is the right formula for time period of oscillations

7. (a, b, c) Reynolds number and coefficient of friction are dimensionless.

Latent heat and gravitational potential both have dimension $[L^2T^{-2}]$.

Curie and frequency of a light wave both have dimension $[T^{-1}]$. But dimensions of Planck's constant is $[ML^2T^{-1}]$ and torque is $[ML^2T^{-2}]$

8. (a) Time $\propto c^x G^y h^z \Rightarrow T = kc^x G^y h^z$

Putting the dimensions in the above relation

$\Rightarrow\ [M^0L^0T^1] = [LT^{-1}]^x [M^{-1}L^3T^{-2}]^y [ML^2T^{-1}]^z$

$\Rightarrow\ [M^0L^0T^1] = [M^{-y+z}L^{x+3y+2z}T^{-x-2y-z}]$

Comparing the powers of M, L and T

$-y + z = 0$ ...(i)

$x + 3y + 2z = 0$ ...(ii)

$-x - 2y - z = 1$ ...(iii)

On solving equations (i) and (ii) and (iii)

$x = \frac{-5}{2}, y = z = \frac{1}{2}$

Hence dimension of time are $[G^{1/2}h^{1/2}c^{-5/2}]$

9. (a) Let radius of gyration $[k] \propto [h]^x[c]^y[G]^z$

By substituting the dimension of $[k] = [L]$

$[h] = [ML^2T^{-1}], [c] = [LT^{-1}], [G] = [M^{-1}L^3T^{-2}]$

and by comparing the power of both sides

we can get $x = 1/2, y = -3/2, z = 1/2$

So dimension of radius of gyration are $[h]^{1/2}[c]^{-3/2}[G]^{1/2}$

10. (d) $Y = \frac{X}{3Z^2} = \frac{M^{-1}L^{-2}T^4A^2}{[MT^{-2}A^{-1}]^2} = [M^{-3}L^{-2}T^8A^4]$

11. (a) In given equation, $\frac{\alpha z}{k\theta}$ should be dimensionless

$\therefore\ \alpha = \frac{k\theta}{z} \Rightarrow [\alpha] = \frac{[ML^2T^{-2}K^{-1} \times K]}{[L]} = [MLT^{-2}]$

and $P = \frac{\alpha}{\beta} \Rightarrow [\beta] = \left[\frac{\alpha}{p}\right] = \frac{[MLT^{-2}]}{[ML^{-1}T^{-2}]} = [M^0L^2T^0]$.

12. (c) $v = \frac{P}{2l}\left[\frac{F}{m}\right]^{1/2} \Rightarrow v^2 = \frac{P^2}{4l^2}\left[\frac{F}{m}\right] \therefore m \propto \frac{F}{l^2v^2}$

$\Rightarrow [m] = \left[\frac{MLT^{-2}}{L^2T^{-2}}\right] = [ML^{-1}T^0]$

13. (a)

14. (d) $\because$ Density, $\rho = \frac{M}{V} = \frac{M}{\pi r^2 L}$

$\Rightarrow \frac{\Delta\rho}{\rho} = \frac{\Delta M}{M} + 2\frac{\Delta r}{r} + \frac{\Delta L}{L}$

$= \frac{0.003}{0.3} + 2 \times \frac{0.005}{0.5} + \frac{0.06}{6}$

$= 0.01 + 0.02 + 0.01 = 0.04$

$\therefore$ Percentage error $= \frac{\Delta\rho}{\rho} \times 100 = 0.04 \times 100 = 4\%$

15. (a)

Assertion and Reason

1. (c) Light year and wavelength both represents the distance, so both has dimension of length not of time.

2. (d) Light year measures distance and year measures time. One light year is the distance traveled by light in one year.

3. (a) Addition and subtraction can be done between quantities having same dimension.

4. (c) Density is not always mass per unit volume.

5. (d) Rate of flow of liquid is expressed as the volume of liquid flowing per second and it has dimension $[L^3T^{-1}]$.

6. (a)

7. (a) As the distance of star increases, the parallax angle decreases, and great degree of accuracy is required for its measurement. Keeping in view the practical limitation in measuring the parallax angle, the maximum distance of a star we can measure is limited to 100 light year.

8. (c) Since zeros placed to the left of the number are never significant, but zeros placed to right of the number are significant.

9. (b) The last number is most accurate because it has greatest significant figure (3).

10. (a) As length, mass and time represent our basic scientific notations, therefore they are called fundamental quantities and they cannot be obtained from each other.

11. (c) Because density can be derived from fundamental quantities.

12. (c) Because representation of standard metre in terms of wavelength of light is most accurate.

13. (a) As radar is most accurate instrument used to detect aeroplane in sky based on principle of reflection of radio waves.

14. (c) As surface tension and surface energy both have different S.I. unit and same dimensional formula.

15. (c) As ω (angular velocity) has the dimension of $[T^{-1}]$ not $[T]$.

16. (e) Radian is the unit of plane angle.

17. (b) A.U. is used (Astronomical units) to measure the average distance of the centre of the sun from the centre of the earth, while angstrom is used for very short distances. 1 A.U. = $1.5\times 10^{11}\,m;\ 1\mathring{A}=10^{-10}\,m.$

18. (c) We know that $Q=n_1u_1=n_2u_2$ are the two units of measurement of the quantity Q and n, n are their respective numerical values. From relation $Q_1=n_1u_1=n_2u_2$, nu = constant $\Rightarrow n\propto 1/u$ *i.e.,* smaller the unit of measurement, greater is its numerical value.

19. (c) Dimensional constants are the quantities whose value are constant and they posses dimensions. For example, velocity of light in vacuum, universal gravitational constant, Boltzman constant, Planck's constant etc.

20. (e) Let us write the dimension of various quantities on two sides of the given relation.

L.H.S. $=T=[T]$, R.H.S. $=2\pi\sqrt{g/l}=\sqrt{\dfrac{LT^{-2}}{L}}=[T^{-1}]$

($\because 2\pi$ has no dimension). As dimensions of L.H.S. is not equal to dimension of R.H.S. therefore according to principle of homogeneity the relation

$T=2\pi\sqrt{g/l}$ is not valid.

21. (b) From, $f=\dfrac{1}{2l}\sqrt{\dfrac{T}{m}}$, $f^2=\dfrac{T}{4l^2m}$

or, $m=\dfrac{T}{4l^2f^2}=\dfrac{[MLT^{-2}]}{L^2T^{-2}}=\dfrac{M}{L}=\dfrac{\text{Mass}}{\text{length}}$ = linear mass density.

22. (a) According to statement of reason, as the graph is a straight line, $P\propto Q$, or P = constant × Q

i.e. $\dfrac{P}{Q}$ = constant

23. (c) Avogadro number (*N*) represents the number of atoms in 1 gram mole of an element, i.e. it has the dimensions of mole.

24. (a) Unit of quantity (L/R) is Henry/ohm.

As Henry = ohm × sec, hence unit of L/R is sec i.e.

[L/R] = [T].

Similarly, unit of product CR is farad × ohm or,

$\dfrac{\text{Coulomb}}{\text{Volt}}\times\dfrac{\text{Volt}}{\text{Amp}}$ or, $\dfrac{\text{Sec}\times\text{Amp}}{\text{Amp}}$ or, *sec* i.e. [CR] = [T] therefore [L/R] and [CR] both have the same dimension.

25. (b) Both assertion and reason are true but reason is not the correct explanation of assertion.

$[\varepsilon_0]=[M^{-1}L^{-3}T^4I^2]$, $[\mu_0]=[MLT^{-2}I^{-2}]$

$$\Rightarrow \frac{1}{\sqrt{(\mu_0/4\pi)\times 4\pi E_0}}=\sqrt{\frac{9\times 10^9}{10^{-7}}}=\sqrt{9\times 10^{16}}$$

$=3\times 10^8\,m/s.$

Therefore $\dfrac{1}{\sqrt{\mu_0\varepsilon_0}}$ has dimension of velocity and numerically equal to velocity of light.

Units, Dimensions and Measurement

SET Self Evaluation Test - 1

1. The surface tension of a liquid is $70\ dyne/cm$. In MKS system its value is

[CPMT 1973, 74; AFMC 1996; BHU 2002]

(a) $70\,N/m$ (b) $7\times10^{-2}\ N/m$

(c) $7\times10^{3}\ N/m$ (d) $7\times10^{2}\ N/m$

2. The SI unit of universal gas constant (R) is

[MP Board 1988; JIPMER 1993; AFMC 1996; MP PMT 1987, 94; CPMT 1984, 87; UPSEAT 1999]

(a) $Watt K^{-1} mol^{-1}$

(b) $Newton\ K^{-1} mol^{-1}$

(c) $Joule\,K^{-1}\ mol^{-1}$

(d) $Erg\,K^{-1}\ mol^{-1}$

3. The unit of permittivity of free space ε_0 is

[MP PET 1993; MP PMT 2003; CBSE PMT 2004]

(a) *Coulomb/Newton-metre*

(b) Newton-$metre^2/Coulomb^2$

(c) Coulomb$^2/(Newton\text{-}metre)^2$

(d) $Coulomb^2/Newton\text{-}metre^2$

4. The temperature of a body on Kelvin scale is found to be $X\ K$. When it is measured by a Fahrenheit thermometer, it is found to be $X^0 F$. Then X is

[UPSEAT 2000]

(a) 301.25

(b) 574.25

(c) 313

(d) 40

5. What are the units of $K = 1/4\pi\varepsilon_0$

[AFMC 2004]

(a) $C^2N^{-1}m^{-2}$ (b) Nm^2C^{-2}

(c) Nm^2C^2 (d) Unitless

6. The SI unit of surface tension is [DCE 2003]

(a) *Dyne/cm* (b) *Newton/cm*

(c) *Newton/metre* (d) *Newton-metre*

7. E, m, l and G denote energy, mass, angular momentum and gravitational constant respectively, then the dimension of $\frac{El^2}{m^5G^2}$ are [AIIMS 1985]

(a) Angle (b) Length

(c) Mass (d) Time

8. From the equation $\tan\theta = \frac{rg}{v^2}$, one can obtain the angle of banking θ for a cyclist taking a curve (the symbols have their usual meanings). Then say, it is

(a) Both dimensionally and numerically correct

(b) Neither numerically nor dimensionally correct

(c) Dimensionally correct only

(d) Numerically correct only

9. A dimensionally consistent relation for the volume V of a liquid of coefficient of viscosity η flowing per *second* through a tube of radius r and length l and having a pressure difference p across its end, is

(a) $V = \frac{\pi p r^4}{8\eta l}$ (b) $V = \frac{\pi\eta l}{8pr^4}$

(c) $V = \frac{8p\eta l}{\pi r^4}$ (d) $V = \frac{\pi p\eta}{8lr^4}$

10. The velocity v (in $cm/\sec$) of a particle is given in terms of time t (in *sec*) by the relation $v = at + \frac{b}{t+c}$; the dimensions of a, b and c are [CPMT 1990]

(a) $a = L^2, b = T, c = LT^2$

(b) $a = LT^2, b = LT, c = L$

(c) $a = LT^{-2}, b = L, c = T$

(d) $a = L, b = LT, c = T^2$

11. From the dimensional consideration, which of the following equation is correct [CPMT 1983]

(a) $T = 2\pi\sqrt{\frac{R^3}{GM}}$ (b) $T = 2\pi\sqrt{\frac{GM}{R^3}}$

(c) $T = 2\pi\sqrt{\frac{GM}{R^2}}$ (d) $T = 2\pi\sqrt{\frac{R^2}{GM}}$

12. The position of a particle at time t is given by the relation $x(t) = \left(\frac{v_0}{\alpha}\right)(1 - c^{-\alpha t})$, where v_0 is a constant and $\alpha > 0$. The dimensions of v_0 and α are respectively

[CBSE PMT 1995]

(a) $M^0L^1T^{-1}$ and T^{-1}

(b) $M^0L^1T^0$ and T^{-1}

(c) $M^0L^1T^{-1}$ and LT^{-2}

(d) $M^0L^1T^{-1}$ and T

13. The equation of state of some gases can be expressed as $\left(P+\frac{a}{V^2}\right)=\frac{R\theta}{V}$ Where P is the pressure, V the volume, θ the absolute temperature and a and b are constants. The dimensional formula of a is

[UPSEAT 2002; Orissa PMT 2004]

(a) $[ML^5T^{-2}]$ (b) $[M^{-1}L^5T^{-2}]$

(c) $[ML^{-1}T^{-2}]$ (d) $[ML^{-5}T^{-2}]$

14. The dimensions of $\frac{a}{b}$ in the equation $P=\frac{a-t^2}{bx}$, where P is pressure, x is distance and t is time, are

[KCET 2003]

(a) MT^{-2} (b) M^2LT^{-3}

(c) ML^3T^{-1} (d) LT^{-3}

15. Dimensions of $\frac{1}{\mu_0\varepsilon_0}$, where symbols have their usual meaning, are [AIEEE 2003]

(a) $[LT^{-1}]$ (b) $[L^{-1}T]$

(c) $[L^{-2}T^2]$ (d) $[L^2T^{-2}]$

16. The dimensions of $e^2/4\pi\varepsilon_0 hc$, where e, ε_0, h and c are electronic charge, electric permittivity, Planck's constant and velocity of light in vacuum respectively [UPSEAT 2004]

(a) $[M^0L^0T^0]$ (b) $[M^1L^0T^0]$

(c) $[M^0L^1T^0]$ (d) $[M^0L^0T^1]$

17. If radius of the sphere is $(5.3\pm 0.1)\,cm$. Then percentage error in its volume will be [Pb. PET 2000]

(a) $3+6.01\times\frac{100}{5.3}$ (b) $\frac{1}{3}\times 0.01\times\frac{100}{5.3}$

(c) $\left(\frac{3\times 0.1}{5.3}\right)\times 100$ (d) $\frac{0.1}{5.3}\times 100$

18. The pressure on a square plate is measured by measuring the force on the plate and the length of the sides of the plate. If the maximum error in the measurement of force and length are respectively 4% and 2%, The maximum error in the measurement of pressure is

[CPMT 1993]

(a) 1% (b) 2%

(c) 6% (d) 8%

19. While measuring the acceleration due to gravity by a simple pendulum, a student makes a positive error of 1% in the length of the pendulum and a negative error of 3% in the value of time period. His percentage error in the measurement of g by the relation $g=4\pi^2\left(l/T^2\right)$ will be

(a) 2% (b) 4%

(c) 7% (d) 10%

20. The length, breadth and thickness of a block are given by $l=12\,cm, b=6\,cm$ and $t=2.45\,cm$

The volume of the block according to the idea of significant figures should be [CPMT 2004]

(a) $1\times 10^2\,cm^3$ (b) $2\times 10^2\,cm^3$

(c) $1.763\times 10^2 cm^3$ (d) None of these

Answers and Solutions

(SET -1)

1. (b) 1 $dyne = 10^{-5}$ $Newton$, $1\,cm = 10^{-2}\,m$

$$70\,\frac{dyne}{cm} = \frac{70\times 10^{-5}}{10^{-2}}\,\frac{N}{m}$$

$= 7\times 10^{-2}\,N/m$.

2. (c) $PV = nRT \Rightarrow R = \dfrac{PV}{nT} = \dfrac{Joule}{mole \times Kelvin} = JK^{-1}mol^{-1}$

3. (d) $F = \dfrac{1}{4\pi\varepsilon_0}\cdot\dfrac{Q_1Q_2}{r^2}$

$$\Rightarrow \varepsilon_0 \propto \frac{Q^2}{F\times r^2}$$

So ε_0 has units of $Coulomb^2/Newton\text{-}m^2$

4. (b) $\dfrac{F-32}{9} = \dfrac{K-273}{5} \Rightarrow \dfrac{x-32}{9} = \dfrac{x-273}{5} \Rightarrow x = 574.25$

5. (b) Unit of $\varepsilon_0 = C^2/N\text{-}m^2$ $\therefore$Unit of K = Nm^2C^{-2}

6. (c)

7. (a) $[E] = [ML^2T^{-2}], [m] = [M], [l] = [ML^2T^{-1}]$ and $[G] = [M^{-1}L^3T^{-2}]$ Substituting the dimension of above quantities in the given formula :

$$\frac{El^2}{m^5G^2}\ \frac{[ML^2T^{-2}][ML^2T^{-1}]^2}{[M^5][M^{-1}L^3T^{-2}]^2} = \frac{M^3L^6T^{-4}}{M^3L^6T^{-4}} = [M^0L^0T^0]$$

8. (c) Given equation is dimensionally correct because both sides are dimensionless but numerically wrong because the correct equation is $\tan\theta = \dfrac{v^2}{rg}$.

9. (a) Formula for viscosity $\eta = \dfrac{\pi pr^4}{8Vl} \Rightarrow V = \dfrac{\pi pr^4}{8\eta l}$

10. (c) From the principle of dimensional homogenity $[v] = [at] \Rightarrow [a] = [LT^{-2}]$. Similarly $[b] = [L]$ and $[c] = [T]$

11. (a) By substituting the dimensions in $T = 2\pi\sqrt{\dfrac{R^3}{GM}}$

we get $\sqrt{\dfrac{L^3}{M^{-1}L^3T^{-2}\times M}} = T$

12. (a) Dimension of αt = $[M^0L^0T^0]$ $\therefore$ $[\alpha]$ = $[T^{-1}]$

Again $\left[\dfrac{v_0}{\alpha}\right] = [L]$ so $[v_0] = [LT^{-1}]$

***

13. (a) By the principle of dimensional homogenity

$$[P] = \left[\frac{a}{V^2}\right] \Rightarrow [a] = [P]\times[V^2] = [ML^{-1}T^{-2}][L^6]$$

$= [ML^5T^{-2}]$

14. (a) $[a] = [T^2]$ and $[b] = \dfrac{[a-t^2]}{[P][x]} = \dfrac{T^2}{[ML^{-1}T^{-2}][L]}$

$\Rightarrow [b] = [M^{-1}T^4]$

So $\left[\dfrac{a}{b}\right] = \dfrac{[T^2]}{[M^{-1}T^4]} = [MT^{-2}]$

15. (d) $C = \dfrac{1}{\sqrt{\mu_0\varepsilon_0}} \Rightarrow \dfrac{1}{\mu_0\varepsilon_0} = c^2 = [L^2T^{-2}]$

16. (a) $[e] = [AT]$, $\epsilon_0 = [M^{-1}L^{-3}T^4A^2]$, $[h] = [ML^2T^{-1}]$

and $[c] = [LT^{-1}]$

$$\therefore \left[\frac{e^2}{4\pi\epsilon_0\,hc}\right] = \left[\frac{A^2T^2}{M^{-1}L^{-3}T^4A^2\times ML^2T^{-1}\times LT^{-1}}\right]$$

$= [M^0L^0T^0]$

17. (c) Volume of sphere $(V) = \dfrac{4}{3}\pi r^3$

% error in volume $= 3\times\dfrac{\Delta r}{r}\times 100 = \left(3\times\dfrac{0.1}{5.3}\right)\times 100$

18. (d) $P = \dfrac{F}{A} = \dfrac{F}{l^2}$, so maximum error in pressure (P)

$$\left(\frac{\Delta P}{P}\times 100\right)_{\max} = \frac{\Delta F}{F}\times 100 + 2\frac{\Delta l}{l}\times 100$$

=4%+2×2%=8%

19. (c) Percentage error in g = (%error in l) + 2(% error in T) = 1% + 2(3%) = 7%

20. (b) Volume $V = l\times b\times t$

$= 12\times 6\times 2.45 = 176.4\ cm^3$

$V = 1.764\times 10^2\ cm^3$

since, the minimum number of significant figure is one in breadth, hence volume will also contain only one significant figure. Hence, $V = 2\times 10^2\ cm^3$.

Chapter

2

Motion In One Dimension

Position

Any object is situated at point O and three observers from three different places are looking at same object, then all three observers will have different observations about the position of point O and no one will be wrong. Because they are observing the object from different positions.

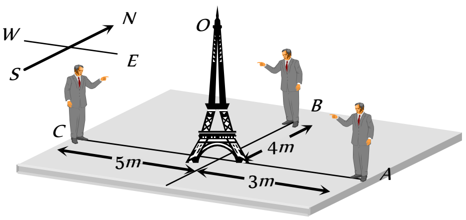


Fig. 2.1

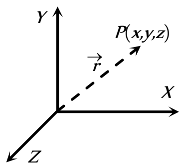
Observer 'A' says : Point O is 3 m away in west direction.

Observer 'B' says : Point O is 4 m away in south direction.

Observer 'C' says : Point O is 5 m away in east direction.

Therefore position of any point is completely expressed by two factors: Its distance from the observer and its direction with respect to observer.

That is why position is characterised by a vector known as position vector.



Consider a point P in xy plane and its coordinates are (x, y). Then position vector $(\vec{r})$ of point will be $x\hat{i} + y\hat{j}$ and if the point P is in space and its coordinates are (x, y, z) then position vector can be expressed as $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.

Rest and Motion

If a body does not change its position as time passes with respect to frame of reference, it is said to be at rest.

And if a body changes its position as time passes with respect to frame of reference, it is said to be in motion.

Frame of Reference : It is a system to which a set of coordinates are attached and with reference to which observer describes any event.

A passenger standing on platform observes that a tree on a platform is at rest. But the same passenger passing away in a train through station, observes that tree is in motion. In both conditions observer is right. But observations are different because in first situation observer stands on a platform, which is reference frame at rest and in second situation observer moving in train, which is reference frame in motion.

So rest and motion are relative terms. It depends upon the frame of references.

Table 2.1 : Types of motion

One dimensional	Two dimensional	Three dimensional
Motion of a body in a straight line is called one dimensional motion.	Motion of body in a plane is called two dimensional motion.	Motion of body in a space is called three dimensional motion.
When only one coordinate of the position of a body changes with time then it is said to be moving one dimensionally.	When two coordinates of the position of a body changes with time then it is said to be moving two dimensionally.	When all three coordinates of the position of a body changes with time then it is said to be moving three dimensionally.
Ex. (i) Motion of car on a straight road. (ii) Motion of freely falling body.	*Ex.* (i) Motion of car on a circular turn. (ii) Motion of billiards ball.	*Ex.* (i) Motion of flying kite. (ii) Motion of flying insect.

Particle or Point Mass or Point object

The smallest part of matter with zero dimension which can be described by its mass and position is defined as a particle or point mass.

If the size of a body is negligible in comparison to its range of motion then that body is known as a particle.

A body (Group of particles) can be treated as a particle, depends upon types of motion. For example in a planetary motion around the sun the different planets can be presumed to be the particles.

In above consideration when we treat body as particle, all parts of the body undergo same displacement and have same velocity and acceleration.

Distance and Displacement

(1) **Distance :** It is the actual length of the path covered by a moving particle in a given interval of time.

(i) If a particle starts from A and reach to C through point B as shown in the figure.

Then distance travelled by particle

$= AB + BC = 7\ m$

(ii) Distance is a scalar quantity.

(iii) Dimension : $[M^0 L T^0]$

(iv) Unit : metre (S.I.)

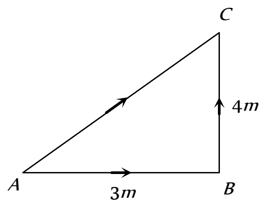


Fig. 2.2

(2) **Displacement :** Displacement is the change in position vector i.e., A vector joining initial to final position.

(i) Displacement is a vector quantity

(ii) Dimension : $[M^0 L T^0]$

(iii) Unit : metre (S.I.)

(iv) In the above figure the displacement of the particle $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} \Rightarrow |\ AC|$

$= \sqrt{(AB)^2 + (BC)^2 + 2(AB)(BC)\cos 90^o} = 5\ m$

(v) If $\vec{S}_1, \vec{S}_2, \vec{S}_3$ .... $\vec{S}_n$ are the displacements of a body then the total (net) displacement is the vector sum of the individuals. $\vec{S} = \vec{S}_1 + \vec{S}_2 + \vec{S}_3 + ... + \vec{S}_n$

(3) **Comparison between distance and displacement :**

(i) The magnitude of displacement is equal to minimum possible distance between two positions.

So distance $\geq$ |Displacement|.

(ii) For a moving particle distance can never be negative or zero while displacement can be.

(zero displacement means that body after motion has came back to initial position)

i.e., Distance > 0 but Displacement > = or < 0

(iii) For motion between two points, displacement is single valued while distance depends on actual path and so can have many values.

(iv) For a moving particle distance can never decrease with time while displacement can. Decrease in displacement with time means body is moving towards the initial position.

(v) In general, magnitude of displacement is not equal to distance. However, it can be so if the motion is along a straight line without change in direction.

(vi) If $\vec{r}_A$ and $\vec{r}_B$ are the position vectors of particle initially and finally.

Then displacement of the particle $\vec{r}_{AB} = \vec{r}_B - \vec{r}_A$

and s is the distance travelled if the particle has gone through the path *APB*.

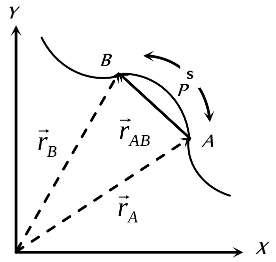


Fig. 2.3

Speed and Velocity

(1) **Speed :** The rate of distance covered with time is called speed.

(i) It is a scalar quantity having symbol υ .

(ii) Dimension : $[M^0 L T^{-1}]$

(iii) Unit : *metre/second* (S.I.), *cm/second* (C.G.S.)

(iv) Types of speed :

(a) **Uniform speed :** When a particle covers equal distances in equal intervals of time, (no matter how small the intervals are) then it is said to be moving with uniform speed. In given illustration motorcyclist travels equal distance (= 5*m*) in each second. So we can say that particle is moving with uniform speed of 5 *m/s*.

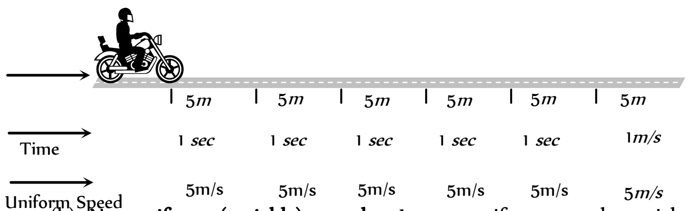


Fig. 2.4

(b) **Non-uniform (variable) speed :** In non-uniform speed particle covers unequal distances in equal intervals of time. In the given illustration motorcyclist travels 5*m* in 1st second, 8*m* in 2nd second, 10*m* in 3rd second, 4*m* in 4th second *etc*.

Therefore its speed is different for every time interval of one second. This means particle is moving with variable speed.

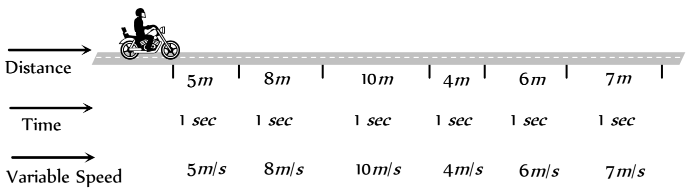


Fig. 2.5

(c) **Average speed :** The average speed of a particle for a given 'Interval of time' is defined as the ratio of total distance travelled to the time taken.

Average speed $= \dfrac{\text{Total distance travelled}}{\text{Time taken}}$; $v_{av} = \dfrac{\Delta s}{\Delta t}$

❑ Time average speed : When particle moves with different uniform speed υ_1, υ_2, υ_3 ... *etc* in different time intervals t_1, t_2, t_3, ... *etc* respectively, its average speed over the total time of journey is given as

$$v_{av} = \frac{\text{Total distance covered}}{\text{Total time elapsed}}$$

$$= \frac{d_1 + d_2 + d_3 + ...}{t_1 + t_2 + t_3 + ...} = \frac{\upsilon_1 t_1 + \upsilon_2 t_2 + \upsilon_3 t_3 + ...}{t_1 + t_2 + t_3 + ...}$$

❑ Distance averaged speed : When a particle describes different distances d_1, d_2, d_3, ... with different time intervals t_1, t_2, t_3, ... with speeds v_1, v_2, v_3...... respectively then the speed of particle averaged over the total distance can be given as

$$\upsilon_{av} = \frac{\text{Total distance covered}}{\text{Total time elapsed}} = \frac{d_1 + d_2 + d_3 + ...}{t_1 + t_2 + t_3 + ...}$$

$$= \frac{d_1 + d_2 + d_3 + ...}{\dfrac{d_1}{\upsilon_1} + \dfrac{d_2}{\upsilon_2} + \dfrac{d_3}{\upsilon_3} + ...}$$

❑ If speed is continuously changing with time then

$$v_{av} = \frac{\int v dt}{\int dt}$$

(d) **Instantaneous speed :** It is the speed of a particle at a particular instant of time. When we say "speed", it usually means instantaneous speed.

The instantaneous speed is average speed for infinitesimally small time interval (*i.e.*, $\Delta t \to 0$). Thus

Instantaneous speed $v = \lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}$

(2) **Velocity :** The rate of change of position *i.e.* rate of displacement with time is called velocity.

(i) It is a vector quantity having symbol $\vec{v}$.

(ii) Dimension : $[M^0LT^{-1}]$

(iii) Unit : *metre/second* (S.I.), *cm/second* (C.G.S.)

(iv) Types of velocity :

(a) **Uniform velocity :** A particle is said to have uniform velocity, if magnitudes as well as direction of its velocity remains same and this is possible only when the particles moves in same straight line without reversing its direction.

(b) **Non-uniform velocity :** A particle is said to have non-uniform velocity, if either of magnitude or direction of velocity changes or both of them change.

(c) **Average velocity** : It is defined as the ratio of displacement to time taken by the body

$$\text{Average velocity} = \frac{\text{Displacement}}{\text{Time taken}}; \quad \vec{v}_{av} = \frac{\Delta \vec{r}}{\Delta t}$$

(d) **Instantaneous velocity :** Instantaneous velocity is defined as rate of change of position vector of particles with time at a certain instant of time.

Instantaneous velocity $\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$

(v) **Comparison between instantaneous speed and instantaneous velocity**

(a) instantaneous velocity is always tangential to the path followed by the particle.

When a stone is thrown from point O then at point of projection the instantaneous velocity of stone is $\vec{v}_1$, at point A the instantaneous velocity of stone is $\vec{v}_2$, similarly at point B and C are $\vec{v}_3$ and $\vec{v}_4$ respectively.

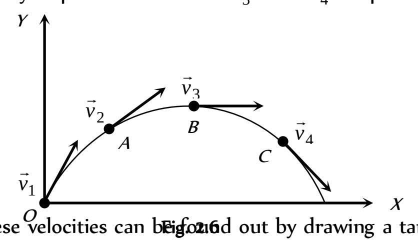


Fig. 2.6

Direction of these velocities can be found out by drawing a tangent on the trajectory at a given point.

(b) A particle may have constant instantaneous speed but variable instantaneous velocity.

Example : When a particle is performing uniform circular motion then for every instant of its circular motion its speed remains constant but velocity changes at every instant.

(c) The magnitude of instantaneous velocity is equal to the instantaneous speed.

(d) If a particle is moving with constant velocity then its average velocity and instantaneous velocity are always equal.

(e) If displacement is given as a function of time, then time derivative of displacement will give velocity.

Let displacement $\vec{x} = A_0 - A_1 t + A_2 t^2$

Instantaneous velocity $\vec{v} = \frac{d\vec{x}}{dt} = \frac{d}{dt}(A_0 - A_1 t + A_2 t^2)$

$\vec{v} = -A_1 + 2A_2 t$

For the given value of t, we can find out the instantaneous velocity.

e.g for $t = 0$,Instantaneous velocity $\vec{v} = -A_1$ and Instantaneous speed $|\vec{v}| = A_1$

(vi) **Comparison between average speed and average velocity**

(a) Average speed is a scalar while average velocity is a vector both having same units (*m/s*) and dimensions $[LT^{-1}]$.

(b) Average speed or velocity depends on time interval over which it is defined.

(c) For a given time interval average velocity is single valued while average speed can have many values depending on path followed.

(d) If after motion body comes back to its initial position then $\vec{v}_{av} = 0$ (as $\Delta \vec{r} = 0$) but $v_{av} > 0$ and finite as $(\Delta s > 0)$.

(e) For a moving body average speed can never be negative or zero (unless $t \to \infty$) while average velocity can be *i.e.* $v_{av} > 0$ while $\vec{v}_{av}$ = or < 0.

(f) As we know for a given time interval

Distance ≥ |displacement|

∴ Average speed ≥ |Average velocity|

Acceleration

The time rate of change of velocity of an object is called acceleration of the object.

(1) It is a vector quantity. It's direction is same as that of change in velocity (Not of the velocity)

Table 2.2 : Possible ways of velocity change

When only direction of velocity changes	When only magnitude of velocity changes	When both magnitude and direction of velocity changes
Acceleration perpendicular to velocity	Acceleration parallel or anti-parallel to velocity	Acceleration has two components one is perpendicular to velocity and another parallel or anti-parallel to velocity
Ex. Uniform circular motion	*Ex.* Motion under gravity	*Ex.* Projectile motion

(2) Dimension : $[M^0LT^{-2}]$

(3) Unit : *metre/second²* (S.I.); *cm/second²* (C.G.S.)

(4) Types of acceleration :

(i) **Uniform acceleration :** A body is said to have uniform acceleration if magnitude and direction of the acceleration remains constant during particle motion.

(ii) **Non-uniform acceleration :** A body is said to have non-uniform acceleration, if either magnitude or direction or both of them change during motion.

(iii) **Average acceleration :** $\vec{a}_{av} = \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}_2 - \vec{v}_1}{\Delta t}$

The direction of average acceleration vector is the direction of the change in velocity vector as $\vec{a} = \frac{\Delta \vec{v}}{\Delta t}$

(iv) **Instantaneous acceleration** = $\vec{a} = \lim_{\Delta t \to 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt}$

(v) For a moving body there is no relation between the direction of instantaneous velocity and direction of acceleration.

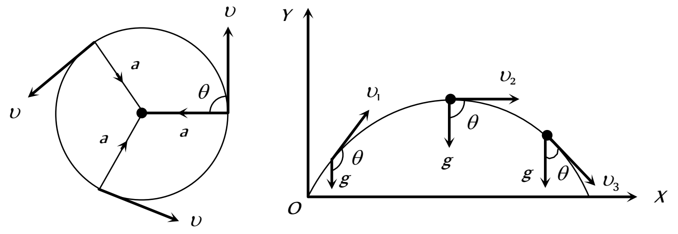


Fig. 2.7

Ex. (a) In uniform circular motion θ = 90° always

(b) In a projectile motion θ is variable for every point of trajectory.

(vi) If a force $\vec{F}$ acts on a particle of mass *m*, by Newton's 2nd law, acceleration $\vec{a} = \frac{\vec{F}}{m}$

(vii) By definition $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{x}}{dt^2}\left[\text{As } \vec{v} = \frac{d\vec{x}}{dt}\right]$

i.e., if *x* is given as a function of time, second time derivative of displacement gives acceleration

(viii) If velocity is given as a function of position, then by chain rule

$$a = \frac{dv}{dt} = \frac{dv}{dx} \times \frac{dx}{dt} = v.\frac{d\upsilon}{dx}\left[\text{as } v = \frac{dx}{dt}\right]$$

(xi) Acceleration can be positive, zero or negative. Positive acceleration means velocity increasing with time, zero acceleration means velocity is uniform constant while negative acceleration (retardation) means velocity is decreasing with time.

(xii) For motion of a body under gravity, acceleration will be equal to "*g*", where g is the acceleration due to gravity. Its value is $9.8\ m/s^2$ or $980\ cm/s^2$ or $32\ feet/s^2$.

Position time Graph

During motion of the particle its parameters of kinematical analysis (*v*, *a*, *s*) changes with time. This can be represented on the graph.

Position time graph is plotted by taking time *t* along *x*-axis and position of the particle on *y*-axis.

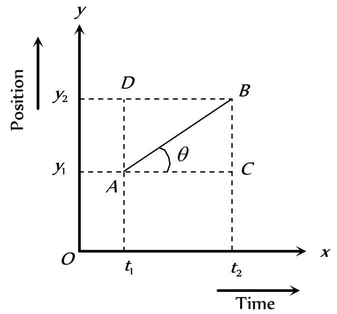


Fig. 2.8

Let *AB* is a position-time graph for any moving particle

As Velocity = $\frac{\text{Change in position}}{\text{Time taken}} = \frac{y_2 - y_1}{t_2 - t_1}$ ....(i)

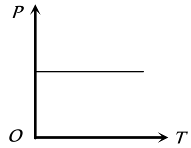
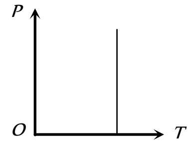
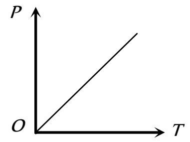
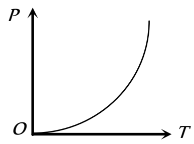
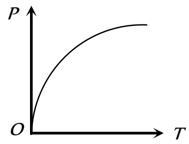
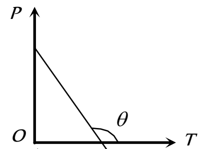
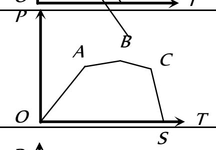
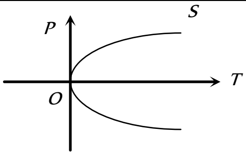
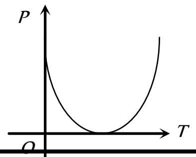
From triangle *ABC*, $\tan\theta = \frac{BC}{AC} = \frac{AD}{AC} = \frac{y_2 - y_1}{t_2 - t_1}$ ....(ii)

By comparing (i) and (ii) Velocity = $\tan\theta$

$v = \tan\theta$

It is clear that slope of tangent on position-time graph represents the velocity of the particle.

Table 2.3 : Various position -time graphs and their interpretation

	$\theta = 0°$ so $v = 0$ *i.e.*, line parallel to time axis represents that the particle is at rest.
	$\theta = 90°$ so $v = \infty$ i.e., line perpendicular to time axis represents that particle is changing its position but time does not changes it means the particle possesses infinite velocity. Practically this is not possible.
	$\theta =$ constant so $v =$ constant, $a = 0$ i.e., line with constant slope represents uniform velocity of the particle.
	θ is increasing so v is increasing, a is positive. *i.e.*, line bending towards position axis represents increasing velocity of particle. It means the particle possesses acceleration.
	θ is decreasing so v is decreasing, a is negative *i.e.*, line bending towards time axis represents decreasing velocity of the particle. It means the particle possesses retardation.
	θ constant but > 90° so v will be constant but negative *i.e.*, line with negative slope represent that particle returns towards the point of reference. (negative displacement).
	Straight line segments of different slopes represent that velocity of the body changes after certain interval of time.
	This graph shows that at one instant the particle has two positions, which is not possible.
	The graph shows that particle coming towards origin initially and after that it is moving away from origin.

Note : ❑ If the graph is plotted between distance and time then it is always an increasing curve and it never comes back towards origin because distance never decrease with time. Hence such type of distance time graph is valid up to point A only, after point *A*, it is not valid as shown in the figure.

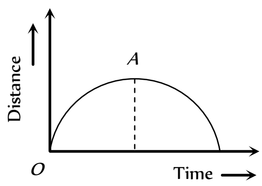


Fig. 2.9

Velocity-time Graph

The graph is plotted by taking time t along x-axis and velocity of the particle on y-axis.

Calculation of Distance and displacement : The area covered between the velocity time graph and time axis gives the displacement and distance travelled by the body for a given time interval.

Total distance $= |A_1| + |A_2| + |A_3|$

= Addition of modulus of different area. *i.e.* $s = \int |v|\, dt$

Total displacement $= A_1 + A_2 + A_3$

= Addition of different area considering their sign.

i.e. $r = \int v\,dt$

Area above time axis is taken as positive, while area below time axis is taken as negative

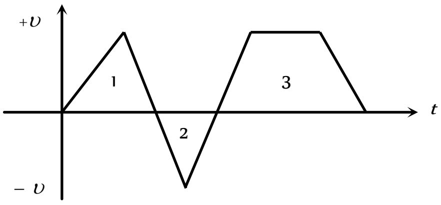


Fig. 2.10

here A_1 and A_2 are area of triangle 1 and 2 respectively and A_3 is the area of trapezium .

Calculation of Acceleration : Let *AB* is a velocity-time graph for any moving particle

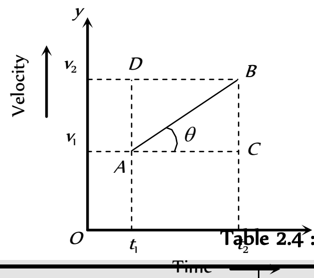


Fig. 2.11

As Acceleration = $\dfrac{\text{Change in velocity}}{\text{Time taken}}$

$$= \frac{v_2 - v_1}{t_2 - t_1} \qquad \text{...(i)}$$

From triangle *ABC*, $\tan\theta = \dfrac{BC}{AC} = \dfrac{AD}{AC}$

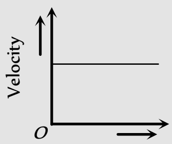
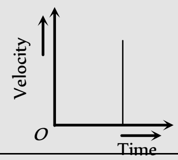
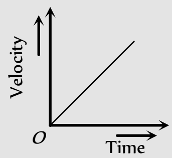
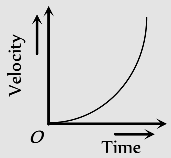
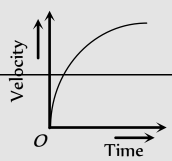
$$= \frac{v_2 - v_1}{t_2 - t_1} \qquad \text{....(ii)}$$

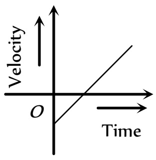
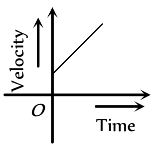
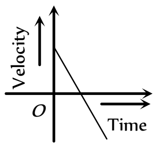
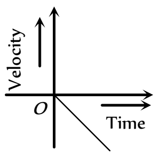
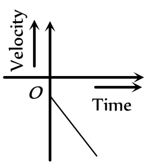
By comparing (i) and (ii)

Acceleration (a) = $\tan\theta$

It is clear that slope of tangent on velocity-time graph represents the acceleration of the particle.

Table 2.4 : Various velocity -time graphs and their interpretation

	$\theta = 0°$, $a = 0$, v = constant *i.e.*, line parallel to time axis represents that the particle is moving with constant velocity.
	$\theta = 90°$, $a = \infty$, v = increasing *i.e.*, line perpendicular to time axis represents that the particle is increasing its velocity, but time does not change. It means the particle possesses infinite acceleration. Practically it is not possible.
	θ = constant, so a = constant and v is increasing uniformly with time *i.e.*, line with constant slope represents uniform acceleration of the particle.
	θ increasing so acceleration increasing *i.e.*, line bending towards velocity axis represent the increasing acceleration in the body.
	θ decreasing so acceleration decreasing

	i.e. line bending towards time axis represents the decreasing acceleration in the body
	Positive constant acceleration because θ is constant and < 90° but initial velocity of the particle is negative.
	Positive constant acceleration because θ is constant and < 90° but initial velocity of particle is positive.
	Negative constant acceleration because θ is constant and > 90° but initial velocity of the particle is positive.
	Negative constant acceleration because θ is constant and > 90° but initial velocity of the particle is zero.
	Negative constant acceleration because θ is constant and > 90° but initial velocity of the particle is negative.

Equation of Kinematics

These are the various relations between u, v, a, t and s for the particle moving with uniform acceleration where the notations are used as :

u = Initial velocity of the particle at time t = 0 *sec*

v = Final velocity at time t *sec*

a = Acceleration of the particle

s = Distance travelled in time t *sec*

s_n = Distance travelled by the body in n *sec*

(1) **When particle moves with zero acceleration**

(i) It is a unidirectional motion with constant speed.

(ii) Magnitude of displacement is always equal to the distance travelled.

(iii) $v = u$, $s = u\,t$ [As a = 0]

(2) **When particle moves with constant acceleration**

(i) Acceleration is said to be constant when both the magnitude and direction of acceleration remain constant.

(ii) There will be one dimensional motion if initial velocity and acceleration are parallel or anti-parallel to each other.

(iii) Equations of motion

Equations of motion (in scalar from)	Equation of motion (in vector from)
$\upsilon = u + at$	$\vec{v} = \vec{u} + \vec{a}t$
$s = ut + \frac{1}{2}at^2$	$\vec{s} = \vec{u}t + \frac{1}{2}\vec{a}t^2$
$\upsilon^2 = u^2 + 2as$	$\vec{v}.\vec{v} - \vec{u}.\vec{u} = 2\vec{a}.\vec{s}$
$s = \left(\frac{u+v}{2}\right)t$	$\vec{s} = \frac{1}{2}(\vec{u} + \vec{v})t$

$$s_n = u + \frac{a}{2}(2n-1) \qquad \vec{s}_n = \vec{u} + \frac{\vec{a}}{2}(2n-1)$$

Motion of Body Under Gravity (Free Fall)

The force of attraction of earth on bodies, is called force of gravity. Acceleration produced in the body by the force of gravity, is called acceleration due to gravity. It is represented by the symbol g.

In the absence of air resistance, it is found that all bodies (irrespective of the size, weight or composition) fall with the same acceleration near the surface of the earth. This motion of a body falling towards the earth from a small altitude ($h << R$) is called free fall.

An ideal example of one-dimensional motion is motion under gravity in which air resistance and the small changes in acceleration with height are neglected.

(1) **If a body is dropped from some height (initial velocity zero)**

(i) Equations of motion : Taking initial position as origin and direction of motion (*i.e.*, downward direction) as a positive, here we have

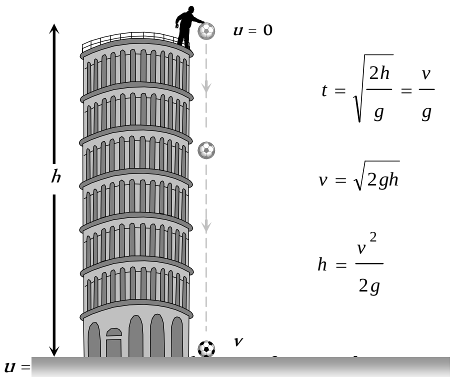


Fig. 2.12

$a = +g$ [As acceleration is in the direction of motion]

$v = g\,t$ ...(i)

$h = \frac{1}{2}gt^2$ ...(ii)

$v^2 = 2gh$ ...(iii)

$h_n = \frac{g}{2}(2n-1)$ ...(iv)

(ii) Graph of distance, velocity and acceleration with respect to time :

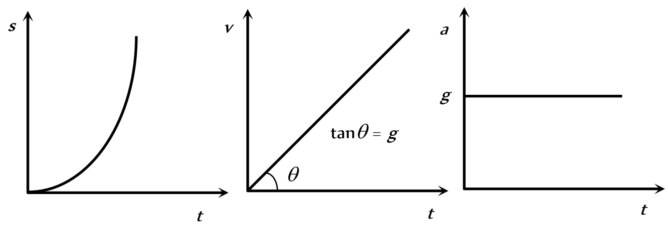


Fig. 2.13

(iii) As $h = (1/2)gt^2$, *i.e.*, $h \propto t^2$, distance covered in time t, $2t$, $3t$, *etc.*, will be in the ratio of $1^2 : 2^2 : 3^2$, *i.e.*, square of integers.

(iv) The distance covered in the *nth sec*, $h_n = \frac{1}{2}g(2n-1)$

So distance covered in 1st, 2nd, 3rd *sec*, *etc.*, will be in the ratio of 1 : 3 : 5, *i.e.*, odd integers only.

(2) **If a body is projected vertically downward with some initial velocity**

Equation of motion : $v = u + g\,t$

$$h = ut + \frac{1}{2}g\,t^2$$

$$v^2 = u^2 + 2gh$$

$$h_n = u + \frac{g}{2}(2n-1)$$

(3) **If a body is projected vertically upward**

(i) Equation of motion : Taking initial position as origin and direction of motion (*i.e.*, vertically up) as positive

$a = -g$ [As acceleration is downwards while motion upwards]

So, if the body is projected with velocity u and after time t it reaches up to height h then

$$v = u - g\,t\,;\, h = ut - \frac{1}{2}g\,t^2\,;\, v^2 = u^2 - 2gh\,;\, h_n = u - \frac{g}{2}(2n-1)$$

(ii) For maximum height $v = 0$

So from above equation $u = gt$,

$$h = \frac{1}{2}gt^2$$

and $u^2 = 2gh$

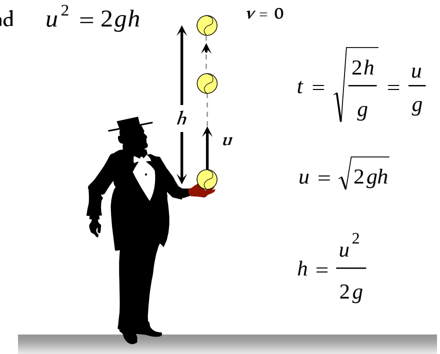


Fig. 2.14

(iii) Graph of displacement, velocity and acceleration with respect to time (for maximum height) :

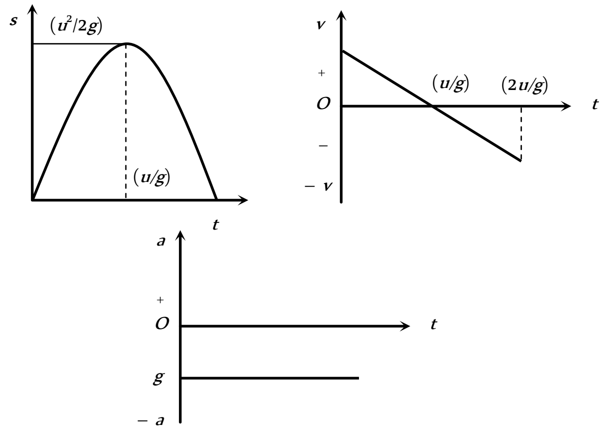


Fig. 2.15

It is clear that both quantities do not depend upon the mass of the body or we can say that in absence of air resistance, all bodies fall on the surface of the earth with the same rate.

(4) The motion is independent of the mass of the body, as in any equation of motion, mass is not involved. That is why a heavy and light body when released from the same height, reach the ground simultaneously and with same velocity *i.e.*, $t = \sqrt{(2h/g)}$ and $v = \sqrt{2gh}$.

(5) In case of motion under gravity, time taken to go up is equal to the time taken to fall down through the same distance. Time of descent (t_2) = time of ascent (t_1) = u/g

$\therefore$ Total time of flight $T = t_1 + t_2 = \dfrac{2u}{g}$

(6) In case of motion under gravity, the speed with which a body is projected up is equal to the speed with which it comes back to the point of projection.

As well as the magnitude of velocity at any point on the path is same whether the body is moving in upwards or downward direction.

(7) A body is thrown vertically upwards. If air resistance is to be taken into account, then the time of ascent is less than the time of descent. $t_2 > t_1$

Let u is the initial velocity of body then time of ascent $t_1 = \dfrac{u}{g+a}$

and $h = \dfrac{u^2}{2(g+a)}$

where g is acceleration due to gravity and a is retardation by air resistance and for upward motion both will work vertically downward.

For downward motion a and g will work in opposite direction because a always work in direction opposite to motion and g always work vertically downward.

So $h = \dfrac{1}{2}(g-a)t_2^2$

$$\Rightarrow \frac{u^2}{2(g+a)} = \frac{1}{2}(g-a)t_2^2$$

$$\Rightarrow t_2 = \frac{u}{\sqrt{(g+a)(g-a)}}$$

Comparing t_1 and t_2 we can say that $t_2 > t_1$

since $(g + a) > (g - a)$

Motion with Variable Acceleration

(i) If acceleration is a function of time

$a = f(t)$ then $v = u + \int_0^t f(t)\,dt$

and $s = ut + \int_0^t \left(\int f(t)\,dt \right) dt$

(ii) If acceleration is a function of distance

$a = f(x)$ then $v^2 = u^2 + 2\int_{x_0}^{x} f(x)\,dx$

(iii) If acceleration is a function of velocity

$a = f(v)$ then $t = \int_u^v \dfrac{dv}{f(v)}$ and $x = x_0 + \int_u^v \dfrac{v\,dv}{f(v)}$

Tips & Tricks

- During translational motion of the body, there is change in the location of the body.
- During rotational motion of the body, there is change in the orientation of the body, while there is no change in the location of the body from the axis of rotation.
- A point object is just a mathematical point. This concept is introduced to study the motion of a body in a simple manner.
- The choice of the origin is purely arbitrary.
- For one dimensional motion the angle between acceleration and velocity is either 0° or 180° and it does not change with time.
- For two dimensional motion, the angle between acceleration and velocity is other than 0° or 180° and also it may change with time.
- If the angle between $\vec{a}$ and $\vec{v}$ is 90°, the path of the particle is a circle.
- The particle speed up, that is the speed of the particle increases when the angle between $\vec{a}$ and $\vec{v}$ lies between –90° and +90°.
- The particle speeds down, that is the speed of the particle decreases, when the angle between $\vec{a}$ and $\vec{v}$ lies between +90° and 270°.
- The speed of the particle remains constant when the angle between $\vec{a}$ and $\vec{v}$ is equal to 90°.
- The distance covered by a particle never decreases with time, it always increases.
- Displacement of a particle is the unique path between the initial and final positions of the particle. It may or may not be the actually travelled path of the particle.
- Displacement of a particle gives no information regarding the nature of the path followed by the particle.
- Magnitude of displacement $\leq$ Distance covered.
- Since distance $\geq$ |Displacement|, so average speed of a body is equal or greater than the magnitude of the average velocity of the body.
- The average speed of a body is equal to its instantaneous speed if the body moves with a constant speed
- No force is required to move the body or an object with uniform velocity.
- Velocity of the body is positive, if it moves to the right side of the origin. Velocity is negative if the body moves to the left side of the origin.
- When a particle returns to the starting point, its displacement is zero but the distance covered is not zero.
- When a body reverses its direction of motion while moving along a straight line, then the distance travelled by the body is greater than the magnitude of the displacement of the body. In this case, average speed of

the body is greater than its average velocity.

- Speedometer measures the instantaneous speed of a vehicle.

- When particle moves with speed v_1 upto half time of its total motion and in rest time it is moving with speed v_2 then $v_{av} = \frac{v_1 + v_2}{2}$

- When particle moves the first half of a distance at a speed of v_1 and second half of the distance at speed v_2 then

$$v_{av} = \frac{2v_1 v_2}{v_1 + v_2}$$

- When particle covers one-third distance at speed v_1 next one third at speed v_2 and last one third at speed v_3 then

$$v_{av} = \frac{3v_1 v_2 v_3}{v_1 v_2 + v_2 v_3 + v_3 v_1}$$

- For two particles having displacement time graph with slopes θ_1 and θ_2 possesses velocities v_1 and v_2 respectively then $\frac{\upsilon_1}{\upsilon_2} = \frac{\tan \theta_1}{\tan \theta_2}$

- Velocity of a particle having uniform motion = slope of displacement–time graph.

- Greater the slope of displacement-time graph, greater is the velocity and vice-versa.

- Area under $v - t$ graph = displacement of the particle.

- Slope of velocity-time graph = acceleration.

- If a particle is accelerated for a time t_1 with acceleration a_1 and for time t_2 with acceleration a_2 then average acceleration is

$$a_{a\upsilon} = \frac{a_1 t_1 + a_2 t_2}{t_1 + t_2}$$

- If same force is applied on two bodies of different masses m_1 and m_2 separately then it produces accelerations a_1 and a_2 respectively. Now these bodies are attached together and form a combined system and same force is applied on that system so that a be the acceleration of the combined system, then

$$a = \frac{a_1 a_2}{a_1 + a_2}$$

- If a body starts from rest and moves with uniform acceleration then distance covered by the body in t *sec* is proportional to t (*i.e.* $s \propto t^2$).

  So we can say that the ratio of distance covered in 1 *sec*, 2 *sec* and 3 *sec* is $1^2 : 2^2 : 3^2$ or 1 : 4 : 9.

- If a body starts from rest and moves with uniform acceleration then distance covered by the body in *n*th sec is proportional to $(2n - 1)$ (*i.e.* $s_n \propto (2n - 1)$)

So we can say that the ratio of distance covered in 1st, 2nd and 3rd is 1 : 3 : 5.

- A body moving with a velocity u is stopped by application of brakes after covering a distance *s*. If the same body moves with velocity nu and same braking force is applied on it then it will come to rest after covering a distance of $n^2 s$.

$$\text{As } \upsilon^2 = u^2 - 2as \Rightarrow 0 = u^2 - 2as \Rightarrow s = \frac{u^2}{2a}, \quad s \propto u^2$$

[since a is constant]

So we can say that if u becomes n times then s becomes n^2 times that of previous value.

- A particle moving with uniform acceleration from A to B along a straight line has velocities υ_1 and υ_2 at A and B respectively. If C is the mid-point between A and B then velocity of the particle at C is equal to

$$\upsilon = \sqrt{\frac{\upsilon_1^2 + \upsilon_2^2}{2}}$$

- The body returns to its point of projection with the same magnitude of the velocity with which it was thrown vertically upward, provided air resistance is neglected.

- All bodies fall freely with the same acceleration.

- The acceleration of the falling bodies does not depend on the mass of the body.

- If two bodies are dropped from the same height, they reach the ground in the same time and with the same velocity.

- If a body is thrown upwards with velocity u from the top of a tower and another body is thrown downwards from the same point and with the same velocity, then both reach the ground with the same speed.

- When a particle returns to the starting point, its average velocity is zero but the average speed is not zero.

- If both the objects A and B move along parallel lines in the same direction, then the relative velocity of A w.r.t. B is given by $v_{AB} = v_A - v_B$

and the relative velocity of B w.r.t. A is given by $v_{BA} = v_B - v_A$

- If both the objects A and B move along parallel lines in the opposite direction, then the relative velocity of A w.r.t. B is given by $v_{AB} = v_A - (-v_B) = v_A + v_B$

and the relative velocity of B w.r.t. A is given by $v_{BA} = -v_B - v_A$

- Suppose a body is projected upwards from the ground and with the velocity u. It is assumed that the friction of the air is negligible. The characteristics of motion of such a body are as follows.

(i) The maximum height attained = $H = u^2/2g$.

(ii) Time taken to go up (ascent) = Time taken to come down (descent) = $t = u/g$.

(iii) Time of flight $T = 2t = 2u/g$.

(iv) The speed of the body on return to the ground = speed with which it was thrown upwards.

(v) When the height attained is not large, that is u is not large, the mass, the weight as well as the acceleration remain constant with time. But its speed, velocity, momentum, potential energy and kinetic energy change with time.

(vi) Let m be the mass of the body. Then in going from the ground to the highest point, following changes take place.

(a) Change in speed = u

(b) Change in velocity = u

(c) Change in momentum = $m\,u$

(d) Change in kinetic energy = Change in potential energy = (1/2) mu.

(vii) On return to the ground the changes in these quantities are as follows

(a) Change in speed = 0

(b) Change in velocity = $2u$

(c) Change in momentum = $2mu$

(d) Change in kinetic energy = Change in potential energy = 0

(viii) If, the friction of air be taken into account, then the motion of the object thrown upwards will have the following properties

(a) Time taken to go up (ascent) < time taken to come down (descent)

(b) The speed of the object on return to the ground is less than the initial speed. Same is true for velocity (magnitude), momentum (magnitude) and kinetic energy.

(c) Maximum height attained is less than $u/2g$.

(d) A part of the kinetic energy is used up in overcoming the friction.

✍ A ball is dropped from a building of height h and it reaches after t seconds on earth. From the same building if two balls are thrown (one upwards and other downwards) with the same velocity u and they reach the earth surface after t_1 and t_2 seconds respectively then

$$t = \sqrt{t_1 t_2}$$

✍ A particle is dropped vertically from rest from a height. The time taken by it to fall through successive distance of $1m$ each will then be in the ratio of the difference in the square roots of the integers *i.e.*

$$\sqrt{1}, (\sqrt{2} - \sqrt{1}), (\sqrt{3} - \sqrt{2}) \ldots\ldots (\sqrt{4} - \sqrt{3}), \ldots\ldots .$$

Ordinary Thinking

Objective Questions

Distance and Displacement

1. A Body moves 6 *m* north. 8 *m* east and 10*m* vertically upwards, what is its resultant displacement from initial position

(a) $10\sqrt{2}m$ (b) $10m$

(c) $\frac{10}{\sqrt{2}}m$ (d) $10\times 2m$

2. A man goes 10*m* towards North, then 20*m* towards east then displacement is [KCET 1999; JIPMER 1999; AFMC 2003]

(a) 22.5*m* (b) 25*m*

(c) 25.5*m* (d) 30*m*

3. A person moves 30 *m* north and then 20 *m* towards east and finally $30\sqrt{2}$ *m* in south-west direction. The displacement of the person from the origin will be [J & K CET 2004]

(a) 10 *m* along north (b) 10 *m* long south

(c) 10 *m* along west (d) Zero

4. An aeroplane flies 400 *m* north and 300 *m* south and then flies 1200 *m* upwards then net displacement is [AFMC 2004]

(a) 1200 *m* (b) 1300 *m*

(c) 1400 *m* (d) 1500 *m*

5. An athlete completes one round of a circular track of radius *R* in 40 *sec*. What will be his displacement at the end of 2 min. 20 *sec* [NCERT 1990; Kerala PMT 2004]

(a) Zero (b) $2R$

(c) $2\pi R$ (d) $7\pi R$

6. A wheel of radius 1 meter rolls forward half a revolution on a horizontal ground. The magnitude of the displacement of the point of the wheel initially in contact with the ground is [BCECE 2005]

(a) 2π (b) $\sqrt{2}\pi$

(c) $\sqrt{\pi^2+4}$ (d) π

Uniform Motion

1. A person travels along a straight road for half the distance with velocity v_1 and the remaining half distance with velocity v_2 The average velocity is given by [MP PMT 2001]

(a) v_1v_2 (b) $\frac{v_2^2}{v_1^2}$

(c) $\frac{v_1+v_2}{2}$ (d) $\frac{2v_1v_2}{v_1+v_2}$

2. The displacement-time graph for two particles *A* and *B* are straight lines inclined at angles of 30^o and 60^o with the time axis. The ratio of velocities of $V_A:V_B$ is [CPMT 1990; MP PET 1999; MP PET 2001; Pb. PET 2003]

(a) $1:2$ (b) $1:\sqrt{3}$

(c) $\sqrt{3}:1$ (d) $1:3$

3. A car travels from *A* to *B* at a speed of $20\ km/hr$ and returns at a speed of $30\ km/hr$. The average speed of the car for the whole journey is [MP PET 1985]

(a) $25\ km/hr$ (b) $24\ km/hr$

(c) $50\ km/hr$ (d) $5\ km/hr$

4. A boy walks to his school at a distance of 6 *km* with constant speed of 2.5 *km/hour* and walks back with a constant speed of 4 *km/hr*. His average speed for round trip expressed in *km/hour*, is [DCE 2000]

(a) 24/13 (b) 40/13

(c) 3 (d) 1/2

5. A car travels the first half of a distance between two places at a speed of 30 *km/hr* and the second half of the distance at 50 *km/hr*. The average speed of the car for the whole journey is [Manipal MEE 1995; A

(a) 42.5 *km/hr* (b) 40.0 *km/hr*

(c) 37.5 *km/hr* (d) 35.0 *km/hr*

6. One car moving on a straight road covers one third of the distance with 20 *km/hr* and the rest with 60 *km/hr*. The average speed is [MP PMT 199

(a) 40 *km/hr* (b) 80 *km/hr*

(c) $46\frac{2}{3}$ *km/hr* (d) 36 *km/hr*

7. A car moves for half of its time at 80 *km/h* and for rest half of time at 40 *km/h*. Total distance covered is 60 *km*. What is the average speed of the car [RPET 1996]

(a) $60\ km/h$ (b) $80\ km/h$

(c) $120\ km/h$ (d) $180\ km/h$

8. A train has a speed of 60 *km/h*. for the first one hour and 40 *km/h* for the next half hour. Its average speed in *km/h* is [JIPMER 1999]

(a) 50 (b) 53.33

(c) 48 (d) 70

9. Which of the following is a one dimensional motion [BHU 2000; CBSE PMT 2001]

(a) Landing of an aircraft

(b) Earth revolving a round the sun

(c) Motion of wheels of a moving trains

(d) Train running on a straight track

10. A 150 *m* long train is moving with a uniform velocity of 45 *km/h*. The time taken by the train to cross a bridge of length 850 meters is [CBSE PMT 2001]

(a) 56 *sec* (b) 68 *sec*

(c) 80 *sec* (d) 92 *sec*

11. A particle is constrained to move on a straight line path. It returns to the starting point after 10 *sec*. The total distance covered by the particle during this time is 30 *m*. Which of the following statements about the motion of the particle is false [CBSE PMT 2000; AFMC 2001]

(a) Displacement of the particle is zero

(b) Average speed of the particle is 3 *m/s*

(c) Displacement of the particle is 30 *m*

(d) Both (a) and (b)

12. A particle moves along a semicircle of radius 10*m* in 5 seconds. The average velocity of the particle is [Kerala (Engg.) 2001]

(a) $2\pi\ ms^{-1}$ (b) $4\pi\ ms^{-1}$

(c) $2\ ms^{-1}$ (d) $4\ ms^{-1}$

13. A man walks on a straight road from his home to a market 2.5 *km* away with a speed of 5 *km/h*. Finding the market closed, he instantly turns and walks back home with a speed of 7.5 *km/h*. The average speed of the man over the interval of time 0 to 40 *min*. is equal to

(a) 5 *km/h* (b) $\frac{25}{4}$ *km/h*

(c) $\frac{30}{4}$ *km/h* (d) $\frac{45}{8}$ *km/h*

14. The ratio of the numerical values of the average velocity and average speed of a body is always [MP PET 2002]

(a) Unity (b) Unity or less

(c) Unity or more (d) Less than unity

15. A person travels along a straight road for the first half time with a velocity v_1 and the next half time with a velocity v_2. The mean velocity V of the man is

[RPET 1999; BHU 2002]

(a) $\frac{2}{V}=\frac{1}{v_1}+\frac{1}{v_2}$ (b) $V=\frac{v_1+v_2}{2}$

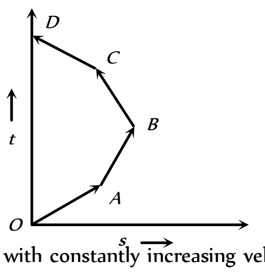
(c) $V=\sqrt{v_1v_2}$ (d) $V=\sqrt{\frac{v_1}{v_2}}$

16. If a car covers 2/5 of the total distance with v_1 speed and 3/5 distance with v_2 then average speed is [MP PMT 2003]

(a) $\frac{1}{2}\sqrt{v_1v_2}$ (b) $\frac{v_1+v_2}{2}$

(c) $\frac{2v_1v_2}{v_1+v_2}$ (d) $\frac{5v_1v_2}{3v_1+2v_2}$

17. Which of the following options is correct for the object having a straight line motion represented by the following graph



(a) The object moves with constantly increasing velocity from O to A and then it moves with constant velocity.

(b) Velocity of the object increases uniformly

(c) Average velocity is zero

(d) The graph shown is impossible

18. The numerical ratio of displacement to the distance covered is always [BHU 2004]

(a) Less than one

(b) Equal to one

(c) Equal to or less than one

(d) Equal to or greater than one

19. A 100 *m* long train is moving with a uniform velocity of 45 *km/hr*. The time taken by the train to cross a bridge of length 1 *km* is

(a) 58 *s* (b) 68 *s*

[AMU (Med) 2002]

(c) 78 *s* (d) 88 *s*

20. A particle moves for 20 *seconds* with velocity 3 *m/s* and then velocity 4 *m/s* for another 20 *seconds* and finally moves with velocity 5 *m/s* for next 20 *seconds*. What is the average velocity of the particle [MH CET 2004]

(a) 3 *m/s* (b) 4 *m/s*

(c) 5 *m/s* (d) Zero

21. The correct statement from the following is [MP PET 1993]

(a) A body having zero velocity will not necessarily have zero acceleration

(b) A body having zero velocity will necessarily have zero acceleration

(c) A body having uniform speed can have only uniform acceleration

(d) A body having non-uniform velocity will have zero acceleration

22. A bullet fired into a fixed target loses half of its velocity after penetrating 3 *cm*. How much further it will penetrate before coming to rest assuming that it faces constant resistance to motion?

(a) 1.5 *cm* (b) 1.0 *cm*

(c) 3.0 *cm* (d) 2.0 *cm*

23. Two boys are standing at the ends A and B of a ground where $AB=a$. The boy at B starts running in a direction perpendicular to AB with velocity v_1. The boy at A starts running simultaneously with velocity v and catches the other boy in a time *t*, where *t* is

(a) $a/\sqrt{v^2+v_1^2}$ (b) $\sqrt{a^2/(v^2-v_1^2)}$

(c) $a/(v-v_1)$ (d) $a/(v+v_1)$

24. A car travels half the distance with constant velocity of 40 *kmph* and the remaining half with a constant velocity of 60 *kmph*. The average velocity of the car in *kmph* is

[DCE 2004] [Kerala PMT 2005]

(a) 40 (b) 45

(c) 48 (d) 50

Non-uniform Motion

1. A particle experiences a constant acceleration for 20 sec after starting from rest. If it travels a distance S_1 in the first 10 sec and a distance S_2 in the next 10 sec, then

[NCERT 1972; CPMT 1997; MP PMT 2002]

(a) $S_1=S_2$ (b) $S_1=S_2/3$

(c) $S_1=S_2/2$ (d) $S_1=S_2/4$

2. The displacement x of a particle along a straight line at time t is given by $x=a_0+a_1t+a_2t^2$. The acceleration of the particle is [NCERT 19

(a) a_0 (b) a_1

(c) $2a_2$ (d) a_2

3. The coordinates of a moving particle at any time are given by $x = at^2$ and $y = bt^2$. The speed of the particle at any moment is [DPMT 1984; CPMT 1997]

(a) $2t(a+b)$ (b) $2t\sqrt{(a^2-b^2)}$

(c) $t\sqrt{a^2+b^2}$ (d) $2t\sqrt{(a^2+b^2)}$

4. An electron starting from rest has a velocity that increases linearly with the time that is $v = kt$, where $k = 2m/\sec^2$. The distance travelled in the first 3 seconds will be [NCERT 1982]

(a) 9 m (b) 16 m

(c) 27 m (d) 36 m

5. The displacement of a body is given to be proportional to the cube of time elapsed. The magnitude of the acceleration of the body is

(a) Increasing with time (b) Decreasing with time

(c) Constant but not zero (d) Zero

6. The instantaneous velocity of a body can be measured

(a) Graphically (b) Vectorially

(c) By speedometer (d) None of these

7. A body is moving from rest under constant acceleration and let S_1 be the displacement in the first $(p-1)$ sec and S_2 be the displacement in the first p sec. The displacement in $(p^2-p+1)^{th}$ sec. will be

(a) S_1+S_2 (b) S_1S_2

(c) S_1-S_2 (d) S_1/S_2

8. A body under the action of several forces will have zero acceleration

(a) When the body is very light

(b) When the body is very heavy

(c) When the body is a point body

(d) When the vector sum of all the forces acting on it is zero

9. A body starts from the origin and moves along the X-axis such that the velocity at any instant is given by $(4t^3-2t)$, where t is in sec and velocity in m/s. What is the acceleration of the particle, when it is 2 m from the origin

(a) $28\ m/s^2$ (b) $22\ m/s^2$

(c) $12\,m/s^2$ (d) $10\ m/s^2$

10. The relation between time and distance is $t = \alpha x^2 + \beta x$, where α and β are constants. The retardation is [NCERT 1982; AIEEE 2005]

(a) $2\alpha v^3$ (b) $2\beta v^3$

(c) $2\alpha\beta v^3$ (d) $2\beta^2 v^3$

11. A point moves with uniform acceleration and v_1, v_2 and v_3 denote the average velocities in the three successive intervals of time t_1, t_2 and t_3. Which of the following relations is correct

(a) $(v_1-v_2):(v_2-v_3)=(t_1-t_2):(t_2+t_3)$

(b) $(v_1-v_2):(v_2-v_3)=(t_1+t_2):(t_2+t_3)$

(c) $(v_1-v_2):(v_2-v_3)=(t_1-t_2):(t_1-t_3)$

(d) $(v_1-v_2):(v_2-v_3)=(t_1-t_2):(t_2-t_3)$

12. The acceleration of a moving body can be found from [DPMT 1981]

(a) Area under velocity-time graph

(b) Area under distance-time graph

(c) Slope of the velocity-time graph

(d) Slope of distance-time graph

13. The initial velocity of a particle is u (at $t=0$) and the acceleration f is given by at. Which of the following relation is valid [CPMT 1981; BHU 1995]

(a) $v = u + at^2$ [NCERT 1990] (b) $v = u + a\frac{t^2}{2}$

(c) $v = u + at$ (d) $v = u$

14. The initial velocity of the particle is $10\ m/\sec$ and its retardation is $2m/\sec^2$. The distance moved by the particle in $5th$ second of its motion is [CPMT 1976]

(a) $1\ m$ (b) $19\ m$

(c) $50\ m$ (d) $75\,m$

15. A motor car moving with a uniform speed of $20\,m/\sec$ comes to stop on the application of brakes after travelling a distance of $10\,m$ Its acceleration is [EAMCET 1979]

(a) $20\,m/\sec^2$ (b) $-20m/\sec^2$

(c) $-40\ m/\sec^2$ (d) $+2m/\sec^2$

16. The velocity of a body moving with a uniform acceleration of $2\ m./\sec^2$ is $10\ m/\sec$. Its velocity after an interval of 4 *sec* is

(a) $12\ m/\sec$ (b) $14\ m/\sec$

(c) $16\ m/\sec$ (d) $18\ m/\sec$

17. A particle starting from rest travels a distance x in first 2 *seconds* and a distance y in next two *seconds*, then [EAMCET 1982]

(a) $y = x$ (b) $y = 2x$

(c) $y = 3x$ (d) $y = 4x$

18. The initial velocity of a body moving along a straight line is 7 m/s. It has a uniform acceleration of $4\,m/s^2$. The distance covered by the body in the 5th second of its motion is

(a) 25 m (b) 35 m

(c) 50 m (d) 85 m

19. The velocity of a body depends on time according to the equation $v = 20 + 0.1t^2$. The body is undergoing [MNR 1995; UPSEAT 2000]

(a) Uniform acceleration

(b) Uniform retardation

(c) Non-uniform acceleration [NCERT 1982]

(d) Zero acceleration

20. Which of the following four statements is false [Manipal MEE 1995]

(a) A body can have zero velocity and still be accelerated

(b) A body can have a constant velocity and still have a varying speed

(c) A body can have a constant speed and still have a varying velocity

(d) The direction of the velocity of a body can change when its acceleration is constant

21. A particle moving with a uniform acceleration travels 24 *m* and 64 *m* in the first two consecutive intervals of 4 *sec* each. Its initial velocity is [MP PET 1995]

(a) 1 *m/sec* (b) $10\,m/\sec$

(c) 5 *m/sec* (d) 2 *m/sec*

22. The position of a particle moving in the *xy*-plane at any time t is given by $x=(3t^2-6t)$ *metres*, $y=(t^2-2t)$ *metres*. Select the correct statement about the moving particle from the following

(a) The acceleration of the particle is zero at $t=0$ second

(b) The velocity of the particle is zero at $t=0$ second

(c) The velocity of the particle is zero at $t=1$ second

(d) The velocity and acceleration of the particle are never zero

23. If body having initial velocity zero is moving with uniform acceleration $8\ m/\sec^2$ the distance travelled by it in fifth second will be [MP PMT 1996; DPMT 2001]

(a) 36 *metres* (b) 40 *metres*

(c) 100 *metres* (d) Zero

24. An alpha particle enters a hollow tube of 4 *m* length with an initial speed of 1 *km/s*. It is accelerated in the tube and comes out of it with a speed of 9 *km/s*. The time for which it remains inside the tube is

(a) $8\times10^{-3}\,s$ (b) $80\times10^{-3}\,s$

(c) $800\times10^{-3}\,s$ (d) $8\times10^{-4}\,s$

25. Two cars A and B are travelling in the same direction with velocities v_1 and v_2 $(v_1>v_2)$. When the car A is at a distance d ahead of the car B, the driver of the car A applied the brake producing a uniform retardation a There will be no collision when

(a) $d<\dfrac{(v_1-v_2)^2}{2a}$ (b) $d<\dfrac{v_1^2-v_2^2}{2a}$

(c) $d>\dfrac{(v_1-v_2)^2}{2a}$ (d) $d>\dfrac{v_1^2-v_2^2}{2a}$

26. A body of mass 10 *kg* is moving with a constant velocity of 10 *m/s*. When a constant force acts for 4 *seconds* on it, it moves with a velocity 2 *m/sec* in the opposite direction. The acceleration produced in it is [MP PET 1997]

(a) $3\,m/\sec^2$ (b) $-3m/\sec^2$

(c) $0.3\,m/\sec^2$ (d) $-0.3\,m/\sec^2$

27. A body starts from rest from the origin with an acceleration of $6\,m/s^2$ along the *x*-axis and $8\,m/s^2$ along the *y*-axis. Its distance from the origin after 4 seconds will be

[MP PMT 1999]

(a) 56 *m* (b) 64 *m*

(c) 80 *m* (d) 128 *m*

28. A car moving with a velocity of 10 *m/s* can be stopped by the application of a constant force *F* in a distance of 20 *m*. If the velocity of the car is 30 *m/s*, it can be stopped by this force in

(a) $\dfrac{20}{3}m$ (b) 20 *m*

(c) 60 *m* (d) 180 *m*

29. The displacement of a particle is given by $y=a+bt+ct^2-dt^4$. The initial velocity and acceleration are respectively [CPMT 1999, 2003]

(a) $b,-4d$ (b) $-b,2c$

(c) $b,2c$ (d) $2c,-4d$

30. A car moving with a speed of 40 *km/h* can be stopped by applying brakes after atleast 2 *m*. If the same car is moving with a speed of 80 *km/h*, what is the minimum stopping distance

[MP PMT 1995][CBSE PMT 1998,1999; AFMC 2000; JIPMER 2001, 02]

(a) 8 *m* (b) 2 *m*

(c) 4 *m* (d) 6 *m*

31. An elevator car, whose floor to ceiling distance is equal to 2.7 *m*, starts ascending with constant acceleration of 1.2 *ms*⁻². 2 *sec* after the start, a bolt begins fallings from the ceiling of the car. The free fall time of the bolt is [KCET 1994]

(a) $\sqrt{0.54}\ s$ (b) $\sqrt{6}\ s$

(c) 0.7 *s* (d) 1 *s*

32. The displacement is given by $x=2t^2+t+5$, the acceleration at $t=2s$ is [EAMCET (Engg.) 1995]

(a) $4\ m/s^2$ (b) $8\ m/s^2$

(c) $10\,m/s^2$ (d) $15\,m/s^2$

33. Two trains travelling on the same track are approaching each other with equal speeds of 40 *m/s*. The drivers of the trains begin to decelerate simultaneously when they are just 2.0 *km* apart. Assuming the decelerations to be uniform and equal, the value of the deceleration to barely avoid collision should be

(a) 11.8 m/s^2 (b) 11.0 m/s^2

(c) 2.1 m/s^2 (d) 0.8 m/s^2

34. A body moves from rest with a constant acceleration of $5\,m/s^2$. [Pb. PET 2004] Its instantaneous speed (in m/s) at the end of 10 *sec* is

(a) 50 (b) 5

(c) 2 (d) 0.5

35. A boggy of uniformly moving train is suddenly detached from train and stops after covering some distance. The distance covered by the boggy and distance covered by the train in the same time has relation [RPET 1997]

(a) Both will be equal

(b) First will be half of second

(c) First will be 1/4 of second

(d) No definite ratio

36. A body starts from rest. What is the ratio of the distance travelled by the body during the 4th and 3rd second

[CBSE PMT 1993]

(a) $\dfrac{7}{5}$ (b) $\dfrac{5}{7}$

(c) $\dfrac{7}{3}$ (d) $\dfrac{3}{7}$

37. The acceleration 'a' in m/s^2 of a particle is given by $a = 3t^2 + 2t + 2$ where t is the time. If the particle starts out with a velocity $u = 2m/s$ at $t = 0$, then the velocity at the end of 2 *second* is [MNR 1994; SCRA 1994]

(a) 12 *m/s* (b) 18 *m/s*
(c) 27 *m/s* (d) 36 *m/s*

38. A particle moves along a straight line such that its displacement at any time t is given by

$S = t^3 - 6t^2 + 3t + 4$ *metres*

The velocity when the acceleration is zero is [CBSE PMT 1994; JIPMER 2001, 02]

(a) $3ms^{-1}$ (b) $-12ms^{-1}$
(c) $42ms^{-1}$ (d) $-9ms^{-1}$

39. For a moving body at any instant of time [NTSE 1995]

(a) If the body is not moving, the acceleration is necessarily zero
(b) If the body is slowing, the retardation is negative
(c) If the body is slowing, the distance is negative
(d) If displacement, velocity and acceleration at that instant are known, we can find the displacement at any given time in future

40. The x and y coordinates of a particle at any time t are given by $x = 7t + 4t^2$ and $y = 5t$, where x and y are in metre and t in seconds. The acceleration of particle at $t = 5$ *s* is

(a) Zero (b) $8\ m/s^2$
(c) $20\ m/s^2$ (d) $40\ m/s^2$

41. The engine of a car produces acceleration $4m/s^2$ in the car. If this car pulls another car of same mass, what will be the acceleration produced [RPET 1996]

(a) $8m/s^2$ (b) $2m/s^2$
(c) $4m/s^2$ (d) $\frac{1}{2}m/s^2$

42. If a body starts from rest and travels 120 *cm* in the 6th second, then what is the acceleration [AFMC 1997]

(a) $0.20\ m/s^2$ (b) $0.027\ m/s^2$
(c) $0.218\ m/s^2$ (d) $0.03\ m/s^2$

43. If a car at rest accelerates uniformly to a speed of 144 *km/h* in 20 *s*. Then it covers a distance of [CBSE PMT 1997]

(a) 20 *m* (b) 400 *m*
(c) 1440 *m* (d) 2880 *m*

44. The position x of a particle varies with time t as $x = at^2 - bt^3$. The acceleration of the particle will be zero at time t equal to [CBSE PMT 1997; BHU 1999; DPMT 2000; KCET 2000]

(a) $\frac{a}{b}$ (b) $\frac{2a}{3b}$
(c) $\frac{a}{3b}$ (d) Zero

45. A truck and a car are moving with equal velocity. On applying the brakes both will stop after certain distance, then [CPMT 1997]

(a) Truck will cover less distance before rest
(b) Car will cover less distance before rest
(c) Both will cover equal distance
(d) None

46. If a train travelling at 72 *kmph* is to be brought to rest in a distance of 200 metres, then its retardation should be [SCRA 1998; MP PMT 2004]

(a) $20\ ms^{-2}$ (b) $10\ ms^{-2}$
(c) $2\ ms^{-2}$ (d) $1\ ms^{-2}$

47. The displacement of a particle starting from rest (at $t = 0$) is given by $s = 6t^2 - t^3$. The time in seconds at which the particle will attain zero velocity again, is [SCRA 1998]

(a) 2 (b) 4
(c) 6 (d) 8

48. What is the relation between displacement, time and acceleration in case of a body having uniform acceleration [DCE 1999]

(a) $S = ut + \frac{1}{2}ft^2$ (b) $S = (u + f)t$
(c) $S = v^2 - 2fs$ (d) None of these

49. Two cars A and B at rest at same point initially. If A starts with uniform velocity of 40 *m/sec* and B starts in the same direction with constant acceleration of $4m/s^2$, then B will catch A after how much time [RPET 1999] [SCRA 1996]

(a) 10 *sec* (b) 20 *sec*
(c) 30 *sec* (d) 35 *sec*

50. The motion of a particle is described by the equation $x = a + bt^2$ where $a = 15$ *cm* and $b = 3$ *cm/s*. Its instantaneous velocity at time 3 *sec* will be [AMU (Med.) 2000]

(a) 36 *cm/sec* (b) 18 *cm/sec*
(c) 16 *cm/sec* (d) 32 *cm/sec*

51. A body travels for 15 *sec* starting from rest with constant acceleration. If it travels distances S_1, S_2 and S_3 in the first five seconds, second five seconds and next five seconds respectively the relation between S_1, S_2 and S_3 is [AMU (Engg.) 2000]

(a) $S_1 = S_2 = S_3$ (b) $5S_1 = 3S_2 = S_3$
(c) $S_1 = \frac{1}{3}S_2 = \frac{1}{5}S_3$ (d) $S_1 = \frac{1}{5}S_2 = \frac{1}{3}S_3$

52. A body is moving according to the equation $x = at + bt^2 - ct^3$ where $x =$ displacement and a, b and c are constants. The acceleration of the body is [BHU 2000]

(a) $a + 2bt$ (b) $2b + 6ct$
(c) $2b - 6ct$ (d) $3b - 6ct^2$

53. A particle travels 10*m* in first 5 *sec* and 10*m* in next 3 *sec*. Assuming constant acceleration what is the distance travelled in next 2 *sec*

(a) 8.3 *m* (b) 9.3 *m*
(c) 10.3 *m* (d) None of above

54. The distance travelled by a particle is proportional to the squares of time, then the particle travels with [RPET 1999; RPMT 2000]

(a) Uniform acceleration (b) Uniform velocity
(c) Increasing acceleration (d) Decreasing velocity

55. Acceleration of a particle changes when [RPMT 2000]
(a) Direction of velocity changes
(b) Magnitude of velocity changes
(c) Both of above
(d) Speed changes

56. The motion of a particle is described by the equation $u = at$. The distance travelled by the particle in the first 4 seconds
(a) $4a$ (b) $12a$
(c) $6a$ (d) $8a$

57. The relation $3t = \sqrt{3x} + 6$ describes the displacement of a particle in one direction where x is in *metres* and t in sec. The displacement, when velocity is zero, is [CPMT 2000]
(a) 24 *metres* (b) 12 *metres*
(c) 5 *metres* (d) Zero

58. A constant force acts on a body of mass 0.9 *kg* at rest for 10*s*. If the body moves a distance of 250 *m*, the magnitude of the force is
(a) $3N$ (b) $3.5N$
(c) $4.0N$ (d) $4.5N$

59. The average velocity of a body moving with uniform acceleration travelling a distance of 3.06 *m* is 0.34 ms^{-1}. If the change in velocity of the body is 0.18ms^{-1} during this time, its uniform acceleration is [EAMCET (Med.) 2000]
(a) 0.01 ms^{-2} (b) 0.02 ms^{-2}
(c) 0.03 ms^{-2} (d) 0.04 ms^{-2}

60. Equation of displacement for any particle is $s = 3t^3 + 7t^2 + 14t + 8m$. Its acceleration at time $t = 1$ sec is [CBSE PMT 2000]
(a) 10 *m/s* (b) 16 *m/s*
(c) 25 *m/s* (d) 32 *m/s*

61. The position of a particle moving along the *x*-axis at certain times is given below :

t (*s*)	0	1	2	3
x (*m*)	–2	0	6	16

Which of the following describes the motion correctly [AMU (Engg.) 2001]
(a) Uniform, accelerated
(b) Uniform, decelerated
(c) Non-uniform, accelerated
(d) There is not enough data for generalization

62. Consider the acceleration, velocity and displacement of a tennis ball as it falls to the ground and bounces back. Directions of which of these changes in the process [AMU (Engg.) 2001]
(a) Velocity only
(b) Displacement and velocity
(c) Acceleration, velocity and displacement
(d) Displacement and acceleration

63. The displacement of a particle, moving in a straight line, is given by $s = 2t^2 + 2t + 4$ where s is in *metres* and t in seconds. The acceleration of the particle is [CPMT 2001]
(a) 2 *m/s* (b) 4 *m/s*
(c) 6 *m/s* (d) 8 *m/s*

64. A body *A* starts from rest with an acceleration a_1. After 2 seconds, another body *B* starts from rest with an acceleration a_2. If they travel equal distances in the 5th second, after the start of A, then the ratio $a_1 : a_2$ is equal to [AIIMS 2001]
(a) 5 : 9 (b) 5 : 7
(c) 9 : 5 (d) 9 : 7

65. The velocity of a bullet is reduced from 200*m/s* to 100*m/s* while travelling through a wooden block of thickness 10*cm*. The retardation, assuming it to be uniform, will be
[DCE 2000] [AIIMS 2001]
(a) 10×10^4 *m/s* (b) 12×10^4 *m/s*
(c) 13.5×10^4 *m/s* (d) 15×10^4 *m/s*

66. A body of 5 *kg* is moving with a velocity of 20 *m/s*. If a force of 100*N* is applied on it for 10*s* in the same direction as its velocity, what will now be the velocity of the body [MP PMT 2000; RPET 2001]
(a) 200 *m/s* (b) 220 *m/s*
(c) 240 *m/s* (d) 260 *m/s*
[EAMCET (Engg.) 2000]

67. A particle starts from rest, accelerates at 2 *m/s* for 10*s* and then goes for constant speed for 30*s* and then decelerates at 4 *m/s* till it stops. What is the distance travelled by it [DCE 2001; AIIMS 2002; DCE 2003]
(a) 750 *m* (b) 800 *m*
(c) 700 *m* (d) 850 *m*

68. The engine of a motorcycle can produce a maximum acceleration 5 *m/s*. Its brakes can produce a maximum retardation 10 *m/s*. What is the minimum time in which it can cover a distance of 1.5 *km*
(a) 30 *sec* (b) 15 *sec*
(c) 10 *sec* (d) 5 *sec*

69. The path of a particle moving under the influence of a force fixed in magnitude and direction is [MP PET 2002]
(a) Straight line (b) Circle
(c) Parabola (d) Ellipse

70. A car, moving with a speed of 50 *km/hr*, can be stopped by brakes after at least 6*m*. If the same car is moving at a speed of 100 *km/hr*, the minimum stopping distance is [AIEEE 2003]
(a) 6*m* (b) 12*m*
(c) 18*m* (d) 24*m*

71. A student is standing at a distance of 50metres from the bus. As soon as the bus begins its motion with an acceleration of 1ms^{-2}, the student starts running towards the bus with a uniform velocity u. Assuming the motion to be along a straight road, the minimum value of u, so that the student is able to catch the bus is [KCET 2003]
(a) 5 ms^{-1} (b) 8 ms^{-1}
(c) 10 ms^{-1} (d) 12 ms^{-1}

72. A body *A* moves with a uniform acceleration a and zero initial velocity. Another body *B*, starts from the same point moves in the same direction with a constant velocity v. The two bodies meet after a time t. The value of t is [MP PET 2003]
(a) $\frac{2v}{a}$ (b) $\frac{v}{a}$
(c) $\frac{v}{2a}$ (d) $\sqrt{\frac{v}{2a}}$

73. A particle moves along X-axis in such a way that its coordinate X varies with time t according to the equation $x = (2 - 5t + 6t^2)m$. The initial velocity of the particle is

[MNR 1987; MP PET 1996; Pb. PET 2004]

(a) $-5\,m/s$ (b) $6\,m/s$

(c) $-3\,m/s$ (d) $3\,m/s$

74. A car starts from rest and moves with uniform acceleration *a* on a straight road from time *t* = 0 to *t* = *T*. After that, a constant deceleration brings it to rest. In this process the average speed of the car is [MP PMT 2004]

(a) $\frac{aT}{4}$ (b) $\frac{3aT}{2}$

(c) $\frac{aT}{2}$ (d) aT

75. An object accelerates from rest to a velocity 27.5 *m/s* in 10 *sec* then find distance covered by object in next 10 *sec* [BCECE 2004]

(a) 550 *m* (b) 137.5 *m*

(c) 412.5 *m* (d) 275 *m*

76. If the velocity of a particle is given by $v = (180 - 16x)^{1/2}$ *m/s*, then its acceleration will be [J & K CET 2004]

(a) Zero (b) 8 *m/s*

(c) – 8 *m/s* (d) 4 *m/s*

77. The displacement of a particle is proportional to the cube of time elapsed. How does the acceleration of the particle depends on time obtained [Pb. PET 2001]

(a) $a \propto t^2$ (b) $a \propto 2t$

(c) $a \propto t^3$ (d) $a \propto t$

78. Starting from rest, acceleration of a particle is $a = 2(t-1)$. The velocity of the particle at $t = 5s$ is [RPET 2002]

(a) 15 *m/sec* (b) 25 *m/sec*

(c) 5 *m/sec* (d) None of these

79. A body is moving with uniform acceleration describes 40 *m* in the first 5 *sec* and 65 *m* in next 5 *sec*. Its initial velocity will be

(a) 4 *m/s* (b) 2.5 *m/s*

(c) 5.5 *m/s* (d) 11 *m/s*

80. Speed of two identical cars are u and $4u$ at a specific instant. The ratio of the respective distances in which the two cars are stopped from that instant is [AIEEE 2002]

(a) 1 : 1 (b) 1 : 4

(c) 1 : 8 (d) 1 : 16

81. The displacement x of a particle varies with time t, $x = ae^{-\alpha t} + be^{\beta t}$, where a, b, α and β are positive constants. The velocity of the particle will [CBSE PMT 2005]

(a) Go on decreasing with time

(b) Be independent of α and β

(c) Drop to zero when $\alpha = \beta$

(d) Go on increasing with time

82. A car, starting from rest, accelerates at the rate *f* through a distance *S*, then continues at constant speed for time *t* and then decelerates at the rate $\frac{f}{2}$ to come to rest. If the total distance traversed is 15 *S*, then [AIEEE 2005]

(a) $S = \frac{1}{2}ft^2$ (b) $S = \frac{1}{4}ft^2$

(c) $S = \frac{1}{72}ft^2$ (d) $S = \frac{1}{6}ft^2$

83. A man is 45 *m* behind the bus when the bus start accelerating from rest with acceleration 2.5 *m/s*. With what minimum velocity should the man start running to catch the bus ?

(a) 12 *m/s* (b) 14 *m/s*

(c) 15 *m/s* (d) 16 *m/s*

84. A particle moves along *x*-axis as

$x = 4(t-2) + a(t-2)^2$

Which of the following is true ? [J&K CET 2005]

(a) The initial velocity of particle is 4

(b) The acceleration of particle is 2*a*

(c) The particle is at origin at *t* = 0

(d) None of these

85. A body starting from rest moves with constant acceleration. The ratio of distance covered by the body during the 5th *sec* to that covered in 5 *sec* is [Kerala PET 2005]

(a) 9/25 (b) 3/5

(c) 25/9 (d) 1/25

86. What determines the nature of the path followed by the particle

(a) Speed (b) Velocity

(c) Acceleration (d) None of these

Relative Motion

1. Two trains, each 50 *m* long are travelling in opposite direction with velocity 10 *m/s* and 15 *m/s*. The time of crossing is [CPMT 1999; JIPMER 2000;

(a) $2s$ (b) $4\,s$

(c) $2\sqrt{3}s$ (d) $4\sqrt{3}\,s$

2. A 120 *m* long train is moving in a direction with speed 20 *m/s*. A train *B* moving with 30 *m/s* in the opposite direction and 130 *m* long crosses the first train in a time

[CPMT 1996; Kerala PET 2002]

(a) 6 *s* (b) 36 *s*

[Pb. PET 2003]

(c) 38 *s* (d) None of these

3. A 210 meter long train is moving due North at a of 25*m/s*. A small bird is flying due South a little above the train with speed 5*m/s*. The time taken by the bird to cross the train is

[AMU (Med.) 2001]

(a) $6s$ (b) $7s$

(c) $9s$ (d) $10s$

4. A police jeep is chasing with, velocity of 45 *km/h* a thief in another jeep moving with velocity 153 *km/h*. Police fires a bullet with muzzle velocity of 180 *m/s*. The velocity it will strike the car of the thief is [BHU 2003

(a) 150 *m/s* (b) 27 *m/s*

(c) 450 *m/s* (d) 250 *m/s*

5. A boat is sent across a river with a velocity of 8 *km/hr*. If the resultant velocity of boat is 10 *km/hr*, then velocity of the river is :

(a) 10 *km/hr* (b) 8 *km/hr*

(c) 6 *km/hr* (d) 4 *km/hr*

6. A train of 150 *meter* length is going towards north direction at a speed of $10m/\sec$. A parrot flies at the speed of $5\ m/\sec$ towards south direction parallel to the railway track. The time taken by the parrot to cross the train is

[CBSE PMT 1992; BHU 1998]

(a) 12 *sec* (b) 8 *sec*

(c) 15 *sec* (d) 10 *sec*

7. A boat is moving with velocity of $3\hat{i}+4\hat{j}$ in river and water is moving with a velocity of $-3\hat{i}-4\hat{j}$ with respect to ground. Relative velocity of boat with respect to water is : [Pb. PET 2002]

(a) $-6\hat{i}-8\hat{j}$ (b) $6\hat{i}+8\hat{j}$

(c) $8\hat{i}$ (d) $6\hat{i}$

8. The distance between two particles is decreasing at the rate of 6 *m/sec*. If these particles travel with same speeds and in the same direction, then the separation increase at the rate of 4 *m/sec*. The particles have speeds as [RPET 1999]

(a) 5 *m/sec* ; 1 *m/sec* (b) 4 *m/sec* ; 1 *m/sec*

(c) 4 *m/sec* ; 2 *m/sec* (d) 5 *m/sec* ; 2 *m/sec*

9. A boat moves with a speed of 5 *km/h* relative to water in a river flowing with a speed of 3 *km/h* and having a width of 1 *km*. The minimum time taken around a round trip is [J&K CET 2005]

(a) 5 min (b) 60 min

(c) 20 min (d) 30 min

10. For a body moving with relativistic speed, if the velocity is doubled, then [Orissa JEE 2005]

(a) Its linear momentum is doubled

(b) Its linear momentum will be less than double

(c) Its linear momentum will be more than double

(d) Its linear momentum remains unchanged

11. A river is flowing from W to E with a speed of 5 *m/min*. A man can swim in still water with a velocity 10 *m/min*. In which direction should the man swim so as to take the shortest possible path to go to the south. [BHU 2005]

(a) 30° with downstream (b) 60° with downstream

(c) 120° with downstream (d) South

12. A train is moving towards east and a car is along north, both with same speed. The observed direction of car to the passenger in the train is [J & K CET 2004]

(a) East-north direction (b) West-north direction

(c) South-east direction (d) None of these

13. An express train is moving with a velocity v_1. Its driver finds another train is moving on the same track in the same direction with velocity v_2. To escape collision, driver applies a retardation a on the train. the minimum time of escaping collision will be

(a) $t=\dfrac{v_1-v_2}{a}$ (b) $t_1=\dfrac{v_1^2-v_2^2}{2}$

(c) None (d) Both

Motion Under Gravity

1. A stone falls from a balloon that is descending at a uniform rate of $12\ m/s$. The displacement of the stone from the point of release after 10 *sec* is

(a) 490 m (b) 510 m

(c) 610 m (d) 725 m

2. A ball is dropped on the floor from a height of 10 *m*. It rebounds to a height of 2.5 *m*. If the ball is in contact with the floor for 0.01 *sec*, the average acceleration during contact is

(a) 2100 $m/\sec^2$ downwards (b) 2100 $m/\sec^2$ upwards

(c) 1400 $m/\sec^2$ (d) 700 $m/\sec^2$

3. A body A is projected upwards with a velocity of $98\,m/s$. The second body B is projected upwards with the same initial velocity but after 4 *sec*. Both the bodies will meet after

(a) 6 sec (b) 8 sec

(c) 10 sec (d) 12 sec

4. Two bodies of different masses m_a and m_b are dropped from two different heights a and b. The ratio of the time taken by the two to cover these distances are [NCERT 1972; MP PMT 1993]

(a) $a:b$ (b) $b:a$

(c) $\sqrt{a}:\sqrt{b}$ (d) $a^2:b^2$

5. A body falls freely from rest. It covers as much distance in the last second of its motion as covered in the first three seconds. The body has fallen for a time of [MNR 1998]

(a) $3\ s$ (b) $5\ s$

(c) $7\ s$ (d) $9\ s$

6. A stone is dropped into water from a bridge $44.1\ m$ above the water. Another stone is thrown vertically downward 1 *sec* later. Both strike the water simultaneously. What was the initial speed of the second stone

(a) $12.25\ m/s$ (b) $14.75\ m/s$

(c) $16.23\ m/s$ (d) $17.15\ m/s$

7. An iron ball and a wooden ball of the same radius are released from the same height in vacuum. They take the same time to reach the ground. The reason for this is

(a) Acceleration due to gravity in vacuum is same irrespective of the size and mass of the body

(b) Acceleration due to gravity in vacuum depends upon the mass of the body

(c) There is no acceleration due to gravity in vacuum

(d) In vacuum there is a resistance offered to the motion of the body and this resistance depends upon the mass of the body

8. A body is thrown vertically upwards. If air resistance is to be taken into account, then the time during which the body rises is [RPET 2000; KCET 2001; DPMT 2001]

(a) Equal to the time of fall

(b) Less than the time of fall

(c) Greater than the time of fall [RPET 2002]

(d) Twice the time of fall

9. A ball P is dropped vertically and another ball Q is thrown horizontally with the same velocities from the same height and at the same time. If air resistance is neglected, then [MNR 1986; BHU 1994]

(a) Ball P reaches the ground first

(b) Ball Q reaches the ground first

(c) Both reach the ground at the same time

(d) The respective masses of the two balls will decide the time

10. A body is released from a great height and falls freely towards the earth. Another body is released from the same height exactly one second later. The separation between the two bodies, two seconds after the release of the second body is [CPMT 1983; Kerala PMT 20

(a) $4.9\,m$ (b) $9.8\,m$

[BHU 1997; CPMT 1997]

(c) $19.6\,m$ (d) $24.5\,m$

11. An object is projected upwards with a velocity of 100 m/s. It will strike the ground after (approximately) [NCERT 1981; AFMC 1995]

(a) 10 sec (b) 20 sec
(c) 15 sec (d) 5 sec

12. A stone dropped from the top of the tower touches the ground in 4 sec. The height of the tower is about [MP PET 1986; AFMC 1994; CPMT 1997; BHU 1998; DPMT 1999; RPET 1999; MH CET 2003]

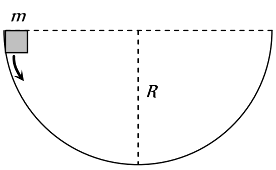
(a) $80\,m$ (b) $40\,m$
(c) $20\ m$ (d) $160\ m$

13. A body is released from the top of a tower of height h. It takes t *sec* to reach the ground. Where will be the ball after time $t/2$ sec [NCERT 1981; MP PMT 2004]

(a) At $h/2$ from the ground
(b) At $h/4$ from the ground
(c) Depends upon mass and volume of the body
(d) At $3h/4$ from the ground

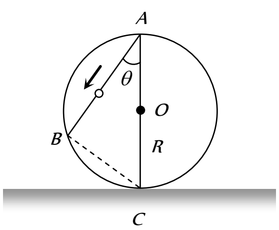
14. A mass m slips along the wall of a semispherical surface of radius R. The velocity at the bottom of the surface is [MP PMT 1993]

(a) $\sqrt{Rg}$
(b) $\sqrt{2Rg}$
(c) $2\sqrt{\pi Rg}$
(d) $\sqrt{\pi Rg}$



15. A frictionless wire *AB* is fixed on a sphere of radius *R*. A very small spherical ball slips on this wire. The time taken by this ball to slip from *A* to *B* is

(a) $\dfrac{2\sqrt{gR}}{g\cos\theta}$
(b) $2\sqrt{gR}.\dfrac{\cos\theta}{g}$
(c) $2\sqrt{\dfrac{R}{g}}$
(d) $\dfrac{gR}{\sqrt{g\cos\theta}}$



16. A body is slipping from an inclined plane of height h and length l. If the angle of inclination is θ, the time taken by the body to come from the top to the bottom of this inclined plane is

(a) $\sqrt{\dfrac{2h}{g}}$ (b) $\sqrt{\dfrac{2l}{g}}$
(c) $\dfrac{1}{\sin\theta}\sqrt{\dfrac{2h}{g}}$ (d) $\sin\theta\sqrt{\dfrac{2h}{g}}$

17. A particle is projected up with an initial velocity of $80\,ft/\sec$. The ball will be at a height of $96\,ft$ from the ground after

(a) 2.0 and 3.0 *sec* (b) Only at 3.0 *sec*
(c) Only at 2.0 *sec* (d) After 1 and 2 *sec*

18. A body falls from rest, its velocity at the end of first second is $(g = 32\,ft/\sec)$ [AFMC 1980]

(a) $16\ ft/\sec$ (b) $32\ ft/\sec$
(c) $64\ ft/\sec$ (d) $24\ ft/\sec$

19. A stone thrown upward with a speed u from the top of the tower reaches the ground with a velocity $3u$. The height of the tower is [EAMCET 1983; RPET 2003]

(a) $3u^2/g$ (b) $4u^2/g$
(c) $6u^2/g$ (d) $9u^2/g$

20. Two stones of different masses are dropped simultaneously from the top of a building [EAMCET 1978]

(a) Smaller stone hit the ground earlier
(b) Larger stone hit the ground earlier
(c) Both stones reach the ground simultaneously
(d) Which of the stones reach the ground earlier depends on the composition of the stone

21. A body thrown with an initial speed of $96\,ft/\sec$ reaches the ground after $(g = 32\,ft/\sec^2)$ [EAMCET 1980]

(a) 3 *sec* (b) 6 *sec*
(c) 12 *sec* (d) 8 *sec*

22. A stone is dropped from a certain height which can reach the ground in 5 *second*. If the stone is stopped after 3 *second* of its fall and then allowed to fall again, then the time taken by the stone to reach the ground for the remaining distance is

(a) 2 *sec* (b) 3 *sec*
(c) 4 *sec* (d) None of these

23. A man in a balloon rising vertically with an acceleration of $4.9\,m/\sec^2$ releases a ball 2 *sec* after the balloon is let go from the ground. The greatest height above the ground reached by the ball is $(g = 9.8\,m/\sec^2)$ [MNR 1986]

(a) 14.7 *m* (b) 19.6 *m*
(c) 9.8 *m* (d) 24.5 *m*

24. A particle is dropped under gravity from rest from a height $h(g = 9.8\,m/\sec^2)$ and it travels a distance $9h/25$ in the last second, the height h is [MNR 1987]

(a) 100 *m* (b) 122.5 *m*
(c) 145 *m* (d) 167.5 *m*

25. A balloon is at a height of 81 *m* and is ascending upwards with a velocity of 12 *m/s*. A body of $2kg$ weight is dropped from it. If $g = 10\,m/s^2$, the body will reach the surface of the earth in

(a) 1.5 *s* (b) 4.025 *s*
(c) 5.4 *s* (d) 6.75 *s*

26. An aeroplane is moving with a velocity u. It drops a packet from a height h. The time t taken by the packet in reaching the ground will be [MP PMT 1985]

(a) $\sqrt{\left(\dfrac{2g}{h}\right)}$ (b) $\sqrt{\left(\dfrac{2u}{g}\right)}$
(c) $\sqrt{\left(\dfrac{h}{2g}\right)}$ (d) $\sqrt{\left(\dfrac{2h}{g}\right)}$

27. Water drops fall at regular intervals from a tap which is $5\ m$ above the ground. The third drop is leaving the tap at the instant the first

drop touches the ground. How far above the ground is the second drop at that instant [CBSE PMT 1995]

(a) 2.50 *m* (b) 3.75 *m*

(c) 4.00 *m* (d) 1.25 *m*

28. A ball is thrown vertically upwards from the top of a tower at $4.9\ ms^{-1}$. It strikes the pond near the base of the tower after 3 *seconds*. The height of the tower is

[Manipal MEE 1995]

(a) 73.5 *m* (b) 44.1 *m*

(c) 29.4 *m* (d) None of these

29. An aeroplane is moving with horizontal velocity u at height h. The velocity of a packet dropped from it on the earth's surface will be (g is acceleration due to gravity)

[MP PET 1995]

(a) $\sqrt{u^2+2gh}$ (b) $\sqrt{2gh}$

(c) $2gh$ (d) $\sqrt{u^2-2gh}$

30. A rocket is fired upward from the earth's surface such that it creates an acceleration of 19.6 *m/sec*. If after 5 *sec* its engine is switched off, the maximum height of the rocket from earth's surface would be

(a) 245 *m* (b) 490 *m*

(c) 980 *m* (d) 735 *m*

31. A bullet is fired with a speed of $1000\ m/\sec$ in order to hit a target 100 *m* away. If $g = 10\ m/s^2$, the gun should be aimed

(a) Directly towards the target

(b) 5 *cm* above the target

(c) 10 *cm* above the target

(d) 15 *cm* above the target

32. A body starts to fall freely under gravity. The distances covered by it in first, second and third *second* are in ratio

[MP PET 1997; RPET 2001]

(a) $1:3:5$ (b) $1:2:3$

(c) $1:4:9$ (d) $1:5:6$

33. P, Q and R are three balloons ascending with velocities $U, 4U$ and $8U$ respectively. If stones of the same mass be dropped from each, when they are at the same height, then

(a) They reach the ground at the same time

(b) Stone from P reaches the ground first

(c) Stone from R reaches the ground first

(d) Stone from Q reaches the ground first

34. A body is projected up with a speed 'u' and the time taken by it is T to reach the maximum height H. Pick out the correct statement [EAMCET (Engg.) 1995]

(a) It reaches $H/2$ in $T/2$ *sec*

(b) It acquires velocity $u/2$ in $T/2\sec$

(c) Its velocity is $u/2$ at $H/2$

(d) Same velocity at $2T$

35. A body falling for 2 seconds covers a distance S equal to that covered in next second. Taking $g = 10\,m/s^2, S =$

[EAMCET (Engg.) 1995]

(a) 30 *m* (b) 10 *m*

(c) 60 *m* (d) 20 *m*

36. A body dropped from a height h with an initial speed zero, strikes the ground with a velocity $3\,km/h$. Another body of same mass is dropped from the same height h with an initial speed $-u' = 4km/h$. Find the final velocity of second body with which it strikes the ground [CBSE PMT 1996]

(a) 3 *km/h* (b) 4 *km/h*

(c) 5 *km/h* (d) 12 *km/h*

37. A ball of mass m_1 and another ball of mass m_2 are dropped from equal height. If time taken by the balls are t_1 and t_2 respectively, then [BHU 1997]

(a) $t_1 = \frac{t_2}{2}$ (b) $t_1 = t_2$

(c) $t_1 = 4t_2$ (d) $t_1 = \frac{t_2}{4}$

38. With what velocity a ball be projected vertically so that the distance covered by it in 5th second is twice the distance it covers in its 6th second $(g = 10\,m/s^2)$

[CPMT 1997; MH CET 2000]

(a) 58 *m/s* [MP PET 1995] (b) 49 *m/s*

(c) 65 *m/s* (d) 19.6 *m/s*

39. A body sliding on a smooth inclined plane requires 4 seconds to reach the bottom starting from rest at the top. How much time does it take to cover one-fourth distance starting from rest at the top

(a) 1 *s* (b) 2 *s*

(c) 4 *s* [MP PET 1996] (d) 16 *s*

40. A ball is dropped downwards. After 1 second another ball is dropped downwards from the same point. What is the distance between them after 3 seconds [BHU 1998]

(a) 25 *m* (b) 20 *m*

(c) 50 *m* (d) 9.8 *m*

41. A stone is thrown with an initial speed of 4.9 *m/s* from a bridge in vertically upward direction. It falls down in water after 2 *sec*. The height of the bridge is [AFMC 1999; Pb. PMT 2003]

(a) 4.9 *m* (b) 9.8 *m*

(c) 19.8 *m* (d) 24.7 *m*

42. A stone is shot straight upward with a speed of 20 *m/sec* from a tower 200 *m* high. The speed with which it strikes the ground is approximately [AMU (Engg.) 1999]

(a) 60 *m/sec* (b) 65 *m/sec*

(c) 70 *m/sec* [ISM Dhanbad 1994] (d) 75 *m/sec*

43. A body freely falling from the rest has a velocity '*v*' after it falls through a height '*h*'. The distance it has to fall down for its velocity to become double, is [BHU 1999]

(a) $2h$ (b) $4h$

(c) $6h$ (d) $8h$

44. The time taken by a block of wood (initially at rest) to slide down a smooth inclined plane 9.8 *m* long (angle of inclination is 30^o) is

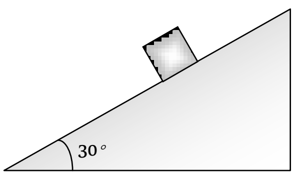
[JIPMER 1999]

(a) $\frac{1}{2}\sec$

(b) 2 sec

(c) 4 sec

(d) 1 sec



45. Velocity of a body on reaching the point from which it was projected upwards, is [AIIMS 1999; Pb. PMT 1999]
(a) $v = 0$ (b) $v = 2u$
(c) $v = 0.5u$ (d) $v = u$

46. A body projected vertically upwards with a velocity u returns to the starting point in 4 seconds. If $g = 10$ *m/sec*, the value of u is [KCET 1999]
(a) 5 *m/sec* (b) 10 *m/sec*
(c) 15 *m/sec* (d) 20 *m/sec*

47. Time taken by an object falling from rest to cover the height of h_1 and h_2 is respectively t_1 and t_2 then the ratio of t_1 to t_2 is [RPMT 1996; RPET 2002]
(a) $h_1 : h_2$ (b) $\sqrt{h_1} : \sqrt{h_2}$
(c) $h_1 : 2h_2$ (d) $2h_1 : h_2$

48. A body is thrown vertically up from the ground. It reaches a maximum height of 100*m* in 5*sec*. After what time it will reach the ground from the maximum height position [Pb. PMT 2000]
(a) 1.2 *sec* (b) 5 *sec*
(c) 10 *sec* (d) 25 *sec*

49. A body thrown vertically upwards with an initial velocity u reaches maximum height in 6 seconds. The ratio of the distances travelled by the body in the first second and the seventh second is [EAMCET (Engg.) 2000]
(a) 1 : 1 (b) 11 : 1
(c) 1 : 2 (d) 1 : 11

50. A particle is thrown vertically upwards. If its velocity at half of the maximum height is 10 *m/s*, then maximum height attained by it is (Take $g = 10$ *m/s*) [CBSE PMT 2001, 2004]
(a) 8 *m* (b) 10 *m*
(c) 12 *m* (d) 16 *m*

51. A body, thrown upwards with some velocity, reaches the maximum height of 20*m*. Another body with double the mass thrown up, with double initial velocity will reach a maximum height of [KCET 2001]
(a) 200 *m* (b) 16 *m*
(c) 80 *m* (d) 40 *m*

52. A balloon starts rising from the ground with an acceleration of 1.25 *m/s* after 8s, a stone is released from the balloon. The stone will ($g = 10$ *m/s*) [KCET 2001]
(a) Reach the ground in 4 second
(b) Begin to move down after being released
(c) Have a displacement of 50 *m*
(d) Cover a distance of 40 *m* in reaching the ground

53. A body is thrown vertically upwards with a velocity u. Find the true statement from the following [Kerala 2001]
(a) Both velocity and acceleration are zero at its highest point
(b) Velocity is maximum and acceleration is zero at the highest point
(c) Velocity is maximum and acceleration is g downwards at its highest point
(d) Velocity is zero at the highest point and maximum height reached is $u^2/2g$

54. A man throws a ball vertically upward and it rises through 20 *m* and returns to his hands. What was the initial velocity (u) of the ball and for how much time (T) it remained in the air $[g = 10m/s^2]$ [MP PET 2001]
(a) u = 10 *m/s*, T = 2*s* (b) u = 10 *m/s*, T = 4*s*
(c) u = 20 *m/s*, T = 2*s* (d) u = 20 *m/s*, T = 4*s*

55. A particle when thrown, moves such that it passes from same height at 2 and 10*s*, the height is [UPSEAT 2001]
(a) g (b) $2g$
(c) $5g$ (d) $10g$

56. Three different objects of masses m_1, m_2 and m_3 are allowed to fall from rest and from the same point '*O*' along three different frictionless paths. The speeds of the three objects, on reaching the ground, will be in the ratio of [AIIMS 2002]
(a) $m_1 : m_2 : m_3$ (b) $m_1 : 2m_2 : 3m_3$
(c) 1 : 1 : 1 (d) $\frac{1}{m_1} : \frac{1}{m_2} : \frac{1}{m_3}$

57. From the top of a tower, a particle is thrown vertically downwards with a velocity of 10 *m/s*. The ratio of the distances, covered by it in the 3rd and 2nd seconds of the motion is (Take $g = 10m/s^2$) [AIIMS 2000; CBSE PMT 2002]
(a) 5 : 7 (b) 7 : 5
(c) 3 : 6 (d) 6 : 3

58. Two balls *A* and *B* of same masses are thrown from the top of the building. *A*, thrown upward with velocity *V* and *B*, thrown downward with velocity *V*, then [AIEEE 2002]
(a) Velocity of *A* is more than *B* at the ground
(b) Velocity of *B* is more than *A* at the ground
(c) Both *A* & *B* strike the ground with same velocity
(d) None of these

59. A ball is dropped from top of a tower of 100*m* height. Simultaneously another ball was thrown upward from bottom of the tower with a speed of 50 *m/s* ($g = 10m/s^2$). They will cross each other after [Orissa JEE 2002]
(a) 1*s* (b) 2*s*
(c) 3*s* (d) 4*s*

60. A cricket ball is thrown up with a speed of 19.6 *ms*⁻¹. The maximum height it can reach is [Kerala PMT 2002]
(a) 9.8 *m* (b) 19.6 *m*
(c) 29.4 *m* (d) 39.2 *m*

61. A very large number of balls are thrown vertically upwards in quick succession in such a way that the next ball is thrown when the previous one is at the maximum height. If the maximum height is 5*m*, the number of ball thrown per minute is (take $g = 10ms^{-2}$) [KCET 2002]
(a) 120 (b) 80
(c) 60 (d) 40

62. A body falling from a high Minaret travels 40 meters in the last 2 seconds of its fall to ground. Height of Minaret in meters is (take $g = 10m/s^2$) [MP PMT 2002]
(a) 60 (b) 45
(c) 80 (d) 50

63. A body falls from a height $h = 200m$ (at New Delhi). The ratio of distance travelled in each 2 *sec* during t = 0 to $t = 6$ second of the journey is [BHU 2003; CPMT 2004]
(a) 1 : 4 : 9 (b) 1 : 2 : 4
(c) 1 : 3 : 5 (d) 1 : 2 : 3

64. A man drops a ball downside from the roof of a tower of height 400 meters. At the same time another ball is thrown upside with a velocity 50 *meter/sec*. from the surface of the tower, then they will meet at which height from the surface of the tower

(a) 100 meters (b) 320 meters

(c) 80 meters (d) 240 meters

65. Two balls are dropped from heights h and $2h$ respectively from the earth surface. The ratio of time of these balls to reach the earth is [CPMT 2003]

(a) $1:\sqrt{2}$ (b) $\sqrt{2}:1$

(c) 2 : 1 (d) 1 : 4

66. The acceleration due to gravity on the planet *A* is 9 times the acceleration due to gravity on planet *B*. A man jumps to a height of $2m$ on the surface of *A*. What is the height of jump by the same person on the planet *B* [CBSE PMT 2003]

(a) $18m$ (b) $6m$

(c) $\frac{2}{3}m$ (d) $\frac{2}{9}m$

67. A body falls from rest in the gravitational field of the earth. The distance travelled in the fifth second of its motion is $(g = 10m/s^2)$ [MP PET 2003]

(a) $25m$ (b) $45m$

(c) $90m$ (d) $125m$

68. If a body is thrown up with the velocity of 15 *m/s* then maximum height attained by the body is (g = 10 *m/s*) [MP PMT 2003]

(a) 11.25 *m* (b) 16.2 *m*

(c) 24.5 *m* (d) 7.62 *m*

69. A balloon is rising vertically up with a velocity of 29 ms^{-1}. A stone is dropped from it and it reaches the ground in 10 seconds. The height of the balloon when the stone was dropped from it is (g = 9.8 ms^{-2})

(a) 100 *m* (b) 200 *m*

(c) 400 *m* (d) 150 *m*

70. A ball is released from the top of a tower of height h meters. It takes T seconds to reach the ground. What is the position of the ball in $T/3$ seconds [AIEEE 2004]

(a) $h/9$ *meters* from the ground

(b) $7h/9$ *meters* from the ground

(c) $8h/9$ *meters* from the ground

(d) $17h/18$ *meters* from the ground

71. Two balls of same size but the density of one is greater than that of the other are dropped from the same height, then which ball will reach the earth first (air resistance is negligible)

(a) Heavy ball

(b) Light ball

(c) Both simultaneously

(d) Will depend upon the density of the balls

72. A packet is dropped from a balloon which is going upwards with the velocity 12 *m/s*, the velocity of the packet after 2 seconds will be

(a) –12 *m/s* (b) 12 *m/s*

(c) –7.6 *m/s* (d) 7.6 *m/s*

73. If a freely falling body travels in the last second a distance equal to the distance travelled by it in the first three second, the time of the travel is [Pb. PMT 2004; MH CET 2003]

(a) 6 *sec* (b) 5 *sec*

(c) 4 *sec* (d) 3 *sec*

74. The effective acceleration of a body, when thrown upwards with acceleration a will be : [Pb. PMT 2004]

(a) $\sqrt{a-g^2}$ (b) $\sqrt{a^2+g^2}$

(c) $(a-g)$ (d) $(a+g)$

75. A body is thrown vertically upwards with velocity u. The distance travelled by it in the fifth and the sixth seconds are equal. The velocity u is given by (g = 9.8 *m/s*) [UPSEAT 2004]

(a) 24.5 *m/s* (b) 49.0 *m/s*

(c) 73.5 *m/s* (d) 98.0 *m/s*

76. A body, thrown upwards with some velocity reaches the maximum height of 100 *m*. Another body with double the mass thrown up with double the initial velocity will reach a maximum height of [CPMT 2003]

(a) 100 *m* (b) 200 *m*

(c) 300 *m* (d) 400 *m*

77. A parachutist after bailing out falls 50 *m* without friction. When parachute opens, it decelerates at 2 *m/s*. He reaches the ground with a speed of 3 *m/s*. At what height, did he bail out ?

(a) 293 *m* (b) 111 *m*

(c) 91 *m* (d) 182 *m*

78. Three particles *A*, *B* and *C* are thrown from the top of a tower with the same speed. *A* is thrown up, *B* is thrown down and *C* is horizontally. They hit the ground with speeds V_A, V_B and V_C respectively. [Orissa JEE 2005]

(a) $V_A = V_B = V_C$ (b) $V_A = V_B > V_C$

(c) $V_B > V_C > V_A$ (d) $V_A > V_B = V_C$

79. From the top of a tower two stones, whose masses are in the ratio 1 : 2 are thrown one straight up with an initial speed u and the second straight down with the same speed u. Then, neglecting air resistance [KCET 2005]

(a) The heavier stone hits the ground with a higher speed

(b) The lighter stone hits the ground with a higher speed

(c) Both the stones will have the same speed when they hit the ground.

(d) The speed can't be determined with the given data.

80. When a ball is thrown up vertically with velocity V_o, it reaches a maximum height of 'h'. If one wishes to triple the maximum height then the ball should be thrown with velocity [KCET 2004]

(a) $\sqrt{3}V_o$ (b) $3V_o$

(c) $9V_o$ (d) $3/2V_o$

81. An object start sliding on a frictionless inclined plane and from same height another object start falling freely [RPET 2000]

(a) Both will reach with same speed

(b) Both will reach with same acceleration

(c) Both will reach in same time

(d) None of above [J & K CET 2004]

CT Critical Thinking

Objective Questions

[Pb PMT 2004]

1. A particle moving in a straight line covers half the distance with speed of 3 *m/s*. The other half of the distance is covered in two equal time intervals with speed of 4.5 *m/s* and 7.5 *m/s* respectively. The average speed of the particle during this motion is

(a) 4.0 *m/s* (b) 5.0 *m/s*

(c) 5.5 *m/s* (d) 4.8 *m/s*

2. The acceleration of a particle is increasing linearly with time t as bt. The particle starts from the origin with an initial velocity v_0 The distance travelled by the particle in time t will be

(a) $v_0 t + \frac{1}{3}bt^2$ (b) $v_0 t + \frac{1}{3}bt^3$

(c) $v_0 t + \frac{1}{6} b t^3$ (d) $v_0 t + \frac{1}{2} b t^2$

3. The motion of a body is given by the equation $\frac{dv(t)}{dt} = 6.0 - 3v(t)$. where $v(t)$ is speed in m/s and t in sec. If body was at rest at $t = 0$ [IIT-JEE 1995]

(a) The terminal speed is 2.0 m/s

(b) The speed varies with the time as $v(t) = 2(1 - e^{-3t}) m/s$

(c) The speed is $0.1 m/s$ when the acceleration is half the initial value

(d) The magnitude of the initial acceleration is $6.0 m/s^2$

4. A particle of mass m moves on the *x*-axis as follows : it starts from rest at $t = 0$ from the point $x = 0$ and comes to rest at $t = 1$ at the point $x = 1$. No other information is available about its motion at intermediate time $(0 < t < 1)$. If α denotes the instantaneous acceleration of the particle, then [IIT-JEE 1993]

(a) α cannot remain positive for all t in the interval $0 \le t \le 1$

(b) $|\alpha|$ cannot exceed 2 at any point in its path

(c) $|\alpha|$ must be ≥ 4 at some point or points in its path

(d) α must change sign during the motion but no other assertion can be made with the information given

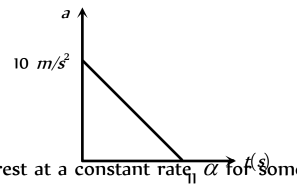
5. A particle starts from rest. Its acceleration (*a*) versus time (*t*) is as shown in the figure. The maximum speed of the particle will be [IIT-JEE (Screening) 2004]

(a) 110 *m/s*

(b) 55 *m/s*

(c) 550 *m/s*

(d) 660 *m/s*



6. A car accelerates from rest at a constant rate α for some time, after which it decelerates at a constant rate β and comes to rest. If the total time elapsed is *t*, then the maximum velocity acquired by the car is [IIT 1978; CBSE PMT 1994]

(a) $\left(\frac{\alpha^2 + \beta^2}{\alpha\beta}\right) t$ (b) $\left(\frac{\alpha^2 - \beta^2}{\alpha\beta}\right) t$

(c) $\frac{(\alpha + \beta) t}{\alpha\beta}$ (d) $\frac{\alpha\beta t}{\alpha + \beta}$

7. A stone dropped from a building of height h and it reaches after t seconds on earth. From the same building if two stones are thrown (one upwards and other downwards) with the same velocity *u* and they reach the earth surface after t_1 and t_2 seconds respectively, then [CPMT 1997; UPSEAT 2002; KCET 2002]

(a) $t = t_1 - t_2$ (b) $t = \frac{t_1 + t_2}{2}$

(c) $t = \sqrt{t_1 t_2}$ (d) $t = t_1^2 t_2^2$

8. A ball is projected upwards from a height h above the surface of the earth with velocity v. The time at which the ball strikes the ground is

(a) $\frac{v}{g} + \frac{2hg}{\sqrt{2}}$ (b) $\frac{v}{g}\left[1 - \sqrt{1 + \frac{2h}{g}}\right]$

(c) $\frac{v}{g}\left[1 + \sqrt{1 + \frac{2gh}{v^2}}\right]$ (d) $\frac{v}{g}\left[1 + \sqrt{v^2 + \frac{2g}{h}}\right]$

9. A particle is dropped vertically from rest from a height. The time taken by it to fall through successive distances of 1 *m* each will then be [Kurukshetra CEE 1996]

(a) All equal, being equal to $\sqrt{2/g}$ second

(b) In the ratio of the square roots of the integers 1, 2, 3.....

(c) In the ratio of the difference in the square roots of the integers *i.e.* $\sqrt{1}, (\sqrt{2} - \sqrt{1}), (\sqrt{3} - \sqrt{2}), (\sqrt{4} - \sqrt{3})$....

(d) In the ratio of the reciprocal of the square roots of the integers *i.e.*, $\frac{1}{\sqrt{1}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{4}}$

10. A man throws balls with the same speed vertically upwards one after the other at an interval of 2 seconds. What should be the speed of the throw so that more than two balls are in the sky at any time (Given $g = 9.8 m/s^2$) [CBSE PMT 2003]

(a) At least 0.8 *m/s*

(b) Any speed less than 19.6 *m/s*

(c) Only with speed 19.6 *m/s*

(d) More than 19.6 *m/s*

11. If a ball is thrown vertically upwards with speed u, the distance covered during the last t seconds of its ascent is [CBSE PMT 2003]

(a) $\frac{1}{2} g t^2$ (b) $ut - \frac{1}{2} g t^2$

(c) $(u - gt)t$ (d) ut d

12. A small block slides without friction down an inclined plane starting from rest. Let S_n be the distance travelled from time $t = n - 1$ to $t = n$. Then $\frac{S_n}{S_{n+1}}$ is [IIT-JEE (Screening) 2004]

(a) $\frac{2n-1}{2n}$ (b) $\frac{2n+1}{2n-1}$

(c) $\frac{2n-1}{2n+1}$ (d) $\frac{2n}{2n+1}$

GQ Graphical Questions

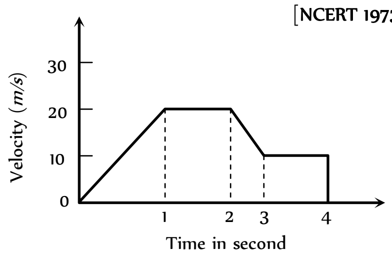
1. The variation of velocity of a particle with time moving along a straight line is illustrated in the following figure. The distance travelled by the particle in four seconds is [NCERT 1973]

(a) 60 m

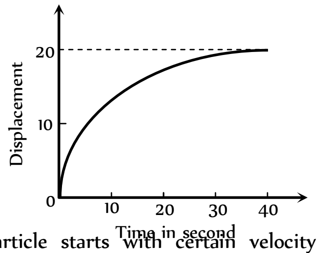
(b) 55 m

(c) 25 m

(d) 30 m



2. The displacement of a particle as a function of time is shown in the figure. The figure shows that [CPMT 1970, 86]



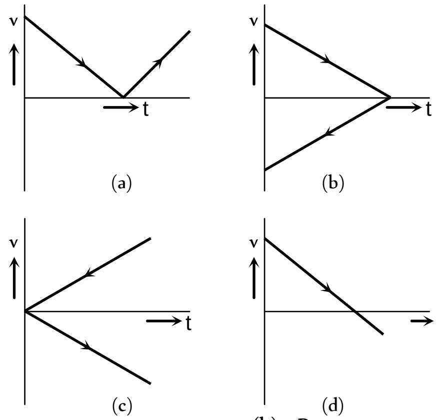
(a) The particle starts with certain velocity but the motion is retarded and finally the particle stops

(b) The velocity of the particle is constant throughout

(c) The acceleration of the particle is constant throughout.

(d) The particle starts with constant velocity, then motion is accelerated and finally the particle moves with another constant velocity

3. A ball is thrown vertically upwards. Which of the following graph/graphs represent velocity-time graph of the ball during its flight (air resistance is neglected) [CPMT 1993; AMU (Engg.) 2000]

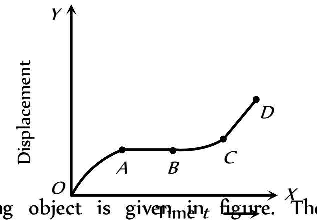


(a) A (b) B

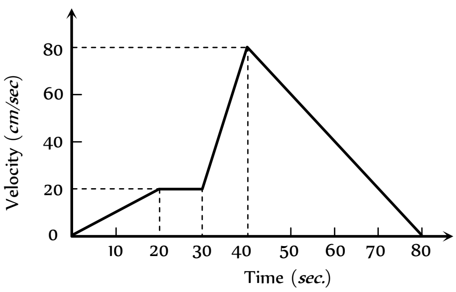
(c) C (d) D

4. The graph between the displacement x and time t for a particle moving in a straight line is shown in figure. During the interval OA, AB, BC and CD, the acceleration of the particle is

	OA,	AB,	BC,	CD
(a)	+	0	+	+
(b)	–	0	+	0
(c)	+	0	–	+
(d)	–	0	–	0



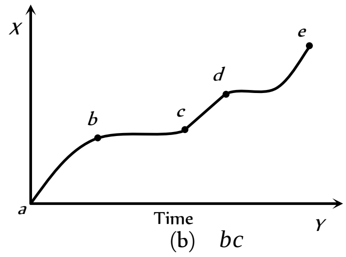
5. The $v-t$ graph of a moving object is given in figure. The maximum acceleration is [NCERT 1972]



(a) $1cm/\sec c^2$ (b) $2cm/\sec^2$

(c) $3\ cm/\sec^2$ (d) $6\ cm/\sec^2$

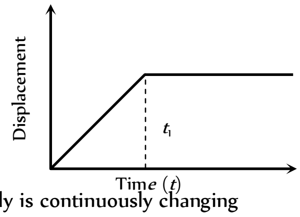
6. The displacement versus time graph for a body moving in a straight line is shown in figure. Which of the following regions represents the motion when no force is acting on the body



(a) ab (b) bc

(c) cd (d) de

7. The $x-t$ graph shown in figure represents [CPMT 1984]



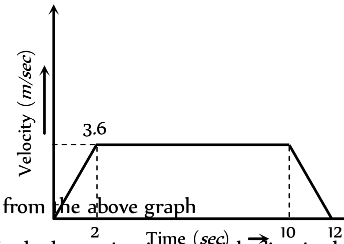
(a) Constant velocity

(b) Velocity of the body is continuously changing

(c) Instantaneous velocity

(d) The body travels with constant speed upto time t_1 and then stops

8. A lift is going up. The variation in the speed of the lift is as given in the graph. What is the height to which the lift takes the passengers [CPMT 1986]



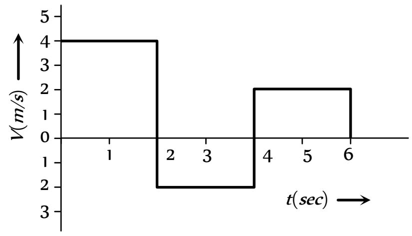
(a) 3.6 m

(b) 28.8 m

(c) 36.0 m

(d) Cannot be calculated from the above graph

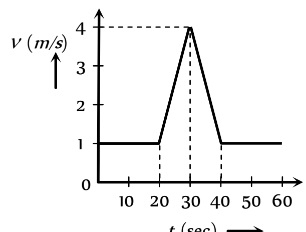
9. The velocity-time graph of a body moving in a straight line is shown in the figure. The displacement and distance travelled by the body in 6 *sec* are respectively [MP PET 1994]



(a) 8 m, 16 m (b) 16 m, 8 m

(c) 16 m, 16 m (d) 8 m, 8 m

10. Velocity-time (v-t) graph for a moving object is shown in the figure. Total displacement of the object during the time interval when there is non-zero acceleration and retardation is



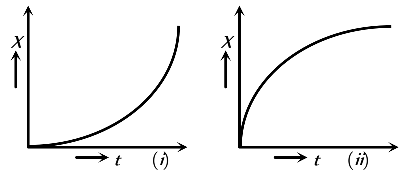
(a) 60 m

(b) 50 m

(c) 30 m

(d) 40 m

11. Figures (i) and (ii) below show the displacement-time graphs of two particles moving along the x-axis. We can say that



(a) Both the particles are having a uniformly accelerated motion

(b) Both the particles are having a uniformly retarded motion

(c) Particle (i) is having a uniformly accelerated motion while particle (ii) is having a uniformly retarded motion

(d) Particle (i) is having a uniformly retarded motion while particle (ii) is having a uniformly accelerated motion

[Kurukshetra CEE 1996]

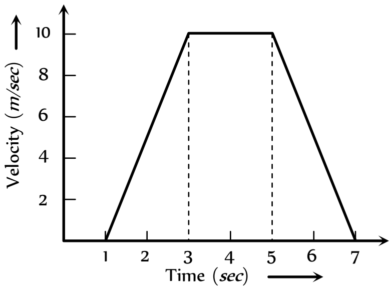
12. For the velocity-time graph shown in figure below the distance covered by the body in last two seconds of its motion is what fraction of the total distance covered by it in all the seven seconds [MP PMT/PET 1998; RPET 2001]

(a) $\frac{1}{2}$

(b) $\frac{1}{4}$

(c) $\frac{1}{3}$

(d) $\frac{2}{3}$



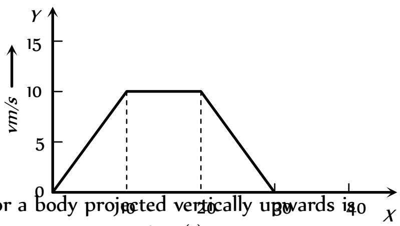
13. In the following graph, distance travelled by the body in metres is

(a) 200

(b) 250

(c) 300

(d) 400

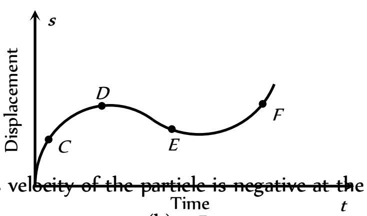


14. Velocity-time curve for a body projected vertically upwards is [EAMCET (Med.) 1995; AIIMS 1999; Pb. PMT 2004; BHU 2004]

(a) Parabola (b) Ellipse

(c) Hyperbola (d) Straight line

15. The displacement-time graph of moving particle is shown below



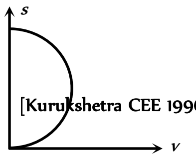
The instantaneous velocity of the particle is negative at the point

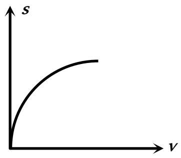
(a) D (b) F

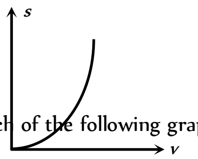
(c) C (d) E

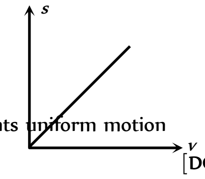
16. An object is moving with a uniform acceleration which is parallel to its instantaneous direction of motion. The displacement (s) – velocity (v) graph of this object is

[SCRA 1998; DCE 2000; AIIMS 2003; Orissa PMT 2004]

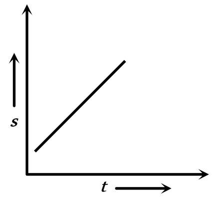
(a) 

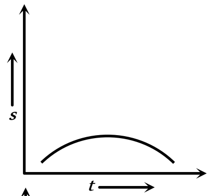
(b) 

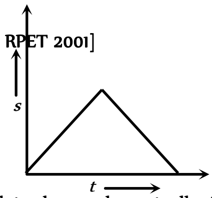
(c) 

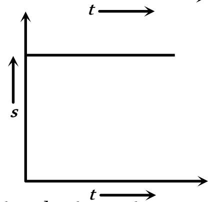
(d) 

17. Which of the following graph represents uniform motion [DCE 1999]

(a) 

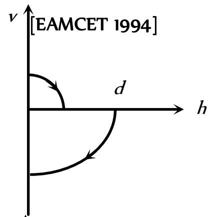
(b) 

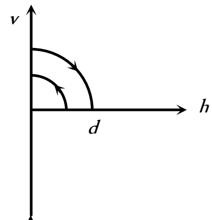
(c) 

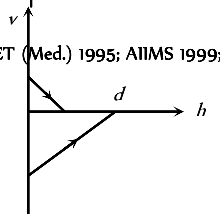
(d) 

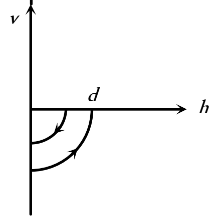
18. A ball is dropped vertically from a height d above the ground. It hits the ground and bounces up vertically to a height $d/2$. Neglecting subsequent motion and air resistance, its velocity v varies with the height h above the ground is

[EAMCET 1994]

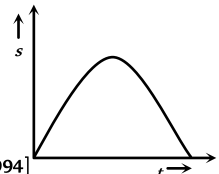
(a) 

(b) 

(c) 

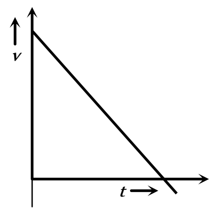
(d) 

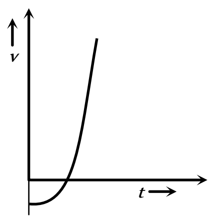
19. The graph of displacement v/s time is



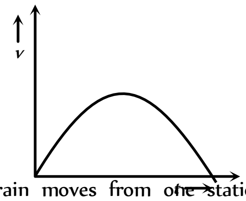
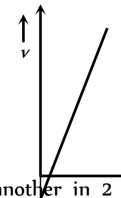
[CBSE PMT 1994]

Its corresponding velocity-time graph will be [DCE 2001]



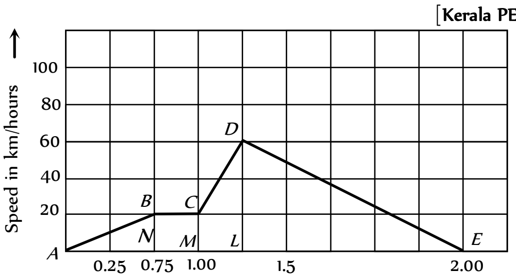


(a) (b)

(c)  (d) 

20. A train moves from one station to another in 2 hours time. Its speed-time graph during this motion is shown in the figure. The maximum acceleration during the journey is

[Kerala PET 2002]



(a) 140 $km\ h^{-1}$ (b) 160 $km\ h^{-1}$
(c) 100 $km\ h^{-1}$ (d) 120 $km\ h^{-1}$

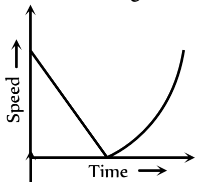
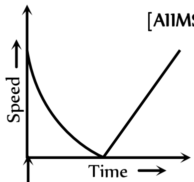
21. The area under acceleration-time graph gives

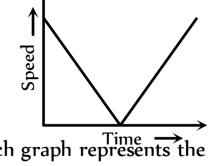
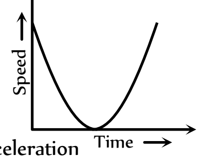
[Kerala PET 2005]

(a) Distance travelled (b) Change in acceleration
(c) Force acting (d) Change in velocity

22. A ball is thrown vertically upwards. Which of the following plots represents the speed-time graph of the ball during its height if the air resistance is not ignored

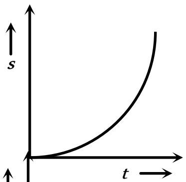
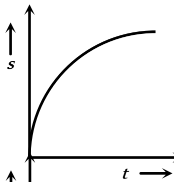
[AIIMS 2003]

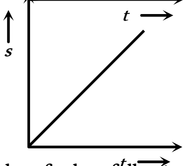
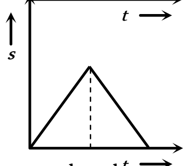
(a)  (b) 

(c)  (d) 

23. Which graph represents the uniform acceleration

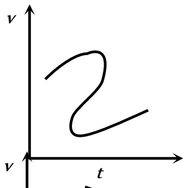
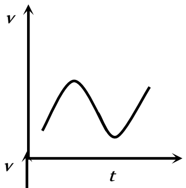
[DCE 2003]

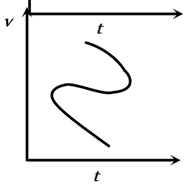
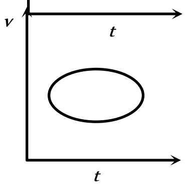
(a)  (b) 

(c)  (d) 

24. Which of the following velocity-time graphs shows a realistic situation for a body in motion

[AIIMS 2004]

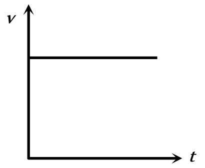
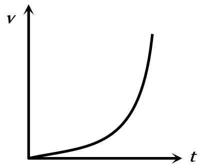
(a)  (b) 

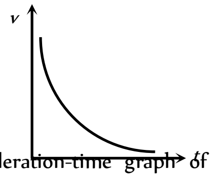
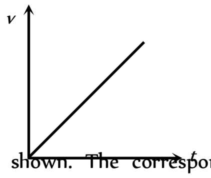
 

(c) (d)

25. Which of the following velocity-time graphs represent uniform motion

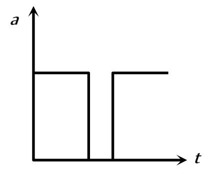
[Kerala PMT 2004]

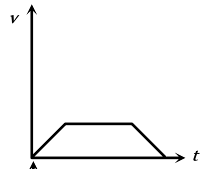
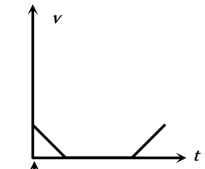
(a)  (b) 

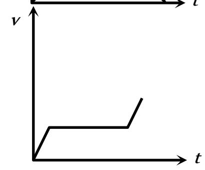
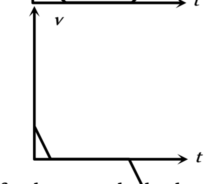
(c)  (d) 

26. Acceleration-time graph of a body is shown. The corresponding velocity-time graph of the same body is

[DPMT 2004]

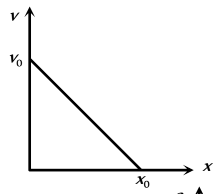


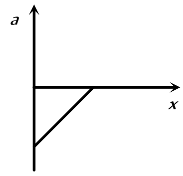
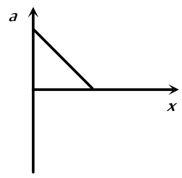
(a)  (b) 

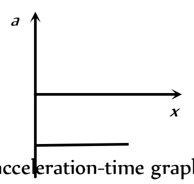
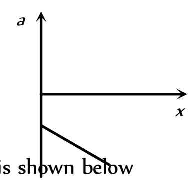
(c)  (d) 

27. The given graph shows the variation of velocity with displacement. Which one of the graph given below correctly represents the variation of acceleration with displacement

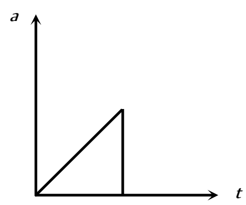
[IIT-JEE (Screening) 2005]



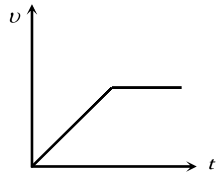
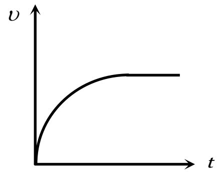
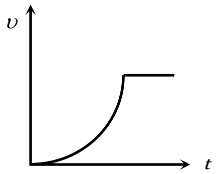
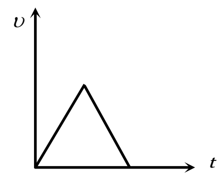
(a)  (b) 

(c)  (d) 

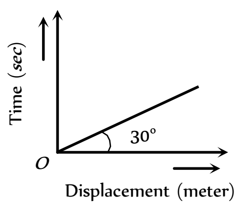
28. The acceleration-time graph of a body is shown below



The most probable velocity-time graph of the body is

(a) 
(b) 
(c) 
(d) 

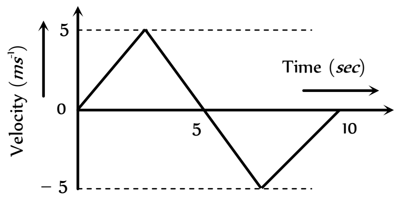
29. From the following displacement-time graph find out the velocity of a moving body



(a) $\frac{1}{\sqrt{3}}$ *m/s* (b) 3 *m/s*

(c) $\sqrt{3}$ *m/s* (d) $\frac{1}{3}$

30. The $\upsilon - t$ plot of a moving object is shown in the figure. The average velocity of the object during the first 10 seconds is



(a) 0 (b) 2.5 ms^{-1}

(c) 5 ms^{-1} (d) 2 ms^{-1}

Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

Read the assertion and reason carefully to mark the correct option out of the options given below:

(a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
(b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
(c) If assertion is true but reason is false.
(d) If the assertion and reason both are false.
(e) If assertion is false but reason is true.

1. Assertion : A body can have acceleration even if its velocity is zero at a given instant of time.
Reason : A body is momentarily at rest when it reverses its direction of motion.

2. Assertion : Two balls of different masses are thrown vertically upward with same speed. They will pass through their point of projection in the downward direction with the same speed.
Reason : The maximum height and downward velocity attained at the point of projection are independent of the mass of the ball.

3. Assertion : If the displacement of the body is zero, the distance covered by it may not be zero.
Reason : Displacement is a vector quantity and distance is a scalar quantity.

4. Assertion : The average velocity of the object over an interval of time is either smaller than or equal to the average speed of the object over the same interval.
Reason : Velocity is a vector quantity and speed is a scalar quantity.

5. Assertion : An object can have constant speed but variable velocity.
Reason : Speed is a scalar but velocity is a vector quantity.

6. Assertion : The speed of a body can be negative.
Reason : If the body is moving in the opposite direction of positive motion, then its speed is negative.

7. Assertion : The position-time graph of a uniform motion in one dimension of a body can have negative slope.
Reason : When the speed of body decreases with time, the position-time graph of the moving body has negative slope.

8. Assertion : A positive acceleration of a body can be associated with a '*slowing down*' of the body.
Reason : Acceleration is a vector quantity.

9. Assertion : A negative acceleration of a body can be associated with a '*speeding up*' of the body.
Reason : Increase in speed of a moving body is independent of its direction of motion.

10. Assertion : When a body is subjected to a uniform acceleration, it always move in a straight line.
Reason : Straight line motion is the natural tendency of the body.

11. Assertion : Rocket in flight is not an illustration of projectile.
Reason : Rocket takes flight due to combustion of fuel and does not move under the gravity effect alone.

12. Assertion : The average speed of a body over a given interval of time is equal to the average velocity of the body in the same interval of time if a body moves in a straight line in one direction.

Reason : Because in this case distance travelled by a body is equal to the displacement of the body.

13. Assertion : Position-time graph of a stationary object is a straight line parallel to time axis.

Reason : For a stationary object, position does not change with time.

14. Assertion : The slope of displacement-time graph of a body moving with high velocity is steeper than the slope of displacement-time graph of a body with low velocity.

Reason : Slope of displacement-time graph = Velocity of the body.

15. Assertion : Distance-time graph of the motion of a body having uniformly accelerated motion is a straight line inclined to the time axis.

Reason : Distance travelled by a body having uniformly accelerated motion is directly proportional to the square of the time taken.

16. Assertion : A body having non-zero acceleration can have a constant velocity.

Reason : Acceleration is the rate of change of velocity.

17. Assertion : A body, whatever its motion is always at rest in a frame of reference which is fixed to the body itself.

Reason : The relative velocity of a body with respect to itself is zero.

18. Assertion : Displacement of a body may be zero when distance travelled by it is not zero.

Reason : The displacement is the longest distance between initial and final position.

19. Assertion : The equation of motion can be applied only if acceleration is along the direction of velocity and is constant.

Reason : If the acceleration of a body is constant then its motion is known as uniform motion.

20. Assertion : A bus moving due north takes a turn and starts moving towards east with same speed. There will be no change in the velocity of bus.

Reason : Velocity is a vector-quantity.

21. Assertion : The relative velocity between any two bodies moving in opposite direction is equal to sum of the velocities of two bodies.

Reason : Sometimes relative velocity between two bodies is equal to difference in velocities of the two.

22. Assertion : The displacement-time graph of a body moving with uniform acceleration is a straight line.

Reason : The displacement is proportional to time for uniformly accelerated motion.

23. Assertion : Velocity-time graph for an object in uniform motion along a straight path is a straight line parallel to the time axis.

Reason : In uniform motion of an object velocity increases as the square of time elapsed.

24. Assertion : A body may be accelerated even when it is moving uniformly.

Reason : When direction of motion of the body is changing then body may have acceleration.

25. Assertion : A body falling freely may do so with constant velocity.

Reason : The body falls freely, when acceleration of a body is equal to acceleration due to gravity.

26. Assertion : Displacement of a body is vector sum of the area under velocity-time graph.

Reason : Displacement is a vector quantity.

27. Assertion : The position-time graph of a body moving uniformly is a straight line parallel to position-axis.

Reason : The slope of position-time graph in a uniform motion gives the velocity of an object.

28. Assertion : The average speed of an object may be equal to arithmetic mean of individual speed.

Reason : Average speed is equal to total distance travelled per total time taken.

29. Assertion : The average and instantaneous velocities have same value in a uniform motion.

Reason : In uniform motion, the velocity of an object increases uniformly.

30. Assertion : The speedometer of an automobile measure the average speed of the automobile.

Reason : Average velocity is equal to total displacement per total time taken.

Answers

Distance and Displacement

1	a	2	a	3	c	4	a	5	b
6	c								

Uniform Motion

1	d	2	d	3	b	4	b	5	c
6	d	7	a	8	b	9	d	10	c
11	c	12	d	13	d	14	b	15	b
16	d	17	c	18	c	19	d	20	b
21	a	22	b	23	b	24	c		

Non-uniform Motion

1	b	2	c	3	d	4	a	5	a
6	ac	7	a	8	d	9	b	10	a
11	b	12	c	13	b	14	a	15	b
16	d	17	c	18	a	19	c	20	b
21	a	22	c	23	a	24	d	25	c
26	b	27	c	28	d	29	c	30	a
31	c	32	a	33	d	34	a	35	b
36	a	37	b	38	d	39	d	40	b
41	b	42	c	43	b	44	c	45	b
46	d	47	b	48	a	49	b	50	b

51	c	52	c	53	a	54	a	55	c
56	d	57	d	58	d	59	b	60	d
61	c	62	b	63	b	64	a	65	d
66	b	67	a	68	a	69	a	70	d
71	c	72	a	73	a	74	c	75	c
76	c	77	d	78	a	79	c	80	d
81	d	82	c	83	c	84	b	85	a
86	d								

Relative Motion

1	b	2	d	3	b	4	a	5	c
6	d	7	b	8	a	9	d	10	c
11	c	12	b	13	a				

Motion Under Gravity

1	c	2	b	3	d	4	c	5	b
6	a	7	a	8	b	9	c	10	d
11	b	12	a	13	d	14	b	15	c
16	c	17	a	18	b	19	b	20	c
21	b	22	c	23	a	24	b	25	c
26	d	27	b	28	c	29	a	30	d
31	b	32	a	33	b	34	b	35	a
36	c	37	b	38	c	39	b	40	a
41	b	42	b	43	b	44	b	45	d
46	d	47	b	48	b	49	b	50	b
51	c	52	a	53	d	54	d	55	d
56	c	57	b	58	c	59	b	60	b
61	c	62	b	63	c	64	c	65	a
66	a	67	b	68	a	69	b	70	c
71	c	72	c	73	b	74	c	75	b
76	b	77	a	78	a	79	c	80	a
81	a								

Critical Thinking Questions

1	a	2	c	3	abd	4	ad	5	b
6	d	7	c	8	c	9	c	10	d
11	a	12	c						

Graphical Questions

1	b	2	a	3	d	4	b	5	d
6	c	7	d	8	c	9	a	10	b
11	c	12	b	13	a	14	d	15	d
16	c	17	a	18	a	19	a	20	b
21	d	22	c	23	a	24	b	25	a
26	c	27	a	28	c	29	c	30	a

Assertion and Reason

1	a	2	a	3	a	4	a	5	a
6	d	7	c	8	b	9	b	10	e
11	a	12	a	13	a	14	a	15	e
16	e	17	a	18	c	19	d	20	e
21	b	22	d	23	c	24	e	25	e
26	a	27	e	28	b	29	c	30	e

Answers and Solutions

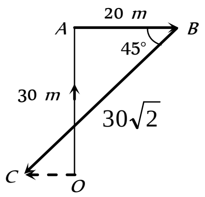
Distance and Displacement

1. (a) $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ $\therefore r = \sqrt{x^2 + y^2 + z^2}$

$r = \sqrt{6^2 + 8^2 + 10^2} = 10\sqrt{2}\ m$

2. (a) $\vec{r} = 20\hat{i} + 10\hat{j}$ $\therefore r = \sqrt{20^2 + 10^2} = 22.5\ m$

3. (c) From figure, $\overrightarrow{OA} = 0\,\hat{i} + 30\,\hat{j}$, $\overrightarrow{AB} = 20\,\hat{i} + 0\,\hat{j}$



$\overrightarrow{BC} = -30\sqrt{2}\ cos45^o\hat{i} - 30\sqrt{2}\ \sin 45^o\hat{j} = -30\,\hat{i} - 30\,\hat{j}$

$\therefore$ Net displacement, $\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AB} + \overrightarrow{BC} = -10\,\hat{i} + 0\,\hat{j}$

$|\overrightarrow{OC}| = 10\ m.$

4. (a) An aeroplane flies 400 m north and 300 m south so the net displacement is 100 m towards north.

Then it flies 1200 m upward so $r = \sqrt{(100)^2 + (1200)^2}$

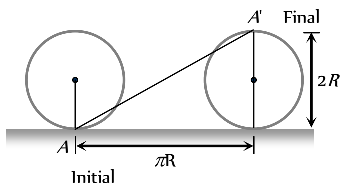
$= 1204m \simeq 1200m$

The option should be 1204 m, because this value mislead one into thinking that net displacement is in upward direction only.

5. (b) Total time of motion is 2 min 20 sec = 140 sec.

As time period of circular motion is 40 sec so in 140 sec. athlete will complete 3.5 revolution *i.e.,* He will be at diametrically opposite point *i.e.,* Displacement = $2R$.

6. (c) Horizontal distance covered by the wheel in half revolution = πR.



So the displacement of the point which was initially in contact with ground = AA' = $\sqrt{(\pi R)^2 + (2R)^2}$

$= R\sqrt{\pi^2 + 4} = \sqrt{\pi^2 + 4}$ $\quad (As\ R = 1m)$

Uniform Motion

1. (d) As the total distance is divided into two equal parts therefore distance averaged speed $= \dfrac{2v_1v_2}{v_1 + v_2}$

2. (d) $\dfrac{v_A}{v_B} = \dfrac{\tan\theta_A}{\tan\theta_B} = \dfrac{\tan 30°}{\tan 60°} = \dfrac{1/\sqrt{3}}{\sqrt{3}} = \dfrac{1}{3}$

3. (b) Distance average speed $= \dfrac{2v_1v_2}{v_1 + v_2} = \dfrac{2 \times 20 \times 30}{20 + 30}$

$= \dfrac{120}{5} = 24\ km/hr$

4. (b) Distance average speed $= \dfrac{2v_1v_2}{v_1 + v_2} = \dfrac{2 \times 2.5 \times 4}{2.5 + 4}$

$= \dfrac{200}{65} = \dfrac{40}{13} km/hr$

5. (c) Distance average speed $= \dfrac{2v_1v_2}{v_1 + v_2} = \dfrac{2 \times 30 \times 50}{30 + 50}$

$= \dfrac{75}{2} = 37.5\ km/hr$

6. (d) Average speed $= \dfrac{\text{Total distance}}{\text{Total time}} = \dfrac{x}{t_1 + t_2}$

$= \dfrac{x}{\dfrac{x/3}{v_1} + \dfrac{2x/3}{v_2}} = \dfrac{1}{\dfrac{1}{3 \times 20} + \dfrac{2}{3 \times 60}} = 36\ km/hr$

7. (a) Time average speed $= \dfrac{v_1 + v_2}{2} = \dfrac{80 + 40}{2} = 60 km/hr$.

8. (b) Distance travelled by train in first 1 *hour* is 60 *km* and distance in next 1/2 *hour* is 20 *km.*

So Average speed $= \dfrac{\text{Total distance}}{\text{Total time}} = \dfrac{60 + 20}{3/2}$

$= 53.33\ km/hour$

9. (d)

10. (c) Total distance to be covered for crossing the bridge

= length of train + length of bridge

$= 150m + 850m = 1000m$

$\text{Time} = \dfrac{\text{Distance}}{\text{Velocity}} = \dfrac{1000}{45 \times \dfrac{5}{18}} = 80\ sec$

11. (c) Displacement of the particle will be zero because it comes back to its starting point

$\text{Average speed} = \dfrac{\text{Total distance}}{\text{Total time}} = \dfrac{30m}{10\ sec} = 3\ m/s$

12. (d) Velocity of particle $= \dfrac{\text{Total diplacement}}{\text{Total time}}$

$= \dfrac{\text{Diameter of circle}}{5} = \dfrac{2 \times 10}{5} = 4\ m/s$

13. (d) A man walks from his home to market with a speed of $5\ km/h$. Distance $= 2.5\ km$ and time $= \dfrac{d}{v} = \dfrac{2.5}{5} = \dfrac{1}{2} hr$.

and he returns back with speed of $7.5\ km/h$ in rest of time of 10 *minutes*.

Distance $= 7.5 \times \frac{10}{60} = 1.25\ km$

So, Average speed $= \frac{\text{Total distance}}{\text{Total time}}$

$= \frac{(2.5 + 1.25)km}{(40/60)hr} = \frac{45}{8} km/hr$.

14. (b) $\frac{|\text{ Average velocity}|}{|\text{ Average speed }|} = \frac{|\text{ displacement}|}{|\text{ distance}|} \le 1$

because displacement will either be equal or less than distance. It can never be greater than distance.

15. (b)

16. (d) Average speed $= \frac{\text{Total distance travelled}}{\text{Total time taken}}$

$= \frac{x}{\frac{2x/5}{v_1} + \frac{3x/5}{v_2}} = \frac{5v_1v_2}{3v_1 + 2v_2}$

17. (c) From given figure, it is clear that the net displacement is zero. So average velocity will be zero.

18. (c) Since displacement is always less than or equal to distance, but never greater than distance. Hence numerical ratio of displacement to the distance covered is always equal to or less than one.

19. (d) Length of train = 100 m

Velocity of train $= 45\ km/hr = 45 \times \frac{5}{18} = 12.5\ m/s$

Length of bridge = 1 km = 1000 m

$\therefore$ Total length covered by train = 1100 m

Time taken by train to cross the bridge $= \frac{1100}{12.5}$ = 88 *sec*

20. (b) Time average velocity $= \frac{v_1 + v_2 + v_3}{3} = \frac{3 + 4 + 5}{3} = 4m/s$

21. (a) When the body is projected vertically upward then at the highest point its velocity is zero but acceleration is not equal to zero $(g = 9.8 m/s^2)$.

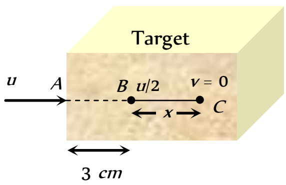
22. (b) Let initial velocity of the bullet = u

After penetrating 3 cm its velocity becomes $\frac{u}{2}$

From $v^2 = u^2 - 2as$

$\left(\frac{u}{2}\right)^2 = u^2 - 2a(3)$

$\Rightarrow 6a = \frac{3u^2}{4} \Rightarrow a = \frac{u^2}{8}$

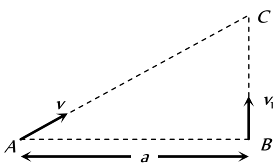


Let further it will penetrate through distance x and stops at point C.

For distance BC, $v = 0, u = u/2, s = x, a = u^2/8$

From $v^2 = u^2 - 2as \Rightarrow 0 = \left(\frac{u}{2}\right)^2 - 2\left(\frac{u^2}{8}\right).x \Rightarrow x = 1\ cm$.

23. (b) Let two boys meet at point C after time 't' from the starting. Then $AC = vt$, $BC = v_1 t$



$(AC)^2 = (AB)^2 + (BC)^2 \Rightarrow v^2t^2 = a^2 + v_1^2 t^2$

By solving we get $t = \sqrt{\frac{a^2}{v^2 - v_1^2}}$

24. (c) $v_{av} = \frac{2v_1v_2}{v_1 + v_2} = \frac{2 \times 40 \times 60}{100} = 48 kmph$.

Non-uniform Motion

1. (b) As $S = ut + \frac{1}{2}at^2 \therefore S_1 = \frac{1}{2}a(10)^2 = 50a$ .....(i)

As $v = u + at \therefore$ velocity acquired by particle in 10 *sec* $v = a \times 10$

For next 10 *sec*, $S_2 = (10a) \times 10 + \frac{1}{2}(a) \times (10)^2$

$S_2 = 150a$ .....(ii)

From (i) and (ii) $S_1 = S_2/3$

2. (c) Acceleration $= \frac{d^2x}{dt^2} = 2a_2$

3. (d) Velocity along X-axis $v_x = \frac{dx}{dt} = 2at$

Velocity along Y-axis $v_y = \frac{dy}{dt} = 2bt$

Magnitude of velocity of the particle,

$v = \sqrt{v_x^2 + v_y^2} = 2t\sqrt{a^2 + b^2}$

4. (a) $S = \int_0^3 v\,dt = \int_0^3 kt\,dt = \left[\frac{1}{2}kt^2\right]_0^3 = \frac{1}{2} \times 2 \times 9 = 9m$

5. (a) $S = kt^3 \therefore a = \frac{d^2S}{dt^2} = 6kt$ *i.e.* $a \propto t$

6. (a,c)

7. (a) From $S = ut + \frac{1}{2}at^2$

$S_1 = \frac{1}{2}a(P-1)^2$ and $S_2 = \frac{1}{2}aP^2$ $[As\ u = 0]$

From $S_n = u + \frac{a}{2}(2n-1)$

$S_{(P^2-P+1)^{th}} = \frac{a}{2}\left[2(P^2 - P + 1) - 1\right] = \frac{a}{2}\left[2P^2 - 2P + 1\right]$

It is clear that $S_{(P^2-P+1)^{th}} = S_1 + S_2$

8. (d) $\vec{a} = \frac{\vec{F}}{m}$. If $\vec{F} = 0$ then $\vec{a} = 0$.

9. (b) $v = 4t^3 - 2t$ (given) $\therefore a = \frac{dv}{dt} = 12t^2 - 2$

and $x = \int_0^t v\,dt = \int_0^t (4t^3 - 2t)\,dt = t^4 - t^2$

When particle is at $2m$ from the origin $t^4 - t^2 = 2$

$\Rightarrow t^4 - t^2 - 2 = 0 \quad (t^2 - 2)(t^2 + 1) = 0 \Rightarrow t = \sqrt{2}$ sec

Acceleration at $t = \sqrt{2}$ sec given by,

$a = 12t^2 - 2 = 12 \times 2 - 2 = 22\ m/s^2$

10. (a) $\frac{dt}{dx} = 2\alpha x + \beta \Rightarrow v = \frac{1}{2\alpha x + \beta}$

$\because \quad a = \frac{dv}{dt} = \frac{dv}{dx} \cdot \frac{dx}{dt}$

$a = v\frac{dv}{dx} = \frac{-v.2\alpha}{(2\alpha x + \beta)^2} = -2\alpha.v.v^2 = -2\alpha v^3$

$\therefore$ Retardation $= 2\alpha v^3$

11. (b) Let u_1, u_2, u_3 and u_4 be velocities at time $t = 0, t_1, (t_1 + t_2)$ and $(t_1 + t_2 + t_3)$ respectively and acceleration is a then $v_1 = \frac{u_1 + u_2}{2}, v_2 = \frac{u_2 + u_3}{2}$ and $v_3 = \frac{u_3 + u_4}{2}$

Also $u_2 = u_1 + at_1$, $u_3 = u_1 + a(t_1 + t_2)$

and $u_4 = u_1 + a(t_1 + t_2 + t_3)$

By solving, we get $\frac{v_1 - v_2}{v_2 - v_3} = \frac{(t_1 + t_2)}{(t_2 + t_3)}$

12. (c) Acceleration $a = \tan\theta$, where θ is the angle of tangent drawn on the graph with the time axis.

13. (b) If acceleration is variable (depends on time) then

$v = u + \int (f)\,dt = u + \int (a\,t)\,dt = u + \frac{a\,t^2}{2}$

14. (a) $S_n = u - \frac{a}{2}(2n - 1) = 10 - \frac{2}{2}(2 \times 5 - 1) = 1\ meter$

15. (b) From $v^2 = u^2 + 2aS \Rightarrow 0 = u^2 + 2aS$

$\Rightarrow a = \frac{-u^2}{2S} = \frac{-(20)^2}{2 \times 10} = -20 m/s^2$

16. (d) $v = u + at = 10 + 2 \times 4 = 18\ m/\text{sec}$

17. (c) If particle starts from rest and moves with constant acceleration then in successive equal interval of time the ratio of distance covered by it will be

$1 : 3 : 5 : 7$ ..... $(2n - 1)$

i.e. ratio of *x* and *y* will be $1 : 3$ *i.e.* $\frac{x}{y} = \frac{1}{3} \Rightarrow y = 3x$

18. (a) $S_n = u + \frac{a}{2}[2n - 1]$

$S_{5^{th}} = 7 + \frac{4}{2}[2 \times 5 - 1] = 7 + 18 = 25m$.

19. (c) Acceleration $a = \frac{dv}{dt} = 0.1 \times 2t = 0.2t$

Which is time dependent *i.e.* non-uniform acceleration.

20. (b) Constant velocity means constant speed as well as same direction throughout.

21. (a) Distance travelled in 4 *sec*

$24 = 4u + \frac{1}{2}a \times 16$ ...(i)

Distance travelled in total 8 *sec*

$88 = 8u + \frac{1}{2}a \times 64$ ...(ii)

After solving (i) and (ii), we get $u = 1$ *m/s.*

22. (c) $v_x = \frac{dx}{dt} = \frac{d}{dt}(3t^2 - 6t) = 6t - 6$. At $t = 1, v_x = 0$

$v_y = \frac{dy}{dt} = \frac{d}{dt}(t^2 - 2t) = 2t - 2$. At $t = 1, v_y = 0$

Hence $v = \sqrt{v_x^2 + v_y^2} = 0$

23. (a) Distance travelled in n^{th} second $= u + \frac{a}{2}(2n - 1)$

Distance travelled in 5^{th} second $= 0 + \frac{8}{2}(2 \times 5 - 1) = 36m$

24. (d) $v^2 = u^2 + 2as \Rightarrow (9000)^2 - (1000)^2 = 2 \times a \times 4$

$\Rightarrow a = 10^7\ m/s^2$ Now $t = \frac{v - u}{a}$

$\Rightarrow t = \frac{9000 - 1000}{10^7} = 8 \times 10^{-4}\ sec$

25. (c) Initial relative velocity $= v_1 - v_2$, Final relative velocity $= 0$

From $v^2 = u^2 - 2as \Rightarrow 0 = (v_1 - v_2)^2 - 2 \times a \times s$

$\Rightarrow s = \frac{(v_1 - v_2)^2}{2a}$

If the distance between two cars is '*s*' then collision will take place. To avoid collision $d > s \quad \therefore d > \frac{(v_1 - v_2)^2}{2a}$

where $d =$ actual initial distance between two cars.

26. (b) $v = u + at \Rightarrow -2 = 10 + a \times 4 \Rightarrow a = -3m/sec^2$

27. (c) $S_x = u_x t + \frac{1}{2}a_x t^2 \Rightarrow S_x = \frac{1}{2} \times 6 \times 16 = 48\ m$

$S_y = u_y t + \frac{1}{2}a_y t^2 \Rightarrow S_y = \frac{1}{2} \times 8 \times 16 = 64\ m$

$S = \sqrt{S_x^2 + S_y^2} = 80m$

28. (d) $S \propto u^2$. If *u* becomes 3 times then *S* will become 9 times *i.e.* $9 \times 20 = 180m$

29. (c) $y = a + bt + ct^2 - dt^4$

$\therefore v = \frac{dy}{dt} = b + 2ct - 4dt^3$ and $a = \frac{dv}{dt} = 2c - 12dt^2$

Hence, at $t = 0$, $v_{initial} = b$ and $a_{initial} = 2c$.

30. (a) $S \propto u^2 \quad \therefore \quad \frac{S_1}{S_2} = \left(\frac{u_1}{u_2}\right)^2 \Rightarrow \frac{2}{S_2} = \frac{1}{4} \Rightarrow S_2 = 8\ m$

31. (c) $t = \sqrt{\frac{2h}{(g + a)}} = \sqrt{\frac{2 \times 2.7}{(9.8 + 1.2)}} = \sqrt{\frac{5.4}{11}} = \sqrt{0.49} = 0.7\,\text{sec}$

As $u = 0$ and lift is moving upward with acceleration

32. (a) Displacement $x = 2t^2 + t + 5$

Velocity $= \frac{dx}{dt} = 4t + 1$

Acceleration $= \frac{d^2x}{dt^2} = 4$ *i.e.* independent of time

Hence acceleration $= 4\ m/s^2$

33. (d) Both trains will travel a distance of 1 *km* before to come in rest. In this case by using $v^2 = u^2 + 2as$

$\Rightarrow 0 = (40)^2 + 2a \times 1000 \Rightarrow a = -0.8\ m/s^2$

34. (a) $v = u + at \Rightarrow v = 0 + 5 \times 10 = 50\ m/s$

35. (b) Let '*a*' *be* the retardation of boggy then distance covered by it be *S*. If *u* is the initial velocity of boggy after detaching from train (*i.e.* uniform speed of train)

$v^2 = u^2 + 2as \Rightarrow 0 = u^2 - 2as \Rightarrow s_b = \dfrac{u^2}{2a}$

Time taken by boggy to stop

$v = u + at \Rightarrow 0 = u - at \Rightarrow t = \dfrac{u}{a}$

In this time *t* distance travelled by train $= s_t = ut = \dfrac{u^2}{a}$

Hence ratio $\dfrac{s_b}{s_t} = \dfrac{1}{2}$

36. (a) $S_n = u + \dfrac{a}{2}(2n-1) = \dfrac{a}{2}(2n-1)$ because $u = 0$

Hence $\dfrac{S_4}{S_3} = \dfrac{7}{5}$

37. (b) $v = u + \int a dt = u + \int (3t^2 + 2t + 2) dt$

$= u + \dfrac{3t^3}{3} + \dfrac{2t^2}{2} + 2t = u + t^3 + t^2 + 2t$

$= 2 + 8 + 4 + 4 = 18\ m/s$ (As $t = 2$ *sec*)

38. (d) $v = \dfrac{ds}{dt} = 3t^2 - 12t + 3$ and $a = \dfrac{dv}{dt} = 6t - 12$

For $a = 0$, we have $t = 2$ and at $t = 2, v = -9\ ms^{-1}$

39. (d)

40. (b) $a = \sqrt{a_x^2 + a_y^2} = \left[\left(\dfrac{d^2x}{dt^2}\right)^2 + \left(\dfrac{d^2y}{dt^2}\right)^2\right]^{\frac{1}{2}}$

Here $\dfrac{d^2y}{dt^2} = 0$. Hence $a = \dfrac{d^2x}{dt^2} = 8m/s^2$

41. (b) $F = m \times a$, If force is constant then $a \propto \dfrac{1}{m}$. So If mass is doubled then acceleration becomes half.

42. (c) $S_n = u + \dfrac{a}{2}(2n-1) \Rightarrow 1.2 = 0 + \dfrac{a}{2}(2 \times 6 - 1)$

$\Rightarrow a = \dfrac{1.2 \times 2}{11} = 0.218\ m/s^2$

43. (b) Here $v = 144\ km/h = 40 m/s$

$v = u + at \Rightarrow 40 = 0 + 20 \times a \Rightarrow a = 2\ m/s^2$

$\therefore s = \dfrac{1}{2}at^2 = \dfrac{1}{2} \times 2 \times (20)^2 = 400\ m$

44. (c) $\dfrac{dx}{dt} = 2at - 3bt^2 \Rightarrow \dfrac{d^2x}{dt^2} = 2a - 6bt = 0 \Rightarrow t = \dfrac{a}{3b}$

45. (b) Stopping distance $= \dfrac{\text{Kinetic energy}}{\text{Retardins force}} = \dfrac{\frac{1}{2}mu^2}{F}$

If retarding force (*F*) and velocity (*v*) are equal then stopping distance $\propto m$ (mass of vehicle)

As $m_{car} < m_{truck}$ therefore car will cover less distance before coming to rest.

46. (d) $u = 72\ kmph = 20 m/s,\ v = 0$

By using $v^2 = u^2 - 2as \Rightarrow a = \dfrac{u^2}{2s} = \dfrac{(20)^2}{2 \times 200} = 1\ m/s^2$

47. (b) $v = \dfrac{ds}{dt} = 12t - 3t^2$

Velocity is zero for $t = 0$ and $t = 4$ sec

48. (a)

49. (b) Let *A* and *B* will meet after time *t sec*. it means the distance travelled by both will be equal.

$S_A = ut = 40t$ and $S_B = \dfrac{1}{2}at^2 = \dfrac{1}{2} \times 4 \times t^2$

$S_A = S_B \Rightarrow 40t = \dfrac{1}{2}4t^2 \Rightarrow t = 20$ *sec*

50. (b) $x = a + bt^2, v = \dfrac{dx}{dt} = 2bt$

Instantaneous velocity $v = 2 \times 3 \times 3 = 18\ cm/\text{sec}$

51. (c) If the body starts from rest and moves with constant acceleration then the ratio of distances in consecutive equal time interval $S_1 : S_2 : S_3 = 1 : 3 : 5$

52. (c) $x = at + bt^2 - ct^3,\ a = \dfrac{d^2x}{dt^2} = 2b - 6ct$

53. (a) Let initial $(t = 0)$ velocity of particle $= u$

For first 5 *sec* motion $s_5 = 10$ *metre*

$s = ut + \dfrac{1}{2}at^2 \Rightarrow 10 = 5u + \dfrac{1}{2}a(5)^2$

$2u + 5a = 4$ ....(i)

For first 8 *sec* of motion $s_8 = 20$ *metre*

$20 = 8u + \dfrac{1}{2}a(8)^2 \Rightarrow 2u + 8a = 5$ ....(ii)

By solving $u = \dfrac{7}{6} m/s$ and $a = \dfrac{1}{3} m/s^2$

Now distance travelled by particle in Total 10 *sec*.

$s_{10} = u \times 10 + \dfrac{1}{2}a(10)^2$

By substituting the value of *u* and *a* we will get $s_{10} = 28.3\ m$

so the distance in last 2 sec $= s_{10} - s_8$

$= 28.3 - 20 = 8.3m$

54. (a) $s \propto t^2$ (given) $\therefore$ $s = Kt^2$

Acceleration $a = \dfrac{d^2s}{dt^2} = 2k$ (constant)

It means the particle travels with uniform acceleration.

55. (c) Because acceleration is a vector quantity

56. (d) $u = at,\ x = \int u\ dt = \int at\ dt = \dfrac{at^2}{2}$

For $t = 4$ sec, $x = 8a$

57. (d) $3t = \sqrt{3x} + 6 \Rightarrow 3x = (3t-6)^2$

$\Rightarrow x = 3t^2 - 12t + 12$

$v = \frac{dx}{dt} = 6t - 12$, for $v = 0, t = 2\sec$

$x = 3(2)^2 - 12\times 2 + 12 = 0$

58. (d) $u = 0, S = 250m, t = 10\sec$

$S = ut + \frac{1}{2}at^2 \Rightarrow 250 = \frac{1}{2}a[10]^2 \Rightarrow a = 5m/s^2$

So, $F = ma = 0.9\times 5 = 4.5N$

59. (b) Time $= \frac{\text{Distance}}{\text{Average velocity}} = \frac{3.06}{0.34} = 9$ sec

Acceleration $= \frac{\text{Change in velocity}}{\text{Time}} = \frac{0.18}{9} = 0.02\, m/s^2$

60. (d) $s = 3t^3 + 7t^2 + 14t + 8\ m$

$a = \frac{d^2s}{dt^2} = 18t + 14$ at $t = 1$ sec $\Rightarrow a = 32m/s^2$

61. (c) Instantaneous velocity $v = \frac{\Delta x}{\Delta t}$

By using the data from the table

$v_1 = \frac{0-(-2)}{1} = 2m/s, \quad v_2 = \frac{6-0}{1} = 6\ m/s$

$v_3 = \frac{16-6}{1} = 10\ m/s$

So, motion is non-uniform but accelerated.

62. (b) Only direction of displacement and velocity gets changed, acceleration is always directed vertically downward.

63. (b) $s = 2t^2 + 2t + 4, a = \frac{d^2s}{dt^2} = 4\, m/s^2$

64. (a) According to problem

Distance travelled by body *A* in 5^{th} *sec* and distance travelled by body *B* in 3^{rd} *sec.* of its motion are equal.

$0 + \frac{a_1}{2}(2\times 5 - 1) = 0 + \frac{a_2}{2}[2\times 3 - 1]$

$9a_1 = 5a_2 \Rightarrow \frac{a_1}{a_2} = \frac{5}{9}$

65. (d) $u = 200\ m/s, v = 100\ m/s, s = 0.1\ m$

$a = \frac{u^2 - v^2}{2s} = \frac{(200)^2 - (100)^2}{2\times 0.1} = 15\times 10^4\ m/s^2$

66. (b) $v = u + at = u + \left(\frac{F}{m}\right)t = 20 + \left(\frac{100}{5}\right)\times 10 = 220\ m/s$

67. (a) Velocity acquired by body in 10 *sec*

$v = 0 + 2\times 10 = 20m/s$

and distance travelled by it in 10 *sec*

$S_1 = \frac{1}{2}\times 2\times (10)^2 = 100\ m$

then it moves with constant velocity (20 *m/s*) for 30 sec

$S_2 = 20\times 30 = 600\ m$

After that due to retardation $(4m/s^2)$ it stops

$S_3 = \frac{v^2}{2a} = \frac{(20)^2}{2\times 4} = 50m$

Total distance travelled $S_1 + S_2 + S_3 = 750m$

68. (a) If a body starts from rest with acceleration α and then retards with retardation β and comes to rest. The total time taken for this journey is t and distance covered is S

then $S = \frac{1}{2}\frac{\alpha\beta t^2}{(\alpha+\beta)} = \frac{1}{2}\frac{5\times 10}{(5+10)}\times t^2$

$\Rightarrow \quad 1500 = \frac{1}{2}\frac{5\times 10}{(5+10)}\times t^2 \quad \Rightarrow t = 30\sec$.

69. (a)

70. (d) $S \propto u^2$. Now speed is two times so distance will be four times $S = 4\times 6 = 24m$

71. (c) Let student will catch the bus after *t sec.* So it will cover distance *ut.*

Similarly distance travelled by the bus will be $\frac{1}{2}at^2$ for the given condition

$ut = 50 + \frac{1}{2}at^2 = 50 + \frac{t^2}{2} \qquad [a = 1\, m/s^2]$

$\Rightarrow u = \frac{50}{t} + \frac{t}{2}$

To find the minimum value of u

$\frac{du}{dt} = 0$, so we get $t = 10\sec$, then $u = 10\ m/s$

72. (a) $\frac{1}{2}at^2 = vt \Rightarrow t = \frac{2v}{a}$

73. (a) The velocity of the particle is

$\frac{dx}{dt} = \frac{d}{dt}(2 - 5t + 6t^2) = (0 - 5 + 12t)$

For initial velocity $t = 0$, hence $v = -5\ m/s$.

74. (c) For First part,

$u = 0, t = T$ and acceleration $= a$

$\therefore v = 0 + aT = aT$ and $S_1 = 0 + \frac{1}{2}aT^2 = \frac{1}{2}aT^2$

For Second part,

$u = aT$, retardation $= a_1$, $v = 0$ and time taken $= T_1$ (let)

$\therefore\ 0 = u - a_1T_1 \Rightarrow aT = a_1T_1$

and from $v^2 = u^2 - 2aS_2 \Rightarrow S_2 = \frac{u^2}{2a_1} = \frac{1}{2}\frac{a^2T^2}{a_1}$

$S_2 = \frac{1}{2}aT\times T_1 \qquad \left(\text{As } a_1 = \frac{aT}{T_1}\right)$

$\therefore\ v_{av} = \frac{S_1 + S_2}{T + T_1} = \frac{\frac{1}{2}aT^2 + \frac{1}{2}aT\times T_1}{T + T_1}$

$= \frac{\frac{1}{2}aT(T + T_1)}{T + T_1} = \frac{1}{2}aT$

75. (c) $u = 0$, $v = 27.5\ m/s$ and $t = 10$ *sec*

$\therefore a = \frac{27.5 - 0}{10} = 2.75\ m/s^2$

Now, the distance traveled in next 10 *sec*,

$S = ut + \frac{1}{2}at^2 = 27.5\times 10 + \frac{1}{2}\times 2.75\times 100$

$= 275 + 137.5 = 412.5\ m$

76. (c) $v = (180 - 16x)^{1/2}$

As $a = \frac{dv}{dt} = \frac{dv}{dx}.\frac{dx}{dt}$

$\therefore a = \frac{1}{2}(180-16x)^{-1/2} \times (-16)\left(\frac{dx}{dt}\right)$

$= -8\,(180-16x)^{-1/2} \times v$

$= -8\,(180-16x)^{-1/2} \times (180-16x)^{1/2} = -8\ m/s^2$

77. (d) $x \propto t^3 \ \therefore\ x = Kt^3$

$\Rightarrow v = \frac{dx}{dt} = 3\ Kt^2$ and $a = \frac{dv}{dt} = 6\ Kt$

i.e. $a \propto t$

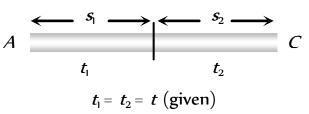
78. (a) $\because a = \frac{dv}{dt} = 2(t-1) \Rightarrow dv = 2(t-1)\,dt$

$\Rightarrow v = \int_0^5 2\,(t-1)dt = 2\left[\frac{t^2}{2} - t\right]_0^5 = 2\left[\frac{25}{2} - 5\right]$ = 15 *m/s*

79. (c) $\because S_1 = ut + \frac{1}{2}at^2$ .....(i)

and velocity after first *t sec*

$v = u + at$



Now, $S_2 = vt + \frac{1}{2}at^2$

$= (u+at)t + \frac{1}{2}at^2$ ..... (ii)

Equation (ii) – (i) $\Rightarrow S_2 - S_1 = at^2$

$\Rightarrow a = \frac{S_2 - S_1}{t^2} = \frac{65-40}{(5)^2} = 1\ m/s^2$

From equation (i), we get,

$S_1 = ut + \frac{1}{2}at^2 \Rightarrow 40 = 5u + \frac{1}{2} \times 1 \times 25$

$\Rightarrow 5u = 27.5 \ \therefore\ u = 5.5\ m/s$

80. (d) $S \propto u^2 \Rightarrow \frac{S_1}{S_2} = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$

81. (d) $x = ae^{-\alpha t} + be^{\beta t}$

Velocity $v = \frac{dx}{dt} = \frac{d}{dt}(ae^{-\alpha t} + be^{\beta t})$

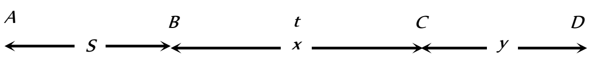
$= a.e^{-\alpha t}(-\alpha) + be^{\beta t}.\beta) = -a\alpha e^{-\alpha t} + b\beta e^{\beta t}$

Acceleration $= -a\alpha e^{-\alpha t}(-\alpha) + b\beta e^{bt}.\beta$

$= a\alpha^2\, e^{-\alpha t} + b\beta^2 e^{\beta t}$

Acceleration is positive so velocity goes on increasing with time.

82. (c) Let car starts from point *A* from rest and moves up to point *B* with acceleration *f*



Velocity of car at point *B*, $v = \sqrt{2fS}$

$[\text{As } v^2 = u^2 + 2as]$

Car moves distance *BC* with this constant velocity in time *t*

$x = \sqrt{2fS}\,.t$ ......(i) [As $s = ut$]

So the velocity of car at point *C* also will be $\sqrt{2fs}$ and finally car stops after covering distance *y*.

Distance $CD \Rightarrow y = \frac{(\sqrt{2fS})^2}{2(f/2)} = \frac{2fS}{f} = 2S$ ....(ii)

$[\text{As} v^2 = u^2 - 2as \Rightarrow s = u^2/2a]$

So, the total distance $AD = AB + BC + CD = 15S$ (given)

$\Rightarrow \quad S + x + 2S = 15S \ \Rightarrow\ x = 12S$

Substituting the value of *x* in equation (i) we get

$x = \sqrt{2fS}\,.t \Rightarrow 12S = \sqrt{2fS}.t \ \Rightarrow\ 144S^2 = 2fS.t^2$

$\Rightarrow\ S = \frac{1}{72}ft^2.$

83. (c) Let man will catch the bus after '*t*' *sec* . So he will cover distance *ut.*

Similarly distance travelled by the bus will be $\frac{1}{2}at^2$. For the given condition

$u\,t = 45 + \frac{1}{2}a\,t^2 = 45 + 1.25\,t^2$ $[As\ a = 2.5 m/s^2]$

$\Rightarrow\ u = \frac{45}{t} + 1.25\,t$

To find the minimum value of *u*

$\frac{du}{dt} = 0$ so we get $t = 6$ sec then,

$u = \frac{45}{6} + 1.25 \times 6 = 7.5 + 7.5 = 15 m/s$

84. (b) $x = 4(t-2) + a(t-2)^2$

At $t = 0, x = -8 + 4a = 4a - 8$

$v = \frac{dx}{dt} = 4 + 2a(t-2)$

At $t = 0,\ v = 4 - 4a = 4(1-a)$

But acceleration, $a = \frac{d^2x}{dt^2} = 2a$

85. (a) Distance covered in 5^{th} second,

$S_{5^{th}} = u + \frac{a}{2}(2n-1) = 0 + \frac{a}{2}(2 \times 5 - 1) = \frac{9a}{2}$

and distance covered in 5 second,

$S_5 = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times a \times 25 = \frac{25a}{2}$

$\therefore\ \frac{S_{5^{th}}}{S_5} = \frac{9}{25}$

86. (d) The nature of the path is decided by the direction of velocity, and the direction of acceleration. The trajectory can be a straight line, circle or a parabola depending on these factors.

Relative Motion

1. (b) Time $= \dfrac{\text{Total length}}{\text{Relativevelocity}} = \dfrac{50+50}{10+15} = \dfrac{100}{25} = 4\ sec$

2. (d) Total distance $= 130 + 120 = 250\ m$

Relative velocity $= 30 - (-20) = 50\ m/s$

Hence $t = 250/50 = 5s$

3. (b) Relative velocity of bird *w.r.t* train $= 25 + 5 = 30\ m/s$

time taken by the bird to cross the train $t = \dfrac{210}{30} = 7\ sec$

4. (a) Effective speed of the bullet
= speed of bullet + speed of police jeep
$= 180\ m/s + 45\ km/h = (180 + 12.5)\ m/s = 192.5\ m/s$

Speed of thief's jeep $= 153 km/h = 42.5 m/s$

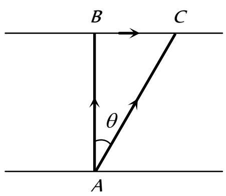
Velocity of bullet *w.r.t* thief's car $= 192.5 - 42.5$ =150*m/s*

5. (c) Given $\overrightarrow{AB}$ = Velocity of boat= 8 *km/hr*

$\overrightarrow{AC}$ = Resultant velocity of boat = 10 *km/hr*

$\overrightarrow{BC}$ = Velocity of river $= \sqrt{AC^2 - AB^2}$

$= \sqrt{(10)^2 - (8)^2} = 6\ km/hr$



6. (d) Relative velocity
$= 10 + 5 = 15\ m/sec$

$\therefore\ t = \dfrac{150}{15} = 10\ sec$

7. (b) The relative velocity of boat *w.r.t.* water

$= v_{\text{boat}} - v_{\text{water}} = (3\,\hat{i} + 4\,\hat{j}) - (-3\,\hat{i} - 4\,\hat{j}) = 6\,\hat{i} + 8\,\hat{j}$

8. (a) When two particles moves towards each other then $v_1 + v_2 = 6$ ...(i)

When these particles moves in the same direction then $v_1 - v_2 = 4$ ...(ii)

By solving $v_1 = 5$ and $v_2 = 1\ m/s$

9. (d) For the round trip he should cross perpendicular to the river

$\therefore$ Time for trip to that side $= \dfrac{1 km}{4 km/hr} = 0.25 hr$

To come back, again he take 0.25 *hr* to cross the river.
Total time is 30 min, he goes to the other bank and come back at the same point.

10. (c) Relativistic momentum $= \dfrac{m_0 v}{\sqrt{1 - v^2/c^2}}$

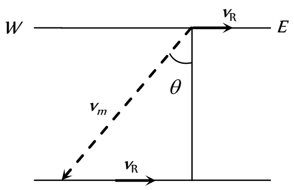
If velocity is doubled then the relativistic mass also increases. Thus value of linear momentum will be more than double.

11. (c) For shortest possible path man should swim with an angle (90+θ) with downstream.
From the fig,

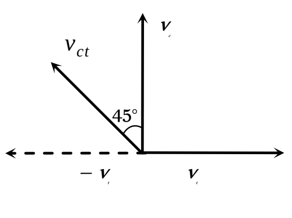
$\sin\theta = \dfrac{v_r}{v_m} = \dfrac{5}{10} = \dfrac{1}{2}$

$\Rightarrow\ \therefore\ \theta = 30°$

So angle with downstream $= 90° + 30° = 120°$



12. (b) $\overrightarrow{v_{ct}} = \overrightarrow{v_c} - \overrightarrow{v_t}$



$\overrightarrow{v_{ct}} = \overrightarrow{v_c} + (-\overrightarrow{v_t})$

Velocity of car *w.r.t.* train (v_{ct}) is towards West – North

13. (a) As the trains are moving in the same direction. So the initial relative speed $(v_1 - v_2)$ and by applying retardation final relative speed becomes zero.

From $v = u - at \Rightarrow 0 = (v_1 - v_2) - at \Rightarrow t = \dfrac{v_1 - v_2}{a}$

Motion Under Gravity

1. (c) $u = 12\ m/s,\ g = 9.8\ m/\sec^2,\ t = 10$ sec

Displacement $= ut + \dfrac{1}{2}gt^2$

$= 12 \times 10 + \dfrac{1}{2} \times 9.8 \times 100 = 610m$

2. (b) Velocity at the time of striking the floor,

$u = \sqrt{2gh_1} = \sqrt{2 \times 9.8 \times 10} = 14 m/s$

Velocity with which it rebounds.

$v = \sqrt{2gh_2} = \sqrt{2 \times 9.8 \times 2.5} = 7\ m/s$

$\therefore$ Change in velocity $\Delta v = 7 - (-14) = 21 m/s$

$\therefore$ Acceleration $= \dfrac{\Delta v}{\Delta t} = \dfrac{21}{0.01} = 2100\ m/s^2$ (upwards)

3. (d) Let t be the time of flight of the first body after meeting, then $(t-4)$ *sec* will be the time of flight of the second body. Since $h_1 = h_2$

$\therefore\ 98t - \dfrac{1}{2}gt^2 = 98(t-4) - \dfrac{1}{2}g(t-4)^2$

On solving, we get $t = 12$ *seconds*

4. (c) $h = \dfrac{1}{2}gt^2 \Rightarrow t = \sqrt{2h/g}$

$t_a = \sqrt{\dfrac{2a}{g}}$ and $t_b = \sqrt{\dfrac{2b}{g}} \Rightarrow \dfrac{t_a}{t_b} = \sqrt{\dfrac{a}{b}}$

5. (b) $\dfrac{1}{2}g(3)^2 = \dfrac{g}{2}(2n-1) \Rightarrow n = 5\ s$

6. (a) Time taken by first stone to reach the water surface from the bridge be t, then

$h = ut + \dfrac{1}{2}gt^2 \Rightarrow 44.1 = 0 \times t + \dfrac{1}{2} \times 9.8t^2$

$t = \sqrt{\dfrac{2 \times 44.1}{9.8}} = 3\ sec$

Second stone is thrown 1 *sec* later and both strikes simultaneously. This means that the time left for second stone $= 3 - 1 = 2\ sec$

Hence $44.1 = u \times 2 + \dfrac{1}{2}9.8(2)^2$

$\Rightarrow 44.1 - 19.6 = 2u \Rightarrow u = 12.25\ m/s$

7. (a)

8. (b) Let the initial velocity of ball be u

Time of rise $t_1 = \dfrac{u}{g+a}$ and height reached $= \dfrac{u^2}{2(g+a)}$

Time of fall t_2 is given by

$\frac{1}{2}(g-a)t_2^2 = \frac{u^2}{2(g+a)}$

$\Rightarrow t_2 = \frac{u}{\sqrt{(g+a)(g-a)}} = \frac{u}{(g+a)}\sqrt{\frac{g+a}{g-a}}$

$\therefore\ t_2 > t_1$ because $\frac{1}{g+a} < \frac{1}{g-a}$

9. (c) Vertical component of velocities of both the balls are same and equal to zero. So $t = \sqrt{\frac{2h}{g}}$

10. (d) The separation between the two bodies, two seconds after the release of second body

$= \frac{1}{2}\times 9.8[(3)^2 - (2)^2] = 24.5\ m$

11. (b) Time of flight $= \frac{2u}{g} = \frac{2\times 100}{10} = 20\ sec$

12. (a) $h = \frac{1}{2}gt^2 = \frac{1}{2}\times 10\times (4)^2 = 80\ m$

13. (d) Let the body after time $t/2$ be at x from the top, then

$x = \frac{1}{2}g\frac{t^2}{4} = \frac{gt^2}{8}$ ...(i)

$h = \frac{1}{2}gt^2$ ...(ii)

Eliminate t from (i) and (ii), we get $x = \frac{h}{4}$

$\therefore$ Height of the body from the ground $= h - \frac{h}{4} = \frac{3h}{4}$

14. (b) By applying law of conservation of energy

$mgR = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2Rg}$

15. (c) Acceleration of body along AB is $g\cos\theta$

Distance travelled in time t $sec = AB = \frac{1}{2}(g\cos\theta)t^2$

From ΔABC, $AB = 2R\cos\theta$; $2R\cos\theta = \frac{1}{2}g\cos\theta t^2$

$\Rightarrow t^2 = \frac{4R}{g}$ or $t = 2\sqrt{\frac{R}{g}}$

16. (c) Force down the plane $= mg\sin\theta$

$\therefore$ Acceleration down the plane $= g\sin\theta$

Since $l = 0 + \frac{1}{2}g\sin\theta t^2$

$\therefore\ t^2 = \frac{2l}{g\sin\theta} = \frac{2h}{g\sin^2\theta} \Rightarrow t = \frac{1}{\sin\theta}\sqrt{\frac{2h}{g}}$

17. (a) $h = ut - \frac{1}{2}gt^2 \Rightarrow 96 = 80t - \frac{32}{2}t^2$

$\Rightarrow t^2 - 5t + 6 = 0 \Rightarrow t = 2$ sec or 3 sec

18. (b) $v = g\times t = 32\times 1 = 32\,ft/\text{sec}$

19. (b) $v^2 = u^2 + 2gh \Rightarrow (3u)^2 = (-u)^2 + 2gh \Rightarrow h = \frac{4u^2}{g}$

20. (c) $t = \sqrt{\frac{2h}{g}}$ and h and g are same.

21. (b) Time of flight $= \frac{2u}{g} = \frac{2\times 96}{32} = 6\ sec$

22. (c) Total distance $= \frac{1}{2}gt^2 = \frac{25}{2}g$

Distance moved in 3 sec $= \frac{9}{2}g$

Remaining distance $= \frac{16}{2}g$

If t is the time taken by the stone to reach the ground for the remaining distance then

$\Rightarrow \frac{16}{2}g = \frac{1}{2}gt^2 \Rightarrow t = 4\text{ sec}$

23. (a) Height travelled by ball (with balloon) in 2 sec

$h_1 = \frac{1}{2}a\,t^2 = \frac{1}{2}\times 4.9\times 2^2 = 9.8\ m$

Velocity of the balloon after 2 sec

$v = a\,t = 4.9\times 2 = 9.8\ m/s$

Now if the ball is released from the balloon then it acquire same velocity in upward direction.

Let it move up to maximum height h_2

$v^2 = u^2 - 2gh_2 \Rightarrow 0 = (9.8)^2 - 2\times(9.8)\times h_2 \therefore h_2 = 4.9m$

Greatest height above the ground reached by the ball $= h_1 + h_2 = 9.8 + 4.9 = 14.7\ m$

24. (b) Let h distance is covered in n sec

$\Rightarrow h = \frac{1}{2}gn^2$ ...(i)

Distance covered in n^{th} $sec = \frac{1}{2}g(2n-1)$

$\Rightarrow \frac{9h}{25} = \frac{g}{2}(2n-1)$ ...(ii)

From (i) and (ii), $h = 122.5\ m$

25. (c) $h = ut + \frac{1}{2}gt^2 \Rightarrow 81 = -12t + \frac{1}{2}\times 10\times t^2 \Rightarrow t = 5.4\ sec$

26. (d) The initial velocity of aeroplane is horizontal, then the vertical component of velocity of packet will be zero.

So $t = \sqrt{\frac{2h}{g}}$

27. (b) Time taken by first drop to reach the ground $t = \sqrt{\frac{2h}{g}}$

$\Rightarrow t = \sqrt{\frac{2\times 5}{10}} = 1\text{ sec}$

As the water drops fall at regular intervals from a tap therefore time difference between any two drops $= \frac{1}{2}$ sec

In this given time, distance of second drop from the tap $= \frac{1}{2}g\left(\frac{1}{2}\right)^2 = \frac{5}{5} = 1.25\,m$

Its distance from the ground $= 5 - 1.25 = 3.75\ m$

28. (c) $h = ut + \frac{1}{2}gt^2$, $t = 3$ sec, $u = -4.9\ m/s$

$\Rightarrow h = -4.9\times 3 + 4.9\times 9 = 29.4\ m$

29. (a) Horizontal velocity of dropped packet $= u$

Vertical velocity $= \sqrt{2gh}$

$\therefore$ Resultant velocity at earth $= \sqrt{u^2 + 2gh}$

30. (d) Given $a = 19.6\ m/s^2 = 2g$

Resultant velocity of the rocket after 5 *sec*

$v = 2g \times 5 = 10g\ m/s$

Height achieved after 5 *sec*, $h_1 = \frac{1}{2} \times 2g \times 25 = 245m$

On switching off the engine it goes up to height h_2 where its velocity becomes zero.

$0 = (10g)^2 - 2gh_2 \Rightarrow h_2 = 490m$

$\therefore$ Total height of rocket $= 245 + 490 = 735\ m$

31. (b) Bullet will take $\frac{100}{1000} = 0.1$ *sec* to reach target.

During this period vertical distance (downward) travelled by the bullet $= \frac{1}{2}gt^2$

$= \frac{1}{2} \times 10 \times (0.1)^2 m = 5\ cm$

So the gun should be aimed 5 *cm* above the target.

32. (a) $S_n = u + \frac{g}{2}(2n-1)$; when $u = 0$, $S_1 : S_2 : S_3 = 1 : 3 : 5$

33. (b) It has lesser initial upward velocity.

34. (b) At maximum height velocity $v = 0$

We know that $v = u + at$, hence

$0 = u - gT \Rightarrow u = gT$

When $v = \frac{u}{2}$, then

$\frac{u}{2} = u - gt \Rightarrow gt = \frac{u}{2} \Rightarrow gt = \frac{gT}{2} \Rightarrow t = \frac{T}{2}$

Hence at $t = \frac{T}{2}$, it acquires velocity $\frac{u}{2}$

35. (a) If u is the initial velocity then distance covered by it in 2 *sec*

$S = ut + \frac{1}{2}at^2 = u \times 2 + \frac{1}{2} \times 10 \times 4 = 2u + 20$ ...(i)

Now distance covered by it in 3 *sec*

$S_{3^{rd}} = u + \frac{g}{2}(2 \times 3 - 1)10 = u + 25$ ...(ii)

From(i) and (ii), $2u + 20 = u + 25 \Rightarrow u = 5$

$\therefore S = 2 \times 5 + 20 = 30\ m$

36. (c) For first case $v^2 - 0^2 = 2gh \Rightarrow (3)^2 = 2gh$

For second case $v^2 = (-u)^2 + 2gh = 4^2 + 3^2 \ \therefore v = 5 km/h$

37. (b) The time of fall is independent of the mass.

38. (c) $h_{n^{th}} = u - \frac{g}{2}(2n-1)$

$h_{5^{th}} = u - \frac{10}{2}(2 \times 5 - 1) = u - 45$

$h_{6^{th}} = u - \frac{10}{2}(2 \times 6 - 1) = u - 55$

Given $h_{5^{th}} = 2 \times h_{6^{th}}$ .By solving we get $u = 65\ m/s$

39. (b) $S = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}at^2$

Hence $t \propto \sqrt{S}$ *i.e.*, if S becomes one-fourth then t will become half *i.e.*, 2 *sec*

40. (a) Distance between the balls = Distance travelled by first ball in 3 seconds –Distance travelled by second ball in 2 seconds

$= \frac{1}{2}g\,(3)^2 - \frac{1}{2}g\,(2)^2 = 45 - 20 = 25\ m$

41. (b) Speed of stone in a vertically upward direction is 4.9 *m/s*. So for vertical downward motion we will consider $u = -4.9\ m/s$

$h = ut + \frac{1}{2}gt^2 = -4.9 \times 2 + \frac{1}{2} \times 9.8 \times (2)^2 = 9.8\ m$

42. (b) Speed of stone in a vertically upward direction is 20 *m/s*. So for vertical downward motion we will consider $u = -20\ m/s$

$v^2 = u^2 + 2gh = (-20)^2 + 2 \times 9.8 \times 200 = 4320\ m/s$

$\therefore v \simeq 65\,m/s$.

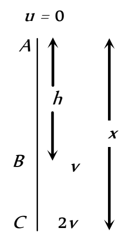
43. (b) Let at point A initial velocity of body is equal to zero

for path AB: $v^2 = 0 + 2gh$ ...(i)

for path AC: $(2v)^2 = 0 + 2gx$

$4v^2 = 2gx$ ...(ii)

Solving (i) and (ii) $x = 4h$



44. (b) For one dimensional motion along a plane

$S = ut + \frac{1}{2}at^2 \Rightarrow 9.8 = 0 + \frac{1}{2}\ g\sin 30^o\, t^2 \Rightarrow t = 2\sec$

45. (d) Body reaches the point of projection with same velocity.

46. (d) Time of flight $T = \frac{2u}{g} = 4\ \sec \Rightarrow u = 20\ m/s$

47. (b) $t = \sqrt{\frac{2h}{g}} \Rightarrow \frac{t_1}{t_2} = \sqrt{\frac{h_1}{h_2}}$

48. (b) Time of ascent = Time of descent = 5 *sec*

49. (b) Time of ascent $= \frac{u}{g} = 6\ \sec \Rightarrow u = 60\ m/s$

Distance in first second $h_{\text{first}} = 60 - \frac{g}{2}(2 \times 1 - 1) = 55\ m$

Distance in seventh second will be equal to the distance in first second of vertical downward motion

$h_{\text{seventh}} = \frac{g}{2}(2 \times 1 - 1) = 5\ m \Rightarrow h_{\text{first}} / h_{\text{seventh}} = 11 : 1$

50. (b) Let particle thrown with velocity u and its maximum height is H then $H = \frac{u^2}{2g}$

When particle is at a height $H/2$, then its speed is 10 *m/s*

From equation $v^2 = u^2 - 2gh$

$(10)^2 = u^2 - 2g\left(\frac{H}{2}\right) = u^2 - 2g\frac{u^2}{4g} \Rightarrow u^2 = 200$

Maximum height $\Rightarrow H = \frac{u^2}{2g} = \frac{200}{2 \times 10} = 10\ m$

51. (c) Mass does not affect on maximum height.

$H = \dfrac{u^2}{2g} \Rightarrow H \propto u^2$, So if velocity is doubled then height will become four times. *i.e.* $H = 20 \times 4 = 80m$

52. (a) When the stone is released from the balloon. Its height $h = \dfrac{1}{2}at^2 = \dfrac{1}{2} \times 1.25 \times (8)^2 = 40\ m$ and velocity

$v = at = 1.25 \times 8 = 10\ m/s$

Time taken by the stone to reach the ground

$$t = \frac{v}{g}\left[1 + \sqrt{1 + \frac{2gh}{v^2}}\right] = \frac{10}{10}\left[1 + \sqrt{1 + \frac{2 \times 10 \times 40}{(10)^2}}\right] = 4\ sec$$

53. (d) At highest point $v = 0$ and $H_{\max} = \dfrac{u^2}{2g}$

54. (d) $u = \sqrt{2gh} = \sqrt{2 \times 10 \times 20} = 20\ m/s$

and $T = \dfrac{2u}{g} = \dfrac{2 \times 20}{10} = 4\ sec$

55. (d) If t_1 and t_2 are the time, when body is at the same height then, $h = \dfrac{1}{2}g\,t_1 t_2 = \dfrac{1}{2} \times g \times 2 \times 10 = 10\ g$

56. (c) Speed of the object at reaching the ground $v = \sqrt{2gh}$

If heights are equal then velocity will also be equal.

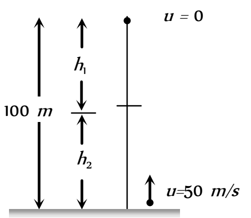
57. (b) $S_{3^{rd}} = 10 + \dfrac{10}{2}(2 \times 3 - 1) = 35\ m$

$S_{2^{nd}} = 10 + \dfrac{10}{2}(2 \times 2 - 1) = 25m \Rightarrow \dfrac{S_{3^{rd}}}{S_{2^{nd}}} = \dfrac{7}{5}$

58. (c) $v^2 = u^2 + 2gh \Rightarrow v = \sqrt{u^2 + 2gh}$

so for both the cases velocity will be equal.

59. (b) $h_1 = \dfrac{1}{2}gt^2$, $h_2 = 50t - \dfrac{1}{2}gt^2$



Given $h_1 + h_2 = 100m \Rightarrow 50t = 100 \Rightarrow t = 2$ sec

60. (b) $H_{\max} = \dfrac{u^2}{2g} = \dfrac{19.6 \times 19.6}{2 \times 9.8} = 19.6\ m$

61. (c) Maximum height of ball = 5 m

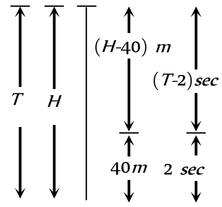
So velocity of projection $\Rightarrow u = \sqrt{2gh} = 10\ m/s$

Time interval between two balls (time of ascent)

$= \dfrac{u}{g} = 1\ sec = \dfrac{1}{60}min$.

So number of ball thrown per min. = 60

62. (b) Let height of minaret is H and body take time T to fall from top to bottom.



$H = \dfrac{1}{2}gT^2$ ...(i)

In last 2 *sec.* body travels distance of 40*meter* so in $(T-2)$ *sec* distance travelled $= (H - 40)\ m$.

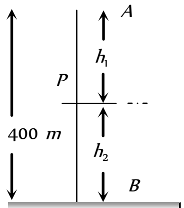
$(H - 40) = \dfrac{1}{2}g(T-2)^2$ ...(ii)

By solving (i) and (ii) $T = 3$ *sec* and $H = 45\ m$.

63. (c) $S_n \propto (2n - 1)$. In equal time interval of 2 *seconds*

Ratio of distance $= 1 : 3 : 5$

64. (c) Let both balls meet at point *P* after time *t*.



The distance travelled by ball *A*, $h_1 = \dfrac{1}{2}gt^2$

The distance travelled by ball *B*, $h_2 = ut - \dfrac{1}{2}gt^2$

$h_1 + h_2 = 400\ m \Rightarrow ut = 400,\ t = 400/50 = 8\ sec$

$\therefore\ h_1 = 320\ m$ and $h_2 = 80\ m$

65. (a) $t = \sqrt{\dfrac{2h}{g}} \Rightarrow \dfrac{t_1}{t_2} = \sqrt{\dfrac{h_1}{h_2}} = \sqrt{\dfrac{1}{2}} = \dfrac{1}{\sqrt{2}}$

66. (a) $H_{\max} = \dfrac{u^2}{2g} \Rightarrow H_{\max} \propto \dfrac{1}{g}$

On planet *B* value of *g* is $1/9$ times to that of *A*. So value of $H_{\max}$ will become 9 times *i.e.* $2 \times 9 = 18\ metre$

67. (b) $h_n = \dfrac{g}{2}(2n - 1) \Rightarrow h_{5^{th}} = \dfrac{10}{2}(2 \times 5 - 1) = 45\ m$.

68. (a) $h_{\max} = \dfrac{u^2}{2g} = \dfrac{(15)^2}{2 \times 10} = 11.25\ m$.

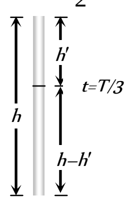
69. (b) For stone to be dropped from rising balloon of velocity 29 *m/s*.

$u = -29\ m/s$, $t = 10$ *sec.*

$\therefore h = -29 \times 10 + \dfrac{1}{2} \times 9.8 \times 100$

$= -290 + 490 = 200$ *m.*

70. (c) $\because\ h = ut + \dfrac{1}{2}gt^2 \Rightarrow h = \dfrac{1}{2}gT^2$



After $\frac{T}{3}$ seconds, the position of ball,

$$h' = 0 + \frac{1}{2}g\left(\frac{T}{3}\right)^2 = \frac{1}{2}\times\frac{g}{9}\times T^2$$

$$h' = \frac{1}{2}\times\frac{g}{9}\times T^2 = \frac{h}{9}m \text{ from top}$$

$\therefore$ Position of ball from ground $= h - \frac{h}{9} = \frac{8\,h}{9}m$.

71. (c) Since acceleration due to gravity is independent of mass, hence time is also independent of mass (or density) of object.

72. (c) When packet is released from the balloon, it acquires the velocity of balloon of value 12 *m/s*. Hence velocity of packet after 2 *sec*, will be

$v = u + gt = 12 - 9.8\times 2 = -7.6$ *m/s*.

73. (b) The distance traveled in last second.

$$S_{\text{Last}} = u + \frac{g}{2}(2t-1) = \frac{1}{2}\times 9.8(2t-1) = 4.9(2t-1)$$

and distance traveled in first three second,

$$S_{\text{Three}} = 0 + \frac{1}{2}\times 9.8\times 9 = 44.1\ m$$

According to problem $S_{\text{Last}} = S_{\text{Three}}$

$\Rightarrow 4.9(2t-1) = 44.1 \Rightarrow 2t - 1 = 9 \Rightarrow t = 5$ *sec*.

74. (c) Net acceleration of a body when thrown upward

= acceleration of body – acceleration due to gravity

$= a - g$

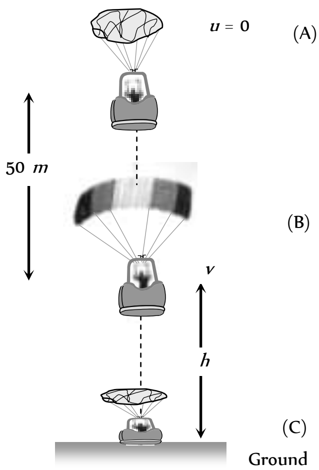
75. (b) The given condition is possible only when body is at its highest position after 5 seconds

It means time of ascent = 5 *sec*

and time of flight $T = \frac{2u}{g} = 10 \Rightarrow u = 50\ m/s$

76. (b) $H_{\max} \propto u^2$, It body projected with double velocity then maximum height will become four times *i.e.* 200 *m*.

77. (a) After bailing out from point *A* parachutist falls freely under gravity. The velocity acquired by it will '*v*'



From $v^2 = u^2 + 2as = 0 + 2\times 9.8\times 50 = 980$

[As $u = 0$, $a = 9.8 m/s^2$, $s = 50\ m$]

At point *B*, parachute opens and it moves with retardation of $2\,m/s^2$ and reach at ground (Point *C*) with velocity of $3m/s$

For the part '*BC*' by applying the equation $v^2 = u^2 + 2as$

$v = 3m/s$, $u = \sqrt{980}\,m/s$, $a = -2m/s^2$, $s = h$

$\Rightarrow (3)^2 = (\sqrt{980})^2 + 2\times(-2)\times h \Rightarrow 9 = 980 - 4h$

$$\Rightarrow h = \frac{980 - 9}{4} = \frac{971}{4} = 242.7 \cong 243\ m.$$

So, the total height by which parachutist bail out $= 50 + 243 = 293\ m$.

78. (a)

79. (c)

80. (a) $H_{\max} \propto u^2 \ \therefore u \propto \sqrt{H_{\max}}$

i.e. to triple the maximum height, ball should be thrown with velocity $\sqrt{3}\,u$.

81. (a)

Critical Thinking Questions

1. (a) If t_1 and $2t_2$ are the time taken by particle to cover first and second half distance respectively.

$$t_1 = \frac{x/2}{3} = \frac{x}{6} \quad \text{...(i)}$$

$x_1 = 4.5\ t_2$ and $x_2 = 7.5\ t_2$

So, $x_1 + x_2 = \frac{x}{2} \Rightarrow 4.5t_2 + 7.5t_2 = \frac{x}{2}$

$$t_2 = \frac{x}{24} \quad \text{...(ii)}$$

Total time $t = t_1 + 2t_2 = \frac{x}{6} + \frac{x}{12} = \frac{x}{4}$

So, average speed $= 4\ m/sec$.

2. (c) $\frac{dv}{dt} = bt \Rightarrow dv = bt\,dt \Rightarrow v = \frac{bt^2}{2} + K_1$

At $t = 0, v = v_0 \Rightarrow K_1 = v_0$

We get $v = \frac{1}{2}bt^2 + v_0$

Again $\frac{dx}{dt} = \frac{1}{2}bt^2 + v_0 \Rightarrow x = \frac{1}{2}\frac{bt^2}{3} + v_0 t + K_2$

At $t = 0, x = 0 \Rightarrow K_2 = 0$

$\therefore x = \frac{1}{6}bt^3 + v_0 t$

3. (a,b,d) $\frac{dv}{dt} = 6 - 3v \Rightarrow \frac{dv}{6-3v} = dt$

Integrating both sides, $\int \frac{dv}{6-3v} = \int dt$

$\Rightarrow \frac{\log_e(6-3v)}{-3} = t + K_1$

$\Rightarrow \log_e(6-3v) = -3t + K_2$ ...(i)

At $t=0, v=0$ $\therefore$ $\log_e 6 = K_2$

Substituting the value of K_2 in equation (i)

$\log_e(6-3v) = -3t + \log_e 6$

$\Rightarrow \log_e\left(\frac{6-3v}{6}\right) = -3t \Rightarrow e^{-3t} = \frac{6-3v}{6}$

$\Rightarrow 6-3v = 6e^{-3t} \Rightarrow 3v = 6(1-e^{-3t})$

$\Rightarrow v = 2(1-e^{-3t})$

$\therefore$ $v_{\text{trminal}} = 2\ m/s$ (When $t=\infty$)

Acceleration $a = \frac{dv}{dt} = \frac{d}{dt}\left[2\left(1-e^{-3t}\right)\right] = 6e^{-3t}$

Initial acceleration = $6\ m/s^2$.

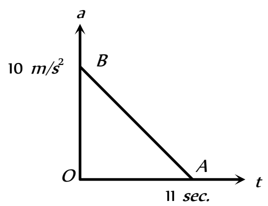
4. (a,d) The body starts from rest at $x=0$ and then again comes to rest at $x=1$. It means initially acceleration is positive and then negative.

So we can conclude that α can not remains positive for all t in the interval $0 \le t \le 1$ *i.e.* α must change sign during the motion.

5. (b) The area under acceleration time graph gives change in velocity. As acceleration is zero at the end of 11 *sec*

i.e. $v_{\max}$ = Area of ΔOAB

$= \frac{1}{2} \times 11 \times 10 = 55\ m/s$



6. (d) Let the car accelerate at rate α for time t_1 then maximum velocity attained, $v = 0 + \alpha t_1 = \alpha t_1$

Now, the car decelerates at a rate β for time $(t-t_1)$ and finally comes to rest. Then,

$0 = v - \beta(t-t_1) \Rightarrow 0 = \alpha t_1 - \beta t + \beta t_1$

$\Rightarrow t_1 = \frac{\beta}{\alpha+\beta} t$

$\therefore$ $v = \frac{\alpha\beta}{\alpha+\beta} t$

7. (c) If a stone is dropped from height h

then $h = \frac{1}{2} g t^2$ ...(i)

If a stone is thrown upward with velocity u then

$h = -u\ t_1 + \frac{1}{2} g\ t_1^2$ ...(ii)

If a stone is thrown downward with velocity u then

$h = ut_2 + \frac{1}{2} g t_2^2$ ...(iii)

From (i) (ii) and (iii) we get

$-ut_1 + \frac{1}{2} g\, t_1^2 = \frac{1}{2} g\, t^2$ ...(iv)

$ut_2 + \frac{1}{2} g\, t_2^2 = \frac{1}{2} g\, t^2$ ...(v)

Dividing (iv) and (v) we get

$\therefore$ $\frac{-ut_1}{ut_2} = \frac{\frac{1}{2} g(t^2 - t_1^2)}{\frac{1}{2} g(t^2 - t_2^2)}$

or $-\frac{t_1}{t_2} = \frac{t^2 - t_1^2}{t^2 - t_2^2}$

By solving $t = \sqrt{t_1 t_2}$

8. (c) Since direction of v is opposite to the direction of g and h so from equation of motion

$h = -vt + \frac{1}{2} gt^2$

$\Rightarrow gt^2 - 2vt - 2h = 0$

$\Rightarrow t = \frac{2v \pm \sqrt{4v^2 + 8gh}}{2g}$

$\Rightarrow t = \frac{v}{g}\left[1 + \sqrt{1 + \frac{2gh}{v^2}}\right]$

9. (c) $h = ut + \frac{1}{2} gt^2 \Rightarrow 1 = 0 \times t_1 + \frac{1}{2} gt_1^2 \Rightarrow t_1 = \sqrt{2/g}$

Velocity after travelling 1*m* distance

$v^2 = u^2 + 2gh \Rightarrow v^2 = (0)^2 + 2g \times 1 \Rightarrow v = \sqrt{2g}$

For second 1 meter distance

$1 = \sqrt{2g} \times t_2 + \frac{1}{2} gt_2^2 \Rightarrow gt_2^2 + 2\sqrt{2g}t_2 - 2 = 0$

$t_2 = \frac{-2\sqrt{2g} \pm \sqrt{8g + 8g}}{2g} = \frac{-\sqrt{2} \pm 2}{\sqrt{g}}$

Taking +ve sign $t_2 = (2-\sqrt{2})/\sqrt{g}$

$\therefore$ $\frac{t_1}{t_2} = \frac{\sqrt{2/g}}{(2-\sqrt{2})/\sqrt{g}} = \frac{1}{\sqrt{2}-1}$ and so on.

10. (d) Interval of ball throw = 2 *sec.*

If we want that minimum three (more than two) ball remain in air then time of flight of first ball must be greater than 4 *sec.*

$T > 4\ sec$

$\frac{2u}{g} > 4\ sec \Rightarrow u > 19.6\ m/s$

for u =19.6. First ball will just strike the ground(in sky)

Second ball will be at highest point (in sky)

Third ball will be at point of projection or at ground (not in sky)

11. (a) The distance covered by the ball during the last t seconds of its upward motion = Distance covered by it in first t seconds of its downward motion

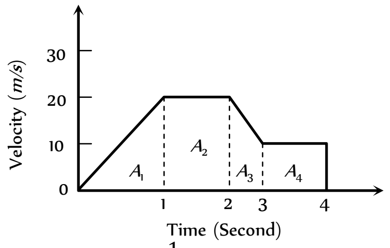
From $h = ut + \frac{1}{2} g t^2$

$h = \frac{1}{2} g\, t^2$ [As u = 0 for it downward motion]

12. (c)

Graphical Questions

1. (b) Distance = Area under $v-t$ graph $= A_1 + A_2 + A_3 + A_4$



$= \frac{1}{2} \times 1 \times 20 + (20 \times 1) + \frac{1}{2}(20 + 10) \times 1 + (10 \times 1)$

$= 10 + 20 + 15 + 10 = 55\ m$

2. (a) The slope of displacement-time graph goes on decreasing, it means the velocity is decreasing *i.e.* It's motion is retarded and finally slope becomes zero *i.e.* particle stops.

3. (d) In the positive region the velocity decreases linearly (during rise) and in the negative region velocity increases linearly (during fall) and the direction is opposite to each other during rise and fall, hence fall is shown in the negative region.

4. (b) Region *OA* shows that graph bending toward time axis *i.e.* acceleration is negative.

Region *AB* shows that graph is parallel to time axis *i.e.* velocity is zero. Hence acceleration is zero.

Region *BC* shows that graph is bending towards displacement axis *i.e.* acceleration is positive.

Region *CD* shows that graph having constant slope *i.e.* velocity is constant. Hence acceleration is zero.

5. (d) Maximum acceleration means maximum change in velocity in minimum time interval.

In time interval $t = 30$ to $t = 40\,\text{sec}$

$a = \frac{\Delta v}{\Delta t} = \frac{80-20}{40-30} = \frac{60}{10} = 6\ cm/\text{sec}^2$

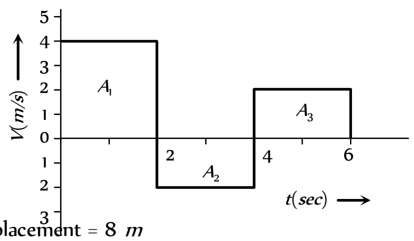
6. (c) In part *cd* displacement-time graph shows constant slope *i.e.* velocity is constant. It means no acceleration or no force is acting on the body.

7. (d) Up to time t_1 slope of the graph is constant and after t_1 slope is zero *i.e.* the body travel with constant speed up to time t_1 and then stops.

8. (c) Area of trapezium $= \frac{1}{2} \times 3.6 \times (12 + 8) = 36.0\ m$

9. (a) Displacement = Summation of all the area with sign

$= (A_1) + (-A_2) + (A_3) = (2 \times 4) + (-2 \times 2) + (2 \times 2)$



$\therefore$ Displacement = 8 *m*

Distance =Summation of all the areas without sign

$= |A_1| + |-A_2| + |A_3| = |8| + |-4| + |4| = 8 + 4 + 4$

$\therefore$ Distance = 16 *m*.

10. (b) Between time interval 20 *sec* to 40 *sec*, there is non-zero acceleration and retardation. Hence distance travelled during this interval

= Area between time interval 20 *sec* to 40 *sec*

$= \frac{1}{2} \times 20 \times 3 + 20 \times 1 = 30 + 20 = 50\ m.$

11. (c)

12. (b) $\frac{(S)_{(last\ 2s)}}{(S)_{7s}} = \frac{\frac{1}{2} \times 2 \times 10}{\frac{1}{2} \times 2 \times 10 + 2 \times 10 + \frac{1}{2} \times 2 \times 10} = \frac{1}{4}$

13. (a) Distance = Area covered between graph and displacement axis $= \frac{1}{2}(30 + 10)10 = 200\ meter$.

14. (d) Because acceleration due to gravity is constant so the slope of line will be constant *i.e.* velocity time curve for a body projected vertically upwards is straight line.

15. (d) Slope of displacement time graph is negative only at point *E*.

16. (c) $v^2 = u^2 + 2aS$, If $u = 0$ then $v^2 \propto S$

i.e. graph should be parabola symmetric to displacement axis.

17. (a) This graph shows uniform motion because line having a constant slope.

18. (a) For the given condition initial height $h = d$ and velocity of the ball is zero. When the ball moves downward its velocity increases and it will be maximum when the ball hits the ground & just after the collision it becomes half and in opposite direction. As the ball moves upward its velocity again decreases and becomes zero at height $d/2$. This explanation match with graph (*A*).

19. (a) We know that the velocity of body is given by the slope of displacement – time graph. So it is clear that initially slope of the graph is positive and after some time it becomes zero (corresponding to the peak of graph) and then it will becomes negative.

20. (b) Maximum acceleration will be represented by *CD* part of the graph

Acceleration $= \frac{dv}{dt} = \frac{(60-20)}{0.25} = 160\ km/h^2$

21. (d)

22. (c) For upward motion

Effective acceleration $= -(g + a)$

and for downward motion

Effective acceleration $= (g - a)$

But both are constants. So the slope of speed-time graph will be constant.

23. (a) Since slope of graph remains constant for velocity-time graph.

24. (b) Other graph shows more than one velocity of the particle at single instant of time which is not practically possible.

25. (a) Slope of velocity-time graph measures acceleration. For graph (a) slope is zero. Hence $a = 0$ *i.e.* motion is uniform.

26. (c) From acceleration time graph, acceleration is constant for first part of motion so, for this part velocity of body increases uniformly with time and as $a = 0$ then the velocity becomes constant. Then again increased because of constant acceleration.

27. (a) Given line have positive intercept but negative slope. So its equation can be written as

$$v = -mx + v_0 \quad .....(i) \qquad [\text{where } m = \tan\theta = \frac{v_0}{x_0}]$$

By differentiating with respect to time we get

$$\frac{dv}{dt} = -m\frac{dx}{dt} = -mv$$

Now substituting the value of v from eq. (i) we get

$$\frac{dv}{dt} = -m[-mx + v_0] = m^2x - mv_0 \quad \therefore \quad a = m^2x - mv_0$$

i.e. the graph between a and x should have positive slope but negative intercept on a-axis. So graph (a) is correct.

28. (c) From given $a - t$ graph it is clear that acceleration is increasing at constant rate

$$\therefore \frac{da}{dt} = k \text{ (constant)} \Rightarrow a = kt \text{ (by integration)}$$

$$\Rightarrow \frac{dv}{dt} = kt \Rightarrow dv = ktdt$$

$$\Rightarrow \int dv = k\int tdt \Rightarrow v = \frac{kt^2}{2}$$

i.e. v is dependent on time parabolically and parabola is symmetric about v-axis.

and suddenly acceleration becomes zero. *i.e.* velocity becomes constant.

Hence (c) is most probable graph.

29. (c) In first instant you will apply $\upsilon = \tan\theta$ and say, $\upsilon = \tan 30^o = \frac{1}{\sqrt{3}}$ *m/s*.

But it is wrong because formula $\upsilon = \tan\theta$ is valid when angle is measured with time axis.

Here angle is taken from displacement axis. So angle from time axis $= 90^o - 30^o = 60^o$

Now $\upsilon = \tan 60^o = \sqrt{3}$

30. (a) Since total displacement is zero, hence average velocity is also zero.

Assertion and Reason

1. (a) When body going vertically upwards, reaches at the highest point, then it is momentarily at rest and it then reverses its direction. At the highest point of motion, its velocity is zero but its acceleration is equal to acceleration due to gravity.

2. (a) As motion is governed by force of gravity and acceleration due to gravity (g) is independent of mass of object.

3. (a) As distance being a scalar quantity is always positive but displacement being a vector may be positive, zero and negative depending on situation.

4. (a) As displacement is either smaller or equal to distance but never be greater than distance.

5. (a) Since velocity is a vector quantity, hence as its direction changes keeping magnitude constant, velocity is said to be changed. But for constant speed in equal time interval distance travelled should be equal.

6. (d) Speed can never be negative because it is a scalar quantity.

7. (c) Negative slope of position time graph represents that the body is moving towards the negative direction and if the slope of the graph decrease with time then it represents the decrease in speed *i.e.* retardation in motion.

8. (b) A body having positive acceleration can be associated with slowing down, as time rate of change of velocity decreases, but velocity increases with time, from graph it is clear that slope with time axis decreases, but velocity increases with time.

9. (b) A body having negative acceleration can be associated with a speeding up, if object moves along negative X-direction with increasing speed.

10. (e) It is not necessary that an object moving under uniform acceleration have straight path. *eg.* projectile motion.

11. (a) Motion of rocket is based on action reaction phenomena and is governed by rate of fuel burning causing the change in momentum of ejected gas.

12. (a) When a body moves on a straight path in one direction value of distance & displacement remains same so that average speed equals the average velocity for a given time interval.

13. (a) Position-time graph for a stationary object is a straight line parallel to time axis showing that no change in position with time.

14. (a) Since slope of displacement-time graph measures velocity of an object.

15. (e) For distance-time graph, a straight line inclined to time axis measures uniform speed for which acceleration is zero and for uniformly accelerated motion $S \propto t^2$.

16. (e) As per definition, acceleration is the rate of change of velocity, *i.e.* $\vec{a} = \frac{d\vec{v}}{dt}$.

If velocity is constant $d\vec{v}/dt = 0, \quad \therefore \vec{a} = 0$.

Therefore, if a body has constant velocity it cannot have non zero acceleration.

17. (a) A body has no relative motion with respect to itself. Hence if a frame of reference of the body is fixed, then the body will be always at relative rest in this frame of reference.

18. (c) The displacement is the shortest distance between initial and final position. When final position of a body coincides with its initial position, displacement is zero, but the distance travelled is not zero.

19. (d) Equation of motion can be applied if the acceleration is in opposite direction to that of velocity and uniform motion mean the acceleration is zero.

20. (e) As velocity is a vector quantity, its value changes with change in direction. Therefore when a bus takes a turn from north to east its velocity will also change.

21. (b) When two bodies are moving in opposite direction, relative velocity between them is equal to sum of the velocity of bodies. But if the bodies are moving in same direction their relative velocity is equal to difference in velocity of the bodies.

22. (d) The displacement of a body moving in straight line is given by, $s = ut + \frac{1}{2}at^2$. This is a equation of a parabola, not straight line. Therefore the displacement-time graph is a parabola. The displacement time graph will be straight line, if acceleration of body is zero or body moving with uniform velocity.

23. (c) In uniform motion the object moves with uniform velocity, the magnitude of its velocity at different instant *i.e.* at $t = 0$, $t = 1 sec$, $t = 2 sec$,.... will always be constant. Thus velocity-time graph for an object in uniform motion along a straight path is a straight line parallel to time axis.

24. (e) The uniform motion of a body means that the body is moving with constant velocity, but if the direction of motion is changing (such as in uniform circular motion), its velocity changes and thus acceleration is produced in uniform motion.

25. (e) When a body falling freely, only gravitational force acts on it in vertically downward direction. Due to this downward acceleration the velocity of a body increases and will be maximum when the body touches the ground.

26. (a) According to definition,displacement = velocity × time Since displacement is a vector quantity so its value is equal to the vector sum of the area under velocity-time graph.

27. (e) If the position-time graph of a body moving uniformly in a straight line parallel to position axis, it means that the position of body is changing at constant time. The statement is abrupt and shows that the velocity of body is infinite.

28. (b) Average speed = Total distance /Total time

Time average speed $= \dfrac{v_1 + v_2 + v_3 + ...}{n}$

29. (c) An object is said to be in uniform motion if it undergoes equal displacement in equal intervals of time.

$$\therefore v_{av} = \frac{s_1 + s_2 + s_3 + ...}{t_1 + t_2 + t_3 + ...} = \frac{s + s + s + ...}{t + t + t + ...} = \frac{ns}{nt} = \frac{s}{t}$$

and $v_{ins} = \dfrac{s}{t}$.

Thus, in uniform motion average and instantaneous velocities have same value and body moves with constant velocity.

30. (e) Speedometer measures instantaneous speed of automobile.

Motion In One Dimension

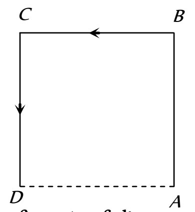
SET Self Evaluation Test - 2

1. A car travels a distance S on a straight road in two hours and then returns to the starting point in the next three hours. Its average velocity is

(a) $S/5$ (b) $2S/5$

(c) $S/2 + S/3$ (d) None of the above

2. A particle moves along the sides AB, BC, CD of a square of side 25 m with a velocity of 15 ms^{-1}. Its average velocity is



(a) 15 ms^{-1}

(b) 10 ms^{-1}

(c) 7.5 ms^{-1}

(d) 5 ms^{-1}

3. A body has speed V, $2V$ and $3V$ in first 1/3 of distance S, seconds 1/3 of S and third 1/3 of S respectively. Its average speed will be

(a) V (b) $2V$

(c) $\frac{18}{11}V$ (d) $\frac{11}{18}V$

4. If the body covers one-third distance at speed v_1, next one third at speed v_2 and last one third at speed v_3, then average speed will be

(a) $\frac{v_1 v_2 + v_2 v_3 + v_3 v_1}{v_1 + v_2 + v_3}$ (b) $\frac{v_1 + v_2 + v_3}{3}$

(c) $\frac{v_1 v_2 v_3}{v_1 v_2 + v_2 v_3 + v_3 v_1}$ (d) $\frac{3 v_1 v_2 v_3}{v_1 v_2 + v_2 v_3 + v_3 v_1}$

5. The displacement of the particle varies with time according to the relation $x = \frac{k}{b}[1 - e^{-bt}]$. Then the velocity of the particle is

(a) $k(e^{-bt})$ (b) $\frac{k}{b^2 e^{-bt}}$

(c) $k b e^{-bt}$ (d) None of these

6. The acceleration of a particle starting from rest, varies with time according to the relation $A = -a\omega \sin\omega t$. The displacement of this particle at a time t will be

(a) $-\frac{1}{2}(a\omega^2 \sin\omega t)t^2$ (b) $a\omega \sin\omega t$

(c) $a\omega \cos\omega t$ (d) $a \sin\omega t$

7. If the velocity of a particle is $(10 + 2t)$ m/s, then the average acceleration of the particle between $2s$ and $5s$ is

(a) 2 m/s^2 (b) 4 m/s^2

(c) 12 m/s^2 (d) 14 m/s^2

8. A bullet moving with a velocity of 200 cm/s penetrates a wooden block and comes to rest after traversing 4 cm inside it. What velocity is needed for travelling distance of 9 cm in same block

(a) 100 cm/s (b) 136.2 cm/s

(c) 300 cm/s (d) 250 cm/s

9. A thief is running away on a straight road in jeep moving with a speed of 9 ms^{-1}. A police man chases him on a motor cycle moving at a speed of 10 ms^{-1}. If the instantaneous separation of the jeep from the motorcycle is 100 m, how long will it take for the police to catch the thief

(a) 1 s (b) 19 s

(c) 90 s (d) 100 s

10. A car A is travelling on a straight level road with a uniform speed of 60 km/h. It is followed by another car B which is moving with a speed of 70 km/h. When the distance between them is 2.5 km, the car B is given a deceleration of 20 km/h^2. After how much time will B catch up with A

(a) 1 hr (b) 1/2 hr

(c) 1/4 hr (d) 1/8 hr

11. The speed of a body moving with uniform acceleration is u. This speed is doubled while covering a distance S. When it covers an additional distance S, its speed would become

(a) $\sqrt{3}\,u$ (b) $\sqrt{5}\,u$

(c) $\sqrt{11}\,u$ (d) $\sqrt{7}\,u$

12. Two trains one of length 100 m and another of length 125 m, are moving in mutually opposite directions along parallel lines, meet each other, each with speed 10 m/s. If their acceleration are 0.3 m/s^2 and 0.2 m/s^2 respectively, then the time they take to pass each other will be

(a) 5 s (b) 10 s

(c) 15 s (d) 20 s

13. A body starts from rest with uniform acceleration. If its velocity after n second is υ, then its displacement in the last two seconds is

(a) $\frac{2\upsilon(n+1)}{n}$ (b) $\frac{\upsilon(n+1)}{n}$

(c) $\frac{\upsilon(n-1)}{n}$ (d) $\frac{2\upsilon(n-1)}{n}$

14. A point starts moving in a straight line with a certain acceleration. At a time t after beginning of motion the acceleration suddenly becomes retardation of the same value. The time in which the point returns to the initial point is

(a) $\sqrt{2t}$

(b) $(2 + \sqrt{2})\,t$

(c) $\frac{t}{\sqrt{2}}$

(d) Cannot be predicted unless acceleration is given

15. A particle is moving in a straight line and passes through a point O with a velocity of $6\ ms^{-1}$. The particle moves with a constant retardation of $2\ ms^{-2}$ for 4 s and there after moves with constant velocity. How long after leaving O does the particle return to O

(a) $3s$ (b) $8s$

(c) Never (d) $4s$

16. A bird flies for 4 s with a velocity of $|t-2|\ m/s$ in a straight line, where t is time in seconds. It covers a distance of

(a) $2\ m$ (b) $4\ m$

(c) $6\ m$ (d) $8\ m$

17. A particle is projected with velocity υ_0 along $x-axis$. The deceleration on the particle is proportional to the square of the distance from the origin i.e., $a = \alpha x^2$. The distance at which the particle stops is

(a) $\sqrt{\frac{3\upsilon_0}{2\alpha}}$ (b) $\left(\frac{3v_o}{2\alpha}\right)^{\frac{1}{3}}$

(c) $\sqrt{\frac{3\upsilon_0^2}{2\alpha}}$ (d) $\left(\frac{3\upsilon_0^2}{2\alpha}\right)^{\frac{1}{3}}$

18. A body is projected vertically up with a velocity v and after some time it returns to the point from which it was projected. The average velocity and average speed of the body for the total time of flight are

(a) $\vec{v}/2$ and $v/2$ (b) 0 and $v/2$

(c) 0 and 0 (d) $\vec{v}/2$ and 0

19. A stone is dropped from a height h. Simultaneously, another stone is thrown up from the ground which reaches a height 4 h. The two stones cross each other after time

(a) $\sqrt{\frac{h}{8g}}$ (b) $\sqrt{8gh}$

(c) $\sqrt{2gh}$ (d) $\sqrt{\frac{h}{2g}}$

20. Four marbles are dropped from the top of a tower one after the other with an interval of one second. The first one reaches the ground after 4 *seconds*. When the first one reaches the ground the distances between the first and second, the second and third and the third and forth will be respectively

(a) 35, 25 and 15 m (b) 30, 20 and 10 m

(c) 20, 10 and 5 m (d) 40, 30 and 20 m

21. A balloon rises from rest with a constant acceleration $g/8$. A stone is released from it when it has risen to height h. The time taken by the stone to reach the ground is

(a) $4\sqrt{\frac{h}{g}}$ (b) $2\sqrt{\frac{h}{g}}$

(c) $\sqrt{\frac{2h}{g}}$ (d) $\sqrt{\frac{g}{h}}$

22. Two bodies are thrown simultaneously from a tower with same initial velocity v_0 : one vertically upwards, the other vertically downwards. The distance between the two bodies after time t is

(a) $2v_0t+\frac{1}{2}gt^2$ (b) $2v_0t$

(c) $v_0t+\frac{1}{2}gt^2$ (d) v_0t

23. A body falls freely from the top of a tower. It covers 36% of the total height in the last second before striking the ground level. The height of the tower is

(a) 50 m (b) 75 m

(c) 100 m (d) 125 m

24. A particle is projected upwards. The times corresponding to height h while ascending and while descending are t_1 and t_2 respectively. The velocity of projection will be

(a) gt_1 (b) gt_2

(c) $g(t_1+t_2)$ (d) $\frac{g(t_1+t_2)}{2}$

25. A projectile is fired vertically upwards with an initial velocity u. After an interval of T seconds a second projectile is fired vertically upwards, also with initial velocity u.

(a) They meet at time $t=\frac{u}{g}$ and at a height $\frac{u^2}{2g}+\frac{gT^2}{8}$

(b) They meet at time $t=\frac{u}{g}+\frac{T}{2}$ and at a height $\frac{u^2}{2g}+\frac{gT^2}{8}$

(c) They meet at time $t=\frac{u}{g}+\frac{T}{2}$ and at a height $\frac{u^2}{2g}-\frac{gT^2}{8}$

(d) They never meet

Answers and Solutions

(SET -2)

1. (d) Average velocity $=\frac{\text{Total displacement}}{\text{Time}}=\frac{0}{2+3}=0$

2. (d) Average velocity $=\frac{\text{Total displacement}}{\text{Time taken}}=\frac{25}{75/15}=5\,m/s$

3. (c) $v_{av}=\frac{\text{Total distance}}{\text{Time taken}}=\frac{x}{\frac{x/3}{v}+\frac{x/3}{2v}+\frac{x/3}{3v}}=\frac{18}{11}v$

4. (d) $v_{av}=\frac{x}{\frac{x/3}{v_1}+\frac{x/3}{v_2}+\frac{x/3}{v_3}}=\frac{3v_1v_2v_3}{v_1v_2+v_2v_3+v_1v_3}$

5. (a) $v=\frac{dx}{dt}=\frac{d}{dt}\left[\frac{k}{b}\left(1-e^{-bt}\right)\right]=\frac{k}{b}\left[0-(-b)e^{-bt}\right]=ke^{-bt}$.

6. (d) Velocity $v=\int A\,dt=\int(-a\omega^2\sin\omega t)\,dt=a\omega\cos\omega t$

Displacement $x=\int v\,dt=\int a\omega\cos\omega t\,dt=a\sin\omega t$

7. (d) Average acceleration $=\frac{\text{Change in velocity}}{\text{Time taken}}=\frac{v_2-v_1}{t_2-t_1}$

$=\frac{\left[10+2(5)^2\right]-\left[10+2(2)^2\right]}{3}=\frac{60-18}{3}=14\ m/s^2$.

8. (c) As $v^2=u^2-2as\Rightarrow u^2=2as\quad(\because v=0)$

$\Rightarrow u^2\propto s\ \Rightarrow\ \frac{u_2}{u_1}=\left(\frac{s_2}{s_1}\right)^{1/2}$

$\Rightarrow u_2=\left(\frac{9}{4}\right)^{1/2}u_1=\frac{3}{2}u_1=300cm/s$.

9. (d) The relative velocity of policeman *w.r.t.* thief $=10-9=1m/s$.

$\therefore$ Time taken by police to catch the thief $=\frac{100}{1}$ =100 *sec*

10. (b) Let car *B* catches, car A after '*t*' sec, then

$60t+2.5=70t-\frac{1}{2}\times 20\times t^2$

$\Rightarrow 10t^2-10t+2.5=0\Rightarrow t^2-t+0.25=0$

$\therefore\ t=\frac{1\pm\sqrt{1-4\times(0.25)}}{2}=\frac{1}{2}hr$

11. (d) As $v^2=u^2+2as\ \Rightarrow\ (2u)^2=u^2+2as\Rightarrow 2as=3u^2$

Now, after covering an additional distance *s*, if velocity becomes *v*, then,

$v^2=u^2+2a(2s)=u^2+4as=u^2+6u^2=7u^2$

$\therefore\ v=\sqrt{7}u$.

12. (b) Relative velocity of one train *w.r.t.* other

=10+10=20 *m/s*.

Relative acceleration =0.3+0.2=0.5 *m/s*

If trains cross each other then from $s=ut+\frac{1}{2}at^2$

$As,\ s=s_1+s_2=100+125=225$

$\Rightarrow 225=20t+\frac{1}{2}\times 0.5\times t^2\ \Rightarrow\ 0.5t^2+40t-450=0$

$\Rightarrow t=-\frac{40\pm\sqrt{1600+4.(005)\times 450}}{1}=-40\pm 50$

$\therefore\ t=10\sec$ (Taking +ve value).

13. (d) $\because v=0+na\Rightarrow a=v/n$

Now, distance travelled in *n sec.* $\Rightarrow S_n=\frac{1}{2}an^2$ and distance travelled in $(n-2)\sec\Rightarrow S_{n-2}=\frac{1}{2}a(n-2)^2$

$\therefore$ Distance travelled in last two seconds,

$=S_n-S_{n-2}=\frac{1}{2}an^2-\frac{1}{2}a(n-2)^2$

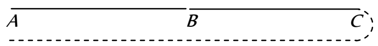
$=\frac{a}{2}\left[n^2-(n-2)^2\right]=\frac{a}{2}[n+(n-2)][n-(n-2)]$

$=a(2n-2)\ =\frac{v}{n}(2n-2)\ =\frac{2v(n-1)}{n}$

14. (b) In this problem point starts moving with uniform acceleration *a* and after time *t* (Position *B*) the direction of acceleration get reversed *i.e.* the retardation of same value works on the point. Due to this velocity of points goes on decreasing and at position *C* its velocity becomes zero. Now the direction of motion of point reversed and it moves from *C* to *A* under the effect of acceleration *a*.

We have to calculate the total time in this motion. Starting velocity at position *A* is equal to zero.

Velocity at position $B\Rightarrow v=$ at [As $u=0$]



Distance between *A* and *B*, $S_{AB}=\frac{1}{2}at^2$

As same amount of retardation works on a point and it comes to rest therefore $S_{BC}=S_{AB}\ =\frac{1}{2}a\,t^2$

$\therefore\ S_{AC}=S_{AB}+S_{BC}=a\,t^2$ and time required to cover this distance is also equal to *t*.

$\therefore$ Total time taken for motion between *A* and $C=2t$

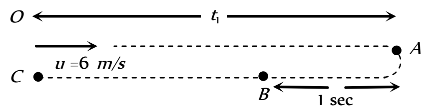
Now for the return journey from *C* to *A* $\left(S_{AC}=a\,t^2\right)$

$S_{AC}=u\,t+\frac{1}{2}at^2\ \Rightarrow\ a\,t^2=0+\frac{1}{2}a\,t_1^2\ \Rightarrow\ t_1=\sqrt{2}\ t$

Hence total time in which point returns to initial point

$T=2t+\sqrt{2}\ t\ =(2+\sqrt{2}\ )t$

15. (b) Let the particle moves toward right with velocity 6 *m/s.* Due to retardation after time t_1 its velocity becomes zero.



From $v=u-a\,t\Rightarrow\ 0=6-2\times t_1\ \Rightarrow\ t_1=3\sec$

But retardation works on it for 4 *sec.* It means after reaching point *A* direction of motion get reversed and acceleration works on the particle for next one second.

$$S_{OA} = ut_1 - \frac{1}{2}at_1^2 = 6\times 3 - \frac{1}{2}(2)(3)^2 = 18 - 9 = 9m$$

$$S_{AB} = \frac{1}{2}\times 2\times (1)^2 = 1m$$

$$\therefore\ S_{BC} = S_{0A} - S_{AB} = 9 - 1 = 8m$$

Now velocity of the particle at point B in return journey

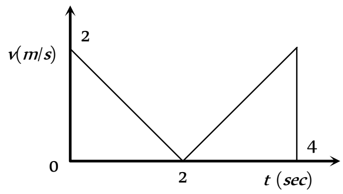
$v = 0 + 2\times 1 = 2m/s$

In return journey from B to C, particle moves with constant velocity 2 m/s to cover the distance $8m$.

Time taken $= \dfrac{\text{Distance}}{\text{Velocity}} = \dfrac{8}{2} = 4$ sec

Total time taken by particle to return at point O is $\Rightarrow T = t_{0A} + t_{AB} + t_{BC} = 3 + 1 + 4 = 8$ sec .

16. (b) The velocity time graph for given problem is shown in the figure.



Distance travelled S = Area under curve= 2+2=4m

17. (d) $a = \dfrac{dv}{dt} = \dfrac{dv}{dx}\dfrac{dx}{dt} = v\dfrac{dv}{dx} = -\alpha x^2$ (given)

$$\Rightarrow \int_{v_0}^{0} v\,dv = -\alpha\int_0^S x^2 dx \Rightarrow \left[\frac{v^2}{2}\right]_{v_0}^{0} = -\alpha\left[\frac{x^3}{3}\right]_0^S$$

$$\Rightarrow \frac{v_0^2}{2} = \frac{\alpha S^3}{3} \Rightarrow S = \left(\frac{3v_0^2}{2\alpha}\right)^{\frac{1}{3}}$$

18. (b) Average velocity =0 because net displacement of the body is zero.

Average speed $= \dfrac{\text{Total distance covered}}{\text{Time of flight}} = \dfrac{2H_{\max}}{2u/g}$

$$\Rightarrow v_{av} = \frac{2u^2/2g}{2u/g} \Rightarrow v_{av} = u/2$$

Velocity of projection = v (given)

$\therefore\ v_{\text{av}} = v/2$

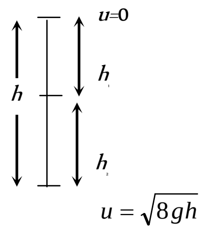
19. (a) For first stone $u = 0$ and

For second stone $\dfrac{u^2}{2g} = 4h \Rightarrow u^2 = 8gh$

$\therefore\ u = \sqrt{8gh}$

Now, $h_1 = \dfrac{1}{2}gt^2$

$h_2 = \sqrt{8gh}t - \dfrac{1}{2}gt^2$

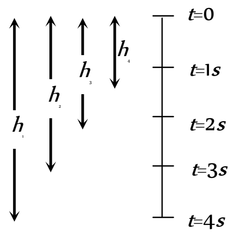


where, t =time to cross each other.

$\because h_1 + h_2 = h$

$$\Rightarrow \frac{1}{2}gt^2 + \sqrt{8gh}t - \frac{1}{2}gt^2 = h \Rightarrow t = \frac{h}{\sqrt{8gh}} = \sqrt{\frac{h}{8g}}$$

20. (a) For first marble, $h_1 = \dfrac{1}{2}g\times 16 = 8g$



For Second marble, $h_2 = \dfrac{1}{2}g\times 9 = 4.5g$

For third marble, $h_3 = \dfrac{1}{2}g\times 4 = 2g$

For fourth marble, $h_4 = \dfrac{1}{2}g\times 1 = 0.5g$

$\therefore\ h_1 - h_2 = 8g - 4.5g = 3.5g = 35m.$

$h_2 - h_3 = 4.5g - 2g = 2.5g = 25m$ and

$h_3 - h_4 = 2g - 0.5g = 1.5g = 15m$.

21. (b) The velocity of balloon at height h, $v = \sqrt{2\left(\dfrac{g}{8}\right)h}$

When the stone released from this balloon, it will go upward with velocity $v = \dfrac{\sqrt{gh}}{2}$ (Same as that of balloon). In this condition time taken by stone to reach the ground

$$t = \frac{v}{g}\left[1 + \sqrt{1 + \frac{2gh}{v^2}}\right] = \frac{\sqrt{gh}/2}{g}\left[1 + \sqrt{1 + \frac{2gh}{gh/4}}\right]$$

$$= \frac{2\sqrt{gh}}{g} = 2\sqrt{\frac{h}{g}}$$

22. (b) For vertically upward motion, $h_1 = v_0 t - \dfrac{1}{2}gt^2$ and for vertically down ward motion, $h_2 = v_0 t + \dfrac{1}{2}gt^2$

$\therefore$ Total distance covered in t sec $h = h_1 + h_2 = 2v_o t$.

23. (d) Let height of tower is h and body takes t time to reach to ground when it fall freely.

$\therefore\ h = \dfrac{1}{2}gt^2$ ...(i)

In last second *i.e.* t^{th} *sec* body travels = 0.36 h

It means in rest of the time *i.e.* in $(t-1)$ *sec* it travels

$= h - 0.36\,h = 0.64\,h$

Now applying equation of motion for $(t-1)$ *sec*

$0.64\,h = \dfrac{1}{2}g(t-1)^2$ ...(ii)

From (i) and (ii) we get, $t = 5\,\text{sec}$ and $h = 125m$

***

24. (d) If t_1 and t_2 are time of ascent and descent respectively then

time of flight $T = t_1 + t_2 = \frac{2u}{g}$

$\Rightarrow u = \frac{g(t_1 + t_2)}{2}$

25. (c) For first projectile, $h_1 = ut - \frac{1}{2}gt^2$

For second projectile, $h_2 = u(t - T) - \frac{1}{2}g(t - T)^2$

When both meet *i.e.* $h_1 = h_2$

$ut - \frac{1}{2}gt^2 = u(t - T) - \frac{1}{2}g(t - T)^2$

$\Rightarrow uT + \frac{1}{2}gT^2 = gtT$

$\Rightarrow t = \frac{u}{g} + \frac{T}{2}$

and $h_1 = u\left(\frac{u}{g} + \frac{T}{2}\right) - \frac{1}{2}g\left(\frac{u}{2} + \frac{T}{2}\right)^2$

$= \frac{u^2}{2g} - \frac{gT^2}{8}.$

Chapter
3
Motion In Two Dimension

The motion of an object is called two dimensional, if two of the three co-ordinates required to specify the position of the object in space, change *w.r.t* time.

In such a motion, the object moves in a plane. For example, a billiard ball moving over the billiard table, an insect crawling over the floor of a room, earth revolving around the sun *etc*.

Two special cases of motion in two dimension are

1. Projectile motion   2. Circular motion

Introduction of Projectile Motion

A hunter aims his gun and fires a bullet directly towards a monkey sitting on a distant tree. If the monkey remains in his position, he will be safe but at the instant the bullet leaves the barrel of gun, if the monkey drops from the tree, the bullet will hit the monkey because the bullet will not follow the linear path.

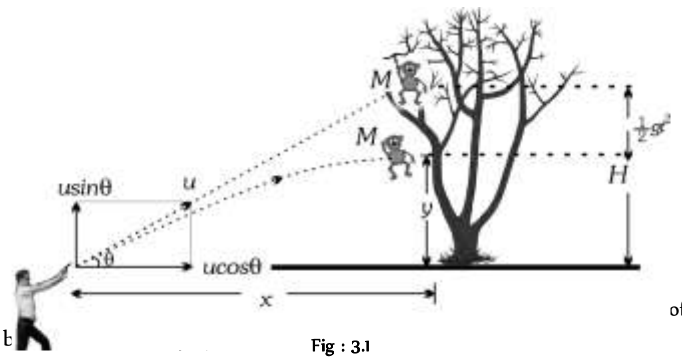


Fig : 3.1

If the force acting on a particle is oblique with initial velocity then the motion of particle is called projectile motion.

Projectile

A body which is in flight through the atmosphere under the effect of gravity alone and is not being propelled by any fuel is called projectile.

Example:

(i) A bomb released from an aeroplane in level flight

(ii) A bullet fired from a gun

(iii) An arrow released from bow

(iv) A Javelin thrown by an athlete

Assumptions of Projectile Motion

(1) There is no resistance due to air.

(2) The effect due to curvature of earth is negligible.

(3) The effect due to rotation of earth is negligible.

(4) For all points of the trajectory, the acceleration due to gravity '*g*' is constant in magnitude and direction.

Principle of Physical Independence of Motions

(1) The motion of a projectile is a two-dimensional motion. So, it can be discussed in two parts. Horizontal motion and vertical motion. These two motions take place independent of each other. This is called the principle of physical independence of motions.

(2) The velocity of the particle can be resolved into two mutually perpendicular components. Horizontal component and vertical component.

(3) The horizontal component remains unchanged throughout the flight. The force of gravity continuously affects the vertical component.

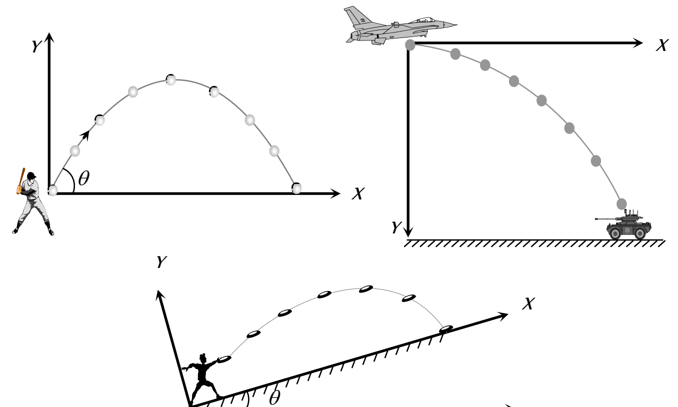
(4) The horizontal motion is a uniform motion and the vertical motion is a uniformly accelerated or retarded motion.

Types of Projectile Motion

(1) Oblique projectile motion

(2) Horizontal projectile motion

(3) Projectile motion on an inclined plane



Oblique Projectile

In projectile motion, horizontal component of velocity (u cosθ), acceleration (g) and mechanical energy remains constant while, speed, velocity, vertical component of velocity (u sin θ), momentum, kinetic energy and potential energy all changes. Velocity, and KE are maximum at the point of projection while minimum (but not zero) at highest point.

(1) **Equation of trajectory :** A projectile is thrown with velocity u at an angle θ with the horizontal. The velocity u can be resolved into two rectangular components.

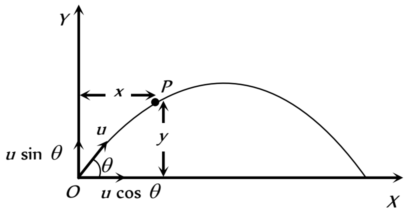


Fig : 3.3

v cos θ component along X–axis and u sin θ component along Y–axis.

For horizontal motion $x = u \cos\theta \times t \;\Rightarrow\; t = \dfrac{x}{u\cos\theta}$ ... (i)

For vertical motion $y = (u\sin\theta)t - \dfrac{1}{2}gt^2$ ...(ii)

From equation (i) and (ii)

$$y = u\sin\theta\left(\frac{x}{u\cos\theta}\right) - \frac{1}{2}g\left(\frac{x^2}{u^2\cos^2\theta}\right)$$

$$y = x\tan\theta - \frac{1}{2}\frac{gx^2}{u^2\cos^2\theta}$$

This equation shows that the trajectory of projectile is parabolic because it is similar to equation of parabola

$y = ax - bx$

Note : ❑ Equation of oblique projectile also can be written as

$$y = x\tan\theta\left[1 - \frac{x}{R}\right] \text{ (where } R = \text{horizontal range} = \frac{u^2\sin 2\theta}{g}\text{)}$$

(2) **Displacement of projectile $(\vec{r})$:** Let the particle acquires a position P having the coordinates (x, y) just after time t from the instant of projection. The corresponding position vector of the particle at time t is $\vec{r}$ as shown in the figure.

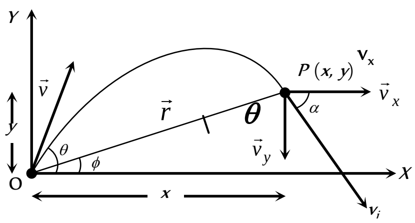


Fig : 3.4

$$\vec{r} = x\hat{i} + y\hat{j} \quad ...(i)$$

The horizontal distance covered during time t is given as

$$x = v_x\, t \Rightarrow x = u\cos\theta\; t \quad ...(ii)$$

The vertical velocity of the particle at time t is given as

$$v_y = (v_0)_y - gt, \quad ...(iii)$$

Now the vertical displacement y is given as

$$y = u\sin\theta\, t - 1/2\, gt^2 \quad ...(iv)$$

Putting the values of x and y from equation (ii) and equation (iv) in equation (i) we obtain the position vector at any time t as

$$\vec{r} = (u\cos\theta)t\,\hat{i} + \left((u\sin\theta)t - \frac{1}{2}gt^2\right)\hat{j}$$

$$\Rightarrow r = \sqrt{(u\,t\cos\theta)^2 + \left((u\,t\sin\theta) - \frac{1}{2}gt^2\right)^2}$$

$$r = u\,t\sqrt{1 + \left(\frac{gt}{2u}\right)^2 - \frac{gt\sin\theta}{u}} \text{ and } \phi = \tan^{-1}(y/x)$$

$$= \tan^{-1}\left(\frac{ut\sin\theta - \frac{1}{2}gt^2}{(u\,t\cos\theta)}\right) \text{ or } \phi = \tan^{-1}\left(\frac{2u\sin\theta - gt}{2u\cos\theta}\right)$$

Note : ❑ The angle of elevation ϕ of the highest point of the projectile and the angle of projection θ are related to each other as

$$\tan\phi = \frac{1}{2}\tan\theta$$

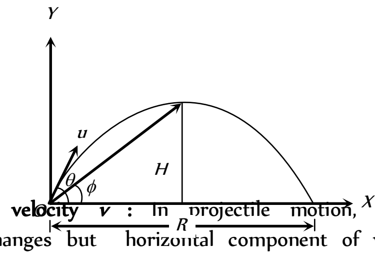


Fig : 3.5

(3) **Instantaneous velocity v :** In projectile motion, vertical component of velocity changes but horizontal component of velocity remains always constant.

Example : When a man jumps over the hurdle leaving behind its skateboard then vertical component of his velocity is changing, but not the horizontal component which matches with the skateboard velocity.

As a result, the skateboard stays underneath him, allowing him to land on it.

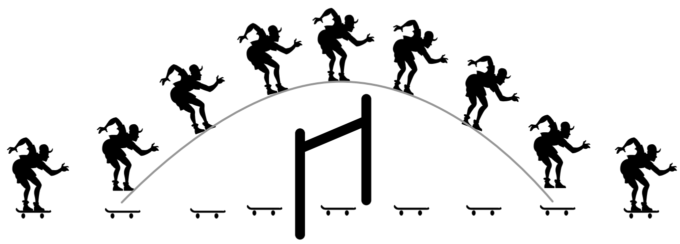


Fig : 3.6

Let v_i be the instantaneous velocity of projectile at time t, direction of this velocity is along the tangent to the trajectory at point P.

$$\vec{v}_i = v_x i + v_y\hat{j} \Rightarrow v_i = \sqrt{v_x^2 + v_y^2}$$

$= \sqrt{u^2 \cos^2 \theta + (u \sin\theta - gt)^2}$

$v_i = \sqrt{u^2 + g^2t^2 - 2u\,gt\sin\theta}$

Direction of instantaneous velocity $\tan\alpha = \dfrac{v_y}{v_x} = \dfrac{u\sin\theta - gt}{u\cos\theta}$

or $\quad \alpha = \tan^{-1}\left[\tan\theta - \dfrac{gt}{u}\sec\theta\right]$

(4) **Change in velocity :** Initial velocity (at projection point) $\vec{u}_i = u\cos\theta\hat{i} + u\sin\theta\hat{j}$

Final velocity (at highest point) $\vec{u}_f = u\cos\theta\hat{i} + 0\hat{j}$

(i) Change in velocity (Between projection point and highest point) $\Delta\vec{u} = \vec{u}_f - \vec{u}_i = -u\sin\theta\hat{j}$

When body reaches the ground after completing its motion then final velocity $\vec{u}_f = u\cos\theta\hat{i} - u\sin\theta\hat{j}$

(ii) Change in velocity (Between complete projectile motion) $\Delta\vec{u} = u_f - u_i = -2u\sin\theta\hat{i}$

(5) **Change in momentum :** Simply by the multiplication of mass in the above expression of velocity (Article-4).

(i) Change in momentum (Between projection point and highest point) $\Delta\vec{p} = \vec{p}_f - \vec{p}_i = -mu\sin\theta\hat{j}$

(ii) Change in momentum (For the complete projectile motion) $\Delta\vec{p} = \vec{p}_f - \vec{p}_i = -2mu\sin\theta\hat{j}$

(6) **Angular momentum :** Angular momentum of projectile at highest point of trajectory about the point of projection is given by

$L = mvr \qquad \left[\text{Here } r = H = \dfrac{u^2\sin^2\theta}{2g}\right]$

$\therefore \quad L = m\,u\cos\theta\dfrac{u^2\sin^2\theta}{2g} = \dfrac{m\,u^3\cos\theta\sin^2\theta}{2g}$

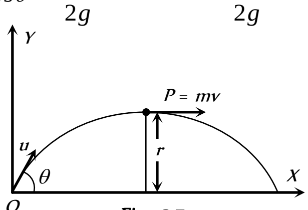


Fig : 3.7

(7) **Time of flight :** The total time taken by the projectile to go up and come down to the same level from which it was projected is called time of flight.

For vertical upward motion $0 = u\sin\theta - gt$

$\Rightarrow \; t = (u\sin\theta/g)$

Now as time taken to go up is equal to the time taken to come down so

Time of flight $T = 2t = \dfrac{2u\sin\theta}{g}$

(i) Time of flight can also be expressed as : $T = \dfrac{2.u_y}{g}$ (where u_y is the vertical component of initial velocity).

(ii) For complementary angles of projection θ and $90^\circ - \theta$

(a) Ratio of time of flight $= \dfrac{T_1}{T_2} = \dfrac{2u\sin\theta/g}{2u\sin(90-\theta)/g}$

$= \tan\theta \;\Rightarrow\; \dfrac{T_1}{T_2} = \tan\theta$

(b) Multiplication of time of flight $= T_1T_2 = \dfrac{2u\sin\theta}{g}\dfrac{2u\cos\theta}{g}$

$\Rightarrow T_1T_2 = \dfrac{2R}{g}$

(iii) If t_1 is the time taken by projectile to rise upto point p and t_2 is the time taken in falling from point p to ground level then

$t_1 + t_2 = \dfrac{2u\sin\theta}{g}$ = time of flight or $u\sin\theta = \dfrac{g(t_1 + t_2)}{2}$

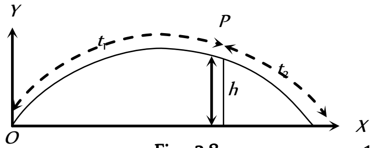


Fig : 3.8

and height of the point p is given by $h = u\sin\theta\, t_1 - \dfrac{1}{2}gt_1^2$

$h = g\dfrac{(t_1 + t_2)}{2}t_1 - \dfrac{1}{2}gt_1^2$

by solving $h = \dfrac{g\,t_1t_2}{2}$

(iv) If B and C are at the same level on trajectory and the time difference between these two points is t_1 similarly A and D are also at the same level and the time difference between these two positions is t_2 then

$t_2^2 - t_1^2 = \dfrac{8h}{g}$

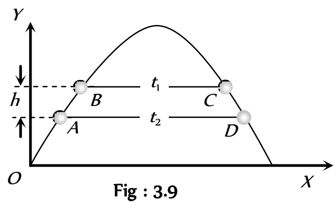


Fig : 3.9

(8) **Horizontal range :** It is the horizontal distance travelled by a body during the time of flight.

So by using second equation of motion in x-direction

$R = u\cos\theta \times T$

$= u\cos\theta \times (2u\sin\theta/g)$

$= \dfrac{u^2\sin 2\theta}{g}$

$R = \dfrac{u^2\sin 2\theta}{g}$

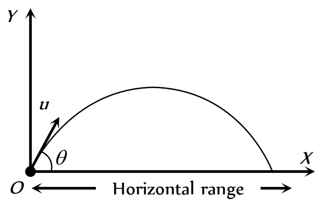


Fig : 3.10

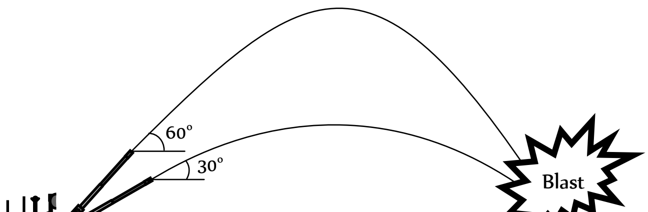
(i) Range of projectile can also be expressed as :

$R = u\cos\theta \times T = u\cos\theta\dfrac{2u\sin\theta}{g}$

$= \dfrac{2u\cos\theta\;u\sin\theta}{g} = \dfrac{2u_xu_y}{g}$

$\therefore \; R = \dfrac{2u_xu_y}{g}$ (where u_x and u_y are the horizontal and vertical component of initial velocity)

(ii) If angle of projection is changed from θ to $\theta' = (90 - \theta)$ then range remains unchanged.



$$R' = \frac{u^2\sin 2\theta'}{g} = \frac{u^2 \sin[2(90^o - \theta)]}{g} = \frac{u^2 \sin 2\theta}{g} = R$$

So a projectile has same range at angles of projection θ and $(90 - \theta)$, though time of flight, maximum height and trajectories are different.

These angles θ and $90^\circ - \theta$ are called complementary angles of projection and for complementary angles of projection, ratio of range

$$\frac{R_1}{R_2} = \frac{u^2\sin 2\theta / g}{u^2 \sin[2(90^o - \theta)]/g} = 1 \Rightarrow \frac{R_1}{R_2} = 1$$

(iii) For angle of projection $\theta_1 = (45 - \alpha)$ and $\theta_2 = (45 + \alpha)$, range will be same and equal to $u \cos 2\alpha/g$.

θ_1 and θ_2 are also the complementary angles.

(iv) Maximum range : For range to be maximum

$$\frac{dR}{d\theta} = 0 \Rightarrow \frac{d}{d\theta}\left[\frac{u^2\sin 2\theta}{g}\right] = 0$$

$\Rightarrow$ $\cos 2\theta = 0$ *i.e.* $2\theta = 90^\circ \Rightarrow \theta = 45^\circ$

and $R_{max} = (u/g)$

i.e., a projectile will have maximum range when it is projected at an angle of 45° to the horizontal and the maximum range will be (u/g).

When the range is maximum, the height H reached by the projectile

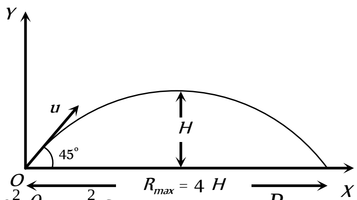


Fig : 3.12

$$H = \frac{u^2\sin^2\theta}{2g} = \frac{u^2\sin^2 45^\circ}{2g} = \frac{u^2}{4g} = \frac{R_{max}}{4}$$

i.e., if a person can throw a projectile to a maximum distance R_{max},

The maximum height during the flight to which it will rise is $\left(\frac{R_{max}}{4}\right)$.

(v) Relation between horizontal range and maximum height : $R = \frac{u^2\sin 2\theta}{g}$ and $H = \frac{u^2\sin^2\theta}{2g}$

$$\therefore \frac{R}{H} = \frac{u^2\sin 2\theta/g}{u^2\sin^2\theta/2g} = 4\cot\theta \Rightarrow R = 4H\cot\theta$$

(vi) If in case of projectile motion range R is n times the maximum height H

i.e. $R = nH \Rightarrow \frac{u^2\sin 2\theta}{g} = n\frac{u^2\sin^2\theta}{2g}$

$\Rightarrow$ $\tan\theta = [4/n]$ or $\theta = \tan^{-1}[4/n]$

The angle of projection is given by $\theta = \tan^{-1}[4/n]$

Note : ❑ If $R = H$ then $\theta = \tan^{-1}(4)$ or $\theta = 76^o$.

If $R = 4H$ then $\theta = \tan^{-1}(1)$ or $\theta = 45^o$.

(9) **Maximum height :** It is the maximum height from the point of projection, a projectile can reach.

So, by using $v^2 = u^2 + 2as$

$$0 = (u\sin\theta)^2 - 2gH$$

$$H = \frac{u^2\sin^2\theta}{2g}$$

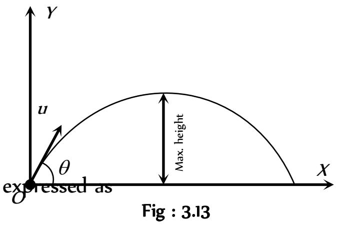


Fig : 3.13

(i) Maximum height can also be expressed as

$H = \frac{u_y^2}{2g}$ (where u_y is the vertical component of initial velocity).

(ii) $H_{max} = \frac{u^2}{2g}$ (when $\sin\theta = max = 1$ *i.e.,* $\theta = 90^\circ$)

i.e., for maximum height body should be projected vertically upward. So it falls back to the point of projection after reaching the maximum height.

(iii) For complementary angles of projection θ and $90^\circ - \theta$

Ratio of maximum height

$$= \frac{H_1}{H_2} = \frac{u^2\sin^2\theta/2g}{u^2\sin^2(90^o - \theta)/2g} = \frac{\sin^2\theta}{\cos^2\theta} = \tan^2\theta$$

$$\therefore \frac{H_1}{H_2} = \tan^2\theta$$

(10) **Projectile passing through two different points on same height at time t_1 and t_2 :** If the particle passes two points situated at equal height y at $t = t_1$ and $t = t_2$, then

(i) **Height (y):** $y = (u\sin\theta)t_1 - \frac{1}{2}gt_1^2$ ...(i)

and $y = (u\sin\theta)t_2 - \frac{1}{2}gt_2^2$ ...(ii)

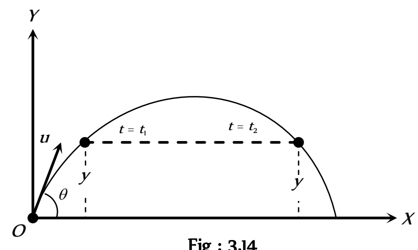


Fig : 3.14

Comparing equation (i) with equation (ii)

$$u\sin\theta = \frac{g(t_1 + t_2)}{2}$$

Substituting this value in equation (i)

$$y = g\left(\frac{t_1 + t_2}{2}\right)t_1 - \frac{1}{2}gt_1^2 \Rightarrow y = \frac{gt_1t_2}{2}$$

(ii) **Time (t_1 and t_2):** $y = u\sin\theta\, t - \frac{1}{2}gt^2$

$$t^2 - \frac{2u\sin\theta}{g}t + \frac{2y}{g} = 0 \Rightarrow \; t = \frac{u\sin\theta}{g}\left[1 \pm \sqrt{1 - \left(\frac{\sqrt{2gy}}{u\sin\theta}\right)^2}\right]$$

$$t_1 = \frac{u\sin\theta}{g}\left[1 + \sqrt{1 - \left(\frac{\sqrt{2gy}}{u\sin\theta}\right)^2}\right]$$

and $t_2 = \dfrac{u\sin\theta}{g}\left[1 - \sqrt{1 - \left(\dfrac{\sqrt{2gy}}{u\sin\theta}\right)^2}\right]$

(11) **Motion of a projectile as observed from another projectile :** Suppose two balls A and B are projected simultaneously from the origin, with initial velocities u_1 and u_2 at angle θ_1 and θ_2, respectively with the horizontal.

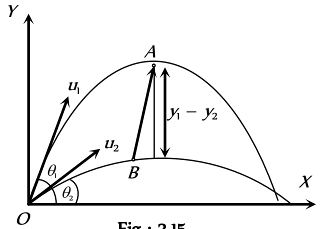


Fig : 3.15

The instantaneous positions of the two balls are given by

Ball A : $x_1 = (u_1\cos\theta_1)t,\; y_1 = (u_1\sin\theta_1)t - \frac{1}{2}gt^2$

Ball B : $x_2 = (u_2\cos\theta_2)t,\; y_2 = (u_2\sin\theta_2)t - \frac{1}{2}gt^2$

The position of the ball A with respect to ball B is given by

$$x = x_1 - x_2 = (u_1\cos\theta_1 - u_2\cos\theta_2)t$$

$$y = y_1 - y_2 = (u_1\sin\theta_1 - u_2\sin\theta_2)t$$

Now $\dfrac{y}{x} = \left(\dfrac{u_1\sin\theta_1 - u_2\sin\theta_2}{u_1\cos\theta_1 - u_2\cos\theta_2}\right) =$ constant

Thus motion of a projectile relative to another projectile is a straight line.

(12) **Energy of projectile :** When a projectile moves upward its kinetic energy decreases, potential energy increases but the total energy always remain constant.

If a body is projected with initial kinetic energy $K(=1/2\, mu^2)$, with angle of projection θ with the horizontal then at the highest point of trajectory

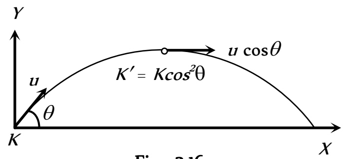


Fig : 3.16

(i) **Kinetic energy** $= \frac{1}{2}m(u\cos\theta)^2 = \frac{1}{2}mu^2\cos^2\theta$

$$\therefore \; K' = K\cos^2\theta$$

(ii) **Potential energy** $= mgH = mg\dfrac{u^2\sin^2\theta}{2g}$

$$= \frac{1}{2}mu^2\sin^2\theta \qquad \left(\text{As } H = \frac{u^2\sin^2\theta}{2g}\right)$$

$$= K\sin^2\theta$$

(iii) **Total energy** = Kinetic energy + Potential energy

$$= \frac{1}{2}mu^2\cos^2\theta + \frac{1}{2}mu^2\sin^2\theta$$

$= \frac{1}{2}mu^2 =$ Energy at the point of projection.

This is in accordance with the law of conservation of energy.

Horizontal Projectile

When a body is projected horizontally from a certain height 'y' vertically above the ground with initial velocity u. If friction is considered to be absent, then there is no other horizontal force which can affect the horizontal motion. The horizontal velocity therefore remains constant and so the object covers equal distance in horizontal direction in equal intervals of time.

The horizontal velocity therefore remains constant and so the object covers equal distance in horizontal direction in equal intervals of time.

(1) **Trajectory of horizontal projectile :** The horizontal displacement x is governed by the equation

$x = ut \Rightarrow t = \dfrac{x}{u}$ .... (i)

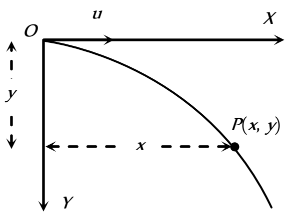


Fig : 3.17

The vertical displacement y is governed by

$y = \frac{1}{2}gt^2$ .... (ii)

(since initial vertical velocity is zero)

By substituting the value of t in equation (ii) $y = \dfrac{1}{2}\dfrac{gx^2}{u^2}$

(2) **Displacement of Projectile** $(\vec{r})$: After time t, horizontal displacement $x = ut$ and vertical displacement $y = \frac{1}{2}gt^2$.

So, the position vector $\vec{r} = ut\,\hat{i} + \frac{1}{2}gt^2\,\hat{j}$

Therefore $\quad r = ut\sqrt{1 + \left(\dfrac{gt}{2u}\right)^2} \quad$ and $\alpha = \tan^{-1}\left(\dfrac{gt}{2u}\right)$

$$\alpha = \tan^{-1}\left(\sqrt{\frac{gy}{2}}\Big/u\right) \qquad \left(\text{as } t = \sqrt{\frac{2y}{g}}\right)$$

(3) **Instantaneous velocity :** Throughout the motion, the horizontal component of the velocity is $v_x = u$.

The vertical component of velocity increases with time and is given by

$v_y = 0 + g\,t = g\,t$ (From $v = u + g\,t$)

So, $\vec{v} = v_x\hat{i} + v_y\hat{j} = u\,\hat{i} + g\,t\,\hat{j}$

i.e. $v=\sqrt{u^2+(gt)^2} \;\; = u\sqrt{1+\left(\frac{gt}{u}\right)^2}$

Again $\vec{v}=u\hat{i}+\sqrt{2gy}\,\hat{j}$

i.e. $v=\sqrt{u^2+2gy}$

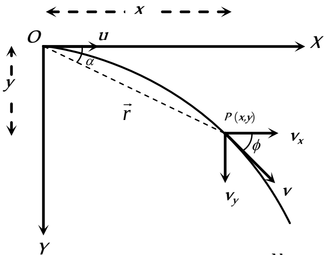


Fig : 3.18

Direction of instantaneous velocity : $\tan\phi=\frac{v_y}{v_x}$

$$\Rightarrow \;\; \phi=\tan^{-1}\left(\frac{v_y}{v_x}\right) = \tan^{-1}\left(\frac{\sqrt{2gy}}{u}\right) \text{ or } \phi=\tan^{-1}\left(\frac{gt}{u}\right)$$

Where ϕ is the angle of instantaneous velocity from the horizontal.

(4) **Time of flight :** If a body is projected horizontally from a height h with velocity u and time taken by the body to reach the ground is T, then

$h=0+\frac{1}{2}gT^2$ (for vertical motion)

$$T=\sqrt{\frac{2h}{g}}$$

(5) **Horizontal range :** Let R is the horizontal distance travelled by the body

$R=uT+\frac{1}{2}\,0\,T^2$ (for horizontal motion)

$$R=u\sqrt{\frac{2h}{g}}$$

(6) If projectiles A and B are projected horizontally with different initial velocity from same height and third particle C is dropped from same point then

(i) All three particles will take equal time to reach the ground.

(ii) Their net velocity would be different but all three particle possess same vertical component of velocity.

(iii) The trajectory of projectiles A and B will be straight line *w.r.t.* particle C.

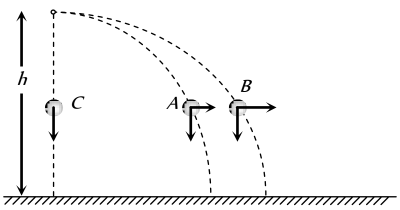


Fig : 3.19

(7) If various particles thrown with same initial velocity but in different direction then

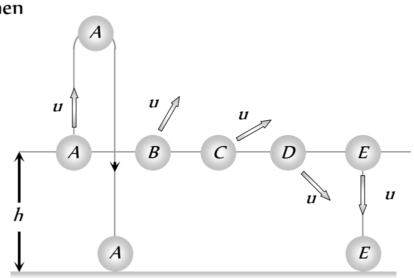


Fig : 3.20

(i) They strike the ground with same speed at different times irrespective of their initial direction of velocities.

(ii) Time would be least for particle E which was thrown vertically downward.

(iii) Time would be maximum for particle A which was thrown vertically upward.

Projectile Motion on An Inclined Plane

Let a particle be projected up with a speed u from an inclined plane which makes an angle α with the horizontal and velocity of projection makes an angle θ with the inclined plane.

We have taken reference x-axis in the direction of plane.

Hence the component of initial velocity parallel and perpendicular to the plane are equal to $u\cos\theta$ and $u\sin\theta$ respectively *i.e.* $u_{\parallel}=u\cos\theta$ and $u_{\perp}=u\sin\theta$.

The component of g along the plane is $g\sin\alpha$ and perpendicular to the plane is $g\cos\alpha$ as shown in the figure *i.e.* $a_{\parallel}=-g\sin\alpha$ and $a_{\perp}=g\cos\alpha$.

Therefore the particle decelerates at a rate of $g\sin\alpha$ as it moves from O to P.

(1) **Time of flight :** We know for oblique projectile motion $T=\frac{2u\sin\theta}{g}$ or we can say $T=\frac{2u_{\perp}}{a_{\perp}}$

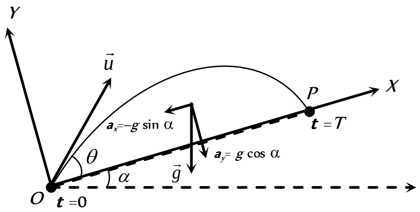


Fig : 3.21

$\therefore$ Time of flight on an inclined plane $T=\frac{2u\sin\theta}{g\cos\alpha}$

(2) **Maximum height :** We know for oblique projectile motion $H=\frac{u^2\sin^2\theta}{2g}$ or we can say $H=\frac{u_{\perp}^2}{2a_{\perp}}$

$\therefore$ Maximum height on an inclined plane $H=\frac{u^2\sin^2\theta}{2g\cos\alpha}$

(3) **Horizontal range :** For one dimensional motion $s=ut+\frac{1}{2}at^2$

Horizontal range on an inclined plane $R=u_{\parallel}T+\frac{1}{2}a_{\parallel}T^2$

$$R=u\cos\theta\,T-\frac{1}{2}g\sin\alpha\,T^2$$

$$R=u\cos\theta\left(\frac{2u\sin\theta}{g\cos\alpha}\right)-\frac{1}{2}g\sin\alpha\left(\frac{2u\sin\theta}{g\cos\alpha}\right)^2$$

By solving $R=\frac{2u^2}{g}\frac{\sin\theta\cos(\theta+\alpha)}{\cos^2\alpha}$

(i) Maximum range occurs when $\theta=\frac{\pi}{4}-\frac{\alpha}{2}$

(ii) The maximum range along the inclined plane when the projectile is thrown upwards is given by

$$R_{\max} = \frac{u^2}{g(1+\sin\alpha)}$$

(iii) The maximum range along the inclined plane when the projectile is thrown downwards is given by

$$R_{\max} = \frac{u^2}{g(1-\sin\alpha)}$$

Circular Motion

Circular motion is another example of motion in two dimensions. To create circular motion in a body it must be given some initial velocity and a force must then act on the body which is always directed at right angles to instantaneous velocity.

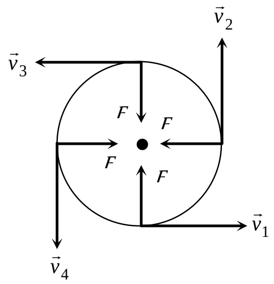


Fig : 3.22

Since this force is always at right angles to the displacement therefore no work is done by the force on the particle. Hence, its kinetic energy and thus speed is unaffected. But due to simultaneous action of the force and the velocity the particle follows resultant path, which in this case is a circle. Circular motion can be classified into two types – Uniform circular motion and non-uniform circular motion.

Variables of Circular Motion

(1) **Displacement and distance :** When particle moves in a circular path describing an angle θ during time t (as shown in the figure) from the position A to the position B, we see that the magnitude of the position vector $\vec{r}$ (that is equal to the radius of the circle) remains constant. *i.e.,* $|\vec{r}_1| = |\vec{r}_2| = r$ and the direction of the position vector changes from time to time.

(i) Displacement : The change of position vector or the displacement $\Delta\vec{r}$ of the particle from position A to the position B is given by referring the figure.

$$\Delta\vec{r} = \vec{r}_2 - \vec{r}_1 \Rightarrow \Delta r = |\Delta\vec{r}| = |\vec{r}_2 - \vec{r}_1|$$

$$\Delta r = \sqrt{r_1^2 + r_2^2 - 2r_1r_2\cos\theta}$$

Putting $r_1 = r_2 = r$ we obtain

$$\Delta r = \sqrt{r^2 + r^2 - 2r.r\cos\theta}$$

$$\Rightarrow \Delta r = \sqrt{2r^2(1-\cos\theta)}$$

$$= \sqrt{2r^2\left(2\sin^2\frac{\theta}{2}\right)}$$

$$\Delta r = 2r\sin\frac{\theta}{2}$$

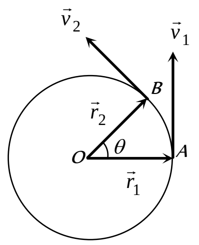


Fig : 3.23

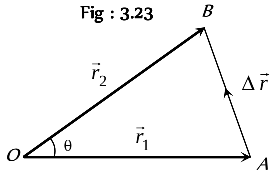


Fig : 3.24

(ii) Distance : The distanced covered by the particle during the time t is given as

d = length of the arc AB = $r\,\theta$

(iii) Ratio of distance and displacement : $\dfrac{d}{\Delta r} = \dfrac{r\theta}{2r\sin\theta/2}$

$$= \frac{\theta}{2}\operatorname{cosec}(\theta/2)$$

(2) **Angular displacement (θ) :** The angle turned by a body moving in a circle from some reference line is called angular displacement.

(i) Dimension = $[M^0L^0T^0]$ (as θ = arc/radius).

(ii) Units = Radian or Degree. It is some time also specified in terms of fraction or multiple of revolution.

(iii) $2\pi\,\text{rad} = 360^o = 1\,\text{Revolution}$

(iv) Angular displacement is a axial vector quantity.

Its direction depends upon the sense of rotation of the object and can be given by Right Hand Rule; which states that if the curvature of the fingers of right hand represents the sense of rotation of the object, then the thumb, held perpendicular to the curvature of the fingers, represents the direction of angular displacement vector.

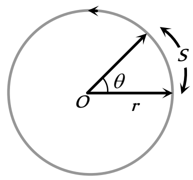


Fig : 3.25

(v) Relation between linear displacement and angular displacement $\vec{s} = \vec{\theta} \times \vec{r}$

or $s = r\theta$

(3) **Angular velocity (ω) :** Angular velocity of an object in circular motion is defined as the time rate of change of its angular displacement.

(i) Angular velocity $\omega = \dfrac{\text{angle traced}}{\text{time taken}} = \underset{\Delta t \to 0}{Lt} \dfrac{\Delta\theta}{\Delta t} = \dfrac{d\theta}{dt}$

$$\therefore\ \omega = \frac{d\theta}{dt}$$

(ii) Dimension : $[M^0L^0T^{-1}]$

(iii) Units : Radians per second ($rad.s^{-1}$) or Degree per second.

(iv) Angular velocity is an axial vector.

Its direction is the same as that of $\Delta\theta$. For anticlockwise rotation of the point object on the circular path, the direction of ω, according to Right hand rule is along the axis of circular path directed upwards. For clockwise rotation of the point object on the circular path, the direction of ω is along the axis of circular path directed downwards.

(v) Relation between angular velocity and linear velocity $\vec{v} = \vec{\omega} \times \vec{r}$

(vi) For uniform circular motion ω remains constant where as for non-uniform motion ω varies with respect to time.

Note : It is important to note that nothing actually moves in the direction of the angular velocity vector $\vec{\omega}$. The direction of $\vec{\omega}$ simply represents that the circular motion is taking place in a plane perpendicular to it.

(4) **Change in velocity :** We want to know the magnitude and direction of the change in velocity of the particle which is performing uniform circular motion as it moves from A to B during time t as shown in figure. The change in velocity vector is given as

$$\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$$

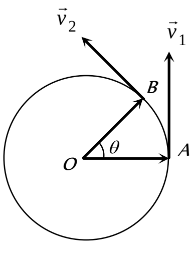


Fig : 3.26

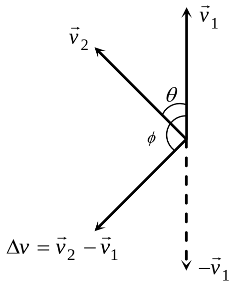


Fig : 3.27

or $|\Delta\vec{v}| = |\vec{v}_2 - \vec{v}_1| \Rightarrow \Delta v = \sqrt{v_1^2 + v_2^2 - 2v_1v_2\cos\theta}$

For uniform circular motion $v_1 = v_2 = v$

So $\Delta v = \sqrt{2v^2(1-\cos\theta)} = 2v\sin\dfrac{\theta}{2}$

The direction of $\Delta\vec{v}$ is shown in figure that can be given as

$$\phi = \frac{180^o - \theta}{2} = (90^o - \theta/2)$$

(5) **Time period (T) :** In circular motion, the time period is defined as the time taken by the object to complete one revolution on its circular path.

(i) Units : second.

(ii) Dimension : $[M^0L^0T]$

(iii) Time period of second's hand of watch = 60 *second*.

(iv) Time period of minute's hand of watch = 60 *minute*

(v) Time period of hour's hand of watch = 12 *hour*

(6) **Frequency (n) :** In circular motion, the frequency is defined as the number of revolutions completed by the object on its circular path in a unit time.

(i) Units : s^{-1} or hertz (Hz).

(ii) Dimension : $[M^0L^0T^{-1}]$

Note : ❑ Relation between time period and frequency : If n is the frequency of revolution of an object in circular motion, then the object completes n revolutions in 1 second. Therefore, the object will complete one revolution in $1/n$ second.

$\therefore T = 1/n$

❑ Relation between angular velocity, frequency and time period : Consider a point object describing a uniform circular motion with frequency n and time period T. When the object completes one revolution, the angle traced at its axis of circular motion is 2π radians. It means, when time $t = T$, $\theta = 2\pi$ radians. Hence, angular velocity $\omega = \dfrac{\theta}{t} = \dfrac{2\pi}{T} = 2\pi n$ ($\because T = 1/n$)

$$\omega = \frac{2\pi}{T} = 2\pi n$$

❑ If two particles are moving on same circle or different coplanar concentric circles in same direction with different uniform angular speeds ω_A and ω_B respectively, the angular velocity of B relative to A will be

$$\omega_{\text{rel}} = \omega_B - \omega_A$$

So the time taken by one to complete one revolution around O with respect to the other (i.e., time in which B complete one revolution around O with respect to the other (*i.e.*, time in which B completes one more or less revolution around O than A)

$$T = \frac{2\pi}{\omega_{\text{rel}}} = \frac{2\pi}{\omega_2 - \omega_1} = \frac{T_1T_2}{T_1 - T_2} \qquad \left[\text{as } T = \frac{2\pi}{\omega}\right]$$

Special case : If $\omega_B = \omega_A, \omega_{\text{rel}} = 0$ and so $T = \infty$., particles will maintain their position relative to each other. This is what actually happens in case of geostationary satellite ($\omega_1 = \omega_2$ = constant)

(7) **Angular acceleration (α) :** Angular acceleration of an object in circular motion is defined as the time rate of change of its angular velocity.

(i) If $\Delta\omega$ be the change in angular velocity of the object in time interval Δt, while moving on a circular path, then angular acceleration of the object will be

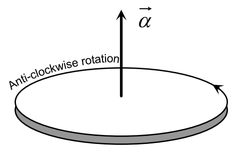


Fig : 3.28

$$\alpha = \underset{\Delta t \to 0}{Lt} \frac{\Delta\omega}{\Delta t} = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$$

(ii) Units : *rad*. s^{-2}

(iii) Dimension : $[M^0L^0T^{-2}]$

(iv) Relation between linear acceleration and angular acceleration $\vec{a} = \vec{\alpha} \times \vec{r}$

(v) For uniform circular motion since ω is constant so $\alpha = \dfrac{d\omega}{dt} = 0$

(vi) For non-uniform circular motion $\alpha \neq 0$

Centripetal Acceleration

(1) Acceleration acting on the object undergoing uniform circular motion is called centripetal acceleration.

(2) It always acts on the object along the radius towards the centre of the circular path.

(3) Magnitude of centripetal acceleration,

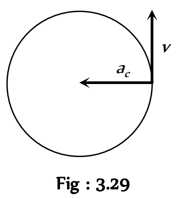


Fig : 3.29

$$a = \frac{v^2}{r} = \omega^2 r = 4\pi^2 n^2 r = \frac{4\pi^2}{T^2} r$$

(4) Direction of centripetal acceleration : It is always the same as that of $\Delta\vec{\upsilon}$. When Δt decreases, $\Delta\theta$ also decreases. Due to which $\Delta\vec{\upsilon}$ becomes more and more perpendicular to $\vec{\upsilon}$. When $\Delta t \to 0$, $\Delta\vec{\upsilon}$ becomes perpendicular to the velocity vector. As the velocity vector of the particle at an instant acts along the tangent to the circular path, therefore $\Delta\vec{\upsilon}$ and hence the centripetal acceleration vector acts along the radius of the circular path at that point and is directed towards the centre of the circular path.

Centripetal force

According to Newton's first law of motion, whenever a body moves in a straight line with uniform velocity, no force is required to maintain this velocity. But when a body moves along a circular path with uniform speed, its direction changes continuously *i.e.* velocity keeps on changing on account of a change in direction. According to Newton's second law of motion, a change in the direction of motion of the body can take place only if some external force acts on the body.

Due to inertia, at every point of the circular path; the body tends to move along the tangent to the circular path at that point (in figure). Since every body has directional inertia, a velocity cannot change by itself and as such we have to apply a force. But this force should be such that it changes the direction of velocity and not its magnitude. This is possible only if the force acts perpendicular to the direction of velocity. Because the velocity is along the tangent, this force must be along the radius (because the radius of a circle at any point is perpendicular to the tangent at that point). Further, as this force is to move the body in a circular path, it must acts towards the centre. This centre-seeking force is called the centripetal force.

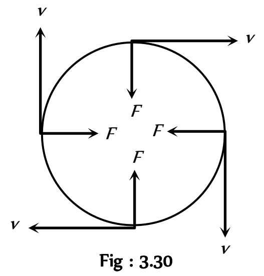


Fig : 3.30

Hence, centripetal force is that force which is required to move a body in a circular path with uniform speed. The force acts on the body along the radius and towards centre.

Formulae for centripetal force :

$$F = \frac{mv^2}{r} = m\omega^2 r = m\,4\pi^2 n^2 r = \frac{m4\pi^2 r}{T^2}$$

Table 3.1 : Centripetal force in different situation

Situation	Centripetal Force
A particle tied to a string and whirled in a horizontal circle	Tension in the string
Vehicle taking a turn on a level road	Frictional force exerted by the road on the tyres
A vehicle on a speed breaker	Weight of the body or a component of weight
Revolution of earth around the sun	Gravitational force exerted by the sun
Electron revolving around the nucleus in an atom	Coulomb attraction exerted by the protons in the nucleus
A charged particle describing a circular path in a magnetic field	Magnetic force exerted by the agent that sets up the magnetic field

Centrifugal Force

It is an imaginary force due to incorporated effects of inertia. When a body is rotating in a circular path and the centripetal force vanishes, the body would leave the circular path. To an observer *A* who is not sharing the motion along the circular path, the body appears to fly off tangentially at the point of release. To another observer *B*, who is sharing the motion along the circular path (*i.e.*, the observer *B* is also rotating with the body with the same velocity), the body appears to be stationary before it is released. When the body is released, it appears to *B*, as if it has been thrown off along the radius away from the centre by some force. In reality no force is actually seen to act on the body. In absence of any real force the body tends to continue its motion in a straight line due to its inertia. The observer *A* easily relates this events to be due to inertia but since the inertia of both the observer *B* and the body is same, the observer *B* can not relate the above happening to inertia. When the centripetal force ceases to act on the body, the body leaves its circular path and continues to move in its straight-line motion but to observer *B* it appears that a real force has actually acted on the body and is responsible for throwing the body radially out-wards. This imaginary force is given a name to explain the effects of inertia to the observer who is sharing the circular motion of the body. This inertial force is called centrifugal force. Thus centrifugal force is a fictitious force which has significance only in a rotating frame of reference.

Work Done by Centripetal Force

The work done by centripetal force is always zero as it is perpendicular to velocity and hence instantaneous displacement.

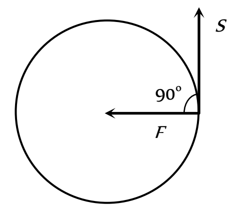


Fig : 3.31

Work done = Increment in kinetic energy of revolving body

Work done = 0

Also $W = \vec{F}.\vec{S} = F \cdot S\cos\theta = FS\cos 90^\circ = 0$

Example : (i) When an electron revolves around the nucleus in hydrogen atom in a particular orbit, it neither absorb nor emit any energy means its energy remains constant.

(ii) When a satellite established once in a orbit around the earth and it starts revolving with particular speed, then no fuel is required for its circular motion.

Skidding of Vehicle on A Level Road

When a vehicle takes a turn on a circular path it requires centripetal force.

If friction provides this centripetal force then vehicle can move in circular path safely if

Friction force $\geq$ Required centripetal force

$$\mu mg \geq \frac{mv^2}{r}$$

$$\therefore\ v_{safe} \leq \sqrt{\mu rg}$$

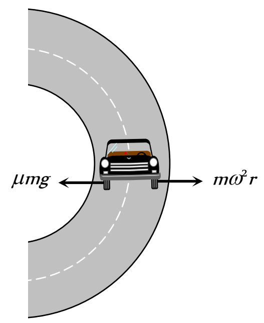


Fig : 3.32

This is the maximum speed by which vehicle can take a turn on a circular path of radius *r*, where coefficient of friction between the road and tyre is μ.

Skidding of Object on A Rotating Platform

On a rotating platform, to avoid the skidding of an object (mass *m*) placed at a distance *r* from axis of rotation, the centripetal force should be provided by force of friction.

Centripetal force $\leq$ Force of friction

$m\omega r \leq \mu mg$

$$\therefore\ \omega_{\max} = \sqrt{(\mu g / r)},$$

Hence maximum angular velocity of rotation of the platform is $\sqrt{(\mu g / r)}$, so that object will not skid on it.

Bending of A Cyclist

A cyclist provides himself the necessary centripetal force by leaning inward on a horizontal track, while going round a curve. Consider a cyclist of weight *mg* taking a turn of radius *r* with velocity *v*. In order to provide the necessary centripetal force, the cyclist leans through angle θ inwards as shown in figure.

The cyclist is under the action of the following forces :

The weight *mg* acting vertically downward at the centre of gravity of cycle and the cyclist.

The reaction *R* of the ground on cyclist. It will act along a line-making angle θ with the vertical.

The vertical component $R\cos\theta$ of the normal reaction *R* will balance the weight of the cyclist, while the horizontal component $R\sin\theta$ will provide the necessary centripetal force to the cyclist.

$$R\sin\theta = \frac{mv^2}{r} \quad ...(i)$$

and $R\cos\theta = mg$ ...(ii)

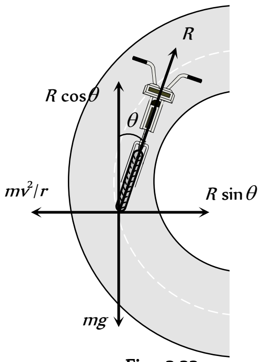


Fig : 3.33

Dividing equation (i) by (ii), we have

$$\frac{R\sin\theta}{R\cos\theta} = \frac{m v^2/r}{mg}$$

or $\tan\theta = \dfrac{v^2}{rg}$ ...(iii)

Therefore, the cyclist should bend through an angle $\theta = \tan^{-1}\left(\dfrac{v^2}{rg}\right)$

It follows that the angle through which cyclist should bend will be greater, if

(i) The radius of the curve is small *i.e.* the curve is sharper

(ii) The velocity of the cyclist is large.

Note : ❑ For the same reasons, an ice skater or an aeroplane has to bend inwards, while taking a turn.

Banking of A Road

For getting a centripetal force, cyclist bend towards the centre of circular path but it is not possible in case of four wheelers.

Therefore, outer bed of the road is raised so that a vehicle moving on it gets automatically inclined towards the centre.

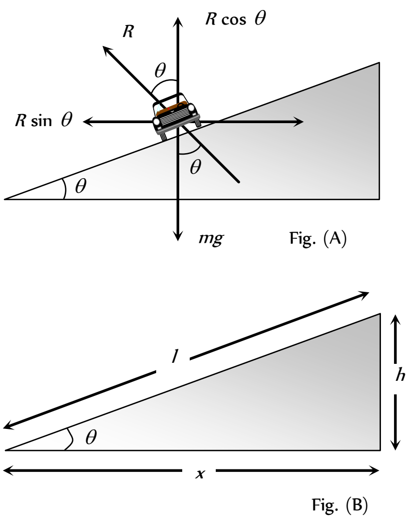


Fig : 3.34

In the figure (A) shown reaction *R* is resolved into two components, the component $R\cos\theta$ balances weight of vehicle

$$\therefore\ R\cos\theta = mg \quad ...(i)$$

and the horizontal component $R\sin\theta$ provides necessary centripetal force as it is directed towards centre of desired circle

Thus $R\sin\theta = \dfrac{mv^2}{r}$ ...(ii)

Dividing (ii) by (i), we have

$$\tan\theta = \frac{v^2}{rg} \quad ...(iii)$$

or $\tan\theta = \dfrac{\omega^2 r}{g} = \dfrac{v\omega}{g}$ ...(iv) [As $v = r\omega$]

If l = width of the road, h = height of the outer edge from the ground level then from the figure (B)

$\tan\theta = \dfrac{h}{x} = \dfrac{h}{l}$ ...(v) [since θ is very small]

From equation (iii), (iv) and (v)

$$\tan\theta = \frac{v^2}{rg} = \frac{\omega^2 r}{g} = \frac{v\omega}{g} = \frac{h}{l}$$

Note : ❑ If friction is also present between the tyres and road then $\dfrac{v^2}{rg} = \dfrac{\mu + \tan\theta}{1 - \mu\tan\theta}$

❑ Maximum safe speed on a banked frictional road $v = \sqrt{\dfrac{rg(\mu + \tan\theta)}{1 - \mu\tan\theta}}$

Overturning of Vehicle

When a car moves in a circular path with speed more than a certain maximum speed then it overturns even if friction is sufficient to avoid skidding and its inner wheel leaves the ground first

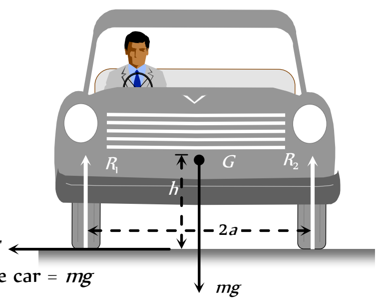


Fig : 3.35

Weight of the car = *mg*

Speed of the car = *v*

Radius of the circular path = *r*

Distance between the centre of wheels of the car = 2*a*

Height of the centre of gravity (*G*) of the car from the road level = *h*

Reaction on the inner wheel of the car by the ground = R_1

Reaction on the outer wheel of the car by the ground = R_2

When a car move in a circular path, horizontal friction force *F* provides the required centripetal force *i.e.*, $F = \dfrac{mv^2}{R}$ ...(i)

For rotational equilibrium, by taking the moment of forces R_1, R_2 and F about G

$$Fh + R_1 a = R_2 a \quad \text{...(ii)}$$

As there is no vertical motion so $R_1 + R_2 = mg$ ...(iii)

By solving (i), (ii) and (iii)

$$R_1 = \frac{1}{2}M\left[g - \frac{v^2 h}{ra}\right] \quad \text{...(iv)}$$

and $$R_2 = \frac{1}{2}M\left[g + \frac{v^2 h}{ra}\right] \quad \text{...(v)}$$

It is clear from equation (iv) that if v increases value of R_1 decreases and for $R_1 = 0$

$$\frac{v^2 h}{ra} = g \text{ or } v = \sqrt{\frac{gra}{h}}$$

i.e. the maximum speed of a car without overturning on a flat road is given by $v = \sqrt{\dfrac{gra}{h}}$

Motion of Charged Particle In Magnetic Field

When a charged particle having mass m, charge q enters perpendicularly in a magnetic field B with velocity v then it describes a circular path.

Because magnetic force (qvB) works in the perpendicular direction of v and it provides required centripetal force

Magnetic force = Centripetal force

$$qvB = \frac{mv^2}{r}$$

$\therefore$ radius of the circular path

$$r = \frac{mv}{qB}$$

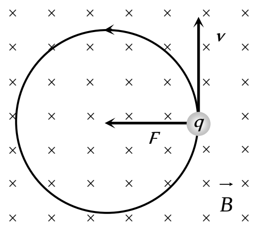
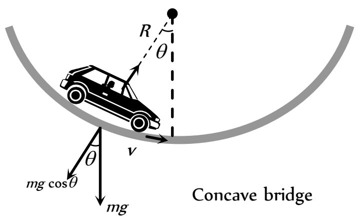


Fig : 3.36

Reaction of Road On Car

(1) When car moves on a concave bridge then



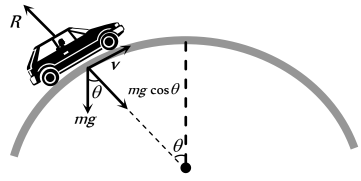
Concave bridge

Fig : 3.37

Centripetal force = $R - mg\cos\theta = \dfrac{mv^2}{r}$

and reaction $R = mg\cos\theta + \dfrac{mv^2}{r}$

(2) When car moves on a convex bridge



Convex bridge

Fig : 3.38

Centripetal force = $mg\cos\theta - R = \dfrac{mv^2}{r}$

and reaction $R = mg\cos\theta - \dfrac{mv^2}{r}$

Non-Uniform Circular Motion

If the speed of the particle in a horizontal circular motion changes with respect to time, then its motion is said to be non-uniform circular motion.

Consider a particle describing a circular path of radius r with centre at O. Let at an instant the particle be at P and $\vec{\upsilon}$ be its linear velocity and $\vec{\omega}$ be its angular velocity.

Then, $\vec{\upsilon} = \vec{\omega} \times \vec{r}$ ...(i)

Differentiating both sides of *w.r.t.* time t we have

$$\frac{d\vec{\upsilon}}{dt} = \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt} \quad \text{...(ii)}$$

Here, $\dfrac{d\vec{v}}{dt} = \vec{a}$, (Resultant acceleration)

$$\vec{a} = \vec{\alpha} \times \vec{r} + \vec{\omega} \times \vec{\upsilon}$$

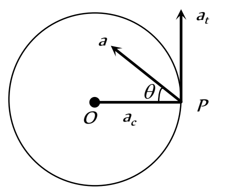


Fig : 3.39

$\dfrac{d\vec{\omega}}{dt} = \vec{\alpha}$ (Angular acceleration)

$$\vec{a} = \vec{a}_t + \vec{a}_c \quad \text{...(iii)}$$

$\dfrac{d\vec{r}}{dt} = \vec{\upsilon}$ (Linear velocity)

Thus the resultant acceleration of the particle at P has two component accelerations

(1) **Tangential acceleration :** $\vec{a_t} = \vec{\alpha} \times \vec{r}$

It acts along the tangent to the circular path at P in the plane of circular path.

According to right hand rule since $\vec{\alpha}$ and $\vec{r}$ are perpendicular to each other, therefore, the magnitude of tangential acceleration is given by

$$|\vec{a_t}| = |\vec{\alpha} \times \vec{r}| = \alpha r \sin 90^o = \alpha r.$$

(2) **Centripetal (Radial) acceleration :** $\vec{a_c} = \vec{\omega} \times \vec{v}$

It is also called centripetal acceleration of the particle at P.

It acts along the radius of the particle at P.

According to right hand rule since $\vec{\omega}$ and $\vec{\upsilon}$ are perpendicular to each other, therefore, the magnitude of centripetal acceleration is given by

$$|\vec{a}_c| = |\vec{\omega} \times \vec{\upsilon}| = \omega\upsilon \sin 90^o = \omega\upsilon = \omega(\omega r) = \omega^2 r = \upsilon^2 / r$$

Table 3.2 : Tangential and centripetal acceleration

Centripetal acceleration	Tangential acceleration	Net acceleration	Type of motion
$a_c = 0$	$a_t = 0$	$a = 0$	Uniform

			translatory motion
$a_c = 0$	$a_t \neq 0$	$a = a_t$	Accelerated translatory motion
$a_c \neq 0$	$a_t = 0$	$a = a_c$	Uniform circular motion
$a_c \neq 0$	$a_t \neq 0$	$a = \sqrt{a_c^2 + a_t^2}$	Non-uniform circular motion

Note : ❑ Here a_t governs the magnitude of $\vec{v}$ while $\vec{a}_c$ its direction of motion.

(3) **Force :** In non-uniform circular motion the particle simultaneously possesses two forces

Centripetal force : $F_c = ma_c = \dfrac{mv^2}{r} = mr\omega^2$

Tangential force : $F_t = ma_t$

Net force : $F_{net} = ma = m\sqrt{a_c^2 + a_t^2}$

Note : ❑ In non-uniform circular motion work done by centripetal force will be zero since $\vec{F}_c \perp \vec{v}$

❑ In non uniform circular motion work done by tangential force will not be zero since $F_t \neq 0$

❑ Rate of work done by net force in non-uniform circular motion = rate of work done by tangential force

i.e. $P = \dfrac{dW}{dt} = \vec{F}_t . \vec{v}$

Equations of Circular Motion

For accelerated motion	For retarded motion
$\omega_2 = \omega_1 + \alpha t$	$\omega_2 = \omega_1 - \alpha t$
$\theta = \omega_1 t + \frac{1}{2}\alpha t^2$	$\theta = \omega_1 t - \frac{1}{2}\alpha t^2$
$\omega_2^2 = \omega_1^2 + 2\alpha\theta$	$\omega_2^2 = \omega_1^2 - 2\alpha\theta$
$\theta_n = \omega_1 + \frac{\alpha}{2}(2n-1)$	$\theta_n = \omega_1 - \frac{\alpha}{2}(2n-1)$

Where

ω_1 = Initial angular velocity of particle

ω_2 = Final angular velocity of particle

α = Angular acceleration of particle

θ = Angle covered by the particle in time t

θ_n = Angle covered by the particle in n^{th} second

Motion in vertical circle

This is an example of non-uniform circular motion. In this motion body is under the influence of gravity of earth. When body moves from lowest point to highest point. Its speed decrease and becomes minimum at highest point. Total mechanical energy of the body remains conserved and *KE* converts into *PE* and vice versa.

(1) **Velocity at any point on vertical loop :** If u is the initial velocity imparted to body at lowest point then velocity of body at height h is given by

$$v = \sqrt{u^2 - 2gh} = \sqrt{u^2 - 2gl(1-\cos\theta)}$$

[As $h = l - l\cos\theta = l(1-\cos\theta)$]

where l is the length of the string

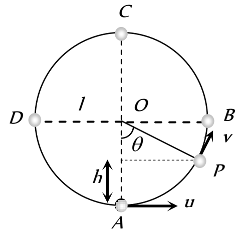


Fig : 3.40

(2) **Tension at any point on vertical loop** : Tension at general point *P*, According to Newton's second law of motion.

Net force towards centre = centripetal force

$T - mg\cos\theta = \dfrac{mv^2}{l}$

or $T = mg\cos\theta + \dfrac{mv^2}{l}$

$T = \dfrac{m}{l}[u^2 - gl(2 - 3\cos\theta)]$

[As $v = \sqrt{u^2 - 2gl(1-\cos\theta)}$]

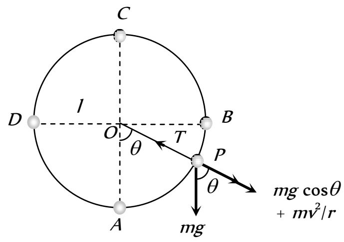


Fig : 3.41

Table 3.3 : Velocity and tension in a vertical loop

Position	Angle	Velocity	Tension
A	0°	u	$\frac{mu^2}{l} + mg$
B	90°	$\sqrt{u^2 - 2gl}$	$\frac{mu^2}{l} - 2mg$
C	180°	$\sqrt{u^2 - 4gl}$	$\frac{mu^2}{l} - 5mg$
D	270°	$\sqrt{u^2 - 2gl}$	$\frac{mu^2}{l} - 2mg$

It is clear from the table that : $T_A > T_B > T_C$ and $T_B = T_D$

$T_A - T_B = 3mg,$

$T_A - T_C = 6mg$

and $T_B - T_C = 3mg$

Table 3.4 : Various conditions for vertical motion

Velocity at lowest point	Condition
$u_A > \sqrt{5gl}$	Tension in the string will not be zero at any of the point and body will continue the circular motion.
$u_A = \sqrt{5gl},$	Tension at highest point C will be zero and body will just complete the circle.
$\sqrt{2gl} < u_A < \sqrt{5gl},$	Particle will not follow circular motion. Tension in string become zero somewhere between points *B* and *C* whereas velocity remain positive. Particle leaves circular path and follow parabolic trajectory.

$u_A = \sqrt{2gl}$	Both velocity and tension in the string becomes zero at B and particle will oscillate along semi-circular path.
$u_A < \sqrt{2gl}$	velocity of particle becomes zero between A and B but tension will not be zero and the particle will oscillate about the point A.

Note : ❑ *K.E.* of a body moving in horizontal circle is same throughout the path but the *K.E.* of the body moving in vertical circle is different at different places.

❑ If body of mass m is tied to a string of length l and is projected with a horizontal velocity u then :

Height at which the velocity vanishes is $h = \dfrac{u^2}{2g}$

Height at which the tension vanishes is $h = \dfrac{u^2 + gl}{3g}$

(3) **Critical condition for vertical looping :** If the tension at C is zero, then body will just complete revolution in the vertical circle. This state of body is known as critical state. The speed of body in critical state is called as critical speed.

From the above table 3.3 $T_c = \dfrac{mu^2}{l} - 5mg = 0$

$\Rightarrow u = \sqrt{5gl}$

It means to complete the vertical circle the body must be projected with minimum velocity of $\sqrt{5gl}$ at the lowest point.

Table 3.5 : Different variables in vertical loop

Quantity	Point A	Point B	Point C	Point D	Point P
Linear velocity (v)	$\sqrt{5gl}$	$\sqrt{3gl}$	$\sqrt{gl}$	$\sqrt{3gl}$	$\sqrt{gl(3+2\cos\theta)}$
Angular velocity (ω)	$\sqrt{\frac{5g}{l}}$	$\sqrt{\frac{3g}{l}}$	$\sqrt{\frac{g}{l}}$	$\sqrt{\frac{3g}{l}}$	$\sqrt{\frac{g}{l}(3+2\cos\theta)}$
Tension in String (T)	6 mg	3 mg	0	3 mg	$3mg(1+\cos\theta)$
Kinetic Energy (KE)	$\frac{5}{2}mgl$	$\frac{3}{2}mgl$	$\frac{1}{2}mgl$	$\frac{3}{2}mgl$	$\frac{mu^2}{l} - 5mg = 0$
Potential Energy (PE)	0	mgl	2 mgl	mgl	$mgl(1-\cos\theta)$
Total Energy (TE)	$\frac{5}{2}mgl$	$\frac{5}{2}mgl$	$\frac{5}{2}mgl$	$\frac{5}{2}mgl$	$\frac{5}{2}mgl$

(4) **Motion of a block on frictionless hemisphere :** A small block of mass m slides down from the top of a frictionless hemisphere of radius r. The component of the force of gravity ($mg\cos\theta$) provides required centripetal force but at point B it's circular motion ceases and the block lose contact with the surface of the sphere.

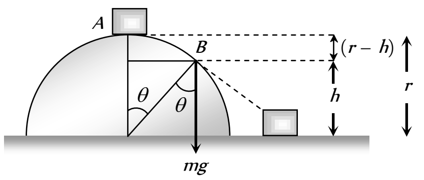


Fig : 3.42

For point B, by equating the forces

$$mg\cos\theta = \frac{mv^2}{r} \qquad \text{...(i)}$$

For point A and B, by law of conservation of energy

Total energy at point A = Total energy at point B

$K.E._A + P.E._A = K.E._B + P.E._B$

$$0 + mgr = \frac{1}{2}mv^2 + mgh \;\Rightarrow\; v = \sqrt{2g(r-h)} \qquad \text{...(ii)}$$

and from the given figure $h = r\cos\theta$ ...(iii)

By substituting the value of v and h from eq (ii) and (iii) in eq (i)

$$mg\left(\frac{h}{r}\right) = \frac{m}{r}\left(\sqrt{2g(r-h)}\right)^2 \Rightarrow h = 2(r-h) \;\Rightarrow h = \frac{2}{3}r$$

i.e. the block lose contact at the height of $\frac{2}{3}r$ from the ground.

and angle from the vertical can be given by $\cos\theta = \dfrac{h}{r} = \dfrac{2}{3}$

$\therefore\; \theta = \cos^{-1}\dfrac{2}{3}$.

Conical Pendulum

This is the example of uniform circular motion in horizontal plane.

A bob of mass m attached to a light and in-extensible string rotates in a horizontal circle of radius r with constant angular speed ω about the vertical. The string makes angle θ with vertical and appears tracing the surface of a cone. So this arrangement is called conical pendulum.

The force acting on the bob are tension and weight of the bob.

From the figure $T\sin\theta = \dfrac{mv^2}{r}$ ....(i)

and $T\cos\theta = mg$ ....(ii)

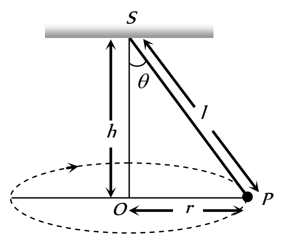


Fig : 3.43

(1) Tension in the string : $T = mg\sqrt{1+\left(\frac{v^2}{rg}\right)^2}$

$$T = \frac{mg}{\cos\theta} = \frac{mgl}{\sqrt{l^2-r^2}} \qquad \left[\text{As } \cos\theta = \frac{h}{l} = \frac{\sqrt{l^2-r^2}}{l}\right]$$

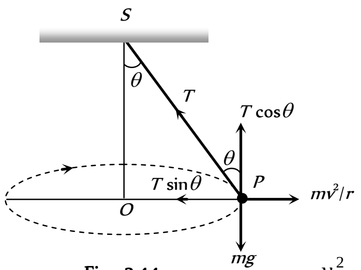


Fig : 3.44

(2) Angle of string from the vertical : $\tan\theta = \frac{v^2}{rg}$

(3) Linear velocity of the bob : $v = \sqrt{gr\tan\theta}$

(4) Angular velocity of the bob :

$$\omega = \sqrt{\frac{g}{r}\tan\theta} = \sqrt{\frac{g}{h}} = \sqrt{\frac{g}{l\cos\theta}}$$

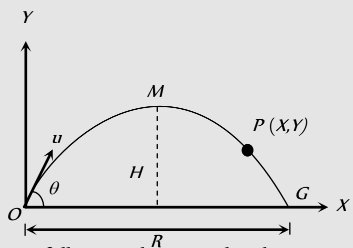
(5) Time period of revolution :

$$T_P = 2\pi\sqrt{\frac{l\cos\theta}{g}} = 2\pi\sqrt{\frac{h}{g}}$$

$$= 2\pi\sqrt{\frac{l^2-r^2}{g}} = 2\pi\sqrt{\frac{r}{g\tan\theta}}$$

Tips & Tricks

✍ Consider a projectile of mass m thrown with velocity u making angle θ with the horizontal. It is projected from the point O and returns to the ground at G. Also M is the highest point attained by it. (See figure).



(i) In going from O to M, following changes take place –

(a) Change in velocity $= u\sin\theta$

(b) Change in speed $= u(1-\cos\theta) = 2u\cos^2(\theta/2)$

(c) Change in momentum $= mu\sin\theta$

(d) Change (loss) in kinetic energy $= 1/2\, mu^2\sin^2\theta$

(e) Change (gain) in potential energy $= 1/2\, mu^2\sin^2\theta$

(f) Change in the direction of motion $= \angle\theta$

(ii) On return to the ground, that is in going from O to G, the following changes take place

(a) Change in speed = zero

(b) Change in velocity = $2u\sin\theta$

(c) Change in momentum = $2mu\sin\theta$

(d) Change in kinetic energy = zero

(e) Change in potential energy = zero

(f) Change in the direction of motion = $\angle 2\theta$

✍ (i) At highest point, the horizontal component of velocity is $v_x = u\cos\theta$ and vertical component of velocity v_y is zero.

(ii) At highest point, linear momentum of a particle

$m v_x = mu\cos\theta$.

(iii) Kinetic energy of the particle at the highest point = $\frac{1}{2}mv_x^2$

$= \frac{1}{2}mu^2\cos^2\theta$.

✍ At highest point, acceleration due to gravity acting vertically downward makes an angle of 90° with the horizontal component of the velocity of the projectile.

✍ At the highest point, momentum of the projectile thrown at an angle θ with horizontal is $p\cos\theta$ and K.E. = $(K.E.)_i\cos^2\theta$.

✍ In projectile motion, horizontal component $u\cos\theta$ of velocity u remains constant throughout, whereas vertical component $u\sin\theta$ changes and becomes zero at the highest point.

✍ The trajectory of a projectile is parabolic.

✍ For a projectile, time of flight and maximum height depend on the vertical component of the velocity of projection.

✍ The range of the projectile is maximum for the angle of projection $\theta = 45°$.

✍ The maximum range of the projectile is :

$$R_{\max} = \frac{u^2}{g}$$

✍ When the range is maximum, the height attained by the projectile is :

$$H = \frac{u^2}{4g} = \frac{R_{\max}}{4}$$

✍ When the range of the projectile is maximum, the time of flight is :

$$T = 2t = \frac{\sqrt{2}u}{g}$$

✍ The height attained by a projectile is maximum, when $\theta = 90°$.

$$H_{\max} = \frac{u^2}{2g}$$

It is twice that of height attained, when the range is maximum.

- The time of flight of the projectile is also largest for $\theta = 90°$.

$$T_{max} = \frac{2u}{g}$$

- The trajectory of the projectile is a symmetric parabola only when g is constant through out the motion and θ is not equal to 0°, 90° or 180°.

- If velocity of projection is made n times, the maximum height attained and the range become n times and the time of flight becomes n times the initial value.

- If the force acting on a particle is always perpendicular to the velocity of the particle, then the path of the particle is a circle. The centripetal force is always perpendicular to the velocity of the particle.

- If circular motion of the object is uniform, the object will possess only centripetal acceleration.

- If circular motion of the object is non-uniform, the object will possess both centripetal and transverse acceleration.

- When the particle moves along the circular path with constant speed, the angular velocity is also constant. But linear velocity, momentum as well as centripetal acceleration change in direction, although their magnitude remains unchanged.

- For circular motion of rigid bodies with uniform speed, the angular speed is same for all particles, but linear speed varies directly as the radius of the circular path described by the particle $(v \propto r)$.

- When a body rotates, all its particles describe circular paths about a line, called axis of rotation.

- The centre of the circle describe by the different particles of the rotating body lie on the axis of rotation.

- Centripetal force $F_c = ma_c$, $m\omega^2 r$ where m = mass of the body.

- Centripetal force is always directed towards the centre of the circular path.

- When a body rotates with uniform velocity, its different particles have centripetal acceleration directly proportional to the radius $(a_c \propto r)$.

- There can be no circular motion without centripetal force.

- Centripetal force can be mechanical, electrical or magnetic force.

- Planets go round the earth in circular orbits due to the centripetal force provided by gravitational force of the sun.

- Gravitational pull of earth provides centripetal force for the orbital motion of the moon and artificial satellites.

- Centripetal force cannot change the kinetic energy of the body.

- In uniform circular motion the magnitude of the centripetal acceleration remains constant whereas its direction changes continuously but always directed towards the centre.

- A pseudo force, that is equal and opposite to the centripetal force is called centrifugal force.

- The $\vec{\theta}$, $\vec{\omega}$ and $\vec{\alpha}$ are directed along the axis of the circular path. Their sense of direction is given by the right hand fist rule as follows : 'If we catch axis of rotation in right hand fist such that the fingers point in the direction of rotation, then the outstretched thumb gives the direction of $\vec{\theta}$, $\vec{\omega}$ and $\vec{\alpha}$

- $\vec{\theta}$, $\vec{\omega}$ and $\vec{\alpha}$ are called pseudo vectors or axial vectors.

- For circular motion we have –

(i) $\vec{r} \perp \vec{v}$ (ii) $\vec{r}$ antiparallel to $\vec{a}_c$

(iii) $\vec{a}_c \perp \vec{v}$ (iv) $\vec{a}_c \perp \vec{a}_t$

(v) $\vec{\theta}$, $\vec{\omega}$, $\vec{\alpha}$ are perpendicular to $\vec{r}, \vec{a}_c, \vec{a}_t, \vec{v}$

(vi) $\vec{r}, \vec{a}_c, \vec{a}_t$ and $\vec{v}$ lie in the same plane

Ordinary Thinking

Objective Questions

Uniform Circular Motion

1. If the body is moving in a circle of radius r with a constant speed v, its angular velocity is [CPMT 1975; RPET 1999]

(a) v^2/r (b) vr

(c) v/r (d) r/v

2. Two racing cars of masses m_1 and m_2 are moving in circles of radii r_1 and r_2 respectively. Their speeds are such that each makes a complete circle in the same duration of time t. The ratio of the angular speed of the first to the second car is

CBSE PMT 1999; UPSEAT 2000]

(a) $m_1 : m_2$ (b) $r_1 : r_2$

(c) 1 : 1 (d) $m_1 r_1 : m_2 r_2$

3. A cyclist turns around a curve at 15 miles/hour. If he turns at double the speed, the tendency to overturn is

[CPMT 1974; AFMC 2003]

(a) Doubled (b) Quadrupled

(c) Halved (d) Unchanged

4. A body of mass m is moving in a circle of radius r with a constant speed v. The force on the body is $\frac{mv^2}{r}$ and is directed towards the centre. What is the work done by this force in moving the body over half the circumference of the circle

(a) $\frac{mv^2}{r} \times \pi r$ (b) Zero

(c) $\frac{mv^2}{r^2}$ (d) $\frac{\pi r^2}{mv^2}$

5. If a particle moves in a circle describing equal angles in equal times, its velocity vector

[CPMT 1972, 74; JIPMER 1997]

(a) Remains constant

(b) Changes in magnitude

(c) Changes in direction

(d) Changes both in magnitude and direction

6. A stone of mass m is tied to a string of length l and rotated in a circle with a constant speed v. If the string is released, the stone flies [NCERT 1977]

(a) Radially outward

(b) Radially inward

(c) Tangentially outward

(d) With an acceleration $\frac{mv^2}{l}$

7. A body is moving in a circular path with a constant speed. It has

(a) A constant velocity

(b) A constant acceleration

(c) An acceleration of constant magnitude

(d) An acceleration which varies with time

8. A motor cyclist going round in a circular track at constant speed has

(a) Constant linear velocity (b) Constant acceleration

(c) Constant angular velocity (d) Constant force

9. A particle P is moving in a circle of radius 'a' with a uniform speed v. C is the centre of the circle and AB is a diameter. When passing through B the angular velocity of P about A and C are in the ratio [NCERT 1982]

(a) 1 : 1 (b) 1 : 2

(c) 2 : 1 (d) 4 : 1

10. A car moving on a horizontal road may be thrown out of the road in taking a turn [NCERT 1983]

(a) By the gravitational force

(b) Due to lack of sufficient centripetal force

(c) Due to rolling frictional force between tyre and road

(d) Due to the reaction of the ground

11. Two particles of equal masses are revolving in circular paths of radii r_1 and r_2 respectively with the same speed. The ratio of their centripetal forces is [NCERT 1980; MNR 1995; [NCERT 1984]

(a) $\frac{r_2}{r_1}$ (b) $\sqrt{\frac{r_2}{r_1}}$

(c) $\left(\frac{r_1}{r_2}\right)^2$ (d) $\left(\frac{r_2}{r_1}\right)^2$

12. A particle moves with constant angular velocity in a circle. During the motion its

(a) Energy is conserved

(b) Momentum is conserved

(c) Energy and momentum both are conserved

(d) None of the above is conserved

13. A stone tied to a string is rotated in a circle. If the string is cut, the stone flies away from the circle because [NCERT 1977; RPET 1999]

(a) A centrifugal force acts on the stone

(b) A centripetal force acts on the stone

(c) Of its inertia

(d) Reaction of the centripetal force

14. A body is revolving with a constant speed along a circle. If its direction of motion is reversed but the speed remains the same, then which of the following statement is true

(a) The centripetal force will not suffer any change in magnitude

(b) The centripetal force will have its direction reversed

(c) The centripetal force will not suffer any change in direction

(d) The centripetal force would be doubled

15. When a body moves with a constant speed along a circle

[CBSE PMT 1994; Orissa PMT 2004]

(a) No work is done on it

(b) No acceleration is produced in the body

(c) No force acts on the body

(d) Its velocity remains constant

16. A body of mass m moves in a circular path with uniform angular velocity. The motion of the body has constant

[MP PET 2003]

[CPMT 1972]

(a) Acceleration (b) Velocity

(c) Momentum (d) Kinetic energy

17. On a railway curve, the outside rail is laid higher than the inside one so that resultant force exerted on the wheels of the rail car by the tops of the rails will [NCERT 1975]

(a) Have a horizontal inward component

(b) Be vertical

(c) Equilibriate the centripetal force

(d) Be decreased

18. If the overbridge is concave instead of being convex, the thrust on the road at the lowest position will be

(a) $mg + \frac{mv^2}{r}$ (b) $mg - \frac{mv^2}{r}$

(c) $\frac{m^2v^2g}{r}$ (d) $\frac{v^2g}{r}$

19. A cyclist taking turn bends inwards while a car passenger taking same turn is thrown outwards. The reason is [NCERT 1972; CPMT 1974]

(a) Car is heavier than cycle

(b) Car has four wheels while cycle has only two

(c) Difference in the speed of the two

(d) Cyclist has to counteract the centrifugal force while in the case of car only the passenger is thrown by this force

20. A car sometimes overturns while taking a turn. When it overturns, it is [AFMC 1988; MP PMT 2003]

(a) The inner wheel which leaves the ground first

(b) The outer wheel which leaves the ground first

(c) Both the wheels leave the ground simultaneously

(d) Either wheel leaves the ground first

21. A tachometer is a device to measure [DPMT 1999]

(a) Gravitational pull (b) Speed of rotation

(c) Surface tension (d) Tension in a spring

22. Two bodies of mass 10 kg and 5 kg moving in concentric orbits of radii R and r such that their periods are the same. Then the ratio between their centripetal acceleration is [CBSE PMT 2001]

(a) R/r (b) r/R

(c) R^2/r^2 (d) r^2/R^2

23. The ratio of angular speeds of minute hand and hour hand of a watch is [MH CET 2002]

(a) 1 : 12 (b) 6 : 1

(c) 12 : 1 (d) 1 : 6

24. A car travels north with a uniform velocity. It goes over a piece of mud which sticks to the tyre. The particles of the mud, as it leaves the ground are thrown

(a) Vertically upwards (b) Vertically inwards

(c) Towards north (d) Towards south

25. An aircraft executes a horizontal loop with a speed of 150 *m/s* with its, wings banked at an angle of 12^o. The radius of the loop is $(g = 10\ m/s^2)$ [Pb. PET 2001]

(a) 10.6 *km* (b) 9.6 *km*

(c) 7.4 *km* (d) 5.8 *km*

26. A particle is moving in a horizontal circle with constant speed. It has constant [MP PMT 1987; AFMC 1993; CPMT 1997; MP PET 2000]

(a) Velocity (b) Acceleration

(c) Kinetic energy (d) Displacement

27. A motor cyclist moving with a velocity of 72 *km/hour* on a flat road takes a turn on the road at a point where the radius of curvature of the road is 20 *meters*. The acceleration due to gravity is 10 *m/sec*. In order to avoid skidding, he must not bend with respect to the vertical plane by an angle greater than

(a) $\theta = \tan^{-1} 6$ (b) $\theta = \tan^{-1} 2$

(c) $\theta = \tan^{-1} 25.92$ (d) $\theta = \tan^{-1} 4$

28. A train is moving towards north. At one place it turns towards north-east, here we observe that [AIIMS 1980]

(a) The radius of curvature of outer rail will be greater than that of the inner rail

(b) The radius of the inner rail will be greater than that of the outer rail

(c) The radius of curvature of one of the rails will be greater

(d) The radius of curvature of the outer and inner rails will be the same

29. The angular speed of a fly wheel making 120 *revolutions/minute* is [CBSE P

(a) $2\pi\ rad/s$ (b) $4\pi^2\ rad/s$

(c) $\pi\ rad/s$ (d) $4\pi\ rad/s$

30. A particle is moving on a circular path with constant speed, then its acceleration will be [RPET 2003]

(a) Zero

(b) External radial acceleration

(c) Internal radial acceleration

(d) Constant acceleration

31. A car is moving on a circular path and takes a turn. If R_1 and R_2 be the reactions on the inner and outer wheels respectively, then

(a) $R_1 = R_2$ (b) $R_1 < R_2$

(c) $R_1 > R_2$ (d) $R_1 \geq R_2$

32. A mass of 100 *gm* is tied to one end of a string 2 *m* long. The body is revolving in a horizontal circle making a maximum of 200 revolutions per min. The other end of the string is fixed at the centre of the circle of revolution. The maximum tension that the string can bear is (approximately) [MP PET 1993]

(a) 8.76 *N* (b) 8.94 *N*

(c) 89.42 *N* (d) 87.64 *N*

33. A road is 10 *m* wide. Its radius of curvature is 50 *m*. The outer edge is above the lower edge by a distance of 1.5 *m*. This road is most suited for the velocity

(a) 2.5 *m/sec* (b) 4.5 *m/sec*

(c) 6.5 *m/sec* (d) 8.5 *m/sec*

34. Certain neutron stars are believed to be rotating at about $1 rev/\sec$. If such a star has a radius of 20 *km*, the acceleration of an object on the equator of the star will be [NCERT 1982]

(a) $20 \times 10^8 m/\sec^2$ (b) $8 \times 10^5 m/\sec^2$

(c) $120 \times 10^5 m/\sec^2$ (d) $4 \times 10^8 m/\sec^2$

35. A particle revolves round a circular path. The acceleration of the particle is [MNR 1986; UPSEAT 1999]

(a) Along the circumference of the circle

(b) Along the tangent

(c) Along the radius

(d) Zero

36. The length of second's hand in a watch is 1 *cm*. The change in velocity of its tip in 15 seconds is [MP PMT 1987, 2003]

(a) Zero (b) $\frac{\pi}{30\sqrt{2}} cm/\sec$

(c) $\frac{\pi}{30} cm/\sec$ (d) $\frac{\pi\sqrt{2}}{30} cm/\sec$

37. A particle moves in a circle of radius 25 *cm* at two revolutions per second. The acceleration of the particle in m/s^2 is [MNR 1991; UPSEAT 2000; DPMT 1999; RPET 2003; Pb. PET 2004]

(a) π^2 (b) $8\pi^2$

(c) $4\pi^2$ (d) $2\pi^2$

38. An electric fan has blades of length 30 *cm* as measured from the axis of rotation. If the fan is rotating at 1200 *r.p.m.* The acceleration of a point on the tip of the blade is about [CBSE PMT 1990]

(a) 1600 $m/\sec^2$ (b) 4740 $m/\sec^2$

(c) 2370 $m/\sec^2$ (d) 5055 $m/\sec^2$

39. The force required to keep a body in uniform circular motion is [EAMCET 1982; AFMC 2003]

(a) Centripetal force (b) Centrifugal force

(c) Resistance (d) None of the above

40. Cream gets separated out of milk when it is churned, it is due to [EAMCET 1981]

(a) Gravitational force (b) Centripetal force

(c) Centrifugal force (d) Frictional force

41. A particle of mass m is executing uniform circular motion on a path of radius r. If p is the magnitude of its linear momentum. The radial force acting on the particle is [MP PET 1994]

(a) pmr (b) $\frac{rm}{p}$

(c) $\frac{mp^2}{r}$ (d) $\frac{p^2}{rm}$

42. A particle moves in a circular orbit under the action of a central attractive force inversely proportional to the distance 'r'. The speed of the particle is [CBSE PMT 1995]

(a) Proportional to r^2 (b) Independent of r

(c) Proportional to r (d) Proportional to $1/r$

43. Two masses M and m are attached to a vertical axis by weightless threads of combined length l. They are set in rotational motion in a horizontal plane about this axis with constant angular velocity ω. If the tensions in the threads are the same during motion, the distance of M from the axis is [MP PET 1995]

(a) $\frac{Ml}{M+m}$ (b) $\frac{ml}{M+m}$

(c) $\frac{M+m}{M}l$ (d) $\frac{M+m}{m}l$

44. A boy on a cycle pedals around a circle of 20 *metres* radius at a speed of $20 metres/\sec$. The combined mass of the boy and the cycle is 90 *kg*. The angle that the cycle makes with the vertical so that it may not fall is $(g = 9.8\, m/\sec^2)$ [MP PMT 1995]

(a) 60.25^o (b) 63.90^o

(c) 26.12^o (d) 30.00^o

45. The average acceleration vector for a particle having a uniform circular motion is [Kurukshetra CEE 1996]

(a) A constant vector of magnitude $\frac{v^2}{r}$

(b) A vector of magnitude $\frac{v^2}{r}$ directed normal to the plane of the given uniform circular motion

(c) Equal to the instantaneous acceleration vector at the start of the motion

(d) A null vector

46. Radius of the curved road on national highway is R. Width of the road is b. The outer edge of the road is raised by h with respect to inner edge so that a car with velocity v can pass safe over it. The value of h is [MP PMT 1996]

(a) $\frac{v^2 b}{Rg}$ (b) $\frac{v}{Rgb}$

(c) $\frac{v^2 R}{g}$ (d) $\frac{v^2 b}{R}$

47. When a particle moves in a uniform circular motion. It has

(a) Radial velocity and radial acceleration

(b) Tangential velocity and radial acceleration

(c) Tangential velocity and tangential acceleration

(d) Radial velocity and tangential acceleration

48. A motorcycle is going on an overbridge of radius R. The driver maintains a constant speed. As the motorcycle is ascending on the overbridge, the normal force on it [MP PET 1997]

(a) Increases (b) Decreases

(c) Remains the same (d) Fluctuates

49. A mass of 2 *kg is* whirled in a horizontal circle by means of a string at an initial speed of 5 *revolutions per minute*. Keeping the radius constant the tension in the string is doubled. The new speed is nearly [MP PMT/PET 1998; JIPMER 2000]

(a) 14 *rpm* (b) 10 *rpm*

(c) 2.25 *rpm* (d) 7 *rpm*

50. The magnitude of the centripetal force acting on a body of mass m executing uniform motion in a circle of radius r with speed v is [AFMC 1998

(a) mvr (b) mv^2/r

(c) v/r^2m (d) v/rm

51. A string breaks if its tension exceeds 10 *newtons*. A stone of mass 250 *gm* tied to this string of length 10 *cm* is rotated in a horizontal circle. The maximum angular velocity of rotation can be

(a) 20 *rad/s* (b) 40 *rad/s*

(c) 100 *rad/s* (d) 200 *rad/s*

52. A 500 *kg* car takes a round turn of radius 50 *m* with a velocity of 36 *km/hr*. The centripetal force is [KCET 2001; CBSE PMT 1999;

JIPMER 2001, 02]

(a) 250 N (b) 750 N

(c) 1000 N (d) 1200 N

53. A ball of mass 0.25 *kg* attached to the end of a string of length 1.96 *m* is moving in a horizontal circle. The string will break if the tension is more than 25 *N*. What is the maximum speed with which the ball can be moved

[CBSE PMT 1998]

(a) 14 *m/s* (b) 3 *m/s*

(c) 3.92 *m/s* (d) 5 *m/s*

54. A body of mass 5 *kg* is moving in a circle of radius $1\,m$ with an angular velocity of 2 *radian/sec*. The centripetal force is

[AIIMS 1998]

(a) 10 *N* (b) 20 *N*

(c) 30 *N* (d) 40 *N*

55. If a particle of mass m is moving in a horizontal circle of radius r with a centripetal force $(-k/r^2)$, the total energy is [EAMCET (Med.) 1995; AMU (Engg.) 2001]

(a) $-\frac{k}{2r}$ (b) $-\frac{k}{r}$

(c) $-\frac{2k}{r}$ (d) $-\frac{4k}{r}$

56. A stone of mass of 16 *kg* is attached to a string 144 *m* long and is whirled in a horizontal circle. The maximum tension the string can withstand is 16 *Newton*. The maximum velocity of revolution that can be given to the stone without breaking it, will be

(a) 20 ms^{-1} (b) 16 ms^{-1}

(c) 14 ms^{-1} (d) 12 ms^{-1}

57. A circular road of radius 1000 *m* has banking angle 45^o. The maximum safe speed of a car having mass 2000 *kg* will be, if the coefficient of friction between tyre and road is 0.5

[RPET 1997]

(a) 172 *m/s* (b) 124 *m/s*

(c) 99 *m/s* (d) 86 *m/s*

58. The second's hand of a watch has length 6 *cm*. Speed of end point and magnitude of difference of velocities at two perpendicular positions will be [RPET 1997]

(a) 6.28 and 0 *mm/s* (b) 8.88 and 4.44 *mm/s*

(c) 8.88 and 6.28 *mm/s* (d) 6.28 and 8.88 *mm/s*

59. A sphere of mass m is tied to end of a string of length l and rotated through the other end along a horizontal circular path with speed v. The work done in full horizontal circle is

[CPMT 1993; JIPMER 2000]

(a) 0 (b) $\left(\frac{mv^2}{l}\right).2\pi l$

(c) $mg\,.2\pi l$ (d) $\left(\frac{mv^2}{l}\right).(l)$

60. A body is whirled in a horizontal circle of radius 20 *cm*. It has angular velocity of 10 *rad/s*. What is its linear velocity at any point on circular path [CBSE PMT 1996]

(a) 10 *m/s* (b) 2 *m/s*

(c) 20 *m/s* (d) $\sqrt{2}$ *m/s*

61. Find the maximum velocity for skidding for a car moved on a circular track of radius 100 *m*. The coefficient of friction between the road and tyre is 0.2

[CPMT 1996; Pb. PMT 2001]

(a) 0.14 *m/s* (b) 140 *m/s*

(c) 1.4 *km/s* (d) 14 *m/s*

62. A car when passes through a convex bridge exerts a force on it which is equal to [AFMC 1997]

(a) $Mg+\frac{Mv^2}{r}$ (b) $\frac{Mv^2}{r}$

(c) Mg (d) None of these

63. The angular speed of seconds needle in a mechanical watch is

[RPMT 1999; CPMT 1997; MH CET 2000, 01; BHU 2000]

(a) $\frac{\pi}{30}$ *rad/s* (b) 2π *rad/s*

(c) π *rad/s* (d) $\frac{60}{\pi}$ *rad/s*

64. The angular velocity of a particle rotating in a circular orbit 100 times per minute is [SCRA 1998; DPMT 2000]

(a) 1.66 *rad/s* (b) 10.47 *rad/s*

(c) 10.47 *deg/s* (d) 60 *deg/s*

65. A body of mass 100 *g* is rotating in a circular path of radius *r* with constant velocity. The work done in one complete revolution is [SCRA 1994]

(a) 100*r J* (b) $(r/100)J$

(c) $(100/r)J$ (d) Zero

66. A particle comes round a circle of radius 1 *m* once. The time taken by it is 10 *sec*. The average velocity of motion is

[JIPMER 1999]

(a) 0.2 $\pi m/s$ (b) 2 $\pi m/s$

(c) 2 m/s (d) Zero

67. An unbanked curve has a radius of $60\,m$. The maximum speed at which a car can make a turn if the coefficient of static friction is 0.75, is [JIPMER 1999]

(a) 2.1 m/s (b) 14 m/s

(c) 21 m/s (d) 7 m/s

68. A wheel completes 2000 revolutions to cover the 9.5 *km*. distance. then the diameter of the wheel is [RPMT 1999]

(a) 1.5 *m* (b) 1.5 *cm*

(c) 7.5 *cm* (d) 7.5 *m*

69. A cycle wheel of radius 0.4 *m* completes one revolution in one *second* then the acceleration of a point on the cycle wheel will be

(a) 0.8 *m/s* (b) 0.4 *m/s*

(c) $1.6\pi^2\,m/s^2$ (d) $0.4\pi^2\,m/s^2$

70. The centripetal acceleration is given by [RPET 1999]

(a) *v/r* (b) *vr*

(c) *vr* (d) *v/r*

71. A cylindrical vessel partially filled with water is rotated about its vertical central axis. It's surface will [RPET 2000]

(a) Rise equally (b) Rise from the sides

(c) Rise from the middle (d) Lowered equally

72. If a particle covers half the circle of radius R with constant speed then [RPMT 2000]
(a) Momentum change is *mvr*
(b) Change in *K.E.* is 1/2 *mv*
(c) Change in *K.E.* is *mv*
(d) Change in *K.E.* is zero

73. An aeroplane is flying with a uniform speed of 100 *m/s* along a circular path of radius 100 *m*. the angular speed of the aeroplane will be [KCET 2000]
(a) 1 *rad/sec* (b) 2 *rad/sec*
(c) 3 *rad/sec* (d) 4 *rad/sec*

74. A body moves with constant angular velocity on a circle. Magnitude of angular acceleration [RPMT 2000]
(a) $r\omega$ (b) Constant
(c) Zero (d) None of the above

75. What is the value of linear velocity, if $\vec{\omega} = 3\hat{i} - 4\hat{j} + \hat{k}$ and $\vec{r} = 5\hat{i} - 6\hat{j} + 6\hat{k}$ [Pb. PMT 2000]
(a) $6\hat{i} + 2\hat{j} - 3\hat{k}$ (b) $-18\hat{i} - 13\hat{j} + 2\hat{k}$
(c) $4\hat{i} - 13\hat{j} + 6\hat{k}$ (d) $6\hat{i} - 2\hat{j} + 8\hat{k}$

76. A stone is tied to one end of a string 50 *cm* long is whirled in a horizontal circle with a constant speed. If the stone makes 10 revolutions in 20 *s*, what is the magnitude of acceleration of the stone [Pb. PMT 2000]
(a) 493 cm/s^2 (b) 720 cm/s^2
(c) 860 cm/s^2 (d) 990 cm/s^2

77. A 100 *kg* car is moving with a maximum velocity of 9 *m/s* across a circular track of radius 30 *m*. The maximum force of friction between the road and the car is [Pb. PMT 2000]
(a) 1000 *N* (b) 706 *N*
(c) 270 *N* (d) 200 *N*

78. The maximum speed of a car on a road–turn of radius 30 *m*, if the coefficient of friction between the tyres and the road is 0.4, will be
(a) 10.84 *m/sec* (b) 9.84 *m/sec*
(c) 8.84 *m/sec* (d) 6.84 *m/sec*

79. The angular velocity of a wheel is 70 *rad/sec*. If the radius of the wheel is 0.5 *m*, then linear velocity of the wheel is [MH CET 2000]
(a) 70 *m/s* (b) 35 *m/s*
(c) 30 *m/s* (d) 20 *m/s*

80. A cyclist goes round a circular path of circumference 34.3 *m* in $\sqrt{22}$ *sec*. the angle made by him, with the vertical, will be
(a) 45° (b) 40°
(c) 42° (d) 48°

81. A particle of mass M is moving in a horizontal circle of radius R with uniform speed V. When it moves from one point to a diametrically opposite point, its [CBSE PMT 1992]
(a) Kinetic energy changes by $MV^2/4$
(b) Momentum does not change
(c) Momentum changes by $2MV$
(d) Kinetic energy changes by MV^2

82. A ball of mass 0.1 *Kg.* is whirled in a horizontal circle of radius 1 *m.* by means of a string at an initial speed of 10 *R.P.M.* Keeping the radius constant, the tension in the string is reduced to one quarter of its initial value. The new speed is
(a) 5 *r.p.m.* (b) 10 *r.p.m.*
(c) 20 *r.p.m.* (d) 14 *r.p.m.*

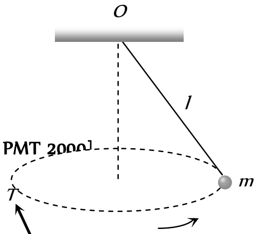
83. A cyclist riding the bicycle at a speed of $14\sqrt{3}$ ms^{-1} takes a turn around a circular road of radius $20\sqrt{3}$ *m* without skidding. Given g = 9.8 ms^{-2}, what is his inclination to the vertical
(a) 30° (b) 90°
(c) 45° (d) 60°

84. If a cycle wheel of radius 4 *m* completes one revolution in two seconds. Then acceleration of a point on the cycle wheel will be
(a) $\pi^2 m/s^2$ (b) $2\pi^2 m/s^2$
(c) $4\pi^2 m/s^2$ (d) $8\pi\, m/s^2$

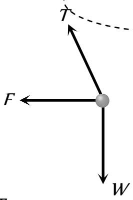
85. A bob of mass 10 *kg* is attached to wire 0.3 *m* long. Its breaking stress is 4.8×10^7 N/m^2. The area of cross section of the wire is 10^{-6} m^2. The maximum angular velocity with which it can be rotated in a horizontal circle [Pb. PMT 2001]
(a) 8 *rad/sec* (b) 4 *rad/sec*
(c) 2 *rad/sec* (d) 1 *rad/sec*

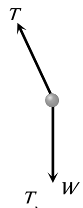
86. In uniform circular motion, the velocity vector and acceleration vector are [DCE 2000, 01, 03]
(a) Perpendicular to each other
(b) Same direction
(c) Opposite direction
(d) Not related to each other

87. A point mass m is suspended from a light thread of length l, fixed at O, is whirled in a horizontal circle at constant speed as shown. From your point of view, stationary with respect to the mass, the forces on the mass are [AMU (Med.) 2001]

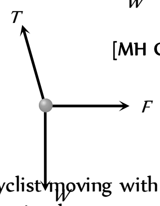


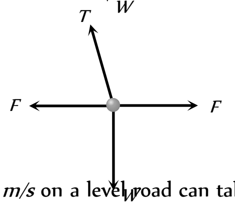
[CBSE PMT 2000]

(a)


(b)


[MH CET 2000]

(c)


(d)


88. If a cyclist moving with a speed of 4.9 *m/s* on a level road can take a sharp circular turn of radius 4 *m*, then coefficient of friction between the cycle tyres and road is [AIIMS 1999; AFMC 2001]
(a) 0.41 (b) 0.51
(c) 0.61 (d) 0.71

89. A car moves on a circular road. It describes equal angles about the centre in equal intervals of time. Which of the following statement about the velocity of the car is true [BHU 2001]

[MP PMT 2001]

(a) Magnitude of velocity is not constant
(b) Both magnitude and direction of velocity change
(c) Velocity is directed towards the centre of the circle
(d) Magnitude of velocity is constant but direction changes

90. A scooter is going round a circular road of radius 100 *m* at a speed of 10 *m/s*. The angular speed of the scooter will be [Pb. PMT 2002]

(a) 0.01 *rad/s* (b) 0.1 *rad/s*
(c) 1 *rad/s* (d) 10 *rad/s*

91. A particle of mass *M* moves with constant speed along a circular path of radius *r* under the action of a force *F*. Its speed is

(a) $\sqrt{\frac{rF}{m}}$ (b) $\sqrt{\frac{F}{r}}$
(c) $\sqrt{Fmr}$ (d) $\sqrt{\frac{F}{mr}}$

92. In an atom for the electron to revolve around the nucleus, the necessary centripetal force is obtained from the following force exerted by the nucleus on the electron [MP PET 2002]

(a) Nuclear force (b) Gravitational force
(c) Magnetic force (d) Electrostatic force

93. A particle moves with constant speed *v* along a circular path of radius *r* and completes the circle in time *T*. The acceleration of the particle is [Orissa JEE 2002]

(a) $2\pi v/T$ (b) $2\pi r/T$
(c) $2\pi r^2/T$ (d) $2\pi v^2/T$

94. The maximum velocity (in ms^{-1}) with which a car driver must traverse a flat curve of radius 150 *m* and coefficient of friction 0.6 to avoid skidding is [AIEEE 2002]

(a) 60 (b) 30
(c) 15 (d) 25

95. A car is moving with high velocity when it has a turn. A force acts on it outwardly because of [AFMC 2002]

(a) Centripetal force (b) Centrifugal force
(c) Gravitational force (d) All the above

96. A motor cycle driver doubles its velocity when he is having a turn. The force exerted outwardly will be [AFMC 2002]

(a) Double (b) Half
(c) 4 times (d) $\frac{1}{4}$ times

97. The coefficient of friction between the tyres and the road is 0.25. The maximum speed with which a car can be driven round a curve of radius 40 *m* without skidding is (assume $g = 10\ ms^{-2}$)

(a) 40 ms^{-1} (b) 20 ms^{-1}
(c) 15 ms^{-1} (d) 10 ms^{-1}

98. An athlete completes one round of a circular track of radius 10 *m* in 40 *sec*. The distance covered by him in 2 *min* 20 *sec* is

(a) 70 *m* (b) 140 *m*
(c) 110 *m* (d) 220 *m*

99. A proton of mass 1.6×10^{-27} *kg* goes round in a circular orbit of radius 0.10 *m* under a centripetal force of 4×10^{-13} *N*. then the frequency of revolution of the proton is about [Kerala (Med.) 2002]

(a) 0.08×10^{8} *cycles per sec*
(b) 4×10^{8} *cycles per sec*
(c) 8×10^{8} *cycles per sec*
(d) 12×10^{8} *cycles per sec*

100. A particle is moving in a circle with uniform speed *v*. In moving from a point to another diametrically opposite point [Orissa JEE 2003]

(a) The momentum changes by *mv*
(b) The momentum changes by 2*mv*
(c) The kinetic energy changes by (1/2)*mv*
(d) The kinetic energy changes by *mv*

101. In uniform circular motion [MP PMT 2002] [MP PMT 1994]

(a) Both the angular velocity and the angular momentum vary
(b) The angular velocity varies but the angular momentum remains constant
(c) Both the angular velocity and the angular momentum stay constant
(d) The angular momentum varies but the angular velocity remains constant

102. When a body moves in a circular path, no work is done by the force since, [KCET 2004]

(a) There is no displacement
(b) There is no net force
(c) Force and displacement are perpendicular to each other
(d) The force is always away from the centre

103. Which of the following statements is false for a particle moving in a circle with a constant angular speed [AIEEE 2004]

(a) The velocity vector is tangent to the circle
(b) The acceleration vector is tangent to the circle
(c) The acceleration vector points to the centre of the circle
(d) The velocity and acceleration vectors are perpendicular to each other

104. If a_r and a_t represent radial and tangential accelerations, the motion of a particle will be uniformly circular if [CPMT 2004]

(a) $a_r = 0$ and $a_t = 0$ (b) $a_r = 0$ but $a_t \neq 0$
(c) $a_r \neq 0$ but $a_t = 0$ (d) $a_r \neq 0$ and $a_t \neq 0$

105. A person with his hands in his pockets is skating on ice at the velocity of 10 *m/s* and describes a circle of radius 50 *m*. What is his inclination with vertical [Pb. PET 2000]

(a) $\tan^{-1}\left(\frac{1}{10}\right)$ [Kerala (Med.) 2002] (b) $\tan^{-1}\left(\frac{3}{5}\right)$
(c) $\tan^{-1}(1)$ (d) $\tan^{-1}\left(\frac{1}{5}\right)$

106. If the radius of curvature of the path of two particles of same masses are in the ratio 1 : 2, then in order to have constant centripetal force, their velocity, should be in the ratio of [Kerala (Med.) 2002] [Pb. PET 2000]

(a) 1 : 4 (b) 4 : 1
(c) $\sqrt{2} : 1$ (d) $1 : \sqrt{2}$

107. An object is moving in a circle of radius 100 *m* with a constant speed of 31.4 *m/s*. What is its average speed for one complete revolution [DCE 2004]

(a) Zero (b) 31.4 *m/s*
(c) 3.14 *m/s* (d) $\sqrt{2} \times 31.4\ m/s$

108. A body of mass 1 *kg* tied to one end of string is revolved in a horizontal circle of radius 0.1 *m* with a speed of 3 *revolution/sec*,

assuming the effect of gravity is negligible, then linear velocity, acceleration and tension in the string will be

(a) $1.88\ m/s, 35.5\ m/s^2, 35.5\ N$

(b) $2.88\ m/s, 45.5\ m/s^2, 45.5\ N$

(c) $3.88\ m/s, 55.5\ m/s^2, 55.5\ N$

(d) None of these

109. The acceleration of a train travelling with speed of 400 *m/s* as it goes round a curve of radius 160 *m*, is [Pb. PET 2003]

(a) $1\ km/s^2$ (b) $100\ m/s^2$

(c) $10\ m/s^2$ (d) $1\ m/s^2$

110. A car of mass 800 *kg* moves on a circular track of radius 40 *m*. If the coefficient of friction is 0.5, then maximum velocity with which the car can move is [MH CET 2004]

(a) 7 *m/s* (b) 14 *m/s*

(c) 8 *m/s* (d) 12 *m/s*

111. A 500 *kg* crane takes a turn of radius 50 *m* with velocity of 36 *km/hr*. The centripetal force is [Pb. PMT 2003]

(a) 1200 *N* (b) 1000 *N*

(c) 750 *N* (d) 250 *N*

112. Two bodies of equal masses revolve in circular orbits of radii R_1 and R_2 with the same period. Their centripetal forces are in the ratio [Kerala PMT 2004]

(a) $\left(\frac{R_2}{R_1}\right)^2$ (b) $\frac{R_1}{R_2}$

(c) $\left(\frac{R_1}{R_2}\right)^2$ (d) $\sqrt{R_1 R_2}$

113. In case of uniform circular motion which of the following physical quantity do not remain constant [Kerala PMT 2004]

(a) Speed (b) Momentum

(c) Kinetic energy (d) Mass

114. What happens to the centripetal acceleration of a revolving body if you double the orbital speed v and half the angular velocity ω

(a) The centripetal acceleration remains unchanged

(b) The centripetal acceleration is halved

(c) The centripetal acceleration is doubled

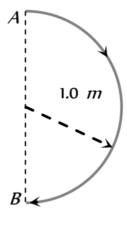
(d) The centripetal acceleration is quadrupled

115. A mass is supported on a frictionless horizontal surface. It is attached to a string and rotates about a fixed centre at an angular velocity ω_0. If the length of the string and angular velocity are doubled, the tension in the string which was initially T_0 is now

(a) T_0 (b) $T_0/2$

(c) $4T_0$ (d) $8T_0$

116. In 1.0 *s*, a particle goes from point *A* to point *B*, moving in a semicircle of radius 1.0 *m* (see figure). The magnitude of the average velocity is [IIT-JEE 1999]

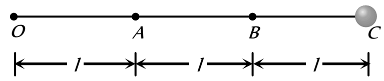


[DPMT 2003]

(a) $3.14 m/s$ (b) $2.0\ m/s$

(c) $1.0\ m/s$ (d) Zero

117. Three identical particles are joined together by a thread as shown in figure. All the three particles are moving in a horizontal plane. If the velocity of the outermost particle is *v*, then the ratio of tensions in the three sections of the string is



(a) 3 : 5 : 7 (b) 3 : 4 : 5

(c) 7 : 11 : 6 (d) 3 : 5 : 6

118. A particle is moving in a circle of radius *R* with constant speed *v*, if radius is double then its centripetal force to keep the same speed should be [BCECE 2005]

(a) Doubled

(b) Halved

(c) Quadrupled

(d) Unchanged

119. A stone ties to the end of a string $1m$ long is whirled in a horizontal circle with a constant speed. If the stone makes 22 revolution in 44 seconds, what is the magnitude and direction of acceleration of the stone [CBSE PMT 2005]

(a) $\frac{\pi^2}{4} ms^{-2}$ and direction along the radius towards the centre

(b) $\pi^2 ms^{-2}$ and direction along the radius away from the centre

(c) $\pi^2 ms^{-2}$ and direction along the radius towards the centre

(d) $\pi^2 ms^{-2}$ and direction along the tangent to the circle

120. A particle describes a horizontal circle in a conical funnel whose inner surface is smooth with speed of 0.5 *m/s*. What is the height of the plane of circle from vertex of the funnel ? [J&K CET 2005]

(a) 0.25 *cm* (b) 2 *cm*

(c) 4 *cm* (d) 2.5 *cm*

121. What is the angular velocity of earth [Orissa JEE 2005]

(a) $\frac{2\pi}{86400} rad/\sec$ (b) $\frac{2\pi}{3600} rad/\sec$

(c) $\frac{2\pi}{24} rad/\sec$ [AIIMS 1985] (d) $\frac{2\pi}{6400} rad/\sec$

122. If the length of the second's hand in a stop clock is 3 *cm* the angular velocity and linear velocity of the tip is [Kerala PET 2005]

(a) 0.2047 *rad/sec.*, 0.0314 *m/sec*

(b) 0.2547 *rad/sec.*, 0.314 *m/sec*

(c) 0.1472 *rad/sec.*, 0.06314 *m/sec*

(d) 0.1047 *rad/sec.*, 0.00314 *m/sec*

Non-uniform Circular Motion

1. In a circus stuntman rides a motorbike in a circular track of radius R in the vertical plane. The minimum speed at highest point of track will be

[CPMT 1979; JIPMER 1997; RPET 1999]

(a) $\sqrt{2gR}$ (b) $2gR$

(c) $\sqrt{3gR}$ (d) $\sqrt{gR}$

2. A block of mass m at the end of a string is whirled round in a vertical circle of radius R. The critical speed of the block at the top of its swing below which the string would slacken before the block reaches the top is [DCE 1999, 2001]

(a) Rg (b) $(Rg)^2$

(c) R/g (d) $\sqrt{Rg}$

3. A sphere is suspended by a thread of length l. What minimum horizontal velocity has to be imparted the ball for it to reach the height of the suspension

[ISM Dhanbad 1994]

(a) gl (b) $2gl$

(c) $\sqrt{gl}$ (d) $\sqrt{2gl}$

4. A bottle of sodawater is grasped by the neck and swing briskly in a vertical circle. Near which portion of the bottle do the bubbles collect

(a) Near the bottom

(b) In the middle of the bottle

(c) Near the neck

(d) Uniformly distributed in the bottle

5. A bucket tied at the end of a 1.6 m long string is whirled in a vertical circle with constant speed. What should be the minimum speed so that the water from the bucket does not spill, when the bucket is at the highest position (Take $g = 10 m/\sec^2$)

(a) 4 *m/sec* (b) 6.25 *m/sec*

(c) 16 *m/sec* (d) None of the above

6. A wheel is subjected to uniform angular acceleration about its axis. Initially its angular velocity is zero. In the first 2 *sec*, it rotates through an angle θ_1. In the next 2 *sec*, it rotates through an additional angle θ_2. The ratio of θ_2/θ_1 is

[AIIMS 1985]

(a) 1 (b) 2

(c) 3 (d) 5

7. A 1 *kg* stone at the end of 1 *m* long string is whirled in a vertical circle at constant speed of 4 *m/sec*. The tension in the string is 6 *N*, when the stone is at (g = 10 *m/sec*)

[AIIMS 1982]

(a) Top of the circle (b) Bottom of the circle

(c) Half way down (d) None of the above

8. A cane filled with water is revolved in a vertical circle of radius 4 *meter* and the water just does not fall down. The time period of revolution will be [CPMT 1985;

RPET 1995; UPSEAT 2002; MH CET 2002]

(a) 1 *sec* (b) 10 *sec*

(c) 8 *sec* (d) 4 *sec*

9. A 2 *kg* stone at the end of a string 1 *m* long is whirled in a vertical circle at a constant speed. The speed of the stone is 4 *m/sec*. The tension in the string will be 52 *N*, when the stone is

(a) At the top of the circle

(b) At the bottom of the circle

(c) Halfway down

(d) None of the above

10. A body slides down a frictionless track which ends in a circular loop of diameter D, then the minimum height h of the body in term of D so that it may just complete the loop, is

(a) $h = \frac{5D}{2}$ (b) $h = \frac{5D}{4}$

(c) $h = \frac{3D}{4}$ (d) $h = \frac{D}{4}$

11. A car is moving with speed 30 $m/\sec$ on a circular path of radius 500 *m*. Its speed is increasing at the rate of $2m/\sec^2$, What is the acceleration of the car

[MP PMT 2003; Roorkee 1982; RPET 1996; MH CET 2002]

(a) $2m/\sec^2$ (b) $2.7m/\sec^2$

(c) $1.8m/\sec^2$ (d) $9.8\,m/\sec^2$

12. The string of pendulum of length l is displaced through 90^o from the vertical and released. Then the minimum strength of the string in order to withstand the tension, as the pendulum passes through the mean position is

[MP PMT 1986]

(a) mg (b) $3mg$

[AIIMS 1987]

(c) $5\,mg$ (d) $6\,mg$

13. A weightless thread can support tension upto 30 *N*. A stone of mass 0.5 *kg* is tied to it and is revolved in a circular path of radius 2 *m* in a vertical plane. If $g = 10m/s^2$, then the maximum angular velocity of the stone will be

[MP PMT 1994]

(a) $5\ rad/s$ (b) $\sqrt{30}\,rad/s$

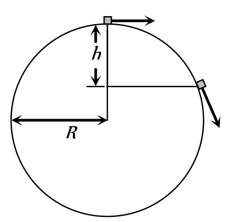
(c) $\sqrt{60}\,rad/s$ (d) $10\,rad/s$

14. A particle originally at rest at the highest point of a smooth vertical circle is slightly displaced. It will leave the circle at a vertical distance h below the highest point such that

(a) $h = R$

(b) $h = \frac{R}{3}$

(c) $h = \frac{R}{2}$



(d) $h = \dfrac{2R}{3}$

15. A heavy mass is attached to a thin wire and is whirled in a vertical circle. The wire is most likely to break

[MP PET 1997]

(a) When the mass is at the highest point of the circle
(b) When the mass is at the lowest point of the circle
(c) When the wire is horizontal
(d) At an angle of $\cos^{-1}(1/3)$ from the upward vertical

16. A weightless thread can bear tension upto 3.7 *kg wt*. A stone of mass 500 *gms* is tied to it and revolved in a circular path of radius 4 *m* in a vertical plane. If $g = 10\,ms^{-2}$, then the maximum angular velocity of the stone will be

[MP PMT/PET 1998]

(a) 4 *radians/sec* (b) 16 *radians/sec*
(c) $\sqrt{21}$ *radians/sec* (d) 2 *radians/sec*

17. The maximum velocity at the lowest point, so that the string just slack at the highest point in a vertical circle of radius l

[CPMT 1999; MH CET 2004]

(a) $\sqrt{gl}$ (b) $\sqrt{3gl}$
(c) $\sqrt{5gl}$ (d) $\sqrt{7gl}$

18. If the equation for the displacement of a particle moving on a circular path is given by $(\theta) = 2t^3 + 0.5$, where θ is in radians and t in seconds, then the angular velocity of the particle after 2 *sec* from its start is [AIIMS 1998]

(a) 8 *rad/sec* (b) 12 *rad/sec*
(c) 24 *rad/sec* (d) 36 *rad/sec*

19. A body of mass m hangs at one end of a string of length *l*, the other end of which is fixed. It is given a horizontal velocity so that the string would just reach where it makes an angle of 60^o with the vertical. The tension in the string at mean position is

(a) $2mg$ (b) mg
(c) $3mg$ (d) $\sqrt{3}mg$

20. In a vertical circle of radius *r*, at what point in its path a particle has tension equal to zero if it is just able to complete the vertical circle

(a) Highest point
(b) Lowest point
(c) Any point
(d) At a point horizontally from the centre of circle of radius *r*

21. The tension in the string revolving in a vertical circle with a mass m at the end which is at the lowest position

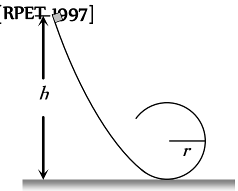
[EAMCET (Engg.) 1995; AIIMS 2001]

(a) $\dfrac{mv^2}{r}$ (b) $\dfrac{mv^2}{r} - mg$
(c) $\dfrac{mv^2}{r} + mg$ (d) mg

22. A hollow sphere has radius 6.4 *m*. Minimum velocity required by a motor cyclist at bottom to complete the circle will be

(a) 17.7 *m/s* (b) 10.2 *m/s*
(c) 12.4 *m/s* (d) 16.0 *m/s*

23. A block follows the path as shown in the figure from height h. If radius of circular path is r, then relation that holds good to complete full circle is [RPET 1997]



(a) $h < 5r/2$
(b) $h > 5r/2$
(c) $h = 5r/2$
(d) $h \geq 5r/2$

24. A pendulum bob on a 2 *m* string is displaced 60° from the vertical and then released. What is the speed of the bob as it passes through the lowest point in its path [JIPMER 1999]

(a) $\sqrt{2}\ m/s$ (b) $\sqrt{9.8}\ m/s$
(c) 4.43 *m/s* (d) $1/\sqrt{2}\ m/s$

25. A fan is making 600 revolutions per *minute*. If after some time it makes 1200 revolutions per *minute*, then increase in its angular velocity is [BHU 1999]

(a) $10\pi\, rad/\sec$ (b) $20\pi\, rad/\sec$
(c) $40\pi\, rad/\sec$ (d) $60\pi\, rad/\sec$

26. A particle is tied to 20*cm* long string. It performs circular motion in vertical plane. What is the angular velocity of string when the tension in the string at the top is zero

[RPMT 1999]

(a) $5\, rad/sec$ (b) $2\, rad/sec$
(c) $7.5\, rad/sec$ (d) $7\, rad/sec$

27. A stone tied with a string, is rotated in a vertical circle. The minimum speed with which the string has to be rotated

[CBSE PMT 1999]

(a) Is independent of the mass of the stone
(b) Is independent of the length of the string
(c) Decreases with increasing mass of the stone
(d) Decreases with increasing in length of the string

28. For a particle in a non-uniform accelerated circular motion

[ISM Dhanbad 1994] [AMU (Med.) 2000]

(a) Velocity is radial and acceleration is transverse only
(b) Velocity is transverse and acceleration is radial only
(c) Velocity is radial and acceleration has both radial and transverse components
(d) Velocity is transverse and acceleration has both radial and transverse components [EAMCET 1994]

29. A fighter plane is moving in a vertical circle of radius '*r*'. Its minimum velocity at the highest point of the circle will be

[MP PET 2000]

(a) $\sqrt{3gr}$ (b) $\sqrt{2gr}$
(c) $\sqrt{gr}$ (d) $\sqrt{gr/2}$

30. A ball is moving to and fro about the lowest point *A* of a smooth hemispherical bowl. If it is able to rise up to a height of 20 *cm* on either side of *A*, its speed at *A* must be (Take = 10 *m/s*, mass of the body 5 *g*) [JIPMER 2000]

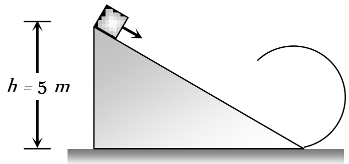
(a) 0.2 *m/s* (b) 2 *m/s*
(c) 4 *m/s* (d) 4.5 *ms*

31. A stone of mass *m* is tied to a string and is moved in a vertical circle of radius *r* making *n* revolutions per *minute*. The total tension in the string when the stone is at its lowest point is [RPET 1997]

(a) *mg* (b) $m(g + \pi n r^2)$

(c) $m(g+\pi n r)$ (d) $m\{g+(\pi^2 n^2 r)/900\}$

32. As per given figure to complete the circular loop what should be the radius if initial height is 5 *m* [RPET 2001]



(a) 4 *m*
(b) 3 *m*
(c) 2.5 *m*
(d) 2 *m*

33. A coin, placed on a rotating turn-table slips, when it is placed at a distance of 9 *cm* from the centre. If the angular velocity of the turn-table is trippled, it will just slip, if its distance from the centre is [CPMT 2001]

(a) 27 *cm* (b) 9 *cm*
(c) 3 *cm* (d) 1 *cm*

34. When a ceiling fan is switched off its angular velocity reduces to 50% while it makes 36 rotations. How many more rotation will it make before coming to rest (Assume uniform angular retardation)

(a) 18 (b) 12
(c) 36 (d) 48

35. A body crosses the topmost point of a vertical circle with critical speed. Its centripetal acceleration, when the string is horizontal will be [MH CET 2002]

(a) 6 *g* (b) 3 *g*
(c) 2 *g* (d) *g*

36. A simple pendulum oscillates in a vertical plane. When it passes through the mean position, the tension in the string is 3 times the weight of the pendulum bob. What is the maximum displacement of the pendulum of the string with respect to the vertical

(a) 30° (b) 45°
(c) 60° (d) 90°

37. A particle is moving in a vertical circle. The tensions in the string when passing through two positions at angles 30° and 60° from vertical (lowest position) are T_1 and T_2 respectively. then

(a) $T_1 = T_2$
(b) $T_2 > T_1$
(c) $T_1 > T_2$
(d) Tension in the string always remains the same

38. A particle is kept at rest at the top of a sphere of diameter 42 *m*. When disturbed slightly, it slides down. At what height '*h*' from the bottom, the particle will leave the sphere [BHU 2003]

(a) 14 *m* (b) 28 *m*
(c) 35 *m* (d) 7 *m*

39. The coordinates of a moving particle at any time '*t*' are given by $x = \alpha t$ and $y = \beta t$. The speed of the particle at time '*t*' is given by

(a) $\sqrt{\alpha^2+\beta^2}$ (b) $3t\sqrt{\alpha^2+\beta^2}$
(c) $3t^2\sqrt{\alpha^2+\beta^2}$ (d) $t^2\sqrt{\alpha^2+\beta^2}$

40. A small disc is on the top of a hemisphere of radius R. What is the smallest horizontal velocity v that should be given to the disc for it to leave the hemisphere and not slide down it ? [There is no friction] [CPMT 1991]

(a) $v=\sqrt{2gR}$ (b) $v=\sqrt{gR}$
(c) $v=\dfrac{g}{R}$ (d) $v=\sqrt{g^2R}$

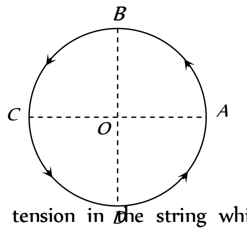
41. A body of mass 0.4 *kg* is whirled in a vertical circle making 2 *rev/sec*. If the radius of the circle is 2 *m*, then tension in the string when the body is at the top of the circle, is [CBSE PMT 1999]

(a) 41.56 *N* (b) 89.86 *N*
(c) 109.86 *N* (d) 115.86 *N*

42. A bucket full of water is revolved in vertical circle of radius 2*m*. What should be the maximum time-period of revolution so that the water doesn't fall off the bucket [AFMC 2004]

(a) 1 *sec* (b) 2 *sec*
(c) 3 *sec* (d) 4 *sec*

43. Figure shows a body of mass *m* moving with a uniform speed *v* along a circle of radius *r*. The change in velocity in going from *A* to *B* is [DPMT 2004]



(a) $v\sqrt{2}$
(b) $v/\sqrt{2}$ [KCET 2001]
(c) v
(d) zero

44. The maximum and minimum tension in the string whirling in a circle of radius 2.5 *m* with constant velocity are in the ratio 5 : 3 then its velocity is [Pb. PET 2003]

(a) $\sqrt{98}\ m/s$ (b) $7\ m/s$
(c) $\sqrt{490}\ m/s$ (d) $\sqrt{4.9}$

45. For a particle in circular motion the centripetal acceleration is [Orissa JEE 2002]

(a) Less than its tangential acceleration
(b) Equal to its tangential acceleration
(c) More than its tangential acceleration
(d) May be more or less than its tangential acceleration

46. A particle moves in a circular path with decreasing speed. Choose the correct statement. [Orissa JEE 2002] [IIT JEE 2005]

(a) Angular momentum remains constant
(b) Acceleration ($\vec{a}$) is towards the center
(c) Particle moves in a spiral path with decreasing radius
(d) The direction of angular momentum remains constant

47. A body of mass 1 *kg* is moving in a vertical circular path of radius 1*m*. The difference between the kinetic energies at its highest and lowest position is

(a) 20*J* (b) 10*J*
(c) $4\sqrt{5}J$ [AIEEE 2003] (d) $10(\sqrt{5}-1)J$

48. The angle turned by a body undergoing circular motion depends on time as $\theta=\theta_0+\theta_1 t+\theta_2 t^2$. Then the angular acceleration of the body is [Orissa JEE 2005]

(a) θ_1 (b) θ_2
(c) $2\theta_1$ (d) $2\theta_2$

Horizontal Projectile Motion

1. The maximum range of a gun on horizontal terrain is 16 *km*. If $g=10m/s^2$. What must be the muzzle velocity of the shell

(a) 200 *m/s* (b) 400 *m/s*

(c) 100 *m/s* (d) 50 *m/s*

2. A stone is just released from the window of a train moving along a horizontal straight track. The stone will hit the ground following [NCERT 1972; AFMC 1996; BHU 2000]

(a) Straight path (b) Circular path

(c) Parabolic path (d) Hyperbolic path

3. A bullet is dropped from the same height when another bullet is fired horizontally. They will hit the ground

(a) One after the other (b) Simultaneously

(c) Depends on the observer (d) None of the above

4. An aeroplane is flying at a constant horizontal velocity of 600 *km/hr* at an elevation of 6 *km* towards a point directly above the target on the earth's surface. At an appropriate time, the pilot releases a ball so that it strikes the target at the earth. The ball will appear to be falling [MP PET 1993]

(a) On a parabolic path as seen by pilot in the plane

(b) Vertically along a straight path as seen by an observer on the ground near the target

(c) On a parabolic path as seen by an observer on the ground near the target

(d) On a zig-zag path as seen by pilot in the plane

5. A bomb is dropped from an aeroplane moving horizontally at constant speed. When air resistance is taken into consideration, the bomb

[EAMCET (Med.) 1995; AFMC 1999]

(a) Falls to earth exactly below the aeroplane

(b) Fall to earth behind the aeroplane

(c) Falls to earth ahead of the aeroplane

(d) Flies with the aeroplane

6. A man projects a coin upwards from the gate of a uniformly moving train. The path of coin for the man will be

[RPET 1997]

(a) Parabolic (b) Inclined straight line

(c) Vertical straight line (d) Horizontal straight line

7. An aeroplane is flying horizontally with a velocity of 600 *km/h* at a height of 1960 *m*. When it is vertically at a point A on the ground, a bomb is released from it. The bomb strikes the ground at point *B*. The distance *AB* is

[CPMT 1996; JIPMER 2001, 02]

(a) 1200 *m* (b) 0.33 *km*

(c) 3.33 *km* (d) 33 *km*

8. A ball is rolled off the edge of a horizontal table at a speed of 4 *m/second*. It hits the ground after 0.4 *second*. Which statement given below is true [AMU (Med.) 1999]

(a) It hits the ground at a horizontal distance 1.6 *m* from the edge of the table

(b) The speed with which it hits the ground is 4.0 *m/second*

(c) Height of the table is 0.8 *m*

(d) It hits the ground at an angle of 60° to the horizontal

9. An aeroplane flying 490 *m* above ground level at 100 *m/s*, releases a block. How far on ground will it strike

[RPMT 2000]

(a) 0.1 *km* (b) 1 *km*

(c) 2 *km* (d) None

10. A body is thrown horizontally from the top of a tower of height 5 *m*. It touches the ground at a distance of 10 *m* from the foot of the tower. The initial velocity of the body is (g = 10 ms^{-2})

(a) 2.5 ms^{-1} (b) 5 ms^{-1}

(c) 10 ms^{-1} (d) 20 ms^{-1}

11. An aeroplane moving horizontally with a speed of 720 *km/h* drops a food pocket, while flying at a height of 396.9 *m*. the time taken by a food pocket to reach the ground and its horizontal range is (Take g = 9.8 *m/sec*) [AFMC 2001]

(a) 3 *sec* and 2000 *m* (b) 5 *sec* and 500 *m*

(c) 8 *sec* and 1500 *m* (d) 9 *sec* and 1800 *m*

12. A particle (*A*) is dropped from a height and another particle (*B*) is thrown in horizontal direction with speed of 5 *m/sec* from the same height. The correct statement is

[CBSE PMT 2002; Orissa JEE 2003]

(a) Both particles will reach at ground simultaneously

(b) Both particles will reach at ground with same speed

(c) Particle (*A*) will reach at ground first with respect to particle (*B*)

(d) Particle (*B*) will reach at ground first with respect to particle (*A*)

13. A particle moves in a plane with constant acceleration in a direction different from the initial velocity. The path of the particle will be [MP PMT 200

(a) A straight line (b) An arc of a circle

(c) A parabola (d) An ellipse

14. At the height 80 *m*, an aeroplane is moving with 150 *m/s*. A bomb is dropped from it so as to hit a target. At what distance from the target should the bomb be dropped (given g = 10 *m/s*)

(a) 605.3 *m* (b) 600 *m*

(c) 80 *m* (d) 230 *m*

15. A bomber plane moves horizontally with a speed of 500 *m/s* and a bomb released from it, strikes the ground in 10 sec. Angle at which it strikes the ground will be $(g = 10\ m/s^2)$

[MH CET 2003]

(a) $\tan^{-1}\left(\frac{1}{5}\right)$ (b) $\tan\left(\frac{1}{5}\right)$

(c) $\tan^{-1}(1)$ (d) $\tan^{-1}(5)$

16. A large number of bullets are fired in all directions with same speed v. What is the maximum area on the ground on which these bullets will spread

(a) $\pi\frac{v^2}{g}$ (b) $\pi\frac{v^4}{g^2}$

(c) $\pi^2\frac{v^4}{g^2}$ (d) $\pi^2\frac{v^2}{g^2}$

Oblique Projectile Motion

1. A projectile fired with initial velocity u at some angle θ has a range R. If the initial velocity be doubled at the same angle of projection, then the range will be

(a) $2R$ (b) $R/2$

(c) R (d) $4R$

2. If the initial velocity of a projectile be doubled, keeping the angle of projection same, the maximum height reached by it will

(a) Remain the same (b) Be doubled

(c) Be quadrupled (d) Be halved

3. In the motion of a projectile freely under gravity, its

(a) Total energy is conserved

(b) Momentum is conserved

(c) Energy and momentum both are conserved

(d) None is conserved

4. The range of a projectile for a given initial velocity is maximum when the angle of projection is 45^o. The range will be minimum, if the angle of projection is

(a) 90^o (b) 180^o

(c) 60^o (d) 75^o

5. The angle of projection at which the horizontal range and maximum height of projectile are equal is

[Kurukshetra CEE 1996; BCECE 2003; Pb. PET 2001]

(a) 45^o (b) $\theta = \tan^{-1}(0.25)$

(c) $\theta = \tan^{-1} 4$ or $(\theta = 76°)$ (d) 60^o

6. A ball is thrown upwards and it returns to ground describing a parabolic path. Which of the following remains constant

[BHU 1999; DPMT 2001; AMU (Engg.) 2000]

(a) Kinetic energy of the ball

(b) Speed of the ball

(c) Horizontal component of velocity

(d) Vertical component of velocity

7. At the top of the trajectory of a projectile, the directions of its velocity and acceleration are

(a) Perpendicular to each other

(b) Parallel to each other

(c) Inclined to each other at an angle of 45^o

(d) Antiparallel to each other

8. An object is thrown along a direction inclined at an angle of 45^o with the horizontal direction. The horizontal range of the particle is equal to [MP PMT 1985]

(a) Vertical height

(b) Twice the vertical height

(c) Thrice the vertical height

(d) Four times the vertical height

9. The height y and the distance x along the horizontal plane of a projectile on a certain planet (with no surrounding atmosphere) are given by $y = (8t - 5t^2)$ *meter* and $x = 6t$ *meter*, where t is in second. The velocity with which the projectile is projected is[CPMT 1981; MP PET

(a) 8 *m/sec*

(b) 6 *m/sec*

(c) 10 *m/sec*

(d) Not obtainable from the data

10. Referring to above question, the angle with the horizontal at which the projectile was projected is [CPMT 1981]

(a) $\tan^{-1}(3/4)$

(b) $\tan^{-1}(4/3)$

(c) $\sin^{-1}(3/4)$

(d) Not obtainable from the given data

11. Referring to the above two questions, the acceleration due to gravity is given by [CPMT 1981]

(a) $10\,m/\sec^2$ (b) $5m/\sec^2$

(c) $20\ m/\sec^2$ (d) $2.5\ m/\sec^2$

12. The range of a particle when launched at an angle of 15^o with the horizontal is 1.5 *km*. What is the range of the projectile when launched at an angle of 45^o to the horizontal

(a) 1.5 *km* (b) 3.0 *km*

(c) 6.0 *km* (d) 0.75 *km*

13. A cricketer hits a ball with a velocity $25\ m/s$ at 60^o above the horizontal. How far above the ground it passes over a fielder 50 m from the bat (assume the ball is struck very close to the ground)

(a) 8.2 *m* (b) 9.0 *m*

(c) 11.6 *m* (d) 12.7 *m*

14. A stone is projected from the ground with velocity $25\,m/s$. Two seconds later, it just clears a wall 5 *m* high. The angle of projection of the stone is $(g = 10m/\sec^2)$

(a) 30^o (b) 45^o

(c) 50.2^o (d) 60^o

15. Galileo writes that for angles of projection of a projectile at angles $(45 + \theta)$ and $(45 - \theta)$, the horizontal ranges described by the projectile are in the ratio of (if $\theta \le 45$)

[MP PET 1993]

(a) 2 : 1 (b) 1 : 2

(c) 1 : 1 (d) 2 : 3

16. A projectile thrown with a speed v at an angle θ has a range R on the surface of earth. For same v and θ, its range on the surface of moon will be

(a) $R/6$ (b) $6R$

(c) $R/36$ (d) $36R$

17. The greatest height to which a man can throw a stone is h. The greatest distance to which he can throw it, will be

(a) $\frac{h}{2}$ (b) h

(c) $2h$ (d) $3h$

18. The horizontal range is four times the maximum height attained by a projectile. The angle of projection is

[MP PET 1994; CBSE PMT 2000; RPET 2001]

(a) 90^o (b) 60^o

(c) 45^o (d) 30^o

19. A ball is projected with kinetic energy E at an angle of 45^o to the horizontal. At the highest point during its flight, its kinetic energy will be

[MP PMT 1994; CBSE PMT 1997, 2001; AIEEE 2002; Pb. PMT 2004; Orissa PMT 2004]

(a) Zero (b) $\frac{E}{2}$

(c) $\frac{E}{\sqrt{2}}$ (d) E

20. A particle of mass m is projected with velocity v making an angle of 45^o with the horizontal. The magnitude of the angular momentum of the particle about the point of projection when the particle is at its maximum height is (where $g =$ acceleration due to gravity)

[MP PMT 1994; MP PET 2001; Pb. PET 2004]

(a) Zero (b) $mv^3/(4\sqrt{2}g)$

(c) $mv^3/(\sqrt{2}g)$ (d) $mv^2/2g$

21. A particle reaches its highest point when it has covered exactly one half of its horizontal range. The corresponding point on the displacement time graph is characterised by

[AIIMS 1995]

(a) Negative slope and zero curvature

(b) Zero slope and negative curvature

(c) Zero slope and positive curvature

(d) Positive slope and zero curvature

22. At the top of the trajectory of a projectile, the acceleration is

(a) Maximum (b) Minimum

(c) Zero (d) g

23. When a body is thrown with a velocity u making an angle θ with the horizontal plane, the maximum distance covered by it in horizontal direction is [MP PMT 1996; RPET 2001]

(a) $\frac{u^2 \sin\theta}{g}$ (b) $\frac{u^2 \sin 2\theta}{2g}$

(c) $\frac{u^2 \sin 2\theta}{g}$ (d) $\frac{u^2 \cos 2\theta}{g}$

24. A football player throws a ball with a velocity of 50 *metre/sec* at an angle 30 degrees from the horizontal. The ball remains in the air for $(g = 10\, m/s^2)$

(a) 2.5 *sec* (b) 1.25 *sec*

(c) 5 *sec* (d) 0.625 *sec*

25. A body of mass 0.5 *kg* is projected under gravity with a speed of 98 *m/s* at an angle of 30^o with the horizontal. The change in momentum (in magnitude) of the body is

[MP PET 1997]

(a) $24.5\, N-s$ (b) $49.0\, N-s$

(c) $98.0\, N-s$ (d) $50.0\, N-s$

26. A body is projected at such an angle that the horizontal range is three times the greatest height. The angle of projection is [AIIM

(a) $25^o 8'$ (b) $33^o 7'$

(c) $42^o 8'$ (d) $53^o 8'$

27. A gun is aimed at a target in a line of its barrel. The target is released and allowed to fall under gravity at the same instant the gun is fired. The bullet will [EAMCET 1994]

(a) Pass above the target (b) Pass below the target

(c) Hit the target (d) Certainly miss the target

28. Two bodies are projected with the same velocity. If one is projected at an angle of 30^o and the other at an angle of 60^o to the horizontal, the ratio of the maximum heights reached is

[AIIMS 2001; EAMCET (Med.) 1995; Pb. PMT 2000]

(a) 3 : 1 (b) 1 : 3

(c) 1 : 2 (d) 2 : 1

29. If the range of a gun which fires a shell with muzzle speed V is R, then the angle of elevation of the gun is [AMU 1995]

(a) $\cos^{-1}\left(\frac{V^2}{Rg}\right)$ (b) $\cos^{-1}\left(\frac{gR}{V^2}\right)$

(c) $\frac{1}{2}\left(\frac{V^2}{Rg}\right)$ (d) $\frac{1}{2}\sin^{-1}\left(\frac{gR}{V^2}\right)$

30. If time of flight of a projectile is 10 seconds. Range is 500 *meters*. The maximum height attained by it will be

[RPMT 1997]

(a) 125 *m* (b) 50 *m*

(c) 100 *m* (d) 150 *m*

31. If a body *A* of mass *M* is thrown with velocity *V* at an angle of 30^o to the horizontal and another body *B* of the same mass is thrown with the same speed at an angle of 60^o to the horizontal. The ratio of horizontal range of *A* to *B* will be

[CBSE PMT 1992]

(a) 1 : 3 [Manipal MEE 1995] (b) 1 : 1

(c) $1:\sqrt{3}$ (d) $\sqrt{3}:1$

32. A bullet is fired from a cannon with velocity 500 *m/s*. If the angle of projection is 15^o and $g = 10 m/s^2$. Then the range is

(a) $25 \times 10^3 m$ (b) $12.5 \times 10^3 m$

(c) $50 \times 10^2 m$ (d) $25 \times 10^2 m$

33. A ball thrown by a boy is caught by another after 2 *sec*. some distance away in the same level. If the angle of projection is 30°, the velocity of projection is [JIPMER 1999]

(a) 19.6 *m/s* (b) 9.8 *m/s*

(c) 14.7 *m/s* (d) None of these

34. A particle covers 50 *m* distance when projected with an initial speed. On the same surface it will cover a distance, when projected with double the initial speed [RPMT 2000]

(a) 100 *m* (b) 150 *m*

(c) 200 *m* (d) 250 *m*

35. A ball is thrown upwards at an angle of 60° to the horizontal. It falls on the ground at a distance of 90 *m*. If the ball is thrown with

the same initial velocity at an angle 30°, it will fall on the ground at a distance of [BHU 2000]

(a) 30 m (b) 60 m
(c) 90 m (d) 120 m

36. Four bodies P, Q, R and S are projected with equal velocities having angles of projection 15°, 30°, 45° and 60° with the horizontal respectively. The body having shortest range is

(a) P (b) Q
(c) R (d) S

37. For a projectile, the ratio of maximum height reached to the square of flight time is (g = 10 ms^{-2})

[EAMCET (Med.) 2000]

(a) 5 : 4 (b) 5 : 2
(c) 5 : 1 (d) 10 : 1

38. A stone projected with a velocity u at an angle θ with the horizontal reaches maximum height H_1. When it is projected with velocity u at an angle $\left(\frac{\pi}{2}-\theta\right)$ with the horizontal, it reaches maximum height H_2. The relation between the horizontal range R of the projectile, H_1 and H_2 is

(a) $R = 4\sqrt{H_1 H_2}$ (b) $R = 4(H_1 - H_2)$

(c) $R = 4(H_1 + H_2)$ (d) $R = \frac{H_1^{\,2}}{H_2^{\,2}}$

39. An object is projected with a velocity of 20 m/s making an angle of 45° with horizontal. The equation for the trajectory is $h = Ax - Bx^2$ where h is height, x is horizontal distance, A and B are constants. The ratio $A : B$ is (g = 10 ms^{-2})

[EAMCET 2001]

(a) 1 : 5 (b) 5 : 1
(c) 1 : 40 (d) 40 : 1

40. Which of the following sets of factors will affect the horizontal distance covered by an athlete in a long–jump event

(a) Speed before he jumps and his weight
(b) The direction in which he leaps and the initial speed
(c) The force with which he pushes the ground and his speed
(d) None of these

41. A ball thrown by one player reaches the other in 2 sec. the maximum height attained by the ball above the point of projection will be about [Pb. PMT 2002]

(a) 10 m (b) 7.5 m
(c) 5 m (d) 2.5 m

42. In a projectile motion, velocity at maximum height is

[AIEEE 2002]

(a) $\frac{u\cos\theta}{2}$ (b) $u\cos\theta$

(c) $\frac{u\sin\theta}{2}$ (d) None of these

43. If two bodies are projected at 30° and 60° respectively, with the same velocity, then [JIPMER 2002; CBSE PMT 2000]

(a) Their ranges are same
(b) Their heights are same
(c) Their times of flight are same
(d) All of these

44. A body is thrown with a velocity of 9.8 m/s making an angle of 30° with the horizontal. It will hit the ground after a time

[JIPMER 2001, 2002; KCET 2001]

(a) 1.5 s (b) 1 s
(c) 3 s (d) 2 s

45. The equation of motion of a projectile are given by x = 36 t *metre* and $2y$ = 96 t – 9.8 t^2 *metre*. The angle of projection is

[EAMCET (Engg.) 2000]

(a) $\sin^{-1}\left(\frac{4}{5}\right)$ (b) $\sin^{-1}\left(\frac{3}{5}\right)$

(c) $\sin^{-1}\left(\frac{4}{3}\right)$ (d) $\sin^{-1}\left(\frac{3}{4}\right)$

46. For a given velocity, a projectile has the same range R for two angles of projection if t_1 and t_2 are the times of flight in the two cases then[KCET 2003;

(a) $t_1 t_2 \propto R^2$ (b) $t_1 t_2 \propto R$

(c) $t_1 t_2 \propto \frac{1}{R}$ (d) $t_1 t_2 \propto \frac{1}{R^2}$

47. A body of mass m is thrown upwards at an angle θ with the horizontal with velocity v. While rising up the velocity of the mass after t *seconds* will be [AMU (Engg.) 1999]

[EAMCET 2000]

(a) $\sqrt{(v\cos\theta)^2 + (v\sin\theta)^2}$

(b) $\sqrt{(v\cos\theta - v\sin\theta)^2 - gt}$

(c) $\sqrt{v^2 + g^2t^2 - (2v\sin\theta)gt}$

(d) $\sqrt{v^2 + g^2t^2 - (2v\cos\theta)gt}$

48. A cricketer can throw a ball to a maximum horizontal distance of 100 m. With the same effort, he throws the ball vertically upwards. The maximum height attained by the ball is

(a) 100 m (b) 80 m
(c) 60 m (d) 50 m

49. A cricketer can throw a ball to a maximum horizontal distance of 100 m. The speed with which he throws the ball is (to the nearest integer) [Kerala (Med.) 2002]

[AMU (Engg.) 2001]

(a) 30 ms^{-1} (b) 42 ms^{-1}
(c) 32 ms^{-1} (d) 35 ms^{-1}

50. A ball is projected with velocity V_o at an angle of elevation 30°. Mark the correct statement [MP PMT 2004]

(a) Kinetic energy will be zero at the highest point of the trajectory
(b) Vertical component of momentum will be conserved
(c) Horizontal component of momentum will be conserved
(d) Gravitational potential energy will be minimum at the highest point of the trajectory

51. Neglecting the air resistance, the time of flight of a projectile is determined by [J & K CET 2004]

(a) $U_{vertical}$

(b) $U_{horizontal}$

(c) $U = U^2{}_{vertical} + U^2{}_{horizontal}$

(d) $U = U(U^2{}_{vertical} + U^2{}_{horizontal})^{1/2}$

52. A ball is thrown from a point with a speed v_o at an angle of projection θ. From the same point and at the same instant a person starts running with a constant speed $v_o/2$ to catch the ball. Will the person be able to catch the ball? If yes, what should be the angle of projection [AIEEE 2004]

(a) Yes, 60^o (b) Yes, 30^o
(c) No (d) Yes, 45^o

53. A stone is thrown at an angle θ to the horizontal reaches a maximum height H. Then the time of flight of stone will be [BCECE 2004]

(a) $\sqrt{\frac{2H}{g}}$ (b) $2\sqrt{\frac{2H}{g}}$

(c) $\frac{2\sqrt{2H\sin\theta}}{g}$ (d) $\frac{\sqrt{2H\sin\theta}}{g}$

54. The horizontal range of a projectile is $4\sqrt{3}$ times its maximum height. Its angle of projection will be [J & K CET 2004; DPMT 2003]

(a) 45^o (b) 60^o

(c) 90^o (d) 30^o

55. A ball is projected upwards from the top of tower with a velocity $50\ ms^{-1}$ making an angle 30^o with the horizontal. The height of tower is 70 *m*. After how many seconds from the instant of throwing will the ball reach the ground [DPMT 2004]

(a) 2 *s* (b) 5 *s*

(c) 7 *s* (d) 9 *s*

56. Two bodies are thrown up at angles of 45 and 60, respectively, with the horizontal. If both bodies attain same vertical height, then the ratio of velocities with which these are thrown is

(a) $\sqrt{\frac{2}{3}}$ (b) $\frac{2}{\sqrt{3}}$

(c) $\sqrt{\frac{3}{2}}$ (d) $\frac{\sqrt{3}}{2}$

57. At what point of a projectile motion acceleration and velocity are perpendicular to each other [Orissa JEE 2005]

(a) At the point of projection

(b) At the point of drop

(c) At the topmost point

(d) Any where in between the point of projection and topmost point

58. An object is projected at an angle of 45° with the horizontal. The horizontal range and the maximum height reached will be in the ratio. [Kerala PET 2005]

(a) 1 : 2 (b) 2 : 1

(c) 1 : 4 (d) 4 : 1

59. The maximum horizontal range of a projectile is 400 *m*. The maximum value of height attained by it will be [AFMC 2005]

(a) 100 *m* (b) 200 *m*

(c) 400 *m* (d) 800 *m*

Critical Thinking

Objective Questions

1. A particle is acted upon by a force of constant magnitude which is always perpendicular to the velocity of the particle. The motion of the particle takes place in a plane. It follows that

(a) Velocity is constant

(b) Acceleration is constant

(c) Kinetic energy is constant

(d) It moves in a circular path

2. A tube of length L is filled completely with an incompressible liquid of mass M and closed at both the ends. The tube is then rotated in a horizontal plane about one of its ends with a uniform angular velocity ω. The force exerted by the liquid at the other end is [IIT 1992]

(a) $\frac{ML\omega^2}{2}$ (b) $ML\omega^2$

(c) $\frac{ML\omega^2}{4}$ (d) $\frac{ML^2\omega^2}{2}$

3. The kinetic energy k of a particle moving along a circle of radius R depends on the distance covered s as $k = as^2$ where a is a constant. The force acting on the particle is [MNR 1992; JIPMER 2001, 02; AMU (Engg.) 1999]

(a) $2a\frac{s^2}{R}$ (b) $2as\left(1+\frac{s^2}{R^2}\right)^{1/2}$

(c) $2as$ [DPMT 2005] (d) $2a\frac{R^2}{s}$

4. A car is moving in a circular horizontal track of radius 10 *m* with a constant speed of 10 *m/sec*. A plumb bob is suspended from the roof of the car by a light rigid rod of length 1.00 *m*. The angle made by the rod with track is [IIT 1992]

(a) Zero (b) 30^o

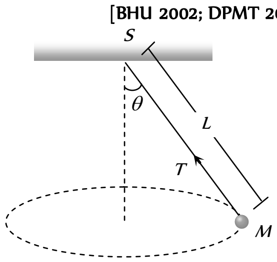
(c) 45^o (d) 60^o

5. A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration a_c is varying with time t as, $a_c = k^2rt^2$, The power delivered to the particle by the forces acting on it is [IIT 1994]

(a) $2\pi mk^2r^2t$ (b) mk^2r^2t

(c) $\frac{mk^4r^2t^5}{3}$ (d) Zero

6. A string of length L is fixed at one end and carries a mass M at the other end. The string makes $2/\pi$ revolutions per *second* around the vertical axis through the fixed end as shown in the figure, then tension in the string is [BHU 2002; DPMT 2004]



(a) *ML*

(b) 2 *ML*

(c) 4 *ML*

(d) 16 *ML*

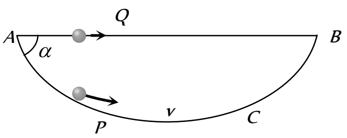
7. A stone of mass 1 *kg* tied to a light inextensible string of length $L = \frac{10}{3}\,m$ is whirling in a circular path of radius L in a vertical plane. [IIT 1987] The ratio of the maximum tension in the string to the

minimum tension in the string is 4 and if g is taken to be $10 m/\sec^2$, the speed of the stone at the highest point of the circle is [CBSE PMT 1990]

(a) $20 m/\sec$ (b) $10\sqrt{3} m/\sec$

(c) $5\sqrt{2}\ m/\sec$ (d) $10 m/\sec$

8. A particle P is sliding down a frictionless hemispherical bowl. It passes the point A at $t = 0$. At this instant of time, the horizontal component of its velocity is v. A bead Q of the same mass as P is ejected from A at $t = 0$ along the horizontal string AB (see figure) with the speed v. Friction between the bead and the string may be neglected. Let t_P and t_Q be the respective time taken by P and Q to reach the point B. Then



(a) $t_P < t_Q$

(b) $t_P = t_Q$

(c) $t_P > t_Q$

(d) All of these

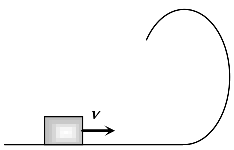
9. A long horizontal rod has a bead which can slide along its length, and initially placed at a distance L from one end A of the rod. The rod is set in angular motion about A with constant angular acceleration α. If the coefficient of friction between the rod and the bead is μ, and gravity is neglected, then the time after which the bead starts slipping is

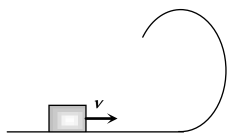
[IIT-JEE Screening 2000]

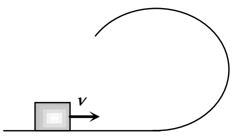
(a) $\sqrt{\frac{\mu}{\alpha}}$ (b) $\frac{\mu}{\sqrt{\alpha}}$

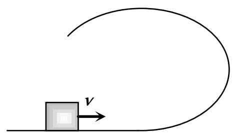
(c) $\frac{1}{\sqrt{\mu\alpha}}$ (d) Infinitesimal

10. A small block is shot into each of the four tracks as shown below. Each of the tracks rises to the same height. The speed with which the block enters the track is the same in all cases. At the highest point of the track, the normal reaction is maximum in

(a) 

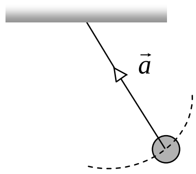
(b) 

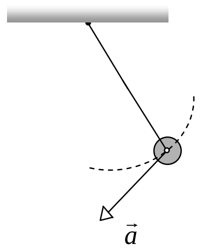
(c) 

(d) 

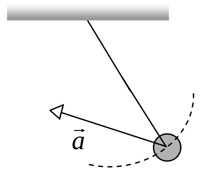
11. A simple pendulum is oscillating without damping. When the displacement of the bob is less than maximum, its acceleration vector $\vec{a}$ is correctly shown in

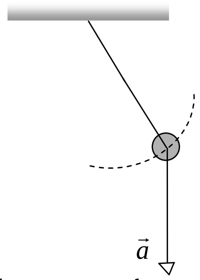
[IIT-JEE Screening 2002]





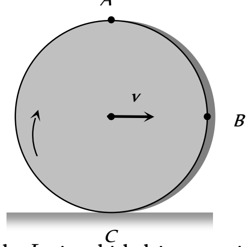
(a) (b)

(c) 

(d) 

[IIT 1993]

12. A solid disc rolls clockwise without slipping over a horizontal path with a constant speed υ. Then the magnitude of the velocities of points A, B and C (see figure) with respect to a standing observer are respectively [UPSEAT 2002]



(a) υ, υ and υ

(b) $2\upsilon, \sqrt{2}\upsilon$ and zero

(c) $2\upsilon, 2\upsilon$ and zero

(d) $2\upsilon, \sqrt{2}\upsilon$ and $\sqrt{2}\upsilon$

13. A stone tied to a string of length L is whirled in a vertical circle with the other end of the string at the centre. At a certain instant of time, the stone is at its lowest position and has a speed u. The magnitude of the change in its velocity as it reaches a position where the string is horizontal is

[IIT 1998; CBSE PMT 2004]

(a) $\sqrt{u^2 - 2gL}$ (b) $\sqrt{2gL}$

(c) $\sqrt{u^2 - gl}$ (d) $\sqrt{2(u^2 - gL)}$

14. The driver of a car travelling at velocity v suddenly see a broad wall in front of him at a distance d. He should

[IIT 1977]

[IIT-JEE Screening 2001]

(a) Brake sharply (b) Turn sharply

(c) (a) and (b) both (d) None of the above

15. Four persons K, L, M and N are initially at the corners of a square of side of length d. If every person starts moving, such that K is always headed towards L, L towards M, M is headed directly towards N and N towards K, then the four persons will meet after [IIT 1984]

(a) $\frac{d}{v}$ *sec* (b) $\frac{\sqrt{2}d}{v}$ *sec*

(c) $\frac{d}{\sqrt{2}v}$ *sec* (d) $\frac{d}{2v}$ *sec*

16. The coordinates of a particle moving in a plane are given by $x(t) = a\cos(pt)$ and $y(t) = b\sin(pt)$ where $a, b(< a)$ and p are positive constants of appropriate dimensions. Then

[IIT-JEE 1999]

(a) The path of the particle is an ellipse

(b) The velocity and acceleration of the particle are normal to each other at $t = \pi/(2p)$

(c) The acceleration of the particle is always directed towards a focus

(d) The distance travelled by the particle in time interval $t=0$ to $t=\pi/(2p)$ is a

17. A particle is moving eastwards with velocity of 5 *m/s*. In 10 *sec* the velocity changes to 5 *m/s* northwards. The average acceleration in this time is

[IIT 1982; AFMC 1999; Pb PET 2000; JIPMER 2001, 02]

(a) Zero

(b) $\frac{1}{\sqrt{2}}\ m/s^2$ toward north-west

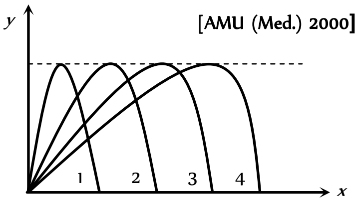
(c) $\frac{1}{\sqrt{2}}\ m/s^2$ toward north-east

(d) $\frac{1}{2}\ m/s^2$ toward north-west

Graphical Questions

1. Figure shows four paths for a kicked football. Ignoring the effects of air on the flight, rank the paths according to initial horizontal velocity component, highest first

[AMU (Med.) 2000]



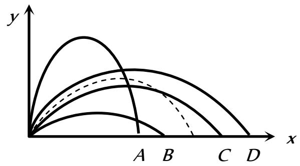
(a) 1, 2, 3, 4

(b) 2, 3, 4, 1

(c) 3, 4, 1, 2

(d) 4, 3, 2, 1

2. The path of a projectile in the absence of air drag is shown in the figure by dotted line. If the air resistance is not ignored then which one of the path shown in the figure is appropriate for the projectile



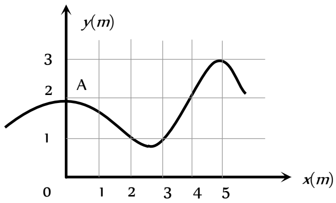
(a) B

(b) A

(c) D

(d) C

3. The trajectory of a particle moving in vast maidan is as shown in the figure. The coordinates of a position *A* are (0,2). The coordinates of another point at which the instantaneous velocity is same as the average velocity between the points are



(a) (1, 4)

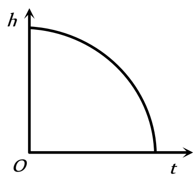
(b) (5, 3)

(c) (3, 4)

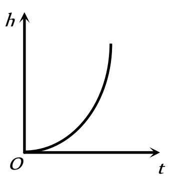
(d) (4, 1)

4. Which of the following is the graph between the height (h) of a projectile and time (t), when it is projected from the ground

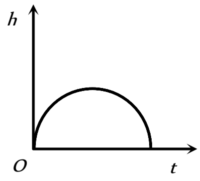
(a)



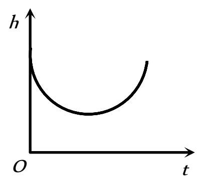
(b)



(c)

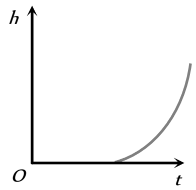


(d)

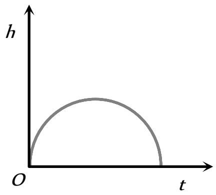


5. Which of the following is the altitude-time graph for a projectile thrown horizontally from the top of the tower

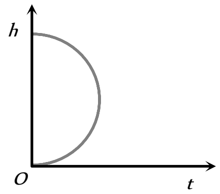
(a)



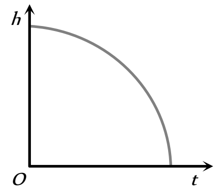
(b)



(c)



(d)



Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

(a) If both assertion and reason are true and the reason is the correct explanation of the assertion.

(b) If both assertion and reason are true but reason is not the correct explanation of the assertion.

(c) If assertion is true but reason is false.

(d) If the assertion and reason both are false.

(e) If assertion is false but reason is true.

1. Assertion : In projectile motion, the angle between the instantaneous velocity and acceleration at the highest point is 180°.

Reason : At the highest point, velocity of projectile will be in horizontal direction only.

2. Assertion : Two particles of different mass, projected with same velocity at same angles. The maximum height attained by both the particle will be same.

Reason : The maximum height of projectile is independent of particle mass.

3. Assertion : The maximum horizontal range of projectile is proportional to square of velocity.

Reason : The maximum horizontal range of projectile is equal to maximum height attained by projectile.

4. Assertion : Horizontal range is same for angle of projection θ and (90 – θ).

Reason : Horizontal range is independent of angle of projection.

5. Assertion : For projection angle $\tan^{-1}(4)$, the horizontal range and the maximum height of a projectile are equal.

Reason : The maximum range of projectile is directly proportional to square of velocity and inversely proportional to acceleration due to gravity.

6. Assertion : The trajectory of projectile is quadratic in y and linear in x.

Reason : y component of trajectory is independent of x-component.

7. Assertion : In javelin throw, the athlete throws the projectile at an angle slightly more than 45°.

Reason : The maximum range does not depends upon angle of projection.

8. Assertion : When a body is dropped or thrown horizontally from the same height, it would reach the ground at the same time.

Reason : Horizontal velocity has no effect on the vertical direction.

9. Assertion : When the velocity of projection of a body is made n times, its time of flight becomes n times.

Reason : Range of projectile does not depend on the initial velocity of a body.

10. Assertion : The height attained by a projectile is twenty five percent of range, when projected for maximum range.

Reason : The height is independent of initial velocity of projectile.

11. Assertion : When range of a projectile is maximum, its angle of projection may be 45° or 135°.

Reason : Whether θ is 45° or 135°, value of range remains the same, only the sign changes.

12. Assertion : In order to hit a target, a man should point his rifle in the same direction as target.

Reason : The horizontal range of the bullet is dependent on the angle of projectile with horizontal direction.

13. Assertion : When a particle moves in a circle with a uniform speed, its velocity and acceleration both changes.

Reason : The centripetal acceleration in circular motion is dependent on angular velocity of the body.

14. Assertion : During a turn, the value of centripetal force should be less than the limiting frictional force.

Reason : The centripetal force is provided by the frictional force between the tyres and the road.

15. Assertion : When a vehicle takes a turn on the road, it travels along a nearly circular path.

Reason : In circular motion, velocity of vehicle remains same.

16. Assertion : As the frictional force increases, the safe velocity limit for taking a turn on an unbanked road also increases.

Reason : Banking of roads will increase the value of limiting velocity.

17. Assertion : If the speed of a body is constant, the body cannot have a path other than a circular or straight line path.

Reason : It is not possible for a body to have a constant speed in an accelerated motion.

18. Assertion : In circular motion, work done by centripetal force is zero.

Reason : In circular motion centripetal force is perpendicular to the displacement.

19. Assertion : In circular motion, the centripetal and centrifugal force acting in opposite direction balance each other.

Reason : Centripetal and centrifugal forces don't act at the same time.

20. Assertion : If both the speed of a body and radius of its circular path are doubled, then centripetal force also gets doubled.

Reason : Centripetal force is directly proportional to both speed of a body and radius of circular path.

21. Assertion : When an automobile while going too fast around a curve overturns, its inner wheels leave the ground first.

Reason : For a safe turn the velocity of automobile should be less than the value of safe limit velocity.

22. Assertion : A safe turn by a cyclist should neither be fast nor sharp.

Reason : The bending angle from the vertical would decrease with increase in velocity.

23. Assertion : Improper banking of roads causes wear and tear of tyres.

Reason : The necessary centripetal force is provided by the force of friction between the tyres and the road.

24. Assertion : Cream gets separated out of milk when it is churned, it is due to gravitational force.

Reason : In circular motion gravitational force is equal to centripetal force.

25. Assertion : Two similar trains are moving along the equatorial line with the same speed but in opposite direction. They will exert equal pressure on the rails.

Reason : In uniform circular motion the magnitude of acceleration remains constant but the direction continuously changes.

26. Assertion : A coin is placed on phonogram turn table. The motor is started, coin moves along the moving table.

Reason : The rotating table is providing necessary centripetal force to the coin.

Answers

Uniform Circular Motion

1	c	2	c	3	b	4	b	5	c
6	c	7	c	8	c	9	b	10	b
11	a	12	a	13	c	14	ac	15	a
16	d	17	a	18	a	19	d	20	a
21	b	22	a	23	c	24	d	25	a
26	c	27	b	28	a	29	d	30	c
31	b	32	d	33	d	34	b	35	c
36	d	37	c	38	b	39	a	40	c
41	d	42	b	43	b	44	b	45	d
46	a	47	b	48	a	49	d	50	b

51	a	52	c	53	a	54	b	55	a
56	d	57	a	58	d	59	a	60	b
61	d	62	d	63	a	64	b	65	d
66	d	67	c	68	a	69	c	70	a
71	b	72	d	73	a	74	c	75	b
76	a	77	c	78	a	79	b	80	a
81	c	82	a	83	d	84	c	85	b
86	a	87	c	88	c	89	d	90	b
91	a	92	d	93	a	94	b	95	b
96	c	97	d	98	d	99	a	100	b
101	c	102	c	103	b	104	c	105	d
106	d	107	b	108	a	109	a	110	b
111	b	112	b	113	b	114	a	115	d
116	b	117	d	118	b	119	c	120	d
121	a	122	d						

Non-uniform Circular Motion

1	d	2	d	3	d	4	c	5	a
6	c	7	a	8	d	9	b	10	b
11	b	12	b	13	a	14	b	15	b
16	a	17	c	18	c	19	a	20	a
21	c	22	a	23	d	24	c	25	b
26	d	27	a	28	d	29	c	30	b
31	d	32	d	33	d	34	b	35	b
36	d	37	c	38	c	39	c	40	b
41	d	42	c	43	a	44	a	45	d
46	d	47	a	48	d				

Horizontal Projectile Motion

1	b	2	c	3	b	4	c	5	b
6	c	7	c	8	ac	9	b	10	c
11	d	12	a	13	c	14	a	15	a
16	b								

Oblique Projectile Motion

1	d	2	c	3	a	4	a	5	c
6	c	7	a	8	d	9	c	10	b
11	a	12	b	13	a	14	a	15	c
16	b	17	c	18	c	19	b	20	b
21	b	22	d	23	c	24	c	25	b
26	d	27	c	28	b	29	d	30	a
31	b	32	b	33	a	34	c	35	c
36	a	37	a	38	a	39	d	40	b

41	c	42	b	43	a	44	b	45	a
46	b	47	c	48	d	49	c	50	c
51	a	52	a	53	b	54	d	55	c
56	c	57	c	58	d	59	b		

Critical Thinking Questions

1	cd	2	a	3	b	4	c	5	b
6	d	7	d	8	a	9	a	10	a
11	c	12	b	13	d	14	a	15	a
16	ab	17	b						

Graphical Questions

1	d	2	a	3	b	4	c	5	d

Assertion and Reason

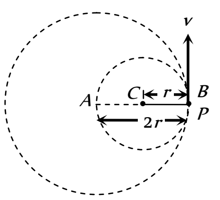
1	e	2	a	3	c	4	c	5	b
6	d	7	d	8	a	9	c	10	c
11	a	12	e	13	b	14	a	15	c
16	b	17	d	18	a	19	d	20	c
21	b	22	c	23	a	24	d	25	e
26	d								

AS Answers and Solutions

Uniform Circular Motion

1. (c) $v = r\omega \Rightarrow \omega = \frac{v}{r}$ = constant [As v and r are constant]
2. (c) As time periods are equal therefore ratio of angular speeds will be same. $\omega = \frac{2\pi}{T}$
3. (b) $F = \frac{mv^2}{r} \Rightarrow F \propto v^2$. If v becomes double then F (tendency to overturn) will become four times.
4. (b) Work done by centripetal force is always zero.
5. (c) It is always directed in a direction of tangent to circle.
6. (c) Stone flies in the direction of instantaneous velocity due to inertia
7. (c) Centripetal acceleration $= \frac{v^2}{r} =$ constant. Direction keeps changing.
8. (c) Linear velocity, acceleration and force varies in direction.
9. (b) Angular velocity of particle P about point A,

$\omega_A = \frac{v}{r_{AB}} = \frac{v}{2r}$



Angular velocity of particle P about point C,

$\omega_C = \frac{v}{r_{BC}} = \frac{v}{r}$

Ratio $\frac{\omega_A}{\omega_C} = \frac{v/2r}{v/r} = \frac{1}{2}$.

10. (b)
11. (a) $F = \frac{mv^2}{r}$. If m and v are constants then $F \propto \frac{1}{r}$

$\therefore \frac{F_1}{F_2} = \left(\frac{r_2}{r_1}\right)$

12. (a) In uniform circular motion (constant angular velocity) kinetic energy remains constant but due to change in velocity of particle its momentum varies.
13. (c)
14. (a,c) Centripetal force $= \frac{mv^2}{r}$ and is directed always towards the centre of circle. Sense of rotation does not affect magnitude and direction of this centripetal force.
15. (a) When speed is constant in circular motion, it means work done by centripetal force is zero.
16. (d)
17. (a) This horizontal inward component provides required centripetal force.
18. (a) Thrust at the lowest point of concave bridge

$= mg + \frac{mv^2}{r}$

19. (d)
20. (a) Because the reaction on inner wheel decreases and becomes zero. So it leaves the ground first.
21. (b)
22. (a) $\frac{a_R}{a_r} = \frac{\omega_R^2 \times R}{\omega_r^2 \times r} = \frac{T_r^2}{T_R^2} \times \frac{R}{r} = \frac{R}{r}$ [As $T_r = T_R$]
23. (c) $\omega_{min} = \frac{2\pi}{60}\frac{Rad}{min}$ and $\omega_{hr} = \frac{2\pi}{12 \times 60}\frac{Rad}{min}$

$\therefore \frac{\omega_{min}}{\omega_{hr}} = \frac{2\pi/60}{2\pi/12 \times 60}$

24. (d) The particle performing circular motion flies off tangentially.
25. (a) The angle of banking, $\tan\theta = \frac{v^2}{rg}$

$\Rightarrow \tan 12° = \frac{(150)^2}{r \times 10} \Rightarrow r = 10.6 \times 10^3\, m = 10.6\, km$

26. (c) K.E. $= \frac{1}{2}mv^2$. Which is scalar, so it remains constant.
27. (b) $v = 72\, km\, /\, hour = 20\, m\, /\, \sec$

$\theta = \tan^{-1}\left(\frac{v^2}{rg}\right) = \tan^{-1}\left(\frac{20 \times 20}{20 \times 10}\right) = \tan^{-1}(2)$

28. (a)
29. (d) $120\, rev\, /\, \min = 120 \times \frac{2\pi}{60}\, rad\, /\, \sec = 4\pi\, rad\, /\, \sec$
30. (c) In uniform circular motion, acceleration causes due to change in direction and is directed radially towards centre.
31. (b) Reaction on inner wheel $R_1 = \frac{1}{2}M\left[g - \frac{v^2h}{ra}\right]$

Reaction on outer wheel $R_2 = \frac{1}{2}M\left[g + \frac{v^2h}{ra}\right]$

where, r = radius of circular path, $2a$ = distance between two wheels and h = height of centre of gravity of car.

32. (d) Maximum tension $= m\omega^2 r = m \times 4\pi^2 \times n^2 \times r$

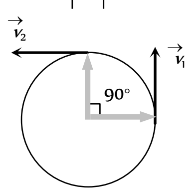
By substituting the values we get $T_{max} = 87.64\, N$

33. (d) $\frac{v^2}{rg} = \frac{h}{l} \Rightarrow v = \sqrt{\frac{rgh}{l}} = \sqrt{\frac{50 \times 1.5 \times 9.8}{10}} = 8.57\, m\, /\, s$
34. (b) $a = \omega^2 r = 4\pi^2 n^2 r = 4\pi^2 \times 1^2 \times 20 \times 10^3$

$\therefore$ a $= 8 \times 10^5\ m/\sec^2$

35. (c)
36. (d) In 15 second's hand rotate through 90°.

Change in velocity $|\overrightarrow{\Delta v}| = 2v\sin(\theta/2)$



$= 2(r\omega)\sin(90^\circ/2) = 2\times 1\times\dfrac{2\pi}{T}\times\dfrac{1}{\sqrt{2}}$

$= \dfrac{4\pi}{60\sqrt{2}} = \dfrac{\pi\sqrt{2}}{30}\dfrac{cm}{\sec}$ [As T = 60 sec]

37. (c) Since $n = 2$, $\omega = 2\pi\times 2 = 4\pi\, rad/s^2$

So acceleration $= \omega^2 r = (4\pi)^2 \times \dfrac{25}{100} m/s^2 = 4\pi^2$

38. (b) $\omega^2 r = 4\pi^2 n^2 r = 4\pi^2\left(\dfrac{1200}{60}\right)^3 \times 30 = 4740\, m/s^2$

39. (a)

40. (c) Particles of cream are lighter so they get deposited near the centre of circular path.

41. (d) Radial force $= \dfrac{mv^2}{r} = \dfrac{m}{r}\left(\dfrac{p}{m}\right)^2 = \dfrac{p^2}{mr}$ [As $p = mv$]

42. (b) $\dfrac{mv^2}{r} \propto \dfrac{K}{r} \Rightarrow v \propto r^o$

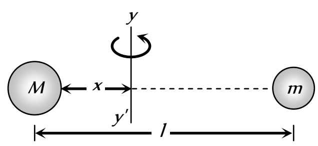
i.e. speed of the particle is independent of *r*.

43. (b) If the both mass are revolving about the axis yy' and tension in both the threads are equal then

$M\omega^2 x = m\omega^2(l - x)$

$\Rightarrow Mx = m(l - x)$

$\Rightarrow x = \dfrac{ml}{M + m}$



44. (b) $\tan\theta = \dfrac{v^2}{rg} = \dfrac{400}{20\times 9.8} \Rightarrow \theta = 63.9^\circ$

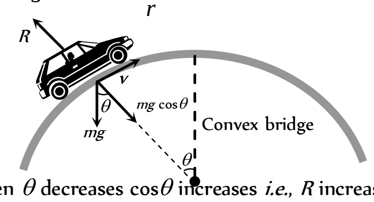
45. (d) In complete revolution change in velocity becomes zero so average acceleration will be zero.

46. (a) We know that $\tan\theta = \dfrac{v^2}{Rg}$ and $\tan\theta = \dfrac{h}{b}$

Hence $\dfrac{h}{b} = \dfrac{v^2}{Rg} \Rightarrow h = \dfrac{v^2 b}{Rg}$

47. (b)

48. (a) $R = mg\cos\theta - \dfrac{mv^2}{r}$



when θ decreases $\cos\theta$ increases *i.e.*, R increases.

49. (d) Tension in the string $T = m\omega^2 r = 4\pi^2 n^2 mr$

$\therefore T \propto n^2 \Rightarrow \dfrac{n_2}{n_1} = \sqrt{\dfrac{T_2}{T_1}} \Rightarrow n_2 = 5\sqrt{\dfrac{2T}{T}} = 7\, rpm$

50. (b)

51. (a) $T = m\omega^2 r \Rightarrow 10 = 0.25\times\omega^2\times 0.1 \Rightarrow \omega = 20\, rad/s$

52. (c) $v = 36\dfrac{km}{h} = 10\dfrac{m}{s}$ $\therefore F = \dfrac{mv^2}{r} = \dfrac{500\times 100}{50} = 1000N.$

53. (a) $T = \dfrac{mv^2}{r} \Rightarrow 25 = \dfrac{0.25\times v^2}{1.96} \Rightarrow v = 14\, m/s$

54. (b) Centripetal force $= mr\omega^2 = 5\times 1\times(2)^2 = 20\, N$

55. (a) $\dfrac{mv^2}{r} = \dfrac{k}{r^2} \Rightarrow mv^2 = \dfrac{k}{r}$ $\therefore$ K.E.$= \dfrac{1}{2}mv^2 = \dfrac{k}{2r}$

P.E.$= \int F\,dr = \int \dfrac{k}{r^2}dr = -\dfrac{k}{r}$

$\therefore$ Total energy = K.E. + P.E. $= \dfrac{k}{2r} - \dfrac{k}{r} = -\dfrac{k}{2r}$

56. (d) Maximum tension $= \dfrac{mv^2}{r} = 16N$

$\Rightarrow \dfrac{16\times v^2}{144} = 16 \Rightarrow v = 12\; m/s$

57. (a) The maximum velocity for a banked road with friction,

$v^2 = gr\left(\dfrac{\mu + \tan\theta}{1 - \mu\tan\theta}\right)$

$\Rightarrow v^2 = 9.8\times 1000\times\left(\dfrac{0.5 + 1}{1 - 0.5\times 1}\right) \Rightarrow v = 172\, m/s$

58. (d) $v = r\omega = \dfrac{r\times 2\pi}{T} = \dfrac{0.06\times 2\pi}{60} = 6.28\, mm/s$

Magnitude of change in velocity $= |\vec{v}_2 - \vec{v}_1|$

$= \sqrt{v_1^2 + v_2^2} = 8.88\, mm/s$ $(\text{As } v_1 = v_2 = 6.28 mm/s)$

59. (a) Work done by centripetal force in uniform circular motion is always equal to zero.

60. (b) $v = r\omega = 20\times 10\, cm/s = 2\, m/s$

61. (d) $v_{\max} = \sqrt{\mu rg} = \sqrt{0.2\times 100\times 9.8} = 14 m/s$

62. (d) $F = mg - \dfrac{mv^2}{r}$

63. (a) $\omega = \dfrac{2\pi}{T} = \dfrac{2\pi}{60} = \dfrac{\pi}{30} rad/s$

64. (b) $\omega = 2\pi n = \dfrac{2\pi\times 100}{60} = 10.47\, rad/s$

65. (d) Work done in circular motion is always zero.

66. (d) In complete revolution total displacement is zero so average velocity is zero

67. (c) $v_{\max} = \sqrt{\mu rg} = \sqrt{0.75\times 60\times 9.8} = 21\, m/s$

68. (a) Distance covered in 'n' revolution $= n\, 2\pi\, r = n\pi D$

$\Rightarrow 2000\pi D = 9500$ [As n = 2000, distance = 9500 m]

$\Rightarrow D = \dfrac{9500}{2000\times\pi} = 1.5\, m$

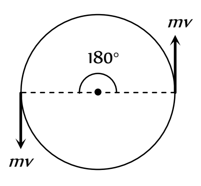
69. (c) Centripetal acceleration = $4\pi n r = 4\pi\times(1)\times 0.4 = 1.6\pi$

70. (a)

71. (b) Due to centrifugal force.

72. (d) As momentum is vector quantity

$\therefore$ change in momentum



$\Delta P = 2mv\sin(\theta/2)$

$= 2mv\sin(90) = 2mv$

But kinetic energy remains always constant so change in kinetic energy is zero.

73. (a) $\omega = \frac{v}{r} = \frac{100}{100} = 1\,rad/s$

74. (c) $\alpha = \frac{d\omega}{dt} = 0$ (As ω = constant)

75. (b) $\vec{v} = \vec{\omega} \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -4 & 1 \\ 5 & -6 & 6 \end{vmatrix} = -18\hat{i} - 13\hat{j} + 2\hat{k}$

76. (a) $a = 4\pi^2 n^2 r = 4\pi^2 \left(\frac{1}{2}\right)^2 \times 50 = 493\,cm/s^2$

77. (c) Maximum force of friction = centripetal force

$\frac{mv^2}{r} = \frac{100 \times (9)^2}{30} = 270\,N$

78. (a) $v = \sqrt{\mu rg} = \sqrt{0.4 \times 30 \times 9.8} = 10.84\,m/s$

79. (b) $v = r\omega = 0.5 \times 70 = 35\,m/s$

80. (a) $2\pi r = 34.3 \Rightarrow r = \frac{34.3}{2\pi}$ and $v = \frac{2\pi r}{T} = \frac{2\pi r}{\sqrt{22}}$

Angle of binding $\theta = \tan^{-1}\left(\frac{v^2}{rg}\right) = 45°$

81. (c)

82. (a) $T = m\omega^2 r \Rightarrow \omega \propto \sqrt{T}\ \therefore \frac{\omega_2}{\omega_1} = \sqrt{\frac{1}{4}} \Rightarrow \omega_2 = \frac{\omega_1}{2} = 5\,rpm$

83. (d) $\theta = \tan^{-1}\left(\frac{v^2}{rg}\right) = \tan^{-1}\left[\frac{(14\sqrt{3})^2}{20\sqrt{3} \times 9.8}\right] = \tan^{-1}[\sqrt{3}] = 60°$

84. (c) Centripetal acceleration $= 4\pi^2 n^2 r = 4\pi^2 \left(\frac{1}{2}\right)^2 \times 4 = 4\pi^2$

85. (b) Centripetal force = breaking force

$\Rightarrow m\omega^2 r =$ breaking stress × cross sectional area

$\Rightarrow m\omega^2 r = p \times A \Rightarrow \omega = \sqrt{\frac{p \times A}{mr}} = \sqrt{\frac{4.8 \times 10^7 \times 10^{-6}}{10 \times 0.3}}$

$\therefore \omega = 4\,rad/\sec$

86. (a) Because velocity is always tangential and centripetal acceleration is radial.

87. (c) T = tension, W = weight and F = centrifugal force.

88. (c) $\mu = \frac{v^2}{rg} = \frac{(4.9)^2}{4 \times 9.8} = 0.61$

89. (d) As body covers equal angle in equal time intervals. its angular velocity and hence magnitude of linear velocity is constant.

90. (b) $\omega = \frac{v}{r} = \frac{10}{100} = 0.1\,rad/s$

91. (a) $F = \frac{mv^2}{r} \Rightarrow v = \sqrt{\frac{rF}{m}}$

92. (d) Electrostatic force provides necessary centripetal force for circular motion of electron.

93. (a) Acceleration $= \omega^2 r = \frac{v^2}{r} = \omega v = \frac{2\pi}{T} v$

94. (b) $v = \sqrt{\mu rg} = \sqrt{0.6 \times 150 \times 10} = 30\,m/s$

95. (b)

96. (c) $F = \frac{mv^2}{r} \Rightarrow F \propto v^2$ *i.e.* force will become 4 times.

97. (d) $v = \sqrt{\mu rg} = \sqrt{0.25 \times 40 \times 10} = 10\,m/s$

98. (d) Time period = 40 *sec*

No. of revolution $= \frac{\text{Total time}}{\text{Time period}} = \frac{140\text{ sec}}{40\text{ sec}} = 3.5\,Rev$.

So, distance $= 3.5 \times 2\pi R = 3.5 \times 2\pi \times 10 = 220\,m$.

99. (a) $m4\pi^2 n^2 r = 4 \times 10^{-13} \Rightarrow n = 0.08 \times 10^8\,cycles/\sec$.

100. (b) Momentum changes by $2mv$ but kinetic energy remains same.

101. (c) $L = I\omega$. In U.C.M. ω=constant ∴ L = constant.

102. (c) $\because W = FS\cos\theta\ \therefore \theta = 90°$

103. (b)

104. (c) In uniform circular motion tangential acceleration remains zero but magnitude of radial acceleration remains constant.

105. (d) The inclination of person from vertical is given by,

$\tan\theta = \frac{v^2}{rg} = \frac{(10)^2}{50 \times 10} = \frac{1}{5}\ \therefore \theta = \tan^{-1}(1/5)$

106. (d) The centripetal force, $F = \frac{mv^2}{r} \Rightarrow r = \frac{mv^2}{F}$

$\therefore r \propto v^2$ or $v \propto \sqrt{r}$ (If m and F are constant),

$\Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{r_1}{r_2}} = \sqrt{\frac{1}{2}}$

107. (b) As the speed is constant throughout the circular motion therefore its average speed is equal to instantaneous speed.

108. (a) Linear velocity,

$v = \omega r = 2\pi nr = 2 \times 3.14 \times 3 \times 0.1 = 1.88\,m/s$

Acceleration, $a = \omega^2 r = (6\pi)^2 \times 0.1 = 35.5\,m/s^2$

Tension in string, $T = m\omega^2 r = 1 \times (6\pi)^2 \times 0.1 = 35.5\,N$

109. (a) $a = \frac{v^2}{r} = \frac{(400)^2}{160} = 10^3\,m/s^2 = 1\,km/s^2$

110. (b) $v_{max.} = \sqrt{\mu rg} = \sqrt{0.5 \times 40 \times 9.8} = 14\,m/s$

111. (b) $F = \frac{mv^2}{r} = \frac{500 \times 100}{50} = 10^3\,N$

112. (b) $F = m\left(\frac{4\pi^2}{T^2}\right)R$. If masses and time periods are same then $F \propto R \;\therefore\; F_1/F_2 = R_1/R_2$

113. (b) It is a vector quantity.

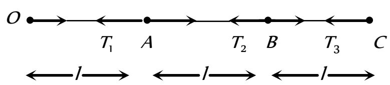
114. (a) $a = \frac{v^2}{r} = v\omega \Rightarrow a' = (2v)\left(\frac{\omega}{2}\right) = a$ *i.e.* remains constant.

115. (d) Tension in the string $T_0 = mR\omega_0^2$

In the second case $T = m(2R)(4\omega_0^2) = 8mR\omega_0^2 = 8T_0$

116. (b) Average velocity $= \frac{\text{Total displacement}}{\text{Total time}} = \frac{2m}{1s} = 2ms^{-1}$

117. (d) Let ω is the angular speed of revolution



$T_3 = m\omega^2 3l$

$T_2 - T_3 = m\omega^2 2l \Rightarrow T_2 = m\omega^2 5l$

$T_1 - T_2 = m\omega^2 l \Rightarrow T_1 = m\omega^2 6l$

$T_3 : T_2 : T_1 = 3 : 5 : 6$

118. (b) $F = \frac{mv^2}{r}$. For same mass and same speed if radius is doubled then force should be halved.

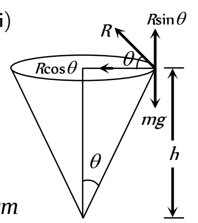
119. (c) $a = \frac{v^2}{r} = \omega^2 r = 4\pi^2 n^2 r = 4\pi^2\left(\frac{22}{44}\right)^2 \times 1 = \pi^2\ m/s^2$

and its direction is always along the radius and towards the centre.

120. (d) The particle is moving in circular path

From the figure, $mg = R\sin\theta$ ...(i)

$\frac{mv^2}{r} = R\cos\theta$ ...(ii)



From equation (i) and (ii) we get

$\tan\theta = \frac{rg}{v^2}$ but $\tan\theta = \frac{r}{h}$

$\therefore h = \frac{v^2}{g} = \frac{(0.5)^2}{10} = 0.025m = 2.5\,cm$

121. (a) Angular velocity $= \frac{2\pi}{T} = \frac{2\pi}{24} rad/hr = \frac{2\pi}{86400} rad/s$

122. (d) $\omega = \frac{2\pi}{T} = \frac{2\pi}{60} = 0.1047\ rad/s$

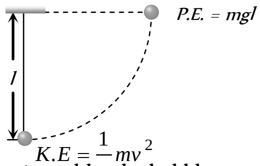
and $v = \omega r = 0.1047 \times 3 \times 10^{-2} = 0.00314\ m/s$

Non-uniform Circular Motion

1. (d) Minimum speed at the highest point of vertical circular path $v = \sqrt{gR}$

2. (d) At highest point $\frac{mv^2}{R} = mg \Rightarrow v = \sqrt{gR}$

3. (d) Kinetic energy given to a sphere at lowest point = potential energy at the height of suspension



$\Rightarrow \frac{1}{2}mv^2 = mgl$

$\therefore v = \sqrt{2gl}$

4. (c) Due to less centrifugal force experienced by the bubbles.

5. (a) Critical velocity at highest point $= \sqrt{gR} = \sqrt{10 \times 1.6} = 4\,m/s$

6. (c) Using relation $\theta = \omega_0 t + \frac{1}{2}at^2$

$\theta_1 = \frac{1}{2}(\alpha)(2)^2 = 2\alpha$ ...(i) (As $\omega_0 = 0, t = 2$ sec)

Now using same equation for $t = 4$ sec, $\omega_0 = 0$

$\theta_1 + \theta_2 = \frac{1}{2}\alpha(4)^2 = 8\alpha$ ...(ii)

From (i) and (ii), $\theta_1 = 2\alpha$ and $\theta_2 = 6\alpha \;\therefore\; \frac{\theta_2}{\theta_1} = 3$

7. (a) $mg = 1 \times 10 = 10N$, $\frac{mv^2}{r} = \frac{1 \times (4)^2}{1} = 16$

Tension at the top of circle $= \frac{mv^2}{r} - mg = 6N$

Tension at the bottom of circle $= \frac{mv^2}{r} + mg = 26N$

8. (d) For critical condition at the highest point $\omega = \sqrt{g/R}$

$\Rightarrow T = \frac{2\pi}{\omega} = 2\pi\sqrt{R/g} = 2 \times 3.14\sqrt{4/9.8} = 4$ sec.

9. (b) $mg = 20N$ and $\frac{mv^2}{r} = \frac{2 \times (4)^2}{1} = 32N$

It is clear that 52 N tension will be at the bottom of the circle. Because we know that $T_{\text{Bottom}} = mg + \frac{mv^2}{r}$

10. (b) $h = \frac{5}{2}R = \frac{5}{2}\left(\frac{D}{2}\right) = \frac{5D}{4}$

11. (b) Net acceleration in nonuniform circular motion,

$a = \sqrt{a_t^2 + a_c^2} = \sqrt{(2)^2 + \left(\frac{900}{500}\right)^2} = 2.7\,m/s^2$

a_t = tangential acceleration

a_c = centripetal acceleration $= \frac{v^2}{r}$

12. (b) $T = mg + \frac{mv^2}{l} = mg + 2mg = 3mg$

where $v = \sqrt{2gl}$ from $\frac{1}{2}mv^2 = mgl$

13. (a) $T_{\max} = m\omega^2_{\max} r + mg \Rightarrow \frac{T_{\max}}{m} = \omega^2 r + g$

$$\Rightarrow \frac{30}{0.5} - 10 = \omega^2{}_{\max} r \Rightarrow \omega_{\max} = \sqrt{\frac{50}{r}} = \sqrt{\frac{50}{2}} = 5\, rad/s$$

14. (b)

15. (b) Because here tension is maximum.

16. (a) Max. tension that string can bear = 3.7 *kgwt* = 37*N*

Tension at lowest point of vertical $\text{loop} = mg + m\omega^2 r$

=

$0.5 \times 10 + 0.5 \times \omega^2 \times 4 = 5 + 2\omega^2$

$\therefore$ 37 = 5 + $2\omega \Rightarrow \omega$ = 4 rad/s.

17. (c)

18. (c) $\omega = \frac{d\theta}{dt} = \frac{d}{dt}(2t^3 + 0.5) = 6t^2$

at t =2 *s*, $\omega = 6 \times (2)^2 = 24 rad/s$

19. (a) When body is released from the position p (inclined at angle θ from vertical) then velocity at mean position

$v = \sqrt{2gl(1 - \cos\theta)}$

$\therefore$ Tension at the lowest point = $mg + \frac{mv^2}{l}$

$$= mg + \frac{m}{l}[2gl(1 - \cos 60)] = mg + mg = 2mg$$

20. (a)

21. (c) Tension = Centrifugal force + weight $= \frac{mv^2}{r} + mg$

22. (a) $v_{\min} = \sqrt{5\,gr} = 17.7\, m/\sec$

23. (d)

24. (c) $v = \sqrt{2gl(1 - \cos\theta)} = \sqrt{2 \times 9.8 \times 2(1 - \cos 60°)} = 4.43\, m/s$

25. (b) Increment in angular velocity $\omega = 2\pi(n_2 - n_1)$

$$\omega = 2\pi(1200 - 600)\frac{rad}{min} = \frac{2\pi \times 600}{60}\frac{rad}{s} = 20\pi\frac{rad}{s}$$

26. (d) $\omega = \sqrt{\frac{g}{r}} = \sqrt{\frac{9.8}{0.2}} = 7\, rad/s$

27. (a)

28. (d) In non-uniform circular motion particle possess both centripetal as well as tangential acceleration.

29. (c)

30. (b) $v = \sqrt{2gh} = \sqrt{2 \times 10 \times 0.2} = 2\, m/s$

31. (d) $T = mg + m\omega^2 r = m\{g + 4\pi^2 n^2 r\}$

$$= m\left\{g + \left(4\pi^2\left(\frac{n}{60}\right)^2 r\right)\right\} = m\left\{g + \left(\frac{\pi^2 n^2 r}{900}\right)\right\}$$

32. (d) $h = \frac{5}{2}r \Rightarrow r = \frac{2}{5} \times h = \frac{2}{5} \times 5 = 2\, metre$

33. (d) In the given condition friction provides the required centripetal force and that is constant. *i.e.* $m\omega r$=constant

$$\Rightarrow r \propto \frac{1}{\omega^2} \therefore r_2 = r_1\left(\frac{\omega_1}{\omega_2}\right)^2 = 9\left(\frac{1}{3}\right)^2 = 1\, cm$$

34. (b) By using equation $\omega^2 = \omega_0^2 - 2\alpha\theta$

$$\left(\frac{\omega_0}{2}\right)^2 = \omega_0^2 - 2\alpha(2\pi n) \Rightarrow \alpha = \frac{3}{4}\frac{\omega_0^2}{4\pi \times 36}, \ (n = 36) \quad ..(i)$$

Now let fan completes total n' revolution from the starting to come to rest

$$0 = \omega_0^2 - 2\alpha(2\pi n') \Rightarrow n' = \frac{\omega_0^2}{4\alpha\pi}$$

substituting the value of α from equation (i)

$$n' = \frac{\omega_0^2}{4\pi}\frac{4 \times 4\pi \times 36}{3\omega_0^2} = 48 \text{ revolution}$$

Number of rotation = 48 – 36 = 12

35. (b) $v = \sqrt{3gr}$ and $a = \frac{v^2}{r} = \frac{3gr}{r} = 3g$

36. (d) Tension at mean position, $mg + \frac{mv^2}{r} = 3mg$

$v = \sqrt{2gl}$ ...(i)

and if the body displaces by angle θ with the vertical then

$v = \sqrt{2gl(1 - \cos\theta)}$ ...(ii)

Comparing (i) and (ii), $\cos\theta = 0 \Rightarrow \theta = 90°$

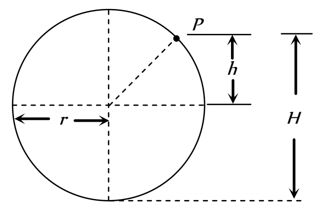
37. (c) Tension, $T = \frac{mv^2}{r} + mg\cos\theta$

For, $\theta = 30°, T_1 = \frac{mv^2}{r} + mg\cos 30°$

$\theta = 60°, T_2 = \frac{mv^2}{r} + mg\cos 60° \therefore T_1 > T_2$

38. (c) As we know for hemisphere the particle will leave the sphere at height $h = 2r/3$

$h = \frac{2}{3} \times 21 = 14\, m$



but from the bottom

$H = h + r = 14 + 21 = 35\, metre$

39. (c) $x = \alpha t^3$ and $y = \beta t^3$ (given)

$v_x = \frac{dx}{dt} = 3\alpha t^2$ and $v_y = \frac{dy}{dt} = 3\beta t^2$

Resultant velocity $= v = \sqrt{v_x^2 + v_y^2} = 3t^2\sqrt{\alpha^2 + \beta^2}$

40. (b)

41. (d) Tension at the top of the circle, $T = m\omega^2 r - mg$

$T = 0.4 \times 4\pi^2 n^2 \times 2 - 0.4 \times 9.8 = 115.86\,N$

42. (c) Minimum angular velocity $\omega_{\min} = \sqrt{g/R}$

$\therefore T_{\max} = \dfrac{2\pi}{\omega_{\min}} = 2\pi\sqrt{\dfrac{R}{g}} = = 2\pi\sqrt{\dfrac{2}{10}} = 2\sqrt{2} \cong 3s$

43. (a) $|\overrightarrow{\Delta v}| = 2v\sin(\theta/2) = 2v\sin\left(\dfrac{90}{2}\right) = 2v\sin 45 = v\sqrt{2}$

44. (a) In this problem it is assumed that particle although moving in a vertical loop but its speed remain constant.

Tension at lowest point $T_{\max} = \dfrac{mv^2}{r} + mg$

Tension at highest point $T_{\min} = \dfrac{mv^2}{r} - mg$

$$\frac{T_{\max}}{T_{\min}} = \frac{\dfrac{mv^2}{r} + mg}{\dfrac{mv^2}{r} - mg} = \frac{5}{3}$$

by solving we get, $v = \sqrt{4gr} = \sqrt{4 \times 9.8 \times 2.5} = \sqrt{98}\ m/s$

45. (d) There is no relation between centripetal and tangential acceleration. Centripetal acceleration is must for circular motion but tangential acceleration may be zero.

46. (d) Angular momentum is a axial vector. It is directed always in a fix direction (perpendicular to the plane of rotation either outward or inward), if the sense of rotation remain same.

47. (a) Difference in kinetic energy = $2mg\,r = 2 \times 1 \times 10 \times 1 = 20\,J$

48. (d) Angular acceleration = $\dfrac{d^2\theta}{dt^2} = 2\theta$

Horizontal Projectile Motion

1. (b) $R_{\max} = \dfrac{u^2}{g} = 16 \times 10^3 \Rightarrow u = 400\,m/s$

2. (c) Due to constant velocity along horizontal and vertical downward force of gravity stone will hit the ground following parabolic path.

3. (b) Because the vertical components of velocities of both the bullets are same and equal to zero and $t = \sqrt{\dfrac{2h}{g}}$.

4. (c) The pilot will see the ball falling in straight line because the reference frame is moving with the same horizontal velocity but the observer at rest will see the ball falling in parabolic path.

5. (b) Due to air resistance, it's horizontal velocity will decrease so it will fall behind the aeroplane.

6. (c) Because horizontal velocity is same for coin and the observer. So relative horizontal displacement will be zero.

7. (c) Horizontal displacement of the bomb

AB = Horizontal velocity × time available

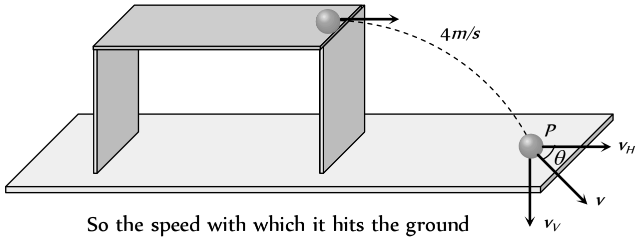
$AB = u \times \sqrt{\dfrac{2h}{g}} = 600 \times \dfrac{5}{18} \times \sqrt{\dfrac{2 \times 1960}{9.8}} = 3.33\,Km.$

8. (a,c) Vertical component of velocity of ball at point P

$v_V = 0 + gt = 10 \times 0.4 = 4\,m/s$

Horizontal component of velocity = initial velocity

$\Rightarrow v_H = 4\,m/s$



So the speed with which it hits the ground

$v = \sqrt{v_H^2 + v_V^2} = 4\sqrt{2}\,m/s$

and $\tan\theta = \dfrac{v_V}{v_H} = \dfrac{4}{4} = 1 \Rightarrow \theta = 45°$

It means the ball hits the ground at an angle of $45°$ to the horizontal.

Height of the table $h = \dfrac{1}{2}gt^2 = \dfrac{1}{2} \times 10 \times (0.4)^2 = 0.8\,m$

Horizontal distance travelled by the ball from the edge of table $h = ut = 4 \times 0.4 = 1.6\,m$

9. (b) $S = u \times \sqrt{\dfrac{2h}{g}} = 100 \times \sqrt{\dfrac{2 \times 490}{9.8}} = 1000\,m = 1\,km$

10. (c) $S = u \times \sqrt{\dfrac{2h}{g}} \Rightarrow 10 = u\sqrt{2 \times \dfrac{5}{10}} \Rightarrow u = 10\,m/s$

11. (d) $t = \sqrt{\dfrac{2h}{g}} = \sqrt{\dfrac{2 \times 396.9}{9.8}} \simeq 9\sec$ and $u = 720\,km/hr = 200\,m/s$

$\therefore R = u \times t = 200 \times 9 = 1800\,m$

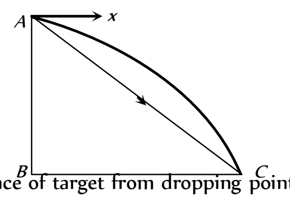
12. (a) For both cases $t = \sqrt{\dfrac{2h}{g}}$ = constant.

Because vertical downward component of velocity will be zero for both the particles.

13. (c)

14. (a) The horizontal distance covered by bomb,

$BC = v_H \times \sqrt{\dfrac{2h}{g}} = 150\sqrt{\dfrac{2 \times 80}{10}} = 660m$

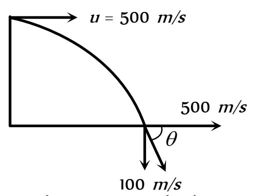


$\therefore$ The distance of target from dropping point of bomb,

$AC = \sqrt{AB^2 + BC^2} = \sqrt{(80)^2 + (600)^2} = 605.3\,m$

15. (a) Horizontal component of velocity $v = 500\ m/s$

and vertical components of velocity while striking the ground.



$v_y = 0 + 10 \times 10 = 100\, m/s$

$\therefore$ Angle with which it strikes the ground.

$$\theta = \tan^{-1}\left(\frac{v_y}{v_x}\right) = \tan^{-1}\left(\frac{100}{500}\right) = \tan^{-1}\left(\frac{1}{5}\right)$$

16. (b) Area in which bullet will spread = πr^2

For maximum area, $r = R_{\max} = \frac{v^2}{g}$ [when $\theta = 45°$]

Maximum area $\pi R_{\max}^2 = \pi\left(\frac{v^2}{g}\right)^2 = \frac{\pi v^4}{g^2}$

Oblique Projectile Motion

1. (d) $R = \frac{u^2 \sin 2\theta}{g}$ $\therefore R \propto u^2$. If initial velocity be doubled then range will become four times.

2. (c) $H = \frac{u^2 \sin^2\theta}{2g}$ $\therefore H \propto u^2$. If initial velocity be doubled then maximum height reached by the projectile will quadrupled.

3. (a) An external force by gravity is present throughout the motion so momentum will not be conserved.

4. (a) Range $= \frac{u^2 \sin 2\theta}{g}$; when $\theta = 90°$, $R = 0$ *i.e.* the body will fall at the point of projection after completing one dimensional motion under gravity.

5. (c) $R = 4H\cot\theta$.

When $R = H$ then $\cot\theta = 1/4 \Rightarrow \theta = \tan^{-1}(4)$

6. (c) Because there is no accelerating or retarding force available in horizontal motion.

7. (a) Direction of velocity is always tangent to the path so at the top of trajectory, it is in horizontal direction and acceleration due to gravity is always in vertically downward direction. It means angle between $\vec{v}$ and $\vec{g}$ are perpendicular to each other.

8. (d) $R = 4H\cot\theta$ if $\theta = 45°$ then $R = 4H\cot(45°) = 4H$

9. (c) $v_y = \frac{dy}{dt} = 8 - 10t$, $v_x = \frac{dx}{dt} = 6$

at the time of projection *i.e.* $v_y = \frac{dy}{dt} = 8$ and $v_x = 6$

$\therefore v = \sqrt{v_x^2 + v_y^2} = \sqrt{6^2 + 8^2} = 10\ m/s$

10. (b) The angle of projection is given by $\theta = \tan^{-1}\left(\frac{v_y}{v_x}\right) = \tan^{-1}\left(\frac{4}{3}\right)$

11. (a) $a_x = \frac{d}{dt}(v_x) = 0$, $a_y = \frac{d}{dt}(v_y) = -10\ m/s^2$

$\therefore$ Net acceleration $a = \sqrt{a_x^2 + a_y^2} = \sqrt{0^2 + 10^2} = 10\,m/s$

12. (b) $R_{15°} = \frac{u^2 \sin(2 \times 15°)}{g} = \frac{u^2}{2g} = 1.5\, km$

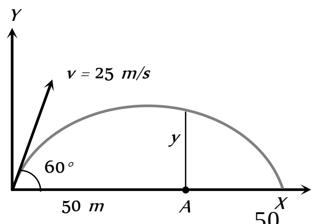
$R_{45°} = \frac{u^2 \sin(2 \times 45°)}{g} = \frac{u^2}{g} = 1.5 \times 2 = 3\, km$

13. (a) Horizontal component of velocity

$v_x = 25\cos 60° = 12.5\, m/s$

Vertical component of velocity

$v_y = 25\sin 60° = 12.5\sqrt{3}\, m/s$



Time to cover 50 *m* distance $t = \frac{50}{12.5} = 4$ sec

The vertical height y is given by

$$y = v_y t - \frac{1}{2}gt^2 = 12.5\sqrt{3} \times 4 - \frac{1}{2} \times 9.8 \times 16 = 8.2\, m$$

14. (a) For vertical upward motion $h = ut - \frac{1}{2}gt^2$

$5 = (25\sin\theta) \times 2 - \frac{1}{2} \times 10 \times (2)^2$

$\Rightarrow 25 = 50 \sin\theta \Rightarrow \sin\theta = \frac{1}{2} \Rightarrow \theta = 30°$

15. (c) For angle $(45° - \theta)$, $R = \frac{u^2 \sin(90° - 2\theta)}{g} = \frac{u^2 \cos 2\theta}{g}$

For angle $(45° + \theta)$, $R = \frac{u^2 \sin(90° + 2\theta)}{g} = \frac{u^2 \cos 2\theta}{g}$

16. (b) Range is given by $R = \frac{u^2 \sin 2\theta}{g}$

On moon $g_m = \frac{g}{6}$. Hence $R_m = 6R$

17. (c) For greatest height $\theta = 90°$

$H_{\max} = \frac{u^2 \sin^2(90°)}{2g} = \frac{u^2}{2g} = h$ (given)

$R_{\max} = \frac{u^2 \sin^2 2(45°)}{g} = \frac{u^2}{g} = 2h$

18. (c) $R = 4H\cot\theta$, if $R = 4H$ then $\cot\theta = 1 \Rightarrow \theta = 45°$

19. (b) $E' = E\cos^2\theta = E\cos^2(45°) = \frac{E}{2}$

20. (b)

21. (b)

22. (d) Acceleration through out the projectile motion remains constant and equal to g.

23. (c)

24. (c) Time of flight $= \frac{2u\sin\theta}{g} = \frac{2 \times 50 \times \sin 30}{10} = 5s$

25. (b) Change in momentum = $2mu\sin\theta$

$= 2\times 0.5\times 98\times \sin 30 = 45 N\text{-}s$

26. (d) $R = 4H\cot\theta$, if $R = 3H$ then $\cot\theta = \frac{3}{4} \Rightarrow \theta = 53°8'$

27. (c) Became vertical downward displacement of both (barrel and bullet) will be equal.

28. (b) As $H = \frac{u^2\sin^2\theta}{2g}$ $\therefore$ $\frac{H_1}{H_2} = \frac{\sin^2\theta_1}{\sin\theta_2} = \frac{\sin^2 30°}{\sin^2 60} = \frac{1/4}{3/4} = \frac{1}{3}$

29. (d) $R = \frac{v^2\sin 2\theta}{g} \Rightarrow \theta = \frac{1}{2}\sin^{-1}\left(\frac{gR}{v^2}\right)$

30. (a) $T = \frac{2u\sin\theta}{g} = 10\sec \Rightarrow u\sin\theta = 50\ m/s$

$\therefore\ H = \frac{u^2\sin^2\theta}{2g} = \frac{(u\sin\theta)^2}{2g} = \frac{50\times 50}{2\times 10} = 125m$

31. (b) For complementary angles range will be equal.

32. (b) $R = \frac{u^2\sin 2\theta}{g} = \frac{(500)^2\times\sin 30°}{10} = 12.5\times 10^3\ m$

33. (a) $T = \frac{2u\sin\theta}{g} \Rightarrow u = \frac{T\times g}{2\sin\theta} = \frac{2\times 9.8}{2\times\sin 30} = 19.6\ m/s$

34. (c) $R = \frac{u^2\sin 2\theta}{g} = R\propto u^2$. So if the speed of projection doubled, the range will becomes four times,

i.e., $4\times 50 = 200\,m$

35. (c) Range will be equal for complementary angles.

36. (a) When the angle of projection is very far from 45° then range will be minimum.

37. (a) $H = \frac{u^2\sin^2\theta}{2g}$ and $T = \frac{2u\sin\theta}{g}$

So $\frac{H}{T^2} = \frac{u^2\sin^2\theta/2g}{4u^2\sin^2\theta/g^2} = \frac{g}{8} = \frac{5}{4}$

38. (a) $H_1 = \frac{u^2\sin^2\theta}{2g}$ and $H_2 = \frac{u^2\sin^2(90-\theta)}{2g} = \frac{u^2\cos^2\theta}{2g}$

$H_1H_2 = \frac{u^2\sin^2\theta}{2g}\times\frac{u^2\cos^2\theta}{2g} = \frac{(u^2\sin 2\theta)^2}{16g^2} = \frac{R^2}{16}$

$\therefore R = 4\sqrt{H_1H_2}$

39. (d) Standard equation of projectile motion

$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$

Comparing with given equation

$A = \tan\theta$ and $B = \frac{g}{2u^2\cos^2\theta}$

So $\frac{A}{B} = \frac{\tan\theta\times 2u^2\cos^2\theta}{g} = 40$

(As $\theta = 45°, u = 20\ m/s, g = 10\ m/s^2$)

40. (b) Range $= \frac{u^2\sin 2\theta}{g}$. It is clear that range is proportional to the direction (angle) and the initial speed.

41. (c) $\frac{2u\sin\theta}{g} = 2\sec \Rightarrow u\sin\theta = 10$

$\therefore\ H = \frac{u^2\sin^2\theta}{2g} = \frac{100}{2g} = 5m$

42. (b) Only horizontal component of velocity $(u\cos\theta)$.

43. (a) For complementary angles range is same.

44. (b) $T = \frac{2u\sin\theta}{g} = \frac{2\times 9.8\times\sin 30}{9.8} = 1\,s$

45. (a) $x = 36t$ $\therefore$ $v_x = \frac{dx}{dt} = 36\,m/s$

$y = 48t - 4.9t^2$ $\therefore$ $v_y = 48 - 9.8t$

at $t = 0$ $v_x = 36$ and $v_y = 48\,m/s$

So, angle of projection $\theta = \tan^{-1}\left(\frac{v_y}{v_x}\right) = \tan^{-1}\left(\frac{4}{3}\right)$

Or $\theta = \sin^{-1}(4/5)$

46. (b) For same range angle of projection should be θ and $90-\theta$

So, time of flights $t_1 = \frac{2u\sin\theta}{g}$ and

$t_2 = \frac{2u\sin(90-\theta)}{g} = \frac{2u\cos\theta}{g}$

By multiplying $= t_1t_2 = \frac{4u^2\sin\theta\cos\theta}{g^2}$

$t_1t_2 = \frac{2}{g}\frac{(u^2\sin 2\theta)}{g} = \frac{2R}{g} \Rightarrow t_1t_2\propto R$

47. (c) Instantaneous velocity of rising mass after t sec will be $v_t = \sqrt{v_x^2 + v_y^2}$

where $v_x = v\cos\theta =$ Horizontal component of velocity

$v_y = v\sin\theta - gt =$ Vertical component of velocity

$v_t = \sqrt{(v\cos\theta)^2 + (v\sin\theta - gt)^2}$

$v_t = \sqrt{v^2 + g^2t^2 - 2v\sin\theta\, gt}$

48. (d) Maximum range $= \frac{u^2}{g} = 100\,m$

Maximum height $= \frac{u^2}{2g} = \frac{100}{2} = 50\,m$

49. (c) $R_{\max} = \frac{u^2}{g} = 100 \Rightarrow u = 10\sqrt{10} = 32\,m/s$

50. (c) Since horizontal component of velocity is constant, hence momentum is constant.

51. (a) Time of flight $= \frac{2u\sin\theta}{g} = \frac{2u_y}{g} = \frac{2\times u_{\text{vertical}}}{g}$

52. (a) Person will catch the ball if its velocity will be equal to horizontal component of velocity of the ball.

$\frac{v_0}{2} = v_0\cos\theta \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = 60°$

53. (b) $H = \frac{u^2\sin^2\theta}{2g}$ and $T = \frac{2u\sin\theta}{g} \Rightarrow T^2 = \frac{4u^2\sin^2\theta}{g^2}$

$\therefore \frac{T^2}{H} = \frac{8}{g} \Rightarrow T = \sqrt{\frac{8H}{g}} = 2\sqrt{\frac{2H}{g}}$

54. (d) $R = 4H\cot\theta$, if $R = 4\sqrt{3}H$ then $\cot\theta = \sqrt{3} \Rightarrow \theta = 30°$

55. (c) The vertical component of velocity of projection $= -50\sin 30° = -25\, m/s$

If t be the time taken to reach the ground,

$h = ut + \frac{1}{2}gt^2 \Rightarrow 70 = -25t + \frac{1}{2}\times 10t^2$

$\Rightarrow 70 = -25t + 5t^2 \Rightarrow t^2 - 5t - 14 = 0 \Rightarrow t = -2s$ and $7s$

Since, $t = -2\ s$ is not valid $\therefore t = 7\ s$

56. (c) $H_{-} = \frac{u^2\sin^2\theta}{2g}$

According to problem $\frac{u_1^{\,2}\sin^2 45°}{2g} = \frac{u_2^{\,2}\sin^2 60°}{2g}$

$\Rightarrow \frac{u_1^{\,2}}{u_2^{\,2}} = \frac{\sin^2 60°}{\sin^2 45°} \Rightarrow \frac{u_1}{u_2} = \frac{\sqrt{3}/2}{1/\sqrt{2}} = \sqrt{\frac{3}{2}}.$

57. (c)

58. (d) $R = 4H\cot\theta$, if $\theta = 45°$ then $R = 4H \Rightarrow \frac{R}{H} = \frac{4}{1}$

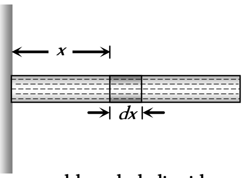
59. (b) $R_{\max} = \frac{u^2}{g} = 400m$ (For $\theta = 45°$)

$H_{\max} = \frac{u^2}{2g} = \frac{400}{2} = 200m$ (For $\theta = 90°$)

Critical Thinking Questions

1. (c,d) In the given condition, the particle undergoes uniform circular motion and for uniform circular motion the velocity and acceleration vector changes continuously but kinetic energy is constant for every point.

2. (a) $dM = \left(\frac{M}{L}\right)dx$



force on 'dM mass is

$dF = (dM)\omega^2 x$

By integration we can get the force exerted by whole liquid

$\Rightarrow F = \int_0^L \frac{M}{L}\omega^2 x\, dx = \frac{1}{2}M\omega^2 L$

3. (b) According to given problem $\frac{1}{2}mv^2 = as^2 \Rightarrow v = s\sqrt{\frac{2a}{m}}$

So $a_R = \frac{v^2}{R} = \frac{2as^2}{mR}$ ...(i)

Further more as $a_t = \frac{dv}{dt} = \frac{dv}{ds}\cdot\frac{ds}{dt} = v\frac{dv}{ds}$ ...(ii)

(By chain rule)

Which in light of equation (i) *i.e.* $v = s\sqrt{\frac{2a}{m}}$ yields

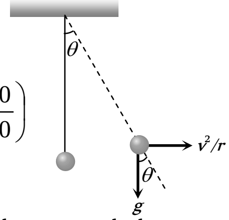
$a_t = \left[s\sqrt{\frac{2a}{m}}\right]\left[\sqrt{\frac{2a}{m}}\right] = \frac{2as}{m}$ ...(iii)

So that $a = \sqrt{a_R^2 + a_t^2} = \sqrt{\left[\frac{2as^2}{mR}\right]^2 + \left[\frac{2as}{m}\right]^2}$

Hence $a = \frac{2as}{m}\sqrt{1 + [s/R]^2}$

$\therefore F = ma = 2as\sqrt{1 + [s/R]^2}$

4. (c) $\tan\theta = \frac{v^2/r}{g} = \frac{v^2}{rg}$



$\therefore \theta = \tan^{-1}\left(\frac{v^2}{rg}\right) = \tan^{-1}\left(\frac{10\times 10}{10\times 10}\right)$

$\therefore \theta = \tan^{-1}(1) = 45°$

5. (b) Here the tangential acceleration also exits which requires power.

Given that $a_C = k^2rt^2$ and $a_C = \frac{v^2}{r}$ $\therefore \frac{v^2}{r} = k^2rt^2$

or $v^2 = k^2r^2t^2$ or $v = krt$

Tangential acceleration $a = \frac{dv}{dt} = kr$

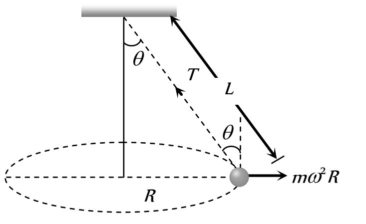
Now force $F = m\times a = mkr$

So power $P = F\times v = mkr\times krt = mk^2r^2t$

6. (d) $T\sin\theta = M\omega^2 R$ ...(i)

$T\sin\theta = M\omega^2 L\sin\theta$ ...(ii)

From (i) and (ii)



$T = M\omega^2 L$

$= M\,4\pi^2 n^2 L$

$= M\,4\pi^2\left(\frac{2}{\pi}\right)^2 L$

$= 16\,ML$

7. (d) Since the maximum tension T_B in the string moving in the vertical circle is at the bottom and minimum tension T_T is at the top.

$\therefore T_B = \frac{mv_B^2}{L} + mg$ and $T_T = \frac{mv_T^2}{L} - mg$

$\therefore \frac{T_B}{T_T} = \frac{\frac{mv_B^2}{L} + mg}{\frac{mv_T^2}{L} - mg} = \frac{4}{1}$ or $\frac{v_B^2 + gL}{v_T^2 - gL} = \frac{4}{1}$

or $v_B^2 + gL = 4v_T^2 - 4gL$ but $v_B^2 = v_T^2 + 4gL$

$\therefore v_T^2 + 4gL + gL = 4v_T^2 - 4gL \Rightarrow 3v_T^2 = 9gL$

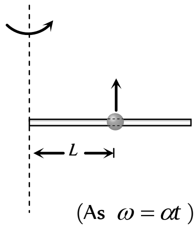
$\therefore\ v_T^2 = 3\times g\times L = 3\times 10\times\frac{10}{3}$ or $v_T = 10\, m/\sec$

8. (a) For particle P, motion between A and C will be an accelerated one while between C and B a retarded one. But in any case horizontal component of it's velocity will be greater than or equal to v on the other hand in case of particle Q, it is always equal to v. horizontal displacement of both the particles are equal, so $t_P < t_Q$.

9. (a) Let the bead starts slipping after time t

For critical condition

Frictional force provides the centripetal force



$$m\omega^2 L = \mu R = \mu m\times a_t = \mu Lm\alpha$$

$$\Rightarrow m(\alpha t)^2 L = \mu mL\alpha \Rightarrow t = \sqrt{\frac{\mu}{\alpha}} \quad (\text{As } \omega = \alpha t)$$

10. (a) Normal reaction at the highest point

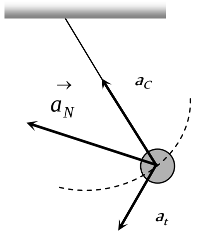
$$R = \frac{mv^2}{r} - mg$$

Reaction is inversely proportional to the radius of the curvature of path and radius is minimum for path depicted in (a).

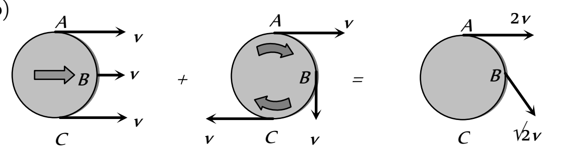
11. (c) a_c = centripetal acceleration

a_t = tangential acceleration

a_N = net acceleration = Resultant of a_c and a_t



12. (b)



Pure translation + Pure Rotation = Rolling without Slipping

13. (d) $\frac{1}{2}mu^2 - \frac{1}{2}mv^2 = mgL$

$$\Rightarrow v = \sqrt{u^2 - 2gL}$$

$$|\vec{v} - \vec{u}| = \sqrt{u^2 + v^2} = \sqrt{u^2 + u^2 - 2gL} = \sqrt{2(u^2 - gL)}$$

14. (a) When driver applies brakes and the car covers distance x before coming to rest, under the effect of retarding force F

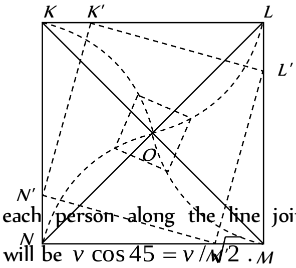
then $\frac{1}{2}mv^2 = Fx \Rightarrow x = \frac{mv^2}{2F}$

But when he takes turn then $\frac{mv^2}{r} = F \Rightarrow r = \frac{mv^2}{F}$

It is clear that $x = r/2$

i.e. by the same retarding force the car can be stopped in a less distance if the driver apply breaks. This retarding force is actually a friction force.

15. (a) It is obvious from considerations of symmetry that at any moment of time all of the persons will be at the corners of square whose side gradually decreases (see fig.) and so they will finally meet at the centre of the square O.



The speed of each person along the line joining his initial position and O will be $v\cos 45 = v/\sqrt{2}$.

As each person has displacement $d\cos 45 = d/\sqrt{2}$ to reach the centre, the four persons will meet at the centre of the square O after time.

$$\therefore\ t = \frac{d/\sqrt{2}}{v/\sqrt{2}} = \frac{d}{v}$$

16. (a,b) $x = a\cos(pt)$ and $y = b\sin(pt)$ (given)

$\therefore\ \cos pt = \frac{x}{a}$ and $\sin pt = \frac{y}{b}$

By squaring and adding

$$\cos^2(pt) + \sin^2(pt) = \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Hence path of the particle is ellipse.

Now differentiating x and y *w.r.t.* time

$$v_x = \frac{dx}{dt} = \frac{d}{dt}(a\cos(pt)) = -ap\sin(pt)$$

$$v_y = \frac{dy}{dt} = \frac{d}{dt}(b\sin(pt)) = bp\cos(pt)$$

$$\therefore\ \vec{v} = v_x\hat{i} + v_y\hat{j} = -ap\sin(pt)\hat{i} + bp\cos(pt)\hat{j}$$

Acceleration $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}[-ap\sin(pt)\hat{i} + bp\cos(pt)\hat{j}]$

$$\vec{a} = -ap^2\cos(pt)\,\hat{i} - bp^2\sin(pt)\hat{j}$$

Velocity at $t = \frac{\pi}{2p}$

$$\vec{v} = -ap\sin p\left(\frac{\pi}{2p}\right)\hat{i} + bp\cos p\left(\frac{\pi}{2p}\right)\hat{j} = -ap\,\hat{i}$$

Acceleration at $t = \frac{\pi}{2p}$

$$\vec{a} = ap^2\cos p\left(\frac{\pi}{2p}\right)\hat{i} - bp^2\sin p\left(\frac{\pi}{2p}\right)\hat{j} = -bp^2\hat{j}$$

As $\vec{v}\,.\,\vec{a} = 0$

Hence velocity and acceleration are perpendicular to each other at $t = \frac{\pi}{2p}$.

17. (b) $\Delta\vec{\upsilon} = \vec{\upsilon}_2 - \vec{\upsilon}_1 = \sqrt{\upsilon_1^2 + \upsilon_2^2 - 2\upsilon_1\upsilon_2\cos 90^o} = \sqrt{5^2 + 5^2} = 5\sqrt{2}$

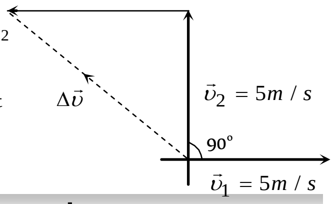
$-\vec{v}_1$

Average acceleration

$= \frac{\Delta v}{\Delta t} = \frac{5\sqrt{2}}{10} = \frac{1}{\sqrt{2}}$ m/s ^{2}

Directed toward north-west

(As clear from the figure).



Graphical Questions

1. (d) $R = \frac{u^2 \sin 2\theta}{g} = \frac{2u_x v_y}{g}$

$\therefore$ Range $\propto$ horizontal initial velocity (u)

In path 4 range is maximum so football possess maximum horizontal velocity in this path.

2. (a) If air resistance is taken into consideration then range and maximum height, both will decrease.

3. (b)

4. (c)

5. (d)

Assertion and Reason

1. (e) At the highest point, vertical component of velocity becomes zero so there will be only horizontal velocity and it is perpendicular to the acceleration due to gravity.

2. (a) $H = \frac{u^2 \sin^2 \theta}{2g}$ *i.e.* it is independent of mass of projectile.

3. (c) $R = \frac{u^2 \sin 2\theta}{g}$ $\therefore R_{\max} = \frac{u^2}{g}$ when $\theta = 45°$ $\therefore R_{\max} \propto u^2$

Height $H = \frac{u^2 \sin^2 \theta}{2g} \Rightarrow H_{\max} = \frac{u^2}{2g}$ when $\theta = 90°$

It is clear that $H_{\max} = \frac{R_{\max}}{2}$

4. (c) Horizontal range depends upon angle of projection and it is same for complementary angles *i.e.* θ and $(90 - \theta)$.

5. (b) We know $R = 4H \cot \theta$

if $R = H$ then $\cot \theta = \left(\frac{1}{4}\right)$ or tan $\theta = (4)$

and $R = \frac{u^2 \sin 2\theta}{g}$ $\therefore$ $R \propto \frac{u^2}{g}$

6. (d) $y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$

7. (d) If a body is projected from a place above the surface of earth, then for the maximum range, the angle of projection should be slightly less than 45^o.

8. (a) Both body will take same time to reach the earth because vertical downward component of velocity for both the bodies will be zero and time of descent $t = \sqrt{\frac{2h}{g}}$. Horizontal velocity has no effect on the vertical direction.

9. (c) $T \propto u$ and $R \propto u^2$

When velocity of projection of a body is made n times, then its time of flight becomes n times and range becomes n times.

10. (c) Range will be maximum when $\theta = 45°$ and in this condition

$R = 4H \Rightarrow H = R/4$ (always)

because $R = 4H \cot \theta$ and $\theta = 45°$

So maximum height is 25% of maximum range.

It does not depends upon the velocity of projection.

11. (a) Range, $R = \frac{u^2 \sin 2\theta}{g}$

when $\theta = 45^o$, $R_{\max} = \frac{u^2}{g} \sin 90^o = \frac{u^2}{g}$

when $\theta = 135^o$, $R_{\max} = \frac{u^2}{g} \sin 270^o = \frac{-u^2}{g}$

Negative sign shows opposite direction.

12. (e) The man should point his rifle at a point higher than the target since the bullet suffers a vertically downward deflection $\left(y = \frac{1}{2} gt^2\right)$ due to gravity.

13. (b) In uniform circular motion, the magnitude of velocity and acceleration remains same, but due to change in direction of motion, the direction of velocity and acceleration changes. Also the centripetal acceleration is given by $a = \omega^2 r$.

14. (a) The body is able to move in a circular path due to centripetal force. The centripetal force in case of vehicle is provided by frictional force. Thus if the value of frictional force μmg is less than centripetal force, then it is not possible for a vehicle to take a turn and the body would overturn.

Thus condition for safe turning of vehicle is, $\mu mg \geq \frac{mv^2}{r}$.

15. (c) In circular motion the frictional force acting towards the centre of the horizontal circular path provides the centripetal force and avoid overturning of vehicle. Due to the change in direction of motion, velocity changes in circular motion.

16. (b) On an unbanked road, friction provides the necessary centripetal force $\frac{mv^2}{r} = \mu mg$ $\therefore v = \sqrt{\mu rg}$.

Thus with increase in friction, safe velocity limit also increases.

When the road is banked with angle of θ then its limiting velocity is given by $v = \sqrt{\frac{rg(\tan \theta + \mu)}{1 - \mu \tan \theta}}$.

Thus limiting velocity increase with banking of road.

17. (d) If the speed of a body is constant, all curved paths are possible.

In uniform circular motion a body has constant speed, but its direction keeps on changing, due to which it has non-zero acceleration.

18. (a) We know that $W = Fs\cos\theta$

in the circular motion if $\theta = 90^o$ then $W = 0$

19. (d) While moving along a circle, the body has a constant tendency to regain its natural straight line path.

This tendency gives rise to a force called centrifugal force. The centrifugal force does not act on the body in motion, the only force acting on the body in motion is centripetal force. The centrifugal force acts on the source of centripetal force to displace it radially outward from centre of the path.

20. (c) Centripetal force is defined from formula

$$F = \frac{mv^2}{r} \Rightarrow F \propto \frac{v^2}{r}$$

If v and r both are doubled then F also gets doubled.

21. (b) When automobile moves in circular path then reaction on inner wheel and outer wheel will be different.

$$R_{\text{inner}} = \frac{M}{2}\left[g - \frac{v^2h}{ra}\right] \text{ and } R_{\text{outer}} = \frac{M}{2}\left[g + \frac{v^2h}{ra}\right]$$

In critical condition $v_{\text{safe}} = \sqrt{\frac{gra}{h}}$

If v is equal or more that thus critical value then reaction on inner wheel becomes zero. So it leaves the ground first.

22. (c) For safe turn $\tan\theta \geq \frac{v^2}{rg}$.

It is clear that for safe turn v should be small and r should be large. Also bending angle from the vertical would increase with increase in velocity.

23. (a) When roads are not properly banked, force of friction between tyres and road provides partially the necessary centripetal force. This cause wear and tear of tyres.

24. (d) When the milk is churned centrifugal force acts on it outward and due to which cream in milk is separated from it.

25. (e) Due to earth's axial rotation, the speed of the trains relative to earth will be different and hence the centripetal forces on them will be different. Thus their effective weights $mg - \frac{mv^2}{r}$ and $mg + \frac{mv^2}{r}$ will be different. So they exert different pressure on the rails.

26. (d) Within a certain speed of the turn table the frictional force between the coin and the turn table supplies the necessary centripetal force required for circular motion. On further increase of speed, the frictional force cannot supply the necessary centripetal force. Therefore the coin flies off tangentially.

Motion In Two Dimension

SET Self Evaluation Test -3

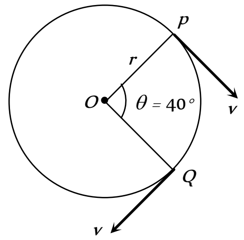
1. Roads are banked on curves so that
(a) The speeding vehicles may not fall outwards
(b) The frictional force between the road and vehicle may be decreased
(c) The wear and tear of tyres may be avoided
(d) The weight of the vehicle may be decreased

2. In uniform circular motion
(a) Both velocity and acceleration are constant
(b) Acceleration and speed are constant but velocity changes
(c) Both acceleration and velocity changes
(d) Both acceleration and speed are constant

3. For a body moving in a circular path, a condition for no skidding if μ is the coefficient of friction, is
(a) $\frac{mv^2}{r} \le \mu mg$ (b) $\frac{mv^2}{r} \ge \mu mg$
(c) $\frac{v}{r} = \mu g$ (d) $\frac{mv^2}{r} = \mu mg$

4. A car is moving with a uniform speed on a level road. Inside the car there is a balloon filled with helium and attached to a piece of string tied to the floor. The string is observed to be vertical. The car now takes a left turn maintaining the speed on the level road. The balloon in the car will
(a) Continue to remain vertical
(b) Burst while taking the curve
(c) Be thrown to the right side
(d) Be thrown to the left side

5. A particle is moving on a circular path of radius r with uniform velocity v. The change in velocity when the particle moves from P to Q is $(\angle POQ = 40^o)$



(a) $2v\cos 40^o$
(b) $2v\sin 40^o$
(c) $2v\sin 20^o$
(d) $2v\cos 20^o$

6. A body is revolving with a uniform speed v in a circle of radius r. The tangential acceleration is
(a) $\frac{v}{r}$ (b) $\frac{v^2}{r}$
(c) Zero (d) $\frac{v}{r^2}$

7. A particle does uniform circular motion in a horizontal plane. The radius of the circle is 20 *cm*. The centripetal force acting on the particle is 10 *N*. It's kinetic energy is
(a) 0.1 *J* (b) 0.2 *J*
(c) 2.0 *J* (d) 1.0 *J*

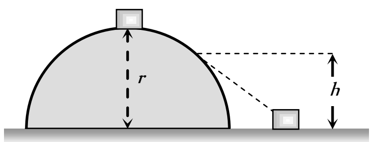
8. A body of mass m is suspended from a string of length l. What is minimum horizontal velocity that should be given to the body in its lowest position so that it may complete one full revolution in the vertical plane with the point of suspension as the centre of the circle
(a) $v = \sqrt{2lg}$ (b) $v = \sqrt{3lg}$
(c) $v = \sqrt{4lg}$ (d) $v = \sqrt{5lg}$

9. A particle moves with constant angular velocity in circular path of certain radius and is acted upon by a certain centripetal force F. If the angular velocity is doubled, keeping radius the same, the new force will be
(a) $2F$ (b) F^2
(c) $4F$ (d) $F/2$

10. In the above question, if the angular velocity is kept same but the radius of the path is halved, the new force will be
(a) $2F$ (b) F^2
(c) $F/2$ (d) $F/4$

11. In above question, if the centripetal force F is kept constant but the angular velocity is doubled, the new radius of the path (original radius R) will be
(a) $2R$ (b) $R/2$
(c) $R/4$ (d) $4R$

12. A small body of mass m slides down from the top of a hemisphere of radius r. The surface of block and hemisphere are frictionless. The height at which the body lose contact with the surface of the sphere is



(a) $\frac{3}{2}r$ (b) $\frac{2}{3}r$
(c) $\frac{1}{2}gt^2$ (d) $\frac{v^2}{2g}$

13. A body of mass m kg is rotating in a vertical circle at the end of a string of length r metre. The difference in the kinetic energy at the top and the bottom of the circle is
(a) $\frac{mg}{r}$ (b) $\frac{2mg}{r}$
(c) $2mgr$ (d) mgr

14. A car is travelling with linear velocity v on a circular road of radius r. If it is increasing its speed at the rate of 'a' $meter/\sec^2$, then the resultant acceleration will be

(a) $\sqrt{\left\{\frac{v^2}{r^2}-a^2\right\}}$ (b) $\sqrt{\left\{\frac{v^4}{r^2}+a^2\right\}}$

(c) $\sqrt{\left\{\frac{v^4}{r^2}-a^2\right\}}$ (d) $\sqrt{\left\{\frac{v^2}{r^2}+a^2\right\}}$

15. A ball of mass 0.1 *kg* is suspended by a string. It is displaced through an angle of 60^o and left. When the ball passes through the mean position, the tension in the string is

(a) 19.6 *N* (b) 1.96 *N*

(c) 9.8 *N* (d) Zero

16. An aeroplane moving horizontally at a speed of 200 *m/s* and at a height of $8.0\times10^3\ m$ is to drop a bomb on a target. At what horizontal distance from the target should the bomb be released

(a) 7.234 *km* (b) 8.081 *km*

(c) 8.714 *km* (d) 9.124 *km*

17. A body is projected horizontally from a height with speed 20 *metres/sec*. What will be its speed after 5 *seconds* $(g=10\ metres/\sec^2)$

(a) 54 *metres/sec* (b) 20 *metres/sec*

(c) 50 *metres/sec* (d) 70 *metres/sec*

18. A man standing on the roof of a house of height h throws one particle vertically downwards and another particle horizontally with the same velocity u. The ratio of their velocities when they reach the earth's surface will be

(a) $\sqrt{2gh+u^2}:u$ (b) $1:2$

(c) $1:1$ (d) $\sqrt{2gh+u^2}:\sqrt{2gh}$

19. (A projectile projected at an angle 30^o from the horizontal has a range R. If the angle of projection at the same initial velocity be 60^o, then the range will be

(a) R (b) $2R$

(c) $R/2$ (d) R^2

20. At the highest point of the path of a projectile, its

(a) Kinetic energy is maximum

(b) Potential energy is minimum

(c) Kinetic energy is minimum

(d) Total energy is maximum

21. A cricket ball is hit at 30^o with the horizontal with kinetic energy K. The kinetic energy at the highest point is

(a) Zero (b) $K/4$

(c) $K/2$ (d) $3K/4$

22. A cannon on a level plane is aimed at an angle θ above the horizontal and a shell is fired with a muzzle velocity v_0 towards a vertical cliff a distance D away. Then the height from the bottom at which the shell strikes the side walls of the cliff is

(a) $D\sin\theta-\frac{gD^2}{2v_0^2\sin^2\theta}$ (b) $D\cos\theta-\frac{gD^2}{2v_0^2\cos^2\theta}$

(c) $D\tan\theta-\frac{gD^2}{2v_0^2\cos^2\theta}$ (d) $D\tan\theta-\frac{gD^2}{2v_0^2\sin^2\theta}$

23. A stone is projected from the ground with velocity 50 *m/s* at an angle of 30^o. It crosses a wall after 3 *sec*. How far beyond the wall the stone will strike the ground $(g=10\,m/\sec^2)$

(a) 90.2 *m* (b) 89.6 *m*

(c) 86.6 *m* (d) 70.2 *m*

24. A body of mass m is projected at an angle of 45^o with the horizontal. If air resistance is negligible, then total change in momentum when it strikes the ground is

(a) $2mv$ (b) $\sqrt{2}\,mv$

(c) mv (d) $mv/\sqrt{2}$

25. A ball of mass m is thrown vertically upwards. Another ball of mass $2m$ is thrown at an angle θ with the vertical. Both of them stay in air for same period of time. The heights attained by the two balls are in the ratio of

(a) 2 : 1 (b) $1:\cos\theta$

(c) 1 : 1 (d) $\cos\theta:1$

26. A particle is projected with a velocity *v* such that its range on the horizontal plane is twice the greatest height attained by it. The range of the projectile is (where g is acceleration due to gravity)

(a) $\frac{4v^2}{5g}$ (b) $\frac{4g}{5v^2}$

(c) $\frac{v^2}{g}$ (d) $\frac{4v^2}{\sqrt{5}g}$

Answers and Solutions

(SET -3)

1. (a) By doing so component of weight of vehicle provides centripetal force.

2. (c) Both changes in direction although their magnitudes remains constant.

3. (a) The value of frictional force should be equal or more than required centripetal force. *i.e.* $\mu mg\geq\frac{mv^2}{r}$

4. (d) Air outside the balloon is heavier so it will have more tendency to move towards right and will keep the balloon towards left side (Here in this question car is supposed to be air tight).

5. (c) Change in velocity = $2v\sin(\theta/2) = 2v\sin 20^\circ$

6. (c) In uniform circular motion only centripetal acceleration works.

7. (d) $\frac{mv^2}{r} = 10 \Rightarrow \frac{1}{2}mv^2 = 10 \times \frac{r}{2} = 1\,J$

8. (d) For looping the loop minimum velocity at the lowest point should be $\sqrt{5gl}$.

9. (c) $F = m\omega^2 R \;\therefore\; F \propto \omega^2$ (m and R are constant)
If angular velocity is doubled force will becomes four times.

10. (c) $F = m\omega^2 R \;\therefore\; F \propto R$ (m and ω are constant)
If radius of the path is halved, then force will also become half.

11. (c) $F = m\omega^2 R \;\therefore\; R \propto \frac{1}{\omega^2}$ (m and F are constant)
If ω is doubled then radius will become 1/4 times *i.e.* $R/4$

12. (b)

13. (c) Difference in K.E. = Difference in P.E. = $2mgr$

14. (b) $a_{\text{resultant}} = \sqrt{a_{\text{radial}}^2 + a_{\text{tangential}}^2} = \sqrt{\frac{v^4}{r^2} + a^2}$

15. (b) $T = mg + \frac{mv^2}{l} = mg + \frac{m}{l}\left[2gl(1-\cos\theta)\right]$

$= mg + 2mg(1-\cos 60^\circ) = 2mg = 2 \times 0.1 \times 9.8 = 1.96\,N$

16. (b) Horizontal distance travelled by the bomb $S = u \times t$

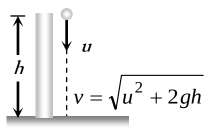
$= 200 \times \sqrt{\frac{2h}{g}} = 200 \times \sqrt{\frac{2 \times 8 \times 10^3}{9.8}} = 8.081\,km$

17. (a) Horizontal velocity $v_x = 20\,m/s$

Vertical velocity $v_y = u + gt = 0 + 10 \times 5 = 50\,m/\text{sec}$

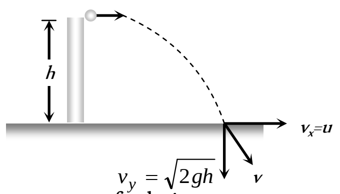
Net velocity $v = \sqrt{v_x^2 + v_y^2} = \sqrt{(20)^2 + (50)^2} = 54$ *m/s.*

18. (c) When particle thrown in vertical downward direction with velocity u then final velocity at the ground level



$v^2 = u^2 + 2gh \;\therefore\; v = \sqrt{u^2 + 2gh}$

Another particle is thrown horizontally with same velocity then*** at the surface of earth.



Horizontal component of velocity $v_x = u$

$\therefore$ Resultant velocity, $v = \sqrt{u^2 + 2gh}$

For both the particle final velocities when they reach the earth's surface are equal.

19. (a) For complementary angles of projection horizontal range is same.

20. (c) At the highest point of the path. Potential energy is maximum, so the kinetic energy will be minimum.

21. (d) Kinetic energy at the highest point

$$K' = K\cos^2\theta = K\cos^2 30 = K\left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3K}{4}$$

22. (c) Equation of trajectory for oblique projectile motion

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$$

Substituting $x = D$ and $u = v_0$

$$h = D\tan\theta - \frac{gD^2}{2u_0^2\cos^2\theta}.$$

23. (c) Total time of flight $= \frac{2u\sin\theta}{g} = \frac{2 \times 50 \times 1}{2 \times 10} = 5\,s$

Time to cross the wall = 3 *sec* (given)
Time in air after crossing the wall = (5 – 3) = 2 *sec*
$\therefore$ Distance travelled beyond the wall $= (u\cos\theta)t$

$= 50 \times \frac{\sqrt{3}}{2} \times 2 = 86.6\,m$

24. (b) Change in momentum= $2mv\sin\theta = 2mv\sin\frac{\pi}{4} = \sqrt{2}mv$

25. (c) The vertical components of velocity of both the balls will be same if they stay in air for the same period of time. Hence vertical height attained will be same.

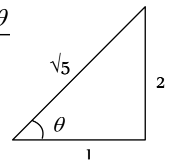
26. (a) $R = 2H$ given

We know $R = 4H\cot\theta \Rightarrow \cot\theta = \frac{1}{2}$

From triangle we can say that $\sin\theta = \frac{2}{\sqrt{5}}$, $\cos\theta = \frac{1}{\sqrt{5}}$

$\therefore$ Range of projectile $R = \frac{2v^2\sin\theta\cos\theta}{g}$

$= \frac{2v^2}{g} \times \frac{2}{\sqrt{5}} \times \frac{1}{\sqrt{5}} = \frac{4v^2}{5g}.$



Chapter

4

Newton's Laws of Motion

Point Mass

(1) An object can be considered as a point object if during motion in a given time, it covers distance much greater than its own size.

(2) Object with zero dimension considered as a point mass.

(3) Point mass is a mathematical concept to simplify the problems.

Inertia

(1) Inherent property of all the bodies by virtue of which they cannot change their state of rest or uniform motion along a straight line by their own is called inertia.

(2) Inertia is not a physical quantity, it is only a property of the body which depends on mass of the body.

(3) Inertia has no units and no dimensions

(4) Two bodies of equal mass, one in motion and another is at rest, possess same inertia because it is a factor of mass only and does not depend upon the velocity.

Linear Momentum

(1) Linear momentum of a body is the quantity of motion contained in the body.

(2) It is measured in terms of the force required to stop the body in unit time.

(3) It is also measured as the product of the mass of the body and its velocity *i.e.*, Momentum = mass × velocity.

If a body of mass m is moving with velocity $\vec{v}$ then its linear momentum $\vec{p}$ is given by $\vec{p} = m\vec{v}$

(4) It is a vector quantity and it's direction is the same as the direction of velocity of the body.

(5) Units : *kg-m/sec* [S.I.], *g-cm/sec* [C.G.S.]

(6) Dimension : $[MLT^{-1}]$

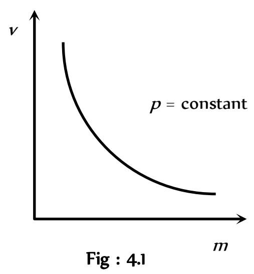


Fig : 4.1

(7) If two objects of different masses have same momentum, the lighter body possesses greater velocity.

$$p = m_1 v_1 = m_2 v_2 = \text{constant} \qquad \therefore \qquad \frac{v_1}{v_2} = \frac{m_2}{m_1}$$

i.e. $v \propto \dfrac{1}{m}$

[As p is constant]

(8) For a given body $p \propto v$

(9) For different bodies moving with same velocities $p \propto m$

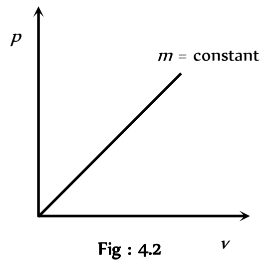


Fig : 4.2

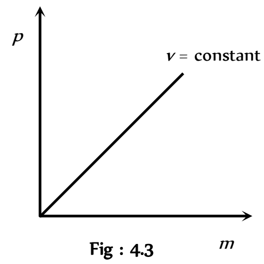


Fig : 4.3

Newton's First Law

A body continue to be in its state of rest or of uniform motion along a straight line, unless it is acted upon by some external force to change the state.

(1) If no net force acts on a body, then the velocity of the body cannot change *i.e.* the body cannot accelerate.

(2) Newton's first law defines inertia and is rightly called the law of inertia. Inertia are of three types :

Inertia of rest, Inertia of motion and Inertia of direction.

(3) **Inertia of rest :** It is the inability of a body to change by itself, its state of rest. This means a body at rest remains at rest and cannot start moving by its own.

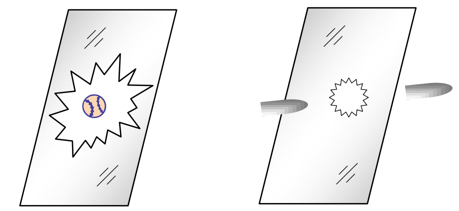
Example : (i) A person who is standing freely in bus, thrown backward, when bus starts suddenly.

When a bus suddenly starts, the force responsible for bringing bus in motion is also transmitted to lower part of body, so this part of the body comes in motion along with the bus. While the upper half of body (say above the waist) receives no force to overcome inertia of rest and so it stays in its original position. Thus there is a relative displacement between the two parts of the body and it appears as if the upper part of the body has been thrown backward.

Note : ❑ (i) If the motion of the bus is slow, the inertia of motion will be transmitted to the body of the person uniformly and so the entire body of the person will come in motion with the bus and the person will not experience any jerk.

(ii) When a horse starts suddenly, the rider tends to fall backward on account of inertia of rest of upper part of the body as explained above.

(iii) A bullet fired on a window pane makes a clean hole through it, while a ball breaks the whole window. The bullet has a speed much greater than the ball. So its time of contact with glass is small. So in case of bullet the motion is transmitted only to a small portion of the glass in that small time. Hence a clear hole is created in the glass window, while in case of ball, the time and the area of contact is large. During this time the motion is transmitted to the entire window, thus creating the cracks in the entire window.



Cracks by the ball   Hole by the bullet

Fig : 4.4

(iv) In the arrangement shown in the figure :

(a) If the string B is pulled with a sudden jerk then it will experience tension while due to inertia of rest of mass M this force will not be transmitted to the string A and so the string B will break.

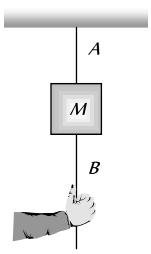


Fig : 4.5

(b) If the string B is pulled steadily the force applied to it will be transmitted from string B to A through the mass M and as tension in A will be greater than in B by Mg (weight of mass M), the string A will break.

(v) If we place a coin on smooth piece of card board covering a glass and strike the card board piece suddenly with a finger. The cardboard slips away and the coin falls into the glass due to inertia of rest.

(vi) The dust particles in a carpet falls off when it is beaten with a stick. This is because the beating sets the carpet in motion whereas the dust particles tend to remain at rest and hence separate.

(4) **Inertia of motion :** It is the inability of a body to change by itself its state of uniform motion *i.e.,* a body in uniform motion can neither accelerate nor retard by its own.

Example : (i) When a bus or train stops suddenly, a passenger sitting inside tends to fall forward. This is because the lower part of his body comes to rest with the bus or train but the upper part tends to continue its motion due to inertia of motion.

(ii) A person jumping out of a moving train may fall forward.

(iii) An athlete runs a certain distance before taking a long jump. This is because velocity acquired by running is added to velocity of the athlete at the time of jump. Hence he can jump over a longer distance.

(5) **Inertia of direction :** It is the inability of a body to change by itself it's direction of motion.

Example : (i) When a stone tied to one end of a string is whirled and the string breaks suddenly, the stone flies off along the tangent to the circle. This is because the pull in the string was forcing the stone to move in a circle. As soon as the string breaks, the pull vanishes. The stone in a bid to move along the straight line flies off tangentially.

(ii) The rotating wheel of any vehicle throw out mud, if any, tangentially, due to directional inertia.

(iii) When a car goes round a curve suddenly, the person sitting inside is thrown outwards.

Newton's Second Law

(1) The rate of change of linear momentum of a body is directly proportional to the external force applied on the body and this change takes place always in the direction of the applied force.

(2) If a body of mass m, moves with velocity $\vec{v}$ then its linear momentum can be given by $\vec{p} = m\vec{v}$ and if force $\vec{F}$ is applied on a body, then

$$\vec{F} \propto \frac{d\vec{p}}{dt} \Rightarrow F = K\frac{d\vec{p}}{dt}$$

or $\vec{F} = \frac{d\vec{p}}{dt}$ (K = 1 in C.G.S. and S.I. units)

or $\vec{F} = \frac{d}{dt}(m\vec{v}) = m\frac{d\vec{v}}{dt} = m\vec{a}$

(As $a = \frac{d\vec{v}}{dt}$ = acceleration produced in the body)

$\therefore \vec{F} = m\vec{a}$

Force = mass × acceleration

Force

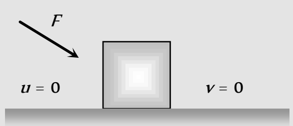
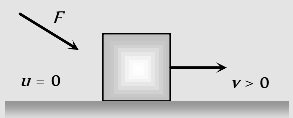
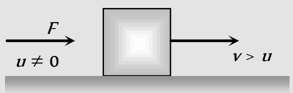
(1) Force is an external effect in the form of a push or pull which

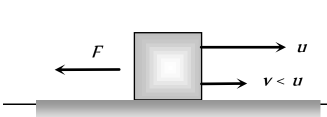
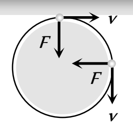
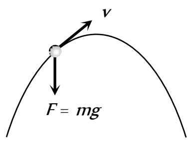
(i) Produces or tries to produce motion in a body at rest.

(ii) Stops or tries to stop a moving body.

(iii) Changes or tries to change the direction of motion of the body.

Table 4.1 : Various condition of force application

	Body remains at rest. Here force is trying to change the state of rest.
	Body starts moving. Here force changes the state of rest.
	In a small interval of time, force increases the magnitude of speed and direction of motion remains same.

	In a small interval of time, force decreases the magnitude of speed and direction of motion remains same.
	In uniform circular motion only direction of velocity changes, speed remains constant. Force is always perpendicular to velocity.
	In non-uniform circular motion, elliptical, parabolic or hyperbolic motion force acts at an angle to the direction of motion. In all these motions. Both magnitude and direction of velocity changes.

(2) Dimension : Force = mass × acceleration

$$[F]=[M][LT^{-2}]=[MLT^{-2}]$$

(3) Units : Absolute units : (i) *Newton* (S.I.) (ii) *Dyne* (C.G.S)

Gravitational units : (i) Kilogram-force (M.K.S.) (ii) Gram-force (C.G.S)

Newton : One Newton is that force which produces an acceleration of $1m/s^2$ in a body of mass 1 *Kilogram*.

$\therefore 1\ Newton\ = 1kg - m/s^2$

Dyne : One dyne is that force which produces an acceleration of $1cm/s^2$ in a body of mass 1 *gram*.

$\therefore$ 1 $Dyne = 1gm\,cm/\sec^2$

Relation between absolute units of force 1 *Newton* $= 10^5$ *Dyne*

Kilogram-force : It is that force which produces an acceleration of $9.8m/s^2$ in a body of mass 1 *kg*.

$\therefore$ 1 *kg-f* = 9.80 *Newton*

Gram-force : It is that force which produces an acceleration of $980cm/s^2$ in a body of mass 1*gm*.

$\therefore$ 1 *gm-f* = 980 *Dyne*

(4) $\vec{F} = m\vec{a}$ formula is valid only if force is changing the state of rest or motion and the mass of the body is constant and finite.

(5) If m is not constant $\vec{F} = \frac{d}{dt}(m\vec{v}) = m\frac{d\vec{v}}{dt} + \vec{v}\frac{dm}{dt}$

(6) If force and acceleration have three component along *x*, *y* and *z* axis, then

$$\vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k} \text{ and } \vec{a} = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$$

From above it is clear that $F_x = ma_x$, $F_y = ma_y$, $F_z = ma_z$

(7) No force is required to move a body uniformly along a straight line with constant speed.

$\vec{F} = m\vec{a}$ $\qquad \therefore \vec{F} = 0$ (As $\vec{a} = 0$)

(8) When force is written without direction then positive force means repulsive while negative force means attractive.

Example : *Positive force* – Force between two similar charges

Negative force – Force between two opposite charges

(9) Out of so many natural forces, for distance 10^{-15} *metre*, nuclear force is strongest while gravitational force weakest. $F_{\text{nuclear}} > F_{\text{electromagnetic}} > F_{\text{gravitational}}$

(10) Ratio of electric force and gravitational force between two electron's $F_e/F_g = 10^{43}$ $\quad \therefore F_e >> F_g$

(11) Constant force : If the direction and magnitude of a force is constant. It is said to be a constant force.

(12) Variable or dependent force :

(i) *Time dependent force* : In case of impulse or motion of a charged particle in an alternating electric field force is time dependent.

(ii) *Position dependent force* : Gravitational force between two bodies $\frac{Gm_1m_2}{r^2}$

or Force between two charged particles $= \frac{q_1q_2}{4\pi\varepsilon_0 r^2}$.

(iii) *Velocity dependent force* : Viscous force $(6\pi\eta rv)$

Force on charged particle in a magnetic field $(qvB\sin\theta)$

(13) Central force : If a position dependent force is directed towards or away from a fixed point it is said to be central otherwise non-central.

Example : Motion of Earth around the Sun. Motion of electron in an atom. Scattering of α-particles from a nucleus.

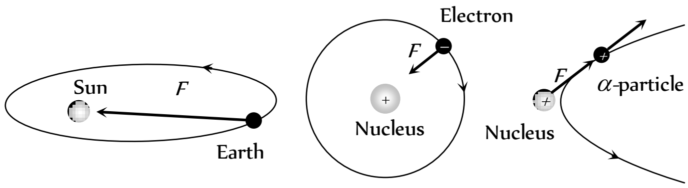


Fig : 4.6

(14) Conservative or non conservative force : If under the action of a force the work done in a round trip is zero or the work is path independent, the force is said to be conservative otherwise non conservative.

Example : Conservative force : Gravitational force, electric force, elastic force.

Non conservative force : Frictional force, viscous force.

(15) Common forces in mechanics :

(i) *Weight* : Weight of an object is the force with which earth attracts it. It is also called the force of gravity or the gravitational force.

(ii) *Reaction or Normal force :* When a body is placed on a rigid surface, the body experiences a force which is perpendicular to the surfaces in contact. Then force is called 'Normal force' or 'Reaction'.

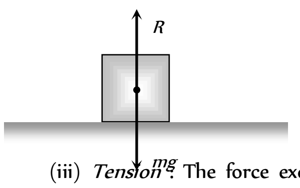


Fig : 4.7

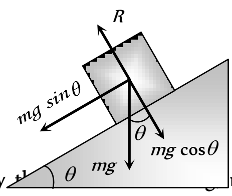


Fig : 4.8

(iii) *Tension* : The force exerted by [illegible] rope or chain against pulling (applied) force is called the tension. The direction of tension is so as to pull the body.

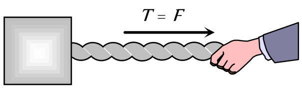


Fig : 4.9

(iv) *Spring force :* Every spring resists any attempt to change its length. This resistive force increases with change in length. Spring force is given by $F = -Kx$; where x is the change in length and K is the spring constant (unit N/m).

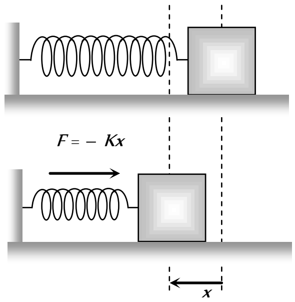


Fig : 4.10

Equilibrium of Concurrent Force

(1) If all the forces working on a body are acting on the same point, then they are said to be concurrent.

(2) A body, under the action of concurrent forces, is said to be in equilibrium, when there is no change in the state of rest or of uniform motion along a straight line.

(3) The necessary condition for the equilibrium of a body under the action of concurrent forces is that the vector sum of all the forces acting on the body must be zero.

(4) Mathematically for equilibrium $\sum \vec{F}_{\text{net}} = 0$ or $\sum F_x = 0$; $\sum F_y = 0$; , $\sum F_z = 0$

(5) Three concurrent forces will be in equilibrium, if they can be represented completely by three sides of a triangle taken in order.

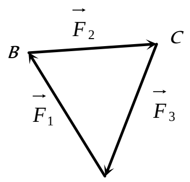


Fig : 4.11

(6) Lami's Theorem : For three concurrent forces in equilibrium $\dfrac{F_1}{\sin\alpha} = \dfrac{F_2}{\sin\beta} = \dfrac{F_3}{\sin\gamma}$

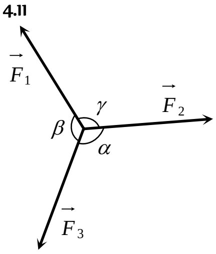


Fig : 4.12

Newton's Third Law

To every action, there is always an equal (in magnitude) and opposite (in direction) reaction.

(1) When a body exerts a force on any other body, the second body also exerts an equal and opposite force on the first.

(2) Forces in nature always occurs in pairs. A single isolated force is not possible.

(3) Any agent, applying a force also experiences a force of equal magnitude but in opposite direction. The force applied by the agent is called '*Action*' and the counter force experienced by it is called '*Reaction*'.

(4) Action and reaction never act on the same body. If it were so, the total force on a body would have always been zero *i.e.* the body will always remain in equilibrium.

(5) If $\vec{F}_{AB}$ = force exerted on body A by body B (Action) and $\vec{F}_{BA}$ = force exerted on body B by body A (Reaction)

Then according to Newton's third law of motion $\vec{F}_{AB} = -\vec{F}_{BA}$

(6) Example : (i) A book lying on a table exerts a force on the table which is equal to the weight of the book. This is the force of action.

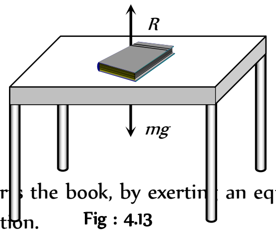


Fig : 4.13

The table supports the book, by exerting an equal force on the book. This is the force of reaction.

As the system is at rest, net force on it is zero. Therefore force of action and reaction must be equal and opposite.

(ii) Swimming is possible due to third law of motion.

(iii) When a gun is fired, the bullet moves forward (action). The gun recoils backward (reaction)

(iv) Rebounding of rubber ball takes place due to third law of motion.

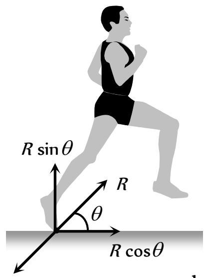


Fig : 4.14

(v) While walking a person presses the ground in the backward direction (action) by his feet. The ground pushes the person in forward direction with an equal force (reaction). The component of reaction in horizontal direction makes the person move forward.

(vi) It is difficult to walk on sand or ice.

(vii) Driving a nail into a wooden block without holding the block is difficult.

Frame of Reference

(1) A frame in which an observer is situated and makes his observations is known as his 'Frame of reference'.

(2) The reference frame is associated with a co-ordinate system and a clock to measure the position and time of events happening in space. We can describe all the physical quantities like position, velocity, acceleration etc. of an object in this coordinate system.

(3) Frame of reference are of two types : (i) Inertial frame of reference (ii) Non-inertial frame of reference.

(i) **Inertial frame of reference :**

(a) A frame of reference which is at rest or which is moving with a uniform velocity along a straight line is called an inertial frame of reference.

(b) In inertial frame of reference Newton's laws of motion holds good.

(c) Inertial frame of reference are also called unaccelerated frame of reference or Newtonian or Galilean frame of reference.

(d) Ideally no inertial frame exist in universe. For practical purpose a frame of reference may be considered as inertial if it's acceleration is negligible with respect to the acceleration of the object to be observed.

(e) To measure the acceleration of a falling apple, earth can be considered as an inertial frame.

(f) To observe the motion of planets, earth can not be considered as an inertial frame but for this purpose the sun may be assumed to be an inertial frame.

Example : The lift at rest, lift moving (up or down) with constant velocity, car moving with constant velocity on a straight road.

(ii) **Non-inertial frame of reference**

(a) Accelerated frame of references are called non-inertial frame of reference.

(b) Newton's laws of motion are not applicable in non-inertial frame of reference.

Example : Car moving in uniform circular motion, lift which is moving upward or downward with some acceleration, plane which is taking off.

Impulse

(1) When a large force works on a body for very small time interval, it is called impulsive force.

An impulsive force does not remain constant, but changes first from zero to maximum and then from maximum to zero. In such case we measure the total effect of force.

(2) Impulse of a force is a measure of total effect of force.

(3) $\vec{I} = \int_{t_1}^{t_2} \vec{F}\, dt$.

(4) Impulse is a vector quantity and its direction is same as that of force.

(5) Dimension : $[MLT^{-1}]$

(6) Units : *Newton-second* or *Kg-m-*s^{-1} (S.I.)

Dyne-second or *gm-cm-*s^{-1} (C.G.S.)

(7) Force-time graph : Impulse is equal to the area under *F-t* curve.

If we plot a graph between force and time, the area under the curve and time axis gives the value of impulse.

$I =$ Area between curve and time axis

$= \frac{1}{2} \times$ Base $\times$ Height

$= \frac{1}{2} F\, t$

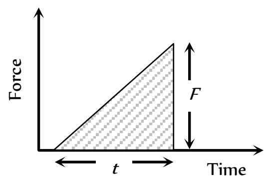


Fig : 4.15

(8) If F_{av} is the average magnitude of the force then

$$I = \int_{t_1}^{t_2} F\, dt = F_{av} \int_{t_1}^{t_2} dt = F_{av} \Delta t$$

(9) From Newton's second law

$$\vec{F} = \frac{d\vec{p}}{dt}$$

or $\int_{t_1}^{t_2} \vec{F}\, dt = \int_{p_1}^{p_2} d\vec{p}$

$\Rightarrow \vec{I} = \vec{p}_2 - \vec{p}_1 = \overrightarrow{\Delta p}$

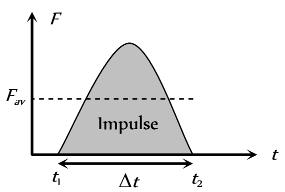


Fig : 4.16

i.e. The impulse of a force is equal to the change in momentum.

This statement is known as *Impulse momentum theorem*.

Examples : Hitting, kicking, catching, jumping, diving, collision *etc.*

In all these cases an impulse acts.

$$I = \int F\, dt = F_{av} . \Delta t = \Delta p = \text{constant}$$

So if time of contact Δt is increased, average force is decreased (or diluted) and vice-versa.

(i) In hitting or kicking a ball we decrease the time of contact so that large force acts on the ball producing greater acceleration.

(ii) In catching a ball a player by drawing his hands backwards increases the time of contact and so, lesser force acts on his hands and his hands are saved from getting hurt.

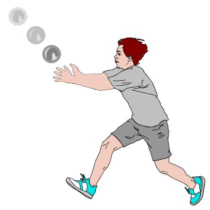


Fig : 4.17

(iii) In jumping on sand (or water) the time of contact is increased due to yielding of sand or water so force is decreased and we are not injured. However if we jump on cemented floor the motion stops in a very short interval of time resulting in a large force due to which we are seriously injured.

(iv) An athlete is advised to come to stop slowly after finishing a fast race, so that time of stop increases and hence force experienced by him decreases.

(v) China wares are wrapped in straw or paper before packing.

Law of Conservation of Linear Momentum

If no external force acts on a system (called isolated) of constant mass, the total momentum of the system remains constant with time.

(1) According to this law for a system of particles $\vec{F} = \frac{d\vec{p}}{dt}$

In the absence of external force $\vec{F} = 0$ then $\vec{p} =$ constant

i.e., $\vec{p} = \vec{p}_1 + \vec{p}_2 + \vec{p}_3 + ... =$ constant.

or $m_1 \vec{v}_1 + m_2 \vec{v}_2 + m_3 \vec{v}_3 + ... =$ constant

This equation shows that in absence of external force for a closed system the linear momentum of individual particles may change but their sum remains unchanged with time.

(2) Law of conservation of linear momentum is independent of frame of reference, though linear momentum depends on frame of reference.

(3) Conservation of linear momentum is equivalent to Newton's third law of motion.

For a system of two particles in absence of external force, by law of conservation of linear momentum.

$\vec{p}_1 + \vec{p}_2 =$ constant.

$\therefore\ m_1 \vec{v}_1 + m_2 \vec{v}_2 =$ constant.

Differentiating above with respect to time

$$m_1 \frac{d\vec{v}_1}{dt} + m_2 \frac{d\vec{v}_2}{dt} = 0 \Rightarrow m_1 \vec{a}_1 + m_2 \vec{a}_2 = 0 \Rightarrow \vec{F}_1 + \vec{F}_2 = 0$$

$\therefore\ \vec{F}_2 = -\vec{F}_1$

i.e. for every action there is an equal and opposite reaction which is Newton's third law of motion.

(4) Practical applications of the law of conservation of linear momentum

(i) When a man jumps out of a boat on the shore, the boat is pushed slightly away from the shore.

(ii) A person left on a frictionless surface can get away from it by blowing air out of his mouth or by throwing some object in a direction opposite to the direction in which he wants to move.

(iii) **Recoiling of a gun :** For bullet and gun system, the force exerted by trigger will be internal so the momentum of the system remains unaffected.

Fig : 4.18

Let $m_G =$ mass of gun, $m_B =$ mass of bullet,

$v_G =$ velocity of gun, $v_B =$ velocity of bullet

Initial momentum of system = 0

Final momentum of system $= m_G\vec{v}_G + m_B\vec{v}_B$

By the law of conservation of linear momentum

$$m_G\vec{v}_G + m_B\vec{v}_B = 0$$

So recoil velocity $\vec{v}_G = -\dfrac{m_B}{m_G}\vec{v}_B$

(a) Here negative sign indicates that the velocity of recoil $\vec{v}_G$ is opposite to the velocity of the bullet.

(b) $v_G \propto \dfrac{1}{m_G}$ *i.e.* higher the mass of gun, lesser the velocity of recoil of gun.

(c) While firing the gun must be held tightly to the shoulder, this would save hurting the shoulder because in this condition the body of the shooter and the gun behave as one body. Total mass become large and recoil velocity becomes too small.

$$v_G \propto \frac{1}{m_G + m_{\text{man}}}$$

(iv) **Rocket propulsion :** The initial momentum of the rocket on its launching pad is zero. When it is fired from the launching pad, the exhaust gases rush downward at a high speed and to conserve momentum, the rocket moves upwards.

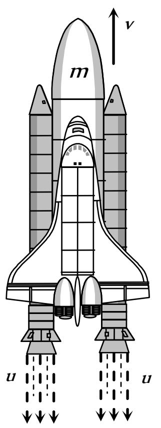


Fig : 4.19

Let $m_0 =$ initial mass of rocket,

m = mass of rocket at any instant 't' (instantaneous mass)

$m_r =$ residual mass of empty container of the rocket

u = velocity of exhaust gases,

v = velocity of rocket at any instant 't' (instantaneous velocity)

$\dfrac{dm}{dt} =$ rate of change of mass of rocket = rate of fuel consumption

= rate of ejection of the fuel.

(a) Thrust on the rocket : $F = -u\dfrac{dm}{dt} - mg$

Here negative sign indicates that direction of thrust is opposite to the direction of escaping gases.

$F = -u\dfrac{dm}{dt}$ (if effect of gravity is neglected)

(b) Acceleration of the rocket : $a = \dfrac{u}{m}\dfrac{dm}{dt} - g$

and if effect of gravity is neglected $a = \dfrac{u}{m}\dfrac{dm}{dt}$

(c) Instantaneous velocity of the rocket :

$$v = u\log_e\left(\frac{m_0}{m}\right) - gt$$

and if effect of gravity is neglected $v = u\log_e\left(\dfrac{m_0}{m}\right)$

$$= 2.303u\log_{10}\left(\frac{m_0}{m}\right)$$

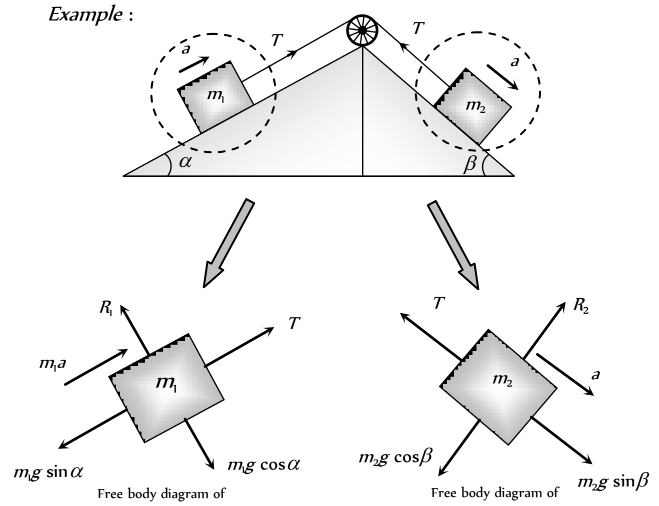
(d) Burnt out speed of the rocket : $v_b = v_{\max} = u\log_e\left(\dfrac{m_0}{m_r}\right)$

The speed attained by the rocket when the complete fuel gets burnt is called burnt out speed of the rocket. It is the maximum speed acquired by the rocket.

Free Body Diagram

In this diagram the object of interest is isolated from its surroundings and the interactions between the object and the surroundings are represented in terms of forces.

Example :



Free body diagram of

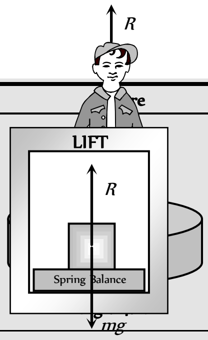
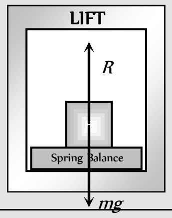
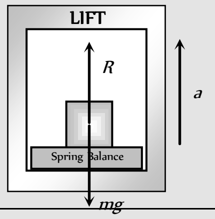
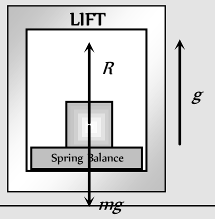
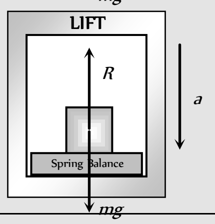
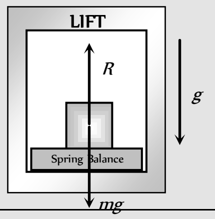
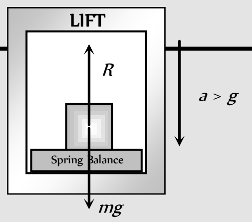
Free body diagram of

Apparent Weight of a Body in a Lift

When a body of mass m is placed on a weighing machine which is placed in a lift, then actual weight of the body is mg.

This acts on a weighing machine which offers a reaction R given by the reading of weighing machine. This reaction exerted by the surface of contact on the body is the *apparent weight* of the body.

Table 4.2 : Apparent weight in a lift

Condition	Figure	Velocity	Acceleration	Reaction	Conclusion
Lift is at rest		$v = 0$	$a = 0$	$R - mg = 0$ $\therefore\ R = mg$	Apparent weight = Actual weight
Lift moving upward or downward with constant velocity		v = constant	$a = 0$	$R - mg = 0$ $\therefore\ R = mg$	Apparent weight = Actual weight
Lift accelerating upward at the rate of 'a'		v = variable	$a < g$	$R - mg = ma$ $\therefore R = m(g + a)$	Apparent weight > Actual weight
Lift accelerating upward at the rate of 'g'		v = variable	$a = g$	$R - mg = mg$ $R = 2mg$	Apparent weight = 2 Actual weight
Lift accelerating downward at the rate of 'a'		v = variable	$a < g$	$mg - R = ma$ $\therefore\ R = m(g - a)$	Apparent weight < Actual weight
Lift accelerating downward at the rate of 'g'		v = variable	$a = g$	$mg - R = mg$ $R = 0$	Apparent weight = Zero (weightlessness)
Lift accelerating downward at the rate		v = variable	$a > g$	$mg - R = ma$ $R = mg - ma$	Apparent weight negative means the body will rise

of a(>g)				$R = -$ *ve*	from the floor of the lift and stick to the ceiling of the lift.

Acceleration of Block on Horizontal Smooth Surface

(1) **When a pull is horizontal**

$R = mg$

and $F = ma$

$\therefore\ a = F/m$

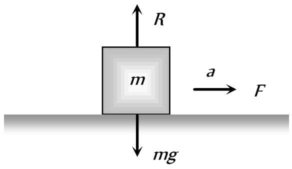


Fig : 4.22

(2) **When a pull is acting at an angle (θ) to the horizontal (upward)**

$R + F\sin\theta = mg$

$\Rightarrow R = mg - F\sin\theta$

and $F\cos\theta = ma$

$$\therefore\ a = \frac{F\cos\theta}{m}$$

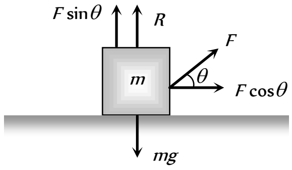


Fig : 4.23

(3) **When a push is acting at an angle (θ) to the horizontal (downward)**

$R = mg + F\sin\theta$

and $F\cos\theta = ma$

$$a = \frac{F\cos\theta}{m}$$

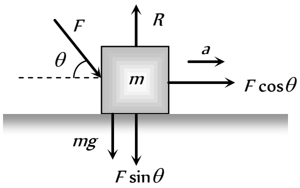


Fig : 4.24

Acceleration of Block on Smooth Inclined Plane

(1) **When inclined plane is at rest**

Normal reaction $R = mg\cos\theta$

Force along a inclined plane

$F = mg\sin\theta$; $ma = mg\ \sin\theta$

$\therefore a = g\sin\theta$

(2) **When a inclined plane given a horizontal acceleration '*b*'**

Since the body lies in an accelerating frame, an inertial force (mb) acts on it in the opposite direction.

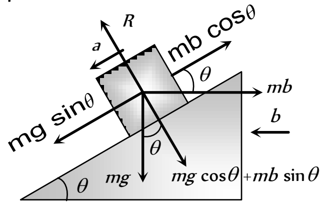


Fig : 4.25

Normal reaction $R = mg\cos\theta + mb\ sin\theta$

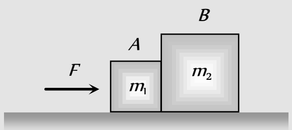
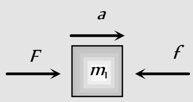
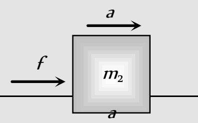
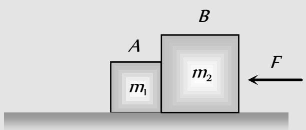
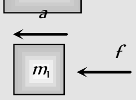
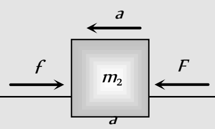
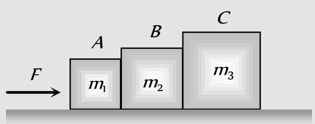
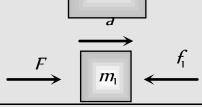
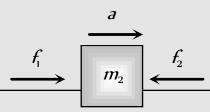
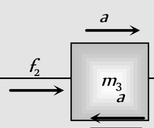
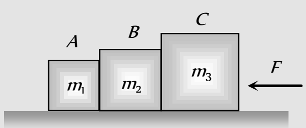
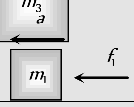
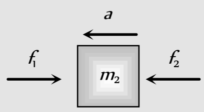
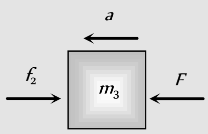
and $ma = mg\ \sin\theta - mb\cos\theta$

$\therefore\quad a = g\sin\theta - b\cos\theta$

Note : ❑ The condition for the body to be at rest relative to the inclined plane : $a = g\sin\theta - b\cos\theta = 0$

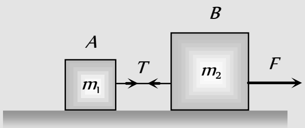
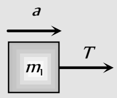
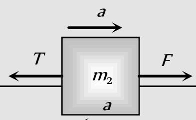
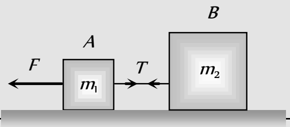
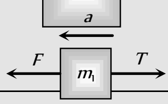
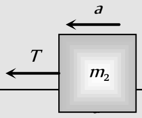
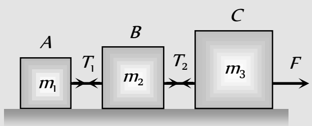
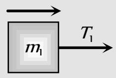
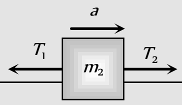
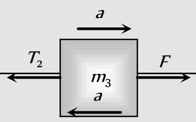
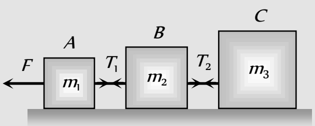
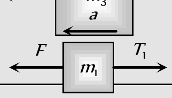
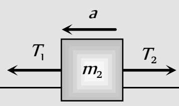
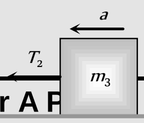
$\therefore\ b = g\tan\theta$

Motion of Blocks In Contact

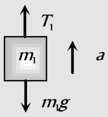
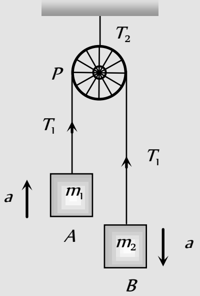
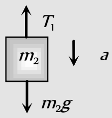
Condition	Free body diagram	Equation	Force and acceleration
		$F - f = m_1 a$	$a = \frac{F}{m_1 + m_2}$
		$f = m_2 a$	$f = \frac{m_2 F}{m_1 + m_2}$
		$f = m_1 a$	$a = \frac{F}{m_1 + m_2}$
		$F - f = m_2 a$	$f = \frac{m_1 F}{m_1 + m_2}$
		$F - f_1 = m_1 a$	$a = \frac{F}{m_1 + m_2 + m_3}$
		$f_1 - f_2 = m_2 a$	$f_1 = \frac{(m_2 + m_3)F}{m_1 + m_2 + m_3}$
		$f_2 = m_3 a$	$f_2 = \frac{m_3 F}{m_1 + m_2 + m_3}$
		$f_1 = m_1 a$	$a = \frac{F}{m_1 + m_2 + m_3}$
		$f_2 - f_1 = m_2 a$	$f_1 = \frac{m_1 F}{m_1 + m_2 + m_3}$
			

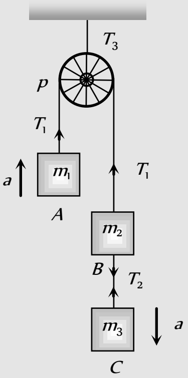
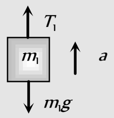
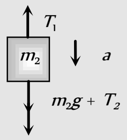
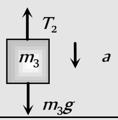
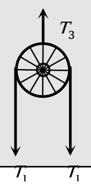
		$F - f_2 = m_3 a$	$f_2 = \frac{(m_1 + m_2)F}{m_1 + m_2 + m_3}$

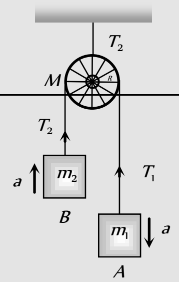
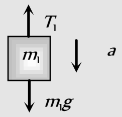
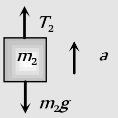
Motion of Blocks Connected by Mass Less String

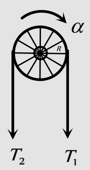
Condition	Free body diagram	Equation	Tension and acceleration
		$T = m_1 a$	$a = \frac{F}{m_1 + m_2}$
		$F - T = m_2 a$	$T = \frac{m_1 F}{m_1 + m_2}$
		$F - T = m_1 a$	$a = \frac{F}{m_1 + m_2}$
		$T = m_2 a$	$T = \frac{m_2 F}{m_1 + m_2}$
		$T_1 = m_1 a$	$a = \frac{F}{m_1 + m_2 + m_3}$
		$T_2 - T_1 = m_2 a$	$T_1 = \frac{m_1 F}{m_1 + m_2 + m_3}$
		$F - T_2 = m_3 a$	$T_2 = \frac{(m_1 + m_2)F}{m_1 + m_2 + m_3}$
		$F - T_1 = m_1 a$	$a = \frac{F}{m_1 + m_2 + m_3}$
		$T_1 - T_2 = m_2 a$	$T_1 = \frac{(m_2 + m_3)F}{m_1 + m_2 + m_3}$
		$T_2 = m_3 a$	$T_2 = \frac{m_3 F}{m_1 + m_2 + m_3}$

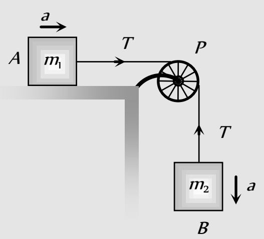
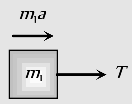
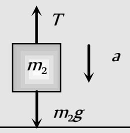
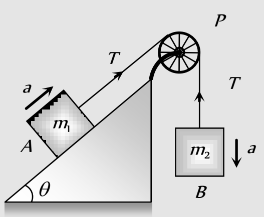
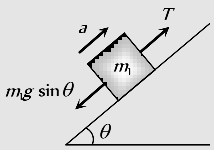
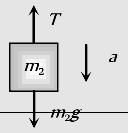
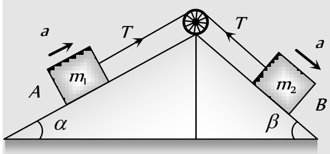
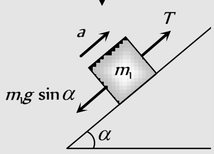
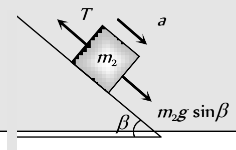
Motion of Connected Block Over A P

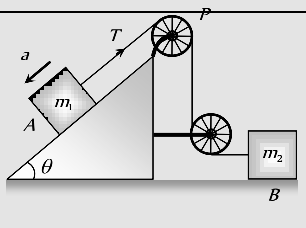
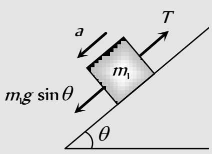
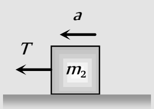
Condition	Free body diagram	Equation	Tension and acceleration
		$m_1 a = T_1 - m_1 g$	$T_1 = \frac{2m_1 m_2}{m_1 + m_2} g$
	 		

		$m_2a = m_2g - T_1$	$T_2 = \dfrac{4m_1m_2}{m_1+m_2}g$
		$T_2 = 2T_1$	$a = \left[\dfrac{m_2-m_1}{m_1+m_2}\right]g$
		$m_1a = T_1 - m_1g$	$T_1 = \dfrac{2m_1[m_2+m_3]}{m_1+m_2+m_3}g$
		$m_2a = m_2g + T_2 - T_1$	$T_2 = \dfrac{2m_1m_3}{m_1+m_2+m_3}g$
		$m_3a = m_3g - T_2$	$T_3 = \dfrac{4m_1[m_2+m_3]}{m_1+m_2+m_3}g$
		$T_3 = 2T_1$	$a = \dfrac{[(m_2+m_3)-m_1]g}{m_1+m_2+m_3}$

Condition	Free body diagram	Equation	Tension and acceleration
When pulley have a finite mass M and radius R then tension in two segments of string are different 		$m_1a = m_1g - T_1$	$a = \dfrac{m_1 - m_2}{m_1+m_2+\dfrac{M}{2}}$
		$m_2a = T_2 - m_2g$	$T_1 = \dfrac{m_1\left[2m_2+\dfrac{M}{2}\right]}{m_1+m_2+\dfrac{M}{2}}g$



		Torque $=(T_1-T_2)R=I\alpha$ $(T_1-T_2)R=I\dfrac{a}{R}$ $(T_1-T_2)R=\dfrac{1}{2}MR^2\dfrac{a}{R}$ $T_1-T_2=\dfrac{Ma}{2}$	$T_2=\dfrac{m_2\left[2m_1+\dfrac{M}{2}\right]}{m_1+m_2+\dfrac{M}{2}}g$
		$T=m_1a$	$a=\dfrac{m_2}{m_1+m_2}g$
		$m_2a=m_2g-T$	$T=\dfrac{m_1m_2}{m_1+m_2}g$
		$m_1a=T-m_1g\sin\theta$	$a=\left[\dfrac{m_2-m_1\sin\theta}{m_1+m_2}\right]g$
		$m_2a=m_2g-T$	$T=\dfrac{m_1m_2(1+\sin\theta)}{m_1+m_2}g$
		$T-m_1g\sin\alpha=m_1a$	$a=\dfrac{(m_2\sin\beta-m_1\sin\alpha)}{m_1+m_2}g$
		$m_2a=m_2g\sin\beta-T$	$T=\dfrac{m_1m_2(\sin\alpha+\sin\beta)}{m_1+m_2}g$

Condition	Free body diagram	Equation	Tension and acceleration
		$m_1g\sin\theta-T=m_1a$	$a=\dfrac{m_1g\sin\theta}{m_1+m_2}$
			

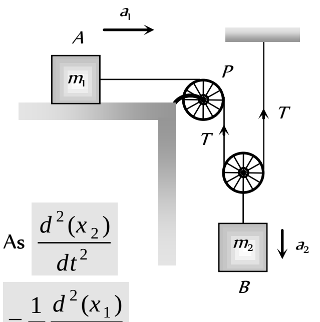
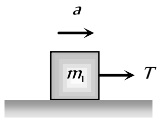
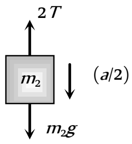
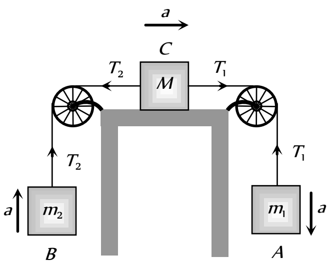
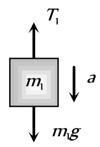
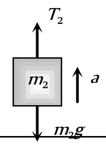
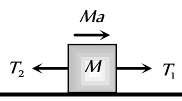
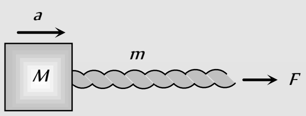
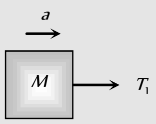
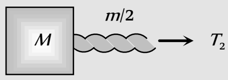
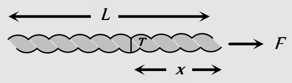
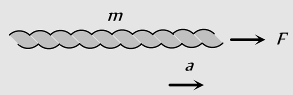
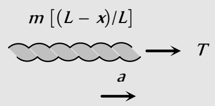
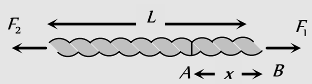
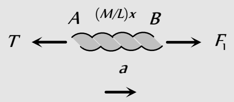
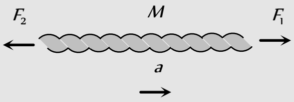
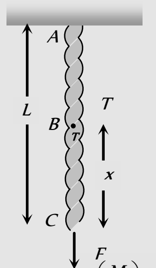
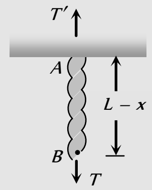
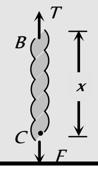
		$T = m_2 a$	$T = \frac{2m_1 m_2}{4m_1 + m_2} g$
 As $\frac{d^2(x_2)}{dt^2}$ $= \frac{1}{2}\frac{d^2(x_1)}{dt^2}$ $\therefore a_2 = \frac{a_1}{2}$ a_1 = acceleration of block A a_2 = acceleration of block B		$T = m_1 a$	$a_1 = a = \frac{2m_2 g}{4m_1 + m_2}$ $a_2 = \frac{m_2 g}{4m_1 + m_2}$ $T = \frac{2m_1 m_2 g}{4m_1 + m_2}$
		$m_2 \frac{a}{2} = m_2 g - 2T$	
		$m_1 a = m_1 g - T_1$	$a = \frac{(m_1 - m_2)}{[m_1 + m_2 + M]} g$
		$m_2 a = T_2 - m_2 g$	$T_1 = \frac{m_1(2m_2 + M)}{[m_1 + m_2 + M]} g$
		$T_1 - T_2 = Ma$	$T_2 = \frac{m_2(2m_2 + M)}{[m_1 + m_2 + M]} g$

Table 4.3 : Motion of massive string

Condition	Free body diagram	Equation	Tension and acceleration
	 T_1 = force applied by the string on the block 	$F = (M + m)a$ $T_1 = Ma$	$a = \frac{F}{M + m}$ $T_1 = M\frac{F}{(M + m)}$

	T_2 = Tension at mid point of the rope	$T_2 = \left(M + \frac{m}{2}\right)a$	$T_2 = \frac{(2M+m)}{2(M+m)}F$
 m = Mass of string T = Tension in string at a distance x from the end where the force is applied		$F = ma$	$a = F/m$
		$T = m\left(\frac{L-x}{L}\right)a$	$T = \left(\frac{L-x}{L}\right)F$
		$F_1 - T = \frac{Mxa}{L}$	$a = \frac{F_1 - F_2}{M}$
M = Mass of uniform string L = Length of string		$F_1 - F_2 = Ma$	$T = F_1\left(1 - \frac{x}{L}\right) + F_2\left(\frac{x}{L}\right)$
 Mass of segment $BC = \left(\frac{M}{L}\right)x$	 	$T' = \frac{M}{L}(L-x)g + T$ $T = F + \frac{M}{L}xg$	$T' = F + Mg$ $T = F + \frac{M}{L}xg$

Spring Balance and Physical Balance

(1) **Spring balance :** When its upper end is fixed with rigid support and body of mass m hung from its lower end. Spring is stretched and the weight of the body can be measured by the reading of spring balance $R = W = mg$

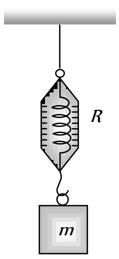


Fig : 4.26

The mechanism of weighing machine is same as that of spring balance.

Effect of frame of reference : In inertial frame of reference the reading of spring balance shows the actual weight of the body but in non-inertial frame of reference reading of spring balance increases or decreases in accordance with the direction of acceleration

(2) **Physical balance :** In physical balance actually we compare the mass of body in both the pans. Here we does not calculate the absolute weight of the body.

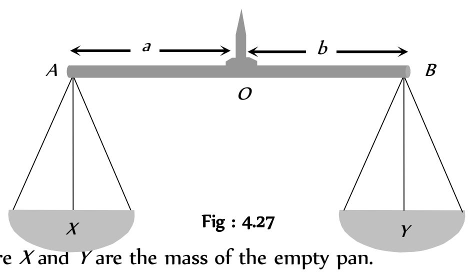


Fig : 4.27

Here X and Y are the mass of the empty pan.

(i) Perfect physical balance :

Weight of the pan should be equal *i.e.* $X = Y$

and the needle must in middle of the beam *i.e.* $a = b$.

Effect of frame of reference : If the physical balance is perfect then there will be no effect of frame of reference (either inertial or non-inertial) on the measurement. It is always errorless.

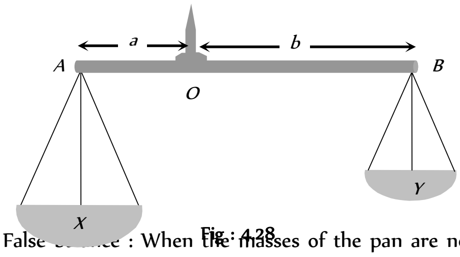


Fig : 4.28

(ii) False balance : When the masses of the pan are not equal then balance shows the error in measurement. False balance may be of two types

(a) If the beam of physical balance is horizontal (when the pans are empty) but the arms are not equal

$X > Y$ and $a < b$

For rotational equilibrium about point '*O*'

$Xa = Yb$ ...(i)

In this physical balance if a body of weight W is placed in pan X then to balance it we have to put a weight W_1 in pan *Y*.

For rotational equilibrium about point '*O*'

$(X + W)a = (Y + W_1)b$ ...(ii)

Now if the pans are changed then to balance the body we have to put a weight W_2 in pan *X*.

For rotational equilibrium about point '*O*'

$(X + W_2)a = (Y + W)b$ ...(iii)

From (i), (ii) and (iii)

True weight $W = \sqrt{W_1 W_2}$

(b) If the beam of physical balance is not horizontal (when the pans are empty) and the arms are equal

i.e. $X > Y$ and $a = b$

In this physical balance if a body of weight W is placed in X Pan then to balance it.

We have to put a weight W_1 in Y Pan

For equilibrium $X + W = Y + W_1$ ...(i)

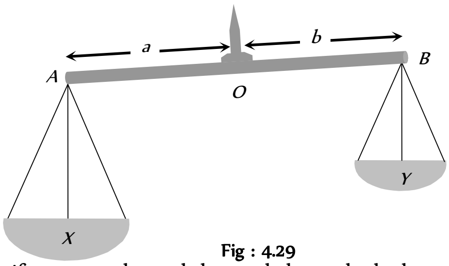


Fig : 4.29

Now if pans are changed then to balance the body we have to put a weight W_2 in X Pan.

For equilibrium $X + W_2 = Y + W$ ...(ii)

From (i) and (ii)

True weight $W = \dfrac{W_1 + W_2}{2}$

Modification of Newton's Laws of motion

According to Newton, time and space are absolute. The velocity of observer has no effect on it. But, according to special theory of relativity Newton's laws are true, as long as we are dealing with velocities which are small compare to velocity of light. Hence the time and space measured by two observers in relative motion are not same. Some conclusions drawn by the special theory of relativity about mass, time and distance which are as follows :

(1) Let the length of a rod at rest with respect to an observer is L_0. If the rod moves with velocity v w.r.t. observer and its length is L, then

$L = L_0\sqrt{1 - v^2 / c^2}$

where, c is the velocity of light.

Now, as v increases L decreases, hence the length will appear shrinking.

(2) Let a clock reads T_0 for an observer at rest. If the clock moves with velocity v and clock reads T with respect to observer, then

$$T = \frac{T_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Hence, the clock in motion will appear slow.

(3) Let the mass of a body is m_0 at rest with respect to an observer. Now, the body moves with velocity v with respect to observer and its mass is m, then $m = \dfrac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$

m_0 is called the rest mass.

Hence, the mass increases with the increases of velocity.

Note : ❑ If $v << c$, *i.e.*, velocity of the body is very small *w.r.t.* velocity of light, then $m = m_0$. *i.e.*, in the practice there will be no change in the mass.

❑ If v is comparable to c, then $m > m_0$ *i.e.*, mass will increase.

❑ If $v = c$, then $m = \dfrac{m_0}{\sqrt{1 - \frac{v^2}{v^2}}}$ or $m = \dfrac{m_0}{0} = \infty$. Hence, the mass becomes infinite, which is not possible, thus the speed cannot be equal to the velocity of light.

❑ The velocity of particles can be accelerated up to a certain limit. Even in cyclotron the speed of charged particles cannot be increased beyond a certain limit.

Tips & Tricks

- Inertia is proportional to mass of the body.
- Force cause acceleration.
- In the absence of the force, a body moves along a straight line path.
- A system or a body is said to be in equilibrium, when the net force acting on it is zero.
- If a number of forces $\vec{F}_1, \vec{F}_2, \vec{F}_3, ...$ act on the body, then it is in equilibrium when $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + ... = \vec{0}$

- A body in equilibrium cannot change the direction of motion.
- Four types of forces exist in nature. They are – gravitation (F_g), electromagnetic (F_{em}), weak force (F_w) and nuclear force (F_n).

$(F_g):(F_w):(F_{em}):(F_n)$: : 1 : 10^{25} : 10^{36} : 10^{38}

- If a body moves along a curved path, then it is certainly acted upon by a force.
- A single isolated force cannot exist.
- Forces in nature always occur in pairs.
- Newton's first law of the motion defines the force.
- Absolute units of force remains the same throughout the universe while gravitational units of force varies from place to place as they depend upon the value of 'g'.
- Newton's second law of motion gives the measure of force i.e. $F = ma$.
- Force is a vector quantity.
- Absolute units of force are dyne in CGS system and newton (N) in *SI*.
- 1 N = 10^5 *dyne.*
- Gravitational units of force are *gf* (or *gwt*) in *CGS* system and *kgf* (or *kgwt*) in *SI*.
- 1 *gf* = 980 *dyne* and 1 *kgf* = 9.8 *N*
- The beam balance compares masses.
- Acceleration of a horse-cart system is

$$a = \frac{H - F}{M + m}$$

where H = Horizontal component of reaction; F = force of friction; M = mass of horse; m = mass of cart.

- The weight of the body measured by the spring balance in a lift is equal to the apparent weight.
- Apparent weight of a freely falling body = ZERO, (state of weightlessness).
- If the person climbs up along the rope with acceleration a, then tension in the rope will be $m(g+a)$
- If the person climbs down along the rope with acceleration, then tension in the rope will be $m(g - a)$
- When the person climbs up or down with uniform speed, tension in the string will be mg.
- A body starting from rest moves along a smooth inclined plane of length l, height h and having angle of inclination θ.

(i) Its acceleration down the plane is $g \sin\theta$.

(ii) Its velocity at the bottom of the inclined plane will be $\sqrt{2gh} = \sqrt{2gl\sin\theta}$.

(iii) Time taken to reach the bottom will be

$$t = \sqrt{\frac{2l}{g\sin\theta}} = \frac{1}{\sin\theta}\sqrt{\frac{2h}{g}}$$

(iv) If the angle of inclination is changed keeping the height constant then

$$\frac{t_1}{t_2} = \frac{\sin\theta_2}{\sin\theta_1}$$

- For an isolated system (on which no external force acts), the total momentum remains conserved (Law of conservation of momentum).
- The change in momentum of a body depends on the magnitude and direction of the applied force and the period of time over which it is applied *i.e.* it depends on its impulse.
- Guns recoil when fired, because of the law of conservation of momentum. The positive momentum gained by the bullet is equal to negative recoil momentum of the gun and so the total momentum before and after the firing of the gun is zero.
- Recoil velocity of the gun is $\vec{V} = \frac{-m}{M}\vec{v}$
- where m = mass of bullet, M = mass of gun and $\vec{v}$ = muzzle velocity of bullet.
- The rocket pushes itself forwards by pushing the jet of exhaust gases backwards.
- Upthrust on the rocket = $u \times \frac{dm}{dt}$.

where u = velocity of escaping gases relative to rocket and $\frac{dm}{dt}$ = rate of consumption of fuel.

- Initial thrust on rocket = $m(g + a)$, where a is the acceleration of the rocket.
- Upward acceleration of rocket = $\frac{u}{m} \times \frac{dm}{dt}$.
- Impulse, $\vec{I} = \vec{F} \times \Delta t =$ change in momentum
- Unit of impulse is *N-s*.
- Action and reaction forces never act on the same body. They act on different bodies. If they act on the same body, the resultant force on the body will be zero i.e., the body will be in equilibrium.
- Action and reaction forces are equal in magnitude but opposite in direction.
- Action and reaction forces act along the line joining the centres of two bodies.
- Newton's third law is applicable whether the bodies are at rest or in motion.
- The non-inertial character of the earth is evident from the fact that a falling object does not fall straight down but slightly deflects to the

east.

Ordinary Thinking

Objective Questions

First law of motion

1. A rider on horse back falls when horse starts running all of a sudden because [MP PMT 1982]
   (a) Rider is taken back
   (b) Rider is suddenly afraid of falling
   (c) Inertia of rest keeps the upper part of body at rest whereas lower part of the body moves forward with the horse
   (d) None of the above
2. When a train stops suddenly, passengers in the running train feel an instant jerk in the forward direction because [MP PMT 1982]
   (a) The back of seat suddenly pushes the passengers forward
   (b) Inertia of rest stops the train and takes the body forward
   (c) Upper part of the body continues to be in the state of motion whereas the lower part of the body in contact with seat remains at rest
   (d) Nothing can be said due to insufficient data
3. Inertia is that property of a body by virtue of which the body is [MGIMS Wardha 1982]
   (a) Unable to change by itself the state of rest
   (b) Unable to change by itself the state of uniform motion
   (c) Unable to change by itself the direction of motion
   (d) Unable to change by itself the state of rest and of uniform linear motion
4. A man getting down a running bus falls forward because [CPMT 1981]
   (a) Due to inertia of rest, road is left behind and man reaches forward
   (b) Due to inertia of motion upper part of body continues to be in motion in forward direction while feet come to rest as soon as they touch the road
   (c) He leans forward as a matter of habit
   (d) Of the combined effect of all the three factors stated in (*a*), (*b*) and (*c*)
5. A boy sitting on the topmost berth in the compartment of a train which is just going to stop on a railway station, drops an apple aiming at the open hand of his brother sitting vertically below his hands at a distance of about 2 *meter*. The apple will fall [CPMT 1986]
   (a) Precisely on the hand of his brother
   (b) Slightly away from the hand of his brother in the direction of motion of the train
   (c) Slightly away from the hand of his brother in the direction opposite to the direction of motion of the train
   (d) None of the above

6. Newton's first law of motion describes the following [MP PMT 1996]

(a) Energy (b) Work
(c) Inertia (d) Moment of inertia

7. A person sitting in an open car moving at constant velocity throws a ball vertically up into air. The ball falls [EAMCET (Med.) 1995; MH CET 2003;BCECE 2004]

(a) Outside the car
(b) In the car ahead of the person
(c) In the car to the side of the person
(d) Exactly in the hand which threw it up

8. A bird weighs 2 kg and is inside a closed cage of 1 kg. If it starts flying, then what is the weight of the bird and cage assembly

(a) 1.5 kg (b) 2.5 kg
(c) 3 kg (d) 4 kg

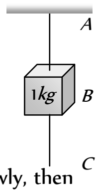
9. A particle is moving with a constant speed along a straight line path. A force is not required to [AFMC 2001]

(a) Increase its speed
(b) Decrease the momentum
(c) Change the direction
(d) Keep it moving with uniform velocity

10. When a bus suddenly takes a turn, the passengers are thrown outwards because of [AFMC 1999; CPMT 2000, 2001]

(a) Inertia of motion (b) Acceleration of motion
(c) Speed of motion (d) Both (b) and (c)

11. A mass of 1 kg is suspended by a string A. Another string C is connected to its lower end (see figure). If a sudden jerk is given to C, then



(a) The portion AB of the string will break
(b) The portion BC of the string will break
(c) None of the strings will break
(d) The mass will start rotating

12. In the above Question, if the string C is stretched slowly, then

(a) The portion AB of the string will break
(b) The portion BC of the string will break
(c) None of the strings will break
(d) None of the above

Second Law of Motion

1. If a bullet of mass 5 gm moving with velocity 100 m/sec, penetrates the wooden block upto 6 cm. Then the average force imposed by the bullet on the block is [MP PMT 2003]

(a) 8300 N (b) 417 N
(c) 830 N (d) Zero

2. Newton's second law gives the measure of [CPMT 1982]

(a) Acceleration (b) Force
(c) Momentum (d) Angular momentum

3. A force of 100 dynes acts on mass of 5 gm for 10 sec. The velocity produced is [MNR 1987]

(a) 2 cm/sec (b) 20 cm/sec
(c) 200 cm/sec (d) 2000 cm/sec

4. An object will continue moving uniformly until [CPMT 1975]

(a) The resultant force acting on it begins to decrease
(b) The resultant force on it is zero
(c) The resultant force is at right angle to its rotation
(d) The resultant force on it is increased continuously

5. A diwali rocket is ejecting 0.05 kg of gases per second at a velocity of 400 m/sec. The accelerating force on the rocket is [NCERT 1979; DPMT 2001; MP PMT 2004]

(a) 20 $dynes$ (b) 20 N
(c) 22 $dynes$ (d) 1000 N

6. A body of mass 2 kg [AFMC 1997] moving on a horizontal surface with an initial velocity of 4 m/sec comes to rest after 2 sec. If one wants to keep this body moving on the same surface with a velocity of 4 m/sec, the force required is [NCERT 1977]

(a) 8 N (b) 4 N
(c) Zero (d) 2 N

7. A body of mass 2 kg is hung on a spring balance mounted vertically in a lift. If the lift descends with an acceleration equal to the acceleration due to gravity 'g', the reading on the spring balance will be [NCERT 1977]

(a) 2 kg (b) $(4 \times g) kg$
(c) $(2 \times g) kg$ (d) Zero

8. In the above problem, if the lift moves up with a constant velocity of 2 m/sec, the reading on the balance will be [NCERT 1977]

(a) 2 kg (b) 4 kg
(c) Zero (d) 1 kg

9. In the above problem if the lift moves up with an acceleration equal to the acceleration due to gravity, the reading on the spring balance will be [NCERT 1977]

(a) 2 kg (b) $(2 \times g) kg$
(c) $(4 \times g) kg$ (d) 4 kg

10. A coin is dropped in a lift. It takes time t_1 to reach the floor when lift is stationary. It takes time t_2 when lift is moving up with constant acceleration. Then

(a) $t_1 > t_2$ (b) $t_2 > t_1$
(c) $t_1 = t_2$ (d) $t_1 >> t_2$

11. If the tension in the cable of 1000 kg elevator is 1000 kg weight, the elevator [NCERT 1971]

(a) Is accelerating upwards
(b) Is accelerating downwards
(c) May be at rest or accelerating
(d) May be at rest or in uniform motion

12. A man weighing 80 kg is standing in a trolley weighing 320 kg. The trolley is resting on frictionless horizontal rails. If the man starts walking on the trolley with a speed of 1 m/s, then after 4 sec his displacement relative to the ground will be

(a) 5 m (b) 4.8 m
(c) 3.2 m (d) 3.0 m

13. In doubling the mass and acceleration of the mass, the force acting on the mass with respect to the previous value
(a) Decreases to half (b) Remains unchanged
(c) Increases two times (d) Increases four times

14. A force of 5 *N* acts on a body of weight 9.8 *N*. What is the acceleration produced in $m/\sec^2$ [NCERT 1990]
(a) 49.00 (b) 5.00
(c) 1.46 (d) 0.51

15. A body of mass 40 *gm* is moving with a constant velocity of 2 *cm/sec* on a horizontal frictionless table. The force on the table is
(a) 39200 *dyne* (b) 160 *dyne*
(c) 80 *dyne* (d) Zero *dyne*

16. When 1 *N* force acts on 1 *kg* body that is able to move freely, the body receives [CPMT 1971]
(a) A speed of 1 *m/sec*
(b) An acceleration of $1\,m/\sec^2$
(c) An acceleration of $980\,cm/\sec^2$
(d) An acceleration of $1\,cm/\sec^2$

17. An object with a mass 10 *kg* moves at a constant velocity of 10 *m/sec*. A constant force then acts for 4 second on the object and gives it a speed of 2 *m/sec* in opposite direction. The acceleration produced in it, is [CPMT 1971]
(a) $3\,m/\sec^2$ (b) $-3\,m/\sec^2$
(c) $0.3\,m/\sec^2$ (d) $-0.3\,m/\sec^2$

18. In the above question, the force acting on the object is [CPMT 1971]
(a) 30 *N* (b) – 30 *N*
(c) 3 *N* (d) – 3 *N*

19. In the above question, the impulse acting on the object is [CPMT 1971]
(a) $120\ newton \times \sec$ (b) $-120\ newtont\sec$
(c) $30\ newton \times \sec$ (d) $-30\ newton \times \sec$

20. A machine gun is mounted on a 2000 *kg* car on a horizontal frictionless surface. At some instant the gun fires bullets of mass 10 *gm* with a velocity of 500 *m/sec* with respect to the car. The number of bullets fired per second is ten. The average thrust on the system is [CPMT 1971]
(a) 550 *N* (b) 50 *N*
(c) 250 *N* (d) 250 *dyne*

21. In the above question, the acceleration of the car will be [CPMT 1971]
(a) $0.25\,m/\sec^2$ (b) $2.5\,m/\sec^2$
(c) $5.0\,m/\sec^2$ (d) $0.025\,m/\sec^2$

22. A person is standing in an elevator. In which situation he finds his weight less than actual when [AIIMS 2005]
(a) The elevator moves upward with constant acceleration
(b) The elevator moves downward with constant acceleration.
(c) The elevator moves upward with uniform velocity
(d) The elevator moves downward with uniform velocity

23. A particle of mass 0.3 *kg* is subjected to a force $F = -kx$ with $k = 15\,N/m$. What will be its initial acceleration if it is released from a point 20 *cm* away from the origin [AIEEE 2005]
(a) 5 *m/s* (b) 10 *m/s*
(c) 3 *m/s* (d) 15 *m/s*

24. A block of metal weighing 2 *kg* is resting on a frictionless plane. It is struck by a jet releasing water at a rate of 1 *kg/sec* and at a speed of 5 *m/sec*. The initial acceleration of the block will be [NCERT 1978]
(a) $2.5\,m/\sec^2$ (b) $5.0\,m/\sec^2$
(c) $10\,m/\sec^2$ (d) None of the above

25. Gravels are dropped on a conveyor belt at the rate of 0.5 *kg/sec*. The extra force required in *newtons* to keep the belt moving at 2 *m/sec* is [EAMCET 1988]
(a) 1 (b) 2
(c) 4 (d) 0.5

26. A parachutist of weight '*w*' strikes the ground with his legs fixed and comes to rest with an upward acceleration of magnitude 3 *g*. Force exerted on him by ground during landing is
(a) w (b) $2w$
(c) $3w$ (d) $4w$

27. At a place where the acceleration due to gravity is $10\,m\,\sec^{-2}$ a force of 5 *kg-wt* acts on a body of mass 10 *kg* initially at rest. The velocity of the body after 4 *second* is [EAMCET 1981]
(a) $5\,m\,\sec^{-1}$ (b) $10\,m\,\sec^{-1}$
(c) $20\,m\,\sec^{-1}$ (d) $50\,m\,\sec^{-1}$

28. In a rocket of mass 1000 *kg* fuel is consumed at a rate of 40 *kg/s*. The velocity of the gases ejected from the rocket is $5 \times 10^4\,m/s$. The thrust on the rocket is [MP PMT 1994]
(a) $2 \times 10^3\,N$ (b) $5 \times 10^4\,N$
(c) $2 \times 10^6\,N$ (d) $2 \times 10^9\,N$

29. A man is standing on a weighing machine placed in a lift. When stationary his weight is recorded as 40 *kg*. If the lift is accelerated upwards with an acceleration of $2\,m/s^2$, then the weight recorded in the machine will be $(g = 10\,m/s^2)$ [MP PMT 1994]
(a) 32 *kg* (b) 40 *kg*
(c) 42 *kg* (d) 48 *kg*

30. A body of mass 4 *kg* weighs 4.8 *kg* when suspended in a moving lift. The acceleration of the lift is [Manipal MEE 1995]
(a) $9.80\,ms^{-2}$ downwards (b) $9.80\,ms^{-2}$ upwards
(c) $1.96\,ms^{-2}$ downwards (d) $1.96\,ms^{-2}$ upwards

31. An elevator weighing 6000 *kg* is pulled upward by a cable with an acceleration of $5\,ms^{-2}$. Taking *g* to be $10\,ms^{-2}$, then the tension in the cable is [Manipal MEE 1995]
(a) 6000 *N* (b) 9000 *N*
(c) 60000 *N* (d) 90000 *N*

32. A ball of mass 0.2 *kg* moves with a velocity of 20 *m/sec* and it stops in 0.1 *sec*; then the force on the ball is [BHU 1995]

(a) 40 N (b) 20 N
(c) 4 N (d) 2 N

33. A vehicle of 100 kg is moving with a velocity of 5 m/sec. To stop it in $\frac{1}{10}$ sec, the required force in opposite direction is [MP PET 1995]

(a) 5000 N (b) 500 N
(c) 50 N (d) 1000 N

34. A boy having a mass equal to 40 *kilograms* is standing in an elevator. The force felt by the feet of the boy will be greatest when the elevator

$(g = 9.8\, metres/\sec^2)$ [MP PMT 1995; BVP 2003]

(a) Stands still
(b) Moves downward at a constant velocity of 4 *metres/sec*
(c) Accelerates downward with an acceleration equal to $4\, metres/\sec^2$
(d) Accelerates upward with an acceleration equal to $4\, metres/\sec^2$

35. A rocket has an initial mass of $20 \times 10^3\, kg$. If it is to blast off with an initial acceleration of $4\, ms^{-2}$, the initial thrust needed is $(g \cong 10\, ms^{-2})$ [Kurukshetra CEE 1996]

(a) $6 \times 10^4\, N$ (b) $28 \times 10^4\, N$
(c) $20 \times 10^4\, N$ (d) $12 \times 10^4\, N$

36. The ratio of the weight of a man in a stationary lift and when it is moving downward with uniform acceleration 'a' is 3 : 2. The value of 'a' is (g-Acceleration due to gravity of the earth)

(a) $\frac{3}{2}g$ (b) $\frac{g}{3}$
(c) $\frac{2}{3}g$ (d) g

37. The mass of a lift is 500 kg. When it ascends with an acceleration of $2\, m/s^2$, the tension in the cable will be $[g = 10\, m/s^2]$

(a) 6000 N (b) 5000 N
(c) 4000 N (d) 50 N

38. If force on a rocket having exhaust velocity of 300 m/sec is 210 N, then rate of combustion of the fuel is [CBSE PMT 1999; MH CET 2003; Pb. PMT 2004]

(a) 0.7 kg/s (b) 1.4 kg/s
(c) 0.07 kg/s (d) 10.7 kg/s

39. In an elevator moving vertically up with an acceleration g, the force exerted on the floor by a passenger of mass M is [CPMT 1999]

(a) Mg (b) $\frac{1}{2}Mg$
(c) Zero (d) 2 Mg

40. A mass 1 kg is suspended by a thread. It is

(i) lifted up with an acceleration $4.9\, m/s^2$

(ii) lowered with an acceleration $4.9\, m/s^2$.

The ratio of the tensions is [CBSE PMT 1998]

(a) 3 : 1 (b) 1 : 3
(c) 1 : 2 (d) 2 : 1

41. A 5000 kg rocket is set for vertical firing. The exhaust speed is $800\, ms^{-1}$. To give an initial upward acceleration of $20\, ms^{-2}$, the amount of gas ejected per second to supply the needed thrust will be $(g = 10\, ms^{-2})$ [CBSE PMT 1998]

(a) $127.5\, kg\, s^{-1}$ (b) $187.5\, kg\, s^{-1}$
(c) $185.5\, kg\, s^{-1}$ (d) $137.5\, kg\, s^{-1}$

42. If a person with a spring balance and a body hanging from it goes up and up in an aeroplane, then the reading of the weight of the body as indicated by the spring balance will [AIIMS 1998; JIPMER 2000]

(a) Go on increasing
(b) Go on decreasing
(c) First increase and then decrease
(d) Remain the same

43. The time period of a simple pendulum measured inside a stationary lift is found to be T. If the lift starts accelerating upwards with an acceleration $g/3$, the time period is [EAMCET 1994; CMEET Bihar 1995; RPMT 2000]

(a) $T\sqrt{3}$ (b) $T\sqrt{3}/2$
(c) $T/\sqrt{3}$ (d) $T/3$

44. A cork is submerged in water by a spring attached to the bottom of a pail. [MP PET 1997] When the pail is kept in a elevator moving with an acceleration downwards, the spring length [EAMCET (Engg.) 1995]

(a) Increases (b) Decreases
(c) Remains unchanged (d) Data insufficient

45. Two trolleys of mass m and $3m$ are connected by a spring. They were compressed and released [MP PMT 1999, 2000] once, they move off in opposite direction and comes to rest after covering distances S_1 and S_2 respectively. Assuming the coefficient of friction to be uniform, the ratio of distances $S_1 : S_2$ is [EAMCET (Engg.) 1995]

(a) 1 : 9 (b) 1 : 3
(c) 3 : 1 (d) 9 : 1

46. A boy of 50 kg is in a lift moving down with an acceleration $9.8\, ms^{-2}$. The apparent weight of the body is $(g = 9.8\, ms^{-2})$ [EAMCET (Med KCET 2000]

(a) $50 \times 9.8\, N$ (b) Zero
(c) 50 N (d) $\frac{50}{9.8}N$

47. A body is imparted motion from rest to move in a straight line. If it is then obstructed by an opposite force, then [NTSE 1995]

(a) The body may necessarily change direction

(b) The body is sure to slow down

(c) The body will necessarily continue to move in the same direction at the same speed

(d) None of these

48. A mass of 10 *gm* is suspended by a string and the entire system is falling with a uniform acceleration of $400\,cm/\sec^2$. The tension in the string will be $(g = 980\,cm/\sec^2)$

(a) 5,800 *dyne* (b) 9,800 *dyne*

(c) 11,800 *dyne* (d) 13,800 *dyne*

49. A second's pendulum is mounted in a rocket. Its period of oscillation decreases when the rocket **[CBSE PMT 1994]**

(a) Comes down with uniform acceleration

(b) Moves round the earth in a geostationary orbit

(c) Moves up with a uniform velocity

(d) Moves up with uniform acceleration

50. Two balls of masses m_1 and m_2 are separated from each other by a powder charge placed between them. The whole system is at rest on the ground. Suddenly the powder charge explodes and masses are pushed apart. The mass m_1 travels a distance s_1 and stops. If the coefficients of friction between the balls and ground are same, the mass m_2 stops after travelling the distance

(a) $s_2 = \frac{m_1}{m_2} s_1$ (b) $s_2 = \frac{m_2}{m_1} s_1$

(c) $s_2 = \frac{m_1^2}{m_2^2} s_1$ (d) $s_2 = \frac{m_2^2}{m_1^2} s_1$

51. A force vector applied on a mass is represented as $\vec{F} = 6\hat{i} - 8\hat{j} + 10\hat{k}$ and accelerates with $1\,m/s^2$. What will be the mass of the body **[CBSE PMT 1996]**

(a) $10\sqrt{2}\,kg$ (b) $2\sqrt{10}\,kg$

(c) 10 *kg* (d) 20 *kg*

52. A cart of mass *M* is tied by one end of a massless rope of length 10 *m*. The other end of the rope is in the hands of a man of mass *M*. The entire system is on a smooth horizontal surface. The man is at *x* = 0 and the cart at *x* = 10 *m*. If the man pulls the cart by the rope, the man and the cart will meet at the point

(a) $x = 0$ (b) $x = 5\,m$

(c) $x = 10\,m$ (d) They will never meet

53. A cricket ball of mass 250 *g* collides with a bat with velocity 10 *m/s* and returns with the same velocity within 0.01 second. The force acted on bat is **[CPMT 1997]**

(a) 25 *N* (b) 50 *N*

(c) 250 *N* (d) 500 *N*

54. A pendulum bob of mass 50 *gm* is suspended from the ceiling of an elevator. The tension in the string if the elevator goes up with uniform velocity is approximately **[AMU (Med.) 1999]**

(a) 0.30 *N* (b) 0.40 *N*

(c) 0.42 *N* (d) 0.50 *N*

55. A train is moving with velocity 20 *m/sec*. on this dust is falling at the rate of 50 *kg/minute*. The extra force required to move this train with constant velocity will be **[RPET 1999]**

(a) 16.66 *N* (b) 1000 *N*

(c) 166.6 *N* (d) 1200 *N*

56. The average force necessary to stop a bullet of mass 20 *g* moving with a speed of 250 *m/s*, as it penetrates into the wood for a distance of 12 *cm* is **[SCRA 1994]** **[CBSE PMT 2000; DPMT 2003]**

(a) $2.2 \times 10^3\,N$ (b) $3.2 \times 10^3\,N$

(c) $4.2 \times 10^3\,N$ (d) $5.2 \times 10^3\,N$

57. The average resisting force that must act on a 5 *kg* mass to reduce its speed from 65 *cm/s* to 15 *cm/s* in 0.2*s* is **[RPET 2000]**

(a) 12.5 *N* (b) 25 *N*

(c) 50 *N* (d) 100 *N*

58. A mass is hanging on a spring balance which is kept in a lift. The lift ascends. The spring balance will show in its reading **[DCE 2000]**

(a) Increase

(b) Decrease

(c) No change

(d) Change depending upon velocity **[BHU 1994]**

59. An army vehicle of mass 1000 *kg* is moving with a velocity of 10 *m/s* and is acted upon by a forward force of 1000 *N* due to the engine and a retarding force of 500 *N* due to friction. What will be its velocity after 10 *s* **[Pb. PMT 2000]**

(a) 5 *m/s* (b) 10 *m/s*

(c) 15 *m/s* (d) 20 *m/s*

60. A body of mass 2 *kg* is moving with a velocity 8 *m/s* on a smooth surface. If it is to be brought to rest in 4 seconds, then the force to be applied is **[Pb. PMT 2000]**

(a) 8 *N* (b) 4 *N*

(c) 2 *N* (d) 1 *N*

61. The apparent weight of the body, when it is travelling upwards with an acceleration of $2m/s^2$ and mass is 10 *kg*, will be

(a) 198 *N* (b) 164 *N*

(c) 140 *N* (d) 118 *N*

62. A man measures time period of a pendulum (T) in stationary lift. If the lift moves upward with acceleration $\frac{g}{4}$, then new time period will be **[CBSE PMT 1997]** **[BHU 2001]**

(a) $\frac{2T}{\sqrt{5}}$ (b) $\frac{\sqrt{5}T}{2}$

(c) $\frac{\sqrt{5}}{2T}$ (d) $\frac{2}{\sqrt{5}T}$

63. A 30 *gm* bullet initially travelling at 120 *m/s* penetrates 12 *cm* into a wooden block. The average resistance exerted by the wooden block is **[AFMC 1999; CPMT 2001]**

(a) 2850*N* (b) 2200 *N*

(c) 2000*N* (d) 1800 *N*

64. A force of 10 Newton acts on a body of mass 20*kg* for 10 seconds. Change in its momentum is [MP PET 2002]

(a) $5\ kg\,m/s$ (b) $100 kg\,m/s$

(c) $200 kg\,m/s$ (d) $1000\ kg\,m/s$

65. A body of mass 1.0*kg* is falling with an acceleration of 10 m/sec^2. Its apparent weight will be $(g = 10 m/\sec^2)$ [MP PET 2002]

(a) $1.0\,kg\,wt$ (b) $2.0\,kg\,wt$

(c) $0.5\,kg\,wt$ (d) Zero

66. A player caught a cricket ball of mass 150 *gm* moving at the rate of 20 *m/sec.* if the catching process be completed in 0.1 *sec* the force of the blow exerted by the ball on the hands of player is

(a) 0.3 *N* (b) 30 *N*

(c) 300 *N* (d) 3000 *N*

67. If rope of lift breaks suddenly, the tension exerted by the surface of lift [AFMC 2002]

(a = acceleration of lift)

(a) mg (b) $m(g+a)$

(c) $m(g-a)$ (d) 0

68. A boy whose mass is 50*kg* stands on a spring balance inside a lift. The lift starts to ascent with an acceleration of $2 ms^{-2}$. The reading of the machine or balance $(g = 10\,ms^{-2})$ is [Kerala PET 2002]

(a) $50 kg$ (b) Zero

(c) $49 kg$ (d) $60 kg$

69. A rocket is ejecting $50 g$ of gases per sec at a speed of $500 m/s$. The accelerating force on the rocket will be [Pb. PMT 2002]

(a) $125\,N$ (b) $25\,N$

(c) $5\,N$ (d) Zero

70. A block of mass $5 kg$ is moving horizontally at a speed of 1.5 *m/s*. A perpendicular force of 5*N* acts on it for 4 sec. What will be the distance of the block from the point where the force started acting

(a) 10 *m* (b) 8 *m*

(c) 6 *m* (d) 2 *m*

71. A lift of mass 1000 *kg* is moving with an acceleration of 1 m/s^2 in upward direction. Tension developed in the string, which is connected to the lift, is [CBSE PMT 2002]

(a) 9,800 *N* (b) 10,000 *N*

(c) 10,800 *N* (d) 11,000 *N*

72. A lift accelerated downward with acceleration '*a*'. A man in the lift throws a ball upward with acceleration $a_0 (a_0 < a)$. Then acceleration of ball observed by observer, which is on earth, is [AIEEE 2002]

(a) $(a + a_0)$ upward (b) $(a - a_0)$ upward

(c) $(a + a_0)$ downward (d) $(a - a_0)$ downward

73. A lift is moving down with acceleration *a*. A man in the lift drops a ball inside the lift. The acceleration of the ball as observed by the man in the lift and a man standing stationary on the ground are respectively [AIEEE 2002]

(a) g, g (b) $g-a, g-a$

(c) $g-a, g$ (d) a, g

74. A man weighs $80 kg$. He stands on a weighing scale in a lift which is moving upwards with a uniform acceleration of $5 m/s^2$. What would be the reading on the scale. $(g = 10 m/s^2)$ [Kerala PET 2005]

(a) 400 *N* (b) 800 *N*

(c) 1200 *N* (d) Zero

75. A monkey of mass 20*kg* is holding a vertical rope. The rope will not break when a mass of 25 *kg* is suspended from it but will break if the mass exceeds 25 *kg*. What is the maximum acceleration with which the monkey can climb up along the rope $(g = 10 m/s^2)$

(a) $10 m/s^2$ (b) $25 m/s^2$

(c) $2.5 m/s^2$ (d) $5 m/s^2$

76. If in a stationary lift, a man is standing with a bucket full of water, having a hole at its bottom. The rate of flow of water through this hole is R_0. If the lift starts to move up and down with same acceleration and then that rates of flow of water are R_u and R_d, then [UPSEAT 2003]

(a) $R_0 > R_u > R_d$ (b) $R_u > R_0 > R_d$

(c) $R_d > R_0 > R_u$ (d) $R_u > R_d > R_0$

77. A rocket with a lift- off mass $3.5 \times 10^4\ kg$ is blasted upwards with an initial acceleration of $10 m/s^2$. Then the initial thrust of the blast is [AIEEE 2003]

(a) $1.75 \times 10^5\,N$ (b) $3.5 \times 10^5\,N$

(c) $7.0 \times 10^5\,N$ (d) $14.0 \times 10^5\,N$

78. A spring balance is attached to the ceiling of a lift. A man hangs his bag on the spring and the spring reads 49 *N*, when the lift is stationary. If the lift moves downward with an acceleration of $5 m/s^2$, the reading of the spring balance will be [Pb. PMT 2002]

(a) 49 *N* (b) 24 *N*

(c) 74 *N* (d) 15 *N*

79. A plumb line is suspended from a ceiling of a car moving with horizontal acceleration of *a*. What will be the angle of inclination with vertical [Orissa JEE 2003]

(a) $\tan^{-1}(a/g)$ (b) $\tan^{-1}(g/a)$

(c) $\cos^{-1}(a/g)$ (d) $\cos^{-1}(g/a)$

80. Mass of a person sitting in a lift is 50 *kg*. If lift is coming down with a constant acceleration of 10 m/sec^2. Then the reading of spring balance will be $(g = 10 m/sec^2)$ [RPET 2003; Kerala PMT 2005]

(a) 0 (b) 1000*N*

(c) 100 *N* (d) 10 *N*

81. A body of mass 2 *kg* has an initial velocity of 3 *meters* per second along *OE* and it is subjected to a force of 4 *N* in a direction perpendicular to *OE*. The distance of the body from *O* after 4 seconds will be **[CPMT 1976]**

(a) 12 *m* (b) 20 *m*

(c) 8 *m* (d) 48 *m*

82. A block of mass *m* is placed on a smooth wedge of inclination θ. The whole system is accelerated horizontally so that the block does not slip on the wedge. The force exerted by the wedge on the block (*g* is acceleration due to gravity) will be

(a) $mg\cos\theta$ (b) $mg\sin\theta$

(c) mg (d) $mg/\cos\theta$

83. A machine gun fires a bullet of mass 40 *g* with a velocity $1200\ ms^{-1}$. The man holding it can exert a maximum force of 144 *N* on the gun. How many bullets can he fire per second at the most **[AIEEE 2004]**

(a) One (b) Four

(c) Two (d) Three

84. An automobile travelling with a speed of $60\ km/h$, can brake to stop within a distance of 20 *m*. If the car is going twice as fast, *i.e.* 120 *km/h*, the stopping distance will be

[AIEEE 2004]

(a) 20 *m* (b) 40 *m*

(c) 60 *m* (d) 80 *m*

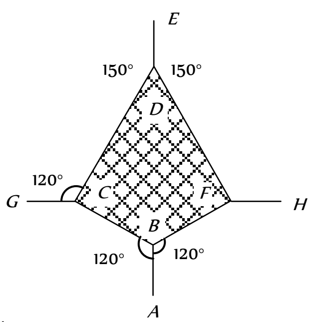
85. A man of weight 75 *kg* is standing in an elevator which is moving with an acceleration of $5\,m/s^2$ in upward direction the apparent weight of the man will be $(g = 10\ m/s^2)$

[Pb. PMT 2004]

(a) 1425 *N* (b) 1375 *N*

(c) 1250 *N* (d) 1125 *N*

86. The adjacent figure is the part of a horizontally stretched net. section *AB* is stretched with a force of 10 *N*. The tensions in the sections *BC* and *BF* are **[KCET 2005]**



(a) 10 *N*, 11 *N*

(b) 10 *N*, 6 *N*

(c) 10 *N*, 10 *N*

(d) Can't calculate due to insufficient data

87. The linear momentum *p* of a body moving in one dimension varies with time according to the equation $p = a + bt^2$ where *a* and *b* are positive constants. The net force acting on the body is

(a) A constant

(b) Proportional to t^2

(c) Inversely proportional to *t*

(d) Proportional to *t*

88. The spring balance inside a lift suspends an object. As the lift begins to ascent, the reading indicated by the spring balance will **[CBSE PMT 2004]**

(a) Increase

(b) Decrease

(c) Remain unchanged

(d) Depend on the speed of ascend

89. There is a simple pendulum hanging from the ceiling of a lift. When the lift is stand still, the time period of the pendulum is *T*. If the resultant acceleration becomes $g/4$, then the new time period of the pendulum is **[DCE 2004]**

(a) 0.8 *T* (b) 0.25 *T*

(c) 2 *T* (d) 4 *T*

90. A man of weight 80 *kg* is standing in an elevator which is moving with an acceleration of $6\ m/s^2$ in upward direction. The apparent weight of the man will be $(g = 10\,m/s^2)$

(a) 1480 *N* (b) 1280 *N*

(c) 1380 *N* (d) None of these

91. A force of 100 dynes acts on a mass of 5 *gram* for 10 *sec*. The velocity produced is **[Pb. PET 2004]**

(a) $2000\ cm/\sec$ (b) $200\ cm/\sec$

(c) $20\ cm/\sec$ (d) $2\ cm/\sec$

92. When the speed of a moving body is doubled

[UPSEAT 2004]

(a) Its acceleration is doubled

(b) Its momentum is doubled

(c) Its kinetic energy is doubled

(d) Its potential energy is doubled

93. A body of mass *m* collides against a wall with a velocity *v* and rebounds with the same speed. Its change of momentum is

(a) 2 *mv* (b) *mv*

(c) – *mv* (d) Zero

94. A thief stole a box full of valuable articles of weight *W* and while carrying it on his back, he jumped down a wall of height '*h*' from the ground. Before he reached the ground he experienced a load of

(a) 2*W* (b) *W*

(c) *W*/2 (d) Zero

95. *N* bullets each of mass *m kg* are fired with a velocity $v\,ms^{-1}$ at the rate of *n* bullets per second upon a wall. The reaction offered by the wall to the bullets is given by

(a) nmv (b) $\dfrac{Nmv}{n}$

(c) $n\dfrac{Nm}{v}$ (d) $n\dfrac{Nv}{m}$

96. If a body of mass m is carried by a lift moving with an upward acceleration a, then the forces acting on the body are (i) the reaction R on the floor of the lift upwards (ii) the weight mg of the body acting vertically downwards. The equation of motion will be given by [MNR 1998]

(a) $R = mg - ma$ (b) $R = mg + ma$

(c) $R = ma - mg$ (d) $R = mg \times ma$

97. With what minimum acceleration can a fireman slides down a rope while breaking strength of the rope is $\frac{2}{3}$ of his weight

(a) $\frac{2}{3}g$ (b) g

(c) $\frac{1}{3}g$ (d) Zero

98. A ball of mass m moves with speed v and it strikes normally with a wall and reflected back normally, if its time of contact with wall is t then find force exerted by ball on wall

[BCECE 2005]

(a) $\frac{2mv}{t}$ (b) $\frac{mv}{t}$

(c) mvt (d) $\frac{mv}{2t}$

99. The velocity of a body at time $t = 0$ is $10\sqrt{2}$ m/s in the north-east direction and it is moving with an acceleration of 2 m/s directed towards the south. The magnitude and direction of the velocity of the body after 5 sec will be

[AMU (Engg.) 1999]

(a) 10 m/s, towards east

(b) 10 m/s, towards north

(c) 10 m/s, towards south

(d) 10 m/s, towards north-east

100. A body of mass 5 kg starts from the origin with an initial velocity $\vec{u} = 30\hat{i} + 40\hat{j}\,ms^{-1}$. If a constant force $\vec{F} = -(\hat{i} + 5\hat{j})N$ acts on the body, the time in which the y-component of the velocity becomes zero is

[EAMCET (Med.) 2000]

(a) 5 *seconds* (b) 20 *seconds*

(c) 40 *seconds* (d) 80 *seconds*

101. A body of mass $8kg$ is moved by a force $F = 3x\,N$, where x is the distance covered. Initial position is $x = 2m$ and the final position is $x = 10$ m. The initial speed is $0.0 m/s$. The final speed is [Orissa JEE 2002]

(a) 6 m/s (b) 12 m/s

(c) 18 m/s (d) 14 m/s

102. The linear momentum p of a body moving in one dimension varies with time according to the equation $p = a + bt^2$, where a and b are positive constants. The net force acting on the body is

(a) Proportional to t^2

(b) A constant

(c) Proportional to t

(d) Inversely proportional to t

103. A ball of mass 0.5 kg moving with a velocity of 2 m/sec strikes a wall normally and bounces back with the same speed. If the time of contact between the ball and the wall is one millisecond, the average force exerted by the wall on the ball is

(a) 2000 N (b) 1000 N

(c) 5000 N (d) 125 N

[CPMT 1979]

104. A particle moves in the xy-plane under the action of a force $\boldsymbol{F}$ such that the components of its linear momentum $\boldsymbol{p}$ at any time t are $p_x = 2\cos t$, $p_y = 2\sin t$. The angle between $\boldsymbol{F}$ and $\boldsymbol{p}$ at time t is

[MP PET 1996; UPSEAT 2000]

(a) 90° (b) 0°

(c) 180° (d) 30°

105. n small balls each of mass m impinge elastically each second on a surface with velocity u. The force experienced by the surface will be

RPET 2001; BHU 2001; MP PMT 2003]

(a) mnu (b) 2 mnu

(c) 4 mnu (d) $\frac{1}{2}mnu$

106. A ball of mass 400 gm is dropped from a height of 5m. A boy on the ground hits the ball vertically upwards with a bat with an average force of 100 *newton* so that it attains a vertical height of 20 m. The time for which the ball remains in contact with the bat is $[g = 10\,m/s^2]$ [MP PMT 1999]

(a) 0.12s (b) 0.08 s

(c) 0.04 s (d) 12 s

107. The time in which a force of 2 N produces a change of momentum of $0.4\,kg - ms^{-1}$ in the body is

[CMEET Bihar 1995]

(a) 0.2 s (b) 0.02 s

(c) 0.5 s (d) 0.05 s

108. A gun of mass $10kg$ fires 4 bullets per second. The mass of each bullet is 20 g and the velocity of the bullet when it leaves the gun is $300\,ms^{-1}$. The force required to hold the gun while firing is

[EAMCET (Med.) 2000]

(a) 6 N (b) 8 N

(c) 24 N (d) 240 N

109. A gardner waters the plants by a pipe of diameter 1mm. The water comes out at the rate or 10 cm/sec. The reactionary force exerted on the hand of the gardner is

[KCET 2000]

(a) Zero (b) $1.27 \times 10^{-2}\,N$

(c) $1.27 \times 10^{-4}\,N$ (d) $0.127N$

110. A solid disc of mass M is just held in air horizontally by throwing 40 stones per sec vertically upwards to strike the disc each with a velocity 6 ms^{-1}. If the mass of each stone is $0.05kg$ what is the mass of the disc $(g = 10ms^{-2})$ [MP PMT 1993]

[Kerala (Engg.) 2001]

(a) $1.2kg$ (b) $0.5kg$

(c) $20kg$ (d) $3kg$

111. A ladder rests against a frictionless vertical wall, with its upper end $6m$ above the ground and the lower end $4m$ away from the wall. The weight of the ladder is 500 *N* and its C. G. at 1/3rd distance from the lower end. Wall's reaction will be, (in *Newton*)

(a) 111 (b) 333

(c) 222 (d) 129

112. A satellite in force-free space sweeps stationary interplanetary dust at a rate $dM/dt = \alpha v$ where *M* is the mass, *v* is the velocity of the satellite and α is a constant. What is the deacceleration of the satellite [CBSE PMT 1994]

(a) $-2\alpha v^2/M$ (b) $-\alpha v^2/M$

(c) $+\alpha v^2/M$ (d) $-\alpha v^2$

113. 10,000 small balls, each weighing 1 *gm*, strike one square *cm* of area per second with a velocity 100 *m/s* in a normal direction and rebound with the same velocity. The value of pressure on the surface will be [MP PMT 1994]

(a) $2\times10^3\ N/m^2$ (b) $2\times10^5\ N/m^2$

(c) $10^7\ N/m^2$ (d) $2\times10^7\ N/m^2$

Third Law of Motion

1. Swimming is possible on account of [AFMC 1998, 2003]

(a) First law of motion

(b) Second law of motion

(c) Third law of motion

(d) Newton's law of gravitation

2. When we jump out of a boat standing in water it moves

(a) Forward (b) Backward

(c) Sideways (d) None of the above

3. You are on a frictionless horizontal plane. How can you get off if no horizontal force is exerted by pushing against the surface

(a) By jumping

(b) By spitting or sneezing

(c) By rolling your body on the surface

(d) By running on the plane

4. On a stationary sail-boat, air is blown at the sails from a fan attached to the boat. The boat will

(a) Remain stationary

(b) Spin around

(c) Move in a direction opposite to that in which air is blown

(d) Move in the direction in which the air is blown

5. A man is at rest in the middle of a pond on perfectly smooth ice. He can get himself to the shore by making use of Newton's

(a) First law (b) Second law

(c) Third law (d) All the laws

6. A cannon after firing recoils due to [EAMCET 1980]

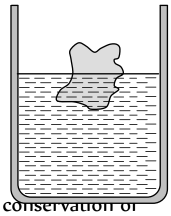
(a) Conservation of energy

(b) Backward thrust of gases produced

(c) Newton's third law of motion

(d) Newton's first law of motion

7. A body floats in a liquid contained in a beaker. If the whole system as shown in figure falls freely under gravity, then the upthrust on the body due to liquid is [Manipal MEE 1995]



(a) Zero [AMU (Med.) 2000]

(b) Equal to the weight of liquid displaced

(c) Equal to the weight of the body in air

(d) None of these

8. Newton's third law of motion leads to the law of conservation of

(a) Angular momentum (b) Energy

(c) Mass (d) Momentum

9. A man is carrying a block of a certain substance (of density 1000 kgm^{-3}) weighing 1 *kg* in his left hand and a bucket filled with water and weighing 10 *kg* in his right hand. He drops the block into the bucket. How much load does he carry in his right hand now

(a) 9 *kg* (b) 10 *kg*

(c) 11 *kg* (d) 12 *kg*

10. A man is standing on a balance and his weight is measured. If he takes a step in the left side, then weight [AFMC 1996]

(a) Will decrease

(b) Will increase

(c) Remains same

(d) First decreases then increases

11. A man is standing at a spring platform. Reading of spring balance is 60 *kg wt*. If man jumps outside platform, then reading of spring balance

[AFMC 1996; AIIMS 2000; Pb. PET 2000]

(a) First increases then decreases to zero

(b) Decreases

(c) Increases

(d) Remains same

12. A cold soft drink is kept on the balance. When the cap is open, then the weight [AFMC 1996]

(a) Increases

(b) Decreases

(c) First increases then decreases

(d) Remains same

13. Action and reaction forces act on

(a) The same body (b) The different bodies

(c) The horizontal surface (d) Nothing can be said

14. A bird is sitting in a large closed cage which is placed on a spring balance. It records a weight of 25 *N*. The bird (mass m = 0.5 *kg*) flies upward in the cage with an acceleration of $2\,m/s^2$. The spring balance will now record a weight of

[MP PMT 1999]

(a) 24 *N* (b) 25 *N*

(c) 26 [CPMT 1981] (d) 27 *N*

15. A light spring balance hangs from the hook of the other light spring balance and a block of mass *M kg* hangs from the former one. Then the true statement about the scale reading is

[AIEEE 2003]

(a) Both the scales read *M*/2 *kg* each

(b) Both the scales read *M kg* each

(c) The scale of the lower one reads M kg and of the upper one zero

(d) The reading of the two scales can be anything but the sum of the reading will be M kg

16. A machine gun fires 20 bullets per second into a target. Each bullet weighs 150 *gms* and has a speed of 800 *m/sec*. Find the force necessary to hold the gun in position [EAMCET 1994]

(a) 800 N (b) 1000 N

(c) 1200 N (d) 2400 N

17. The tension in the spring is [AMU (Engg.) 2001]

(a) Zero (b) $2.5\,N$

(c) $5\,N$ (d) $10\,N$

18. A book is lying on the table. What is the angle between the action of the book on the table and the reaction of the table on the book

(a) 0° (b) 30°

(c) 45° (d) 180°

19. When a horse pulls a wagon, the force that causes the horse to move forward is the force [Pb. PET 2004]

(a) The ground exerts on it (b) It exerts on the ground

(c) The wagon exerts on it (d) It exerts on the wagon

20. A student attempts to pull himself up by tugging on his hair. He will not succeed [KCET 2005]

(a) As the force exerted is small

(b) The frictional force while gripping, is small.

(c) Newton's law of inertia is not applicable to living beings.

(d) As the force applied is internal to the system.

21. A man is standing at the centre of frictionless pond of ice. How can he get himself to the shore [J&K CET 2005]

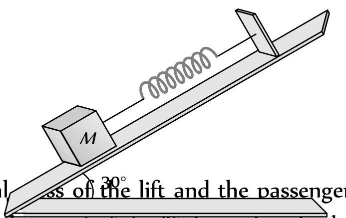
(a) By throwing his shirt in vertically upward direction

(b) By spitting horizontally

(c) He will wait for the ice to melt in pond

(d) Unable to get at the shore

22. A body of mass 5kg is suspended by a spring balance on an inclined plane as shown in figure. The spring balance measure



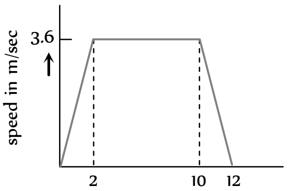
(a) 50 N

(b) 25 N

(c) 500 N

(d) 10 N

23. A lift is going up. The total mass of the lift and the passenger is 1500 kg. The variation in the speed of the lift is as given in the graph. The tension in the rope pulling the lift at t = 11th *sec* will be



(a) 17400 N

(b) 14700 N

(c) 12000 N

(d) Zero

24. In the above ques., the height to which the lift takes the passenger is

(a) 3.6 *meters* (b) 8 *meters*

(c) 1.8 *meters* (d) 36 *meters*

Conservation of Linear Momentum and Impulse

1. A jet plane flies in the air because [NCERT 1971]

(a) The gravity does not act on bodies moving with high speeds

(b) The thrust of the jet compensates for the force of gravity

(c) The flow of air around the wings causes an upward force, which compensates for the force of gravity

(d) The weight of air whose volume is equal to the volume of the plane is more than the weight of the plane

2. A player caught a cricket ball of mass 150 *gm* moving at a rate of 20 *m/s*. If the catching process be completed in 0.1 *s*, then the force of the blow exerted by the ball on the hands of the player is[AFMC 1993; CBSE PMT [Kerala PMT 2005]

(a) 0.3 N (b) 30 N

(c) 300 N (d) 3000 N

3. A rocket has a mass of 100 *kg*. 90% of this is fuel. It ejects fuel vapours at the rate of 1 *kg/sec* with a velocity of 500 *m/sec* relative to the rocket. It is supposed that the rocket is outside the gravitational field. The initial upthrust on the rocket when it just starts moving upwards is [NCERT 1978]

(a) Zero (b) 500 N

(c) 1000 N (d) 2000 N

4. In which of the following cases forces may not be required to keep the [AIIMS 1983]

(a) Particle going in a circle

(b) Particle going along a straight line

(c) The momentum of the particle constant

(d) Acceleration of the particle constant

5. A wagon weighing 1000 *kg* is moving with a velocity 50*km/h* on smooth horizontal rails. A mass of 250 *kg* is dropped into it. The velocity with which it moves now is [MP PMT 1994]

(a) 2.5 *km/hour* (b) 20 *km/hour*

(c) 40 *km/hour* (d) 50 *km/hour*

6. If a force of 250 N act on body, the momentum acquired is 125 *kg-m/s*. What is the period for which force acts on the body

(a) 0.5 *sec* (b) 0.2 *sec*

(c) 0.4 *sec* (d) 0.25 *sec*

7. A 100 *g* iron ball having velocity 10 *m/s* collides with a wall at an angle 30° and rebounds with the same angle. If the period of contact between the ball and wall is 0.1 *second*, then the force experienced by the wall is [CPMT 1997]

(a) 10 N (b) 100 N

(c) 1.0 N (d) 0.1 N

8. A ball of mass 150g starts moving with an acceleration of $20 m/s^2$. When hit by a force, which acts on it for 0.1 sec. The impulsive force is [AFMC 1999; Pb. PMT 2003]
(a) 0.5 *N-s* (b) 0.1 *N-s*
(c) 0.3 *N-s* (d) 1.2 *N-s*

9. A body, whose momentum is constant, must have constant [AIIMS 2000]
(a) Force (b) Velocity
(c) Acceleration (d) All of these

10. The motion of a rocket is based on the principle of conservation of
(a) Mass (b) Kinetic energy
(c) Linear momentum (d) Angular momentum

11. A rope of length 5m is kept on frictionless surface and a force of 5N is applied to one of its end. Find tension in the rope at 1m from this end [RPET 2000]
(a) 1 *N* (b) 3 *N*
(c) 4 *N* (d) 5 *N*

12. An aircraft is moving with a velocity of $300 ms^{-1}$. If all the forces acting on it are balanced, then [Kerala PMT 2004]
(a) It still moves with the same velocity
(b) It will be just floating at the same point in space
(c) It will fall down instantaneously
(d) It will lose its velocity gradually
(e) It will explode

13. A rocket of mass 1000 *kg* exhausts gases at a rate of 4 *kg/sec* with a velocity 3000 *m/s*. The thrust developed on the rocket is
(a) 12000 *N* (b) 120 *N*
(c) 800 *N* (d) 200 *N*

14. The momentum is most closely related to [DCE 2001]
(a) Force (b) Impulse
(c) Power (d) K.E.

15. Rocket engines lift a rocket from the earth surface because hot gas with high velocity [AIIMS 1998; JIPMER 2001, 02]
(a) Push against the earth
(b) Push against the air
(c) React against the rocket and push it up
(d) Heat up the air which lifts the rocket

16. A man fires a bullet of mass 200 *g* at a speed of 5 *m/s*. The gun is of one *kg* mass. by what velocity the gun rebounds backwards[CBSE PMT 1996; JIPMER 2000]
(a) 0.1 *m/s* (b) 10 *m/s*
(c) 1 *m/s* (d) 0.01 *m/s*

17. A bullet of mass 5 *g* is shot from a gun of mass 5 *kg*. The muzzle velocity of the bullet is 500 *m/s*. The recoil velocity of the gun is
(a) 0.5 *m/s* (b) 0.25 *m/s*
(c) 1 *m/s* (d) Data is insufficient

18. A force of 50 *dynes* is acted on a body of mass 5 *g* which is at rest for an interval of 3 seconds, then impulse is [AFMC 1998]
(a) $0.15 \times 10^{-3}\ Ns$ (b) $0.98 \times 10^{-3}\ Ns$
(c) $1.5 \times 10^{-3}\ Ns$ (d) $2.5 \times 10^{-3}\ Ns$

19. A body of mass *M* at rest explodes into three pieces, two of which of mass *M*/4 each are thrown off in perpendicular directions with velocities of 3 *m/s* and 4 *m/s* respectively. The third piece will be thrown off with a velocity of [CPMT 1990]
(a) 1.5 *m/s* (b) 2.0 *m/s*
(c) 2.5 *m/s* (d) 3.0 *m/s*

20. The momentum of a system is conserved [CPMT 1982]
(a) Always
(b) Never
(c) In the absence of an external force on the system
(d) None of the above [AFMC 2000]

21. A body of mass 0.25 *kg* is projected with muzzle velocity $100 ms^{-1}$ from a tank of mass 100 *kg*. What is the recoil velocity of the tank
(a) $5 ms^{-1}$ (b) $25 ms^{-1}$
(c) $0.5 ms^{-1}$ (d) $0.25 ms^{-1}$

22. A bullet is fired from a gun. The force on the bullet is given by $F = 600 - 2 \times 10^5 t$, where F is in *newtons* and t in seconds. The force on the bullet becomes zero as soon as it leaves the barrel. What is the average impulse imparted to the bullet
(a) 9 *Ns* (b) Zero
(c) 0.9 *Ns* (d) 1.8 *Ns*

23. A bullet of mass 0.1 *kg* is fired with a speed of 100 *m/sec*, the mass of gun is 50 *kg*. The velocity of recoil is [AFMC 1995; JIPMER 2000; Pb.PMT 2002]
(a) 0.2 *m/sec* (b) 0.1 *m/sec*
(c) 0.5 *m/sec* [Orissa JEE 2005] (d) 0.05 *m/sec*

24. A bullet mass 10 *gm* is fired from a gun of mass 1*kg*. If the recoil velocity is 5 m/s, the velocity of the muzzle is [Orissa JEE 2002]
(a) 0.05 *m/s* (b) 5 *m/s*
(c) 50 *m/s* (d) 500 *m/s*

25. A rocket can go vertically upwards in earth's atmosphere because
(a) It is lighter than air
(b) Of gravitational pull of the sun
(c) It has a fan which displaces more air per unit time than the weight of the rocket
(d) Of the force exerted on the rocket by gases ejected by it

26. At a certain instant of time the mass of a rocket going up vertically is 100 kg. If it is ejecting 5 kg of gas per second at a speed of 400 m/s, the acceleration of the rocket would be (taking $g = 10 m/s^2$)
(a) $20 m/s^2$ (b) $10 m/s^2$
(c) $2 m/s^2$ [DCE 2004] (d) $1 m/s^2$

27. A jet engine works on the principle of [CPMT 1973; MP PMT 1996]
(a) Conservation of mass
(b) Conservation of energy
(c) Conservation of linear momentum
(d) Conservation of angular momentum

Equilibrium of Forces

1. The weight of an aeroplane flying in the air is balanced by [NCERT 1974]

(a) Vertical component of the thrust created by air currents striking the lower surface of the wings
(b) Force due to reaction of gases ejected by the revolving propeller
(c) Upthrust of the air which will be equal to the weight of the air having the same volume as the plane
(d) Force due to the pressure difference between the upper and lower surfaces of the wings created by different air speeds on the surfaces

2. When a body is stationary [NCERT 1978]
(a) There is no force acting on it
(b) The force acting on it is not in contact with it
(c) The combination of forces acting on it balances each other
(d) The body is in vacuum

3. Two forces of magnitude *F* have a resultant of the same magnitude *F*. The angle between the two forces is [CBSE PMT 1990]
(a) 45° (b) 120°
(c) 150° (d) 60°

4. Two forces with equal magnitudes *F* act on a body and the magnitude of the resultant force is *F*/3. The angle between the two forces is [MP PMT 1999]
(a) $\cos^{-1}\left(-\frac{17}{18}\right)$ (b) $\cos^{-1}\left(-\frac{1}{3}\right)$
(c) $\cos^{-1}\left(\frac{2}{3}\right)$ (d) $\cos^{-1}\left(\frac{8}{9}\right)$

5. An object is subjected to a force in the north-east direction. To balance this force, a second force should be applied in the direction
(a) North-East (b) South
(c) South-West (d) West

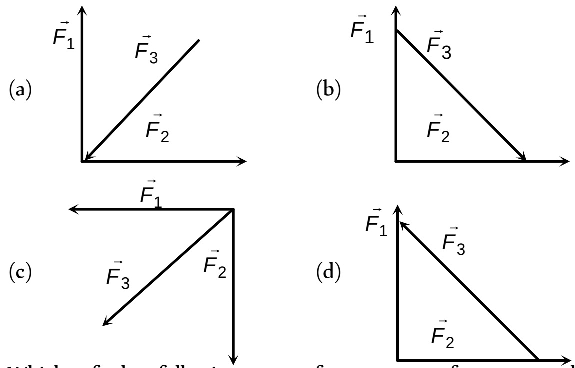
6. The resultant force of 5 *N* and 10 *N* can not be [RPET 2000]
(a) 12 *N* (b) 8 *N*
(c) 4 *N* (d) 5 *N*

7. The resultant of two forces $3P$ and $2P$ is *R*. If the first force is doubled then the resultant is also doubled. The angle between the two forces is [KCET 2001]
(a) 60^o (b) 120^o
(c) 70^o (d) 180^o

8. The resultant of two forces, one double the other in magnitude, is perpendicular to the smaller of the two forces. The angle between the two forces is [KCET 2002]
(a) 60^0 (b) 120^0
(c) 150^0 (d) 90^0

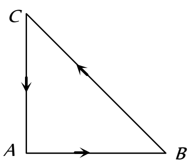
9. Two forces are such that the sum of their magnitudes is 18 *N* and their resultant is perpendicular to the smaller force and magnitude of resultant is 12 N. Then the magnitudes of the forces are
(a) 12 *N*, 6 *N* (b) 13 *N*, 5*N*
(c) 10 *N*, 8 *N* (d) 16 *N*, 2 *N*

10. Which of the four arrangements in the figure correctly shows the vector addition of two forces $\vec{F_1}$ and $\vec{F_2}$ to yield the third force $\vec{F_3}$ [Orissa JEE 2003]



11. Which of the following sets of concurrent forces may be in equilibrium [KCET 2003]
(a) $F_1 = 3N,\ F_2 = 5N,\ F_3 = 9N$
(b) $F_1 = 3N,\ F_2 = 5N,\ F_3 = 1N$
(c) $F_1 = 3N, F_2 = 5N, F_3 = 15N$
(d) $F_1 = 3N,\ F_2 = 5N,\ F_3 = 6N$

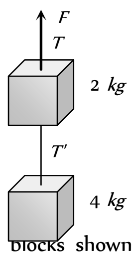
12. Three forces starts acting simultaneously on a particle moving with velocity $\vec{v}$. These forces are represented in magnitude and direction by the three sides of a triangle *ABC* (as shown). The particle will now move with velocity [AIEEE 2003]



(a) $\vec{v}$ remaining unchanged
(b) Less than $\vec{v}$ [KCET 1994]
(c) Greater than $\vec{v}$
(d) $\vec{v}$ in the direction of the largest force *BC*

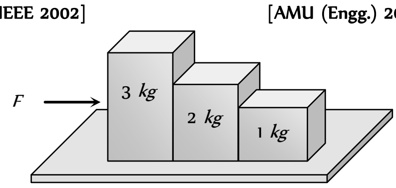
13. Which of the following groups of forces could be in equibrium
(a) 3 *N*, 4 *N*, 5 *N* (b) 4*N*, 5 *N*, 10 *N*
(c) 30*N*, 40 *N*, 80 *N* (d) 1*N*, 3 *N*, 5 *N*

14. Two blocks are connected by a string as shown in the diagram. The upper block is hung by another string. A force *F* applied on the upper string produces an acceleration of $2m/s^2$ in the upward direction in both the blocks. If *T* and T' be the tensions in the two parts of the string, then [AMU (Engg.) 2000]



(a) $T = 70.8N$ and $T' = 47.2N$
(b) $T = 58.8N$ and $T' = 47.2N$
(c) $T = 70.8N$ and $T' = 58.8N$
(d) $T = 70.8N$ and $T' = 0$

15. Consider the following statements about the blocks shown in the diagram that are being pushed by a constant force on a frictionless table [AIEEE 2002] [AMU (Engg.) 2001]



A. All blocks move with the same acceleration

B. The net force on each block is the same Which of these statements are/is correct

(a) A only (b) B only

(c) Both A and B (d) Neither A nor B

16. If two forces of 5 N each are acting along X and Y axes, then the magnitude and direction of resultant is [DCE 2004]

(a) $5\sqrt{2}, \pi/3$ (b) $5\sqrt{2}, \pi/4$

(c) $-5\sqrt{2}, \pi/3$ (d) $-5\sqrt{2}, \pi/4$

17. Which of the following is the correct order of forces [AIEEE 2002]

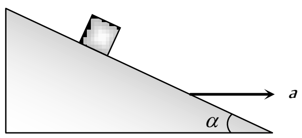
(a) Weak < gravitational forces < strong forces (nuclear) < electrostatic

(b) Gravitational < weak < (electrostatic) < strong force

(c) Gravitational < electrostatic < weak < strong force

(d) Weak < gravitational < electrostatic < strong forces

18. A block is kept on a frictionless inclined surface with angle of inclination 'α'. The incline is given an acceleration 'a' to keep the block stationary. Then a is equal to [AIEEE 2005]



(a) g

(b) $g \tan \alpha$

(c) $g / \tan \alpha$

(d) $g \operatorname{cosec} \alpha$

Motion of Connected Bodies

1. A block of mass M is pulled along a horizontal frictionless surface by a rope of mass m. If a force P is applied at the free end of the rope, the force exerted by the rope on the block will be [CBSE PMT 1993; CPMT 1972, 75, 82; MP PMT 1996; AIEEE 2003]

(a) P (b) $\frac{Pm}{M+m}$

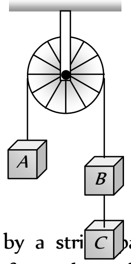
(c) $\frac{PM}{M+m}$ (d) $\frac{Pm}{M-m}$

2. A rope of length L is pulled by a constant force F. What is the tension in the rope at a distance x from the end where the force is applied [MP PET 1996, 97, 2000]

(a) $\frac{FL}{x}$ (b) $\frac{F(L-x)}{L}$

(c) $\frac{FL}{L-x}$ (d) $\frac{Fx}{L-x}$

3. Three equal weights A, B and C of mass 2 kg each are hanging on a string passing over a fixed frictionless pulley as shown in the figure The tension in the string connecting weights B and C is[MP PET 1985; SCRA 1996]



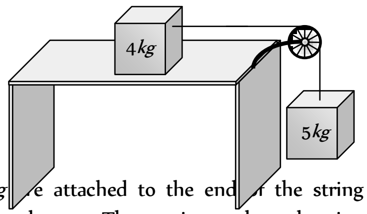
(a) Zero

(b) 13 N

(c) 3.3 N

(d) 19.6 N

4. Two masses of 4 kg and 5 kg are connected by a string passing through a frictionless pulley and are kept on a frictionless table as shown in the figure. The acceleration of 5 kg mass is



(a) $49\, m/s^2$

(b) $5.44\, m/s^2$

(c) $19.5\, m/s^2$

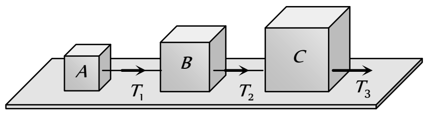
(d) $2.72\, m/s^2$

5. Two masses 2 kg and 3 kg are attached to the end of the string passed over a pulley fixed at the top. The tension and acceleration are

(a) $\frac{7g}{8}; \frac{g}{8}$ (b) $\frac{21g}{8}; \frac{g}{8}$

(c) $\frac{21g}{8}; \frac{g}{5}$ (d) $\frac{12g}{5}; \frac{g}{5}$

6. Three blocks A, B and C weighing 1, 8 and 27 kg respectively are connected as shown in the figure with an inextensible string and are moving on a smooth surface. T_3 is equal to 36 N. Then T_2 is



(a) 18 N

(b) 9 N

(c) 3.375 N

(d) 1.25 N

7. Two bodies of mass 3 kg and 4 kg are suspended at the ends of massless string passing over a frictionless pulley. The acceleration of the system is $(g = 9.8\, m/s^2)$ [MP PET 1994; CBSE PMT 2001]

(a) $4.9\, m/s^2$ (b) $2.45\, m/s^2$

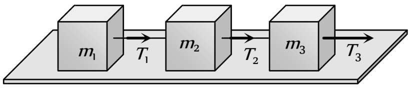
(c) $1.4\, m/s^2$ (d) $9.5\, m/s^2$

8. Three solids of masses m_1, m_2 and m_3 are connected with weightless string in succession and are placed on a frictionless table. If the mass m_3 is dragged with a force T, the tension in the string between m_2 and m_3 is [MP PET 1995]

(a) $\frac{m_2}{m_1+m_2+m_3}T$ (b) $\frac{m_3}{m_1+m_2+m_3}T$

(c) $\frac{m_1+m_2}{m_1+m_2+m_3}T$ (d) $\frac{m_2+m_3}{m_1+m_2+m_3}T$

9. Three blocks of masses m_1, m_2 and m_3 are connected by massless strings as shown on a frictionless table. They are pulled with a force $T_3 = 40\, N$. If $m_1 = 10\, kg, m_2 = 6\, kg$ and $m_3 = 4\, kg$, the tension T_2 will be [MP PMT/PET 1998]



(a) 20 N (b) 40 N

(c) 10 N (d) 32 N

10. A block of mass m_1 rests on a horizontal table. A string tied to the block is passed on a frictionless pulley fixed at the end of the table and to the other end of string is hung another block of mass m_2. The acceleration of the system is

[EAMCET (Med.) 1995; DPMT 2000]

(a) $\frac{m_2 g}{(m_1+m_2)}$ (b) $\frac{m_1 g}{(m_1+m_2)}$

(c) g (d) $\frac{m_2 g}{m_1}$

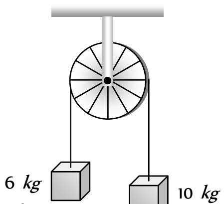
11. A 2 kg block is lying on a smooth table which is connected by a body of mass 1 kg by a string which passes through a pulley. The 1 kg mass is hanging vertically. The acceleration of block and tension in the string will be [RPMT 1997]

(a) $3.27\,m/s^2, 6.54\,N$ (b) $4.38\,m/s^2, 6.54\,N$

(c) $3.27\,m/s^2, 9.86\,N$ (d) $4.38\,m/s^2, 9.86\,N$

12. A light string passes over a frictionless pulley. To one of its ends a mass of 6 kg is attached. To its other end a mass of 10 kg is attached. The tension in the thread will be

[RPET 1996; JIPMER 2001, 02]



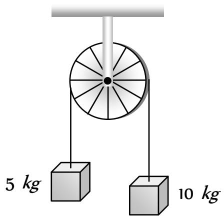
(a) 24.5 N

(b) 2.45 N

(c) 79 N

(d) 73.5 N

13. USS 150) Two masses of 5kg and 10kg are connected to a pulley as shown. What will be the acceleration of the system (g = acceleration due to gravity) [CBSE PMT 2000]



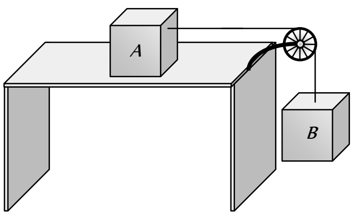
(a) g

(b) $\frac{g}{2}$

(c) $\frac{g}{3}$

(d) $\frac{g}{4}$

14. A block A of mass 7 kg is placed on a frictionless table. A thread tied to it passes over a frictionless pulley and carries a body B of mass 3 kg at the other end. The acceleration of the system is (given $g = 10\,ms^{-2}$) [Kerala (Engg.) 2000]



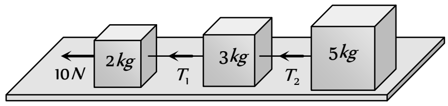
(a) $100\,ms^{-2}$

(b) $3ms^{-2}$

(c) $10ms^{-2}$

(d) $30ms^{-2}$

15. Three blocks of masses 2 kg, 3 kg and 5 kg are connected to each other with light string and are then placed on a frictionless surface as shown in the figure. The system is pulled by a force $F = 10N$, then tension $T_1 =$ [Orissa JEE 2002]



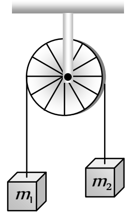
(a) 1N

(b) 5 N

(c) 8 N

(d) 10 N

16. Two masses m_1 and m_2 are attached to a string which passes over a frictionless smooth pulley. When $m_1 = 10kg$, $m_2 = 6kg$, the acceleration of masses is [Orissa JEE 2002]



(a) 20 m/s^2

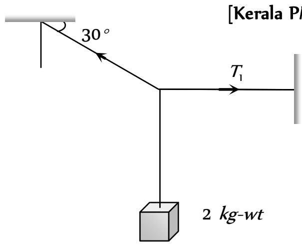
(b) $5m/s^2$

(c) 2.5 m/s^2

(d) $10m/s^2$

17. A body of weight 2kg is suspended as shown in the figure. The tension T_1 in the horizontal string (in kg wt) is

[Kerala PMT 2002]



(a) $2/\sqrt{3}$

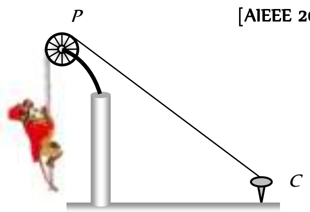
(b) $\sqrt{3}/2$

(c) $2\sqrt{3}$

(d) 2

18. One end of a massless rope, which passes over a massless and frictionless pulley P is tied to a hook C while the other end is free. Maximum tension that the rope can bear is 360 N. with what value of minimum safe acceleration (in ms^{-2}) can a monkey of $60kg$ move down on the rope

[AIEEE 2002]



(a) 16

(b) 6

(c) 4

(d) 8

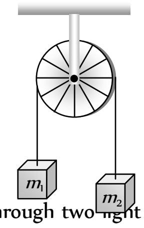
19. A light string passing over a smooth light pulley connects two blocks of masses m_1 and m_2 (vertically). If the acceleration of the system is $g/8$ then the ratio of the masses is

[AIEEE 2002]

(a) 8 : 1 (b) 9 : 7

(c) 4 : 3 (d) 5 : 3

20. Two masses $m_1 = 5\,kg$ and $m_2 = 4.8\,kg$ tied to a string are hanging over a light frictionless pulley. What is the acceleration of the masses when they are free to move $(g = 9.8\ m/s^2)$



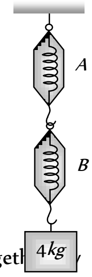
(a) $0.2\ m/s^2$

(b) $9.8\ m/s^2$

(c) $5\ m/s^2$

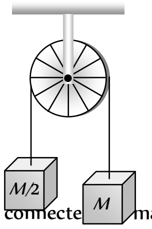
(d) $4.8\ m/s^2$

21. A block of mass 4 *kg* is suspended through two light spring balances *A* and *B*. Then *A* and *B* will read respectively [AIIMS 1995]



(a) 4 *kg* and zero *kg*
(b) Zero *kg* and 4 *kg*
(c) 4 *kg* and 4 *kg*
(d) 2 *kg* and 2 *kg*

22. Two masses *M* and *M*/2 are joint together by means of a light inextensible string passes over a frictionless pulley as shown in figure. When bigger mass is released the small one will ascend with an acceleration of [Kerala PET 2005]



(a) *g*/3
(b) 3*g*/2
(c) *g*/2
(d) *g*

23. Two masses m_1 and m_2 $(m_1 > m_2)$ are connected by massless flexible and inextensible string passed over massless and frictionless pulley. The acceleration of centre of mass is [AIEEE 2004]

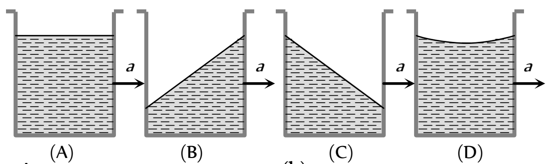
(a) $\left(\dfrac{m_1 - m_2}{m_1 + m_2}\right)^2 g$ (b) $\dfrac{m_1 - m_2}{m_1 + m_2} g$

(c) $\dfrac{m_1 + m_2}{m_1 - m_2} g$ (d) Zero

Critical Thinking

Objective Questions

1. A vessel containing water is given a constant acceleration *a* towards the right, along a straight horizontal path. Which of the following diagram represents the surface of the liquid [IIT 1981]



(a) A (b) B

(c) C (d) D

2. A closed compartment containing gas is moving with some acceleration in horizontal direction. Neglect effect of gravity. Then the pressure in the compartment is [IIT-JEE 1999]

(a) Same everywhere (b) Lower in front side

(c) Lower in rear side (d) Lower in upper side

3. A ship of mass $3 \times 10^7\ kg$ initially at rest is pulled by a force of $5 \times 10^4\ N$ through a distance of 3 *m*. Assume that the resistance due to water is negligible, the speed of the ship is [IIT 1980; MP PMT 2000]

(a) 1.5 *m/s* (b) 60 *m/s*

(c) 0.1 *m/s* (d) 5 *m/s*

4. The mass of a body measured by a physical balance in a lift at rest is found to be *m*. If the lift is going up with an acceleration *a*, its mass will be measured as [MP PET 1994]

(a) $m\left(1 - \dfrac{a}{g}\right)$ (b) $m\left(1 + \dfrac{a}{g}\right)$

(c) m (d) Zero

5. Three weights *W*, 2*W* and 3*W* are connected to identical springs suspended from a rigid horizontal rod. The assembly of the rod and the weights fall freely. The positions of the weights from the rod are such that [Roorkee 1999]

(a) 3*W* will be farthest
(b) *W* will be farthest
(c) All will be at the same distance
(d) 2*W* will be farthest

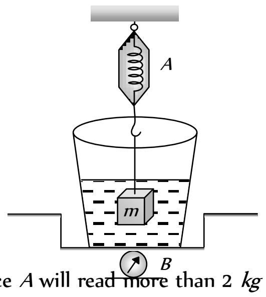
6. When forces F_1, F_2, F_3 are acting on a particle of mass *m* such that F_2 and F_3 are mutually perpendicular, then the particle remains stationary. If the force F_1 is now removed then the acceleration of the particle is [AIEEE 2002]

[J&K CET 2005]

(a) F_1/m (b) F_2F_3/mF_1

(c) $(F_2 - F_3)/m$ (d) F_2/m

7. The spring balance *A* reads 2 *kg* with a block *m* suspended from it. A balance *B* reads 5 *kg* when a beaker filled with liquid is put on the pan of the balance. The two balances are now so arranged that the hanging mass is inside the liquid as shown in figure. In this situation



(a) The balance *A* will read more than 2 *kg*

(b) The balance *B* will read more than 5 *kg*

(c) The balance *A* will read less than 2 *kg* and *B* will read more than 5 *kg*

(d) The balances *A* and *B* will read 2 *kg* and 5 *kg* respectively

8. A rocket is propelled by a gas which is initially at a temperature of 4000 *K*. The temperature of the gas falls to 1000 *K* as it leaves the exhaust nozzle. The gas which will acquire the largest momentum while leaving the nozzle, is

[SCRA 1994]

(a) Hydrogen (b) Helium

(c) Nitrogen (d) Argon

9. Consider the following statement: When jumping from some height, you should bend your knees as you come to rest, instead of keeping your legs stiff. Which of the following relations can be useful in explaining the statement

[AMU (Engg.) 2001]

(a) $\Delta \overrightarrow{P_1} = -\Delta \overrightarrow{P_2}$

(b) $\Delta E = -\Delta(PE + KE) = 0$

(c) $\vec{F}\Delta t = m\Delta \vec{v}$

(d) $\Delta \vec{x} \propto \Delta \vec{F}$

Where symbols have their usual meaning

10. A false balance has equal arms. An object weigh X when placed in one pan and Y when placed in other pan, then the weight W of the object is equal to [AFMC 1994]

(a) $\sqrt{XY}$

(b) $\dfrac{X+Y}{2}$

(c) $\dfrac{X^2+Y^2}{2}$

(d) $\dfrac{2}{\sqrt{X^2+Y^2}}$

11. The vector sum of two forces is perpendicular to their vector differences. In that case, the force [CBSE PMT 2003]

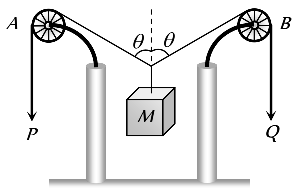
(a) Are equal to each other in magnitude

(b) Are not equal to each other in magnitude

(c) Cannot be predicted

(d) Are equal to each other

12. In the arrangement shown in figure the ends P and Q of an unstretchable string move downwards with uniform speed U. Pulleys A and B are fixed. Mass M moves upwards with a speed



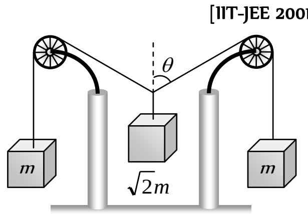
(a) $2U\cos\theta$

(b) $U\cos\theta$

(c) $\dfrac{2U}{\cos\theta}$

(d) $\dfrac{U}{\cos\theta}$

13. The pulleys and strings shown in the figure are smooth and of negligible mass. For the system to remain in equilibrium, the angle θ should be [IIT-JEE 2001]



(a) 0^o

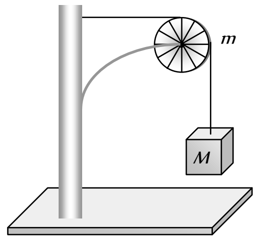
(b) 30^o

(c) 45^o

(d) 60^o

14. A string of negligible mass going over a clamped pulley of mass m supports a block of mass M as shown in the figure. The force on the pulley by the clamp is given by

[IIT-JEE 2001]



(a) $\sqrt{2}Mg$

(b) $\sqrt{2}mg$

(c) $\sqrt{(M+m)^2 + m^2}\,g$

(d) $\sqrt{(M+m)^2 + M^2}\,g$

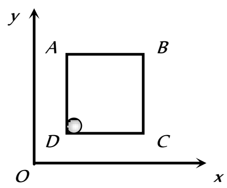
15. A pulley fixed to the ceilling carries a string with blocks of mass m and 3 m attached to its ends. The masses of string and pulley are negligible. When the system is released, its centre of mass moves with what acceleration

[UPSEAT 2002]

(a) 0 (b) $g/4$

(c) $g/2$ (d) $-g/2$

16. A solid sphere of mass 2 kg is resting inside a cube as shown in the figure. The cube is moving with a velocity $v = (5t\hat{i} + 2t\hat{j})m/s$. Here t is the time in second. All surface are smooth. The sphere is at rest with respect to the cube. What is the total force exerted by the sphere on the cube. (Take g = 10 m/s)



(a) $\sqrt{29}N$ (b) 29 N

(c) 26 N (d) $\sqrt{89}\ N$

17. A stick of 1 m is moving with velocity of $2.7\times10^8\,ms^{-1}$. What is the apparent length of the stick $(c = 3\times10^8\,ms^{-1})$ [IIT 1982]

[BHU 1995]

(a) 10 m (b) 0.22 m

(c) 0.44 m (d) 2.4 m

18. One day on a spacecraft corresponds to 2 days on the earth. The speed of the spacecraft relative to the earth is

[CBSE PMT 1993]

(a) $1.5\times10^8\,ms^{-1}$ (b) $2.1\times10^8\,ms^{-1}$

(c) $2.6\times10^8\,ms^{-1}$ (d) $5.2\times10^8\,ms^{-1}$

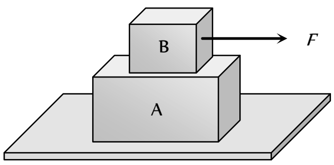
19. A flat plate moves normally with a speed v_1 towards a horizontal jet of water of uniform area of cross-section. The jet discharges water at the rate of volume V per second at a speed of v_2. The density of water is ρ. Assume that water splashes along the surface of the plate at right angles to the original motion. The magnitude of the force acting on the plate due to the jet of water is

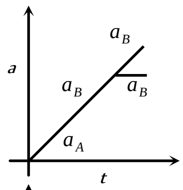
(a) $\rho V v_1$ (b) $\rho V(v_1+v_2)$

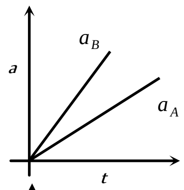
(c) $\frac{\rho V}{v_1+v_2}v_1^2$ (d) $\rho\left[\frac{V}{v_2}\right](v_1+v_2)^2$

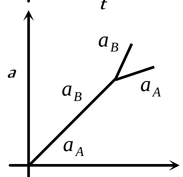
GQ Graphical Questions

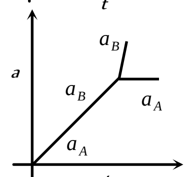
1. A block B is placed on block A. The mass of block B is less than the mass of block A. Friction exists between the blocks, whereas the ground on which the block A is placed is taken to be smooth. A horizontal force F, increasing linearly with time begins to act on B. The acceleration a_A and a_B of blocks A and B respectively are plotted against t. The correctly plotted graph is



(a)


(b)


(c)


(d)


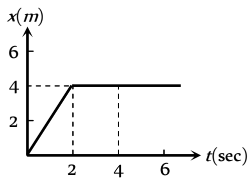
2. In the figure given below, the position-time graph of a particle of mass 0.1 Kg is shown. The impulse at $t = 2\sec$ is

(a) $0.2\,kg\,m\,\sec^{-1}$

(b) $-0.2kg\,m\,\sec^{-1}$

(c) $0.1kg\,m\,\sec^{-1}$

(d) $-0.4kg\,m\,\sec^{-1}$



3. The force-time $(F-t)$ curve of a particle executing linear motion is as shown in the figure. The momentum acquired by the particle in time interval from zero to 8 *second* will be

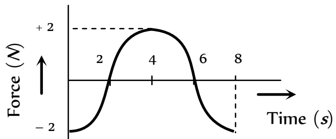
[CPMT 1989]

(a) – 2 N-s

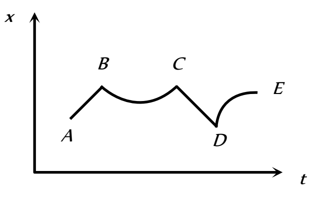
(b) + 4 N-s

(c) 6 N-s

(d) Zero



4. Figure shows the displacement of a particle going along the X-axis as a function of time. The force acting on the particle is zero in the region



(a) AB

(b) BC

(c) CD

(d) DE

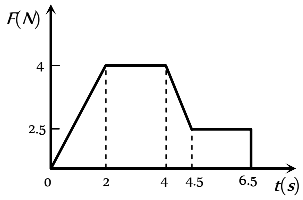
5. A body of 2 kg has an initial speed 5ms. A force acts on it for some time in the direction of motion. The force time graph is shown in figure. The final speed of the body. [IIT 1995]

(a) $9.25\,ms^{-1}$

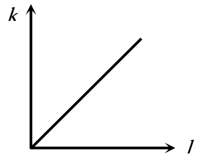
(b) $5\,ms^{-1}$

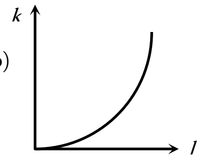
(c) 14.25 ms

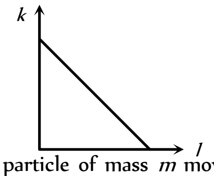
(d) 4.25 ms

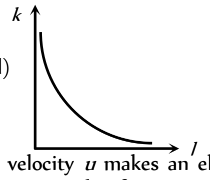


6. Which of the following graph depicts spring constant k versus length l of the spring correctly

(a)


(b)


(c)


(d)


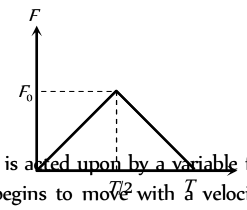
7. A particle of mass m moving with velocity u makes an elastic one dimensional collision with a stationary particle of mass m. They are in contact for a very short time T. Their force of interaction increases from zero to F linearly in time $T/2$, and decreases linearly to zero in further time $T/2$. The magnitude of F is

(a) mu / T

(b) $2mu / T$

(c) $mu / 2T$

(d) None of these



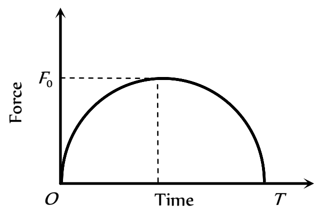
8. A particle of mass m, initially at rest, is acted upon by a variable force F for a brief interval of time T. It begins to move with a velocity u after the force stops acting. F is shown in the graph as a function of time. The curve is a semicircle. [AIIMS 2005]

(a) $u = \frac{\pi F_0^2}{2m}$

(b) $u = \frac{\pi T^2}{8m}$

(c) $u = \frac{\pi F_0 T}{4m}$

(d) $u = \frac{F_0 T}{2m}$



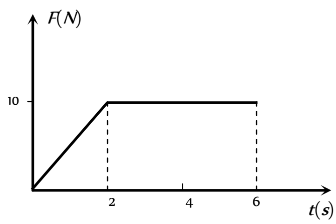
9. A body of mass 3kg is acted on by a force which varies as shown in the graph below. The momentum acquired is given by

(a) Zero

(b) 5 N-s

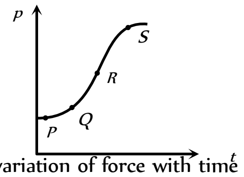
(c) 30 N-s

(d) 50 N-s

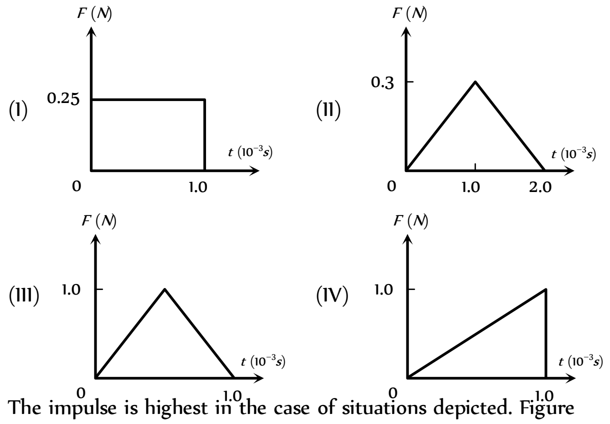


10. The variation of momentum with time of one of the body in a two body collision is shown in fig. The instantaneous force is maximum corresponding to point

(a) P
(b) Q
(c) R
(d) S



11. Figures I, II, III and IV depict variation of force with time



The impulse is highest in the case of situations depicted. Figure

(a) I and II (b) III and I
(c) III and IV (d) IV only

AR Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

(a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
(b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
(c) If assertion is true but reason is false.
(d) If the assertion and reason both are false.
(e) If assertion is false but reason is true.

1. Assertion : Inertia is the property by virtue of which the body is unable to change by itself the state of rest only.
Reason : The bodies do not change their state unless acted upon by an unbalanced external force.

2. Assertion : If the net external force on the body is zero, then its acceleration is zero.
Reason : Acceleration does not depend on force.

3. Assertion : Newton's second law of motion gives the measurement of force.
Reason : According to Newton's second law of motion, force is directly proportional to the rate of change of momentum.

4. Assertion : Force is required to move a body uniformly along a circle.
Reason : When the motion is uniform, acceleration is zero.

5. Assertion : If two objects of different masses have same momentum, the lighter body possess greater velocity.
Reason : For all bodies momentum always remains same.

6. Assertion : Aeroplanes always fly at low altitudes.
Reason : According to Newton's third law of motion, for every action there is an equal and opposite reaction.

7. Assertion : No force is required by the body to remain in any state.
Reason : In uniform linear motion, acceleration has a finite value.

8. Assertion : Mass is a measure of inertia of the body in linear motion.
Reason : Greater the mass, greater is the force required to change its state of rest or of uniform motion.

9. Assertion : The slope of momentum versus time curve give us the acceleration.
Reason : Acceleration is given by the rate of change of momentum.

10. Assertion : A cyclist always bends inwards while negotiating a curve.
Reason : By bending, cyclist lowers his centre of gravity.

11. Assertion : The work done in bringing a body down from the top to the base along a frictionless incline plane is the same as the work done in bringing it down the vertical side.
Reason : The gravitational force on the body along the inclined plane is the same as that along the vertical side.

12. Assertion : Linear momentum of a body changes even when it is moving uniformly in a circle.
Reason : Force required to move a body uniformly along a straight line is zero.

13. Assertion : A bullet is fired from a rifle. If the rifle recoils freely, the kinetic energy of rifle is more than that of the bullet.
Reason : In the case of rifle bullet system the law of conservation of momentum violates.

14. Assertion : A rocket works on the principle of conservation of linear momentum.
Reason : Whenever there is a change in momentum of one body, the same change occurs in the momentum of the second body of the same system but in the opposite direction.

15. Assertion : The apparent weight of a body in an elevator moving with some downward acceleration is less than the actual weight of body.
Reason : The part of the weight is spent in producing downward acceleration, when body is in elevator.

16. Assertion : When the lift moves with uniform velocity the man in the lift will feel weightlessness.
Reason : In downward accelerated motion of lift, apparent weight of a body decreases.

17. Assertion : In the case of free fall of the lift, the man will feel weightlessness.
Reason : In free fall, acceleration of lift is equal to acceleration due to gravity.

18. Assertion : A player lowers his hands while catching a cricket ball and suffers less reaction force.
Reason : The time of catch increases when cricketer lowers its hand while catching a ball.

19. Assertion : The acceleration produced by a force in the motion of a body depends only upon its mass.

Reason : Larger is the mass of the body, lesser will be the acceleration produced.

20. Assertion : Linear momentum of a body changes even when it is moving uniformly in a circle.

Reason : In uniform circular motion velocity remain constant.

21. Assertion : Newton's third law of motion is applicable only when bodies are in motion.

Reason : Newton's third law applies to all types of forces, *e.g.* gravitational, electric or magnetic forces etc.

22. Assertion : A reference frame attached to earth is an inertial frame of reference.

Reason : The reference frame which has zero acceleration is called a non inertial frame of reference.

23. Assertion : A table cloth can be pulled from a table without dislodging the dishes.

Reason : To every action there is an equal and opposite reaction.

24. Assertion : A body subjected to three concurrent forces cannot be in equilibrium.

Reason : If large number of concurrent forces acting on the same point, then the point will be in equilibrium, if sum of all the forces is equal to zero.

25. Assertion : Impulse and momentum have different dimensions.

Reason : From Newton's second law of motion, impulse is equal to change in momentum.

Answers

First Law of Motion

1	c	2	c	3	d	4	b	5	b
6	c	7	d	8	c	9	d	10	a
11	b	12	a						

Second Law of Motion

1	b	2	b	3	c	4	b	5	b
6	b	7	d	8	a	9	d	10	a
11	d	12	c	13	d	14	b	15	a
16	b	17	b	18	b	19	b	20	b
21	d	22	b	23	b	24	a	25	a
26	d	27	c	28	c	29	d	30	d
31	d	32	a	33	a	34	d	35	b
36	b	37	a	38	a	39	d	40	a
41	b	42	c	43	b	44	b	45	d
46	b	47	b	48	a	49	d	50	c
51	a	52	b	53	d	54	d	55	a
56	d	57	a	58	d	59	c	60	b
61	d	62	a	63	d	64	b	65	d
66	b	67	d	68	d	69	b	70	a
71	c	72	d	73	c	74	c	75	c
76	b	77	c	78	b	79	a	80	a
81	b	82	d	83	d	84	d	85	d
86	c	87	d	88	a	89	c	90	b
91	b	92	b	93	a	94	d	95	a
96	b	97	c	98	a	99	a	100	c
101	a	102	c	103	a	104	a	105	b
106	a	107	a	108	c	109	d	110	a
111	a	112	c	113	d				

Third Law of Motion

1	c	2	b	3	b	4	a	5	c
6	c	7	a	8	d	9	c	10	c
11	a	12	c	13	b	14	b	15	b
16	d	17	c	18	d	19	a	20	d
21	b	22	b	23	c	24	d		

Conservation of Linear Momentum Impulse

1	b	2	b	3	b	4	c	5	c
6	a	7	a	8	c	9	b	10	c
11	c	12	a	13	a	14	b	15	c
16	c	17	a	18	c	19	c	20	c
21	d	22	c	23	a	24	d	25	d
26	b	27	c						

Equilibrium of Forces

1	d	2	c	3	b	4	a	5	c
6	c	7	b	8	b	9	b	10	c
11	d	12	a	13	a	14	a	15	a
16	b	17	b	18	b				

Motion of Connected Bodies

1	c	2	b	3	b	4	b	5	d
6	b	7	c	8	c	9	d	10	a
11	a	12	d	13	c	14	b	15	c
16	c	17	c	18	c	19	b	20	a
21	c	22	a	23	a				

Critical Thinking Questions

1	c	2	b	3	c	4	c	5	c
6	a	7	bc	8	d	9	c	10	b
11	a	12	d	13	c	14	d	15	b
16	c	17	c	18	c	19	d		

Graphical Questions

1	d	2	b	3	d	4	ac	5	c
6	d	7	b	8	c	9	d	10	c
11	c								

Assertion & Reason

1	e	2	c	3	a	4	b	5	c
6	a	7	c	8	a	9	d	10	c
11	c	12	b	13	d	14	a	15	c
16	e	17	a	18	a	19	b	20	c
21	e	22	d	23	b	24	e	25	e

AS Answers and Solutions

First Law of Motion

1. (c)
2. (c)
3. (d)
4. (b)
5. (b) Horizontal velocity of apple will remain same but due to retardation of train, velocity of train and hence velocity of boy *w.r.t.* ground decreases, so apple falls away from the hand of boy in the direction of motion of the train.
6. (c) Newton's first law of motion defines the inertia of body. It states that every body has a tendency to remain in its state (either rest or motion) due to its inerta.
7. (d) Horizontal velocity of ball and person are same so both will cover equal horizontal distance in a given interval of time and after following the parabolic path the ball falls exactly in the hand which threw it up.
8. (c) When the bird flies, it pushes air down to balance its weight. So the weight of the bird and closed cage assembly remains unchanged.
9. (d) Particle will move with uniform velocity due to inertia.
10. (a)
11. (b) When a sudden jerk is given to C, an impulsive tension exceeding the breaking tension develops in C first, which breaks before this impulse can reach A as a wave through block.
12. (a) When the spring C is stretched slowly, the tension in A is greater than that of C, because of the weight mg and the former reaches breaking point earlier.

Second Law of Motion

1. (b) $u = 100\ m/s,\ v = 0,\ s = 0.06\ m$

   Retardation $= a = \dfrac{u^2}{2s} = \dfrac{(100)^2}{2 \times 0.06} = \dfrac{1 \times 10^6}{12}$

   $\therefore$ Force $= ma = \dfrac{5 \times 10^{-3} \times 1 \times 10^6}{12} = \dfrac{5000}{12} = 417\ N$

2. (b) $\vec{F} = m\vec{a}$
3. (c) Acceleration $a = \dfrac{F}{m} = \dfrac{100}{5} = 20\ cm/s^2$

   Now $v = at = 20 \times 10 = 200\ cm/s$

4. (b)
5. (b) $F = u\left(\dfrac{dm}{dt}\right) = 400 \times 0.05 = 20\ N$
6. (b) $u = 4\ m/s,\ v = 0,\ t = 2\sec$

   $v = u + at \Rightarrow 0 = 4 + 2a \Rightarrow a = -2\ m/s^2$

   $\therefore$ Retarding force $= ma = 2 \times 2 = 4\ N$

This force opposes the motion. If the same amount of force is applied in forward direction, then the body will move with constant velocity.

7. (d) Reading on the spring balance $= m\,(g-a)$

and since $a=g$ $\therefore$ Force = 0

8. (a) The lift is not accelerated, hence the reading of the balance will be equal to the true weight.

$R = mg = 2g\ Newton$ or $2\ kg$

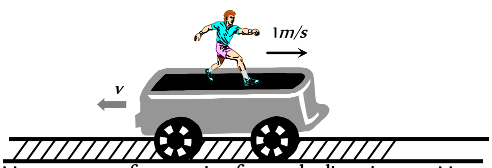
9. (d) When lift moves upward then reading of the spring balance, $R = m(g+a) = 2(g+g) = 4g\ N = 4\ kg$ [As $a=g$]

10. (a) For stationary lift $t_1 = \sqrt{\dfrac{2h}{g}}$

and when the lift is moving up with constant acceleration $t_2 = \sqrt{\dfrac{2h}{g+a}}$ $\therefore$ $t_1 > t_2$

11. (d) Since $T = mg$, it implies that elevator may be at rest or in uniform motion.

12. (c) If the man starts walking on the trolley in the forward direction then whole system will move in backward direction with same momentum.



Momentum of man in forward direction = Momentum of system (man + trolley) in backward direction

$\Rightarrow 80\times 1 = (80+320)\times v \Rightarrow v = 0.2\ m/s$

So the velocity of man *w.r.t.* ground $1.0-0.2 = 0.8\ m/s$

$\therefore$ Displacement of man *w.r.t.* ground $= 0.8\times 4 = 3.2\ m$

13. (d) Force = Mass × Acceleration. If mass and acceleration both are doubled then force will become four times.

14. (b) As weight = 9.8 N $\therefore$ Mass = 1 kg

Acceleration $= \dfrac{\text{Force}}{\text{Mass}} = \dfrac{5}{1} = 5\ m/s^2$

15. (a) Force on the table = $mg = 40\times 980 = 39200\ dyne$

16. (b) $a = \dfrac{F}{m} = \dfrac{1\ N}{1\ kg} = 1\ m/s^2$

17. (b) $\vec{a} = \dfrac{\vec{v}_2 - \vec{v}_1}{t} = \dfrac{(-2)-(+10)}{4} = \dfrac{-12}{4} = -3\ m/s^2$

18. (b) $F = ma = 10\times(-3) = -30\ N$

19. (b) Impulse = Force × Time = $-30\times 4 = -120\ N\text{-}s$

20. (b) u = velocity of bullet

$\dfrac{dm}{dt}$ = Mass thrown per second by the machine gun

= Mass of bullet × Number of bullet fired per second

$= 10\ g\times 10\ bullet/\sec = 100\ g/\sec = 0.1\ kg/\sec$

$\therefore$ Thrust $= \dfrac{udm}{dt} = 500\times 0.1 = 50\ N$

21. (d) Acceleration of the car $= \dfrac{\text{Thrust on the car}}{\text{Mass of the car}}$

$= \dfrac{50}{2000} = \dfrac{1}{40} = 0.025\ m/s^2$

22. (b)

23. (b) Force on particle at 20 cm away $F = kx$

$F = 15\times 0.2 = 3\ N$ $[\text{As } k = 15\ N/m]$

$\therefore$ Acceleration $= \dfrac{\text{Force}}{\text{Mass}} = \dfrac{3}{0.3} = 10\ m/s^2$

24. (a) Force on the block $F = u\left(\dfrac{dm}{dt}\right) = 5\times 1 = 5\ N$

$\therefore$ Acceleration of block $a = \dfrac{F}{m} = \dfrac{5}{2} = 2.5\ m/s^2$

25. (a) Opposing force $F = u\left(\dfrac{dm}{dt}\right) = 2\times 0.5 = 1\ N$

So same amount of force is required to keep the belt moving at 2 m/s

26. (d) Resultant force is $w + 3w = 4w$

27. (c) Acceleration $= \dfrac{\text{Force}}{\text{Mass}} = \dfrac{50\ N}{10\ kg} = 5\ m/s^2$

From $v = u + at = 0 + 5\times 4 = 20\ m/s$

28. (c) Thrust $F = u\left(\dfrac{dm}{dt}\right) = 5\times 10^4 \times 40 = 2\times 10^6\ N$

29. (d) In stationary lift man weighs 40 kg *i.e.* 400 N.

When lift accelerates upward it's apparent weight $= m(g+a) = 40(10+2) = 480\ N$ *i.e.* 48 kg

For the clarity of concepts in this problem *kg-wt* can be used in place of *kg*.

30. (d) As the apparent weight increase therefore we can say that acceleration of the lift is in upward direction.

$R = m(g+a) \Rightarrow 4.8\ g = 4(g+a)$

$\Rightarrow a = 0.2g = 1.96\ m/s^2$

31. (d) $T = m\,(g+a) = 6000\,(10+5) = 90000\ N$

32. (a) $F = ma = \dfrac{m\Delta v}{\Delta t} = \dfrac{0.2\times 20}{0.1} = 40\ N$

33. (a) $F = m\left(\dfrac{dv}{dt}\right) = \dfrac{100\times 5}{0.1} = 5000\ N$

34. (d)

35. (b) $F = m\,(g+a) = 20\times 10^3\times(10+4) = 28\times 10^4\ N$

36. (b) $\dfrac{mg}{m(g-a)} = \dfrac{3}{2} \Rightarrow a = g/3$

37. (a) $T = m\,(g+a) = 500(10+2) = 6000\,N$

38. (a) $F = u\left(\dfrac{dm}{dt}\right) \Rightarrow \dfrac{dm}{dt} = \dfrac{F}{u} = \dfrac{210}{300} = 0.7\ kg/s$

39. (d) $R = m\,(g+a) = m\,(g+g) = 2mg$

40. (a) $T_1 = m\,(g+a) = 1\times\left(g + \dfrac{g}{2}\right) = \dfrac{3g}{2}$

$T_2 = m(g-a) = 1\times\left(g-\frac{g}{2}\right) = \frac{g}{2}$ $\therefore$ $\frac{T_1}{T_2} = \frac{3}{1}$

41. (b) $F = \frac{udm}{dt} = m(g+a)$

$\Rightarrow \frac{dm}{dt} = \frac{m(g+a)}{u} = \frac{5000\times(10+20)}{800} = 187.5\ kg/s$

42. (c) Initially due to upward acceleration apparent weight of the body increases but then it decreases due to decrease in gravity.

43. (b) $T = 2\pi\sqrt{\frac{l}{g}}$ and $T' = 2\pi\sqrt{\frac{l}{4g/3}}$

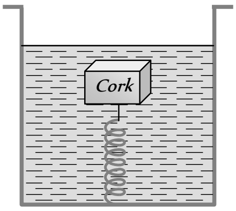
$[As\ g' = g+a = g+\frac{g}{3} = \frac{4g}{3}]$

$\therefore T' = \frac{\sqrt{3}}{2}T$

44. (b) Density of cork = d, Density of water = ρ

Resultant upward force on cork $= V(\rho - d)g$

This causes elongation in the spring. When the lift moves down with acceleration a, the resultant upward force on cork $= V(\rho-d)(g-a)$ which is less than the previous value. So the elongation decreases.



45. (d) When trolley are released then they posses same linear momentum but in opposite direction. Kinetic energy acquired by any trolley will dissipate against friction.

$\therefore$ $\mu mg\ s = \frac{p^2}{2m} \Rightarrow s \propto \frac{1}{m^2}$ [As P and u are constants]

$\Rightarrow \frac{s_1}{s_2} = \left(\frac{m_2}{m_1}\right)^2 = \left(\frac{3}{1}\right)^2 = \frac{9}{1}$

46. (b) Apparent weight $= m(g-a) = 50(9.8-9.8) = 0$

47. (b) Opposite force causes retardation.

48. (a) $T = m(g-a)$ = 10(980 – 400) = 5800 dyne

49. (d) $T = 2\pi\sqrt{\frac{l}{g}}$. T will decrease, If g increases.

It is possible when rocket moves up with uniform acceleration.

50. (c) We know that in the given condition $s \propto \frac{1}{m^2}$

$\therefore$ $\frac{s_2}{s_1} = \left(\frac{m_1}{m_2}\right)^2 \Rightarrow s_2 = \left(\frac{m_1}{m_2}\right)^2 \times s_1$

51. (a) $m = \frac{F}{a} = \frac{\sqrt{6^2+8^2+10^2}}{1} = \sqrt{200} = 10\sqrt{2}\ kg$

52. (b) In the absence of external force, position of centre of mass remain same therefore they will meet at their centre of mass.

53. (d)Force $= m\left(\frac{dv}{dt}\right) = \frac{0.25\times[(10)-(-10)]}{0.01} = 25\times 20 = 500\ N$

54. (d) $T = mg = 50\times 10^{-3}\times 10 = 0.5\ N$

55. (a) $F = u\left(\frac{dm}{dt}\right) = 20\times\frac{50}{60} = 16.66\ N$

56. (d) $u = 250\ m/s$, $v = 0$, $s = 0.12\ metre$

$F = ma = m\left(\frac{u^2-v^2}{2s}\right) = \frac{20\times 10^{-3}\times(250)^2}{2\times 0.12}$

$\therefore$ $F = 5.2\times 10^3\ N$

57. (a) $F = m\left(\frac{v-u}{t}\right) = \frac{5(65-15)\times 10^{-2}}{0.2} = 12.5\ N$

58. (d)

59. (c) $v = u + \frac{F}{m}t = 10 + \left(\frac{1000-500}{1000}\right)\times 10 = 15\ m/s$

60. (b) $F = ma = \frac{m(u-v)}{t} = \frac{2\times(8-0)}{4} = 4N$

61. (d) $R = m(g+a) = 10\times(9.8+2) = 118\ N$

62. (a) $T = 2\pi\sqrt{\frac{l}{g}} \Rightarrow \frac{T'}{T} = \sqrt{\frac{g}{g'}} = \sqrt{\frac{g}{g+\frac{g}{4}}} = \sqrt{\frac{4}{5}} = \frac{2}{\sqrt{5}}$

63. (d) $F = \frac{m(u^2-v^2)}{2S} = \frac{30\times 10^{-3}\times(120)^2}{2\times 12\times 10^{-2}} = 1800\ N$

64. (b) $dp = F\times dt = 10\times 10 = 100\ kg\ m/s$

65. (d) $R = m(g-a) = m(10-10) =$ zero

66. (b) Force exerted by the ball

$\Rightarrow F = m\left(\frac{dv}{dt}\right) = 0.15\times\frac{20}{0.1} = 30\ N$

67. (d) If rope of lift breaks suddenly, acceleration becomes equal to g so that tension, $T = m(g-g) = 0$

68. (d) $R = m(g+a) = 50\times(10+2) = 600\ N = 60\ kg\ wt$

69. (b) $F = u\left(\frac{dm}{dt}\right) = 500\times 50\times 10^{-3} = 25\ N$

70. (a) $S_{\text{Horizontal}} = ut = 1.5\times 4 = 6\ m$

$S_{\text{Vertical}} = \frac{1}{2}at^2 = \frac{1}{2}\frac{F}{m}t^2 = \frac{1}{2}\times 1\times 16 = 8\ m$

$S_{\text{Net}} = \sqrt{6^2+8^2} = 10\ m$

71. (c) $T = m(g+a) = 1000(9.8+1) = 10800\ N$

72. (d) The effective acceleration of ball observed by observer on earth = $(a - a)$

As $a_0 < a$, hence net acceleration is in downward direction.

73. (c) Due to relative motion, acceleration of ball observed by observer in lift = $(g - a)$ and for man on earth the acceleration remains g.

74. (c) For accelerated upward motion

$R = m(g+a) = 80(10+5) = 1200\ N$

75. (c) Tension the string $= m(g+a) =$ Breaking force

$\Rightarrow 20(g+a) = 25\times g \Rightarrow a = g/4 = 2.5\ m/s^2$

76. (b) Rate of flow will be more when lift will move in upward direction with some acceleration because the net downward pull will be more and vice-versa.

$F_{\text{upward}} = m(g+a)$ and $F_{\text{downward}} = m(g-a)$

77. (c) Initial thrust must be

$m[g+a] = 3.5\times10^4(10+10) = 7\times10^5\,N$

78. (b) When the lift is stationary $W = mg$

$\Rightarrow 49 = m\times9.8 \Rightarrow m = 5\,kg.$

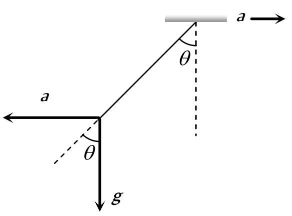
When the lift is moving downward with an acceleration $R = m(9.8-a) = 5[9.8-5] = 24\,N$

79. (a) When car moves towards right with acceleration a then due to pseudo force the plumb line will tilt in backward direction making an angle θ with vertical.

From the figure,

$\tan\theta = a/g$

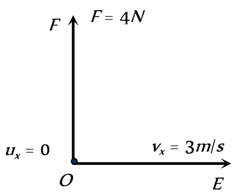
$\therefore\ \theta = \tan^{-1}(a/g)$



80. (a) $R = m(g-a) = 0$

81. (b) Displacement of body in 4 *sec* along *OE*

$s_x = v_x t = 3\times4 = 12\ m$



Force along *OF* (perpendicular to *OE*) = 4 *N*

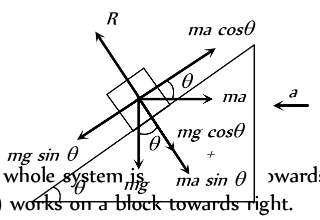
$\therefore\ a_y = \frac{F}{m} = \frac{4}{2} = 2\ m/s^2$

Displacement of body in 4 *sec* along *OF*

$\Rightarrow s_y = u_y t + \frac{1}{2}a_y t^2 = \frac{1}{2}\times2\times(4)^2 = 16\ m$ [As $u_y = 0$]

$\therefore$ Net displacement $s = \sqrt{s_x^2 + s_y^2} = \sqrt{(12)^2 + (16)^2} = 20\ m$

82. (d)



When the whole system is moving towards left then pseudo force (ma) works on a block towards right.

For the condition of equilibrium

$mg\sin\theta = ma\cos\theta \Rightarrow a = \frac{g\sin\theta}{\cos\theta}$

$\therefore$ Force exerted by the wedge on the block

$R = mg\cos\theta + ma\sin\theta$

$R = mg\cos\theta + m\left(\frac{g\sin\theta}{\cos\theta}\right)\sin\theta = \frac{mg(\cos^2\theta + \sin^2\theta)}{\cos\theta}$

$R = \frac{mg}{\cos\theta}$

83. (d) u = velocity of bullet

$\frac{dm}{dt}$ = Mass fired per second by the gun

$\frac{dm}{dt}$ = Mass of bullet (m) × Bullets fired per sec (N)

Maximum force that man can exert $F = u\left(\frac{dm}{dt}\right)$

$\therefore\ F = u\times m_B\times N$

$\Rightarrow N = \frac{F}{m_B\times u} = \frac{144}{40\times10^{-3}\times1200} = 3$

84. (d) The stopping distance, $S \propto u^2$ $(\because v^2 = u^2 - 2as)$

$\Rightarrow \frac{S_2}{S_1} = \left(\frac{u_2}{u_1}\right)^2 = \left(\frac{120}{60}\right)^2 = 4$

$\Rightarrow S_2 = 4\times S_1 = 4\times20 = 80\,m$

85. (d) The apparent weight,

$R = m(g+a) = 75(10+5) = 1125\ N$

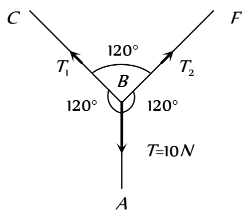
86. (c) By drawing the free body diagram of point B

Let the tension in the section BC and BF are T_1 and T_2 respectively.

From Lami's theorem

$\frac{T_1}{\sin120°} = \frac{T_2}{\sin120°} = \frac{T}{\sin120°}$

$\Rightarrow T = T_1 = T_2 = 10\ N.$



87. (d) $F = \frac{dp}{dt} \equiv \frac{d}{dt}(a+bt^2) = 2bt \ \therefore\ F \propto t$

88. (a) When the lift moves upwards, the apparent weight,

$= m(g+a)$. Hence reading of spring balance increases.

89. (c) When lift is at rest, $T = 2\pi\sqrt{l/g}$

If acceleration becomes $g/4$ then

$T' = 2\pi\sqrt{\frac{l}{g/4}} = 2\pi\sqrt{\frac{4l}{g}} = 2\times T$

90. (b) The apparent weight of man,

$R = m(g+a) = 80(10+6) = 1280\,N$

91. (b) $v = u + at = 0 + \left(\frac{F}{m}\right)t = \left(\frac{100}{5}\right)\times10 = 200\ cm/\sec$

92. (b)

93. (a) $\Delta p = p_i - p_f = mv - (-mv) = 2\,mv$

94. (d) In the condition of free fall apparent weight becomes zero.

95. (a) Total mass of bullets = Nm, time $t = \frac{N}{n}$

Momentum of the bullets striking the wall = Nmv

Rate of change of momentum (Force) = $\frac{Nmv}{t}$ = nmv.

96. (b)

97. (c) If man slides down with some acceleration then its apparent weight decreases. For critical condition rope can bear only 2/3 of his weight. If a is the minimum acceleration then,

Tension in the rope $= m(g-a) =$ Breaking strength

$$\Rightarrow m(g-a) = \frac{2}{3}mg \Rightarrow a = g - \frac{2g}{3} = \frac{g}{3}$$

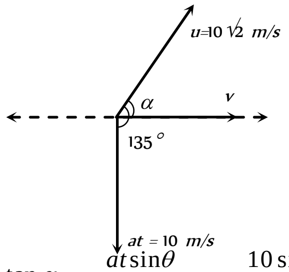
98. (a) For exerted by ball on wall

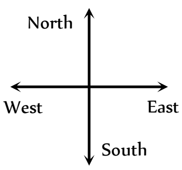
= rate of change in momentum of ball

$$= \frac{mv-(-mv)}{t} = \frac{2mv}{t}$$

99. (a) $\vec{v} = \vec{u} + \vec{a}t \therefore v = \sqrt{u^2 + a^2t^2 + 2u\,at\cos\theta}$

$$v = \sqrt{200 + 100 + 2\times 10\sqrt{2}\times 10\times\cos 135} = 10\,m/s$$





$$\tan\alpha = \frac{at\sin\theta}{u + at\cos\theta}\,\frac{10\sin 135}{10\sqrt{2} + 10\cos 135} = 1 \therefore \alpha = 45°$$

i.e. resultant velocity is 10 *m/s* towards East.

100. (c) $u_y = 40\,m/s$, $F_y = -5N$, $m = 5\,kg$.

So $a_y = \frac{F_y}{m} = -1\,m/s^2$ (As $v = u + at$)

$\therefore v_y = 40 - 1\times t = 0 \Rightarrow t = 40\sec$.

101. (a) Increment in kinetic energy = work done

$$\Rightarrow \frac{1}{2}m(v^2 - u^2) = \int_{x_1}^{x_2} F.dx = \int_2^{10}(3x)\,dx$$

$$\Rightarrow \frac{1}{2}mv^2 = \frac{3}{2}[x^2]_2^{10} = \frac{3}{2}[100-4]$$

$$\Rightarrow \frac{1}{2}\times 8\times v^2 = \frac{3}{2}\times 96 \Rightarrow v = 6m/s$$

102. (c) $\vec{F} = \frac{d\vec{p}}{dt} = \frac{d}{dt}(a + bt^2) = 2bt$ *i.e.* $F \propto t$

103. (a) $F_{av} = \frac{\Delta p}{\Delta t} = \frac{mv-(-mv)}{\Delta t} = \frac{2mv}{\Delta t} = \frac{2\times 0.5\times 2}{10^{-3}} = 2000\ N$

104. (a) Given that $\vec{p} = p_x\hat{i} + p_y\hat{j} = 2\cos t\,\hat{i} + 2\sin t\,\hat{j}$

$$\therefore \vec{F} = \frac{d\vec{p}}{dt} = -2\sin t\,\hat{i} + 2\cos t\,\hat{j}$$

Now, $\vec{F}.\vec{p} = 0$ *i.e.* angle between $\vec{F}$ and $\vec{p}$ is 90°.

105. (b) $\vec{F} = \frac{d\vec{p}}{dt} =$ Rate of change of momentum

As balls collide elastically hence, rate of change of momentum of ball $= n[mu - (mu)] = 2mnu$

i.e. $F = 2mnu$

106. (a) Velocity by which the ball hits the bat

$v_1 = \sqrt{2gh_1} = \sqrt{2\times 10\times 5}$ or $\vec{v_1} = +10\,m/s = 10\,m/s$

velocity of rebound

$v_2 = \sqrt{2gh_2} = \sqrt{2\times 10\times 20} = 20\,m/s$ or $\vec{v_2} = -20\,m/s$

$$F = m\frac{dv}{dt} = \frac{m(\vec{v_2} - \vec{v_1})}{dt} = \frac{0.4(-20-10)}{dt} = 100\,N$$

by solving $dt = 0.12\sec$

107. (a) $\vec{F} = \frac{\Delta\vec{p}}{\Delta t} \Rightarrow \Delta t = \frac{|\Delta\vec{p}|}{|\vec{F}|} = \frac{0.4}{2} = 0.2\,s$

108. (c) Rate of change of momentum of the bullet in forward direction = Force required to hold the gun.

$$F = nmv = 4\times 20\times 10^{-3}\times 300 = 24\,N$$

109. (d) Rate of flow of water $\frac{V}{t} = \frac{10\,cm^3}{\sec} = 10\times 10^{-6}\,\frac{m^3}{\sec}$

Density of water $\rho = \frac{10^3 kg}{m^3}$

Cross-sectional area of pipe $A = \pi(0.5\times 10^{-3})^2$

$$\text{Force} = m\frac{dv}{dt} = \frac{mv}{t} = \frac{V\rho v}{t} = \frac{\rho V}{t}\times\frac{V}{At} = \left(\frac{V}{t}\right)^2\frac{\rho}{A}$$

$$\left(\because v = \frac{V}{At}\right)$$

By substituting the value in the above formula we get

$F = 0.127\,N$

110. (a) Weight of the disc will be balanced by the force applied by the bullet on the disc in vertically upward direction.

$$F = nmv = 40\times 0.05\times 6 = Mg$$

$$\Rightarrow M = \frac{40\times 0.05\times 6}{10} = 1.2\,kg$$

111. (a)

112. (c) $F = \frac{dp}{dt} = v\left(\frac{dM}{dt}\right) = \alpha v^2 \therefore a = \frac{F}{M} = \frac{\alpha v^2}{M}$

113. (d) $P = \frac{F}{A} = \frac{n[mv - (-mv)]}{A} = \frac{2mnv}{A}$

$$= \frac{2\times 10^{-3}\times 10^4\times 10^2}{10^{-4}} = 2\times 10^7\,N/m^2$$

Third Law of Motion

1. (c) Swimming is a result of pushing water in the opposite direction of the motion.

2. (b) Because for every action there is an equal and opposite reaction takes place.

3. (b)

4. (a) The force exerted by the air of fan on the boat is internal and for motion external force is required.

5. (c)

6. (c)

7. (a) Up thrust on the body $= v\sigma g$. For freely falling body effective g becomes zero. So up thrust becomes zero

8. (d)

9. (c) Total weight in right hand = 10 + 1 = 1 *kg*

10. (c)

11. (a) For jumping he presses the spring platform, so the reading of spring balance increases first and finally it becomes zero.

12. (c) Gas will come out with sufficient speed in forward direction, so reaction of this forward force will change the reading of the spring balance.

13. (b)

14. (b) Since the cage is closed and we can treat bird, cage and the air as a closed (isolated) system. In this condition the force applied by the bird on cage is an internal force, due to this the reading of spring balance will not change.

15. (b) As the spring balance are massless therefore both the scales read M kg each.

16. (d) $F = mnv = 150\times10^{-3}\times20\times800 = 2400\ N$.

17. (c) $5N$ force will not produce any tension in spring without support of other $5N$ force. So here the tension in the spring will be $5N$ only.

18. (d) Since action and reaction acts in opposite direction on same line, hence angle between them is 180°.

19. (a)

20. (d) As by an internal force momentum of the system can not be changed.

21. (b)

22. (b) Since downward force along the inclined plane

$= mg\sin\theta = 5\times10\times\sin30° = 25N$

23. (c) At 11th second lift is moving upward with acceleration

$$a = \frac{0-3.6}{2} = -1.8 m/s^2$$

Tension in rope, $T = m(g-a)$

$= 1500(9.8-1.8) = 12000N$

24. (d) Distance travelled by the lift

$=$ Area under velocity time graph

$$= \left(\frac{1}{2}\times2\times3.6\right) + (8\times3.6) + \left(\frac{1}{2}\times2\times3.6\right) = 36m$$

Conservation of Linear Momentum and Impulse

1. (b)

2. (b) Force exerted by the ball on hands of the player

$$= \frac{mdv}{dt} = \frac{0.15\times20}{0.1} = 30\ N$$

3. (b) $F = u\left(\frac{dm}{dt}\right) = 500\times1 = 500\ N$

4. (c) If momentum remains constant then force will be zero because $F = \frac{dP}{dt}$

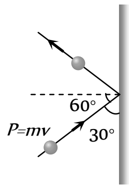
5. (c) According to principle of conservation of linear momentum $1000\times50 = 1250\times v \Rightarrow v = 40\, km/hr$

6. (a) Change in momentum = Impulse

$$\Rightarrow \Delta p = F\times\Delta t \Rightarrow \Delta t = \frac{\Delta p}{F} = \frac{125}{250} = 0.5\ \text{sec}$$

7. (a) During collision of ball with the wall horizontal momentum changes (vertical momentum remains constant)

$$\therefore F = \frac{\text{Change in horizontal momentum}}{\text{Time of contact}}$$



$$= \frac{2P\cos\theta}{0.1} = \frac{2mv\cos\theta}{0.1}$$

$$= \frac{2\times0.1\times10\times\cos60°}{0.1} = 10\ N$$

8. (c) Impulse = Force × time = $m\ a\ t$

$= 0.15\times20\times0.1 = 0.3\ N\text{-}s$

9. (b) For a given mass $P\propto v$. If the momentum is constant then it's velocity must have constant.

10. (c)

11. (c) $T = \frac{F(L-x)}{L} = \frac{5(5-1)}{5} = 4N$

12. (a)

13. (a) $F = u\left(\frac{dm}{dt}\right) = 3000\times4 = 12000\ N$

14. (b)

15. (c) It works on the principle of conservation of momentum.

16. (c) $v_G = \frac{m_B v_B}{m_G} = \frac{0.2\times5}{1} = 1\ m/s$

17. (a) By the conservation of linear momentum $m_B v_B = m_a v_a$

$$\Rightarrow v_G = \frac{m_B\times v_B}{m_G} = \frac{5\times10^{-3}\times500}{5} = 0.5\ m/s$$

18. (c) Impulse, $I = F\times\Delta t = 50\times10^{-5}\times3 = 1.5\times10^{-3}\ N\text{-}s$

19. (c) Momentum of one piece $= \frac{M}{4}\times3$

Momentum of the other piece $= \frac{M}{4}\times4$

$\therefore$ Resultant momentum $= \sqrt{\frac{9M^2}{16}+M^2} = \frac{5M}{4}$

The third piece should also have the same momentum. Let its velocity be v, then

$$\frac{5M}{4} = \frac{M}{2}\times v \text{ or } v = \frac{5}{2} = 2.5\ m/sec$$

20. (c)

21. (d) Using law of conservation of momentum, we get

$100\times v = 0.25\times100 \Rightarrow v = 0.25\ m/s$

22. (c) $F = 600 - 2\times10^5 t = 0 \Rightarrow t = 3\times10^{-3}$ sec

Impulse $I = \int_0^t F\,dt = \int_0^{3\times10^{-3}} (600 - 2\times10^3 t)dt$

$= [600t - 10^5 t^2]_0^{3\times10^{-3}} = 0.9\ N\times\text{sec}$

23. (a) According to principle of conservation of linear momentum, $m_G v_G = m_B v_B$

$$\Rightarrow v_G = \frac{m_B v_B}{m_G} = \frac{0.1\times10^2}{50} = 0.2 m/s$$

24. (d) $m_G v_G = m_B v_B \Rightarrow v_B = \dfrac{m_G v_G}{m_B} = \dfrac{1\times 5}{10\times 10^{-3}} = 500 m/s$

25. (d)

26. (b) The acceleration of a rocket is given by

$$a = \frac{v}{m}\left(\frac{\Delta m}{\Delta t}\right) - g = \frac{400}{100}\left(\frac{5}{1}\right) - 10$$

$$= (20-10) = 10\ m/s^2$$

27. (c)

Equilibrium of Forces

1. (d) Application of Bernoulli's theorem.

2. (c)

3. (b) $F = \sqrt{(F)^2 + (F)^2 + 2F.F\cos\theta} \Rightarrow \theta = 120°$

4. (a) $F_{net}^2 = F_1^2 + F_2^2 + 2F_1F_2\cos\theta$

$$\Rightarrow \left(\frac{F}{3}\right)^2 = F^2 + F^2 + 2F^2\cos\theta \Rightarrow \cos\theta = \left(-\frac{17}{18}\right)$$

5. (c) Direction of second force should be at 180°.

6. (c) $F_{max} = 5 + 10 = 15N$ and $F_{min} = 10 - 5 = 5N$

Range of resultant $5 \le F \le 15$

7. (b) $R^2 = (3P)^2 + (2P)^2 + 2\times 3P \times 2P \times \cos\theta$ ...(i)

$(2R)^2 = (6P)^2 + (2P)^2 + 2\times 6P \times 2P \times \cos\theta$ ...(ii)

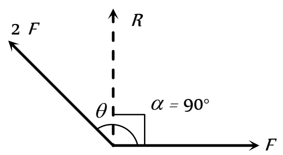
by solving (i) and (ii), $\cos\theta = -1/2 \Rightarrow \theta = 120°$

8. (b) $\tan\alpha = \dfrac{2F\sin\theta}{F + 2F\cos\theta} = \infty$ (as α = 90°)

$\Rightarrow F + 2F\cos\theta = 0$

$\Rightarrow \cos\theta = -\dfrac{1}{2}$

$\theta = 120°$



9. (b) $A + B = 18$ ...(i)

$12 = \sqrt{A^2 + B^2 + 2AB\cos\theta}$ ...(ii)

$\tan\alpha = \dfrac{B\sin\theta}{A + B\cos\theta} = \tan 90° \Rightarrow \cos\theta = -\dfrac{A}{B}$ ...(iii)

By solving (i), (ii) and (iii), $A = 13N$ and $B = 5N$

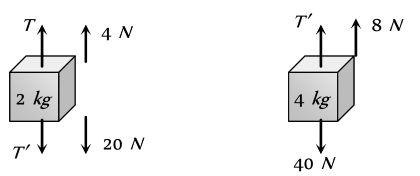
10. (c)

11. (d) Range of resultant of F_1 and F_2 varies between (3+5)=8N and $(5-3) = 2N$. It means for some value of angle (θ), resultant 6 can be obtained. So, the resultant of 3N, 5N and 6N may be zero and the forces may be in equilibrium.

12. (a) Net force on the particle is zero so the $\vec{v}$ remains unchanged.

13. (a) For equilibrium of forces, the resultant of two (smaller) forces should be equal and opposite to third one.

14. (a) FBD of mass 2 kg FBD of mass 4kg



$T - T' - 20 = 4$ ....(i) $T' - 40 = 8$ ...(ii)

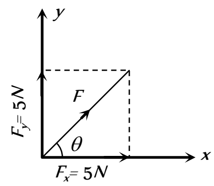
By solving (i) and (ii) $T' = 47.23\ N$ and $T = 70.8\ N$

15. (a)

16. (b) $|\vec{F}| = \sqrt{5^2 + 5^2} = 5\sqrt{2}\ N.$

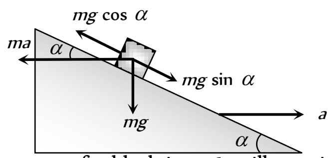
and $\tan\theta = \dfrac{5}{5} = 1$

$\Rightarrow \theta = \pi/4.$



17. (b)

18. (b)

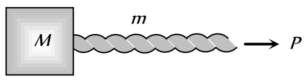


Let the mass of a block is m. It will remains stationary if forces acting on it are in equilibrium *i.e,* $ma\cos\alpha = mg\sin\alpha \Rightarrow a = g\tan\alpha$

Here ma = Pseudo force on block, mg = Weight.

Motion of Connected Bodies

1. (c)



Acceleration of the system $= \dfrac{P}{m+M}$

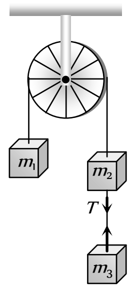
The force exerted by rope on the mass $= \dfrac{MP}{m+M}$

2. (b)

3. (b) Tension between m_2 and m_3 is given by

$$T = \frac{2m_1m_3}{m_1 + m_2 + m_3}\times g$$

$$= \frac{2\times 2\times 2}{2+2+2}\times 9.8 = 13\ N$$



4. (b) $a = \dfrac{m_2}{m_1 + m_2}\times g = \dfrac{5}{4+5}\times 9.8 = \dfrac{49}{9} = 5.44\ m/s^2$

5. (d) $T = \dfrac{2m_1m_2}{m_1 + m_2}g = \dfrac{2\times 2\times 3}{2+3}g = \dfrac{12}{5}g$

$$a = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)g = \left(\frac{3-2}{3+2}\right)g = \frac{g}{5}$$

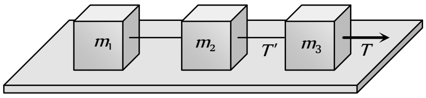
6. (b) $T_2 = (m_A + m_B)\times \dfrac{T_3}{m_A + m_B + m_C}$

$$T_2 = (1+8) \times \frac{36}{(1+8+27)} = 9\ N$$

7. (c) Acceleration $= \frac{(m_2 - m_1)}{(m_2 + m_1)} g$

$$= \frac{4-3}{4+3} \times 9.8 = \frac{9.8}{7} = 1.4\, m/\sec^2$$

8. (c)



$$T' = (m_1 + m_2) \times \frac{T}{m_1 + m_2 + m_3}$$

9. (d) $T_2 = (m_1 + m_2) \times \frac{T_3}{m_1 + m_2 + m_3} = \frac{(10+6) \times 40}{20} = 32\ N$

10. (a)

11. (a) Acceleration $= \frac{m_2}{m_1 + m_2} \times g = \frac{1}{2+1} \times 9.8 = 3.27\ m/s^2$

and $T = m_1 a = 2 \times 3.27 = 6.54\ N$

12. (d) $T = \frac{2m_1 m_2}{m_1 + m_2} g = \frac{2 \times 10 \times 6}{10+6} \times 9.8 = 73.5 N$

13. (c) $a = \frac{m_2 - m_1}{m_1 + m_2} g = \frac{10-5}{10+5} g = \frac{g}{3}$

14. (b) $a = \frac{m_2}{m_1 + m_2} g = \frac{3}{7+3} 10 = 3\ m/s^2$

15. (c) $T_1 = \left(\frac{m_2 + m_3}{m_1 + m_2 + m_3} \right) g = \frac{3+5}{2+3+5} \times 10 = 8 N$

16. (c) $a = \left(\frac{m_2 - m_1}{m_1 + m_2} \right) g = \left(\frac{10-6}{10+6} \right) \times 10 = 2.5\ m/s^2$

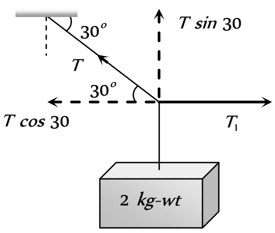
17. (c) $T \sin 30 = 2kg\ wt$

$\Rightarrow T = 4\ kg\ wt$

$T_1 = T \cos 30°$

$= 4 \cos 30°$

$= 2\sqrt{3}$



18. (c) If monkey move downward with acceleration a then its apparent weight decreases. In that condition

Tension in string $= m(g-a)$

This should not be exceed over breaking strength of the rope *i.e.* $360 \geq m(g-a) \Rightarrow 360 \geq 60(10-a)$

$\Rightarrow a \geq 4\ m/s^2$

19. (b) $a = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) g \Rightarrow \frac{g}{8} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) g \Rightarrow \frac{m_1}{m_2} = \frac{9}{7}$

20. (a) $a = \left[\frac{m_1 - m_2}{m_1 + m_2} \right] g = \left[\frac{5-4.8}{5+4.8} \right] \times 9.8 = 0.2\ m/s^2$

21. (c) As the spring balances are massless therefore the reading of both balance should be equal.

22. (a) $a = \left(\frac{m_2 - m_1}{m_1 + m_2} \right) g = \left(\frac{m - m/2}{m + m/2} \right) g = \frac{g}{3}$

23. (a) Acceleration of each mass $= a = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) g$

Now acceleration of centre of mass of the system

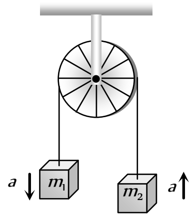
$$A_{cm} = \frac{m_1 \vec{a_1} + m_1 \vec{a_2}}{m_1 + m_2}$$

As both masses move with same acceleration but in opposite direction so $\vec{a_1} = -\vec{a_2} = a$ (let)

$$\therefore A_{cm} = \frac{m_1 a - m_2 a}{m_1 + m_2}$$

$$= \left(\frac{m_1 - m_2}{m_1 + m_2} \right) \times \left(\frac{m_1 - m_2}{m_1 + m_2} \right) \times g$$

$$= \left(\frac{m_1 - m_2}{m_1 + m_2} \right)^2 \times g$$



Critical Thinking Questions

1. (c) Due to acceleration in forward direction, vessel is an accelerated frame therefore a Pseudo force will be exerted in backward direction. Therefore water will be displaced in backward direction.

2. (b) The pressure on the rear side would be more due to fictitious force (acting in the opposite direction of acceleration) on the rear face. Consequently the pressure in the front side would be lowered.

3. (c) $v^2 = 2as = 2\left(\frac{F}{m} \right) s \qquad [As\ u = 0]$

$$\Rightarrow v^2 = 2\left(\frac{5 \times 10^4}{3 \times 10^7} \right) \times 3 = \frac{1}{100}$$

$\Rightarrow v = 0.1\ m/s$

4. (c) Mass measured by physical balance remains unaffected due to variation in acceleration due to gravity.

5. (c) For W, $2W$, $3W$ apparent weight will be zero because the system is falling freely. So the distances of the weight from the rod will be same.

6. (a) For equilibrium of system, $F_1 = \sqrt{F_2^2 + F_3^2}$ As $\theta = 90°$

In the absence of force F_1, Acceleration $= \frac{\text{Net force}}{\text{Mass}}$

$$= \frac{\sqrt{F_2^2 + F_3^2}}{m} = \frac{F_1}{m}$$

7. (b,c) Force of upthrust will be there on mass m shown in figure, so A weighs less than 2 kg. Balance will show sum of load of beaker and reaction of upthrust so it reads more than 5 kg.

8. (d) Heavier gas will acquire largest momentum *i.e.* Argon.

9. (c) $\vec{F}\Delta t = m\Delta\vec{v} \Rightarrow F = \dfrac{m\Delta\vec{v}}{t}$

By doing so time of change in momentum increases and impulsive force on knees decreases.

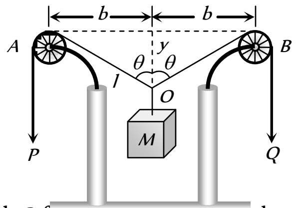
10. (b) When false balance has equal arms then, $W = \dfrac{X+Y}{2}$

11. (a) Let two vectors be $\vec{A}$ and $\vec{B}$ then $(\vec{A}+\vec{B}).(\vec{A}-\vec{B}) = 0$

$$\vec{A}.\vec{A} - \vec{B}.\vec{B} + \vec{B}.\vec{A} - \vec{B}.\vec{B} = 0$$

$$A^2 - B^2 = 0 \Rightarrow A^2 = B^2 \therefore A = B$$

12. (d)



As P and Q fall down, the length l decreases at the rate of U m/s.

From the figure, $l^2 = b^2 + y^2$

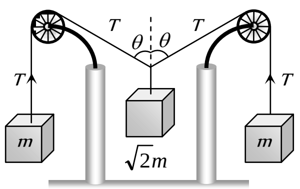
Differentiating with respect to time

$$2l \times \frac{dl}{dt} = 2b \times \frac{db}{dt} + 2y \times \frac{dy}{dt}\left(\text{As} \frac{db}{dt} = 0, \frac{dl}{dt} = U\right)$$

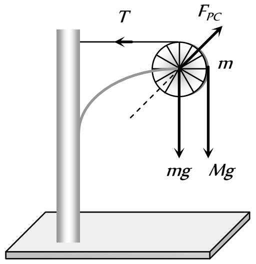
$$\Rightarrow \frac{dy}{dt} = \left(\frac{l}{y}\right) \times \frac{dl}{dt} \Rightarrow \frac{dy}{dt} = \left(\frac{1}{\cos\theta}\right) \times U = \frac{U}{\cos\theta}$$

13. (c) From the figure for the equilibrium of the system

$$2T\cos\theta = \sqrt{2}mg \Rightarrow \cos\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45°$$



14. (d) Force on the pulley by the clamp



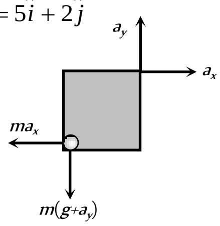
$$F_{pc} = \sqrt{T^2 + [(M+m)g]^2}$$

$$F_{pc} = \sqrt{(Mg)^2 + [(M+m)g]^2}$$

$$F_{pc} = \sqrt{M^2 + (M+m)^2}\, g$$

15. (b) $a_{cm} = \left(\dfrac{m_1 - m_2}{m_1 + m_2}\right)^2 g = \left(\dfrac{3m - m}{3m + m}\right)^2 g = \dfrac{g}{4}$

16. (c) As $\vec{v} = 5t\hat{i} + 2t\hat{j} \therefore \vec{a} = a_x\hat{i} + a_y\hat{j} = 5\hat{i} + 2\hat{j}$



$$\vec{F} = ma_x\hat{i} + m(g + a_y)\hat{j}$$

$$\therefore |\vec{F}| = m\sqrt{a_x^2 + (g + a_y)^2} = 26\ N$$

17. (c) $l = l_0\sqrt{1 - \dfrac{v^2}{c^2}} = 1\sqrt{1 - \left(\dfrac{2.7\times10^8}{3\times10^8}\right)^2} \Rightarrow l = 0.44\ m$

18. (c) $T = \dfrac{T_0}{[1 - (v^2/c^2)]^{1/2}}$

By substituting $T_0 = 1$ day and $T = 2$ days we get

$v = 2.6\times10^8\ ms^{-1}$

19. (d) Force acting on plate, $F = \dfrac{dp}{dt} = v\left(\dfrac{dm}{dt}\right)$

Mass of water reaching the plate per $sec = \dfrac{dm}{dt}$

$$= Av\rho = A(v_1 + v_2)\rho = \frac{V}{v_2}(v_1 + v_2)\rho$$

($v = v_1 + v_2 =$ velocity of water coming out of jet *w.r.t.* plate)

($A =$ Area of cross section of jet $= \dfrac{V}{v_2}$)

$$\therefore F = \frac{dm}{dt}v = \frac{V}{v_2}(v_1 + v_2)\rho \times (v_1 + v_2) = \rho\left[\frac{V}{v_2}\right](v_1 + v_2)^2$$

Graphical Questions

1. (d) If the applied force is less than limiting friction between block A and B, then whole system move with common acceleration

i.e. $a_A = a_B = \dfrac{F}{m_A + m_B}$

But the applied force increases with time, so when it becomes more than limiting friction between A and B, block B starts moving under the effect of net force $F - F_k$

Where $F_k =$ Kinetic friction between block A and B

$\therefore$ Acceleration of block B, $a_B = \dfrac{F - F_k}{m_B}$

As F is increasing with time so a_B will increase with time

Kinetic friction is the cause of motion of block A

$\therefore$ Acceleration of block A, $a_A = \dfrac{F_k}{m_A}$

It is clear that $a_B > a_A$. *i.e.* graph (d) correctly represents the variation in acceleration with time for block A and B.

2. (b) Velocity between $t = 0$ and $t = 2\,\text{sec}$

$$\Rightarrow v_i = \frac{dx}{dt} = \frac{4}{2} = 2\ m/s$$

Velocity at $t = 2\sec$, $v_f = 0$

Impulse = Change in momentum $= m(v_f - v_i)$

$= 0.1(0-2) = -0.2\ kg\ m\ \sec^{-1}$

3. (d) Momentum acquired by the particle is numerically equal to area enclosed between the *F-t* curve and time axis. For the given diagram area in upper half is positive and in lower half is negative (and equal to upper half), so net area is zero. Hence the momentum acquired by the particle will be zero.

4. (a,c) In region *AB* and *CD*, slope of the graph is constant *i.e.* velocity is constant. It means no force acting on the particle in this region.

5. (c) Impulse = Change in momentum $= m(v_2 - v_1)$ ....(i)

Again impulse = Area between the graph and time axis

$= \frac{1}{2}\times 2\times 4 + 2\times 4 + \frac{1}{2}(4+2.5)\times 0.5 + 2\times 2.5$

$= 4 + 8 + 1.625 + 5 = 18.625$ ....(ii)

From (i) and (ii), $m(v_2 - v_1) = 18.625$

$\Rightarrow v_2 = \frac{18.625}{m} + v_1 = \frac{18.625}{2} + 5 = 14.25\ m/s$

6. (d) $K = \frac{F}{x}$ and increment in length is proportional the original length *i.e.* $x \propto l$ $\therefore$ $K \propto \frac{1}{l}$

It means graph between *K* and *l* should be hyperbolic in nature.

7. (b) In elastic one dimensional collision particle rebounds with same speed in opposite direction

i.e. change in momentum $= 2mu$

But Impulse $= F \times T =$ Change in momentum

$\Rightarrow F_0 \times T = 2mu \Rightarrow F_0 = \frac{2mu}{T}$

8. (c) Initially particle was at rest. By the application of force its momentum increases.

Final momentum of the particle = Area of *F - t* graph

$\Rightarrow mu =$ Area of semi circle

$mu = \frac{\pi r^2}{2} = \frac{\pi r_1 r_2}{2} = \frac{\pi (F_0)(T/2)}{2} \Rightarrow u = \frac{\pi F_0 T}{4m}$

9. (d) momentum acquired = Area of force-time graph

$= \frac{1}{2}\times(2)\times(10) + 4\times 10 = 10 + 40 = 50\ N\text{-}S$

10. (c) $F = \frac{dp}{dt}$, so the force is maximum when slope of graph is maximum

11. (c) Impulse = Area between force and time graph and it is maximum for graph (III) and (IV)

Assertion and Reason

1. (e) Inertia is the property by virtue of which the body is unable to change by itself not only the state of rest, but also the state of motion.

2. (c) According to Newton's second law

Acceleration = $\frac{\text{Force}}{\text{Mass}}$ *i.e.* if net external force on the body is zero then acceleration will be zero

3. (a) According to second law $F = \frac{dp}{dt} = ma$.

If we know the values of *m* and *a*, the force acting on the body can be calculated and hence second law gives that how much force is applied on the body.

4. (b) When a body is moving in a circle, its speed remains same but velocity changes due to change in the direction of motion of body. According to first law of motion, force is required to change the state of a body. As in circular motion the direction of velocity of body is changing so the acceleration cannot be zero. But for a uniform motion acceleration is zero (for rectilinear motion).

5. (c) According to definition of momentum

$P = mv$ if P = constant then mv = constant or $v \propto \frac{1}{m}$.

As velocity is inversely proportional to mass, therefore lighter body possess greater velocity.

6. (a) The wings of the aeroplane pushes the external air backward and the aeroplane move forward by reaction of pushed air. At low altitudes. density of air is high and so the aeroplane gets sufficient force to move forward.

7. (c) Force is required to change the state of the body. In uniform motion body moves with constant speed so acceleration should be zero.

8. (a) According to Newton's second law of motion $a = \frac{F}{m}$ *i.e.* magnitude of the acceleration produced by a given force is inversely proportional to the mass of the body. Higher is the mass of the body, lesser will be the acceleration produced *i.e.* mass of the body is a measure of the opposition offered by the body to change a state, when the force is applied *i.e.* mass of a body is the measure of its inertia.

9. (d) $F = \frac{dp}{dt}$ = Slope of momentum-time graph

i.e. Rate of change of momentum = Slope of momentum- time graph = force.

10. (c) The purpose of bending is to acquire centripetal force for circular motion. By doing so component of normal reaction will counter balance the centrifugal force.

11. (c) Work done in moving an object against gravitational force (conservative force) depends only on the initial and final position of the object, not upon the path taken. But gravitational force on the body along the inclined plane is not same as that along the vertical and it varies with the angle of inclination.

12. (b) In uniform circular motion of a body the speed remains constant but velocity changes as direction of motion changes.

As linear momentum = mass × velocity, therefore linear momentum of a body changes in a circle.

On the other hand, if the body is moving uniformly along a straight line then its velocity remains constant and hence acceleration is equal to zero. So force is equal to zero.

13. (d) Law of conservation of linear momentum is correct when no external force acts . When bullet is fired from a rifle then both should possess equal momentum but different kinetic energy. E $=\frac{P^2}{2m}$ $\therefore$ Kinetic energy of the rifle is less than that of bullet because $E \propto 1/m$

14. (a) As the fuel in rocket undergoes combustion, the gases so produced leave the body of the rocket with large velocity and give upthrust to the rocket. If we assume that the fuel is burnt at a constant rate, then the rate of change of momentum of the rocket will be constant. As more and more fuel gets burnt, the mass of the rocket goes on decreasing and it leads to increase of the velocity of rocket more and more rapidly.

15. (c) The apparent weight of a body in an elevator moving with downward acceleration a is given by $W = m(g - a)$.

16. (e) For uniform motion apparent weight = Actual weight
For downward accelerated motion,
Apparent weight < Actual weight

17. (a)

18. (a) By lowering his hand player increases the time of catch, by doing so he experience less force on his hand because $F \propto 1/dt$.

19. (b) According to Newton's second law,
$F = ma \Rightarrow a = F/m$
For constant F, acceleration is inversely proportional to mass *i.e.* acceleration produced by a force depends only upon the mass of the body and for larger mass acceleration will be less.

20. (c) In uniform circular motion, the direction of motion changes, therefore velocity changes.
As $P = mv$ therefore momentum of a body also changes in uniform circular motion.

21. (e) According to third law of motion it is impossible to have a single force out of mutual interaction between two bodies, whether they are moving or at rest. While, Newton's third law is applicable for all types of forces.

22. (d) An inertial frame of reference is one which has zero acceleration and in which law of inertia hold good i.e. Newton's law of motion are applicable equally. Since earth is revolving around the sun and earth is rotating about its own axis also, the forces are acting on the earth and hence there will be acceleration of earth due to these factors. That is why earth cannot be taken as inertial frame of reference.

23. (b) According to law of inertia (Newton's first law), when cloth is pulled from a table, the cloth come in state of motion but dishes remains stationary due to inertia. Therefore when we pull the cloth from table the dishes remains stationary.

24. (e) A body subjected to three concurrent forces is found to in equilibrium if sum of these force is equal to zero.
i.e. $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + ... = 0.$

25. (e) From Newton's second law
Impulse = Change of momentum.
So they have equal dimensions

Newton's Laws of Motion

SET Self Evaluation Test -4

1. A car is moving with uniform velocity on a rough horizontal road. Therefore, according to Newton's first law of motion

(a) No force is being applied by its engine

(b) A force is surely being applied by its engine

(c) An acceleration is being produced in the car

(d) The kinetic energy of the car is increasing

2. A person is sitting in a travelling train and facing the engine. He tosses up a coin and the coin falls behind him. It can be concluded that the train is [SCRA 1994]

(a) Moving forward and gaining speed

(b) Moving forward and losing speed

(c) Moving forward with uniform speed

(d) Moving backward with uniform speed

3. A block can slide on a smooth inclined plane of inclination θ kept on the floor of a lift. When the lift is descending with a retardation *a*, the acceleration of the block relative to the incline is

(a) $(g+a)\sin\theta$ (b) $(g-a)$

(c) $g\sin\theta$ (d) $(g-a)\sin\theta$

4. A 60 *kg* man stands on a spring scale in the lift. At some instant he finds, scale reading has changed from 60 *kg* to 50*kg* for a while and then comes back to the original mark. What should we conclude ?

(a) The lift was in constant motion upwards

(b) The lift was in constant motion downwards

(c) The lift while in constant motion upwards, is stopped suddenly

(d) The lift while in constant motion downwards, is suddenly stopped

5. When a body is acted by a constant force, then which of the following quantities remains constant

(a) Velocity (b) Acceleration

(c) Momentum (d) None of these

6. A man of weight *mg* is moving up in a rocket with acceleration 4 *g*. The apparent weight of the man in the rocket is

(a) Zero (b) 4 *mg*

(c) 5 *mg* (d) *mg*

7. A spring balance and a physical balance are kept in a lift. In these balances equal masses are placed. If now the lift starts moving upwards with constant acceleration, then

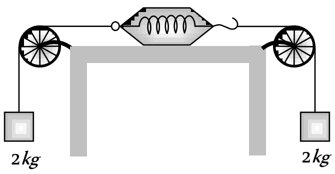
(a) The reading of spring balance will increase and the equilibrium position of the physical balance will disturb

(b) The reading of spring balance will remain unchanged and physical balance will remain in equilibrium

(c) The reading of spring balance will decrease and physical balance will remain in equilibrium

(d) The reading of spring balance will increase and the physical balance will remain in equilibrium

8. As shown in the figure, two equal masses each of 2 *kg* are suspended from a spring balance. The reading of the spring balance will be



(a) Zero

(b) 2 *kg*

(c) 4 *kg*

(d) Between zero and 2 *kg*

9. A player kicks a football of mass 0.5 *kg* and the football begins to move with a velocity of 10 *m/s*. If the contact between the leg and the football lasts for $\frac{1}{50}$ *sec*, then the force acted on the football should be

(a) 2500 *N* (b) 1250 *N*

(c) 250 *N* (d) 625 *N*

10. The engine of a jet aircraft applies a thrust force of 10^5 *N* during take off and causes the plane to attain a velocity of 1 *km/sec* in 10 *sec*. The mass of the plane is

(a) 10^2 *kg* (b) 10^3 *kg*

(c) 10^4 *kg* (d) 10^5 *kg*

11. A force of 50 *dynes* is acted on a body of mass 5 *g* which is at rest for an interval of 3 seconds, then impulse is [AFMC 1998]

(a) 0.15×10^{-3} *N-s* (b) 0.98×10^{-3} *N-s*

(c) 1.5×10^{-3} *N-s* (d) 2.5×10^{-3} *N-s*

12. Two weights w_1 and w_2 are suspended from the ends of a light string passing over a smooth fixed pulley. If the pulley is pulled up at an acceleration *g*, the tension in the string will be

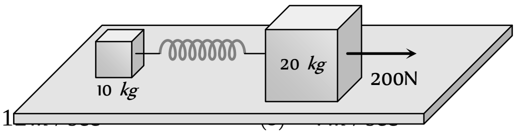
(a) $\frac{4w_1w_2}{w_1+w_2}$

(b) $\frac{2w_1w_2}{w_1+w_2}$

(c) $\frac{w_1w_2}{w_1+w_2}$

(d) $\dfrac{w_1 w_2}{2(w_1 + w_2)}$

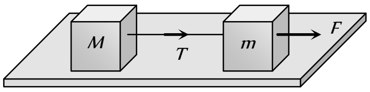
13. The masses of 10 *kg* and 20 *kg* respectively are connected by a massless spring as shown in figure. A force of 200 *N* acts on the 20 *kg* mass. At the instant shown, the 10 *kg* mass has acceleration $12\,m/\sec^2$. What is the acceleration of 20 *kg* mass



(a) 1 (b)

(c) $10\,m/\sec^2$ (d) Zero

14. Two masses *M* and *m* are connected by a weightless string. They are pulled by a force *F* on a frictionless horizontal surface. The tension in the string will be



(a) $\dfrac{FM}{m+M}$ (b) $\dfrac{F}{M+m}$

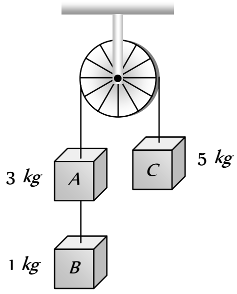
(c) $\dfrac{FM}{m}$ (d) $\dfrac{Fm}{M+m}$

15. In the above question, the acceleration of mass *m* is

(a) $\dfrac{F}{m}$ (b) $\dfrac{F-T}{m}$

(c) $\dfrac{F+T}{m}$ (d) $\dfrac{F}{M}$

16. Three weight *A*, *B* and *C* are connected by string as shown in the figure. The system moves over a frictionless pulley. The tension in the string connecting *A* and *B* is (where *g* is acceleration due to gravity)



(a) g

(b) $\dfrac{g}{9}$

(c) $\dfrac{8g}{9}$

(d) $\dfrac{10g}{9}$

Answers and Solutions

(SET –4)

1. (b) Since, force needed to overcome frictional force.

2. (a) The coin falls behind him it means the velocity of train was increasing otherwise the coin fall directly into the hands of thrower.

3. (a) Acceleration of block in a stationary lift = $g\sin\theta$

If lift is descending with acc. then it will be $(g-a)\sin\theta$.

but in the problem acceleration = – *a* (retardation)

$\therefore$ Acceleration of block = $[g-(-a)]\sin\theta$ = $(g+a)\sin\theta$

4. (c) For upward acceleration apparent weight $= m(g+a)$

If lift suddenly stops during upward motion then apparent weight = $m(g-a)$ because instead of acceleration, we will consider retardation.

In the problem it is given that scale reading initially was 60 *kg* and due to sudden jerk reading decreasing and finally comes back to the original mark *i.e.,* 60 *kg*.

So, we can conclude that lift was moving upward with constant speed and suddenly stops.

5. (b) $F = ma$ for a given body if $F =$ constant then a = constant.

6. (c) $R = m(g+a) = m(g+4g) = 5mg$

7. (d) The fictitious force will act downwards. So the reading of spring balance will increase. In case of physical balance, the fictitious force will act on both the pans, so the equilibrium is not affected.

8. (b) In this case, one 2 *kg wt* on the left will act as the support for the spring balance. Hence its reading will be 2 *kg*.

9. (c) Force on the football $F = m\dfrac{dv}{dt}$

$$F = \frac{m(v_2 - v_1)}{dt} = \frac{0.5 \times (10-0)}{1/50} = 250N.$$

10. (b) Acceleration produced in jet = $\dfrac{\text{Change in velocity}}{\text{Time}}$

$$a = \frac{(10^3 - 0)}{10} = 100 m/s^2$$

$$\therefore \text{ Mass} = \frac{\text{Force}}{\text{Acceleration}} = \frac{10^5}{10^2} = 10^3 kg.$$

11. (c) Impulse = Force × Time = $50 \times 10^{-5} \times 3$

$= 1.5 \times 10^{-3}$ *N-s*

12. (a) $T = \dfrac{2m_1 m_2}{(m_1+m_2)}(g+a) = \dfrac{2m_1 m_2 (g+g)}{m_1 + m_2}$

$$\Rightarrow T = \frac{4m_1 m_2}{m_1 + m_2} g = \frac{4w_1 w_2}{w_1 + w_2}$$

13. (b) As the mass of 10 *kg* has acceleration 12 *m/s* therefore it apply 120*N* force on mass 20*kg* in a backward direction.

$\therefore$ Net forward force on 20 *kg* mass = 200 – 120 = 80*N*

$$\therefore \text{ Acceleration } = \frac{80}{20} = 4 m/s^2.$$

14. (a) $T = M \times a = M \times \left(\dfrac{F}{m+M}\right)$

15. (b) Net force on mass *m*, $ma = F - T \therefore a = \dfrac{F-T}{m}$

16. (d) $T = \dfrac{2 \times m_B m_C}{m_A + m_B + m_C} \times g = \dfrac{2 \times 1 \times 5}{3+1+5} \times g = \dfrac{10}{9} g.$

***

Chapter 5

Friction

Introduction

If we slide or try to slide a body over a surface, the motion is resisted by a bonding between the body and the surface. This resistance is represented by a single force and is called friction force.

The force of friction is parallel to the surface and opposite to the direction of intended motion.

Types of Friction

(1) **Static friction :** The opposing force that comes into play when one body tends to move over the surface of another, but the actual motion has yet not started is called static friction.

(i) If applied force is P and the body remains at rest then static friction F = P.

(ii) If a body is at rest and no pulling force is acting on it, force of friction on it is zero.

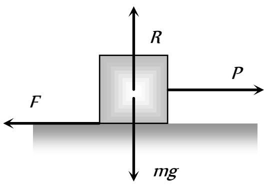


Fig. 5.1

(iii) Static friction is a self-adjusting force because it changes itself in accordance with the applied force and is always equal to net external force.

(2) **Limiting friction :** If the applied force is increased, the force of static friction also increases. If the applied force exceeds a certain (maximum) value, the body starts moving. This maximum value of static friction upto which body does not move is called limiting friction.

(i) The magnitude of limiting friction between any two bodies in contact is directly proportional to the normal reaction between them.

$F_l \propto R$ or $F_l = \mu_s R$

(ii) Direction of the force of limiting friction is always opposite to the direction in which one body is at the verge of moving over the other

(iii) Coefficient of static friction : (a) μ_s is called coefficient of static friction and is defined as the ratio of force of limiting friction and normal reaction $\mu_s = \dfrac{F}{R}$

(b) Dimension : $[M^0L^0T^0]$

(c) Unit : It has no unit.

(d) Value of μ depends on material and nature of surfaces in contact that means whether dry or wet ; rough or smooth polished or non-polished.

(e) Value of μ does not depend upon apparent area of contact.

(3) **Kinetic or dynamic friction :** If the applied force is increased further and sets the body in motion, the friction opposing the motion is called kinetic friction.

(i) Kinetic friction depends upon the normal reaction.

$F_k \propto R$ or $F_k = \mu_k R$ where μ_k is called the coefficient of kinetic friction

(ii) Value of μ_k depends upon the nature of surface in contact.

(iii) Kinetic friction is always lesser than limiting friction $F_k < F_l$

$\therefore \; \mu_k < \mu_s$

i.e. coefficient of kinetic friction is always less than coefficient of static friction. Thus we require more force to start a motion than to maintain it against friction. This is because once the motion starts actually ; inertia of rest has been overcome. Also when motion has actually started, irregularities of one surface have little time to get locked again into the irregularities of the other surface.

(iv) Kinetic friction does not depend upon the velocity of the body.

(v) Types of kinetic friction

(a) **Sliding friction** : The opposing force that comes into play when one body is actually sliding over the surface of the other body is called sliding friction. *e.g.* A flat block is moving over a horizontal table.

(b) **Rolling friction** : When objects such as a wheel (disc or ring), sphere or a cylinder rolls over a surface, the force of friction that comes into play is called rolling friction.

❑ Rolling friction is directly proportional to the normal reaction (*R*) and inversely proportional to the radius (*r*) of the rolling cylinder or wheel.

$$F_{rolling} = \mu_r \frac{R}{r}$$

μ_r is called coefficient of rolling friction. It would have the dimensions of length and would be measured in *metre*.

❑ Rolling friction is often quite small as compared to the sliding friction. That is why heavy loads are transported by placing them on carts with wheels.

❑ In rolling the surfaces at contact do not rub each other.

❑ The velocity of point of contact with respect to the surface remains zero all the times although the centre of the wheel moves forward.

Graph Between Applied Force and Force of Friction

(1) Part OA of the curve represents static friction (F_s). Its value increases linearly with the applied force

(2) At point A the static friction is maximum. This represent limiting friction (F_l).

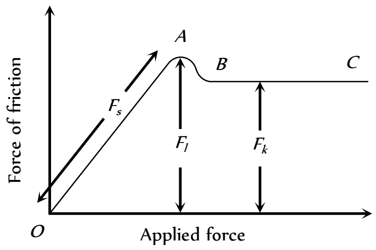


Fig. 5.2

(3) Beyond A, the force of friction is seen to decrease slightly. The portion BC of the curve represents the kinetic friction (F_k).

(4) As the portion BC of the curve is parallel to x-axis therefore kinetic friction does not change with the applied force, it remains constant, whatever be the applied force.

Friction is a Cause of Motion

It is a general misconception that friction always opposes the motion. No doubt friction opposes the motion of a moving body but in many cases it is also the cause of motion. For example :

(1) While moving, a person or vehicle pushes the ground backwards (action) and the rough surface of ground reacts and exerts a forward force due to friction which causes the motion. If there had been no friction there will be slipping and no motion.

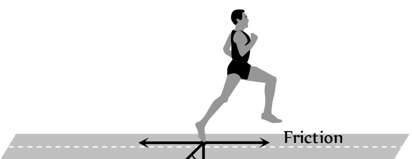
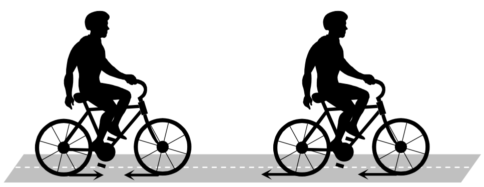


Fig. 5.3

(2) During cycling, the rear wheel moves by the force communicated to it by pedalling while front wheel moves by itself. So, when pedalling a bicycle, the force exerted by rear wheel on ground makes force of friction act on it in the forward direction (like walking). Front wheel moving by itself experience force of friction in backward direction (like rolling of a ball). [However, if pedalling is stopped both wheels move by themselves and so experience force of friction in backward direction].



While pedalling Pedalling is stoped

Fig. 5.4

(3) If a body is placed in a vehicle which is accelerating, the force of friction is the cause of motion of the body along with the vehicle (*i.e.*, the body will remain at rest in the accelerating vehicle until $ma < \mu_s mg$). If there had been no friction between body and vehicle, the body will not move along with the vehicle.

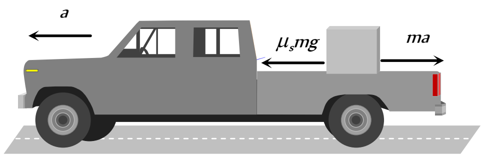


Fig. 5.5

From these examples it is clear that without friction motion cannot be started, stopped or transferred from one body to the other.

Advantages and Disadvantages of Friction

(1) **Advantages of friction**

(i) Walking is possible due to friction.

(ii) Two body sticks together due to friction.

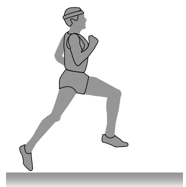


Fig. 5.6

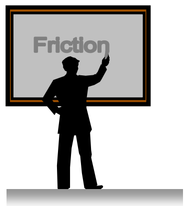


Fig. 5.7

(iii) Brake works on the basis of friction.

(iv) Writing is not possible without friction.

(v) The transfer of motion from one part of a machine to other part through belts is possible by friction.

(2) **Disadvantages of friction**

(i) Friction always opposes the relative motion between any two bodies in contact. Therefore extra energy has to be spent in over coming friction. This reduces the efficiency of machine.

(ii) Friction causes wear and tear of the parts of machinery in contact. Thus their lifetime reduces.

(iii) Frictional force result in the production of heat, which causes damage to the machinery.

Methods of Changing Friction

We can reduce friction

(1) By polishing.

(2) By lubrication.

(3) By proper selection of material.

(4) By streamlining the shape of the body.

(5) By using ball bearing.

Also we can increase friction by throwing some sand on slippery ground. In the manufacturing of tyres, synthetic rubber is preferred because its coefficient of friction with the road is larger.

Angle of Friction

Angle of friction may be defined as the angle which the resultant of limiting friction and normal reaction makes with the normal reaction.

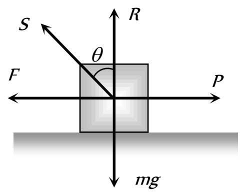


Fig. 5.8

By definition angle θ is called the angle of friction

$$\tan\theta = \frac{F_l}{R}$$

$\therefore$ $\tan\theta = \mu_s$ [As we know $\frac{F_l}{R} = \mu_s$]

or $\theta = \tan^{-1}(\mu_L)$

Hence coefficient of static friction is equal to tangent of the angle of friction.

Resultant Force Exerted by Surface on Block

In the above figure resultant force $S = \sqrt{F^2 + R^2}$

$$S = \sqrt{(\mu mg)^2 + (mg)^2}$$

$$S = mg\sqrt{\mu^2 + 1}$$

when there is no friction $(\mu = 0)$ S will be minimum

i.e. $S = mg$

Hence the range of S can be given by,

$$mg \le S \le mg\sqrt{\mu^2 + 1}$$

Angle of Repose

Angle of repose is defined as the angle of the inclined plane with horizontal such that a body placed on it is just begins to slide.

By definition, α is called the angle of repose.

In limiting condition $F = mg\sin\alpha$ and $R = mg\cos\alpha$

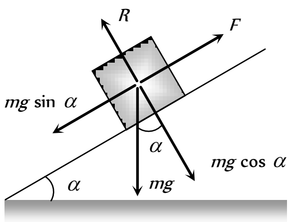


Fig. 5.9

So $\frac{F}{R} = \tan\alpha$

$\therefore$ $\frac{F}{R} = \mu_s = \tan\theta = \tan\alpha$ [As we know $\frac{F}{R} = \mu_s = \tan\theta$]

Thus the coefficient of limiting friction is equal to the tangent of angle of repose.

As well as $\alpha = \theta$ *i.e.* angle of repose = angle of friction.

Calculation of Required Force in Different Situation

If W = weight of the body, θ = angle of friction, $\mu = \tan\theta =$ coefficient of friction

Then we can calculate required force for different situation in the following manner :

(1) **Minimum pulling force P at an angle α from the horizontal**

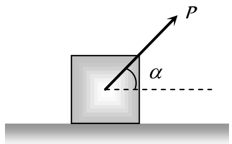


Fig. 5.10

By resolving P in horizontal and vertical direction (as shown in figure)

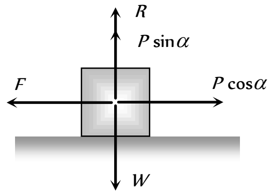


Fig. 5.11

For the condition of equilibrium

$F = P\cos\alpha$ and $R = W - P\sin\alpha$

By substituting these value in $F = \mu R$

$P\cos\alpha = \mu(W - P\sin\alpha)$

$\Rightarrow$ $P\cos\alpha = \frac{\sin\theta}{\cos\theta}(W - P\sin\alpha)$ [As $\mu = \tan\theta$]

$\Rightarrow$ $P = \frac{W\sin\theta}{\cos(\alpha - \theta)}$

(2) **Minimum pushing force P at an angle α from the horizontal**

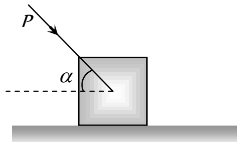


Fig. 5.12

By Resolving P in horizontal and vertical direction (as shown in the figure)

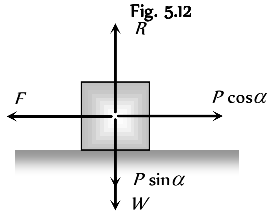


Fig. 5.13

For the condition of equilibrium

$F = P\cos\alpha$ and $R = W + P\sin\alpha$

By substituting these value in $F = \mu R$

$\Rightarrow$ $P\cos\alpha = \mu(W + P\sin\alpha)$

$\Rightarrow$ $P\cos\alpha = \frac{\sin\theta}{\cos\theta}(W + P\sin\alpha)$ [As $\mu = \tan\theta$]

$\Rightarrow$ $P = \frac{W\sin\theta}{\cos(\alpha + \theta)}$

(3) **Minimum pulling force P to move the body up on an inclined plane**

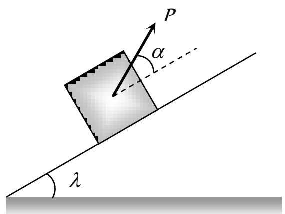


Fig. 5.14

By Resolving P in the direction of the plane and perpendicular to the plane (as shown in the figure)

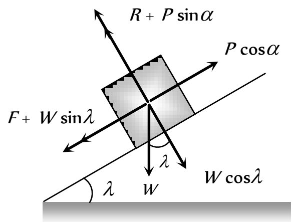


Fig. 5.15

For the condition of equilibrium

$R + P\sin\alpha = W\cos\lambda$

$\therefore \quad R = W\cos\lambda - P\sin\alpha$ and $F + W\sin\lambda = P\cos\alpha$

$\therefore \quad F = P\cos\alpha - W\sin\lambda$

By substituting these values in $F = \mu R$ and solving we get

$$P = \frac{W\sin(\theta + \lambda)}{\cos(\alpha - \theta)}$$

(4) **Minimum force to move a body in downward direction along the surface of inclined plane**

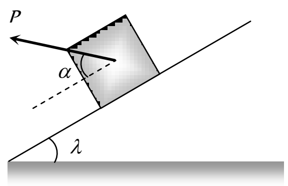


Fig. 5.16

By Resolving P in the direction of the plane and perpendicular to the plane (as shown in the figure)

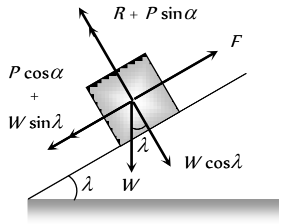


Fig. 5.17

For the condition of equilibrium

$R + P\sin\alpha = W\cos\lambda$

$\therefore \ R = W\cos\lambda - P\sin\alpha$ and $F = P\cos\alpha + W\sin\lambda$

By substituting these values in $F = \mu R$ and solving we get

$$P = \frac{W\sin(\theta - \lambda)}{\cos(\alpha - \theta)}$$

(5) **Minimum force to avoid sliding of a body down on an inclined plane**

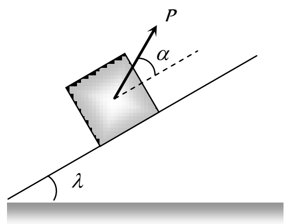


Fig. 5.18

By Resolving P in the direction of the plane and perpendicular to the plane (as shown in the figure)

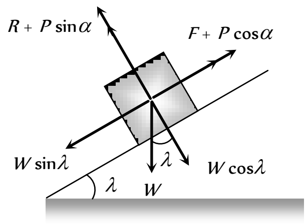


Fig. 5.19

For the condition of equilibrium

$R + P\sin\alpha = W\cos\lambda$

$\therefore \ R = W\cos\lambda - P\sin\alpha$ and $P\cos\alpha + F = W\sin\lambda$

$\therefore \ F = W\sin\lambda - P\cos\alpha$

By substituting these values in $F = \mu R$ and solving we get

$$P = W\left[\frac{\sin(\lambda - \theta)}{\cos(\theta + \alpha)}\right]$$

(6) **Minimum force for motion along horizontal surface and its direction**

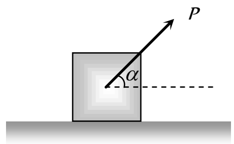


Fig. 5.20

Let the force P be applied at an angle α with the horizontal.

By resolving P in horizontal and vertical direction (as shown in figure)

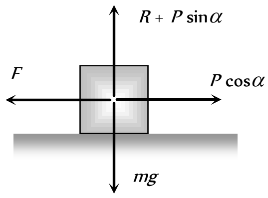


Fig. 5.21

For vertical equilibrium

$R + P\sin\alpha = mg$

$\therefore \ R = mg - P\sin\alpha$ ...(i)

and for horizontal motion

$P\cos\alpha \geq F$

i.e. $P\cos\alpha \geq \mu R$ ...(ii)

Substituting value of R from (i) in (ii)

$P\cos\alpha \geq \mu(mg - P\sin\alpha)$

$$P \geq \frac{\mu mg}{\cos\alpha + \mu\sin\alpha} \quad \text{...(iii)}$$

For the force P to be minimum $(\cos\alpha + \mu\sin\alpha)$ must be maximum *i.e.*

$\frac{d}{d\alpha}[\cos\alpha + \mu\sin\alpha] = 0$

$\Rightarrow \ -\sin\alpha + \mu\cos\alpha = 0$

$\therefore \quad \tan\alpha = \mu$

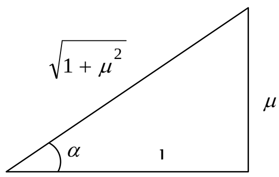


Fig. 5.22

or $\quad \alpha = \tan^{-1}(\mu) = \text{angle of friction}$

i.e. For minimum value of P its angle from the horizontal should be equal to angle of friction

As $\tan\alpha = \mu$ so from the figure, $\sin\alpha = \frac{\mu}{\sqrt{1+\mu^2}}$

and $\cos\alpha = \frac{1}{\sqrt{1+\mu^2}}$

By substituting these value in equation (iii)

$$P \geq \frac{\mu mg}{\frac{1}{\sqrt{1+\mu^2}} + \frac{\mu^2}{\sqrt{1+\mu^2}}} \geq \frac{\mu mg}{\sqrt{1+\mu^2}}$$

$$\therefore \quad P_{\min} = \frac{\mu mg}{\sqrt{1+\mu^2}}$$

Acceleration of a Block Against Friction

(1) **Acceleration of a block on horizontal surface**

When body is moving under application of force *P*, then kinetic friction opposes its motion.

Let a is the net acceleration of the body

From the figure

$$ma = P - F_k$$

$$\therefore \ a = \frac{P - F_k}{m}$$

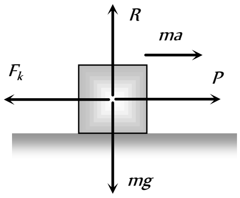


Fig. 5.23

(2) **Acceleration of a block sliding down over a rough inclined plane**

When angle of inclined plane is more than angle of repose, the body placed on the inclined plane slides down with an acceleration *a*.

From the figure $ma = mg\sin\theta - F$

$$\Rightarrow \ ma = mg\sin\theta - \mu R$$

$$\Rightarrow \ ma = mg\sin\theta - \mu mg\cos\theta$$

$\therefore$ Acceleration $a = g[\sin\theta - \mu\cos\theta]$

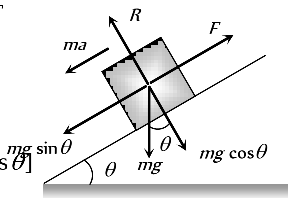


Fig. 5.24

Note : ❑ For frictionless inclined plane $\mu = 0 \ \therefore a = g\sin\theta$.

(3) **Retardation of a block sliding up over a rough inclined plane**

When angle of inclined plane is less than angle of repose, then for the upward motion

$$ma = mg\sin\theta + F$$

$$ma = mg\sin\theta + \mu mg\cos\theta$$

Retardation $a = g[\sin\theta + \mu\cos\theta]$

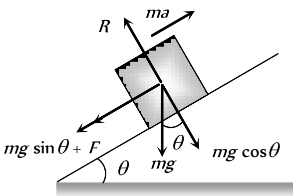


Fig. 5.25

Note : ❑ For frictionless inclined plane $\mu = 0 \ \therefore a = g\sin\theta$

Work done against friction

(1) **Work done over a rough inclined surface**

If a body of mass *m* is moved up slowly on a rough inclined plane through distance *s*, then

Work done = force × distance

= *ma* × *s* = *mg* $[\sin\theta + \mu\cos\theta]s = mg\,s[\sin\theta + \mu\cos\theta]$

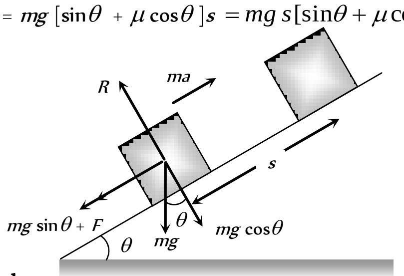


Fig. 5.26

(2) **Work done over a horizontal surface**

In the above expression if we put $\theta = 0$ then

Work done = force × distance = $F \times s = \mu\, mg\, s$

It is clear that work done depends upon

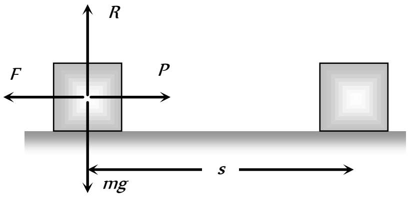


Fig. 5.27

(i) Weight of the body.

(ii) Material and nature of surface in contact.

(iii) Distance moved.

Motion of Two Bodies one Resting on the Other

When a body *A* of mass *m* is resting on a body *B* of mass *M* then two conditions are possible

(1) A force *F* is applied to the upper body, (2) A force *F* is applied to the lower body

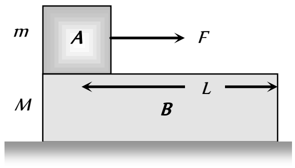


Fig. 5.28

We will discuss above two cases one by one in the following manner :

(1) **A force *F* is applied to the upper body, then following four situations are possible**

(i) **When there is no friction**

(a) The body *A* will move on body *B* with acceleration (*F*/*m*).

$$a_A = F/m$$

(b) The body *B* will remain at rest

$$a_B = 0$$

(c) If *L* is the length of *B* as shown in figure, *A* will fall from *B* after time *t*

$$t = \sqrt{\frac{2L}{a}} = \sqrt{\frac{2mL}{F}} \qquad \left[\text{As } s = \frac{1}{2}at^2 \text{ and } a = F/m\right]$$

(ii) **If friction is present between *A* and *B* only and applied force is less than limiting friction (*F* < F_l)**

(*F* = Applied force on the upper body, F_l = limiting friction between *A* and *B*, F_k = Kinetic friction between *A* and *B*)

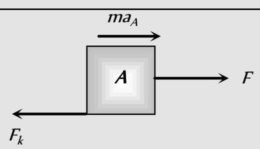
(a) The body *A* will not slide on body *B* till $F < F_l$ *i.e.* $F < \mu_s mg$

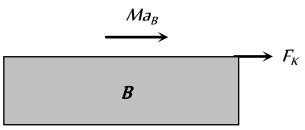
(b) Combined system (*m* + *M*) will move together with common acceleration $a_A = a_B = \dfrac{F}{M+m}$

(iii) **If friction is present between *A* and *B* only and applied force is greater than limiting friction (*F* > F_l)**

In this condition the two bodies will move in the same direction (*i.e.* of applied force) but with different acceleration. Here force of kinetic friction $\mu_k mg$ will oppose the motion of *A* while cause the motion of *B*.

$F - F_k = m a_A$	Free body diagram of *A*



i.e.	$a_A = \dfrac{F - F_k}{m}$	
	$a_A = \dfrac{(F - \mu_k mg)}{m}$	
	$F_k = M a_B$	Free body diagram of B
i.e.	$a_B = \dfrac{F_k}{M}$	
$\therefore$	$a_B = \dfrac{\mu_k mg}{M}$	

Note : ❑ As both the bodies are moving in the same direction.

Acceleration of body A relative to B will be

$$a = a_A - a_B = \frac{MF - \mu_k mg\,(m+M)}{mM}$$

So, A will fall from B after time

$$t = \sqrt{\frac{2L}{a}} = \sqrt{\frac{2m\,ML}{MF - \mu_k mg\,(m+M)}}$$

(iv) **If there is friction between B and floor**

(where $F_l' = \mu'(M+m)g$ = limiting friction between B and floor, F_k = kinetic friction between A and B)

B will move only if $F_k > F_l'$ and then $F_k - F_l' = M a_B$

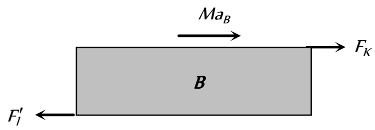


Fig. 5.29

However if B does not move then static friction will work (not limiting friction) between body B and the floor *i.e.* friction force = applied force (= F) not F_l'.

(2) **A force F is applied to the lower body, then following four situations are possible**

(i) **When there is no friction**

(a) B will move with acceleration (F/M) while A will remain at rest (relative to ground) as there is no pulling force on A.

$$a_B = \left(\frac{F}{M}\right) \text{ and } a_A = 0$$

(b) As relative to B, A will move backwards with acceleration (F/M) and so will fall from it in time t.

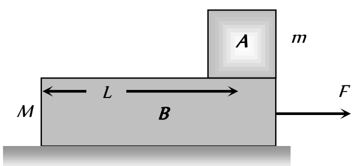


Fig. 5.30

$$\therefore \quad t = \sqrt{\frac{2L}{a}} = \sqrt{\frac{2ML}{F}}$$

(ii) **If friction is present between A and B only and $F' < F_l$**

(where F' = Pseudo force on body A and F_l = limiting friction between body A and B)

(a) Both the body will move together with common acceleration $a = \dfrac{F}{M+m}$

(b) Pseudo force on the body A,

$$F' = ma = \frac{mF}{m+M} \text{ and } F_l = \mu_s mg$$

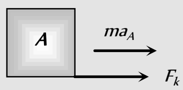
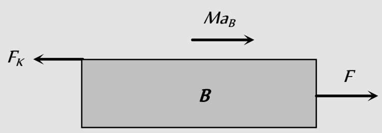
(c) $F' < F_l \;\Rightarrow\; \dfrac{mF}{m+M} < \mu_s mg \;\Rightarrow\; F < \mu_s (m+M)g$

So both bodies will move together with acceleration $a_A = a_B = \dfrac{F}{m+M}$ if $F < \mu_s[m+M]g$

(iii) **If friction is present between A and B only and $F > F_l'$**

(where $F_l = \mu_s mg$ = limiting friction between body A and B)

Both the body will move with different acceleration. Here force of kinetic friction $\mu_k mg$ will oppose the motion of B while will cause the motion of A.

$ma_A = \mu_k mg$	Free body diagram of A
i.e. $a_A = \mu_k g$	
$F - F_k = Ma_B$	Free body diagram of B
i.e. $a_B = \dfrac{[F - \mu_k mg]}{M}$	

Note : ❑ As both the bodies are moving in the same direction

Acceleration of body A relative to B will be

$$a = a_A - a_B = -\left[\frac{F - \mu_k g(m+M)}{M}\right]$$

Negative sign implies that relative to B, A will move backwards and will fall it after time

$$t = \sqrt{\frac{2L}{a}} = \sqrt{\frac{2ML}{F - \mu_k g(m+M)}}$$

(iv) **If there is friction between B and floor and $F > F_l''$:**

(where F_l'' = $\mu(m+M)g$ = limiting friction between body B and surface)

The system will move only if $F > F_l''$ then replacing F by $F - F_l''$. The entire case (iii) will be valid.

However if $F < F_1''$ the system will not move and friction between B and floor will be F while between A and B is zero.

Motion of an Insect in the Rough Bowl

The insect crawl up the bowl, up to a certain height h only till the component of its weight along the bowl is balanced by limiting frictional force.

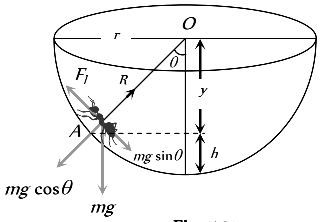


Fig. 5.31

Let m = mass of the insect, r = radius of the bowl, μ = coefficient of friction

for limiting condition at point A

$R = mg\cos\theta$ ......(i) and $F_l = mg\sin\theta$ ......(ii)

Dividing (ii) by (i)

$$\tan\theta = \frac{F_l}{R} = \mu \qquad [\text{As} F_l = \mu R]$$

$$\therefore \frac{\sqrt{r^2 - y^2}}{y} = \mu \quad \text{or} \quad y = \frac{r}{\sqrt{1+\mu^2}}$$

So $h = r - y = r\left[1 - \frac{1}{\sqrt{1+\mu^2}}\right]$, $\therefore h = r\left[1 - \frac{1}{\sqrt{1+\mu^2}}\right]$

Minimum Mass Hung from the String to Just Start the Motion

(1) **When a mass m_1 placed on a rough horizontal plane** Another mass m_2 hung from the string connected by frictionless pulley, the tension (T) produced in string will try to start the motion of mass m_1.

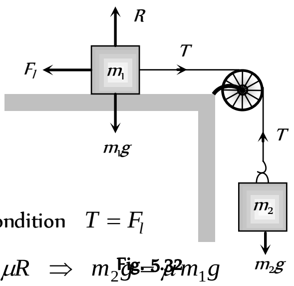


Fig. 5.32

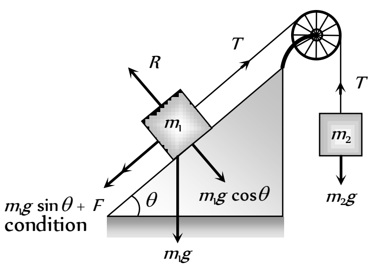
At limiting condition $T = F_l$

$\Rightarrow m_2 g = \mu R \Rightarrow m_2 g = \mu m_1 g$

$\therefore m_2 = \mu m_1$ this is the minimum value of m_2 to start the motion.

Note : ❑ In the above condition Coefficient of friction $\mu = \frac{m_2}{m_1}$

(2) **When a mass m_1 placed on a rough inclined plane** Another mass m_2 hung from the string connected by frictionless pulley, the tension (T) produced in string will try to start the motion of mass m_1.



At limiting condition

Fig. 5.33

For m_2 $T = m_2 g$ ...(i)

For m_1 $T = m_1 g\sin\theta + F$

$\Rightarrow T = m_1 g\sin\theta + \mu R$

$\Rightarrow T = m_1 g\sin\theta + \mu m_1 g\cos\theta$ ...(ii)

From equation (i) and (ii) $m_2 = m_1[\sin\theta + \mu\cos\theta]$

this is the minimum value of m_2 to start the motion

Note : ❑ In the above condition Coefficient of friction

$$\mu = \left[\frac{m_2}{m_1\cos\theta} - \tan\theta\right]$$

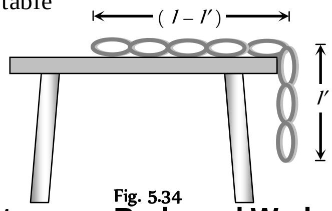
Maximum Length of Hung Chain

A uniform chain of length l is placed on the table in such a manner that its l' part is hanging over the edge of table without sliding. Since the chain have uniform linear density therefore the ratio of mass and ratio of length for any part of the chain will be equal.

We know $\mu = \frac{m_2}{m_1} = \frac{\text{mass hanging from the table}}{\text{mass lying on the table}}$

$\therefore$ For this case we can rewrite above expression in the following manner

$\mu = \frac{\text{length hanging from the table}}{\text{length lying on the table}}$ [As chain have uniform linear density]



$\therefore \quad \mu = \frac{l'}{l - l'}$

by solving $l' = \frac{\mu l}{(\mu + 1)}$

Fig. 5.34

Coefficient of Friction Between a Body and Wedge

A body slides on a smooth wedge of angle θ and its time of descent is t.

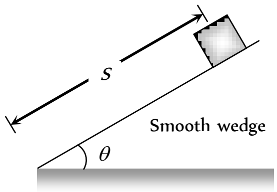


Fig. 5.35

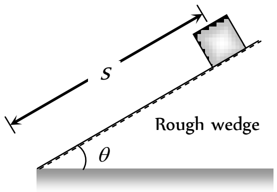


Fig. 5.36

If the same wedge made rough then time taken by it to come down becomes n times more (*i.e.* nt)

The length of path in both the cases are same.

For smooth wedge, $S = ut + \frac{1}{2}at^2$

$S = \frac{1}{2}(g\sin\theta)t^2$ ...(i)

$[\text{As} u = 0 \text{ and } a = g\sin\theta]$

For rough wedge, $S = ut + \frac{1}{2}at^2$

$S = \frac{1}{2}g(\sin\theta - \mu\cos\theta)(nt)^2$ ...(ii)

$[\text{As} u = 0 \text{ and } a = g(\sin\theta - \mu\cos\theta)]$

From equation (i) and (ii)

$$\frac{1}{2}(g\sin\theta)t^2 = \frac{1}{2}g(\sin\theta - \mu\cos\theta)(nt)^2$$

$$\Rightarrow \sin\theta = (\sin\theta - \mu\cos\theta)n^2$$

$$\Rightarrow \mu = \tan\theta\left[1 - \frac{1}{n^2}\right]$$

Stopping of Block Due to Friction

(1) **On horizontal road**

(i) **Distance travelled before coming to rest :** A block of mass m is moving initially with velocity u on a rough surface and due to friction, it comes to rest after covering a distance S.

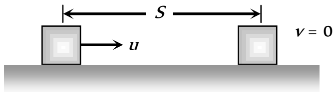


Fig. 5.37

Retarding force $F = ma = \mu R \Rightarrow ma = \mu mg$

$\therefore a = \mu g$

From $v^2 = u^2 - 2aS \Rightarrow 0 = u^2 - 2\mu g S$

[As $v = 0,\ a = \mu g$]

$$\therefore\ S = \frac{u^2}{2\mu g} \quad \text{or} \quad S = \frac{P^2}{2\mu m^2 g}$$

[As momentum $P = mu$]

(ii) **Time taken to come to rest**

From equation $v = u - at \Rightarrow 0 = u - \mu g t$

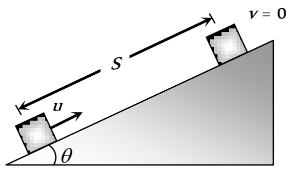
[As $v = 0, a = \mu g$]

$$\therefore\ t = \frac{u}{\mu g}$$

(2) **On inclined road :** When block starts with velocity u its kinetic energy will be converted into potential energy and some part of it goes against friction and after travelling distance S it comes to rest *i.e.* $v = 0$.

We know that retardation $a = g[\sin\theta + \mu\cos\theta]$

By substituting the value of v and a in the following equation



$v^2 = u^2 - 2aS$

Fig. 5.38

$$\Rightarrow\ 0 = u^2 - 2g[\sin\theta + \mu\cos\theta]S$$

$$\therefore\ S = \frac{u^2}{2g(\sin\theta + \mu\cos\theta)}$$

Stopping of Two Blocks Due to Friction

When two masses compressed towards each other and suddenly released then energy acquired by each block will be dissipated against friction and finally block comes to rest

i.e., $F \times S = E$ [Where F = Friction, S = Distance covered by block, E = Initial kinetic energy of the block]

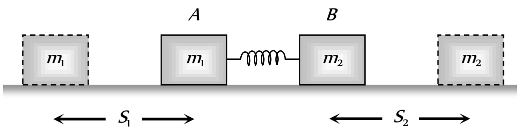


Fig. 5.39

$$\Rightarrow\ F \times S = \frac{P^2}{2m} \qquad \text{[Where } P = \text{momentum of block]}$$

$$\Rightarrow\ \mu mg \times S = \frac{P^2}{2m} \qquad \text{[As } F = \mu mg\text{]}$$

$$\Rightarrow\ S = \frac{P^2}{2\mu m^2 g}$$

In the given condition P and μ are same for both the blocks.

So, $S \propto \frac{1}{m^2}$; $\therefore \frac{S_1}{S_2} = \left[\frac{m_2}{m_1}\right]^2$

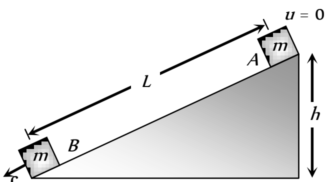
Velocity at the Bottom of Rough Wedge

A body of mass m which is placed at the top of the wedge (of height h) starts moving downward on a rough inclined plane.

Loss of energy due to friction = FL (Work against friction)

PE at point $A = mgh$

KE at point $B = \frac{1}{2}mu^2$



By the law of conservation of energy

Fig. 5.40

i.e. $\frac{1}{2}mv^2 = mgh - FL$

$$v = \sqrt{\frac{2}{m}(mgh - FL)}$$

Sticking of a Block With Accelerated Cart

When a cart moves with some acceleration toward right then a pseudo force (ma) acts on block toward left.

This force (ma) is action force by a block on cart.

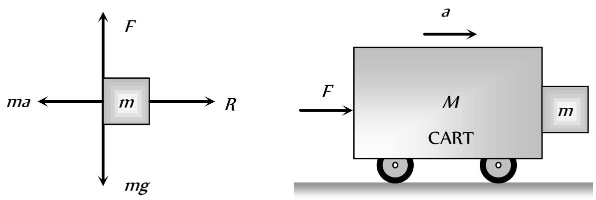


Fig. 5.41

Now block will remain static *w.r.t.* cart. If friction force $\mu R \geq mg$

$\Rightarrow \mu ma \geq mg$ [As $R = ma$]

$$\Rightarrow\ a \geq \frac{g}{\mu}$$

$$\therefore\ a_{min} = \frac{g}{\mu}$$

This is the minimum acceleration of the cart so that block does not fall.

and the minimum force to hold the block together

$$F_{min} = (M + m)a_{min}$$

$$F_{min} = (M + m)\frac{g}{\mu}$$

Sticking of a Person with the Wall of Rotor

A person with a mass m stands in contact against the wall of a cylindrical drum (rotor). The coefficient of friction between the wall and the clothing is μ.

If Rotor starts rotating about its axis, then person thrown away from the centre due to centrifugal force at a particular speed ω, the person stuck to the wall even the floor is removed, because friction force balances its weight in this condition.

From the figure.

Friction force (F) = weight of person (mg)

$\Rightarrow \mu R = mg \Rightarrow \mu F_c = mg$

[Here, F_c = centrifugal force]

$\Rightarrow \mu m \omega_{\min}^2 r = mg$

$\therefore \omega_{\min} = \sqrt{\dfrac{g}{\mu r}}$

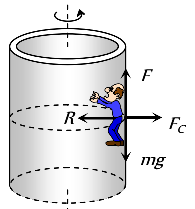


Fig. 5.42

Tips & Tricks

✍ Force of friction is non-conservative force.

✍ Force of friction always acts in a direction opposite to that of the relative motion between the surfaces.

✍ Rolling friction is much less than the sliding friction. This knowledge was used by man to invent the wheels.

✍ The friction between two surfaces increases (rather than to decrease), when the surfaces are made highly smooth.

✍ The atomic and molecular forces of attraction between the two surfaces at the point of contact give rise to friction between the surfaces.

Ordinary Thinking

Objective Questions

Static and limiting friction

1. The coefficient of friction μ and the angle of friction λ are related as

(a) $\sin\lambda = \mu$ (b) $\cos\lambda = \mu$

(c) $\tan\lambda = \mu$ (d) $\tan\mu = \lambda$

2. A force of 98 N is required to just start moving a body of mass 100 kg over ice. The coefficient of static friction is

(a) 0.6 (b) 0.4

(c) 0.2 (d) 0.1

3. A block weighs W is held against a vertical wall by applying a horizontal force F. The minimum value of F needed to hold the block is [MP PMT 1993]

(a) Less than W (b) Equal to W

(c) Greater than W (d) Data is insufficient

4. The maximum static frictional force is

(a) Equal to twice the area of surface in contact

(b) Independent of the area of surface in contact

(c) Equal to the area of surface in contact

(d) None of the above

5. Maximum value of static friction is called [BHU 1995; RPET 2000]

(a) Limiting friction (b) Rolling friction

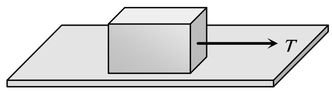
(c) Normal reaction (d) Coefficient of friction

6. Pulling force making an angle θ to the horizontal is applied on a block of weight W placed on a horizontal table. If the angle of friction is α, then the magnitude of force required to move the body is equal to [EAMCET 1987]

(a) $\dfrac{W\sin\alpha}{g\tan(\theta-\alpha)}$ (b) $\dfrac{W\cos\alpha}{\cos(\theta-\alpha)}$

(c) $\dfrac{W\sin\alpha}{\cos(\theta-\alpha)}$ (d) $\dfrac{W\tan\alpha}{\sin(\theta-\alpha)}$

7. In the figure shown, a block of weight 10 N resting on a horizontal surface. The coefficient of static friction between the block and the surface $\mu_s = 0.4$. A force of 3.5 N will keep the block in uniform motion, once it has been set in motion. A horizontal force of 3 N is applied to the block, then the block will



(a) Move over the surface with constant velocity

(b) Move having accelerated motion over the surface

(c) Not move

(d) First it will move with a constant velocity for some time and then will have accelerated motion

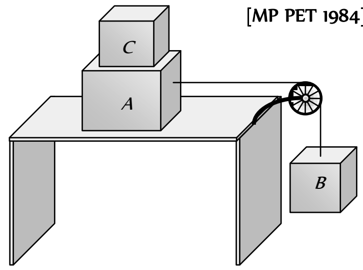
8. Two masses A and B of 10 kg and 5 kg respectively are connected with a string passing over a frictionless pulley fixed at the corner of a table as shown. The coefficient of static friction of A with table is 0.2. The minimum mass of C that may be placed on A to prevent it from moving is [MP PET 1984]

(a) 15 kg

(b) 10 kg

(c) 5 kg

(d) 12 kg



9. The limiting friction is

(a) Always greater than the dynamic friction

(b) Always less than the dynamic friction

(c) Equal to the dynamic friction

(d) Sometimes greater and sometimes less than the dynamic friction

10. Which is a suitable method to decrease friction

(a) Ball and bearings (b) Lubrication

(c) Polishing (d) All the above

11. A uniform rope of length l lies on a table. If the coefficient of friction is μ, then the maximum length l_1 of the part of this rope which can overhang from the edge of the table without sliding down is **[DPMT 2001]**

(a) $\frac{l}{\mu}$ (b) $\frac{l}{\mu+l}$

(c) $\frac{\mu l}{1+\mu}$ (d) $\frac{\mu l}{\mu-1}$

12. Which of the following statements is not true **[CMC Vellore 1989]**

(a) The coefficient of friction between two surfaces increases as the surface in contact are made rough

(b) The force of friction acts in a direction opposite to the applied force

(c) Rolling friction is greater than sliding friction

(d) The coefficient of friction between wood and wood is less than 1

13. A block of 1 *kg* is stopped against a wall by applying a force F perpendicular to the wall. If $\mu = 0.2$ then minimum value of F will be **[MP PMT 2003]**

(a) 980 *N* (b) 49 *N*

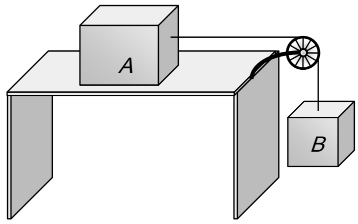
(c) 98 *N* (d) 490 *N*

14. A heavy uniform chain lies on a horizontal table-top. If the coefficient of friction between the chain and table surface is 0.25, then the maximum fraction of length of the chain, that can hang over one edge of the table is **[CBSE PMT 1990]**

(a) 20% (b) 25%

(c) 35% (d) 15%

15. The blocks A and B are arranged as shown in the figure. The pulley is frictionless. The mass of A is 10 *kg*. The coefficient of friction of A with the horizontal surface is 0.20. The minimum mass of B to start the motion will be **[MP PET 1994]**



(a) 2 *kg*

(b) 0.2 *kg*

(c) 5 *kg*

(d) 10 *kg*

16. Work done by a frictional force is

(a) Negative (b) Positive

(c) Zero (d) All of the above

17. A uniform chain of length L changes partly from a table which is kept in equilibrium by friction. The maximum length that can withstand without slipping is l, then coefficient of friction between the table and the chain is **[EAMCET (Engg.) 1995]**

(a) $\frac{l}{L}$ (b) $\frac{l}{L+l}$

(c) $\frac{l}{L-l}$ (d) $\frac{L}{L+l}$

18. When two surfaces are coated with a lubricant, then they **[AFMC 1998, 99; AIIMS 2001]**

(a) Stick to each other (b) Slide upon each other

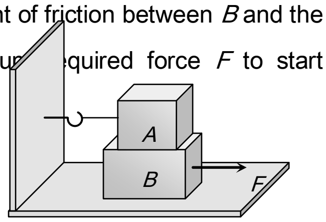
(c) Roll upon each other (d) None of these

19. A 20 *kg* block is initially at rest on a rough horizontal surface. A horizontal force of 75 *N* is required to set the block in motion. After it is in motion, a horizontal force of 60 *N* is required to keep the block moving with constant speed. The coefficient of static friction is **[AMU 1999]**

(a) 0.38 (b) 0.44

(c) 0.52 (d) 0.60

20. A block A with mass 100 *kg* is resting on another block B of mass 200 *kg*. As shown in figure a horizontal rope tied to a wall holds it. The coefficient of friction between A and B is 0.2 while coefficient of friction between B and the ground is 0.3. The minimum required force F to start moving B will be



[RPET 1999]

(a) 900 N

(b) 100 N

(c) 1100 N

(d) 1200 N

21. To avoid slipping while walking on ice, one should take smaller steps because of the [BHU 1999; BCECE 2004]

(a) Friction of ice is large

(b) Larger normal reaction

(c) Friction of ice is small

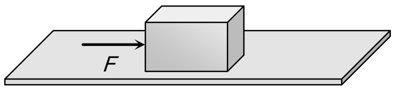
(d) Smaller normal reaction

22. A box is lying on an inclined plane what is the coefficient of static friction if the box starts sliding when an angle of inclination is 60° [KCET 2000]

(a) 1.173 (b) 1.732

(c) 2.732 (d) 1.677

23. A block of mass 2 kg is kept on the floor. The coefficient of static friction is 0.4. If a force F of 2.5 *Newtons* is applied on the block as shown in the figure, the frictional force between the block and the floor will be [MP PET 2000]



(a) 2.5 N

(b) 5 N

(c) 7.84 N

(d) 10 N

24. Which one of the following is not used to reduce friction

[Kerala (Engg.) 2001]

(a) Oil (b) Ball bearings

(c) Sand (d) Graphite

25. If a ladder weighing 250N is placed against a smooth vertical wall having coefficient of friction between it and floor is 0.3, then what is the maximum force of friction available at the point of contact between the ladder and the floor [AIIMS 2002]

(a) 75 N (b) 50 N

(c) 35 N (d) 25 N

26. A body of mass 2 kg is kept by pressing to a vertical wall by a force of 100 N. The coefficient of friction between wall and body is 0.3. Then the frictional force is equal to

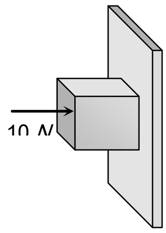
[Orissa JEE 2003]

(a) 6 N (b) 20 N

(c) 600 N (d) 700 N

27. A horizontal force of 10 N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.2. the weight of the block is

[AIEEE 2003]



(a) 2 N

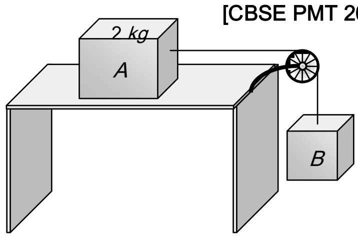
(b) 20 N

(c) 50 N

(d) 100 N

28. The coefficient of static friction, μ_s, between block A of mass 2 kg and the table as shown in the figure is 0.2. What would be the maximum mass value of block B so that the two blocks do not move? The string and the pulley are assumed to be smooth and massless. $(g = 10\, m/s^2)$

[CBSE PMT 2004]



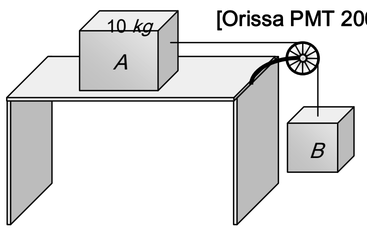
(a) 2.0 kg

(b) 4.0 kg

(c) 0.2 kg

(d) 0.4 kg

29. If mass of $A = 10\ kg$, coefficient of static friction = 0.2, coefficient of kinetic friction = 0.2. Then mass of B to start motion is [Orissa PMT 2004]



(a) 2 kg

(b) 2.2 *kg*

(c) 4.8 *kg*

(d) 200 *gm*

30. A uniform metal chain is placed on a rough table such that one end of chain hangs down over the edge of the table. When one-third of its length hangs over the edge, the chain starts sliding. Then, the coefficient of static friction is

[Kerala PET 2005]

(a) $\frac{3}{4}$ (b) $\frac{1}{4}$

(c) $\frac{2}{3}$ (d) $\frac{1}{2}$

31. A lift is moving downwards with an acceleration equal to acceleration due to gravity. A body of mass *m* kept on the floor of the lift is pulled horizontally. If the coefficient of friction is μ, then the frictional resistance offered by the body is **[DPMT 2004]**

(a) *mg* (b) μmg

(c) $2\mu mg$ (d) Zero

32. If a ladder weighing 250 *N* is placed against a smooth vertical wall having coefficient of friction between it and floor is 0.3, then what is the maximum force of friction available at the point of contact between the ladder and the floor **[BHU 2004]**

(a) 75 *N* (b) 50 *N*

(c) 35 *N* (d) 25 *N*

Kinetic Friction

1. Which one of the following statements is correct

(a) Rolling friction is greater than sliding friction

(b) Rolling friction is less than sliding friction

(c) Rolling friction is equal to sliding friction

(d) Rolling friction and sliding friction are same

2. The maximum speed that can be achieved without skidding by a car on a circular unbanked road of radius *R* and coefficient of static friction μ , is

(a) μRg (b) $Rg\sqrt{\mu}$

(c) $\mu\sqrt{Rg}$ (d) $\sqrt{\mu Rg}$

3. A car is moving along a straight horizontal road with a speed v_0. If the coefficient of friction between the tyres and the road is μ , the shortest distance in which the car can be stopped is **[MP PET 1985; BHU 2002]**

(a) $\frac{v_0^2}{2\mu g}$ (b) $\frac{v_0}{\mu g}$

(c) $\left(\frac{v_0}{\mu g}\right)^2$ (d) $\frac{v_0}{\mu}$

4. A block of mass 5 *kg* is on a rough horizontal surface and is at rest. Now a force of 24 *N* is imparted to it with negligible impulse. If the coefficient of kinetic friction is 0.4 and $g = 9.8\, m/s^2$, then the acceleration of the block is

(a) $0.26\, m/s^2$ (b) $0.39\, m/s^2$

(c) $0.69\, m/s^2$ (d) $0.88\, m/s^2$

5. A body of mass 2 *kg* is being dragged with uniform velocity of 2 *m/s* on a rough horizontal plane. The coefficient of friction between the body and the surface is 0.20. The amount of heat generated in 5 sec is

$(J = 4.2\, joule/cal$ and $g = 9.8\, m/s^2)$

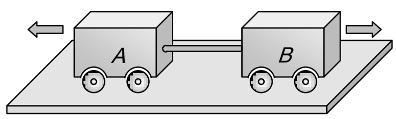
[MH CET (Med.) 2001]

(a) 9.33 *cal* (b) 10.21 *cal*

(c) 12.67 *cal* (d) 13.34 *cal*

6. Two carts of masses 200 *kg* and 300 *kg* on horizontal rails are pushed apart. Suppose the coefficient of friction between the carts and the rails are same. If the 200 *kg* cart travels a distance of 36 *m* and stops, then the distance travelled by the cart weighing 300 *kg* is

[CPMT 1989; DPMT 2002]



(a) 32 *m* (b) 24 *m*

(c) 16 *m* (d) 12 *m*

[NCERT 1990]

7. A body *B* lies on a smooth horizontal table and another body *A* is placed on *B*. The coefficient of friction between

A and B is μ. What acceleration given to B will cause slipping to occur between A and B

(a) μg (b) g/μ

(c) μ/g (d) $\sqrt{\mu g}$

8. A 60 *kg* body is pushed with just enough force to start it moving across a floor and the same force continues to act afterwards. The coefficient of static friction and sliding friction are 0.5 and 0.4 respectively. The acceleration of the body is

(a) $6\,m/s^2$ (b) $4.9\,m/s^2$

(c) $3.92\,m/s^2$ (d) $1\,m/s^2$

9. A car turns a corner on a slippery road at a constant speed of $10\,m/s$. If the coefficient of friction is 0.5, the minimum radius of the arc in meter in which the car turns is

(a) 20 (b) 10

(c) 5 (d) 4

10. A motorcyclist of mass *m* is to negotiate a curve of radius *r* with a speed *v*. The minimum value of the coefficient of friction so that this negotiation may take place safely, is **[Haryana CEE 1996]**

(a) v^2rg (b) $\frac{v^2}{gr}$

(c) $\frac{gr}{v^2}$ (d) $\frac{g}{v^2r}$

11. On a rough horizontal surface, a body of mass 2 *kg* is given a velocity of 10 *m/s*. If the coefficient of friction is 0.2 and $g=10\,m/s^2$, the body will stop after covering a distance of **[MP PMT 1999]**

(a) 10 *m* (b) 25 *m*

(c) 50 *m* (d) 250 *m*

12. A block of mass 50 *kg* can slide on a rough horizontal surface. The coefficient of friction between the block and the surface is 0.6. The least force of pull acting at an angle of 30° to the upward drawn vertical which causes the block to just slide is **[ISM Dhanbad 1994]**

(a) 29.43 *N* (b) 219.6 *N*

(c) 21.96 *N* (d) 294.3 *N*

13. A body of 10 *kg* is acted by a force of 129.4 *N* if $g=9.8\,m/\sec^2$. The acceleration of the block is $10\,m/s^2$. What is the coefficient of kinetic friction **[EAMCET 1994]**

(a) 0.03 (b) 0.01

(c) 0.30 (d) 0.25

14. Assuming the coefficient of friction between the road and tyres of a car to be 0.5, the maximum speed with which the car can move round a curve of 40.0 *m* radius without slipping, if the road is unbanked, should be **[AMU 1995]**

(a) 25 *m/s* (b) 19 *m/s*

(c) 14 *m/s* (d) 11 *m/s*

15. Consider a car moving along a straight horizontal road with a speed of 72 *km/h*. If the coefficient of kinetic friction between the tyres and the road is 0.5, the shortest distance in which the car can be stopped is $[g=10\,ms^{-2}]$ **[CBSE PMT 1992]**

(a) 30 *m* (b) 40 *m*

(c) 72 *m* (d) 20 *m*

16. A 500 *kg* horse pulls a cart of mass 1500 *kg* along a level road with an acceleration of $1\,ms^{-2}$. If the coefficient of sliding friction is 0.2, then the force exerted by the horse in forward direction is **[SCRA 1998]**

(a) 3000 *N* (b) 4000 *N*

(c) 5000 *N* (d) 6000 *N*

17. The maximum speed of a car on a road turn of radius 30*m*, if the coefficient of friction between the tyres and the road is 0.4; will be **[MH CET (Med.) 1999]**

(a) 9.84 *m/s* (b) 10.84 *m/s*

(c) 7.84 *m/s* (d) 5.84 *m/s*

18. A block of mass 50 *kg* slides over a horizontal distance of 1 *m*. If the coefficient of friction between their surfaces is 0.2, then work done against friction is

[BHU 2001; CBSE PMT 1999, 2000; AIIMS 2000]

(a) 98 *J* (b) 72*J*

(c) 56 *J* (d) 34 *J*

19. On the horizontal surface of a truck (μ = 0.6), a block of mass 1 *kg* is placed. If the truck is accelerating at the rate of 5*m/sec*^{2} then frictional force on the block will be

[CBSE PMT 2001]

(a) 5 *N* (b) 6 *N*

(c) 5.88 *N* (d) 8 *N*

20. A vehicle of mass *m* is moving on a rough horizontal road with momentum *P*. If the coefficient of friction between the tyres and the road be μ, then the stopping distance is

[CBSE PMT 2001]

(a) $\frac{P}{2\mu m g}$ (b) $\frac{P^2}{2\mu m g}$

(c) $\frac{P}{2\mu m^2 g}$ (d) $\frac{P^2}{2\mu m^2 g}$

21. A body of weight 64 *N* is pushed with just enough force to start it moving across a horizontal floor and the same force continues to act afterwards. If the coefficients of static and dynamic friction are 0.6 and 0.4 respectively, the acceleration of the body will be (Acceleration due to gravity = *g*)

[EAMCET 2001]

(a) $\frac{g}{6.4}$ (b) 0.64 *g*

(c) $\frac{g}{32}$ (d) 0.2 *g*

22. When a body is moving on a surface, the force of friction is called **[MP PET 2002]**

(a) Static friction (b) Dynamic friction

(c) Limiting friction (d) Rolling friction

23. A block of mass 10 *kg* is placed on a rough horizontal surface having coefficient of friction μ = 0.5. If a horizontal force of 100 *N* is acting on it, then acceleration of the block will be **[AIIMS 2002]**

(a) 0.5 *m/s*^{2} (b) 5 *m/s*^{2}

(c) 10 *m/s*^{2} (d) 15 *m/s*^{2}

24. It is easier to roll a barrel than pull it along the road. This statement is **[BVP 2003]**

(a) False (b) True

(c) Uncertain (d) Not possible

25. A marble block of mass 2 *kg* lying on ice when given a velocity of 6 *m/s* is stopped by friction in 10*s*. Then the coefficient of friction is **[AIEEE 2003]**

(a) 0.01 (b) 0.02

(c) 0.03 (d) 0.06

26. A horizontal force of 129.4 *N* is applied on a 10 *kg* block which rests on a horizontal surface. If the coefficient of friction is 0.3, the acceleration should be

(a) $9.8\, m/s^2$ (b) $10\, m/s^2$

(c) $12.6\, m/s^2$ (d) $19.6\, m/s^2$

27. A 60 *kg* weight is dragged on a horizontal surface by a rope upto 2 *metres*. If coefficient of friction is $\mu = 0.5$, the angle of rope with the surface is 60° and $g = 9.8\, m/\sec^2$, then work done is **[MP PET 1995]**

(a) 294 *joules* (b) 315 *joules*

(c) 588 *joules* (d) 197 *joules*

28. A car having a mass of 1000 *kg* is moving at a speed of 30 *metres/sec*. Brakes are applied to bring the car to rest. If the frictional force between the tyres and the road surface is 5000 *newtons*, the car will come to rest in **[MP PMT 1995]**

(a) 5 *seconds* (b) 10 *seconds*

(c) 12 *seconds* (d) 6 *seconds*

29. If μ_s, μ_k and μ_r are coefficients of static friction, sliding friction and rolling friction, then **[EAMCET (Engg.) 1995]**

(a) $\mu_s < \mu_k < \mu_r$ (b) $\mu_k < \mu_r < \mu_s$

(c) $\mu_r < \mu_k < \mu_s$ (d) $\mu_r = \mu_k = \mu_s$

30. A body of mass 5*kg* rests on a rough horizontal surface of coefficient of friction 0.2. The body is pulled through a distance of 10*m* by a horizontal force of 25 *N*. The kinetic energy acquired by it is (g = 10 ms^{2})

[EAMCET (Med.) 2000]

(a) 330 *J* (b) 150 *J*

(c) 100 *J* (d) 50 *J*

31. A motorcycle is travelling on a curved track of radius 500*m*. If the coefficient of friction between road and tyres is 0.5, the speed avoiding skidding will be

(a) 50 *m/s* (b) 75 *m/s*

(c) 25 *m/s* (d) 35 *m/s*

32. A fireman of mass 60 *kg* slides down a pole. He is pressing the pole with a force of 600 *N*. The coefficient of friction between the hands and the pole is 0.5, with what acceleration will the fireman slide down (g = 10 m/s^2)

[Pb. PMT 2002]

(a) 1 m/s^2 (b) 2.5 m/s^2

(c) 10 m/s^2 (d) 5 m/s^2

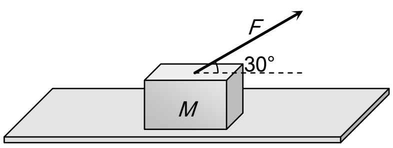
33. A block of mass $M = 5\,kg$ is resting on a rough horizontal surface for which the coefficient of friction is 0.2. When a force $F = 40\ N$ is applied, the acceleration of the block will be $(g = 10\ m/s^2)$ **[MP PMT 2004]**

(a) $5.73\ m/\sec^2$

(b) $8.0\ m/\sec^2$

(c) $3.17\ m/\sec^2$

(d) $10.0\ m/\sec^2$



34. A body is moving along a rough horizontal surface with an initial velocity $6\ m/s$. If the body comes to rest after travelling 9 *m*, then the coefficient of sliding friction will be

[BCECE 2004]

(a) 0.4 (b) 0.2

(c) 0.6 (d) 0.8

35. Consider a car moving on a straight road with a speed of 100 *m/s*. The distance at which car can be stopped is $[\mu_k = 0.5]$ **[AIEEE 2005]**

(a) 100 *m* (b) 400 *m*

(c) 800 *m* (d) 1000 *m*

36. A cylinder of 10 *kg* is sliding in a plane with an initial velocity of 10 *m/s*. If the coefficient of friction between the surface and cylinder is 0.5 then before stopping, it will cover. $(g = 10\ m/s^2)$ **[Pb. PMT 2004]**

[MH CET (Med.) 2001]

(a) 2.5 *m* (b) 5 *m*

(c) 7.5 *m* (d) 10 *m*

Motion on Inclined Surface

1. When a body is lying on a rough inclined plane and does not move, the force of friction

(a) is equal to μR (b) is less than μR

(c) is greater than μR (d) is equal to *R*

2. When a body is placed on a rough plane inclined at an angle θ to the horizontal, its acceleration is

(a) $g(\sin\theta - \cos\theta)$ (b) $g(\sin\theta - \mu\cos\theta)$

(c) $g(\mu\sin\theta 1 - \cos\theta)$ (d) $g\mu(\sin\theta - \cos\theta)$

3. A block is at rest on an inclined plane making an angle α with the horizontal. As the angle α of the incline is increased, the block starts slipping when the angle of inclination becomes θ. The coefficient of static friction between the block and the surface of the inclined plane is

or

A body starts sliding down at an angle θ to horizontal. Then coefficient of friction is equal to **[CBSE PMT 1993]**

(a) $\sin\theta$ (b) $\cos\theta$

(c) $\tan\theta$ (d) Independent of θ

4. A given object takes *n* times as much time to slide down a 45° rough incline as it takes to slide down a perfectly smooth 45° incline. The coefficient of kinetic friction between the object and the incline is given by

(a) $\left(1-\frac{1}{n^2}\right)$ (b) $\frac{1}{1-n^2}$

(c) $\sqrt{\left(1-\frac{1}{n^2}\right)}$ (d) $\sqrt{\frac{1}{1-n^2}}$

5. The force required just to move a body up an inclined plane is double the force required just to prevent the body sliding down. If the coefficient of friction is 0.25, the angle of inclination of the plane is

(a) 36.8° (b) 45°

(c) 30° (d) 42.6°

6. Starting from rest, a body slides down a 45° inclined plane in twice the time it takes to slide down the same distance in the absence of friction. The coefficient of friction between the body and the inclined plane is

(a) 0.33 (b) 0.25

(c) 0.75 (d) 0.80

7. The coefficient of friction between a body and the surface of an inclined plane at 45° is 0.5. If $g=9.8\,m/s^2$, the acceleration of the body downwards in m/s^2 is

[EAMCET 1994]

(a) $\frac{4.9}{\sqrt{2}}$ (b) $4.9\sqrt{2}$

(c) $19.6\sqrt{2}$ (d) 4.9

8. A box is placed on an inclined plane and has to be pushed down. The angle of inclination is

(a) Equal to angle of friction

(b) More than angle of friction

(c) Equal to angle of repose

(d) Less than angle of repose

9. A force of 750 *N* is applied to a block of mass 102 *kg* to prevent it from sliding on a plane with an inclination angle 30° with the horizontal. If the coefficients of static friction and kinetic friction between the block and the plane are 0.4 and 0.3 respectively, then the frictional force acting on

[RPET 1999; AMU 2000]

the block is [SCRA 1994]

(a) 750 *N* (b) 500 *N*

(c) 345 *N* (d) 250 *N*

10. A block is lying on an inclined plane which makes 60° with the horizontal. If coefficient of friction between block and plane is 0.25 and $g=10\,m/s^2$, then acceleration of the block when it moves along the plane will be [RPET 1997]

(a) $2.50\,m/s^2$ (b) $5.00\,m/s^2$

(c) $7.4\,m/s^2$ (d) $8.66\,m/s^2$

11. A body of mass 100 *g* is sliding from an inclined plane of inclination 30°. What is the frictional force experienced if $\mu=1.7$ [BHU 1998]

(a) $1.7\times\sqrt{2}\times\frac{1}{\sqrt{3}}\,N$ (b) $1.7\times\sqrt{3}\times\frac{1}{2}\,N$

[CBSE PMT 1990]

(c) $1.7\times\sqrt{3}\,N$ (d) $1.7\times\sqrt{2}\times\frac{1}{3}\,N$

12. A body takes just twice the time as long to slide down a plane inclined at 30° to the horizontal as if the plane were frictionless. The coefficient of friction between the body and the plane is [JIPMER 1999]

(a) $\frac{\sqrt{3}}{4}$ (b) $\sqrt{3}$

(c) $\frac{4}{3}$ (d) $\frac{3}{4}$

13. A brick of mass 2 *kg* begins to slide down on a plane inclined at an angle of 45° with the horizontal. The force of friction will be [CPMT 2000]

[EAMCET 1994]

(a) 19.6 *sin* 45° (b) 19.6 cos 45°

(c) 9.8 *sin* 45° (d) 9.8 cos 45°

14. The upper half of an inclined plane of inclination θ is perfectly smooth while the lower half is rough. A body starting from the rest at top comes back to rest at the bottom if the coefficient of friction for the lower half is given by

[Pb. PMT 2000]

(a) μ = sin θ (b) μ = cot θ

(c) μ = 2 cos θ (d) μ = 2 tan θ

15. A body is sliding down an inclined plane having coefficient of friction 0.5. If the normal reaction is twice that of the resultant downward force along the incline, the angle between the inclined plane and the horizontal is **[EAMCET (Engg.) 2000]**

(a) 15° (b) 30° (c) 45° (d) 60°

16. A body of mass 10 *kg* is lying on a rough plane inclined at an angle of 30° to the horizontal and the coefficient of friction is 0.5. the minimum force required to pull the body up the plane is **[JIPMER 2000]**

(a) 914 *N* (b) 91.4 *N* (c) 9.14 *N* (d) 0.914 *N*

17. A block of mass 1 *kg* slides down on a rough inclined plane of inclination 60° starting from its top. If the coefficient of kinetic friction is 0.5 and length of the plane is 1 *m*, then work done against friction is (Take *g* = 9.8 m/s^2) **[AFMC 2000; KCET 2001]**

(a) 9.82 *J* (b) 4.94 *J* (c) 2.45*J* (d) 1.96 *J*

18. A block of mass 10 *kg* is placed on an inclined plane. When the angle of inclination is 30°, the block just begins to slide down the plane. The force of static friction is **[Kerala (Engg.) 2001]**

(a) 10 *kg wt* (b) 89 *kg w* (c) 49 *kg wt* (d) 5 *kg wt*

19. A body of 5 *kg* weight kept on a rough inclined plane of angle 30° starts sliding with a constant velocity. Then the coefficient of friction is (assume *g* = 10 m/s^2)

(a) $1/\sqrt{3}$ (b) $2/\sqrt{3}$ (c) $\sqrt{3}$ (d) $2\sqrt{3}$

20. 300 *Joule* of work is done in sliding up a 2 *kg* block on an inclined plane to a height of 10 *metres*. Taking value of acceleration due to gravity '*g*' to be 10 m/s^2, work done against friction is **[MP PMT 2002]**

(a) 100 *J* (b) 200 *J* (c) 300 *J* (d) Zero

21. A 2 *kg* mass starts from rest on an inclined smooth surface with inclination 30° and length 2 *m*. How much will it travel before coming to rest on a frictional surface with frictional coefficient of 0.25

(a) 4 *m* (b) 6 *m* (c) 8 *m* (d) 2 *m*

22. A block rests on a rough inclined plane making an angle of 30^o with the horizontal. The coefficient of static friction between the block and the plane is 0.8. If the frictional force on the block is 10 *N*, the mass of the block (in *kg*) is (take $g = 10\ m/s^2$) **[AIEEE 2004]**

(a) 2.0 (b) 4.0 (c) 1.6 (d) 2.5

23. A body takes time *t* to reach the bottom of an inclined plane of angle θ with the horizontal. If the plane is made rough, time taken now is 2*t*. The coefficient of friction of the rough surface is

(a) $\frac{3}{4}\tan\theta$ (b) $\frac{2}{3}\tan\theta$ (c) $\frac{1}{4}\tan\theta$ (d) $\frac{1}{2}\tan\theta$

24. A block is kept on an inclined plane of inclination θ of length *l*. The velocity of particle at the bottom of inclined is (the coefficient of friction is μ)

(a) $\sqrt{2gl(\mu\cos\theta - \sin\theta)}$ (b) $\sqrt{2gl(\sin\theta - \mu\cos\theta)}$

(c) $\sqrt{2gl(\sin\theta + \mu\cos\theta)}$ (d) $\sqrt{2gl(\cos\theta + \mu\sin\theta)}$

Critical Thinking

Objective Questions

[JIPMER 2002]

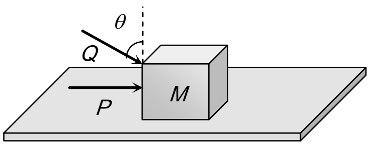
1. A block of mass *m* lying on a rough horizontal plane is acted upon by a horizontal force *P* and another force *Q* inclined at an angle θ to the vertical. The block will remain in equilibrium, if the coefficient of friction between it and the surface is **[Haryana CEE 1996]**

(a) $\frac{(P+Q\sin\theta)}{(mg+Q\cos\theta)}$

(b) $\frac{(P\cos\theta+Q)}{(mg-Q\sin\theta)}$

(c) $\frac{(P+Q\cos\theta)}{(mg+Q\sin\theta)}$

(d) $\frac{(P\sin\theta-Q)}{(mg-Q\cos\theta)}$



2. Which of the following is correct, when a person walks on a rough surface **[IIT 1981]**

(a) The frictional force exerted by the surface keeps him moving

(b) The force which the man exerts on the floor keeps him moving

(c) The reaction of the force which the man exerts on floor keeps him moving

(d) None of the above

3. A block of mass 0.1 *kg* is held against a wall by applying a horizontal force of 5 *N* on the block. If the coefficient of friction between the block and the wall is 0.5, the magnitude of the frictional force acting on the block is **[IIT 1994]**

(a) 2.5 *N* (b) 0.98 *N*

(c) 4.9 *N* (d) 0.49 *N*

4. A body of mass *M* is kept on a rough horizontal surface (friction coefficient μ). A person is trying to pull the body by applying a horizontal force but the body is not moving. The force by the surface on the body is *F*, where

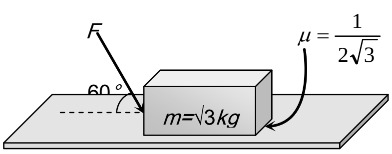
[MP PET 1997]

(a) $F = Mg$ (b) $F = \mu Mgf$

(c) $Mg \le F \le Mg\sqrt{1+\mu^2}$ (d) $Mg \ge F \ge Mg\sqrt{1+\mu^2}$

5. What is the maximum value of the force *F* such that the block shown in the arrangement, does not move

[IIT-JEE Screening 2003]



(a) 20 *N* (b) 10 *N*

(c) 12 *N* (d) 15 *N*

6. A block *P* of mass *m* is placed on a frictionless horizontal surface. Another block *Q* of same mass is kept on *P* and connected to the wall with the help of a spring of spring constant *k* as shown in the figure. μ_s is the coefficient of friction between *P* and *Q*. The blocks move together performing *SHM* of amplitude *A*. The maximum value of the friction force between *P* and *Q* is

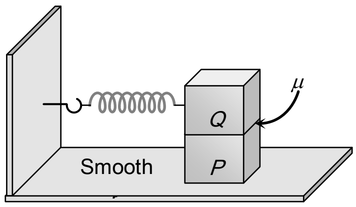
[IIT-JEE (Screening) 2004]

(a) kA

(b) $\frac{kA}{2}$

(c) Zero

(d) $\mu_s mg$



7. A body of mass *m* rests on horizontal surface. The coefficient of friction between the body and the surface is μ. If the mass is pulled by a force *P* as shown in the figure, the limiting friction between body and surface will be

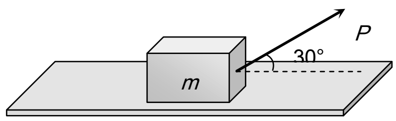
[BHU 2004]

(a) μmg

(b) $\mu\left[mg+\left(\frac{P}{2}\right)\right]$

(c) $\mu\left[mg-\left(\frac{P}{2}\right)\right]$

(d) $\mu\left[mg-\left(\frac{\sqrt{3}\,P}{2}\right)\right]$



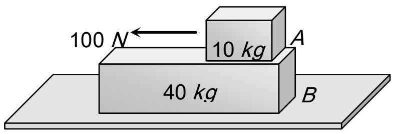
8. A 40 *kg* slab rests on a frictionless floor as shown in the figure. A 10 *kg* block rests on the top of the slab. The static coefficient of friction between the block and slab is 0.60 while the kinetic friction is 0.40. The 10 *kg* block is acted upon by a horizontal force 100 *N*. If $g = 9.8\,m/s^2$, the resulting acceleration of the slab will be **[NCERT 1982]**

(a) $0.98\, m/s^2$

(b) $1.47\, m/s^2$

(c) $1.52\, m/s^2$

(d) $6.1\, m/s^2$



9. A block of mass 2 *kg* rests on a rough inclined plane making an angle of 30° with the horizontal. The coefficient of static friction between the block and the plane is 0.7. The frictional force on the block is **[IIT 1980; J & K CET 2004]**

(a) 9.8 *N*

(b) $0.7 \times 9.8 \times \sqrt{3}\, N$

(c) $9.8 \times \sqrt{3}\, N$

(d) $0.8 \times 9.8\, N$

10. When a bicycle is in motion, the force of friction exerted by the ground on the two wheels is such that it acts

[IIT 1990; Manipal MEE 1995; MP PET 1996]

(a) In the backward direction on the front wheel and in the forward direction on the rear wheel

(b) In the forward direction on the front wheel and in the backward direction on the rear wheel

(c) In the backward direction on both front and the rear wheels

(d) In the forward direction on both front and the rear wheels

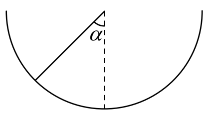
11. An insect crawls up a hemispherical surface very slowly (see the figure). The coefficient of friction between the insect and the surface is 1/3. If the line joining the centre of the hemispherical surface to the insect makes an angle α with the vertical, the maximum possible value of α is given by **[IIT-JEE 2001]**

(a) $\cot\alpha = 3$

(b) $\tan\alpha = 3$

(c) $\sec\alpha = 3$

(d) $\operatorname{cosec}\alpha = 3$



AR Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

(a) If both assertion and reason are true and the reason is the correct explanation of the assertion.

(b) If both assertion and reason are true but reason is not the correct explanation of the assertion.

(c) If assertion is true but reason is false.

(d) If the assertion and reason both are false.

(e) If assertion is false but reason is true.

1. Assertion : On a rainy day, it is difficult to drive a car or bus at high speed.

Reason : The value of coefficient of friction is lowered due to wetting of the surface.

2. Assertion : When a bicycle is in motion, the force of friction exerted by the ground on the two wheels is always in forward direction.

Reason : The frictional force acts only when the bodies are in contact.

3. Assertion : Pulling a lawn roller is easier than pushing it.

Reason : Pushing increases the apparent weight and hence the force of friction.

4. Assertion : Angle of repose is equal to angle of limiting friction.

Reason : When the body is just at the point of motion, the force of friction in this stage is called as limiting friction.

5. Assertion : Two bodies of masses *M* and *m* (*M* > *m*) are allowed to fall from the same height if the air resistance for each be the same then both the bodies will reach the earth simultaneously.

Reason : For same air resistance, acceleration of both the bodies will be same.

6. Assertion : Friction is a self adjusting force.

Reason : Friction does not depend upon mass of the body.

7. Assertion : The value of dynamic friction is less than the limiting friction.

Reason : Once the motion has started, the inertia of rest has been overcome.

8. Assertion : The acceleration of a body down a rough inclined plane is greater than the acceleration due to gravity.

Reason : The body is able to slide on a inclined plane only when its acceleration is greater than acceleration due to gravity.

Answers

Static and Limiting Friction

1	c	2	d	3	c	4	b	5	a
6	c	7	c	8	a	9	a	10	d
11	c	12	c	13	b	14	a	15	a
16	d	17	c	18	b	19	a	20	c
21	c	22	b	23	a	24	c	25	a
26	b	27	a	28	d	29	a	30	d
31	d	32	a						

Kinetic Friction

1	b	2	d	3	a	4	d	5	a
6	c	7	a	8	d	9	a	10	b
11	b	12	d	13	c	14	c	15	b
16	d	17	b	18	a	19	a	20	d
21	d	22	b	23	b	24	b	25	d
26	b	27	b	28	d	29	c	30	b
31	a	32	d	33	a	34	b	35	d
36	d								

Motion on Inclined Surface

1	b	2	b	3	c	4	a	5	a
6	c	7	a	8	d	9	d	10	c
11	b	12	a	13	a	14	d	15	c
16	b	17	c	18	d	19	a	20	a
21	a	22	a	23	a	24	b		

Critical Thinking Questions

1	a	2	c	3	b	4	c	5	a
6	b	7	c	8	a	9	a	10	ac
11	a								

Assertion & Reason

1	a	2	e	3	a	4	b	5	d
6	d	7	a	8	d				

Answers and Solutions

Static and Limiting Friction

1. (c)

2. (d) $\mu = \frac{F}{R} = \frac{F}{mg} = \frac{98}{100 \times 9.8} = \frac{1}{10} = 0.1$

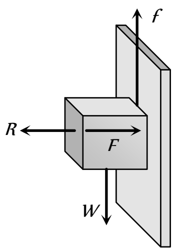
3. (c) Here applied horizontal force F acts as normal reaction.

For holding the block

Force of friction = Weight of block

$f = W \Rightarrow \ \mu R = W \Rightarrow \mu F = W$

$\Rightarrow \ F = \frac{W}{\mu}$



As $\mu < 1 \ \therefore \ \ F > W$

4. (b)

5. (a)

6. (c)

7. (c) $F_l = \mu_s R = 0.4 \times mg = 0.4 \times 10 = 4\,N$ *i.e.* minimum 4*N* force is required to start the motion of a body. But applied force is only 3*N*. So the block will not move.

8. (a) For limiting condition $\mu = \frac{m_B}{m_A + m_C} \Rightarrow 0.2 = \frac{5}{10 + m_C}$

$\Rightarrow \ 2 + 0.2 m_C = 5 \Rightarrow \ m_C = 15 kg$

9. (a)

10. (d) Ball and bearing produce rolling motion for which force of friction is low. Lubrication and polishing reduce roughness of surface.

11. (c) For given condition we can apply direct formula

$l_1 = \left(\frac{\mu}{\mu + 1} \right) l$

12. (c) Sliding friction is greater than rolling friction.

13. (b) $F = \frac{W}{\mu} = \frac{1 \times 9.8}{0.2} = 49\,N$

14. (a) $l' = \left(\frac{\mu}{\mu + 1} \right) l = \left(\frac{0.25}{0.25 + 1} \right) l = \frac{l}{5} = 20\%$ of *l*.

15. (a) $\mu = \frac{m_B}{m_A} \Rightarrow 0.2 = \frac{m_B}{10} \Rightarrow m_B = 2kg$

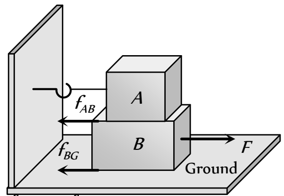
16. (d) Work done by friction can be positive, negative and zero depending upon the situation.

17. (c) $\mu = \frac{\text{Lenght of chain hanging from the table}}{\text{Lenght of chain lying on the table}} = \frac{l}{L - l}$

18. (b) Surfaces always slide over each other.

19. (a) Coefficient of friction $\mu_s = \frac{F_l}{R} = \frac{75}{mg} = \frac{75}{20 \times 9.8} = 0.38$

20. (c)



$F = f_{AB} + f_{BG}$

$= \mu_{AB} m_a g + \mu_{BG}(m_A + m_B) g$

$= 0.2 \times 100 \times 10$

$+ 0.3(300) \times 10$

$= 200 + 900 = 1100\ N$

21. (c)

22. (b) $\mu = \tan$ (Angle of repose) $= \tan 60° = 1.732$

23. (a) Applied force $= 2.5\ N$

Limiting friction $= \mu mg = 0.4 \times 2 \times 9.8 = 7.84\ N$

For the given condition applied force is very smaller than limiting friction.

$\therefore$ Static friction on a body = Applied force = 2.5 *N*

24. (c) Sand is used to increase the friction.

25. (a) $F = \mu R = 0.3 \times 250 = 75\ N$

26. (b) For the given condition, Static friction

= Applied force = Weight of body $= 2 \times 10 = 20\ N$

27. (a) $F = \frac{W}{\mu} \quad \therefore \ W = \mu F = 0.2 \times 10 = 2N$

28. (d) $\mu_s = \frac{m_B}{m_A} \Rightarrow 0.2 = \frac{m_B}{2} \Rightarrow m_B = 0.4\ kg$

29. (a) $\mu_s = \frac{m_B}{m_A} \Rightarrow 0.2 = \frac{m_B}{10} \Rightarrow m_B = 2\ kg$

30. (d) $\mu_s = \frac{\text{Lenght of the chain hanging from the table}}{\text{Length of the chain lying on the table}}$

$= \frac{l/3}{l - l/3} = \frac{l/3}{2l/3} = \frac{1}{2}$

31. (d)

32. (a)

Kinetic Friction

1. (b)

2. (d) In the given condition the required centripetal force is provided by frictional force between the road and tyre.

$\frac{mv^2}{R} = \mu mg \quad \therefore \quad v = \sqrt{\mu Rg}$

3. (a) Retarding force $F = ma = \mu R = \mu mg \ \therefore \ a = \mu g$

Now from equation of motion $v^2 = u^2 - 2as$

$\Rightarrow 0 = u^2 - 2as \Rightarrow \ s = \frac{u^2}{2a} = \frac{u^2}{2\mu g} \ \therefore \ = \frac{v_0^2}{2\mu g}$

4. (d) Net force = Applied force – Friction force

$ma = 24 - \mu\, mg = 24 - 0.4 \times 5 \times 9.8 = 24 - 19.6$

$\Rightarrow a = \dfrac{4.4}{5} = 0.88\ m/s^2$

5. (a) Work done = Force × Displacement $= \mu\, mg \times (v \times t)$

$W = (0.2) \times 2 \times 9.8 \times 2 \times 5\ joule$

Heat generated $Q = \dfrac{W}{J} = \dfrac{0.2 \times 2 \times 9.8 \times 2 \times 5}{4.2} = 9.33\ cal$

6. (c) For given condition $s \propto \dfrac{1}{m^2}$ $\therefore \dfrac{s_2}{s_1} = \left(\dfrac{m_1}{m_2}\right)^2 = \left(\dfrac{200}{300}\right)^2$

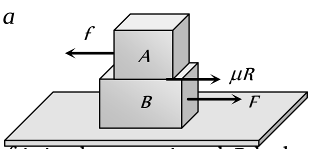
$\Rightarrow s_2 = s_1 \times \dfrac{4}{9} = 36 \times \dfrac{4}{9} = 16\ m$

7. (a) There is no friction between the body B and surface of the table. If the body B is pulled with force F then

$$F = (m_A + m_B)a$$

Due to this force upper body A will feel the pseudo force in a backward direction.

$f = m_A \times a$



But due to friction between A and B, body will not move. The body A will start moving when pseudo force is more than friction force.

i.e. for slipping, $m_A a = \mu\, m_A g$ $\therefore a = \mu g$

8. (d) Limiting friction $= \mu_s R = \mu_s mg = 0.5 \times 60 \times 10 = 300\ N$

Kinetic friction $= \mu_k R = \mu_k mg = 0.4 \times 60 \times 10 = 240\ N$

Force applied on the body = 300 N and if the body is moving then, Net accelerating force

=Applied force – Kinetic friction

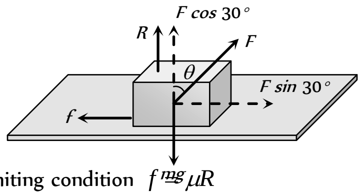
$\Rightarrow ma = 300 - 240 = 60$ $\therefore a = \dfrac{60}{60} = 1\ m/s^2$

9. (a) $v = \sqrt{\mu\, g\, r} \Rightarrow r = \dfrac{v^2}{\mu g} = \dfrac{100}{0.5 \times 10} = 20$

10. (b)

11. (b) $S = \dfrac{u^2}{2\mu g} = \dfrac{(10)^2}{2 \times 0.2 \times 10} = 25\ m$

12. (d)



For limiting condition $f = \mu R$

$F \sin 30° = \mu(mg - F \cos 30°)$, By solving $F = 294.3\ N$

13. (c) Net force on the body = Applied force – Friction

$ma = F - \mu_k mg \Rightarrow \mu_k = \dfrac{F - ma}{mg} = \dfrac{129.4 - 10 \times 10}{10 \times 9.8} = 0.3$

14. (c) $v = \sqrt{\mu g r} = \sqrt{0.5 \times 9.8 \times 40} = \sqrt{196} = 14\ m/s$

15. (b) $s = \dfrac{u^2}{2\mu\, g} = \dfrac{(20)^2}{2 \times 0.5 \times 10} = 40\ m$

16. (d) Net force in forward direction = Accelerating force + Friction

$= ma + \mu\, mg = m(a + \mu\, g) = (1500 + 500)(1 + 0.2 \times 10)$

$= 2000 \times 3 = 6000\ N$

17. (b) $v = \sqrt{\mu r g} = \sqrt{0.4 \times 30 \times 9.8} = 10.84\ m/s$

18. (a) $W = \mu mgS = 0.2 \times 50 \times 9.8 \times 1 = 98\ J$

19. (a) $F_l = \mu mg = 0.6 \times 1 \times 9.8 = 5.88\ N$

Pseudo force on the block = $ma = 1 \times 5 = 5\ N$

Pseudo is less then limiting friction hence static force of friction = 5 N.

20. (d) $S = \dfrac{u^2}{2\mu g} = \dfrac{m^2 u^2}{2\mu g m^2} = \dfrac{P^2}{2\mu m^2 g}$

21. (d) Weight of the body = 64N

so mass of the body $m = 6.4\ kg$, $\mu_s = 0.6$, $\mu_k = 0.4$

Net acceleration $= \dfrac{\text{Applied force - Kinetic friction}}{\text{Mass of the body}}$

$= \dfrac{\mu_s mg - \mu_k mg}{m} = (\mu_s - \mu_k)g = (0.6 - 0.4)g = 0.2\, g$

22. (b)

23. (b) $a = \dfrac{\text{Applied force} - \text{Kinetic friction}}{\text{mass}}$

$= \dfrac{100 - 0.5 \times 10 \times 10}{10} = 5 m/s^2$

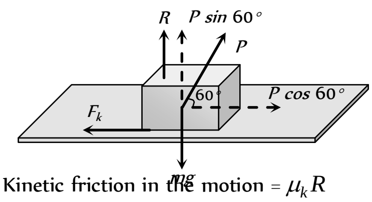
24. (b)

25. (d) $v = u - at \Rightarrow u - \mu g t = 0$ $\therefore \mu = \dfrac{u}{gt} = \dfrac{6}{10 \times 10} = 0.06$

26. (b) From the relation $F - \mu mg = ma$

$a = \dfrac{F - \mu mg}{m} = \dfrac{129.4 - 0.3 \times 10 \times 9.8}{10} = 10\ m/s^2$

27. (b) Let body is dragged with force P, making an angle 60° with the horizontal.



F_k = Kinetic friction in the motion $= \mu_k R$

From the figure $F_k = P \cos 60°$ and $R = mg - P \sin 60°$

$\therefore P \cos 60° = \mu_k (mg - P \sin 60°)$

$\Rightarrow \dfrac{P}{2} = 0.5\left(60 \times 10 - \dfrac{P\sqrt{3}}{2}\right) \Rightarrow P = 315.1\ N$

$\therefore F_k = P \cos 60° = \dfrac{315.1}{2} N$

Work done $= F_k \times s = \dfrac{315.1}{2} \times 2 = 315\ Joule$

28. (d) $v = u - at \Rightarrow t = \frac{u}{a}$ [As $v = 0$]

$t = \frac{u \times m}{F} = \frac{30 \times 1000}{5000} = 6$ sec

29. (c)

30. (b) Kinetic energy acquired by body

= (Total work done on the body) – (work against friction)

$= F \times S - \mu mgS = 25 \times 10 - 0.2 \times 5 \times 10 \times 10$

$= 250 - 100 = 150$ *Joule*

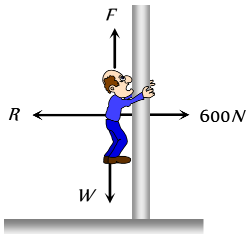
31. (a) $v = \sqrt{\mu rg} = \sqrt{0.5 \times 500 \times 10} = 50\ m/s$

32. (d) Net downward acceleration $= \frac{\text{Weight- Friction force}}{\text{Mass}}$

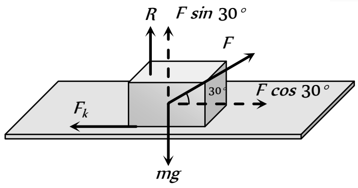
$= \frac{(mg - \mu R)}{m}$

$= \frac{60 \times 10 - 0.5 \times 600}{60}$

$= \frac{300}{60} = 5\ m/s^2$



33. (a)



Kinetic friction = $\mu_k R = 0.2(mg - F\sin 30°)$

$= 0.2\left(5 \times 10 - 40 \times \frac{1}{2}\right) = 0.2(50 - 20) = 6\ N$

Acceleration of the block $= \frac{F\cos 30° - \text{Kinetic friction}}{\text{Mass}}$

$= \frac{40 \times \frac{\sqrt{3}}{2} - 6}{5} = 5.73\ m/s^2$

34. (b) We know $s = \frac{u^2}{2\mu g}$ $\therefore$ $\mu = \frac{u^2}{2gs} = \frac{(6)^2}{2 \times 10 \times 9} = 0.2$

35. (d) $s = \frac{u^2}{2\mu g} = \frac{(100)^2}{2 \times 0.5 \times 10} = 1000\ m$

36. (d) Kinetic energy of the cylinder will go against friction

$\therefore \frac{1}{2}mv^2 = \mu\, mgs \Rightarrow s = \frac{u^2}{2\mu g} = \frac{(10)^2}{2 \times (0.5) \times 10} = 10\,m$

Motion on Inclined Surface

1. (b) When the body is at rest then static friction works on it, which is less than limiting friction (μR).

2. (b)

3. (c) Coefficient of friction = Tangent of angle of repose

$\therefore \mu = \tan\theta$

4. (a) $\mu = \tan\theta\left(1 - \frac{1}{n^2}\right) = 1 - \frac{1}{n^2}$ [As $\theta = 45°$]

5. (a) Retardation in upward motion $= g(\sin\theta + \mu\cos\theta)$

$\therefore$ Force required just to move up $F_{up} = mg(\sin\theta + \mu\cos\theta)$

Similarly for down ward motion $a = g(\sin\theta - \mu\cos\theta)$

$\therefore$ Force required just to prevent the body sliding down

$F_{dn} = mg(\sin\theta - \mu\cos\theta)$

According to problem $F_{up} = 2F_{dn}$

$\Rightarrow mg(\sin\theta + \mu\cos\theta) = 2mg(\sin\theta - \mu\cos\theta)$

$\Rightarrow \sin\theta + \mu\cos\theta = 2\sin\theta - 2\mu\cos\theta$

$\Rightarrow 3\mu\cos\theta = \sin\theta \Rightarrow \tan\theta = 3\mu$

$\Rightarrow \theta = \tan^{-1}(3\mu) = \tan^{-1}(3 \times 0.25) = \tan^{-1}(0.75) = 36.8°$

6. (c) $\mu = \tan\theta\left(1 - \frac{1}{n^2}\right)$

$\theta = 45°$ and $n = 2$ (Given)

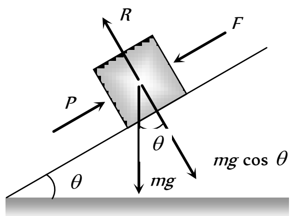
$\therefore \mu = \tan 45°\left(1 - \frac{1}{2^2}\right) = 1 - \frac{1}{4} = \frac{3}{4} = 0.75$

7. (a) $a = g(\sin\theta - \mu\cos\theta) = 9.8(\sin 45^o - 0.5\cos 45^o)$

$= \frac{4.9}{\sqrt{2}}\ m/\sec^2$

8. (d) Because if the angle of inclination is equal to or more than angle of repose then box will automatically slides down the plane.

9. (d)



Net force along the plane

$= P - mg\sin\theta = 750 - 500 = 250\ N$

Limiting friction = $F_l = \mu_s R = \mu_s mg\cos\theta$

= 0.4 × 102 × 9.8 × cos 30 = 346 *N*

As net external force is less than limiting friction therefore friction on the body will be 250 *N*.

10. (c) $a = g(\sin\theta - \mu\cos\theta) = 10(\sin 60° - 0.25\cos 60°)$

$a = 7.4\ m/s^2$

11. (b) $F_k = \mu_k R = \mu_k mg\cos\theta$

$F_k = 1.7 \times 0.1 \times 10 \times \cos 30° = 1.7 \times \frac{\sqrt{3}}{2} N$

12. (a) $\mu = \tan\theta\left(1 - \frac{1}{n^2}\right) = \tan 30\left(1 - \frac{1}{2^2}\right) = \frac{\sqrt{3}}{4}$

13. (a) For angle of repose,

Friction =Component of weight along the plane

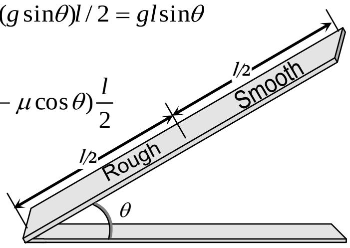
$= mg\sin\theta = 2\times 9.8\times\sin 45^o = 19.6\sin 45^o$

14. (d) For upper half

$v^2 = u^2 + 2al/2 = 2(g\sin\theta)l/2 = gl\sin\theta$

For lower half

$\Rightarrow 0 = u^2 + 2g(\sin\theta - \mu\cos\theta)\dfrac{l}{2}$



$\Rightarrow -gl\sin\theta = gl(\sin\theta - \mu\cos\theta)$

$\Rightarrow \mu\cos\theta = 2\sin\theta \Rightarrow \mu = 2\tan\theta$

15. (c) Resultant downward force along the incline

$= mg(\sin\theta - \mu\cos\theta)$

Normal reaction $= mg\cos\theta$

Given : $mg\cos\theta = 2mg(\sin\theta - \mu\cos\theta)$

By solving $\theta = 45^o$.

16. (b) $F = mg(\sin\theta + \mu\cos\theta)$

$= 10\times 9.8(\sin 30° + 0.5\cos 30°) = 91.4\ N$.

17. (c) $W = \mu mg\cos\theta S = 0.5\times 1\times 9.8\times\dfrac{1}{2}\times 1 = 2.45\ J$

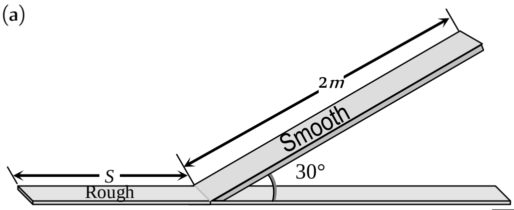
18. (d) $F = mg\sin 30° = 50N = 5kg\text{-}wt$.

19. (a) $\mu = \tan 30° = \dfrac{1}{\sqrt{3}}$.

20. (a) Work done against gravity $= mgh \quad = 2\times 10\times 10 = 200\ J$

Work done against friction = (Total work done – work done against gravity) $= 300 - 200 = 100J$

21. (a)



$v^2 = u^2 + 2as = 0 + 2\times g\sin 30\times 2 \Rightarrow v = \sqrt{20}$

Let it travel distance 'S' before coming to rest

$S = \dfrac{v^2}{2\mu g} = \dfrac{20}{2\times 0.25\times 10} = 4m$

22. (a) Angle of repose $\alpha = \tan^{-1}(\mu) = \tan^{-1}(0.8) = 38.6°$

Angle of inclined plane is given $\theta = 30°$. It means block is at rest therefore,

Static friction = component of weight in downward direction

$= mg\sin\theta = 10\ N \quad \therefore\ m = \dfrac{10}{9\times\sin 30°} = 2\ kg$

23. (a) $\mu = \tan\theta\left(1 - \dfrac{1}{n^2}\right) = \tan\theta\left(1 - \dfrac{1}{2^2}\right) = \dfrac{3}{4}\tan\theta$

24. (b) Acceleration (a) $= g(\sin\theta - \mu\cos\theta)$ and $s = l$

$v = \sqrt{2as} = \sqrt{2gl(\sin\theta - \mu\cos\theta)}$

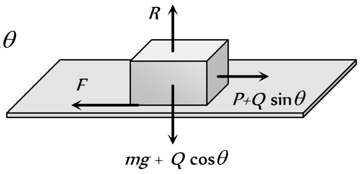
Critical Thinking Questions

1. (a) By drawing the free body diagram of the block for critical condition

$F = \mu R \Rightarrow P + Q\sin\theta$

$= \mu(mg + Q\cos\theta)$

$\therefore\ \mu = \dfrac{P + Q\sin\theta}{mg + Q\cos\theta}$

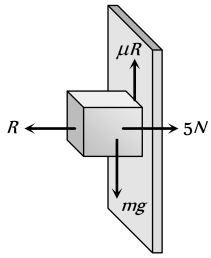


2. (c)

3. (b) Limiting friction

$F_l = \mu_s R = 0.5\times(5) = 2.5\ N$

Since downward force is less than limiting friction therefore block is at rest so the static force of friction will work on it.



F_s = downward force = Weight

$= 0.1\times 9.8 = 0.98\ N$

4. (c) Maximum force by surface when friction works

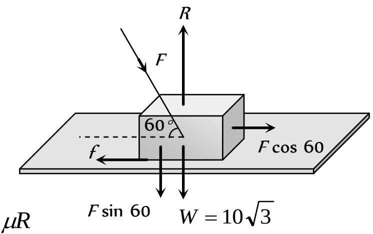
$F = \sqrt{f^2 + R^2} = \sqrt{(\mu R)^2 + R^2} = R\sqrt{\mu^2 + 1}$

Minimum force $= R$ when there is no friction

Hence ranging from R to $R\sqrt{\mu^2 + 1}$

We get, $Mg \le F \le Mg\sqrt{\mu^2 + 1}$

5. (a)



$f = \mu R$

$F\cos 60° = \mu(W + F\sin 60°)$

Substituting $\mu = \dfrac{1}{2\sqrt{3}}$ & $W = 10\sqrt{3}$ we get $F = 20N$

6. (b) When two blocks performs simple harmonic motion together then at the extreme position (at amplitude $= A$)

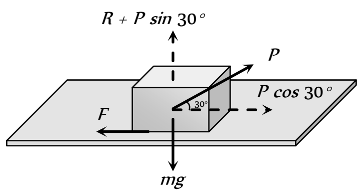
Restoring force $F = KA = 2ma \Rightarrow a = \dfrac{KA}{2m}$

There will be no relative motion between P and Q if pseudo force on block P is less than or just equal to limiting friction between P and Q.

i.e. $m\left(\dfrac{KA}{2m}\right) =$ Limiting friction

$\therefore$ Maximum friction $= \dfrac{KA}{2}$

7. (c) Normal reaction $R = mg - P\sin 30^o = mg - \dfrac{P}{2}$



$\therefore$ Limiting friction between body and surface is given by, $F = \mu R = \mu\left(mg - \frac{P}{2}\right)$.

8. (a) Limiting friction between block and slab $= \mu_s m_A g$

$$= 0.6 \times 10 \times 9.8 = 58.8 N$$

But applied force on block A is 100 N. So the block will slip over a slab.

Now kinetic friction works between block and slab $F_k = \mu_k m_A g = 0.4 \times 10 \times 9.8 = 39.2\ N$

This kinetic friction helps to move the slab

$\therefore$ Acceleration of slab $= \frac{39.2}{m_B} = \frac{39.2}{40} = 0.98\ m/s^2$

9. (a) Limiting friction $F_l = \mu\, mg \cos\theta$

$F_l = 0.7 \times 2 \times 10 \times \cos 30° = 12\ N$ (approximately)

But when the block is lying on the inclined plane then component of weight down the plane $= mg \sin\theta$

$$= 2 \times 9.8 \times \sin 30° = 9.8\ N$$

It means the body is stationary, so static friction will work on it

$\therefore$ Static friction = Applied force = 9.8 N

10. (a,c) In cycling, the rear wheel moves by the force communicated to it by pedalling while front wheel moves by it self. So, while pedalling a bicycle, the force exerted by rear wheel on ground makes force of friction act on it in the forward direction (like walking). Front wheel moving by itself experience force of friction in backward direction (like rolling of a ball). [However, if pedalling is stopped both wheels move by themselves and so experience force of friction in backward direction].

11. (a)

Assertion & Reason

1. (a) On a rainy day, the roads are wet. Wetting of roads lowers the coefficient of friction between the tyres and the road. Therefore, grip of car on the road reduces and thus chances of skidding increases.

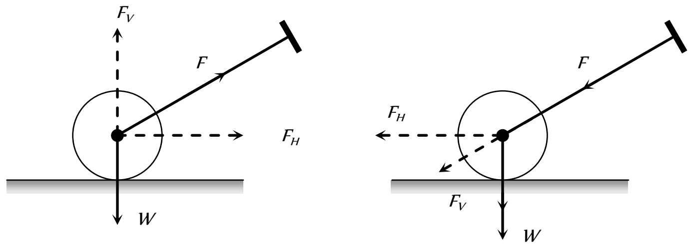
2. (e) When a bicycle is in motion, two cases may arise :

(i) When the bicycle is being pedalled. In this case, the applied force has been communicated to rear wheel. Due to which the rear wheel pushes the earth backwards. Now the force of friction acts in the forward direction on the rear wheel but front wheel move forward due to inertia, so force of friction works on it in backward direction

(ii) When the bicycle is not being pedalled :

In this case both the wheels move in forward direction, due to inertia. Hence force of friction on both the wheels acts in backward direction.

3. (a) Suppose the roller is pushed as in figure (b). The force F is resolved into two components, horizontal component F_H which helps the roller to move forward, and the vertical component acting downwards adds to the weight. Thus weight is increased. But in the case of pull [fig (a)] the vertical component is opposite to its weight. Thus weight is reduced. So pulling is easier than pushing the lawn roller.



(a) (b)

4. (b)

5. (d) The force acting on the body of mass M are its weight Mg acting vertically downwards and air resistance F acting vertically upward.

$\therefore$ Acceleration of the body, $a = \frac{Mg - F}{M} = g - \frac{F}{M}$

Now, $M > m$, therefore, the body with larger mass will have greater acceleration and it will reach the ground first.

6. (d) Only static friction is a self adjusting force. This is because force of static friction is equal and opposite to applied force (so long as actual motion does not start). Frictional force = μmg *i.e.* friction depends on mass.

7. (a)

8. (d) Acceleration down a rough inclined plane

$a = g(\sin\theta - \mu \cos\theta)$ and this is less than g.

Friction

SET Self Evaluation Test -5

1. A force of 19.6 N when applied parallel to the surface just moves a body of mass 10 kg kept on a horizontal surface. If a 5 kg mass is kept on the first mass, the force applied parallel to the surface to just move the combined body is

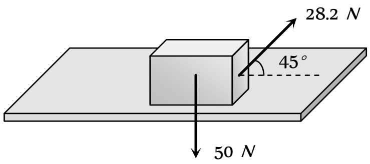
(a) 29.4 N (b) 39.2 N
(c) 18.6 N (d) 42.6 N

2. If the normal force is doubled, the coefficient of friction is

(a) Not changed (b) Halved
(c) Doubled (d) Tripled

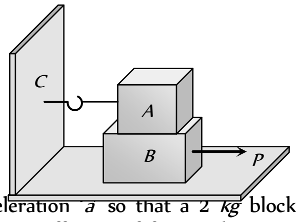
3. A body of weight 50 N placed on a horizontal surface is just moved by a force of 28.2 N. The frictional force and the normal reaction are

(a) 10 N, 15 N
(b) 20 N, 30 N
(c) 2 N, 3 N
(d) 5 N, 6 N



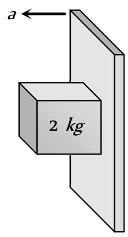
4. Block A weighing 100 kg rests on a block B and is tied with a horizontal string to the wall at C. Block B weighs 200 kg. The coefficient of friction between A and B is 0.25 and between B and the surface is 1/3. The horizontal force P necessary to move the block B should be $(g = 10\,m/s^2)$

(a) 1150 N
(b) 1250 N
(c) 1300 N
(d) 1420 N



5. A rough vertical board has an acceleration 'a' so that a 2 kg block pressing against it does not fall. The coefficient of friction between the block and the board should be

(a) $> g/a$
(b) $< g/a$
(c) $= g/a$
(d) $> a/g$



6. A stone weighing 1 kg and sliding on ice with a velocity of 2 m/s is stopped by friction in 10 sec. The force of friction (assuming it to be constant) will be

(a) $-20\,N$ (b) $-0.2\,N$
(c) 0.2 N (d) 20 N

7. A body of mass 10 kg slides along a rough horizontal surface. The coefficient of friction is $1/\sqrt{3}$. Taking $g = 10\,m/s^2$, the least force which acts at an angle of 30° to the horizontal is

(a) 25 N (b) 100 N
(c) 50 N (d) $\frac{50}{\sqrt{3}}\,N$

8. A lift is moving downwards with an acceleration equal to acceleration due to gravity. A body of mass M kept on the floor of the lift is pulled horizontally. If the coefficient of friction is μ , then the frictional resistance offered by the body is

(a) Mg (b) μMg
(c) $2\mu Mg$ (d) Zero

9. In the above question, if the lift is moving upwards with a uniform velocity, then the frictional resistance offered by the body is

(a) Mg (b) μMg
(c) $2\mu Mg$ (d) Zero

10. A body of mass 2 kg is moving on the ground comes to rest after some time. The coefficient of kinetic friction between the body and the ground is 0.2. The retardation in the body is

(a) $9.8\,m/s^2$ (b) $4.73\,m/s^2$
(c) $2.16\,m/s^2$ (d) $1.96\,m/s^2$

11. A cyclist moves in a circular track of radius 100 m. If the coefficient of friction is 0.2, then the maximum velocity with which the cyclist can take the turn with leaning inwards is

(a) 9.8 m/s (b) 1.4 m/s
(c) 140 m/s (d) 14 m/s

12. A block of mass 5 kg lies on a rough horizontal table. A force of 19.6 N is enough to keep the body sliding at uniform velocity. The coefficient of sliding friction is

(a) 0.5 (b) 0.2
(c) 0.4 (d) 0.8

13. A motor car has a width 1.1 m between wheels. Its centre of gravity is 0.62 m above the ground and the coefficient of friction between the wheels and the road is 0.8. What is the maximum possible speed, if the centre of gravity inscribes a circle of radius 15 m ? (Road surface is horizontal)

(a) 7.64 m/s (b) 6.28 m/s
(c) 10.84 m/s (d) 11.23 m/s

14. A child weighing 25 kg slides down a rope hanging from the branch of a tall tree. If the force of friction acting against him is 2 N, what is the acceleration of the child (Take $g = 9.8\,m/s^2$)

(a) $22.5\,m/s^2$ (b) $8\,m/s^2$
(c) $5\,m/s^2$ (d) $9.72\,m/s^2$

AS Answers and Solutions

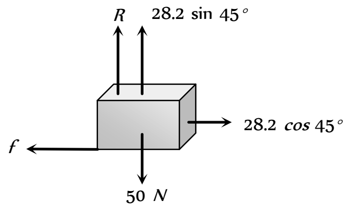
(SET -5)

1. (a) $F_l \propto R \quad \therefore \quad F_l \propto m$ *i.e.* limiting friction depends upon the mass of body. So, $\frac{(F_l)'}{(F_l)} = \frac{m'}{m} = \frac{10+5}{10}$

$$\Rightarrow (F_l)' = \frac{3}{2} \times F_l = \frac{3}{2} \times 19.6 = 29.4\ N$$

2. (a) Coefficient of friction is constant for two given surface in contact. It does not depend upon the weight or normal reaction.

3. (b)



Frictional force = $f = 28.2\cos 45° = 28.2 \times \frac{1}{\sqrt{2}} = 20\ N$

Normal reaction $R = 50 - 28.2\sin 45° = 30\ N$

4. (b) Friction between block A and block B & between block B and surface will oppose the P

$$\therefore\ P = F_{AB} + F_{BS} = \mu_{AB} m_A g + \mu_{BS}(m_A + m_B)g$$

$$= 0.25 \times 100 \times 10 + \frac{1}{3}(100 + 200) \times 10 = 1250\ N$$

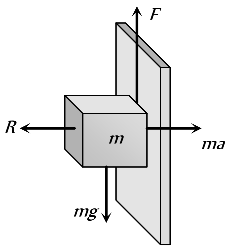
5. (a) For the limiting condition upward friction force between board and block will balance the weight of the block.

i.e. $F > mg$

$\Rightarrow \mu(R) > mg$

$\Rightarrow \mu(ma) > mg$

$\Rightarrow \mu > \frac{g}{a}$



6. (b) $u = 2\ m/s,\ v = 0,\ t = 10$ sec

$$\therefore\ a = \frac{v-u}{t} = \frac{0-2}{10} = -\frac{2}{10} = -\frac{1}{5} = -0.2\ m/s^2$$

$\therefore$ Friction force $= ma = 1 \times (-0.2) = -0.2\ N$

7. (c) Let P force is acting at an angle 30° with the horizontal.

For the condition of motion $F = \mu R$

$$P\cos 30° = \mu(mg - P\sin 30°)$$

$$\Rightarrow P\frac{\sqrt{3}}{2} = \frac{1}{\sqrt{3}}\left(100 - P\frac{1}{2}\right) \Rightarrow \frac{3P}{2} = \left(100 - \frac{P}{2}\right)$$

$$\Rightarrow 2P = 100 \quad \therefore\ P = 50\ N$$

8. (d) $R = m(g-a)$ for downward motion of lift

If $a = g$ then $R = 0 \quad \therefore\ F = \mu R = 0$

9. (b) When the lift is moving upward with constant velocity then, $R = mg \quad \therefore\ F = \mu R = \mu mg$

10. (d) We know that $a = \mu g = 0.2 \times 9.8 = 1.96\ m/s^2$

11. (d) $v = \sqrt{\mu r g} = \sqrt{0.2 \times 100 \times 10} = 10\sqrt{2} = 14\ m/s$

12. (c) $\mu_k = \frac{F}{R} = \frac{19.6}{5 \times 9.8} = \frac{2}{5} = 0.4$

13. (c) $v = \sqrt{\mu g r} = \sqrt{0.8 \times 9.8 \times 15} = 10.84\ m/s$

14. (d) Net downward force = Weight – Friction

$$\therefore\ ma = 25 \times 9.8 - 2 \Rightarrow a = \frac{25 \times 9.8 - 2}{25} = 9.72\ m/s^2$$

***

Chapter

6

Work, Energy, Power and Collision

Introduction

The terms 'work', 'energy' and 'power' are frequently used in everyday language. A farmer clearing weeds in his field is said to be working hard. A woman carrying water from a well to her house is said to be working. In a drought affected region she may be required to carry it over large distances. If she can do so, she is said to have a large stamina or energy. Energy is thus the capacity to do work. The term power is usually associated with speed. In karate, a powerful punch is one delivered at great speed. In physics we shall define these terms very precisely. We shall find that there is a loose correlation between the physical definitions and the physiological pictures these terms generate in our minds.

Work is said to be done when a force applied on the body displaces the body through a certain distance in the direction of force.

Work Done by a Constant Force

Let a constant force $\vec{F}$ be applied on the body such that it makes an angle θ with the horizontal and body is displaced through a distance s

By resolving force $\vec{F}$ into two components :

(i) $F\cos\theta$ in the direction of displacement of the body.

(ii) $F\sin\theta$ in the perpendicular direction of displacement of the body.

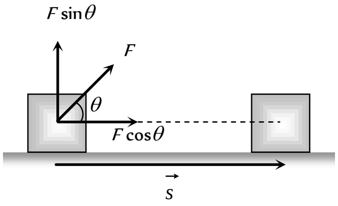


Fig. 6.1

Since body is being displaced in the direction of $F\cos\theta$, therefore work done by the force in displacing the body through a distance s is given by

$$W = (F\cos\theta)s = Fs\cos\theta$$

or $W = \vec{F}.\vec{s}$

Thus work done by a force is equal to the scalar (or dot product) of the force and the displacement of the body.

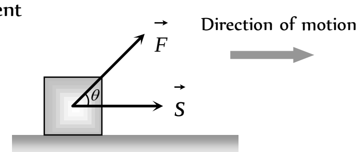
If a number of forces $\vec{F}_1, \vec{F}_2, \vec{F}_3 ...... \vec{F}_n$ are acting on a body and it shifts from position vector $\vec{r}_1$ to position vector $\vec{r}_2$ then

$$W = (\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + ... \vec{F}_n).(\vec{r}_2 - \vec{r}_1)$$

Nature of Work Done

Positive work

Positive work means that force (or its component) is parallel to displacement



$$0^o \le \theta < 90^o$$

Fig. 6.2

The positive work signifies that the external force favours the motion of the body.

Example: (i) When a person lifts a body from the ground, the work done by the (upward) lifting force is positive

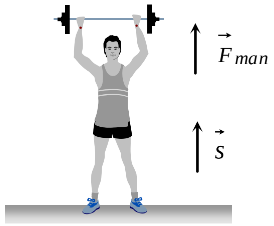


Fig. 6.3

(ii) When a lawn roller is pulled by applying a force along the handle at an acute angle, work done by the applied force is positive.

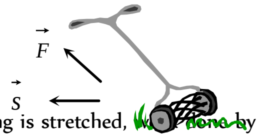


Fig. 6.4

(iii) When a spring is stretched, work done by the external (stretching) force is positive.

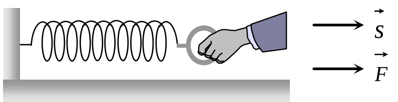
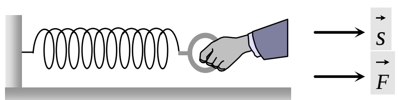


Fig. 6.5



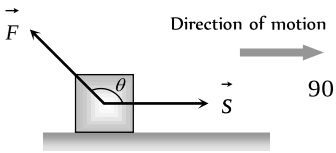
Maximum work : $W_{\max} = F\,s$

When $\cos\theta = \text{maximum} = 1$ *i.e.* $\theta = 0^o$

It means force does maximum work when angle between force and displacement is zero.

Negative work

Negative work means that force (or its component) is opposite to displacement *i.e.*



$90^o < \theta \leq 180^o$

Fig. 6.6

The negative work signifies that the external force opposes the motion of the body.

Example: (i) When a person lifts a body from the ground, the work done by the (downward) force of gravity is negative.

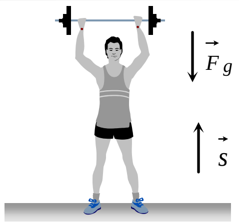


Fig. 6.7

(ii) When a body is moved over a rough surface, the work done by the frictional force is negative.

Minimum work : $W_{\min} = -F\,s$

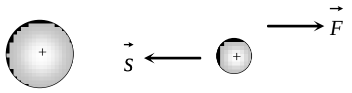


Fig. 6.8

When $\cos\theta = \text{minimum} = -1$ *i.e* $\theta = 180^o$

It means force does minimum [maximum negative] work when angle between force and displacement is 180°.

(iii) When a positive charge is moved towards another positive charge. The work done by electrostatic force between them is negative.

Zero work

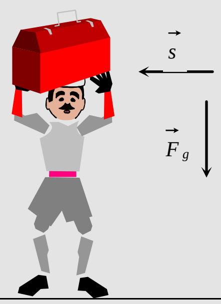
Under three condition, work done becomes zero $W = Fs\cos\theta = 0$

(1) **If the force is perpendicular to the displacement** $[\vec{F} \perp \vec{s}\,]$

Example: (i) When a coolie travels on a horizontal platform with a load on his head, work done against gravity by the coolie is zero.

(ii) When a body moves in a circle the work done by the centripetal force is always zero.

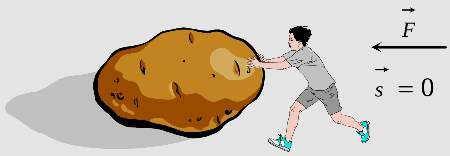
(iii) In case of motion of a charged particle in a magnetic field as force $[\vec{F} = q(\vec{v} \times \vec{B})]$ is always perpendicular to motion, work done by this force is always zero.



(2) **If there is no displacement** [s = 0]

Example: (i) When a person tries to displace a wall or heavy stone by applying a force and it does not move, then work done is zero.

(ii) A weight lifter does work in lifting the weight off the ground but does not work in holding it up.



(3) **If there is no force acting on the body** [F = 0]

Example: Motion of an isolated body in free space.

Work Done by a Variable Force

When the magnitude and direction of a force varies with position, the work done by such a force for an infinitesimal displacement is given by $dW = \vec{F}.\vec{ds}$

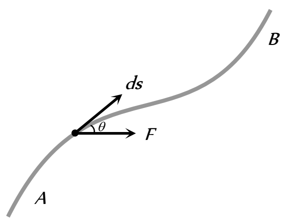


Fig. 6.9

The total work done in going from A to B as shown in the figure is

$$W = \int_A^B \vec{F}.\vec{ds} = \int_A^B (F\cos\theta)ds$$

In terms of rectangular component $\vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$

$$\vec{ds} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

$$\therefore\ W = \int_A^B (F_x\hat{i} + F_y\hat{j} + F_z\hat{k}).(dx\hat{i} + dy\hat{j} + dz\hat{k})$$

or $W = \int_{x_A}^{x_B} F_x\,dx + \int_{y_A}^{y_B} F_y\,dy + \int_{z_A}^{z_B} F_z\,dz$

Dimension and Units of Work

Dimension : As work = Force × displacement

$$[W]=[MLT^{-2}]\times[L]=[ML^2T^{-2}]$$

Units : The units of work are of two types

Absolute units	Gravitational units
Joule [S.I.]: Work done is said to be one *Joule*, when 1 *Newton* force displaces the body through 1 *metre* in its own direction. From, $W = F.s$ 1 *Joule* = 1 *Newton* ×1 *m*	*kg-m* [S.I.]: 1 *kg-m* of work is done when a force of 1*kg-wt.* displaces the body through 1*m* in its own direction. From $W = F s$ 1 *kg-m* = 1 *kg-wt* × 1 *m* = 9.81 *N* × 1 *metre* = 9.81 *Joule*
erg [C.G.S.] : Work done is said to be one *erg* when 1 *dyne* force displaces the body through 1 *cm* in its own direction. From $W = F s$ $1\,erg = 1 dyne \times 1cm$ **Relation between Joule and erg** 1 *Joule* = 1 *N* × 1 *m* = 10^5 *dyne* × 10^2 *cm* = 10^7 *dyne* × *cm* = 10^7 *erg*	*gm-cm* [C.G.S.] : 1 *gm-cm* of work is done when a force of 1*gm-wt* displaces the body through 1*cm* in its own direction. From $W = F s$ 1 *gm-cm* = 1*gm-wt* × 1*cm.* = 981 *dyne* × 1*cm* = 981 *erg*

Work Done Calculation by Force Displacement Graph

Let a body, whose initial position is x_i, is acted upon by a variable force (whose magnitude is changing continuously) and consequently the body acquires its final position x_f.

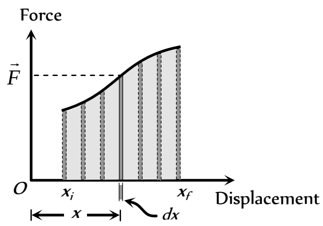


Fig. 6.10

Let F be the average value of variable force within the interval dx from position x to $(x + dx)$ *i.e.* for small displacement dx. The work done will be the area of the shaded strip of width dx. The work done on the body in displacing it from position x_i to x_f will be equal to the sum of areas of all the such strips

$$dW = \vec{F}\,dx$$

$$\therefore W = \int_{x_i}^{x_f} dW = \int_{x_i}^{x_f} F\,dx$$

$$\therefore W = \int_{x_i}^{x_f} (\text{Area of strip of width } dx)$$

$$\therefore W = \text{Area under curve between } x_i \text{ and } x_f$$

i.e. Area under force-displacement curve with proper algebraic sign represents work done by the force.

Work Done in Conservative and Non-conservative Field

(1) In conservative field, work done by the force (line integral of the force *i.e.* $\int \vec{F}.\vec{dl}$) is independent of the path followed between any two points.

$$\underset{\text{Path I}}{W_{A\to B}} = \underset{\text{Path II}}{W_{A\to B}} = \underset{\text{Path III}}{W_{A\to B}}$$

or $$\underset{\text{Path I}}{\int \vec{F}.\vec{dl}} = \underset{\text{Path II}}{\int \vec{F}.\vec{dl}} = \underset{\text{Path III}}{\int \vec{F}.\vec{dl}}$$

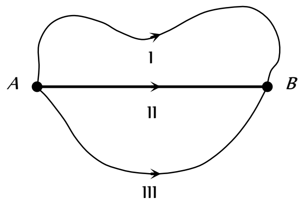


Fig. 6.11

(2) In conservative field work done by the force (line integral of the force *i.e.* $\int \vec{F}.\vec{dl}$) over a closed path/loop is zero.

$$W_{A\to B} + W_{B\to A} = 0$$

or $$\oint \vec{F}.\vec{dl} = 0$$

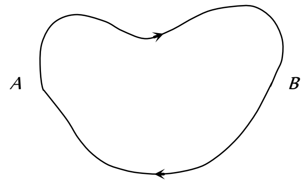


Fig. 6.12

Conservative force : The forces of these type of fields are known as conservative forces.

Example : Electrostatic forces, gravitational forces, elastic forces, magnetic forces *etc* and all the central forces are conservative in nature.

If a body of mass m lifted to height h from the ground level by different path as shown in the figure

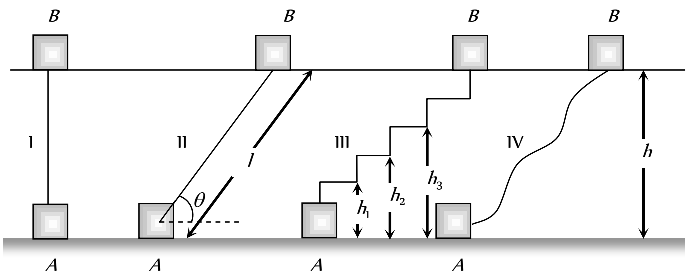


Fig. 6.13

Work done through different paths

$$W_I = F.s = mg \times h = mgh$$

$$W_{II} = F.s = mg\sin\theta \times l = mg\sin\theta \times \frac{h}{\sin\theta} = mgh$$

$$W_{III} = mgh_1 + 0 + mgh_2 + 0 + mgh_3 + 0 + mgh_4$$

$$= mg(h_1 + h_2 + h_3 + h_4) = mgh$$

$$W_{IV} = \int \vec{F}.\vec{ds} = mgh$$

It is clear that $W_I = W_{II} = W_{III} = W_{IV} = mgh$.

Further if the body is brought back to its initial position A, similar amount of work (energy) is released from the system, it means $W_{AB} = mgh$ and $W_{BA} = -mgh$.

Hence the net work done against gravity over a round trip is zero.

$$W_{Net} = W_{AB} + W_{BA} = mgh + (-mgh) = 0$$

i.e. the gravitational force is conservative in nature.

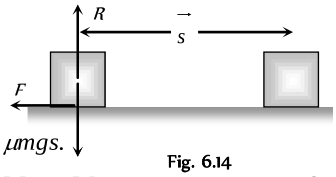
Non-conservative forces : A force is said to be non-conservative if work done by or against the force in moving a body from one position to another, depends on the path followed between these two positions and for complete cycle this work done can never be zero.

Example: Frictional force, Viscous force, Airdrag *etc.*

If a body is moved from position A to another position B on a rough table, work done against frictional force shall depend on the length of the path between A and B and not only on the position A and B.

$$W_{AB} = \mu mgs$$

Further if the body is brought back to its initial position A, work has to be done against the frictional force, which opposes the motion. Hence the net work done against the friction over a round trip is not zero.



$W_{BA} = \mu mgs.$

Fig. 6.14

$\therefore W_{Net} = W_{AB} + W_{BA} = \mu mgs + \mu mgs = 2\mu mgs \neq 0.$

i.e. the friction is a non-conservative force.

Work Depends on Frame of Reference

With change of frame of reference (inertial), force does not change while displacement may change. So the work done by a force will be different in different frames.

Examples : (1) If a porter with a suitcase on his head moves up a staircase, work done by the upward lifting force relative to him will be zero (as displacement relative to him is zero) while relative to a person on the ground will be *mgh.*

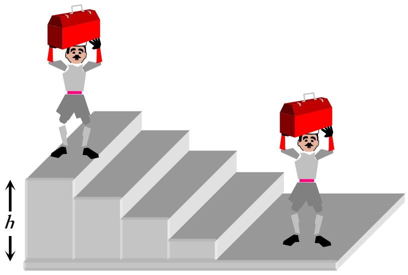


Fig. 6.15

(2) If a person is pushing a box inside a moving train, the work done in the frame of train will $\vec{F}.\vec{s}$ while in the frame of earth will be $\vec{F}.(\vec{s} + \vec{s}_0)$ where $\vec{s}_0$ is the displacement of the train relative to the ground.

Energy

The energy of a body is defined as its capacity for doing work.

(1) Since energy of a body is the total quantity of work done, therefore it is a scalar quantity.

(2) Dimension: $[ML^2T^{-2}]$ it is same as that of work or torque.

(3) Units : *Joule* [S.I.], *erg* [C.G.S.]

Practical units : *electron volt* (*eV*), Kilowatt hour (*KWh*), Calories (*cal*)

Relation between different units:

1 *Joule* = 10^7 *erg*

1 *eV* = 1.6×10^{-19} *Joule*

1 *kWh* = 3.6×10^6 *Joule*

1 *calorie* = 4.18 *Joule*

(4) Mass energy equivalence : Einstein's special theory of relativity shows that material particle itself is a form of energy.

The relation between the mass of a particle m and its equivalent energy is given as

$E = mc^2$ where c = velocity of light in vacuum.

If $m = 1\,amu = 1.67 \times 10^{-27}\,kg$

then $E = 931\,MeV = 1.5 \times 10^{-10}\,Joule$.

If $m = 1kg$ then $E = 9 \times 10^{16}\,Joule$

Examples : (i) Annihilation of matter when an electron (e^-) and a positron (e^+) combine with each other, they annihilate or destroy each other. The masses of electron and positron are converted into energy. This energy is released in the form of γ -rays.

$$e^- + e^+ \to \gamma + \gamma$$

Each γ photon has energy = 0.51 *MeV.*

Here two γ photons are emitted instead of one γ photon to conserve the linear momentum.

(ii) Pair production : This process is the reverse of annihilation of matter. In this case, a photon (γ) having energy equal to 1.02 *MeV* interacts with a nucleus and give rise to electron (e^-) and positron (e^+). Thus energy is converted into matter.

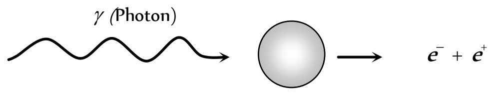


Fig. 6.16

(iii) Nuclear bomb : When the nucleus is split up due to mass defect (The difference in the mass of nucleons and the nucleus), energy is released in the form of γ -radiations and heat.

(5) Various forms of energy

(i) Mechanical energy (Kinetic and Potential)

(ii) Chemical energy

(iii) Electrical energy

(iv) Magnetic energy

(v) Nuclear energy

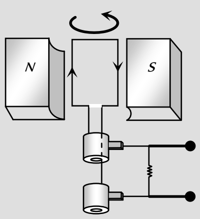
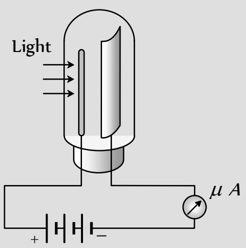
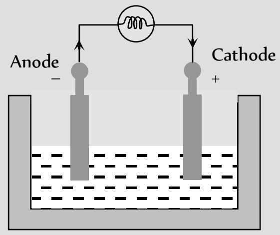
(vi) Sound energy

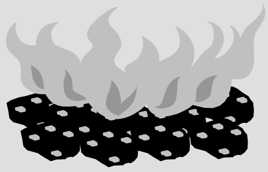
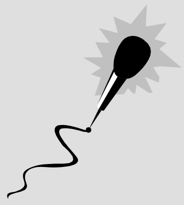
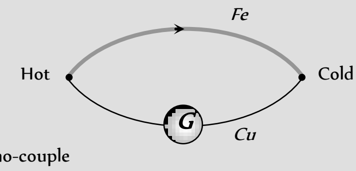
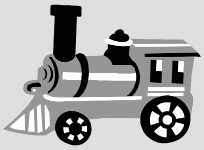
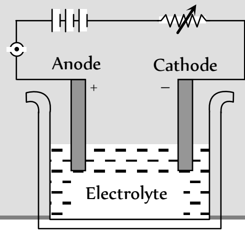
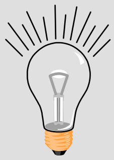
(vii) Light energy

(viii) Heat energy

(6) Transformation of energy : Conversion of energy from one form to another is possible through various devices and processes.

Table : 6.1 Various devices for energy conversion from one form to another

Mechanical $\to$ electrical	Light $\to$ Electrical	Chemical $\to$ electrical
Dynamo 	Photoelectric cell 	Primary cell 

Chemical → heat	Sound → Electrical	Heat → electrical
Coal Burning 	Microphone 	Thermo-couple 
Heat → Mechanical	**Electrical → Mechanical**	**Electrical → Heat**
Engine 	Motor 	Heater 
Electrical → Sound	**Electrical → Chemical**	**Electrical → Light**
Speaker 	Voltameter 	Bulb 

Kinetic Energy

The energy possessed by a body by virtue of its motion, is called kinetic energy.

Examples : (i) Flowing water possesses kinetic energy which is used to run the water mills.

(ii) Moving vehicle possesses kinetic energy.

(iii) Moving air (*i.e.* wind) possesses kinetic energy which is used to run wind mills.

(iv) The hammer possesses kinetic energy which is used to drive the nails in wood.

(v) A bullet fired from the gun has kinetic energy and due to this energy the bullet penetrates into a target.

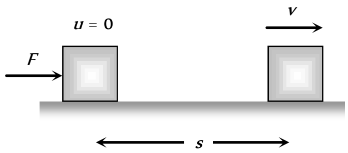


Fig. 6.17

(1) **Expression for kinetic energy :**

Let m = mass of the body,

u = Initial velocity of the body (= 0)

F = Force acting on the body,

a = Acceleration of the body,

s = Distance travelled by the body,

v = Final velocity of the body

From $v^2 = u^2 + 2as$

$$\Rightarrow \quad v^2 = 0 + 2as \quad \therefore s = \frac{v^2}{2a}$$

Since the displacement of the body is in the direction of the applied force, then work done by the force is

$$W = F \times s = ma \times \frac{v^2}{2a}$$

$$\Rightarrow W = \frac{1}{2}mv^2$$

This work done appears as the kinetic energy of the body $KE = W = \frac{1}{2}mv^2$

(2) **Calculus method :** Let a body is initially at rest and force $\vec{F}$ is applied on the body to displace it through small displacement $d\vec{s}$ along its own direction then small work done

$$dW = \vec{F}.d\vec{s} = F\,ds$$

$$\Rightarrow \quad dW = m\,a\,ds \qquad [\text{As } F = ma]$$

$$\Rightarrow \quad dW = m\frac{dv}{dt}ds \qquad \left[\text{As } a = \frac{dv}{dt}\right]$$

$$\Rightarrow \quad dW = mdv.\frac{ds}{dt}$$

$$\Rightarrow \quad dW = m\,v\,dv \qquad \text{...(i)}$$

$$\left[As\ \frac{ds}{dt}=v\right]$$

Therefore work done on the body in order to increase its velocity from zero to v is given by

$$W=\int_0^v mv\,dv=m\int_0^v v\,dv=m\left[\frac{v^2}{2}\right]_0^v=\frac{1}{2}mv^2$$

This work done appears as the kinetic energy of the body $KE=\frac{1}{2}mv^2$.

In vector form $KE=\frac{1}{2}m(\vec{v}.\vec{v})$

As m and $\vec{v}.\vec{v}$ are always positive, kinetic energy is always positive scalar *i.e.* kinetic energy can never be negative.

(3) **Kinetic energy depends on frame of reference :** The kinetic energy of a person of mass m, sitting in a train moving with speed v, is zero in the frame of train but $\frac{1}{2}mv^2$ in the frame of the earth.

(4) **Kinetic energy according to relativity :** As we know $E=\frac{1}{2}mv^2$.

But this formula is valid only for ($v << c$) If v is comparable to c (speed of light in free space = $3\times10^8\ m/s$) then according to Einstein theory of relativity

$$E=\frac{mc^2}{\sqrt{1-(v^2/c^2)}}-mc^2$$

(5) **Work-energy theorem:** From equation (i) $dW=mv\,dv$.

Work done on the body in order to increase its velocity from u to v is given by

$$W=\int_u^v mv\,dv=m\int_u^v v\,dv=m\left[\frac{v^2}{2}\right]_u^v$$

$$\Rightarrow W=\frac{1}{2}m[v^2-u^2]$$

Work done = change in kinetic energy

$W=\Delta E$

This is work energy theorem, it states that work done by a force acting on a body is equal to the change in the kinetic energy of the body.

This theorem is valid for a system in presence of all types of forces (external or internal, conservative or non-conservative).

If kinetic energy of the body increases, work is positive *i.e.* body moves in the direction of the force (or field) and if kinetic energy decreases, work will be negative and object will move opposite to the force (or field).

Examples : (i) In case of vertical motion of body under gravity when the body is projected up, force of gravity is opposite to motion and so kinetic energy of the body decreases and when it falls down, force of gravity is in the direction of motion so kinetic energy increases.

(ii) When a body moves on a rough horizontal surface, as force of friction acts opposite to motion, kinetic energy will decrease and the decrease in kinetic energy is equal to the work done against friction.

(6) **Relation of kinetic energy with linear momentum:** As we know

$$E=\frac{1}{2}mv^2=\frac{1}{2}\left[\frac{P}{v}\right]v^2 \qquad [\text{As } P=mv]$$

$$\therefore\ E=\frac{1}{2}Pv$$

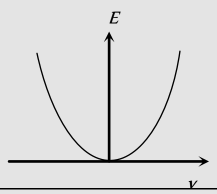
or $$E=\frac{P^2}{2m} \qquad \left[\text{As } v=\frac{P}{m}\right]$$

So we can say that kinetic energy $E=\frac{1}{2}mv^2=\frac{1}{2}Pv=\frac{p^2}{2m}$

and Momentum $P=\frac{2E}{v}=\sqrt{2mE}$

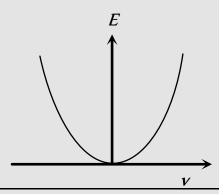
From above relation it is clear that a body can not have kinetic energy without having momentum and vice-versa.

(7) **Various graphs of kinetic energy**



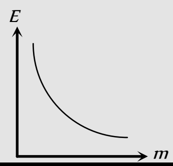
$E\propto v$

m = constant



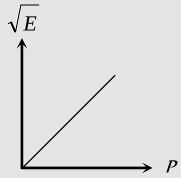
$E\propto P^2$

m = constant



$E\propto\frac{1}{m}$

P = constant



$P\propto\sqrt{E}$

m = constant

Stopping of Vehicle by Retarding Force

If a vehicle moves with some initial velocity and due to some retarding force it stops after covering some distance after some time.

(1) **Stopping distance :** Let m = Mass of vehicle,

v = Velocity, P = Momentum, E = Kinetic energy

F = Stopping force, x = Stopping distance,

t = Stopping time

Then, in this process stopping force does work on the vehicle and destroy the motion.

By the work- energy theorem

$$W = \Delta K = \frac{1}{2}mv^2$$

Initial velocity = v Final velocity = 0

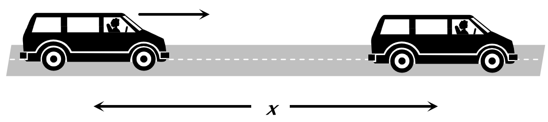


Fig. 6.18

$\Rightarrow$ Stopping force (F) $\times$ Distance (x) = Kinetic energy (E)

$\Rightarrow$ Stopping distance (x) $= \dfrac{\text{Kinetic energy }(E)}{\text{Stopping force }(F)}$

$$\Rightarrow \; x = \frac{mv^2}{2F} \qquad \text{...(i)}$$

(2) **Stopping time :** By the impulse-momentum theorem

$$F \times \Delta t = \Delta P \Rightarrow F \times t = P$$

$$\therefore \; t = \frac{P}{F}$$

$$\text{or } \; t = \frac{mv}{F} \qquad \text{...(ii)}$$

(3) **Comparison of stopping distance and time for two vehicles :** Two vehicles of masses m_1 and m_2 are moving with velocities v_1 and v_2 respectively. When they are stopped by the same retarding force (F).

The ratio of their stopping distances $\dfrac{x_1}{x_2} = \dfrac{E_1}{E_2} = \dfrac{m_1 v_1^2}{m_2 v_2^2}$

and the ratio of their stopping time $\dfrac{t_1}{t_2} = \dfrac{P_1}{P_2} = \dfrac{m_1 v_1}{m_2 v_2}$

(i) If vehicles possess same velocities

$v_1 = v_2$

$$\frac{x_1}{x_2} = \frac{m_1}{m_2} \; ; \; \frac{t_1}{t_2} = \frac{m_1}{m_2}$$

(ii) If vehicle possess same kinetic momentum

$P_1 = P_2$

$$\frac{x_1}{x_2} = \frac{E_1}{E_2} = \left(\frac{P_1^2}{2m_1}\right)\left(\frac{2m_2}{P_2^2}\right) = \frac{m_2}{m_1}$$

$$\frac{t_1}{t_2} = \frac{P_1}{P_2} = 1$$

(iii) If vehicle possess same kinetic energy

$$\frac{x_1}{x_2} = \frac{E_1}{E_2} = 1$$

$$\frac{t_1}{t_2} = \frac{P_1}{P_2} = \frac{\sqrt{2m_1E_1}}{\sqrt{2m_2E_2}} = \sqrt{\frac{m_1}{m_2}}$$

Note : ❑ If vehicle is stopped by friction then

Stopping distance $x = \dfrac{\frac{1}{2}mv^2}{F} = \dfrac{\frac{1}{2}mv^2}{ma} = \dfrac{v^2}{2\mu g}$

[As $a = \mu g$]

Stopping time $t = \dfrac{mv}{F} = \dfrac{mv}{m\mu g} = \dfrac{v}{\mu g}$

Potential Energy

Potential energy is defined only for conservative forces. In the space occupied by conservative forces every point is associated with certain energy which is called the energy of position or potential energy. Potential energy generally are of three types : Elastic potential energy, Electric potential energy and Gravitational potential energy.

(1) **Change in potential energy :** Change in potential energy between any two points is defined in the terms of the work done by the associated conservative force in displacing the particle between these two points without any change in kinetic energy.

$$U_2 - U_1 = -\int_{r_1}^{r_2} \vec{F}.d\vec{r} = -W \qquad \text{...(i)}$$

We can define a unique value of potential energy only by assigning some arbitrary value to a fixed point called the reference point. Whenever and wherever possible, we take the reference point at infinity and assume potential energy to be zero there, *i.e.* if we take $r_1 = \infty$ and $r_2 = r$ then from equation (i)

$$U = -\int_{\infty}^{r} \vec{F}.d\vec{r} = -W$$

In case of conservative force (field) potential energy is equal to negative of work done by conservative force in shifting the body from reference position to given position.

This is why, in shifting a particle in a conservative field (say gravitational or electric), if the particle moves opposite to the field, work done by the field will be negative and so change in potential energy will be positive *i.e.* potential energy will increase. When the particle moves in the direction of field, work will be positive and change in potential energy will be negative *i.e.* potential energy will decrease.

(2) **Three dimensional formula for potential energy:** For only conservative fields $\vec{F}$ equals the negative gradient $(-\vec{\nabla})$ of the potential energy.

So $\vec{F} = -\vec{\nabla}U$ ($\vec{\nabla}$ read as Del operator or Nabla operator and $\vec{\nabla} = \dfrac{\partial}{\partial x}\hat{i} + \dfrac{\partial}{\partial y}\hat{j} + \dfrac{\partial}{\partial z}\hat{k}$)

$$\Rightarrow \vec{F} = -\left[\frac{\partial U}{\partial x}\hat{i} + \frac{\partial U}{\partial y}\hat{j} + \frac{\partial U}{\partial z}\hat{k}\right]$$

where,

$\dfrac{\partial U}{\partial x}$ = Partial derivative of U *w.r.t.* x (keeping y and z constant)

$\dfrac{\partial U}{\partial y}$ = Partial derivative of U *w.r.t.* y (keeping x and z constant)

$\dfrac{\partial U}{\partial z}$ = Partial derivative of U *w.r.t.* z (keeping x and y constant)

(3) **Potential energy curve :** A graph plotted between the potential energy of a particle and its displacement from the centre of force is called potential energy curve.

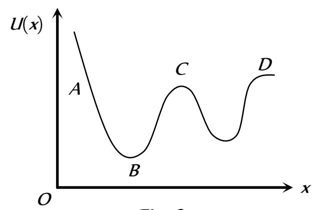


Fig. 6.19

Figure shows a graph of potential energy function $U(x)$ for one dimensional motion.

As we know that negative gradient of the potential energy gives force.

$$\therefore\ -\frac{dU}{dx}=F$$

(4) **Nature of force**

(i) Attractive force :

On increasing x, if U increases,

$\frac{dU}{dx}=\text{positive}$, then F is in negative direction

i.e. force is attractive in nature.

In graph this is represented in region *BC*.

(ii) Repulsive force :

On increasing x, if U decreases,

$\frac{dU}{dx}=\text{negative}$, then F is in positive direction

i.e. force is repulsive in nature.

In graph this is represented in region *AB*.

(iii) Zero force :

On increasing x, if U does not change,

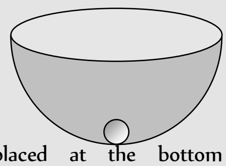
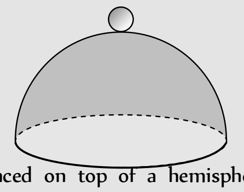
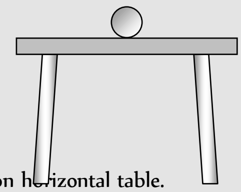
$\frac{dU}{dx}=0$ then F is zero

i.e. no force works on the particle.

Point *B*, *C* and *D* represents the point of zero force or these points can be termed as position of equilibrium.

(5) **Types of equilibrium :** If net force acting on a particle is zero, it is said to be in equilibrium.

For equilibrium $\frac{dU}{dx}=0$, but the equilibrium of particle can be of three types :

Stable	Unstable	Neutral
When a particle is displaced slightly from its present position, then a force acting on it brings it back to the initial position, it is said to be in stable equilibrium position.	When a particle is displaced slightly from its present position, then a force acting on it tries to displace the particle further away from the equilibrium position, it is said to be in unstable equilibrium.	When a particle is slightly displaced from its position then it does not experience any force acting on it and continues to be in equilibrium in the displaced position, it is said to be in neutral equilibrium.
Potential energy is minimum.	Potential energy is maximum.	Potential energy is constant.
$F=-\frac{dU}{dx}=0$	$F=-\frac{dU}{dx}=0$	$F=-\frac{dU}{dx}=0$
$\frac{d^2U}{dx^2}=\text{positive}$ *i.e.* rate of change of $\frac{dU}{dx}$ is positive.	$\frac{d^2U}{dx^2}=\text{negative}$ *i.e.* rate of change of $\frac{dU}{dx}$ is negative.	$\frac{d^2U}{dx^2}=0$ *i.e.* rate of change of $\frac{dU}{dx}$ is zero.
Example :  A marble placed at the bottom of a hemispherical bowl.	*Example* :  A marble balanced on top of a hemispherical bowl.	*Example* :  A marble placed on horizontal table.

Elastic Potential Energy

(1) **Restoring force and spring constant :** When a spring is stretched or compressed from its normal position (x = 0) by a small distance x, then a restoring force is produced in the spring to bring it to the normal position.

According to Hooke's law this restoring force is proportional to the displacement x and its direction is always opposite to the displacement.

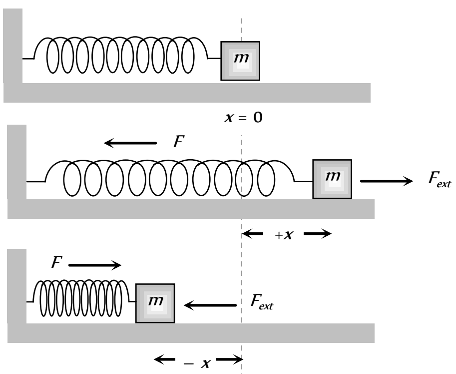


Fig. 6.20

i.e. $\vec{F}\propto-\vec{x}$

or $\vec{F}=-k\,\vec{x}$ ...(i)

where k is called spring constant.

If x = 1, F = k (Numerically)

or k = F

Hence spring constant is numerically equal to force required to produce unit displacement (compression or extension) in the spring. If required force is more, then spring is said to be more stiff and vice-versa.

Actually k is a measure of the stiffness/softness of the spring.

Dimension : As $k=\frac{F}{x}$

$$\therefore\ [k]=\frac{[F]}{[x]}=\frac{[MLT^{-2}]}{L}=[MT^{-2}]$$

Units : S.I. unit *Newton/metre*, C.G.S unit *Dyne/cm*.

Note : ❑ Dimension of force constant is similar to surface tension.

(2) **Expression for elastic potential energy :** When a spring is stretched or compressed from its normal position (x = 0), work has to be done by external force against restoring force. $\vec{F}_{ext}=-\vec{F}_{restoring}=k\vec{x}$

Let the spring is further stretched through the distance *dx*, then work done

$$dW=\vec{F}_{ext}.d\vec{x}=F_{ext}.dx\cos 0^o=kx\ dx\quad [\text{As }\cos 0^\circ=1]$$

Therefore total work done to stretch the spring through a distance x from its mean position is given by

$$W=\int_0^x dW=\int_0^x kx\ dx=k\left[\frac{x^2}{2}\right]_0^x=\frac{1}{2}kx^2$$

This work done is stored as the potential energy in the stretched spring.

$\therefore$ Elastic potential energy $U=\frac{1}{2}kx^2$

$$U=\frac{1}{2}Fx\qquad\left[\text{As}\,k=\frac{F}{x}\right]$$

$$U=\frac{F^2}{2k}\qquad\left[\text{As}\,x=\frac{F}{k}\right]$$

$\therefore$ Elastic potential energy $U=\frac{1}{2}kx^2=\frac{1}{2}Fx=\frac{F^2}{2k}$

Note : ❑ If spring is stretched from initial position x_1 to final position x_2 then work done

= Increment in elastic potential energy

$$=\frac{1}{2}k(x_2^2-x_1^2)$$

❑ Work done by the spring-force on the block in various situation are shown in the following table

Table : 6.2 Work done for spring

Initial state of the spring	Final state of the spring	Initial position (x_1)	Final position (x_2)	Work done (W)
Natural	Compressed	0	$-x$	$-1/2\ kx^2$
Natural	Elongated	0	x	$-1/2\ kx^2$
Elongated	Natural	x	0	$1/2\ kx^2$
Compressed	Natural	$-x$	0	$1/2\ kx^2$
Elongated	Compressed	x	$-x$	0
Compressed	Elongated	$-x$	x	0

(3) **Energy graph for a spring :** If the mass attached with spring performs simple harmonic motion about its mean position then its potential energy at any position (x) can be given by

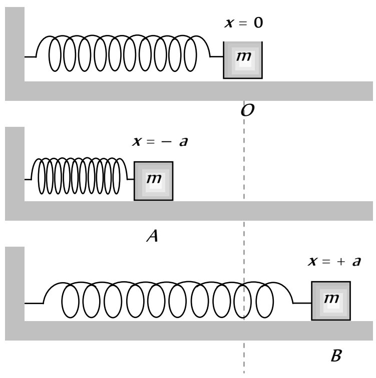


Fig. 6.21

$$U=\frac{1}{2}kx^2\qquad ...(i)$$

So for the extreme position

$$U=\frac{1}{2}ka^2\qquad [\text{As } x=\pm a \text{ for extreme}]$$

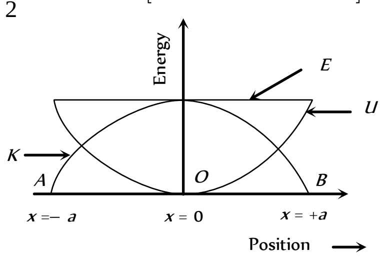


Fig. 6.22

This is maximum potential energy or the total energy of mass.

$$\therefore\ \text{Total energy } E=\frac{1}{2}ka^2\qquad ...(ii)$$

[Because velocity of mass is zero at extreme position]

$$\therefore\ K=\frac{1}{2}mv^2=0\,]$$

Now kinetic energy at any position

$$K=E-U=\frac{1}{2}k\,a^2-\frac{1}{2}k\,x^2$$

$$K=\frac{1}{2}k(a^2-x^2)\qquad ...(iii)$$

From the above formula we can check that

$$U_{\max} = \frac{1}{2}ka^2 \qquad [\text{At extreme } x = \pm a]$$

and $\quad U_{\min} = 0 \qquad [\text{At mean } x = 0]$

$$K_{\max} = \frac{1}{2}ka^2 \qquad [\text{At mean } x = 0]$$

and $\quad K_{\min} = 0 \qquad [\text{At extreme } x = \pm a]$

$$E = \frac{1}{2}ka^2 = \text{constant (at all positions)}$$

It means kinetic energy and potential energy changes parabolically *w.r.t.* position but total energy remain always constant irrespective to position of the mass

Electrical Potential Energy

It is the energy associated with state of separation between charged particles that interact via electric force. For two point charge q_1 and q_2, separated by distance r.

$$U = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q_1 q_2}{r}$$

While for a point charge q at a point in an electric field where the potential is V

$U = qV$

As charge can be positive or negative, electric potential energy can be positive or negative.

Gravitational Potential Energy

It is the usual form of potential energy and this is the energy associated with the state of separation between two bodies that interact via gravitational force.

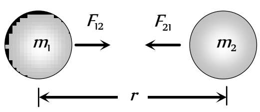


Fig. 6.23

For two particles of masses m_1 and m_2 separated by a distance r

Gravitational potential energy $U = -\dfrac{G m_1 m_2}{r}$

(1) If a body of mass m at height h relative to surface of earth then

Gravitational potential energy $U = \dfrac{mgh}{1 + \dfrac{h}{R}}$

Where R = radius of earth, g = acceleration due to gravity at the surface of the earth.

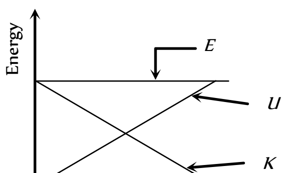
(2) If $h << R$ then above formula reduces to $U = mgh$.

(3) If V is the gravitational potential at a point, the potential energy of a particle of mass m at that point will be

$U = mV$

(4) Energy height graph : When a body projected vertically upward from the ground level with some initial velocity then it possess kinetic energy but its initial potential energy is zero.

As the body moves upward its potential energy increases due to increase in height but kinetic energy decreases (due to decrease in velocity). At maximum height its kinetic energy becomes zero and potential energy maximum but through out the complete motion, total energy remains constant as shown in the figure.



Work Done in Pulling the Chain Against Gravity

A chain of length L and mass M is held on a frictionless table with $(1/n)^{th}$ of its length hanging over the edge.

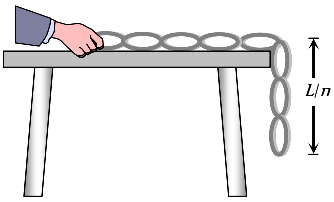


Fig. 6.25

Let $m = \dfrac{M}{L} =$ mass per unit length of the chain and y is the length of the chain hanging over the edge. So the mass of the chain of length y will be ym and the force acting on it due to gravity will be mgy.

The work done in pulling the dy length of the chain on the table.

$dW = F(-dy) \qquad [\text{As } y \text{ is decreasing}]$

i.e. $dW = mgy\,(-dy)$

So the work done in pulling the hanging portion on the table.

$$W = -\int_{L/n}^{0} mgy\,dy = -mg\left[\frac{y^2}{2}\right]_{L/n}^{0} = \frac{mg\,L^2}{2n^2}$$

$$\therefore\ W = \frac{MgL}{2n^2} \qquad [\text{As } m = M/L]$$

Alternative method :

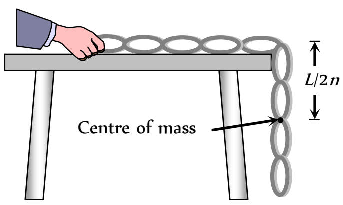


Fig. 6.26

If point mass m is pulled through a height h then work done $W = mgh$

Similarly for a chain we can consider its centre of mass at the middle point of the hanging part *i.e.* at a height of $L/(2n)$ from the lower end and mass of the hanging part of chain $= \dfrac{M}{n}$

So work done to raise the centre of mass of the chain on the table is given by

$$W = \frac{M}{n} \times g \times \frac{L}{2n} \qquad [\text{As } W = mgh]$$

or $\quad W = \dfrac{MgL}{2n^2}$

Velocity of Chain While Leaving the Table

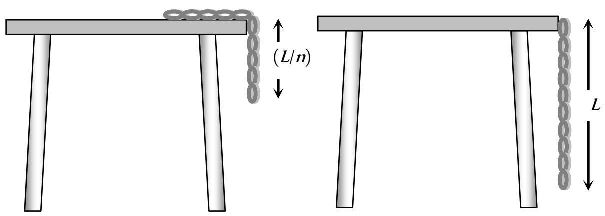


Fig. 6.27

Taking surface of table as a reference level (zero potential energy)

Potential energy of chain when $1/n$ length hanging from the edge $= \frac{-MgL}{2n^2}$

Potential energy of chain when it leaves the table $= -\frac{MgL}{2}$

Kinetic energy of chain = loss in potential energy

$$\Rightarrow \frac{1}{2}Mv^2 = \frac{MgL}{2} - \frac{MgL}{2n^2}$$

$$\Rightarrow \frac{1}{2}Mv^2 = \frac{MgL}{2}\left[1 - \frac{1}{n^2}\right]$$

$\therefore$ Velocity of chain $v = \sqrt{gL\left(1 - \frac{1}{n^2}\right)}$

Law of Conservation of Energy

(1) **Law of conservation of energy**

For a body or an isolated system by work-energy theorem we have

$K_2 - K_1 = \int \vec{F}.d\vec{r}$ ...(i)

But according to definition of potential energy in a conservative field

$U_2 - U_1 = -\int \vec{F}.d\vec{r}$ ...(ii)

So from equation (i) and (ii) we have

$$K_2 - K_1 = -(U_2 - U_1)$$

or $K_2 + U_2 = K_1 + U_1$

i.e. $K + U$ = constant.

For an isolated system or body in presence of conservative forces, the sum of kinetic and potential energies at any point remains constant throughout the motion. It does not depend upon time. This is known as the law of conservation of mechanical energy.

$$\Delta(K + U) = \Delta E = 0$$

[As E is constant in a conservative field]

$$\therefore \Delta K + \Delta U = 0$$

i.e. if the kinetic energy of the body increases its potential energy will decrease by an equal amount and vice-versa.

(2) **Law of conservation of total energy :** If some non-conservative force like friction is also acting on the particle, the mechanical energy is no more constant. It changes by the amount equal to work done by the frictional force.

$$\Delta(K + U) = \Delta E = W_f$$

[where W_f is the work done against friction]

The lost energy is transformed into heat and the heat energy developed is exactly equal to loss in mechanical energy.

We can, therefore, write $\Delta E + Q = 0$

[where Q is the heat produced]

This shows that if the forces are conservative and non-conservative both, it is not the mechanical energy which is conserved, but it is the total energy, may be heat, light, sound or mechanical *etc.*, which is conserved.

In other words : "*Energy may be transformed from one kind to another but it cannot be created or destroyed. The total energy in an isolated system remain constant*". This is the law of conservation of energy.

Power

Power of a body is defined as the rate at which the body can do the work.

Average power $(P_{av.}) = \frac{\Delta W}{\Delta t} = \frac{W}{t}$

Instantaneous power $(P_{inst.}) = \frac{dW}{dt} = \frac{\vec{F}.d\vec{s}}{dt}$ [As $dW = \vec{F}.d\vec{s}$]

$P_{inst} = \vec{F}.\vec{v}$ [As $\vec{v} = \frac{d\vec{s}}{dt}$]

i.e. power is equal to the scalar product of force with velocity.

Important Points

(1) Dimension : $[P] = [F][v] = [MLT^{-2}][LT^{-1}]$

$\therefore [P] = [ML^2T^{-3}]$

(2) Units : *Watt* or *Joule/sec* [S.I.]

Erg/sec [C.G.S.]

Practical units : *Kilowatt* (*KW*), *Mega watt* (*MW*) and Horse power (*hp*)

Relations between different units :

$1\,Watt = 1\,Joule/\sec = 10^7\,erg/\sec$

$1\,hp = 746\,Watt$

$1\,MW = 10^6\,Watt$

$1\,KW = 10^3\,Watt$

(3) If work done by the two bodies is same then power $\propto \frac{1}{\text{time}}$

i.e. the body which perform the given work in lesser time possess more power and vice-versa.

(4) As power = work/time, any unit of power multiplied by a unit of time gives unit of work (or energy) and not power, *i.e.* Kilowatt-hour or watt-day are units of work or energy.

$$1\,KWh = 10^3\,\frac{J}{sec} \times (60 \times 60\,sec) = 3.6 \times 10^6\,Joule$$

(5) The slope of work time curve gives the instantaneous power. As $P = dW/dt = \tan\theta$

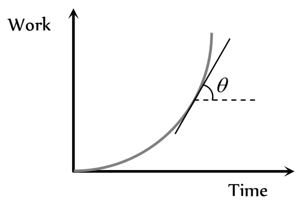


Fig. 6.28

(6) Area under power-time curve gives the work done as $P = \frac{dW}{dt}$

$$\therefore W = \int P\,dt$$

$\therefore$ W = Area under P-t curve

Position and Velocity of an Automobile w.r.t Time

An automobile of mass m accelerates, starting from rest, while the engine supplies constant power P, its position and velocity changes *w.r.t* time.

(1) **Velocity :** As $Fv = P$ = constant

i.e. $m\frac{dv}{dt}v = P$ $\left[\text{As}\,F = \frac{mdv}{dt}\right]$

or $\int v\,dv = \int \frac{P}{m} dt$

By integrating both sides we get $\frac{v^2}{2} = \frac{P}{m}t + C_1$

As initially the body is at rest *i.e.* $v = 0$ at $t = 0$, so $C_1 = 0$

$\therefore\ v = \left(\frac{2Pt}{m}\right)^{1/2}$

(2) **Position :** From the above expression $v = \left(\frac{2Pt}{m}\right)^{1/2}$

or $\frac{ds}{dt} = \left(\frac{2Pt}{m}\right)^{1/2}$ $\qquad \left[\text{As } v = \frac{ds}{dt}\right]$

i.e. $\int ds = \int \left(\frac{2Pt}{m}\right)^{1/2} dt$

By integrating both sides we get

$s = \left(\frac{2P}{m}\right)^{1/2} . \frac{2}{3} t^{3/2} + C_2$

Now as at $t = 0$, $s = 0$, so $C_2 = 0$

$s = \left(\frac{8P}{9m}\right)^{1/2} t^{3/2}$

Collision

Collision is an isolated event in which a strong force acts between two or more bodies for a short time as a result of which the energy and momentum of the interacting particle change.

In collision particles may or may not come in real touch e.g. in collision between two billiard balls or a ball and bat, there is physical contact while in collision of alpha particle by a nucleus (*i.e.* Rutherford scattering experiment) there is no physical contact.

(1) **Stages of collision :** There are three distinct identifiable stages in collision, namely, before, during and after. In the before and after stage the interaction forces are zero. Between these two stages, the interaction forces are very large and often the dominating forces governing the motion of bodies. The magnitude of the interacting force is often unknown, therefore, Newton's second law cannot be used, the law of conservation of momentum is useful in relating the initial and final velocities.

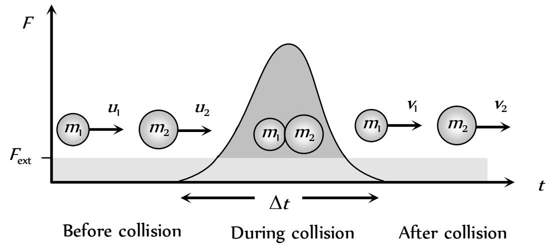


Fig. 6.29

(2) **Momentum and ene n in collision**

(i) Momentum conservation : In a collision, the effect of external forces such as gravity or friction are not taken into account as due to small duration of collision (Δt) average impulsive force responsible for collision is much larger than external force acting on the system and since this impulsive force is 'Internal' therefore the total momentum of system always remains conserved.

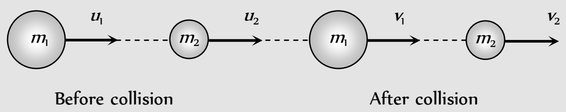
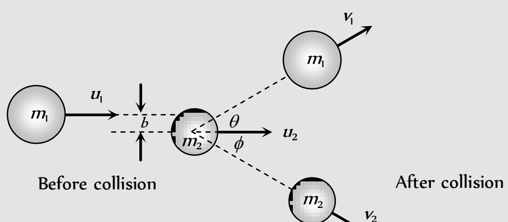
(ii) Energy conservation : In a collision 'total energy' is also always conserved. Here total energy includes all forms of energy such as mechanical energy, internal energy, excitation energy, radiant energy or even mass energy.

These laws are the fundamental laws of physics and applicable for any type of collision but this is not true for conservation of kinetic energy.

(3) **Types of collision :** (i) On the basis of conservation of kinetic energy.

Perfectly elastic collision	Inelastic collision	Perfectly inelastic collision
If in a collision, kinetic energy after collision is equal to kinetic energy before collision, the collision is said to be perfectly elastic.	If in a collision kinetic energy after collision is not equal to kinetic energy before collision, the collision is said to inelastic.	If in a collision two bodies stick together or move with same velocity after the collision, the collision is said to be perfectly inelastic.
Coefficient of restitution $e = 1$	Coefficient of restitution $0 < e < 1$	Coefficient of restitution $e = 0$
$(KE)_{final} = (KE)_{initial}$	Here kinetic energy appears in other forms. In some cases $(KE)_{final} < (KE)_{initial}$ such as when initial KE is converted into internal energy of the product (as heat, elastic or excitation) while in other cases $(KE)_{final} > (KE)_{initial}$ such as when internal energy stored in the colliding particles is released	The term 'perfectly inelastic' does not necessarily mean that all the initial kinetic energy is lost, it implies that the loss in kinetic energy is as large as it can be. (Consistent with momentum conservation).
Examples : (1) Collision between atomic particles (2) Bouncing of ball with same velocity after the collision with earth.	*Examples* : (1) Collision between two billiard balls. (2) Collision between two automobile on a road. In fact all majority of collision belong to this category.	*Example* : Collision between a bullet and a block of wood into which it is fired. When the bullet remains embedded in the block.

(ii) On the basis of the direction of colliding bodies

Head on or one dimensional collision	Oblique collision
In a collision if the motion of colliding particles before and after the collision is along the same line, the collision is said to be head on or one dimensional.	If two particle collision is 'glancing' *i.e.* such that their directions of motion after collision are not along the initial line of motion, the collision is called oblique. If in oblique collision the particles before and after collision are in same plane, the collision is called 2-dimensional otherwise 3-dimensional.
Impact parameter b is zero for this type of collision.	Impact parameter b lies between 0 and $(r_1 + r_2)$ *i.e.* $0 < b < (r_1 + r_2)$ where r_1 and r_2 are radii of colliding bodies.
 	

Example : collision of two gliders on an air track.	*Example* : Collision of billiard balls.

Perfectly elastic head on collision

Let two bodies of masses m_1 and m_2 moving with initial velocities u_1 and u_2 in the same direction and they collide such that after collision their final velocities are v_1 and v_2 respectively.

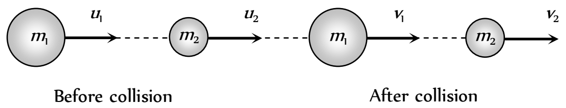


Fig. 6.30

According to law of conservation of momentum

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2 \quad \text{... (i)}$$

$$\Rightarrow \quad m_1(u_1 - v_1) = m_2(v_2 - u_2) \quad \text{...(ii)}$$

According to law of conservation of kinetic energy

$$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \quad \text{...(iii)}$$

$$\Rightarrow \quad m_1(u_1^2 - v_1^2) = m_2(v_2^2 - u_2^2) \quad \text{...(iv)}$$

Dividing equation (iv) by equation (ii)

$$v_1 + u_1 = v_2 + u_2 \quad \text{...(v)}$$

$$\Rightarrow \quad u_1 - u_2 = v_2 - v_1 \quad \text{...(vi)}$$

Relative velocity of separation is equal to relative velocity of approach.

Note : ❑ The ratio of relative velocity of separation and relative velocity of approach is defined as coefficient of restitution.

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

or $\quad v_2 - v_1 = e(u_1 - u_2)$

❑ For perfectly elastic collision, $e = 1$

$\therefore \quad v_2 - v_1 = u_1 - u_2 \qquad$ [As shown in eq. (vi)]

❑ For perfectly inelastic collision, $e = 0$

$\therefore \quad v_2 - v_1 = 0$ or $v_2 = v_1$

It means that two body stick together and move with same velocity.

❑ For inelastic collision, $0 < e < 1$

$\therefore \quad v_2 - v_1 = e(u_1 - u_2)$

In short we can say that e is the degree of elasticity of collision and it is dimensionless quantity.

Further from equation (v) we get

$$v_2 = v_1 + u_1 - u_2$$

Substituting this value of v_2 in equation (i) and rearranging

we get, $\quad v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1 + \frac{2m_2u_2}{m_1 + m_2} \quad$ ...(vii)

Similarly we get,

$$v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)u_2 + \frac{2m_1u_1}{m_1 + m_2} \quad \text{...(viii)}$$

(1) **Special cases of head on elastic collision**

(i) **If projectile and target are of same mass *i.e.* $m_1 = m_2$**

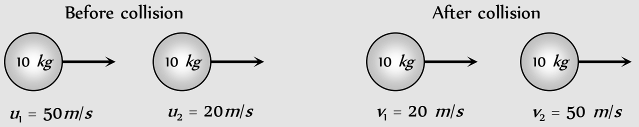
Since $v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1 + \frac{2m_2}{m_1 + m_2}u_2 \quad$ and $\quad v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)u_2 + \frac{2m_1u_1}{m_1 + m_2}$

Substituting $m_1 = m_2$ we get

$v_1 = u_2$ and $v_2 = u_1$

It means when two bodies of equal masses undergo head on elastic collision, their velocities get interchanged.

Example : Collision of two billiard balls



Sub case : $u_2 = 0$ *i.e.* target is at rest

$v_1 = 0$ and $v_2 = u_1$

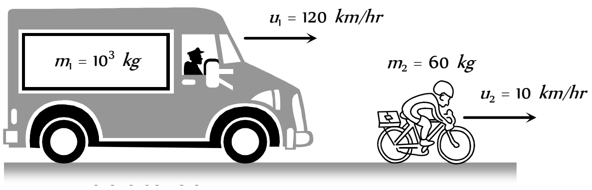
(ii) **If massive projectile collides with a light target *i.e.* $m_1 >> m_2$**

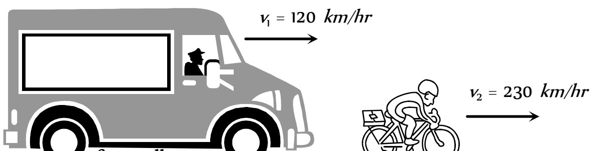
Since $v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1 + \frac{2m_2u_2}{m_1 + m_2}$ and $v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)u_2 + \frac{2m_1u_1}{m_1 + m_2}$

Substituting $m_2 = 0$, we get

$v_1 = u_1$ and $v_2 = 2u_1 - u_2$

Example : Collision of a truck with a cyclist





After collision

Sub case : $u_2 = 0$ *i.e.* target is at rest

$v_1 = u_1$ and $v_2 = 2u_1$

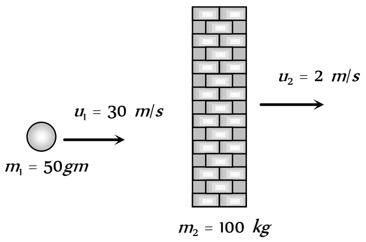
(iii) If light projectile collides with a very heavy target *i.e.* $m_1 << m_2$

Since $v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1 + \frac{2m_2u_2}{m_1 + m_2}$ and $v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)u_2 + \frac{2m_1u_1}{m_1 + m_2}$

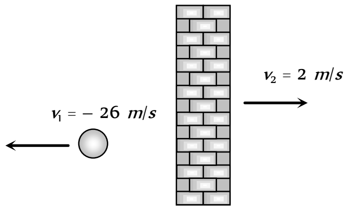
Substituting $m_1 = 0$, we get

$v_1 = -u_1 + 2u_2$ and $v_2 = u_2$

Example : Collision of a ball with a massive wall.



Before collision



After collision

Sub case : $u_2 = 0$ i.e. target is at rest

$v_1 = -u_1$ and $v_2 = 0$

i.e. the ball rebounds with same speed in opposite direction when it collide with stationary and very massive wall.

(2) **Kinetic energy transfer during head on elastic collision**

Kinetic energy of projectile before collision $K_i = \frac{1}{2}m_1u_1^2$

Kinetic energy of projectile after collision $K_f = \frac{1}{2}m_1v_1^2$

Kinetic energy transferred from projectile to target ΔK = decrease in kinetic energy in projectile

$$\Delta K = \frac{1}{2}m_1u_1^2 - \frac{1}{2}m_1v_1^2 = \frac{1}{2}m_1(u_1^2 - v_1^2)$$

Fractional decrease in kinetic energy

$$\frac{\Delta K}{K} = \frac{\frac{1}{2}m_1(u_1^2 - v_1^2)}{\frac{1}{2}m_1u_1^2} = 1 - \left(\frac{v_1}{u_1}\right)^2 \quad \text{...(i)}$$

We can substitute the value of v_1 from the equation

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1 + \frac{2m_2u_2}{m_1 + m_2}$$

If the target is at rest *i.e.* u_2 = 0 then $v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1$

From equation (i) $\frac{\Delta K}{K} = 1 - \left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2$ ...(ii)

or $\frac{\Delta K}{K} = \frac{4m_1m_2}{(m_1 + m_2)^2}$ ...(iii)

or $\frac{\Delta K}{K} = \frac{4m_1m_2}{(m_1 - m_2)^2 + 4m_1m_2}$ ...(iv)

Note : ❑ Greater the difference in masses, lesser will be transfer of kinetic energy and vice versa

❑ Transfer of kinetic energy will be maximum when the difference in masses is minimum

i.e. $m_1 - m_2 = 0$ or $m_1 = m_2$ then

$$\frac{\Delta K}{K} = 1 = 100\%$$

So the transfer of kinetic energy in head on elastic collision (when target is at rest) is maximum when the masses of particles are equal i.e. mass ratio is 1 and the transfer of kinetic energy is 100%.

❑ If $m_2 = n m_1$ then from equation (iii) we get

$$\frac{\Delta K}{K} = \frac{4n}{(1+n)^2}$$

❑ Kinetic energy retained by the projectile

$$\left(\frac{\Delta K}{K}\right)_{\text{Retained}} = 1 - \text{kinetic energy transferred by projectile}$$

$$\Rightarrow \left(\frac{\Delta K}{K}\right)_{\text{Retained}} = 1 - \left[1 - \left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2\right] = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2$$

(3) **Velocity, momentum and kinetic energy of stationary target after head on elastic collision**

(i) Velocity of target : We know

$$v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)u_2 + \frac{2m_1 u_1}{m_1 + m_2}$$

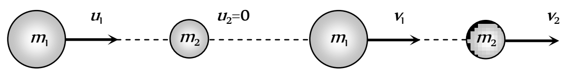


Fig. 6.31

$$\Rightarrow v_2 = \frac{2m_1 u_1}{m_1 + m_2}$$

$$= \frac{2u_1}{1 + m_2/m_1} \quad \text{As } u_2 = 0 \text{ and}$$

Assuming $\dfrac{m_2}{m_1} = n$

$$\therefore v_2 = \frac{2u_1}{1+n}$$

(ii) Momentum of target : $P_2 = m_2 v_2 = \dfrac{2nm_1 u_1}{1+n}$

$$\left[\text{As } m_2 = m_1 n \text{ and } v_2 = \frac{2u_1}{1+n}\right]$$

$$\therefore P_2 = \frac{2m_1 u_1}{1 + (1/n)}$$

(iii) Kinetic energy of target :

$$K_2 = \frac{1}{2} m_2 v_2^2 = \frac{1}{2} n m_1 \left(\frac{2u_1}{1+n}\right)^2 = \frac{2 m_1 u_1^2 n}{(1+n)^2}$$

$$= \frac{4(K_1)n}{(1-n)^2 + 4n} \qquad \left[\text{As } K_1 = \frac{1}{2} m_1 u_1^2\right]$$

(iv) Relation between masses for maximum velocity, momentum and kinetic energy

Velocity	$v_2 = \dfrac{2u_1}{1+n}$	For v_2 to be maximum n must be minimum *i.e.* $n = \dfrac{m_2}{m_1} \to 0 \therefore m_2 << m_1$	Target should be very light.
Momentum	$P_2 = \dfrac{2m_1 u_1}{(1 + 1/n)}$	For P_2 to be maximum, $(1/n)$ must be minimum or n must be maximum. *i.e.* $n = \dfrac{m_2}{m_1} \to \infty \therefore m_2 >> m_1$	Target should be massive.
Kinetic energy	$K_2 = \dfrac{4K_1 n}{(1-n)^2 + 4n}$	For K_2 to be maximum $(1-n)^2$ must be minimum. *i.e.* $1 - n = 0 \Rightarrow n = 1 = \dfrac{m_2}{m_1} \therefore m_2 = m_1$	Target and projectile should be of equal mass.

Perfectly Elastic Oblique Collision

Let two bodies moving as shown in figure.

By law of conservation of momentum

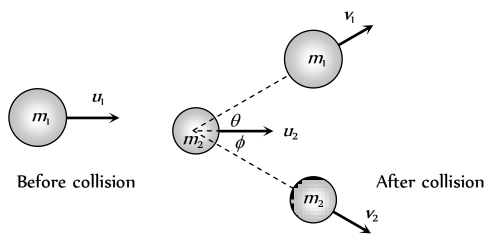


Fig. 6.32

Along x-axis, $m_1 u_1 + m_2 u_2 = m_1 v_1 \cos\theta + m_2 v_2 \cos\phi$ ...(i)

Along y-axis, $0 = m_1 v_1 \sin\theta - m_2 v_2 \sin\phi$ ...(ii)

By law of conservation of kinetic energy

$$\frac{1}{2}m_1u_1^2+\frac{1}{2}m_2u_2^2=\frac{1}{2}m_1v_1^2+\frac{1}{2}m_2v_2^2 \quad \text{...(iii)}$$

In case of oblique collision it becomes difficult to solve problem unless some experimental data is provided, as in these situations more unknown variables are involved than equations formed.

Special condition : If $m_1=m_2$ and $u_2=0$ substituting these values in equation (i), (ii) and (iii) we get

$$u_1=v_1\cos\theta+v_2\cos\phi \quad \text{...(iv)}$$

$$0=v_1\sin\theta-v_2\sin\phi \quad \text{...(v)}$$

and $u_1^2=v_1^2+v_2^2$ ...(vi)

Squaring (iv) and (v) and adding we get

$$u_1^2=v_1^2+v_2^2+2v_1v_2\cos(\theta+\phi) \quad \text{...(vii)}$$

Using (vi) and (vii) we get $\cos(\theta+\phi)=0$

$\therefore\ \theta+\phi=\pi/2$

i.e. after perfectly elastic oblique collision of two bodies of equal masses (if the second body is at rest), the scattering angle $\theta+\phi$ would be 90^o.

Head on Inelastic Collision

(1) **Velocity after collision :** Let two bodies *A* and *B* collide inelastically and coefficient of restitution is *e*.

Where

$$e=\frac{v_2-v_1}{u_1-u_2}=\frac{\text{Relative velocity of separation}}{\text{Relative velocity of approach}}$$

$\Rightarrow v_2-v_1=e(u_1-u_2)$

$$\therefore\ v_2-v_1=e(u_1-u_2) \quad \text{...(i)}$$

From the law of conservation of linear momentum

$$m_1u_1+m_2u_2=m_1v_1+m_2v_2 \quad \text{...(ii)}$$

By solving (i) and (ii) we get

$$v_1=\left(\frac{m_1-em_2}{m_1+m_2}\right)u_1+\left(\frac{(1+e)m_2}{m_1+m_2}\right)u_2$$

Similarly $v_2=\left[\frac{(1+e)m_1}{m_1+m_2}\right]u_1+\left(\frac{m_2-e\,m_1}{m_1+m_2}\right)u_2$

By substituting e = 1, we get the value of v_1 and v_2 for perfectly elastic head on collision.

(2) **Ratio of velocities after inelastic collision :** A sphere of mass *m* moving with velocity *u* hits inelastically with another stationary sphere of same mass.

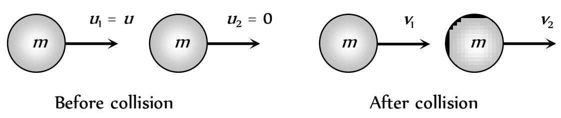


Fig. 6.33

$$\therefore\ e=\frac{v_2-v_1}{u_1-u_2}=\frac{v_2-v_1}{u-0}$$

$$\Rightarrow\ v_2-v_1=eu \quad \text{...(i)}$$

By conservation of momentum :

Momentum before collision = Momentum after collision

$$mu=mv_1+mv_2$$

$$\Rightarrow v_1+v_2=u \quad \text{...(ii)}$$

Solving equation (i) and (ii) we get $v_1=\frac{u}{2}(1-e)$

and $v_2=\frac{u}{2}(1+e)$

$$\therefore \frac{v_1}{v_2}=\frac{1-e}{1+e}$$

(3) **Loss in kinetic energy**

Loss in K.E. (ΔK) = Total initial kinetic energy

– Total final kinetic energy

$$=\left(\frac{1}{2}m_1u_1^2+\frac{1}{2}m_2u_2^2\right)-\left(\frac{1}{2}m_1v_1^2+\frac{1}{2}m_2v_2^2\right)$$

Substituting the value of v_1 and v_2 from the above expressions

Loss (ΔK) = $\frac{1}{2}\left(\frac{m_1m_2}{m_1+m_2}\right)(1-e^2)(u_1-u_2)^2$

By substituting e = 1 we get ΔK = 0 *i.e.* for perfectly elastic collision, loss of kinetic energy will be zero or kinetic energy remains same before and after the collision.

Rebounding of Ball After Collision With Ground

If a ball is dropped from a height *h* on a horizontal floor, then it strikes with the floor with a speed.

$v_0=\sqrt{2gh_0}$ [From $v^2=u^2+2gh$]

and it rebounds from the floor with a speed

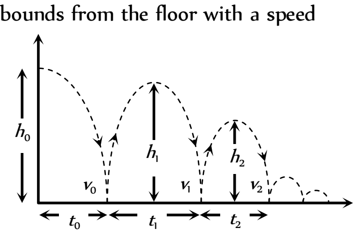


Fig. 6.34

$$v_1=e\,v_0=e\sqrt{2gh_0}$$

$$\left[\text{As } e=\frac{\text{velocity after collision}}{\text{velocity before collision}}\right]$$

(1) **First height of rebound :** $h_1=\frac{v_1^2}{2g}=e^2h_0$

$\therefore\ h_1=e^2h_0$

(2) **Height of the ball after n^{th} rebound :** Obviously, the velocity of ball after n^{th} rebound will be

$$v_n = e^n v_0$$

Therefore the height after n rebound will be

$$h_n = \frac{v_n^2}{2g} = e^{2n} h_0$$

$$\therefore\ h_n = e^{2n} h_0$$

(3) **Total distance travelled by the ball before it stops bouncing**

$$H = h_0 + 2h_1 + 2h_2 + 2h_3 + \ldots = h_0 + 2e^2 h_0 + 2e^4 h_0 + 2e^6 h_0 + \ldots$$

$$H = h_0[1 + 2e^2(1 + e^2 + e^4 + e^6 \ldots.)]$$

$$= h_0\left[1 + 2e^2\left(\frac{1}{1-e^2}\right)\right]$$

$$\left[\text{As } 1 + e^2 + e^4 + \ldots. = \frac{1}{1-e^2}\right]$$

$$\therefore\ H = h_0\left[\frac{1+e^2}{1-e^2}\right]$$

(4) **Total time taken by the ball to stop bouncing**

$$T = t_0 + 2t_1 + 2t_2 + 2t_3 + .. = \sqrt{\frac{2h_0}{g}} + 2\sqrt{\frac{2h_1}{g}} + 2\sqrt{\frac{2h_2}{g}} + ..$$

$$= \sqrt{\frac{2h_0}{g}}\ [1 + 2e + 2e^2 + \ldots\ldots]\quad [\text{As } h_1 = e^2 h_0\ ;\ h_2 = e^4 h_0\ ]$$

$$= \sqrt{\frac{2h_0}{g}}\ [1 + 2e(1 + e + e^2 + e^3 + \ldots\ldots)]$$

$$= \sqrt{\frac{2h_0}{g}}\left[1 + 2e\left(\frac{1}{1-e}\right)\right] = \sqrt{\frac{2h_0}{g}}\left(\frac{1+e}{1-e}\right)$$

$$\therefore\ T = \left(\frac{1+e}{1-e}\right)\sqrt{\frac{2h_0}{g}}$$

Perfectly Inelastic Collision

In such types of collisions, the bodies move independently before collision but after collision as a one single body.

(1) **When the colliding bodies are moving in the same direction**

By the law of conservation of momentum

$$m_1u_1 + m_2u_2 = (m_1 + m_2)v_{\text{comb}}$$

$$\Rightarrow\ v_{\text{comb}} = \frac{m_1u_1 + m_2u_2}{m_1 + m_2}$$

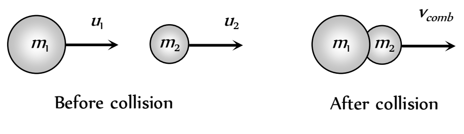


Fig. 6.35

Loss in kinetic energy

$$\Delta K = \left(\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2\right) - \frac{1}{2}(m_1 + m_2)v_{comb}^2$$

$$\Delta K = \frac{1}{2}\left(\frac{m_1m_2}{m_1 + m_2}\right)(u_1 - u_2)^2$$

[By substituting the value of v_{comb}]

(2) **When the colliding bodies are moving in the opposite direction**

By the law of conservation of momentum

$$m_1u_1 + m_2(-u_2) = (m_1 + m_2)v_{\text{comb}}$$

(Taking left to right as positive)

$$\therefore\ v_{\text{comb}} = \frac{m_1u_1 - m_2u_2}{m_1 + m_2}$$

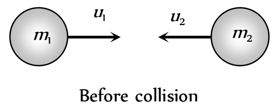


Fig. 3.36

when $m_1u_1 > m_2u_2$ then $v_{\text{comb}} > 0$ (positive)

i.e. the combined body will move along the direction of motion of mass m_1.

when $m_1u_1 < m_2u_2$ then $v_{\text{comb}} < 0$ (negative)

i.e. the combined body will move in a direction opposite to the motion of mass m_1.

(3) **Loss in kinetic energy**

ΔK = Initial kinetic energy – Final kinetic energy

$$= \left(\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2\right) - \left(\frac{1}{2}(m_1 + m_2)v_{\text{comb}}^2\right)$$

$$= \frac{1}{2}\frac{m_1m_2}{m_1 + m_2}(u_1 - u_2)^2$$

Collision Between Bullet and Vertically Suspended Block

A bullet of mass m is fired horizontally with velocity u in block of mass M suspended by vertical thread.

After the collision bullet gets embedded in block. Let the combined system raised upto height h and the string makes an angle θ with the vertical.

(1) **Velocity of system**

Let v be the velocity of the system (block + bullet) just after the collision.

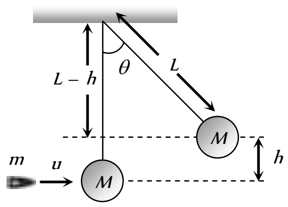


Fig. 3.37

$\text{Momentum}_{\text{bullet}} + \text{Momentum}_{\text{block}} = \text{Momentum}_{\text{bullet and block system}}$

$$mu + 0 = (m + M)v$$

$$\therefore\ v = \frac{mu}{(m + M)} \qquad \ldots\text{(i)}$$

(2) **Velocity of bullet :** Due to energy which remains in the bullet-block system, just after the collision, the system (bullet + block) rises upto height h.

By the conservation of mechanical energy

$$\frac{1}{2}(m + M)v^2 = (m + M)gh \Rightarrow v = \sqrt{2gh}$$

Now substituting this value in the equation (i) we get $\sqrt{2gh} = \frac{mu}{m+M}$

$$\therefore \quad u = \left[\frac{(m+M)\sqrt{2gh}}{m}\right]$$

(3) **Loss in kinetic energy :** We know that the formula for loss of kinetic energy in perfectly inelastic collision

$$\Delta K = \frac{1}{2}\frac{m_1 m_2}{m_1 + m_2}(u_1 - u_2)^2$$ (When the bodies are moving in same direction.)

$$\therefore \quad \Delta K = \frac{1}{2}\frac{mM}{m+M}u^2$$

[As $u_1 = u$, $u_2 = 0$, $m_1 = m$ and $m_2 = M$]

(4) **Angle of string from the vertical**

From the expression of velocity of bullet $u = \left[\frac{(m+M)\sqrt{2gh}}{m}\right]$ we can get $h = \frac{u^2}{2g}\left(\frac{m}{m+M}\right)^2$

From the figure $\cos\theta = \frac{L-h}{L} = 1 - \frac{h}{L} = 1 - \frac{u^2}{2gL}\left(\frac{m}{m+M}\right)^2$

or $\quad \theta = \cos^{-1}\left[1 - \frac{1}{2gL}\left(\frac{mu}{m+M}\right)^2\right]$

Tips & Tricks

✍ The area under the force-displacement graph is equal to the work done.

✍ Work done by gravitation or electric force does not depend on the path followed. It depends on the initial and final positions of the body. Such forces are called conservative. When a body returns to the starting point under the action of conservative force, the net work done is zero that is $\oint dW = 0$.

✍ Work done against friction depends on the path followed. Viscosity and friction are not conservative forces. For non conservative forces, the work done on a closed path is not zero. That is $\oint dW \neq 0$.

✍ Work done is path independent only for a conservative field.

✍ Work done depends on the frame of reference.

✍ Work done by a centripetal force is always zero.

✍ Energy is a promise of work to be done in future. It is the stored ability to do work.

✍ Energy of a body is equal to the work done by the body and it has nothing to do with the time taken to perform the work. On the other hand, the power of the body depends on the time in which the work is done.

✍ When work is done on a body, its kinetic or potential energy increases.

✍ When the work is done by the body, its potential or kinetic energy decreases.

✍ According to the work energy theorem, the work done is equal to the change in energy. That is $W = \Delta E$.

✍ Work energy theorem is particularly useful in calculation of minimum stopping force or minimum stopping distance. If a body is brought to a halt, the work done to do so is equal to the kinetic energy lost.

✍ Potential energy of a system increases when a conservative force does work on it.

✍ The kinetic energy of a body is always positive.

✍ When the momentum of a body increases by a factor *n*, then its kinetic energy is increased by factor *n*.

✍ If the speed of a vehicle is made *n* times, then its stopping distance becomes *n* times.

✍ The total energy (including mass energy) of the universe remains constant.

✍ One form of energy can be changed into other form according to the law of conservation of energy. That is amount of energy lost of one form should be equal to energy or energies produced of other forms.

✍ Kinetic energy can change into potential energy and vice versa. When a body falls, potential energy is converted into kinetic energy.

✍ Pendulum oscillates due to conversion of kinetic energy into potential energy and vice versa. Same is true for the oscillations of mass attached to the spring.

✍ Conservation laws can be used to describe the behaviour of a mechanical system even when the exact nature of the forces involved is not known.

✍ Although the exact nature of the nuclear forces is not known, yet we can solve problems regarding the nuclear forces with the help of the conservation laws.

✍ Violation of the laws of conservation indicates that the event cannot take place.

✍ The gravitational potential energy of a mass *m* at a height *h* above the surface of the earth (radius *R*) is given by $U = \frac{mgh}{1 + h/R}$. When *h* << *R*, we find *U=mgh*.

✍ Electrostatic energy in capacitor - $U = \frac{1}{2}CV^2$, where C is capacitance, V = potential difference between the plates.

✍ Electric potential energy of a test charge q_0 at a place where electric potential is *V*, is given by : $U = q_0 V$.

✍ Electric potential energy between two charges (q_1 and q_2) separated by a distance *r* is given by $U = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r}$. Here ε_0 is permittivity of vacuum and $1/4\pi\varepsilon_0 = 9 \times 10^9 \ Nm^2C^{-2}$.

✍ Magnetic energy stored in an inductor –

$U = \frac{1}{2}LI^2$, where L = inductance, I = current.

- Energy gained by a body of mass *m*, specific heat *C*, when its temperature changes by $\Delta\theta$ is given by : $Q = mC\Delta\theta$.
- The Potential energy associated with a spring of constant *k* when extended or compressed by distance *x* is given by $U = \frac{1}{2}kx^2$.
- Kinetic energy of a particle executing *SHM* is given by : $K = \frac{1}{2}m\omega^2(a^2 - y^2)$ where *m* = mass, ω = angular frequency, *a*= amplitude, *y* = displacement.
- Potential energy of a particle executing SHM is given by : $U = \frac{1}{2}m\omega^2 y^2$.
- Total energy of a particle executing SHM is given by : $E = K + U = \frac{1}{2}m\omega^2 a^2$.
- Energy density associated with a wave $= \frac{1}{2}\rho\omega^2 a^2$ where ρ =density of medium, ω = angular frequency, *a* = amplitude of the of the wave.
- Energy associated with a photon : $E = h\nu = hc/\lambda$, where *h* = planck's constant, ν = frequency of the light wave, *c* = velocity of light, λ = wave length.
- Mass and energy are interconvertible. That is mass can be converted into energy and energy can be converted into mass.
- A mass *m* (in *kg*) is equivalent to energy (in *J*) which is equal to *mc* where *c* = speed of light.
- A stout spring has a large value of force constant, while for a delicate spring, the value of spring constant is low.
- The term energy is different from power. Whereas energy refers to the capacity to perform the work, power determines the rate of performing the work. Thus, in determining power, time taken to perform the work is significant but it is of no importance for measuring energy of a body.
- Collision is the phenomenon in which two bodies exert mutual force on each other.
- The collision generally occurs for very small interval of time.
- Physical contact between the colliding bodies is not essential for the collision.
- The mutual forces between the colliding bodies are action and reaction pair. In accordance with the Newton's third law of motion, they are equal and opposite to each other.
- The collision is said to be elastic when the kinetic energy is conserved.
- In the elastic collisions the forces involved are conservative.
- In the elastic collisions, the kinetic or mechanical energy is not converted into any other form of energy.
- Elastic collisions produce no sound or heat.
- There is no difference between the elastic and perfectly elastic collisions.
- In the elastic collisions, the relative velocity before collision is equal to the relative velocity after the collision. That is $\vec{u}_1 - \vec{u}_2 = \vec{v}_2 - \vec{v}_1$ where $\vec{u}_1$ and $\vec{u}_2$ are initial velocities and $\vec{v}_1$ and $\vec{v}_2$ are the velocities of the colliding bodies after the collision. This is called Newton's law of impact.
- The collision is said to be inelastic when the kinetic energy is not conserved.
- In the perfectly inelastic collision, the colliding bodies stick together. That is the relative velocity of the bodies after the collision is zero.
- In an elastic collision of two equal masses, their kinetic energies are exchanged.
- If a body of mass *m* moving with velocity *v*, collides elastically with a rigid wall, then the change in the momentum of the body is 2*mv*.
- $e = \frac{\vec{v}_2 - \vec{v}_1}{\vec{u}_1 - \vec{u}_2}$ is called coefficient of restitution. Its value is 1 for elastic collisions. It is less than 1 for inelastic collisions and zero for perfectly inelastic collision.
- During collision, velocity of the colliding bodies changes.
- Linear momentum is conserved in all types of collisions.
- Perfectly elastic collision is a rare physical phenomenon.
- Collisions between two ivory or steel or glass balls are nearly elastic.
- The force of interaction in an inelastic collision is non-conservative in nature.
- In inelastic collision, the kinetic energy is converted into heat energy, sound energy, light energy etc.
- In head on collisions, the colliding bodies move along the same straight line before and after collision.
- Head on collisions are also called one dimensional collisions.
- In the oblique collisions the colliding bodies move at certain angles before and/or after the collisions.
- The oblique collisions are two dimensional collisions.
- When a heavy body collides head-on elastically with a lighter body, then the lighter body begins to move with a velocity nearly double the velocity of the heavier body.
- When a light body collides with a heavy body, the lighter body returns almost with the same speed.
- If a light and a heavy body have equal momenta, then lighter body has greater kinetic energy.
- Suppose, a body is dropped form a height h_0 and it strikes the ground with velocity v_0. After the (inelastic) collision let it rise to a height h_1. If v_1 be the velocity with which the body rebounds, then

$$e = \frac{v_1}{v_0} = \left[\frac{2gh_1}{2gh_0}\right]^{1/2} = \left[\frac{h_1}{h_0}\right]^{1/2}$$

- If after *n* collisions with the ground, the velocity is v_n and the height to which it rises be h_n, then

$$e^n = \frac{v_n}{v_0} = \left[\frac{h_n}{h_0}\right]^{1/2}$$

- $P = \vec{F}.\vec{v} = Fv\cos\theta$ where $\vec{v}$ is the velocity of the body and θ is the angle between $\vec{F}$ and $\vec{v}$.

✍ Area under the $F-v$ graph is equal to the power dissipated.

✍ Power dissipated by a conservative force (gravitation, electric force etc.) does not depend on the path followed. It depends on the initial and final positions of the body. That is $\oint dP = 0$.

✍ Power dissipated against friction depends on the path followed. That is $\oint dP \neq 0$.

✍ Power is also measured in horse power (*hp*). It is the *fps* unit of power. 1 *hp* = 746 *W*.

✍ An engine pulls a train of mass *m* with constant velocity. If the rails are on a plane surface and there is no friction, the power dissipated by the engine is zero.

✍ In the above case if the coefficient of friction for the rail is μ, the power of the engine is $P = \mu mgv$.

✍ In the above case if the engine pulls on a smooth track on an inclined plane (inclination θ), then its power $P = (mg\sin\theta)v$.

✍ In the above case if the engine pulls upwards on a rough inclined plane having coefficient of friction μ, then power of the engine is

$P = (\mu\cos\theta + \sin\theta)mg\,v$.

✍ If the engine pulls down on the inclined plane then power of the engine is

$P = (\mu\cos\theta - \sin\theta)mg\,v$.

Ordinary Thinking

Objective Questions

Work Done by Constant Force

1. A body of mass m is moving in a circle of radius r with a constant speed v. The force on the body is $\frac{mv^2}{r}$ and is directed towards the centre. What is the work done by this force in moving the body over half the circumference of the circle

(a) $\frac{mv^2}{\pi r^2}$ (b) Zero

(c) $\frac{mv^2}{r^2}$ (d) $\frac{\pi r^2}{mv^2}$

2. If the unit of force and length each be increased by four times, then the unit of energy is increased by [CPMT 1987]

(a) 16 times (b) 8 times

(c) 2 times (d) 4 times

3. A man pushes a wall and fails to displace it. He does [CPMT 1992]

(a) Negative work

(b) Positive but not maximum work

(c) No work at all

(d) Maximum work

4. The same retarding force is applied to stop a train. The train stops after 80 *m*. If the speed is doubled, then the distance will be

(a) The same (b) Doubled

(c) Halved (d) Four times

5. A body moves a distance of 10 *m* along a straight line under the action of a force of 5 *N*. If the work done is 25 *joules*, the angle which the force makes with the direction of motion of the body is [NCERT 1980; JIPMER 1997; CBSE PMT 1999; BHU 2000; RPMT 2000; Orissa JEE 2002]

(a) 0° (b) 30°

(c) 60° (d) 90°

6. You lift a heavy book from the floor of the room and keep it in the book-shelf having a height 2 *m*. In this process you take 5 seconds. The work done by you will depend upon [MP PET 1993]

(a) Mass of the book and time taken

(b) Weight of the book and height of the book-shelf

(c) Height of the book-shelf and time taken

(d) Mass of the book, height of the book-shelf and time taken

7. A body of mass *m kg* is lifted by a man to a height of one metre in 30 *sec*. Another man lifts the same mass to the same height in 60 *sec*. The work done by them are in the ratio

(a) $1:2$ (b) 1 : 1

(c) 2 : 1 (d) 4 : 1

8. A force $\boldsymbol{F} = (5\hat{\boldsymbol{i}} + 3\hat{\boldsymbol{j}})$ *newton* is applied over a particle which displaces it from its origin to the point $\boldsymbol{r} = (2\hat{\boldsymbol{i}} - 1\hat{\boldsymbol{j}})$ *metres*. The work done on the particle is [MP PMT 1995; RPET 2003]

(a) – 7 *joules* (b) + 13 *joules*

(c) + 7 *joules* (d) + 11 *joules*

9. A force acts on a 30 *gm* particle in such a way that the position of the particle as a function of time is given by $x = 3t - 4t^2 + t^3$, where *x* is in *metres* and *t* is in seconds. The work done during the first 4 seconds is [NCERT 1977] [CBSE PMT 1998]

(a) 5.28 *J* (b) 450 *mJ*

(c) 490 *mJ* (d) 530 *mJ*

10. A body of mass 10 *kg* is dropped to the ground from a height of 10 *metres*. The work done by the gravitational force is $(g = 9.8\, m/\sec^2)$ [SCRA 1994]

(a) – 490 *Joules* (b) + 490 *Joules*

(c) – 980 *Joules* (d) + 980 *Joules*

11. Which of the following is a scalar quantity [AFMC 1998]

(a) Displacement (b) Electric field

(c) Acceleration (d) Work

12. The work done in pulling up a block of wood weighing 2 *kN* for a length of 10*m* on a smooth plane inclined at an angle of 15° with the horizontal is [AFMC 1999; Pb PMT 2003] [CPMT 1984]

(a) 4.36 *kJ* (b) 5.17 *kJ*

(c) 8.91 *kJ* (d) 9.82 *kJ*

13. A force $\vec{F} = 5\hat{i} + 6\hat{j} - 4\hat{k}$ acting on a body, produces a displacement $\vec{s} = 6\vec{i} + 5\vec{k}$. Work done by the force is [KCET 1999]

(a) 18 units (b) 15 units

(c) 12 units (d) 10 units

14. A force of 5 *N* acts on a 15 *kg* body initially at rest. The work done by the force during the first second of motion of the body is

(a) 5 *J* (b) $\frac{5}{6}J$

(c) 6 *J* (d) $75\,J$

15. A force of 5 *N*, making an angle θ with the horizontal, acting on an object displaces it by $0.4m$ along the horizontal direction. If the object gains kinetic energy of 1*J*, the horizontal component of the force is [EAMCET (Engg.) 2000] [MP PMT 1993]

(a) 1.5 *N* (b) 2.5 *N*

(c) 3.5 *N* (d) 4.5 *N*

16. The work done against gravity in taking 10 *kg* mass at 1*m* height in 1*sec* will be [RPMT 2000]

(a) 49 *J* (b) 98 *J*

(c) 196 *J* (d) None of these

17. The energy which an e^- acquires when accelerated through a potential difference of 1 volt is called [UPSEAT 2000]
(a) 1 *Joule* (b) 1 *Electron volt*
(c) 1 *Erg* (d) 1 *Watt.*

18. A body of mass 6*kg* is under a force which causes displacement in it given by $S = \frac{t^2}{4}$ *metres* where *t* is time. The work done by the force in 2 seconds is [EAMCET 2001]
(a) 12 *J* (b) 9 *J*
(c) 6 *J* (d) 3 *J*

19. A body of mass 10*kg* at rest is acted upon simultaneously by two forces 4 *N* and 3*N* at right angles to each other. The kinetic energy of the body at the end of 10 *sec* is [Kerala (Engg.) 2001]
(a) 100 *J* (b) 300 *J*
(c) 50 *J* (d) 125 *J*

20. A cylinder of mass 10*kg* is sliding on a plane with an initial velocity of 10*m/s*. If coefficient of friction between surface and cylinder is 0.5, then before stopping it will describe [Pb. PMT 2001]
(a) 12.5 *m* (b) 5 *m*
(c) 7.5 *m* (d) 10 *m*

21. A force of $(3\hat{i} + 4\hat{j})$ *Newton* acts on a body and displaces it by $(3\hat{i} + 4\hat{j})m$. The work done by the force is [AIIMS 2001]
(a) 10 *J* (b) 12 *J*
(c) 16 *J* (d) 25 *J*

22. A 50*kg* man with 20*kg* load on his head climbs up 20 steps of 0.25*m* height each. The work done in climbing is [JIPMER 2002]
(a) 5 *J* (b) 350 *J*
(c) 100 *J* (d) 3430 *J*

23. A force $\vec{F} = 6\hat{i} + 2\hat{j} - 3\hat{k}$ acts on a particle and produces a displacement of $\vec{s} = 2\hat{i} - 3\hat{j} + x\hat{k}$. If the work done is zero, the value of *x* is [Kerala PMT 2002]
(a) – 2 (b) 1/2
(c) 6 (d) 2

24. A particle moves from position $\vec{r}_1 = 3\hat{i} + 2\hat{j} - 6\hat{k}$ to position $\vec{r}_2 = 14\hat{i} + 13\hat{j} + 9\hat{k}$ under the action of force $4\hat{i} + \hat{j} + 3\hat{k}\,N$. The work done will be [Pb. PMT 2002,03]
(a) 100 *J* (b) 50 *J*
(c) 200 *J* (d) 75 *J*

25. A force $(\vec{F}) = 3\hat{i} + c\hat{j} + 2\hat{k}$ acting on a particle causes a displacement: $(\vec{s}) = -4\hat{i} + 2\hat{j} + 3\hat{k}$ in its own direction. If the work done is $6\,J$, then the value of 'c' is [CBSE PMT 2002]
(a) 0 (b) 1
(c) 6 (d) 12

26. In an explosion a body breaks up into two pieces of unequal masses. In this [MP PET 2002]
(a) Both parts will have numerically equal momentum
(b) Lighter part will have more momentum
(c) Heavier part will have more momentum
(d) Both parts will have equal kinetic energy

27. Which of the following is a unit of energy [AFMC 2002]
(a) Unit (b) Watt
(c) Horse Power (d) None

28. If force and displacement of particle in direction of force are doubled. Work would be [AFMC 2002]
(a) Double (b) 4 times
(c) Half (d) $\frac{1}{4}$ times

29. A body of mass 5 *kg* is placed at the origin, and can move only on the x-axis. A force of 10 *N* is acting on it in a direction making an angle of 60^o with the x-axis and displaces it along the x-axis by 4 *metres*. The work done by the force is
(a) 2.5 *J* (b) 7.25 *J*
(c) 40 *J* (d) 20 *J*

30. A force $\vec{F} = (5\hat{i} + 4\hat{j})$ *N* acts on a body and produces a displacement $\vec{S} = (6\hat{i} - 5\hat{j} + 3\hat{k})\,m$. The work done will be [CPMT 2003]
(a) 10 *J* (b) 20 *J*
(c) 30 *J* (d) 40 *J*

31. A uniform chain of length 2*m* is kept on a table such that a length of 60*cm* hangs freely from the edge of the table. The total mass of the chain is 4*kg*. What is the work done in pulling the entire chain on the table [AIEEE 2004]
(a) 7.2 *J* (b) 3.6 *J*
(c) 120 *J* (d) 1200 *J*

32. A particle is acted upon by a force of constant magnitude which is always perpendicular to the velocity of the particle, the motion of the particle takes place in a plane. It follows that
(a) Its velocity is constant
(b) Its acceleration is constant
(c) Its kinetic energy is constant
(d) It moves in a straight line

33. A ball of mass *m* moves with speed *v* and strikes a wall having infinite mass and it returns with same speed then the work done by the ball on the wall is [BCECE 2004]
(a) Zero (b) *mv J*
(c) *m/v.J* (d) *v/m J*

34. A force $\vec{F} = (5\hat{i} + 3\hat{j} + 2\hat{k})N$ is applied over a particle which displaces it from its origin to the point $\vec{r} = (2\hat{i} - \hat{j})m$. The work done on the particle in joules is [AIEEE 2004]
(a) – 7 (b) +7
(c) +10 (d) +13

35. The kinetic energy acquired by a body of mass *m* is travelling some distance *s*, starting from rest under the actions of a constant force, is directly proportional to [Pb. PET 2000]
(a) m^0 (b) m
(c) m^2 (d) $\sqrt{m}$

36. If a force $\vec{F} = 4\hat{i} + 5\hat{j}$ causes a displacement $\vec{s} = 3\hat{i} + 6\hat{k}$, work done is [Pb. PET 2002]
(a) 4×6 unit (b) 6×3 unit

(c) 5×6 unit (d) 4×3 unit

37. A man starts walking from a point on the surface of earth (assumed smooth) and reaches diagonally opposite point. What is the work done by him [DCE 2004]

(a) Zero (b) Positive

(c) Negative (d) Nothing can be said

38. It is easier to draw up a wooden block along an inclined plane than to haul it vertically, principally because [CPMT 1977; JIPMER 1997]

(a) The friction is reduced

(b) The mass becomes smaller

(c) Only a part of the weight has to be overcome

(d) 'g' becomes smaller

39. Two bodies of masses 1 *kg* and 5 *kg* are dropped gently from the top of a tower. At a point 20 *cm* from the ground, both the bodies will have the same [SCRA 1998]

(a) Momentum (b) Kinetic energy

(c) Velocity (d) Total energy

40. Due to a force of $(6\hat{i} + 2\hat{j})N$ the displacement of a body is $(3\hat{i} - \hat{j})m$, then the work done is [Orissa JEE 2005]

(a) 16 J (b) 12 J

(c) 8 J (d) Zero

41. A ball is released from the top of a tower. The ratio of work done by force of gravity in first, second and third second of the motion of the ball is [Kerala PET 2005]

(a) 1 : 2 : 3 (b) 1 : 4 : 9

(c) 1 : 3 : 5 (d) 1 : 5 : 3

Work Done by Variable Force

1. A particle moves under the effect of a force $F = Cx$ from $x = 0$ to $x = x_1$. The work done in the process is [CPMT 1982; DCE 2002;Orissa JEE 2005]

(a) Cx_1^2 (b) $\frac{1}{2}Cx_1^2$

(c) Cx_1 (d) Zero

2. A cord is used to lower vertically a block of mass M by a distance d with constant downward acceleration $\frac{g}{4}$. Work done by the cord on the block is [CPMT 1972]

(a) $Mg\frac{d}{4}$ (b) $3Mg\frac{d}{4}$

(c) $-3Mg\frac{d}{4}$ (d) Mgd

3. Two springs have their force constant as k_1 and $k_2 (k_1 > k_2)$. When they are stretched by the same force [EAMCET 1981]

(a) No work is done in case of both the springs

(b) Equal work is done in case of both the springs

(c) More work is done in case of second spring

(d) More work is done in case of first spring

4. A spring of force constant 10 N/m has an initial stretch 0.20 m. In changing the stretch to 0.25 m, the increase in potential energy is about [CPMT 1977]

(a) 0.1 *joule* (b) 0.2 *joule*

(c) 0.3 *joule* (d) 0.5 *joule*

5. The potential energy of a certain spring when stretched through a distance 'S' is 10 *joule*. The amount of work (in joule) that must be done on this spring to stretch it through an additional distance 'S' will be [MNR 1991; CPMT 2002; UPSEAT 2000; Pb. PET 2004]

(a) 30 (b) 40

(c) 10 (d) 20

6. Two springs of spring constants 1500 N/m and 3000 N/m respectively are stretched with the same force. They will have potential energy in the ratio [MP PMT/PET 1998; Pb. PMT 2002]

(a) 4 : 1 (b) 1 : 4

(c) 2 : 1 (d) 1 : 2

7. A spring 40 *mm* long is stretched by the application of a force. If 10 N force required to stretch the spring through 1 *mm*, then work done in stretching the spring through 40 *mm* is

(a) 84 J (b) 68 J

(c) 23 J (d) 8 J

8. A position dependent force $F = 7 - 2x + 3x^2\,newton$ acts on a small body of mass 2 *kg* and displaces it from $x = 0$ to $x = 5\,m$. The work done in *joules* is [CBSE PMT 1994]

(a) 70 (b) 270

(c) 35 (d) 135

9. A body of mass 3 *kg* is under a force, which causes a displacement in it is given by $S = \frac{t^3}{3}$ (in m). Find the work done by the force in first 2 seconds [BHU 1998]

(a) 2 J (b) 3.8 J

(c) 5.2 J (d) 24 J

10. The force constant of a wire is k and that of another wire is $2k$. When both the wires are stretched through same distance, then the work done [MH CET 2000]

(a) $W_2 = 2W_1^2$ (b) $W_2 = 2W_1$

(c) $W_2 = W_1$ (d) $W_2 = 0.5W_1$

11. A body of mass 0.1 *kg* moving with a velocity of 10 *m/s* hits a spring (fixed at the other end) of force constant 1000 N/m and comes to rest after compressing the spring. The compression of the spring is

(a) $0.01m$ (b) $0.1m$

(c) $0.2m$ (d) $0.5m$

12. When a 1.0*kg* mass hangs attached to a spring of length 50 *cm*, the spring stretches by 2 *cm*. The mass is pulled down until the length of the spring becomes 60 *cm*. What is the amount of elastic energy stored in the spring in this condition, if g = 10 *m/s*

(a) 1.5 *Joule* (b) 2.0 *Joule*

(c) 2.5 *Joule* (d) 3.0 *Joule*

13. A spring of force constant 800 N/m has an extension of 5*cm*. The work done in extending it from 5*cm* to 15 *cm* is [AIEEE 2002]

(a) 16 J (b) 8 J

(c) 32 J (d) 24 J

14. When a spring is stretched by 2 *cm*, it stores 100 *J* of energy. If it is stretched further by 2 *cm*, the stored energy will be increased by

(a) 100 *J* (b) 200 *J*

(c) 300 *J* (d) 400 *J*

15. A spring when stretched by 2 mm its potential energy becomes 4 J. If it is stretched by 10 mm, its potential energy is equal to

(a) 4 *J* (b) 54 *J*

(c) 415 *J* (d) None

16. A spring of spring constant 5×10^3 *N/m* is stretched initially by 5*cm* from the unstretched position. Then the work required to stretch it further by another $5cm$ is

[AIEEE 2003]

(a) 6.25 *N-m* (b) 12.50 *N-m*

(c) 18.75 *N-m* (d) 25.00 *N-m*

17. A mass of 0.5*kg* moving with a speed of 1.5 *m/s* on a horizontal smooth surface, collides with a nearly weightless spring of force constant $k = 50\ N/m$. The maximum compression of the spring would be [CBSE PMT 2004]

(a) 0.15 *m* (b) 0.12 *m*

(c) 1.5 *m* (d) 0.5 *m*

18. A particle moves in a straight line with retardation proportional to its displacement. Its loss of kinetic energy for any displacement *x* is proportional to [AIEEE 2004]

(a) x^2 (b) e^x

(c) x (d) $\log_e x$

19. A spring with spring constant *k* when stretched through 1 *cm*, the potential energy is *U*. If it is stretched by 4 *cm*. The potential energy will be [Orissa PMT 2004]

(a) 4U (b) 8U

(c) 16 U (d) 2U

20. A spring with spring constant *k* is extended from $x = 0$ to $x = x_1$. The work done will be [Orissa PMT 2004]

(a) kx_1^2 (b) $\frac{1}{2}kx_1^2$

(c) $2kx_1^2$ (d) $2kx_1$

21. If a long spring is stretched by 0.02 *m*, its potential energy is *U*. If the spring is stretched by 0.1 *m*, then its potential energy will be

[MP PMT 2002; CBSE PMT 2003; UPSEAT 2004]

(a) $\frac{U}{5}$ (b) U

(c) 5U (d) 25U

22. Natural length of a spring is 60 *cm*, and its spring constant is 4000 *N/m*. A mass of 20 *kg* is hung from it. The extension produced in the spring is, (Take $g = 9.8\ m/s^2$) [DCE 2004]

(a) 4.9 *cm* (b) 0.49 *cm*

(c) 9.4 *cm* (d) 0.94 *cm*

23. The spring extends by *x* on loading, then energy stored by the spring is :

(if *T* is the tension in spring and *k* is spring constant)

[Pb. PMT 2003]

(a) $\frac{T^2}{2k}$ [Orissa JEE 2002] (b) $\frac{T^2}{2k^2}$

(c) $\frac{2k}{T^2}$ (d) $\frac{2T^2}{k}$

24. The potential energy of a body is given by, [BCECE 2003] $U = A - Bx^2$ (Where *x* is the displacement). The magnitude of force acting on the particle is [BHU 2002]

(a) Constant

(b) Proportional to *x*

(c) Proportional to x^2

(d) Inversely proportional to *x*

25. The potential energy between two atoms in a molecule is given by $U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$; where *a* and *b* are positive constants and *x* is the distance between the atoms. The atom is in stable equilibrium when [CBSE PMT 1995]

(a) $x = \sqrt[6]{\frac{11a}{5b}}$ (b) $x = \sqrt[6]{\frac{a}{2b}}$

(c) $x = 0$ (d) $x = \sqrt[6]{\frac{2a}{b}}$

26. Which one of the following is not a conservative force

[Kerala PMT 2005]

(a) Gravitational force

(b) Electrostatic force between two charges

(c) Magnetic force between two magnetic dipoles

(d) Frictional force

Conservation of Energy and Momentum

1. Two bodies of masses m_1 and m_2 have equal kinetic energies. If p_1 and p_2 are their respective momentum, then ratio $p_1 : p_2$ is equal to [MP PMT 1985; CPMT 1990]

(a) $m_1 : m_2$ (b) $m_2 : m_1$

(c) $\sqrt{m_1} : \sqrt{m_2}$ (d) $m_1^2 : m_2^2$

2. Work done in raising a box depends on

(a) How fast it is raised

(b) The strength of the man

(c) The height by which it is raised

(d) None of the above

3. A light and a heavy body have equal momenta. Which one has greater K.E

[MP PMT 1985; CPMT 1985; Kerala PMT 2004]

(a) The light body (b) The heavy body

(c) The K.E. are equal (d) Data is incomplete

4. A body at rest may have

(a) Energy (b) Momentum

(c) Speed (d) Velocity

5. The kinetic energy possessed by a body of mass *m* moving with a velocity *v* is equal to $\frac{1}{2}mv^2$, provided

(a) The body moves with velocities comparable to that of light

(b) The body moves with velocities negligible compared to the speed of light

(c) The body moves with velocities greater than that of light

(d) None of the above statement is correcst

6. If the momentum of a body is increased n times, its kinetic energy increases

(a) n times (b) $2n$ times

(c) $\sqrt{n}$ times (d) n^2 times

7. When work is done on a body by an external force, its

(a) Only kinetic energy increases

(b) Only potential energy increases

(c) Both kinetic and potential energies may increase

(d) Sum of kinetic and potential energies remains constant

8. The bob of a simple pendulum (mass m and length l) dropped from a horizontal position strikes a block of the same mass elastically placed on a horizontal frictionless table. The K.E. of the block will be

(a) 2 mgl (b) $mgl/2$

(c) mgl (d) 0

9. From a stationary tank of mass 125000 *pound* a small shell of mass 25 *pound* is fired with a muzzle velocity of 1000 *ft/sec*. The tank recoils with a velocity of [NCERT 1973]

(a) 0.1 *ft/sec* (b) 0.2 *ft/sec*

(c) 0.4 *ft*/sec (d) 0.8 *ft/sec*

10. A bomb of 12 *kg* explodes into two pieces of masses 4 *kg* and 8 *kg*. The velocity of 8*kg* mass is 6 *m/sec*. The kinetic energy of the other mass is

[MNR 1985; CPMT 1991; Manipal MEE 1995; Pb. PET 2004]

(a) 48 *J* (b) 32 *J*

(c) 24 *J* (d) 288 *J*

11. A rifle bullet loses 1/20th of its velocity in passing through a plank. The least number of such planks required just to stop the bullet is[EAMCET 1987; AFMC 2004]

(a) 5 (b) 10

(c) 11 (d) 20

12. A body of mass 2 *kg* is thrown up vertically with K.E. of 490 joules. If the acceleration due to gravity is $9.8\,m/s^2$, then the height at which the K.E. of the body becomes half its original value is given by [EAMCET 1986]

(a) 50 *m* (b) 12.5 *m*

(c) 25 *m* (d) 10 *m*

13. Two masses of 1 *gm* and 4 *gm* are moving with equal kinetic energies. The ratio of the magnitudes of their linear momenta is

[AIIMS 1987; NCERT 1983; MP PMT 1993; IIT 1980; RPET 1996; CBSE PMT 1997; Orissa JEE 2003; KCET 1999; DCE 2004]

(a) 4 : 1 (b) $\sqrt{2}:1$

(c) 1 : 2 (d) 1 : 16

14. If the *K.E.* of a body is increased by 300%, its momentum will increase by [JIPMER 1978; AFMC 1993; RPET 1999; CBSE PMT 2002]

(a) 100% (b) 150%

(c) $\sqrt{300}\%$ (d) 175%

15. A light and a heavy body have equal kinetic energy. Which one has a greater momentum ?

[NCERT 1974; CPMT 1997; DPMT 2001]

(a) The light body

(b) The heavy body

(c) Both have equal momentum

(d) It is not possible to say anything without additional information

16. If the linear momentum is increased by 50%, the kinetic energy will increase by

[CPMT 1983; MP PMT 1994; MP PET 1996, 99; UPSEAT 2001]

(a) 50% (b) 100%

(c) 125% (d) 25%

17. A free body of mass 8 *kg* is travelling at 2 *meter* per second in a straight line. At a certain instant, the body splits into two equal parts due to internal explosion which releases 16 *joules* of energy. Neither part leaves the original line of motion finally

(a) Both parts continue to move in the same direction as that of the original body

(b) One part comes to rest and the other moves in the same direction as that of the original body

(c) One part comes to rest and the other moves in the direction opposite to that of the original body

(d) One part moves in the same direction and the other in the direction opposite to that of the original body

18. If the K.E. of a particle is doubled, then its momentum will

[EAMCET 1979; CPMT 2003: Kerala PMT 2005]

(a) Remain unchanged (b) Be doubled

(c) Be quadrupled (d) Increase $\sqrt{2}$ times

19. If the stone is thrown up vertically and return to ground, its potential energy is maximum [EAMCET 1979]

(a) During the upward journey

(b) At the maximum height

(c) During the return journey

(d) At the bottom

20. A body of mass 2 *kg* is projected vertically upwards with a velocity of $2\,m\,sec^{-1}$. The K.E. of the body just before striking the ground is [EAMCET 1980]

(a) 2 *J* (b) 1 *J*

(c) 4 *J* (d) 8 *J*

21. The energy stored in wound watch spring is

[EAMCET 1982]

(a) K.E. (b) P.E.

(c) Heat energy (d) Chemical energy

22. Two bodies of different masses m_1 and m_2 have equal momenta. Their kinetic energies E_1 and E_2 are in the ratio

[EAMCET 1990]

(a) $\sqrt{m_1}:\sqrt{m_2}$ (b) $m_1 : m_2$

(c) $m_2 : m_1$ (d) $m_1^2 : m_2^2$

23. A car travelling at a speed of 30 *km/hour* is brought to a halt in 8 *m* by applying brakes. If the same car is travelling at 60 *km/hour*, it can be brought to a halt with the same braking force in

(a) 8 *m* (b) 16 *m*

(c) 24 *m* (d) 32 *m*

24. Tripling the speed of the motor car multiplies the distance needed for stopping it by [NCERT 1978]

(a) 3 (b) 6

(c) 9 (d) Some other number

25. If the kinetic energy of a body increases by 0.1%, the percent increase of its momentum will be [MP PMT 1994]

(a) 0.05% (b) 0.1%

(c) 1.0% (d) 10%

26. If velocity of a body is twice of previous velocity, then kinetic energy will become [AFMC 1996]

(a) 2 times (b) $\frac{1}{2}$ times

(c) 4 times (d) 1 times

27. Two bodies A and B having masses in the ratio of 3 : 1 possess the same kinetic energy. The ratio of their linear momenta is then

(a) 3 : 1 (b) 9 : 1

(c) 1 : 1 (d) $\sqrt{3}:1$

28. In which case does the potential energy decrease [MP PET 1996]

(a) On compressing a spring

(b) On stretching a spring

(c) On moving a body against gravitational force

(d) On the rising of an air bubble in water

29. A sphere of mass m, moving with velocity V, enters a hanging bag of sand and stops. If the mass of the bag is M and it is raised by height h, then the velocity of the sphere was

(a) $\frac{M+m}{m}\sqrt{2gh}$ (b) $\frac{M}{m}\sqrt{2gh}$

(c) $\frac{m}{M+m}\sqrt{2gh}$ (d) $\frac{m}{M}\sqrt{2gh}$

30. Two bodies of masses m and $2m$ have same momentum. Their respective kinetic energies E_1 and E_2 are in the ratio [MP PET 1997; KCET 2004]

(a) 1 : 2 (b) 2 : 1

(c) $1:\sqrt{2}$ (d) 1 : 4

31. If a lighter body (mass M_1 and velocity V_1) and a heavier body (mass M_2 and velocity V_2) have the same kinetic energy, then

(a) $M_2V_2 < M_1V_1$ (b) $M_2V_2 = M_1V_1$

(c) $M_2V_1 = M_1V_2$ (d) $M_2V_2 > M_1V_1$

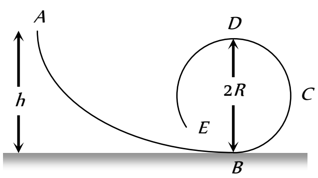
32. A frictionless track $ABCDE$ ends in a circular loop of radius R. A body slides down the track from point A which is at a height h = 5 cm. Maximum value of R for the body to successfully complete the loop is [MP PMT/PET 1998]

(a) 5 cm

(b) $\frac{15}{4}$ cm

(c) $\frac{10}{3}$ cm

(d) 2 cm



33. The force constant of a weightless spring is 16 N/m. A body of mass 1.0 kg suspended from it is pulled down through 5 cm and then released. The maximum kinetic energy of the system (spring + body) will be [MP PET 1999; DPMT 2000]

(a) $2\times10^{-2}\ J$ (b) $4\times10^{-2}\ J$

(c) $8\times10^{-2}\ J$ (d) $16\times10^{-2}\ J$

34. Two bodies with kinetic energies in the ratio of 4 : 1 are moving with equal linear momentum. The ratio of their masses is

(a) 1 : 2 (b) 1 : 1

(c) 4 : 1 (d) 1 : 4

35. If the kinetic energy of a body becomes four times of its initial value, then new momentum will [AIIMS 1998; AIIMS 2002; KCET 2000; J & K CET 2004]

(a) Becomes twice its initial value

[Haryana CEE 1996]

(b) Become three times its initial value

(c) Become four times its initial value

(d) Remains constant

36. A bullet is fired from a rifle. If the rifle recoils freely, then the kinetic energy of the rifle is [AIIMS 1998; JIPMER 2001; UPSEAT 2000]

(a) Less than that of the bullet

(b) More than that of the bullet

(c) Same as that of the bullet

(d) Equal or less than that of the bullet [MP PET 1997]

37. If the water falls from a dam into a turbine wheel 19.6 m below, then the velocity of water at the turbine is $(g = 9.8\ m/s^2)$

(a) 9.8 m/s (b) 19.6 m/s

(c) 39.2 m/s (d) 98.0 m/s

38. Two bodies of masses $2m$ and m have their K.E. in the ratio 8 : 1, then their ratio of momenta is [EAMCET (Engg.) 1995]

(a) 1 : 1 (b) 2 : 1

(c) 4 : 1 (d) 8 : 1

39. A bomb of 12 kg divides in two parts whose ratio of masses is 1 : 3. If kinetic energy of smaller part is 216 J, then momentum of bigger part in kg-m/sec will be [MP PMT 1997] [RPET 1997]

(a) 36 (b) 72

(c) 108 (d) Data is incomplete

40. A 4 kg mass and a 1 kg mass are moving with equal kinetic energies. The ratio of the magnitudes of their linear momenta is [CBSE PMT 1993; Orissa JE

(a) 1 : 2 (b) 1 : 1

(c) 2 : 1 (d) 4 : 1

41. Two identical cylindrical vessels with their bases at same level each contains a liquid of density ρ. The height of the liquid in one vessel is h_1 and that in the other vessel is h_2. The area of either base is A. The work done by gravity in equalizing the levels when the two vessels are connected, is [SCRA 1996]

(a) $(h_1-h_2)g\rho$ (b) $(h_1-h_2)gA\rho$

(c) $\frac{1}{2}(h_1-h_2)^2gA\rho$ (d) $\frac{1}{4}(h_1-h_2)^2gA\rho$

42. If the increase in the kinetic energy of a body is 22%, then the increase in the momentum will be

[RPET 1996; DPMT 2000]

(a) 22% (b) 44%

(c) 10% (d) 300%

43. If a body of mass 200 *g* falls from a height 200 *m* and its total P.E. is converted into K.E. at the point of contact of the body with earth surface, then what is the decrease in P.E. of the body at the contact $(g = 10\, m/s^2)$ [AFMC 1997]

(a) 200 *J* (b) 400 *J*

(c) 600 *J* (d) 900 *J*

44. If momentum is increased by 20%, then K.E. increases by

[AFMC 1997; MP PMT 2004]

(a) 44% (b) 55%

(c) 66% (d) 77%

45. The kinetic energy of a body of mass 2 *kg* and momentum of 2 *Ns* is

(a) 1 *J* (b) 2 *J*

(c) 3 *J* (d) 4 *J*

46. The decrease in the potential energy of a ball of mass 20 *kg* which falls from a height of 50 *cm* is [AIIMS 1997]

(a) 968 *J* (b) 98 *J*

(c) 1980 *J* (d) None of these

47. An object of 1 *kg* mass has a momentum of 10 *kg m/sec* then the kinetic energy of the object will be [RPMT 1999]

(a) 100 *J* (b) 50 *J*

(c) 1000 *J* (d) 200 *J*

48. A ball is released from certain height. It loses 50% of its kinetic energy on striking the ground. It will attain a height again equal to

(a) One fourth the initial height

(b) Half the initial height

(c) Three fourth initial height

(d) None of these

49. A 0.5 *kg* ball is thrown up with an initial speed 14 *m/s* and reaches a maximum height of 8.0*m*. How much energy is dissipated by air drag acting on the ball during the ascent

[AMU (Med.) 2000]

(a) $19.6\, Joule$ (b) $4.9\, Joule$

(c) $10\, Joule$ (d) $9.8\, Joule$

50. An ice cream has a marked value of 700 *kcal*. How many kilowatt-hour of energy will it deliver to the body as it is digested

(a) $0.81 kWh$ (b) $0.90 kWh$

(c) $1.11 kWh$ (d) $0.71 kWh$

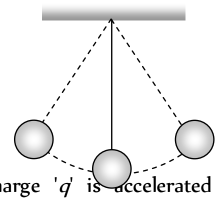
51. What is the velocity of the bob of a simple pendulum at its mean position, if it is able to rise to vertical height of $10 cm$ (Take $g = 9.8\, m/s^2$) [BHU 2000]

(a) 0.6 *m/s*

(b) 1.4 *m/s*

(c) 1.8 *m/s*

(d) 2.2 *m/s*



52. A particle of mass '*m*' and charge '*q*' is accelerated through a potential difference of 'V' volt. Its energy is [UPSEAT 2001]

(a) qV (b) mqV

(c) $\left(\frac{q}{m}\right)V$ (d) $\frac{q}{mV}$

53. A running man has half the kinetic energy of that of a boy of half of his mass. The man speeds up by 1*m/s* so as to have same *K.E.* as that of the boy. The original speed of the man will be

(a) $\sqrt{2}\, m/s$ (b) $(\sqrt{2}-1) m/s$

(c) $\frac{1}{(\sqrt{2}-1)} m/s$ (d) $\frac{1}{\sqrt{2}} m/s$

54. The mass of two substances are 4*gm* and $9gm$ respectively. If their kinetic energies are same, then the ratio of their momenta will be

(a) 4 : 9 (b) 9 : 4

(c) 3 : 2 (d) 2 : 3

55. If the momentum of a body is increased by 100%, then the percentage increase in the kinetic energy is

[AFMC 1998; DPMT 2000]

[BHU 1999; Pb. PMT 1999; CPMT 2000; CBSE PMT 2001; BCECE 2004]

(a) 150% (b) 200%

(c) 225% (d) 300%

56. If a body looses half of its velocity on penetrating 3 *cm* in a wooden block, then how much will it penetrate more before coming to rest

(a) 1 *cm* (b) 2 *cm*

(c) 3 *cm* (d) 4 *cm*

57. A bomb of mass 9*kg* explodes into 2 pieces of mass 3*kg* and 6*kg*. The velocity of mass 3*kg* is 1.6 *m/s*, the K.E. of mass 6*kg* is

(a) 3.84 *J* (b) 9.6 *J*

(c) 1.92 *J* (d) 2.92 *J*

58. Two masses of 1*kg* and 16*kg* are moving with equal K.E. The ratio of magnitude of the linear momentum is [RPMT 2000] [AIEEE 2002]

(a) 1 : 2 (b) 1 : 4

(c) $1 : \sqrt{2}$ (d) $\sqrt{2} : 1$

59. A machine which is 75 percent efficient, uses 12 joules of energy in lifting up a 1 *kg* mass through a certain distance. The mass is then allowed to fall through that distance. The velocity at the end of its fall is (in ms^{-1}) [Kerala PMT 2002]

(a) $\sqrt{24}$ (b) $\sqrt{32}$

(c) $\sqrt{18}$ (d) $\sqrt{9}$

60. Two bodies moving towards each other collide and move away in opposite directions. There is some rise in temperature of bodies because a part of the kinetic energy is converted into [AMU (Med.) 2000]

(a) Heat energy (b) Electrical energy

(c) Nuclear energy (d) Mechanical energy

61. A particle of mass m at rest is acted upon by a force *F* for a time *t*. Its Kinetic energy after an interval *t* is

[Kerala PET 2002]

(a) $\frac{F^2t^2}{m}$ (b) $\frac{F^2t^2}{2m}$

(c) $\frac{F^2t^2}{3m}$ (d) $\frac{Ft}{2m}$

62. The potential energy of a weight less spring compressed by a distance a is proportional to [MP PET 2003]

(a) a (b) a^2

(c) a^{-2} (d) a^0

63. Two identical blocks A and B, each of mass 'm' resting on smooth floor are connected by a light spring of natural length L and spring constant K, with the spring at its natural length. A third identical block 'C' (mass m) moving with a speed v along the line joining A and B collides with A. the maximum compression in the spring is [EAMCET 2003]

(a) $v\sqrt{\frac{m}{2k}}$ (b) $m\sqrt{\frac{v}{2k}}$

(c) $\sqrt{\frac{mv}{k}}$ (d) $\frac{mv}{2k}$

64. Two bodies of masses m and 4 m are moving with equal K.E. The ratio of their linear momentums is

[Orissa JEE 2003; AIIMS 1999]

(a) 4 : 1 (b) 1 : 1

(c) 1 : 2 (d) 1 : 4

65. A stationary particle explodes into two particles of a masses m and m which move in opposite directions with velocities v_1 and v_2. The ratio of their kinetic energies E_1/E_2 is

[CBSE PMT 2003]

(a) m_1/m_2 (b) 1

(c) m_1v_2/m_2v_1 (d) m_2/m_1

66. The kinetic energy of a body of mass 3 *kg* and momentum 2 *Ns* is

(a) 1 *J* (b) $\frac{2}{3}J$

(c) $\frac{3}{2}J$ (d) 4 *J*

67. A bomb of mass 3.0 *Kg* explodes in air into two pieces of masses 2.0 *kg* and 1.0 *kg*. The smaller mass goes at a speed of 80 *m/s*.The total energy imparted to the two fragments is

[AIIMS 2004]

(a) 1.07 *kJ* (b) 2.14 *kJ*

(c) 2.4 *kJ* (d) 4.8 *kJ*

68. A bullet moving with a speed of 100 ms^{-1} can just penetrate two planks of equal thickness. Then the number of such planks penetrated by the same bullet when the speed is doubled will be

(a) 4 (b) 8

(c) 6 (d) 10

69. A particle of mass m_1 is moving with a velocity v_1 and another particle of mass m_2 is moving with a velocity v_2. Both of them have the same momentum but their different kinetic energies are E_1 and E_2 respectively. If $m_1 > m_2$ then

(a) $E_1 < E_2$ (b) $\frac{E_1}{E_2} = \frac{m_1}{m_2}$

(c) $E_1 > E_2$ (d) $E_1 = E_2$

70. A ball of mass 2*kg* and another of mass 4*kg* are dropped together from a 60 *feet* tall building. After a fall of 30 *feet* each towards earth, their respective kinetic energies will be in the ratio of

(a) $\sqrt{2}:1$ (b) $1:4$

(c) $1:2$ (d) $1:\sqrt{2}$

71. Four particles given, have same momentum which has maximum kinetic energy [Orissa PMT 2004]

(a) Proton (b) Electron

(c) Deutron (d) α -particles

72. A body moving with velocity v has momentum and kinetic energy numerically equal. What is the value of v

[Pb. PMT 2002; J&K CET 2004]

(a) 2*m/s* (b) $\sqrt{2}m/s$

(c) 1*m/s* (d) $0.2\ m/s$

73. If a man increase his speed by 2 *m/s* , his K.E. is doubled, the original speed of the man is [Pb. PET 2002]

(a) $(1+2\sqrt{2})\ m/s$ (b) 4 *m/s*

(c) $(2+2\sqrt{2})m/s$ (d) $(2+\sqrt{2})\ m/s$

74. An object of mass 3m splits into three equal fragments. Two fragments have velocities $v\hat{j}$ and $v\hat{i}$. The velocity of the third fragment is [UPSEAT 2004]

(a) $v(\hat{j}-\hat{i})$ (b) $v(\hat{i}-\hat{j})$

(c) $-v(\hat{i}+\hat{j})$ (d) $\frac{v(\hat{i}+\hat{j})}{\sqrt{2}}$

[MP PET 2004]

75. A bomb is kept stationary at a point. It suddenly explodes into two fragments of masses 1 g and 3 g. The total K.E. of the fragments is $6.4\times10^4\ J$. What is the K.E. of the smaller fragment

(a) $2.5\times10^4\ J$ (b) $3.5\times10^4\ J$

(c) $4.8\times10^4\ J$ (d) $5.2\times10^4\ J$

76. Which among the following, is a form of energy [DCE 2004]

(a) Light (b) Pressure

(c) Momentum (d) Power

77. A body is moving with a velocity *v*, breaks up into two equal parts. One of the part retraces back with velocity *v*. Then the velocity of the other part is [DCE 2004]

[KCET 2004]

(a) v in forward direction (b) 3v in forward direction

(c) v in backward direction (d) 3v in backward direction

78. If a shell fired from a cannon, explodes in mid air, then

[Pb. PET 2004]

(a) Its total kinetic energy increases

(b) Its total momentum increases

[CBSE PMT 2004]

(c) Its total momentum decreases

(d) None of these

79. A particle of mass m moving with velocity V_0 strikes a simple pendulum of mass m and sticks to it. The maximum height attained by the pendulum will be [RPET 2002]

[CBSE PMT 2004]

(a) $h=\frac{V_0^2}{8g}$ (b) $\sqrt{V_0 g}$

(c) $2\sqrt{\frac{V_0}{g}}$ (d) $\frac{V_0^2}{4g}$

80. Masses of two substances are 1 *g* and 9 *g* respectively. If their kinetic energies are same, then the ratio of their momentum will be

(a) 1 : 9 (b) 9 : 1

(c) 3 : 1 (d) 1 : 3

81. A body of mass 5 *kg* is moving with a momentum of 10 *kg-m/s*. A force of 0.2 *N* acts on it in the direction of motion of the body for 10 *seconds*. The increase in its kinetic energy is

[MP PET 1999]

(a) 2.8 *Joule* (b) 3.2 *Joule*

(c) 3.8 *Joule* (d) 4.4 *Joule*

82. If the momentum of a body increases by 0.01%, its kinetic energy will increase by **[MP PET 2001]**

(a) 0.01% (b) 0.02%

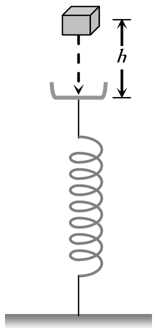
(c) 0.04% (d) 0.08%

83. 1 *a.m.u.* is equivalent to **[UPSEAT 2001]**

(a) 1.6×10^{-12} *Joule* (b) 1.6×10^{-19} *Joule*

(c) 1.5×10^{-10} *Joule* (d) 1.5×10^{-19} *Joule*

84. A block of mass *m* initially at rest is dropped from a height *h* on to a spring of force constant *k*. the maximum compression in the spring is *x* then **[BCECE 2005]**



(a) $mgh=\frac{1}{2}kx^2$

(b) $mg(h+x)=\frac{1}{2}kx^2$

(c) $mgh=\frac{1}{2}k(x+h)^2$

(d) $mg(h+x)=\frac{1}{2}k(x+h)^2$

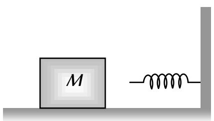
85. A spherical ball of mass 20 *kg* is stationary at the top of a hill of height 100 *m*. It slides down a smooth surface to the ground, then climbs up another hill of height 30 *m* and finally slides down to a horizontal base at a height of 20 *m* above the ground. The velocity attained by the ball is

[AIEEE 2005]

(a) 10 *m/s* (b) $10\sqrt{30}$ *m/s*

(c) 40 *m/s* (d) 20 *m/s*

86. The block of mass *M* moving on the frictionless horizontal surface collides with the spring of spring constant *K* and compresses it by length *L*. The maximum momentum of the block after collision is



(a) Zero

(b) $\frac{ML^2}{K}$

(c) $\sqrt{MK}\,L$

(d) $\frac{KL^2}{2M}$

87. A bomb of mass $30kg$ at rest explodes into two pieces of masses $18\ kg$ and $12\ kg$. The velocity of $18\ kg$ mass is $6\ ms^{-1}$. The kinetic energy of the other mass is

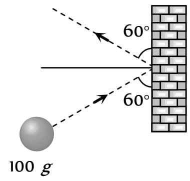
[CBSE PMT 2005]

(a) 256 *J* (b) 486 *J*

(c) 524 *J* (d) 324 *J*

[BHU 2004]

88. A mass of 100*g* strikes the wall with speed 5*m/s* at an angle as shown in figure and it rebounds with the same speed. If the contact time is 2×10^{-3} sec, what is the force applied on the mass by the wall **[Orissa JEE 2005]**



(a) $250\sqrt{3}$ *N* to right

(b) 250 *N* to right

(c) $250\sqrt{3}$ *N* to left

(d) 250 *N* to left

Power

1. If a force F is applied on a body and it moves with a velocity *v*, the power will be

[CPMT 1985, 97; DCE 1999; UPSEAT 2004]

(a) $F\times v$ (b) F/v

(c) F/v^2 (d) $F\times v^2$

2. A body of mass *m* accelerates uniformly from rest to v_1 in time t_1. As a function of time *t*, the instantaneous power delivered to the body is **[AIEEE 2004]**

(a) $\frac{mv_1t}{t_1}$ (b) $\frac{mv_1^2t}{t_1}$

(c) $\frac{mv_1t^2}{t_1}$ (d) $\frac{mv_1^2t}{t_1^2}$

3. A man is riding on a cycle with velocity 7.2 *km/hr* up a hill having a slope 1 in 20. The total mass of the man and cycle is 100 *kg*. The power of the man is

(a) 200 *W* (b) 175 *W*

(c) 125 *W* (d) 98 *W*

4. A 12 *HP* motor has to be operated 8 *hours/day*. How much will it cost at the rate of 50 *paisa/kWh* in 10 days

(a) Rs. 350/- (b) Rs. 358/-

(c) Rs. 375/- (d) Rs. 397/-

5. A motor boat is travelling with a speed of 3.0 *m/sec*. If the force on it due to water flow is 500 *N*, the power of the boat is

(a) 150 *kW* (b) 15 *kW*

(c) 1.5 *kW* (d) 150 *W*

[AIEEE 2005]

6. An electric motor exerts a force of 40 *N* on a cable and pulls it by a distance of 30 *m* in one minute. The power supplied by the motor (in *Watts*) is **[EAMCET 1984]**

(a) 20 (b) 200

(c) 2 (d) 10

7. An electric motor creates a tension of 4500 *newton* in a hoisting cable and reels it in at the rate of 2 *m/sec*. What is the power of electric motor **[MNR 1984]**

(a) 15 *kW* (b) 9 *kW*

(c) 225 *W* (d) 9000 *HP*

8. A weight lifter lifts 300 *kg* from the ground to a height of 2 *meter* in 3 *second*. The average power generated by him is

[CPMT 1989; JIPMER 2001,02]

(a) 5880 *watt* (b) 4410 *watt*

(c) 2205 *watt* (d) 1960 *watt*

9. Power of a water pump is 2 *kW*. If $g = 10\, m/\sec^2$, the amount of water it can raise in one minute to a height of 10 *m* is

(a) 2000 *litre* (b) 1000 *litre*
(c) 100 *litre* (d) 1200 *litre*

10. An engine develops 10 *kW* of power. How much time will it take to lift a mass of 200 *kg* to a height of 40 *m*. $(g = 10\, m/\sec^2)$

(a) 4 *sec* (b) 5 *sec*
(c) 8 *sec* (d) 10 *sec*

11. A car of mass '*m*' is driven with acceleration '*a*' along a straight level road against a constant external resistive force '*R*'. When the velocity of the car is '*V*', the rate at which the engine of the car is doing work will be

[MP PMT/PET 1998; JIPMER 2000]

(a) RV (b) maV
(c) $(R + ma)V$ (d) $(ma - R)V$

12. The average power required to lift a 100 *kg* mass through a height of 50 *metres* in approximately 50 *seconds* would be

[SCRA 1994; MH CET 2000]

(a) 50 *J/s* (b) 5000 *J/s*
(c) 100 *J/s* (d) 980 *J/s*

13. From a waterfall, water is falling down at the rate of 100 *kg/s* on the blades of turbine. If the height of the fall is 100 *m*, then the power delivered to the turbine is approximately equal to [KCET 1994; BHU 1997; MP PET 2000]

(a) 100 *kW* (b) 10 *kW*
(c) 1 *kW* (d) 1000 *kW*

14. The power of a pump, which can pump 200*kg* of water to a height of 200*m* in 10*sec* is $(g = 10\, m/s^2)$ [CBSE PMT 2000]

(a) 40 *kW* (b) 80 *kW*
(c) 400 *kW* (d) 960 *kW*

15. A 10 *H.P.* motor pumps out water from a well of depth 20*m* and fills a water tank of volume 22380 *litres* at a height of 10*m* from the ground. the running time of the motor to fill the empty water tank is $(g = 10 ms^{-2})$

[EAMCET (Engg.) 2000]

(a) 5 *minutes* (b) 10 *minutes*
(c) 15 *minutes* (d) 20 *minutes*

16. A car of mass 1250 *kg* is moving at 30*m/s*. Its engine delivers 30 *kW* while resistive force due to surface is 750*N*. What max acceleration can be given in the car

[RPET 2000]

(a) $\frac{1}{3} m/s^2$ (b) $\frac{1}{4} m/s^2$
(c) $\frac{1}{5} m/s^2$ (d) $\frac{1}{6} m/s^2$

17. A force applied by an engine of a train of mass $2.05 \times 10^6 kg$ changes its velocity from $5 m/s$ to 25 m/s in 5 minutes. The power of the engine is [EAMCET 2001]

(a) $1.025 MW$ (b) $2.05 MW$
(c) $5 MW$ (d) $6 MW$

18. A truck of mass 30,000*kg* moves up an inclined plane of slope 1 in 100 at a speed of 30 *kmph*. The power of the truck is (given $g = 10 ms^{-1}$) [Kerala (Engg.) 2001]

(a) 25 *kW* (b) 10 *kW*
(c) 5 *kW* (d) 2.5 *kW*

19. A 60 *kg* man runs up a staircase in 12 seconds while a 50 *kg* man runs up the same staircase in 11, seconds, the ratio of the rate of doing their work is [CBSE PMT 1990; Kerala PMT 2004] [AMU (Engg.) 2001]

(a) 6 : 5 (b) 12 : 11
(c) 11 : 10 (d) 10 : 11

20. A pump motor is used to deliver water at a certain rate from a given pipe. To obtain twice as much water from the same pipe in the same time, power of the motor has to be increased to [CPMT 1992]

(a) 16 times (b) 4 times
(c) 8 times (d) 2 times

21. What average horsepower is developed by an 80 *kg* man while climbing in 10 *s* a flight of stairs that rises 6 *m* vertically

(a) 0.63 *HP* (b) 1.26 *HP*
(c) 1.8 *HP* (d) 2.1 *HP*

22. A car of mass 1000 *kg* accelerates uniformly from rest to a velocity of 54 km/hour in 5*s*. The average power of the engine during this period in watts is (neglect friction)

[Kerala PET 2002]

(a) 2000 *W* (b) 22500 *W*
(c) 5000 *W* (d) 2250 *W*

23. A quarter horse power motor runs at a speed of 600 *r.p.m.* Assuming 40% efficiency the work done by the motor in one rotation will be [Kerala PET 2002]

(a) 7.46 *J* (b) 7400 *J*
(c) 7.46 *ergs* (d) 74.6 *J*

24. An engine pumps up 100 *kg* of water through a height of 10 *m* in 5 *s*. Given that the efficiency of the engine is 60% . If $g = 10 ms^{-2}$, the power of the engine is [DPMT 2004]

(a) $3.3 kW$ (b) $0.33 kW$
(c) $0.033 kW$ (d) $33 kW$

25. A force of $2\hat{i} + 3\hat{j} + 4\hat{k}\ N$ acts on a body for 4 second, produces a displacement of $(3\hat{i} + 4\hat{j} + 5\hat{k}) m$. The power used is [Pb. PET 2001; CBSE PMT 2

(a) 9.5 *W* (b) 7.5 *W*
(c) 6.5 *W* (d) 4.5 *W*

26. The power of pump, which can pump 200 *kg* of water to a height of 50 *m* in 10 *sec*, will be [DPMT 2003]

(a) 10×10^3 *watt* (b) 20×10^3 *watt*
(c) 4×10^3 *watt* (d) 60×10^3 *watt*

27. From an automatic gun a man fires 360 bullet per minute with a speed of 360 *km/hour*. If each weighs 20 *g*, the power of the gun is

(a) $600 W$ (b) $300 W$
(c) $150 W$ (d) $75 W$

28. An engine pump is used to pump a liquid of density ρ continuously through a pipe of cross-sectional area *A*. If the speed of flow of the liquid in the pipe is *v*, then the rate at which kinetic energy is being imparted to the liquid is

(a) $\frac{1}{2} A\rho v^3$ (b) $\frac{1}{2} A\rho v^2$
(c) $\frac{1}{2} A\rho v$ (d) $A\rho v$

29. If the heart pushes 1 cc of blood in one second under pressure 20000 N/m^2 the power of heart is [J&K CET 2005]

(a) 0.02 W (b) 400 W

(c) 5×10^{-10} W (d) 0.2 W

30. A man does a given amount of work in 10 sec. Another man does the same amount of work in 20 sec. The ratio of the output power of first man to the second man is

[J&K CET 2005]

(a) 1 (b) 1/2

(c) 2/1 (d) None of these

Elastic and Inelastic Collision

1. The coefficient of restitution e for a perfectly elastic collision is

(a) 1 (b) 0

(c) ∞ (d) – 1

2. The principle of conservation of linear momentum can be strictly applied during a collision between two particles provided the time of impact is

(a) Extremely small

(b) Moderately small

(c) Extremely large

(d) Depends on a particular case

3. A shell initially at rest explodes into two pieces of equal mass, then the two pieces will

[CPMT 1982; EAMCET 1988; Orissa PMT 2004]

(a) Be at rest

(b) Move with different velocities in different directions

(c) Move with the same velocity in opposite directions

(d) Move with the same velocity in same direction

4. A sphere of mass m moving with a constant velocity u hits another stationary sphere of the same mass. If e is the coefficient of restitution, then the ratio of the velocity of two spheres after collision will be [RPMT 1996; BHU 1997]

(a) $\frac{1-e}{1+e}$ (b) $\frac{1+e}{1-e}$

(c) $\frac{e+1}{e-1}$ (d) $\frac{e-1}{e+1}t^2$

5. Two solid rubber balls A and B having masses 200 and 400 gm respectively are moving in opposite directions with velocity of A equal to 0.3 m/s. After collision the two balls come to rest, then the velocity of B is [CPMT 1978, 86, 88]

(a) 0.15 m/sec (b) 1.5 m/sec

(c) – 0.15 m/sec (d) None of the above

6. Two perfectly elastic particles P and Q of equal mass travelling along the line joining them with velocities 15 m/sec and 10 m/sec. After collision, their velocities respectively (in m/sec) will be

(a) 0, 25 (b) 5, 20

(c) 10, 15 (d) 20, 5

7. A cannon ball is fired with a velocity 200 m/sec at an angle of 60° with the horizontal. At the highest point of its flight it explodes into 3 equal fragments, one going vertically upwards with a velocity 100 m/sec, the second one falling vertically downwards with a velocity 100 m/sec. The third fragment will be moving with a velocity

[NCERT 1983; AFMC 1997]

(a) 100 m/s in the horizontal direction

(b) 300 m/s in the horizontal direction

(c) 300 m/s in a direction making an angle of 60° with the horizontal

(d) 200 m/s in a direction making an angle of 60° with the horizontal

8. A lead ball strikes a wall and falls down, a tennis ball having the same mass and velocity strikes the wall and bounces back. Check the correct statement

(a) The momentum of the lead ball is greater than that of the tennis ball

(b) The lead ball suffers a greater change in momentum compared with the tennis ball [CBSE PMT 1988]

(c) The tennis ball suffers a greater change in momentum as compared with the lead ball

(d) Both suffer an equal change in momentum

9. When two bodies collide elastically, then

[CPMT 1974; MP PMT 2001; RPET 2000; Kerala PET 2005]

(a) Kinetic energy of the system alone is conserved

(b) Only momentum is conserved

(c) Both energy and momentum are conserved

(d) Neither energy nor momentum is conserved

10. Two balls at same temperature collide. What is conserved

[NCERT 1974; CPMT 1983; DCE 2004]

(a) Temperature (b) Velocity

(c) Kinetic energy (d) Momentum

11. A body of mass 5 kg explodes at rest into three fragments with masses in the ratio 1 : 1 : 3. The fragments with equal masses fly in mutually perpendicular directions with speeds of 21 m/s. The velocity of the heaviest fragment will be

[CBSE PMT 1991]

(a) 11.5 m/s (b) 14.0 m/s

(c) 7.0 m/s (d) 9.89 m/s

12. A heavy steel ball of mass greater than 1 kg moving with a speed of $2\,m\,\sec^{-1}$ collides head on with a stationary ping-pong ball of mass less than 0.1 gm. The collision is elastic. After the collision the ping-pong ball moves approximately with speed

(a) $2\,m\,\sec^{-1}$ (b) $4\,m\,\sec^{-1}$

(c) $2\times10^4\,m\,\sec^{-1}$ (d) $2\times10^3\,m\,\sec^{-1}$

13. A body of mass 'M' collides against a wall with a velocity v and retraces its path with the same speed. The change in momentum is (take initial direction of velocity as positive)

[EAMCET 1982]

(a) Zero (b) 2Mv

(c) Mv (d) – 2 Mv

14. A gun fires a bullet of mass 50 gm with a velocity of $30\,m\,\sec^{-1}$. [CPMT 1988; MP PMT 1994] Because of this the gun is pushed back with a velocity of $1\,m\,\sec^{-1}$. The mass of the gun is

[EAMCET 1989; AIIMS 2001]

(a) 15 kg (b) 30 kg

(c) 1.5 kg (d) 20 kg

15. In an elastic collision of two particles the following is conserved[MP PET 1994; D

(a) Momentum of each particle

(b) Speed of each particle

(c) Kinetic energy of each particle

(d) Total kinetic energy of both the particles

16. A ^{238}U nucleus decays by emitting an alpha particle of speed $v\,ms^{-1}$. The recoil speed of the residual nucleus is (in ms^{-1})[CBSE PMT 1995; AIEEE 2003]

(a) $-4v/234$ (b) $v/4$

(c) $-4v/238$ (d) $4v/238$

17. A smooth sphere of mass M moving with velocity u directly collides elastically with another sphere of mass m at rest. After collision their final velocities are V and v respectively. The value of v is

(a) $\frac{2uM}{m}$ (b) $\frac{2um}{M}$

(c) $\frac{2u}{1+\frac{m}{M}}$ (d) $\frac{2u}{1+\frac{M}{m}}$

18. A body of mass m having an initial velocity v, makes head on collision with a stationary body of mass M. After the collision, the body of mass m comes to rest and only the body having mass M moves. This will happen only when

[MP PMT 1995]

(a) $m >> M$ (b) $m << M$

(c) $m = M$ (d) $m = \frac{1}{2}M$

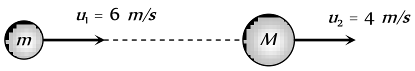
19. A particle of mass m moving with a velocity $\vec{V}$ makes a head on elastic collision with another particle of same mass initially at rest. The velocity of the first particle after the collision will be

[MP PMT 1997; MP PET 2001; UPSEAT 2001]

(a) $\vec{V}$ (b) $-\vec{V}$

(c) $-2\vec{V}$ (d) Zero

20. A particle of mass m moving with horizontal speed 6 m/sec as shown in figure. If $m << M$ then for one dimensional elastic collision, the speed of lighter particle after collision will be



(a) 2m/sec in original direction

(b) 2 m/sec opposite to the original direction

(c) 4 m/sec opposite to the original direction

(d) 4 m/sec in original direction

21. A shell of mass m moving with velocity v suddenly breaks into 2 pieces. The part having mass $m/4$ remains stationary. The velocity of the other shell will be [CPMT 1999]

(a) v (b) $2v$

(c) $\frac{3}{4}v$ (d) $\frac{4}{3}v$

22. Two equal masses m_1 and m_2 moving along the same straight line with velocities + 3 m/s and − 5 m/s respectively collide elastically. Their velocities after the collision will be respectively[CBSE PMT 1994, 98; AIIMS 2000]

(a) + 4 m/s for both (b) − 3 m/s and +5 m/s

(c) − 4 m/s and + 4 m/s (d) − 5 m/s and + 3 m/s

23. A rubber ball is dropped from a height of 5 m on a planet where the acceleration due to gravity is not known. On bouncing, it rises to 1.8 m. The ball loses its velocity on bouncing by a factor of

(a) 16/25 (b) 2/5

(c) 3/5 (d) 9/25

24. A metal ball falls from a height of 32 *metre* on a steel plate. If the coefficient of restitution is 0.5, to what height will the ball rise after second bounce [EAMCET 1994]

(a) 2 m (b) 4 m

(c) 8 m (d) 16 m

25. At high altitude, a body explodes at rest into two equal fragments with one fragment receiving horizontal velocity of 10 m/s. Time taken by the two radius vectors connecting point of explosion to fragments to make 90° is

[EAMCET (Engg.) 1995; DPMT 2000]

(a) 10[MP PET 1995] (b) 4 s

(c) 2 s (d) 1 s

26. A ball of mass 10 kg is moving with a velocity of 10 m/s. It strikes another ball of mass 5 kg which is moving in the same direction with a velocity of 4 m/s. If the collision is elastic, their velocities after the collision will be, respectively

[CMEET Bihar 1995]

(a) 6 m/s, 12 m/s (b) 12 m/s, 6 m/s

(c) 12 m/s, 10 m/s (d) 12 m/s, 25 m/s

27. A body of mass 2 kg collides with a wall with speed 100 m/s and rebounds with same speed. If the time of contact was 1/50 second, the force exerted on the wall is [CPMT 1993]

(a) 8 N (b) 2×10^4 N

(c) 4 N (d) 10^4 N

28. A body falls on a surface of coefficient of restitution 0.6 from a height of 1 m. Then the body rebounds to a height of

[CPMT 1993; Pb. PET 2001]

(a) 0.6 m (b) 0.4 m

(c) 1 m (d) 0.36 m

29. A ball is dropped from a height h. If the coefficient of restitution be e, then to what height will it rise after jumping twice from the ground[MP PMT 2003] [RPMT 1996; Pb. PET 2001]

(a) $eh/2$ (b) $2eh$

(c) eh (d) e^4h

30. A ball of weight 0.1 kg coming with speed 30 m/s strikes with a bat and returns in opposite direction with speed 40 m/s, then the impulse is (Taking final velocity as positive)

[AFMC 1997]

(a) $-0.1\times(40)-0.1\times(30)$ (b) $0.1\times(40)-0.1\times(-30)$

(c) $0.1\times(40)+0.1\times(-30)$ (d) $0.1\times(40)-0.1\times(20)$

31. A billiard ball moving with a speed of 5 m/s collides with an identical ball originally at rest. If the first ball stops after collision, then the second ball will move forward with a speed of

(a) $10\,ms^{-1}$ (b) $5\,ms^{-1}$

(c) $2.5\,ms^{-1}$ (d) $1.0\,ms^{-1}$

32. If two balls each of mass 0.06 kg moving in opposite directions with speed 4 m/s collide and rebound with the same speed, then the impulse imparted to each ball due to other is
[CBSE PMT 1998]

(a) 0.48 kg-m/s (b) 0.24 kg-m/s

(c) 0.81 *kg-m/s* (d) Zero

33. A ball of mass *m* falls vertically to the ground from a height h_1 and rebound to a height h_2. The change in momentum of the ball on striking the ground is [AMU (Engg.) 1999]

(a) $mg(h_1 - h_2)$ (b) $m(\sqrt{2gh_1} + \sqrt{2gh_2})$

(c) $m\sqrt{2g(h_1 + h_2)}$ (d) $m\sqrt{2g}(h_1 + h_2)$

34. A body of mass 50 *kg* is projected vertically upwards with velocity of 100 *m/sec*. 5 seconds after this body breaks into 20 *kg* and 30 *kg*. If 20 *kg* piece travels upwards with 150 *m/sec*, then the velocity of other block will be [RPMT 1999]

(a) 15 *m/sec* downwards (b) 15 *m/sec* upwards

(c) 51 *m/sec* downwards (d) 51 *m/sec* upwards

35. A steel ball of radius 2 *cm* is at rest on a frictionless surface. Another ball of radius 4*cm* moving at a velocity of 81 *cm/sec* collides elastically with first ball. After collision the smaller ball moves with speed of [RPMT 1999]

(a) 81 *cm/sec* (b) 63 *cm/sec*

(c) 144 *cm/sec* (d) None of these

36. A space craft of mass *M* is moving with velocity *V* and suddenly explodes into two pieces. A part of it of mass *m* becomes at rest, then the velocity of other part will be

[RPMT 1999]

(a) $\frac{MV}{M-m}$ (b) $\frac{MV}{M+m}$

(c) $\frac{mV}{M-m}$ (d) $\frac{(M+m)V}{m}$

37. A ball hits a vertical wall horizontally at 10*m/s* bounces back at 10 *m/s* [JIPMER 1999]

(a) There is no acceleration because $10\frac{m}{s} - 10\frac{m}{s} = 0$

(b) There may be an acceleration because its initial direction is horizontal

(c) There is an acceleration because there is a momentum change

(d) Even though there is no change in momentum there is a change in direction. Hence it has an acceleration

38. A bullet of mass 50 *gram* is fired from a 5 *kg* gun with a velocity of 1*km/s*. the speed of recoil of the gun is

[JIPMER 1999]

(a) $5 m/s$ (b) $1 m/s$

(c) $0.5 m/s$ (d) $10 m/s$

39. A body falling from a height of 10m rebounds from hard floor. If it loses 20% energy in the impact, then coefficient of restitution is

(a) 0.89 (b) 0.56

(c) 0.23 (d) 0.18

40. A body of mass m_1 moving with a velocity 3 *ms* collides with another body at rest of mass m_2. After collision the velocities of the two bodies are 2 *ms* and 5*ms* respectively along the direction of motion of m_1 The ratio m_1/m_2 is

[EAMCET (Engg.) 2000]

(a) $\frac{5}{12}$ (b) 5

(c) $\frac{1}{5}$ (d) $\frac{12}{5}$

41. 100 *g* of a iron ball having velocity 10 *m/s* collides with a wall at an angle 30^o and rebounds with the same angle. If the period of contact between the ball and wall is 0.1 second, then the force experienced by the ball is

[DPMT 2000]

(a) 100*N* (b) 10 *N*

(c) 0.1 *N* (d) 1.0 *N*

42. Two bodies having same mass 40 *kg* are moving in opposite directions, one with a velocity of $10 m/s$ and the other with $7m/s$. If they collide and move as one body, the velocity of the combination is [Pb. PMT 2000]

(a) $10 m/s$ (b) $7 m/s$

(c) $3 m/s$ (d) $1.5 m/s$

43. A body at rest breaks up into 3 parts. If 2 parts having equal masses fly off perpendicularly each after with a velocity of 12*m/s*, then the velocity of the third part which has 3 times mass of each part is

(a) $4\sqrt{2}\, m/s$ at an angle of 45^o from each body

(b) $24\sqrt{2}\, m/s$ at an angle of 135^o from each body

(c) $6\sqrt{2}\, m/s$ at 135^o from each body

(d) $4\sqrt{2}\, m/s$ at 135^o from each body

44. A particle falls from a height *h* upon a fixed horizontal plane and rebounds. If *e* is the coefficient of restitution, the total distance travelled before rebounding has stopped is

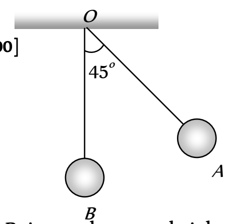
[EAMCET 2001]

(a) $h\left(\frac{1+e^2}{1-e^2}\right)$ (b) $h\left(\frac{1-e^2}{1+e^2}\right)$

(c) $\frac{h}{2}\left(\frac{1-e^2}{1+e^2}\right)$ (d) $\frac{h}{2}\left(\frac{1+e^2}{1-e^2}\right)$

45. The bob A of a simple pendulum is released when the string makes an angle of 45^o with the vertical. It hits another bob *B* of the same material and same mass kept at rest on the table. If the collision is elastic [Kerala (Engg.) 2001]

[AIIMS 2000]



(a) Both *A* and *B* rise to the same height

(b) Both *A* and *B* come to rest at *B*

(c) Both *A* and *B* move with the same velocity of *A*

(d) A comes to rest and B moves with the velocity of A

46. A big ball of mass M, moving with velocity u strikes a small ball of mass m, which is at rest. Finally small ball obtains velocity u and big ball v. Then what is the value of v [RPET 2001]

(a) $\frac{M-m}{M+m}u$ (b) $\frac{m}{M+m}u$

(c) $\frac{2m}{M+m}u$ (d) $\frac{M}{M+m}u$

47. A body of mass 5 kg moving with a velocity 10m/s collides with another body of the mass 20 kg at, rest and comes to rest. The velocity of the second body due to collision is

[Pb. PMT 1999; KCET 2001]

(a) 2.5 m/s (b) 5 m/s

(c) 7.5 m/s (d) 10 m/s

48. A ball of mass m moving with velocity V, makes a head on elastic collision with a ball of the same mass moving with velocity $2V$ towards it. Taking direction of V as positive velocities of the two balls after collision are [MP PMT 2002]

(a) $-V$ and $2V$ (b) $2V$ and $-V$

(c) V and $-2V$ (d) $-2V$ and V

49. A body of mass M_1 collides elastically with another mass M_2 at rest. There is maximum transfer of energy when

[Orissa JEE 2002; DCE 2001, 02]

(a) $M_1 > M_2$

(b) $M_1 < M_2$

(c) $M_1 = M_2$

(d) Same for all values of M_1 and M_2

50. A body of mass $2kg$ makes an elastic collision with another body at rest and continues to move in the original direction with one fourth of its original speed. The mass of the second body which collides with the first body is [Kerala PET 2002]

(a) 2 kg (b) 1.2 kg

(c) 3 kg (d) 1.5 kg

51. In the elastic collision of objects [RPET 2003]

(a) Only momentum remains constant

(b) Only K.E. remains constant

(c) Both remains constant

(d) None of these

52. Two particles having position vectors $\vec{r}_1 = (3\hat{i} + 5\hat{j})$ *metres* and $\vec{r}_2 = (-5\hat{i} - 3\hat{j})$ *metres* are moving with velocities $\vec{v}_1 = (4\hat{i} + 3\hat{j})m/s$ and $\vec{v}_2 = (\alpha\hat{i} + 7\hat{j})$ m/s. If they collide after 2 seconds, the value of 'α' is [EAMCET 2003]

(a) 2 (b) 4

(c) 6 (d) 8

53. A neutron makes a head-on elastic collision with a stationary deuteron. The fractional energy loss of the neutron in the collision is

(a) 16/81 (b) 8/9

(c) 8/27 (d) 2/3

54. A body of mass m is at rest. Another body of same mass moving with velocity V makes head on elastic collision with the first body. After collision the first body starts to move with velocity

(a) V (b) $2V$

(c) Remain at rest (d) No predictable

55. A body of mass M moves with velocity v and collides elastically with a another body of mass m (M>>m) at rest then the velocity of body of mass m is [BCECE 2004]

(a) v (b) $2v$

(c) $v/2$ (d) Zero

56. Four smooth steel balls of equal mass at rest are free to move along a straight line without friction. The first ball is given a velocity of 0.4 m/s. It collides head on with the second elastically, the second one similarly with the third and so on. The velocity of the last ball is[UPSEAT 2

(a) $0.4m/s$ (b) $0.2m/s$

(c) $0.1m/s$ (d) $0.05m/s$

57. A space craft of mass 'M' and moving with velocity 'v' suddenly breaks in two pieces of same mass m. After the explosion one of the mass 'm' becomes stationary. What is the velocity of the other part of craft [DCE 2003]

(a) $\frac{Mv}{M-m}$ (b) v

(c) $\frac{Mv}{m}$ (d) $\frac{M-m}{m}v$

58. Two masses m_A and m_B moving with velocities v_A and v_B in opposite directions collide elastically. After that the masses m_A and m_B move with velocity v_B and v_A respectively. The ratio (m_A/m_B) is

[RPMT 2003, AFMC 2002]

(a) 1 (b) $\frac{v_A - v_B}{v_A + v_B}$

(c) $(m_A + m_B)/m_A$ (d) v_A/v_B

59. A ball is allowed to fall from a height of 10 m. If there is 40% loss of energy due to impact, then after one impact ball will go up to

(a) 10 m (b) 8 m

(c) 4 m (d) 6 m

60. Which of the following statements is true [NCERT 1984]

(a) In elastic collisions, the momentum is conserved but not in inelastic collisions

(b) Both kinetic energy and momentum are conserved in elastic as well as inelastic collisions

(c) Total kinetic energy is not conserved but momentum is conserved in inelastic collisions

(d) Total kinetic energy is conserved in elastic collisions but momentum is not conserved in elastic collisions

61. A tennis ball dropped from a height of 2 m rebounds only 1.5 m after hitting the ground. What fraction of its energy is lost in the impact

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$

(c) $\frac{1}{3}$ (d) $\frac{1}{8}$

62. A body of mass m moving with velocity v makes a head-on collision with another body of mass 2 m which is initially at rest. The loss of kinetic energy of the colliding body (mass m) is [AIIMS 2003]

(a) $\frac{1}{2}$ of its initial kinetic energy

[Orissa PMT 2004]

(b) $\frac{1}{9}$ of its initial kinetic energy

(c) $\frac{8}{9}$ of its initial kinetic energy

(d) $\frac{1}{4}$ of its initial kinetic energy

63. The quantities remaining constant in a collision are

(a) Momentum, kinetic energy and temperature

(b) Momentum and kinetic energy but not temperature

(c) Momentum and temperature but not kinetic energy

(d) Momentum but neither kinetic energy nor temperature

64. An inelastic ball is dropped from a height of 100 *m*. Due to earth, 20% of its energy is lost. To what height the ball will rise

(a) 80 *m* (b) 40 *m*

(c) 60 *m* (d) 20 *m*

65. A ball is projected vertically down with an initial velocity from a height of 20 *m* onto a horizontal floor. During the impact it loses 50% of its energy and rebounds to the same height. The initial velocity of its projection is

[EAMCET (Engg.) 2000]

(a) $20\,ms^{-1}$ (b) $15\,ms^{-1}$

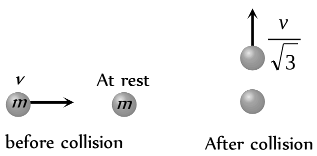
(c) $10\,ms^{-1}$ (d) $5\,ms^{-1}$

66. A tennis ball is released from height *h* above ground level. If the ball makes inelastic collision with the ground, to what height will it rise after third collision [RPET 2002]

(a) he^6 (b) e^2h

(c) e^3h (d) None of these

67. A mass '*m*' moves with a velocity '*v*' and collides inelastically with another identical mass. After collision the 1st mass moves with velocity $\frac{v}{\sqrt{3}}$ in a direction perpendicular to the initial direction of motion. Find the speed of the 2nd mass after collision



(a) $\frac{2}{\sqrt{3}}v$

(b) $\frac{v}{\sqrt{3}}$

(c) v

(d) $\sqrt{3}\,v$

68. A sphere collides with another sphere of identical mass. After collision, the two spheres move. The collision is inelastic. Then the angle between the directions of the two spheres is

(a) 90° (b) 0°

(c) 45° (d) Different from 90°

Perfectly Inelastic Collision

1. A particle of mass *m* moving eastward with a speed *v* collides with another particle of the same mass moving northward with the same speed *v*. The two particles coalesce on collision. The new particle of mass 2*m* will move in the north-easterly direction with a velocity [NCERT 1980; CPMT 1991; MP PET 1999; DPMT 1999, 2005]

(a) $v/2$ (b) $2v$

(c) $v/\sqrt{2}$ (d) v

2. The coefficient of restitution *e* for a perfectly inelastic collision is

(a) 1 (b) 0

(c) ∞ (d) – 1

3. When two bodies stick together after collision, the collision is said to be [MP PET 1997]

(a) Partially elastic (b) Total elastic

(c) Total inelastic (d) None of the above

4. A bullet of mass *a* and velocity *b* is fired into a large block of mass *c*. The final velocity of the system is

[AFMC 1981, 94, 2000; NCERT 1971; MNR 1998]

(a) $\frac{c}{a+b}\cdot b$ [RPMT 1996] (b) $\frac{a}{a+c}\cdot b$

(c) $\frac{a+b}{c}.a$ (d) $\frac{a+c}{a}\cdot b$

5. A mass of 10 *gm* moving with a velocity of 100 *cm/s* strikes a pendulum bob of mass 10 *gm*. The two masses stick together. The maximum height reached by the system now is $(g = 10\,m/s^2)$

(a) Zero (b) 5 *cm*

(c) 2.5 *cm* (d) 1.25 *cm*

6. A completely inelastic collision is one in which the two colliding particles

(a) Are separated after collision

(b) Remain together after collision

(c) Split into small fragments flying in all directions

(d) None of the above

7. A bullet hits and gets embedded in a solid block resting on a horizontal frictionless table. What is conserved ?

[NCERT 1973; CPMT 1970; AFMC 1996; BHU 2001]

(a) Momentum and kinetic energy

(b) Kinetic energy alone

(c) Momentum alone [AIEEE 2005]

(d) Neither momentum nor kinetic energy

8. A body of mass 2 *kg* moving with a velocity of 3 *m/sec* collides head on with a body of mass 1 *kg* moving in opposite direction with a velocity of 4 *m/sec*. After collision, two bodies stick together and move with a common velocity which in *m/sec* is equal to

[NCERT 1984; MNR 1995, 98; UPSEAT 2000]

(a) 1/4 (b) 1/3

(c) 2/3 (d) 3/4

9. A body of mass *m* moving with a constant velocity *v* hits another body of the same mass moving with the same velocity *v* but in the opposite direction and sticks to it. The velocity of the compound body after collision is [KCET 1994]

[NCERT 1977; RPMT 1999]

(a) *v* (b) 2*v*

(c) Zero (d) *v*/2

10. In the above question, if another body is at rest, then velocity of the compound body after collision is

(a) *v*/2 (b) 2*v*

(c) *v* (d) Zero

11. A bag (mass *M*) hangs by a long thread and a bullet (mass *m*) comes horizontally with velocity *v* and gets caught in the bag. Then for the combined (bag + bullet) system

[CPMT 1989; Kerala PMT 2002]

(a) Momentum is $\frac{mvM}{M+m}$

(b) Kinetic energy is $\frac{mv^2}{2}$

(c) Momentum is $\frac{mv(M+m)}{M}$

(d) Kinetic energy is $\frac{m^2v^2}{2(M+m)}$

12. A 50 *g* bullet moving with velocity 10 *m/s* strikes a block of mass 950 *g* at rest and gets embedded in it. The loss in kinetic energy will be [MP PET 1994]

(a) 100% (b) 95%
(c) 5% (d) 50%

13. Two putty balls of equal mass moving with equal velocity in mutually perpendicular directions, stick together after collision. If the balls were initially moving with a velocity of $45\sqrt{2}\,ms^{-1}$ each, the velocity of their combined mass after collision is [Haryana CEE 1996; BVP 2003]

(a) $45\sqrt{2}\,ms^{-1}$ (b) $45\,ms^{-1}$
(c) $90\,ms^{-1}$ (d) $22.5\sqrt{2}\,ms^{-1}$

14. A particle of mass *m* moving with velocity *v* strikes a stationary particle of mass 2*m* and sticks to it. The speed of the system will be [MP PMT/PET 1998; AIIMS 1999; JIPMER 2001, 02]

(a) $v/2$ (b) $2v$
(c) $v/3$ (d) $3v$

15. A moving body of mass *m* and velocity 3 *km/h* collides with a rest body of mass 2*m* and sticks to it. Now the combined mass starts to move. What will be the combined velocity [CBSE PMT 1996; JIPMER 2001, 02]

(a) 3 *km/h* (b) 2 *km/h*
(c) 1 *km/h* (d) 4 *km/h*

16. If a skater of weight 3 *kg* has initial speed 32 *m/s* and second one of weight 4 *kg* has 5 *m/s*. After collision, they have speed (couple) 5 *m/s*. Then the loss in K.E. is [CPMT 1996]

(a) 48 *J* (b) 96 *J*
(c) Zero (d) None of these

17. A ball is dropped from height 10 *m*. Ball is embedded in sand 1 *m* and stops, then [AFMC 1996]

(a) Only momentum remains conserved
(b) Only kinetic energy remains conserved
(c) Both momentum and K.E. are conserved
(d) Neither K.E. nor momentum is conserved

18. A metal ball of mass 2 *kg* moving with a velocity of 36 *km/h* has an head on collision with a stationary ball of mass 3 *kg*. If after the collision, the two balls move together, the loss in kinetic energy due to collision is [CBSE PMT 1997; AIIMS 2001]

(a) 40 *J* (b) 60 *J*
(c) 100 *J* (d) 140 *J*

19. A body of mass 2*kg* is moving with velocity 10 *m/s* towards east. Another body of same mass and same velocity moving towards north collides with former and coalsces and moves towards north-east. Its velocity is [CPMT 1997; JIPMER 2000]

(a) 10 *m/s* (b) 5 *m/s*
(c) 2.5 *m/s* (d) $5\sqrt{2}\,m/s$

20. Which of the following is not a perfectly inelastic collision [BHU 1998; JIPMER 2001, 02; BHU 2005]

(a) Striking of two glass balls
(b) A bullet striking a bag of sand
(c) An electron captured by a proton
(d) A man jumping onto a moving cart

21. A mass of 20 *kg* moving with a speed of 10*m/s* collides with another stationary mass of 5*kg*. As a result of the collision, the two masses stick together. The kinetic energy of the composite mass will be

(a) 600 *Joule* (b) 800 *Joule*
(c) 1000 *Joule* (d) 1200 *Joule*

22. A neutron having mass of $1.67\times10^{-27}\,kg$ and moving at $10^8\,m/s$ collides with a deutron at rest and sticks to it. If the mass of the deutron is $3.34\times10^{-27}\,kg$ then the speed of the combination is [CBSE PMT 2000]

(a) $2.56\times10^3\,m/s$ (b) $2.98\times10^5\,m/s$
(c) $3.33\times10^7\,m/s$ (d) $5.01\times10^9\,m/s$

23. The quantity that is not conserved in an inelastic collision is [Pb. PMT 2000]

(a) Momentum (b) Kinetic energy
(c) Total energy (d) All of these

24. A body of mass 40*kg* having velocity 4 *m/s* collides with another body of mass 60*kg* having velocity 2 *m/s*. If the collision is inelastic, then loss in kinetic energy will be [Pb. PMT 2001]

(a) 440 *J* (b) 392 *J*
(c) 48 *J* (d) 144 *J*

25. A body of mass m_1 is moving with a velocity V. It collides with another stationary body of mass m_2. They get embedded. At the point of collision, the velocity of the system

(a) Increases
(b) Decreases but does not become zero
(c) Remains same
(d) Become zero

26. A bullet of mass *m* moving with velocity *v* strikes a block of mass *M* at rest and gets embedded into it. The kinetic energy of the composite block will be [MP PET 2002]

(a) $\frac{1}{2}mv^2\times\frac{m}{(m+M)}$ (b) $\frac{1}{2}mv^2\times\frac{M}{(m+M)}$
(c) $\frac{1}{2}mv^2\times\frac{(M+m)}{M}$ (d) $\frac{1}{2}Mv^2\times\frac{m}{(m+M)}$

27. In an inelastic collision, what is conserved [DCE 2004]

(a) Kinetic energy (b) Momentum
(c) Both (a) and (b) (d) Neither (a) nor (b)

28. Two bodies of masses 0.1 *kg* and 0.4 *kg* move towards each other with the velocities 1 *m/s* and 0.1 *m/s* respectively, After collision they stick together. In 10 *sec* the combined mass travels

(a) 120 *m* (b) 0.12 *m*
(c) 12 *m* (d) 1.2 *m*

29. A body of mass 4 *kg* moving with velocity 12 *m/s* collides with another body of mass 6 *kg* at rest. If two bodies stick together after collision, then the loss of kinetic energy of system is

(a) Zero (b) 288 J

(c) 172.8 J (d) 144 J

30. Which of the following is not an example of perfectly inelastic collision [AFMC 2005]

(a) A bullet fired into a block if bullet gets embedded into block

(b) Capture of electrons by an atom

(c) A man jumping on to a moving boat

(d) A ball bearing striking another ball bearing

Critical Thinking

Objective Questions

1. A ball hits the floor and rebounds after inelastic collision. In this case [IIT 1986]

(a) The momentum of the ball just after the collision is the same as that just before the collision

(b) The mechanical energy of the ball remains the same in the collision

(c) The total momentum of the ball and the earth is conserved

(d) The total energy of the ball and the earth is conserved

2. A uniform chain of length L and mass M is lying on a smooth table and one third of its length is hanging vertically down over the edge of the table. If g is acceleration due to gravity, the work required to pull the hanging part on to the table is [IIT 1985; MNR 1990; AIEEE 2002; MP PMT 1994, 97, 2000; JIPMER 2000]

(a) MgL (b) $MgL/3$

(c) $MgL/9$ (d) $MgL/18$

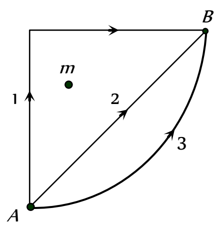
3. If W_1, W_2 and W_3 represent the work done in moving a particle from A to B along three different paths 1, 2 and 3 respectively (as shown) in the gravitational field of a point mass m, find the correct relation between W_1, W_2 and W_3

(a) $W_1 > W_2 > W_3$

(b) $W_1 = W_2 = W_3$

(c) $W_1 < W_2 < W_3$

(d) $W_2 > W_1 > W_3$



4. A particle of mass m is moving in a horizontal circle of radius r under a centripetal force equal to $-K/r^2$, where K is a constant. The total energy of the particle is [IIT 1977]

(a) $\frac{K}{2r}$ (b) $-\frac{K}{2r}$

(c) $-\frac{K}{r}$ (d) $\frac{K}{r}$

5. The displacement x of a particle moving in one dimension under the action of a constant force is related to the time t by the equation $t = \sqrt{x} + 3$, where x is in meters and t is in seconds. The work done by the force in the first 6 seconds is [IIT 1979]

(a) 9 *J* (b) 6 *J*

(c) 0 *J* (d) 3 *J*

6. A force $\mathbf{F} = -K(y\mathbf{i} + x\mathbf{j})$ (where K is a positive constant) acts on a particle moving in the xy-plane. Starting from the origin, the particle is taken along the positive x-axis to the point $(a, 0)$ and then parallel to the y-axis to the point (a, a). The total work done by the force F on the particles is [J&K CET 2005] [IIT 1998]

(a) $-2Ka^2$ (b) $2Ka^2$

(c) $-Ka^2$ (d) Ka^2

7. If g is the acceleration due to gravity on the earth's surface, the gain in the potential energy of an object of mass m raised from the surface of earth to a height equal to the radius of the earth R, is

(a) $\frac{1}{2}mgR$ (b) $2\ mgR$

(c) mgR (d) $\frac{1}{4}mgR$

8. A lorry and a car moving with the same K.E. are brought to rest by applying the same retarding force, then [IIT 1973; MP PMT 2003]

(a) Lorry will come to rest in a shorter distance

(b) Car will come to rest in a shorter distance

(c) Both come to rest in a same distance

(d) None of the above

9. A particle free to move along the x-axis has potential energy given by $U(x) = k[1 - \exp(-x)^2]$ for $-\infty \le x \le +\infty$, where k is a positive constant of appropriate dimensions. Then [IIT-JEE 1999; UPSEAT 2003]

(a) At point away from the origin, the particle is in unstable equilibrium

(b) For any finite non-zero value of x, there is a force directed away from the origin

(c) If its total mechanical energy is $k/2$, it has its minimum kinetic energy at the origin

(d) For small displacements from $x = 0$, the motion is simple harmonic [IIT-JEE Screening 2003]

10. The kinetic energy acquired by a mass m in travelling a certain distance d starting from rest under the action of a constant force is directly proportional to [CBSE PMT 1994]

(a) $\sqrt{m}$ (b) Independent of m

(c) $1/\sqrt{m}$ (d) m

11. An open knife edge of mass 'm' is dropped from a height 'h' on a wooden floor. If the blade penetrates upto the depth 'd' into the wood, the average resistance offered by the wood to the knife edge is [BHU 2002]

(a) mg (b) $mg\left(1 - \frac{h}{d}\right)$

(c) $mg\left(1 + \frac{h}{d}\right)$ (d) $mg\left(1 + \frac{h}{d}\right)^2$

12. Consider the following two statements

1. Linear momentum of a system of particles is zero
2. Kinetic energy of a system of particles is zero

Then [AIEEE 2003]

(a) 1 implies 2 and 2 implies 1

(b) 1 does not imply 2 and 2 does not imply 1

(c) 1 implies 2 but 2 does not imply 1

(d) 1 does not imply 2 but 2 implies 1

13. A body is moved along a straight line by a machine delivering constant power. The distance moved by the body in time t is proportional to

[IIT 1984; BHU 1984, 95; MP PET 1996; JIPMER 2000; AMU (Med.) 1999]

(a) $t^{1/2}$ (b) $t^{3/4}$

(c) $t^{3/2}$ (d) t^2

14. A shell is fired from a cannon with velocity v *m/sec* at an angle θ with the horizontal direction. At the highest point in its path it explodes into two pieces of equal mass. One of the pieces retraces its path to the cannon and the speed in *m/sec* of the other piece immediately after the explosion is

[IIT 1984; RPET 1999, 2001; UPSEAT 2002]

(a) $3v\cos\theta$ (b) $2v\cos\theta$

(c) $\frac{3}{2}v\cos\theta$ (d) $\frac{\sqrt{3}}{2}v\cos\theta$

15. A vessel at rest explodes into three pieces. Two pieces having equal masses fly off perpendicular to one another with the same velocity 30 *meter* per *second*. The third piece has three times mass of each of other piece. The magnitude and direction of the velocity of the third piece will be

[AMU (Engg.) 1999]

(a) $10\sqrt{2}\ m/second$ and 135° from either

(b) $10\sqrt{2}\ m/second$ and 45° from either

(c) $\frac{10}{\sqrt{2}}\ m/second$ and 135° from either

(d) $\frac{10}{\sqrt{2}}\ m/second$ and 45° from either

16. Two particles of masses m_1 and m_2 in projectile motion have velocities $\vec{v}_1$ and $\vec{v}_2$ respectively at time $t = 0$. They collide at time t_0. Their velocities become $\vec{v}_1{}'$ and $\vec{v}_2{}'$ at time $2t_0$ while still moving in air. The value of $|(m_1\overrightarrow{v_1}' + m_2\overrightarrow{v_2}') - (m_1\overrightarrow{v_1} + m_2\overrightarrow{v_2})|$ is

[IIT-JEE Screening 2001]

(a) Zero (b) $(m_1+m_2)gt_0$

(c) $2(m_1+m_2)gt_0$ (d) $\frac{1}{2}(m_1+m_2)gt_0$

17. Consider elastic collision of a particle of mass m moving with a velocity u with another particle of the same mass at rest. After the collision the projectile and the struck particle move in directions making angles θ_1 and θ_2 respectively with the initial direction of motion. The sum of the angles. $\theta_1+\theta_2$, is

(a) 45° (b) 90°

(c) 135° (d) 180°

18. A body of mass m moving with velocity v collides head on with another body of mass $2m$ which is initially at rest. The ratio of K.E. of colliding body before and after collision will be

(a) 1 : 1 (b) 2 : 1

(c) 4 : 1 (d) 9 : 1

19. A particle P moving with speed v undergoes a head -on elastic collision with another particle Q of identical mass but at rest. After the collision [Roorkee 2000]

(a) Both P and Q move forward with speed $\frac{v}{2}$

(b) Both P and Q move forward with speed $\frac{v}{\sqrt{2}}$

(c) P comes to rest and Q moves forward with speed v

(d) P and Q move in opposite directions with speed $\frac{v}{\sqrt{2}}$

20. A set of n identical cubical blocks lies at rest parallel to each other along a line on a smooth horizontal surface. The separation between the near surfaces of any two adjacent blocks is L. The block at one end is given a speed v towards the next one at time $t = 0$. All collisions are completely inelastic, then

(a) The last block starts moving at $t = \frac{(n-1)L}{v}$

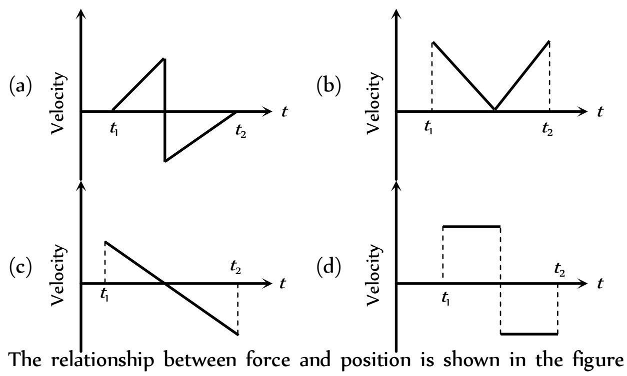
(b) The last block starts moving at $t = \frac{n(n-1)L}{2v}$

(c) The centre of mass of the system will have a final speed v

(d) The centre of mass of the system will have a final speed $\frac{v}{n}$

GQ Graphical Questions

1. A batsman hits a sixer and the ball touches the ground outside the cricket ground. Which of the following graph describes the variation of the cricket ball's vertical velocity v with time between the time t_1 as it hits the bat and time t_2 when it touches the ground



2. The relationship between force and position is shown in the figure given (in one dimensional case). The work done by the force in displacing a body from $x = 1\ cm$ to $x = 5\ cm$ is

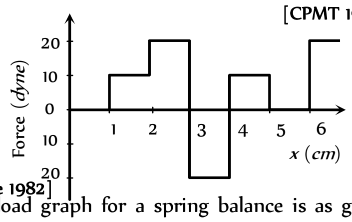
[UPSEAT 2004] [CPMT 1976]

(a) 20 *ergs*

(b) 60 *ergs*

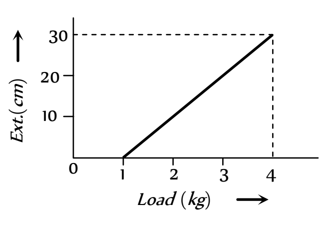
(c) 70 *ergs*

(d) 700 *ergs*



[Roorkee 1982]

3. The pointer reading *v/s* load graph for a spring balance is as given in the figure. The spring constant is



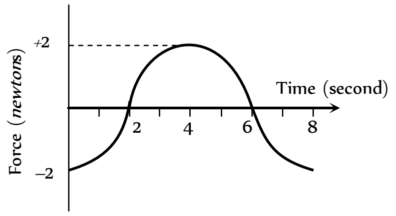
(a) 0.1 *kg/cm*

(b) 5 *kg/cm*

(c) 0.3 *kg/cm*

(d) 1 *kg/cm*

4. A force-time graph for a linear motion is shown in figure where the segments are circular. The linear momentum gained between zero and 8 *second* is [CPMT 1989]



(a) $-2\pi\, newton \times second$ (b) $Zero\ newton \times second$

(c) $+4\pi\, newton \times second$ (d) $-6\pi\, newton \times second$

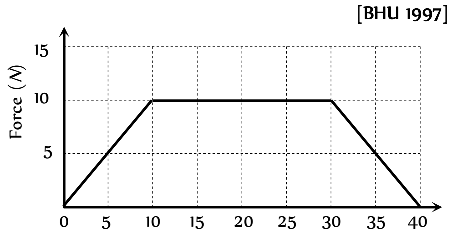
5. Adjacent figure shows the force-displacement graph of a moving body, the work done in displacing body from $x = 0$ to $x = 35\,m$ is equal to [BHU 1997]

(a) 50 *J*

(b) 25 *J*

(c) 287.5 *J*

(d) 200 *J*



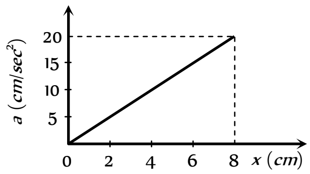
6. A 10*kg* mass moves along *x*-axis. Its acceleration as a function of its position is shown in the figure. What is the total work done on the mass by the force as the mass moves from $x = 0$ to $x = 8\,cm$

(a) $8 \times 10^{-2}\, joules$

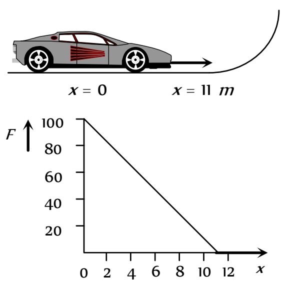
(b) $16 \times 10^{-2}\, joules$

(c) $4 \times 10^{-4}\, joules$

(d) $1.6 \times 10^{-3}\, joules$



7. A toy car of mass 5 *kg* moves up a ramp under the influence of force *F* plotted against displacement *x*. The maximum height attained is given by



(a) $y_{max} = 20m$

(b) $y_{max} = 15m$

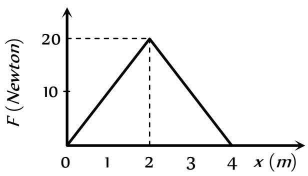
(c) $y_{max} = 11m$

(d) $y_{max} = 5m$

8. The graph between the resistive force *F* acting on a body and the distance covered by the body is shown in the figure. The mass of the body is 25 *kg* and initial velocity is 2 *m/s*. When the distance covered by the body is $4m$, its kinetic energy would be

(a) 50 *J*

(b) 40 *J*



(c) 20 *J*

(d) 10 *J*

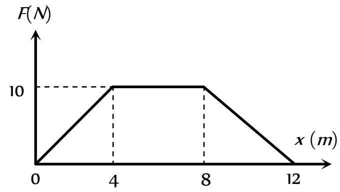
9. A particle of mass 0.1 *kg* is subjected to a force which varies with distance as shown in fig. If it starts its journey from rest at $x = 0$, its velocity at $x = 12\,m$ is [AIIMS 1995]

(a) 0 *m/s*

(b) $20\sqrt{2}\ m/s$

(c) $20\sqrt{3}\ m/s$

(d) $40\ m/s$



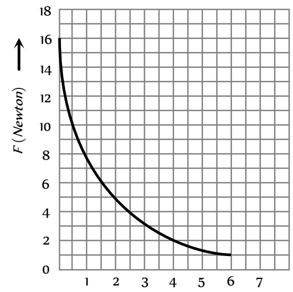
10. The relation between the displacement *X* of an object produced by the application of the variable force *F* is represented by a graph shown in the figure. If the object undergoes a displacement from $X = 0.5\,m$ to $X = 2.5\,m$ the work done will be approximately equal to [CPMT 1986]

(a) 16 *J*

(b) 32 *J*

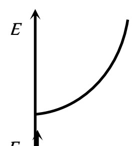
(c) 1.6 *J*

(d) 8 *J*

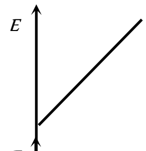


11. A particle is dropped from a height *h*. A constant horizontal velocity is given to the particle. Taking *g* to be constant every where, kinetic energy *E* of the particle *w. r. t.* time *t* is correctly shown in [AMU (Med.) 2000]

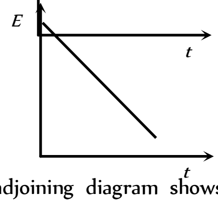
(a)



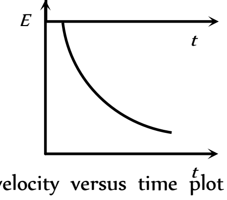
(b)



(c)



(d)



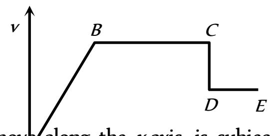
12. The adjoining diagram shows the velocity versus time plot for a particle. The work done by the force on the particle is positive from

(a) *A* to *B*

(b) *B* to *C*

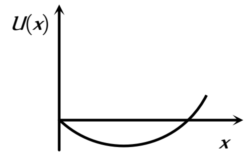
(c) *C* to *D*

(d) *D* to *E*

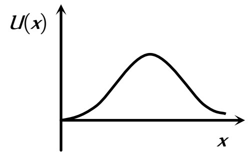


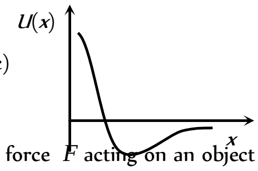
13. A particle which is constrained to move along the *x*-axis, is subjected to a force in the same direction which varies with the distance *x* of the particle from the origin as $F(x) = -kx + ax^3$. Here *k* and *a* are positive constants. For $x \geq 0$, the functional from of the potential energy $U(x)$ of the particle is [IIT-JEE (Screening) 2002]

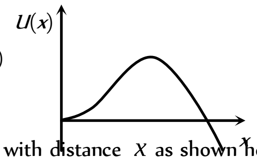
(a)



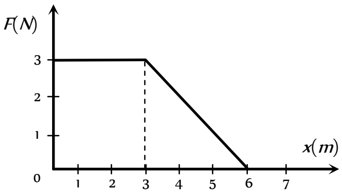
(b)



(c) 

(d) 

14. A force F acting on an object varies with distance x as shown here. The force is in *newton* and x in *metre*. The work done by the force in moving the object from $x = 0$ to $x = 6m$ is



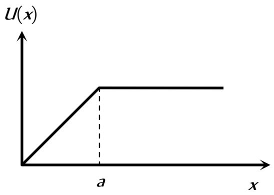
(a) 4.5 J

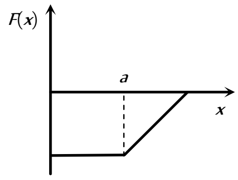
(b) 13.5 J

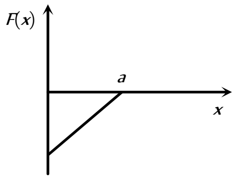
(c) 9.0 J

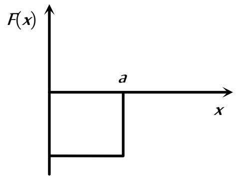
(d) 18.0 J

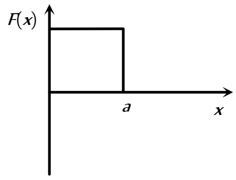
15. The potential energy of a system is represented in the first figure. the force acting on the system will be represented by



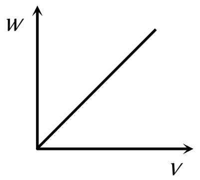
(a) 

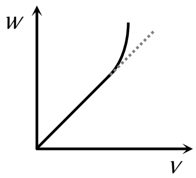
(b) 

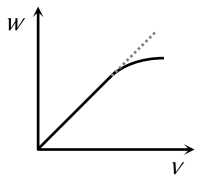
(c) 

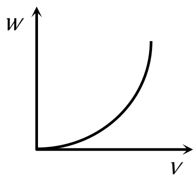
(d) 

16. A particle, initially at rest on a frictionless horizontal surface, is acted upon by a horizontal force which is constant in size and direction. A graph is plotted between the work done (W) on the particle, against the speed of the particle, (v). If there are no other horizontal forces acting on the particle the graph would look like

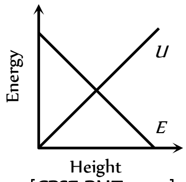
(a) 

(b) 

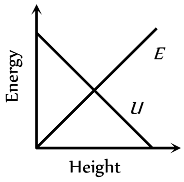
(c) 

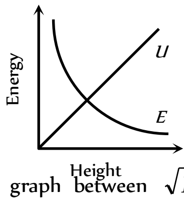
(d) 

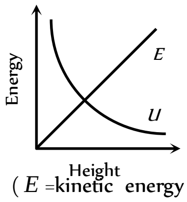
17. Which of the following graphs is correct between kinetic energy (E), potential energy (U) and height (h) from the ground of the particle

(a) 

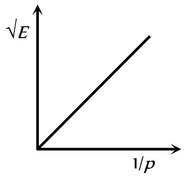
[CBSE PMT 2005]

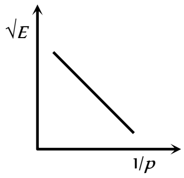
(b) 

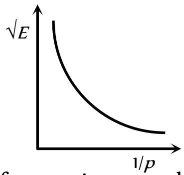
(c) 

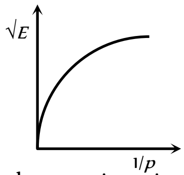
(d) 

18. The graph between $\sqrt{E}$ and $\frac{1}{p}$ is (E =kinetic energy and p = momentum)

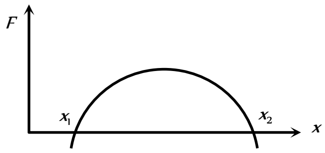
(a) 

(b) 

(c) 

(d) 

19. The force acting on a body moving along x-axis varies with the position of the particle as shown in the fig.



The body is in stable equilibrium at

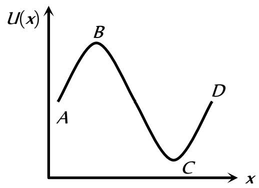
(a) $x = x_1$

(b) $x = x_2$

(c) both x_1 and x_2

(d) neither x_1 nor x_2

20. The potential energy of a particle varies with distance x as shown in the graph.



The force acting on the particle is zero at

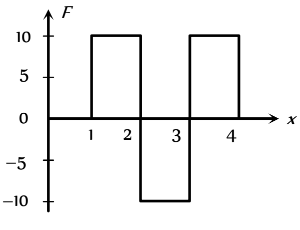
(a) C

(b) B

(c) B and C

(d) A and D

21. Figure shows the F-x graph. Where F is the force applied and x is the distance covered



by the body along a straight line path. Given that F is in *newton* and x in *metre*, what is the work done ?

(a) 10 J (b) 20 J

(c) 30 J (d) 40 J

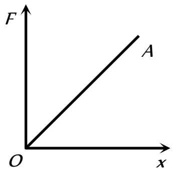
22. The force required to stretch a spring varies with the distance as shown in the figure. If the experiment is performed with the above spring of half length, the line OA will

(a) Shift towards F-axis

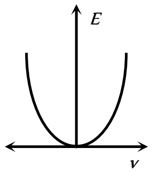
(b) Shift towards X-axis

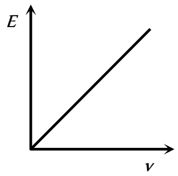
(c) Remain as it is

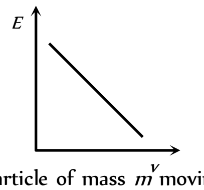
(d) Become double in length

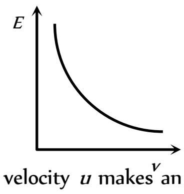


23. The graph between E and v is

(a)


(b)


(c)


(d)


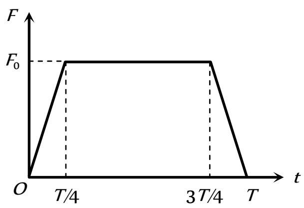
24. A particle of mass m moving with a velocity u makes an elastic one dimensional collision with a stationary particle of mass m establishing a contact with it for extremely small time T. Their force of contact increases from zero to F linearly in time $\frac{T}{4}$, remains constant for a further time $\frac{T}{2}$ and decreases linearly from F to zero in further time $\frac{T}{4}$ as shown. The magnitude possessed by F is

(a) $\frac{mu}{T}$

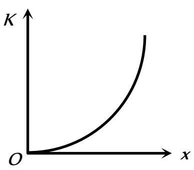
(b) $\frac{2mu}{T}$

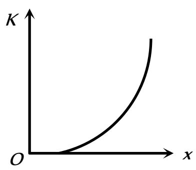
(c) $\frac{4mu}{3T}$

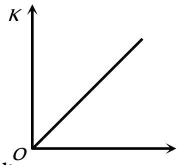
(d) $\frac{3mu}{4T}$

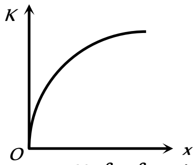


25. A body moves from rest with a constant acceleration. Which one of the following graphs represents the variation of its kinetic energy K with the distance travelled x ?

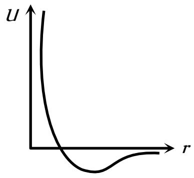
(a)


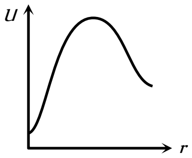
(b)


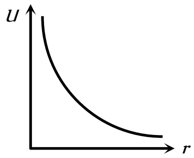
(c)


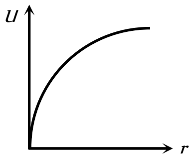
(d)


26. The diagrams represent the potential energy U of a function of the inter-atomic distance r. Which diagram corresponds to stable molecules found in nature.

(a)


(b)


(c)


(d)


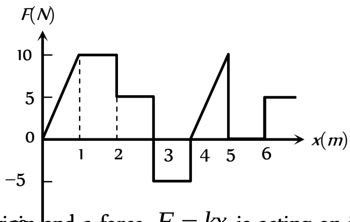
27. The relationship between the force F and position x of a body is as shown in figure. The work done in displacing the body from $x = 1$ m to $x = 5$ m will be **[KCET 2005]**

(a) 30 J

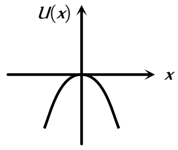
(b) 15 J

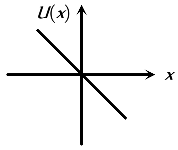
(c) 25 J

(d) 20 J



28. A particle is placed at the origin and a force $F = kx$ is acting on it (where k is positive constant). If $U(0) = 0$, the graph of $U(x)$ versus x will be (where U is the potential energy function)**[IIT-JEE (Screening) 20**

(a)


(b)


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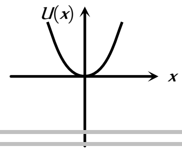

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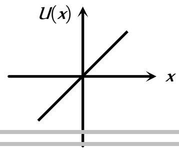


AR Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

(a) If both assertion and reason are true and the reason is the correct explanation of the assertion.

(b) If both assertion and reason are true but reason is not the correct explanation of the assertion.

(c) If assertion is true but reason is false.

(d) If the assertion and reason both are false.

(e) If assertion is false but reason is true.

1. Assertion : A person working on a horizontal road with a load on his head does no work.

Reason : No work is said to be done, if directions of force and displacement of load are perpendicular to each other.

2. Assertion : The work done during a round trip is always zero.
Reason : No force is required to move a body in its round trip.

3. Assertion : Work done by friction on a body sliding down an inclined plane is positive.
Reason : Work done is greater than zero, if angle between force and displacement is acute or both are in same direction.

4. Assertion : When a gas is allowed to expand, work done by gas is positive.
Reason : Force due to gaseous pressure and displacement (of piston) are in the same direction.

5. Assertion : A light body and heavy body have same momentum. Then they also have same kinetic energy.
Reason : Kinetic energy does not depend on mass of the body.

6. Assertion : The instantaneous power of an agent is measured as the dot product of instantaneous velocity and the force acting on it at that instant.
Reason : The unit of instantaneous power is watt.

7. Assertion : The change in kinetic energy of a particle is equal to the work done on it by the net force.
Reason : Change in kinetic energy of particle is equal to the work done only in case of a system of one particle.

8. Assertion : A spring has potential energy, both when it is compressed or stretched.
Reason : In compressing or stretching, work is done on the spring against the restoring force.

9. Assertion : Comets move around the sun in elliptical orbits. The gravitational force on the comet due to sun is not normal to the comet's velocity but the work done by the gravitational force over every complete orbit of the comet is zero.
Reason : Gravitational force is a non conservative force.

10. Assertion : The rate of change of total momentum of a many particle system is proportional to the sum of the internal forces of the system.
Reason : Internal forces can change the kinetic energy but not the momentum of the system.

11. Assertion : Water at the foot of the water fall is always at different temperature from that at the top.
Reason : The potential energy of water at the top is converted into heat energy during falling.

12. Assertion : The power of a pump which raises 100 kg of water in 10sec to a height of 100 m is 10 KW.
Reason : The practical unit of power is horse power.

13. Assertion : According to law of conservation of mechanical energy change in potential energy is equal and opposite to the change in kinetic energy.
Reason : Mechanical energy is not a conserved quantity.

14. Assertion : When the force retards the motion of a body, the work done is zero.
Reason : Work done depends on angle between force and displacement.

15. Assertion : In an elastic collision of two bodies, the momentum and energy of each body is conserved.
Reason : If two bodies stick to each other, after colliding, the collision is said to be perfectly elastic.

16. Assertion : A body cannot have energy without having momentum but it can have momentum without having energy.
Reason : Momentum and energy have same dimensions.

17. Assertion : Power developed in circular motion is always zero.
Reason : Work done in case of circular motion is zero.

18. Assertion : A kinetic energy of a body is quadrupled, when its velocity is doubled.
Reason : Kinetic energy is proportional to square of velocity.

19. Assertion : A quick collision between two bodies is more violent than slow collision, even when initial and final velocities are identical.
Reason : The rate of change of momentum determine that force is small or large.

20. Assertion : Work done by or against gravitational force in moving a body from one point to another is independent of the actual path followed between the two points.
Reason : Gravitational forces are conservative forces.

21. Assertion : Wire through which current flows gets heated.
Reason : When current is drawn from a cell, chemical energy is converted into heat energy.

22. Assertion : Graph between potential energy of a spring versus the extension or compression of the spring is a straight line.
Reason : Potential energy of a stretched or compressed spring, proportional to square of extension or compression.

23. Assertion : Heavy water is used as moderator in nuclear reactor.
Reason : Water cool down the fast neutron.

24. Assertion : Mass and energy are not conserved separately, but are conserved as a single entity called mass-energy.
Reason : Mass and energy conservation can be obtained by Einstein equation for energy.

25. Assertion : If two protons are brought near one another, the potential energy of the system will increase.
Reason : The charge on the proton is $+1.6\times10^{-19}\ C$.

26. Assertion : In case of bullet fired from gun, the ratio of kinetic energy of gun and bullet is equal to ratio of mass of bullet and gun.
Reason : In firing, momentum is conserved.

27. Assertion : Power of machine gun is determined by both, the number of bullet fired per second and kinetic energy of bullets.
Reason : Power of any machine is defined as work done (by it) per unit time.

28. Assertion : A work done in moving a body over a closed loop is zero for every force in nature.
Reason : Work done does not depend on nature of force.

29. Assertion : Mountain roads rarely go straight up the slope.

Reason : Slope of mountains are large therefore more chances of vehicle to slip from roads.

30. Assertion : Soft steel can be made red hot by continued hammering on it, but hard steel cannot.

Reason : Energy transfer in case of soft iron is large as in hard steel.

Answers

Work Done by Constant Force

1	b	2	a	3	c	4	d	5	c
6	b	7	b	8	c	9	a	10	d
11	d	12	b	13	d	14	b	15	b
16	b	17	b	18	d	19	d	20	d
21	d	22	d	23	d	24	a	25	c
26	a	27	d	28	b	29	d	30	a
31	b	32	c	33	a	34	b	35	a
36	d	37	a	38	c	39	c	40	a
41	c								

Work Done by Variable Force

1	b	2	c	3	c	4	a	5	a
6	c	7	d	8	d	9	d	10	b
11	b	12	c	13	b	14	c	15	d
16	c	17	a	18	a	19	c	20	b
21	d	22	a	23	a	24	b	25	d
26	d								

Conservation of Energy and Momentum

1	c	2	c	3	a	4	a	5	b
6	d	7	c	8	c	9	b	10	d
11	c	12	b	13	c	14	a	15	b
16	c	17	b	18	d	19	b	20	c
21	b	22	c	23	d	24	c	25	a
26	c	27	d	28	d	29	a	30	b
31	d	32	d	33	a	34	d	35	a
36	a	37	b	38	c	39	a	40	c
41	d	42	c	43	b	44	a	45	a
46	b	47	b	48	b	49	d	50	a
51	b	52	a	53	c	54	d	55	d
56	a	57	c	58	b	59	c	60	a
61	b	62	b	63	a	64	c	65	d
66	b	67	d	68	b	69	a	70	c
71	b	72	a	73	c	74	c	75	c
76	a	77	b	78	a	79	a	80	d
81	d	82	b	83	c	84	b	85	c
86	c	87	b	88	c				

Power

1	a	2	d	3	d	4	b	5	c
6	a	7	b	8	d	9	d	10	c
11	c	12	d	13	a	14	a	15	c
16	c	17	b	18	a	19	c	20	c
21	a	22	b	23	a	24	a	25	a
26	a	27	a	28	a	29	a	30	c

Elastic and Inelastic collision

1	a	2	a	3	c	4	a	5	c
6	c	7	b	8	c	9	c	10	d
11	d	12	b	13	d	14	c	15	d
16	a	17	c	18	c	19	d	20	a
21	d	22	d	23	b	24	a	25	c
26	a	27	b	28	d	29	d	30	b
31	b	32	a	33	b	34	a	35	c
36	a	37	c	38	d	39	a	40	b
41	b	42	d	43	d	44	a	45	d
46	a	47	a	48	d	49	c	50	b
51	c	52	d	53	b	54	a	55	b
56	a	57	a	58	a	59	d	60	c
61	a	62	c	63	d	64	a	65	a
66	a	67	a	68	d				

Perfectly Inelastic Collision

1	c	2	b	3	c	4	b	5	d
6	b	7	c	8	c	9	c	10	a
11	d	12	b	13	b	14	c	15	c
16	d	17	a	18	b	19	d	20	a
21	b	22	c	23	b	24	c	25	b
26	a	27	b	28	d	29	c	30	d

Critical Thinking Questions

1	c	2	d	3	b	4	b	5	c
6	c	7	a	8	c	9	d	10	b
11	c	12	d	13	c	14	a	15	a
16	c	17	b	18	d	19	c	20	bd

Graphical Questions

1	c	2	a	3	a	4	b	5	c
6	a	7	c	8	d	9	d	10	a
11	a	12	a	13	d	14	b	15	c
16	d	17	a	18	c	19	b	20	c
21	a	22	a	23	a	24	c	25	c
26	a	27	b	28	a				

Assertion and Reason

1	a	2	d	3	e	4	a	5	d
6	b	7	c	8	a	9	c	10	e
11	a	12	b	13	c	14	e	15	d
16	d	17	e	18	a	19	a	20	a
21	c	22	e	23	c	24	a	25	b
26	a	27	a	28	d	29	a	30	a

Answers and Solutions

Work Done by Constant Force

1. (b) Work done by centripetal force is always zero, because force and instantaneous displacement are always perpendicular.

$$W = \vec{F}.\vec{s} = Fs\cos\theta = Fs\cos(90°) = 0$$

2. (a) Work = Force × Displacement (length)

If unit of force and length be increased by four times then the unit of energy will increase by 16 times.

3. (c) No displacement is there.

4. (d) Stopping distance $S \propto u^2$. If the speed is doubled then the stopping distance will be four times.

5. (c) $W = Fs\cos\theta \Rightarrow \cos\theta = \frac{W}{Fs} = \frac{25}{50} = \frac{1}{2} \Rightarrow \theta = 60°$

6. (b) Work done = Force × displacement

= Weight of the book × Height of the book shelf

7. (b) Work done does not depend on time.

8. (c) $W = \vec{F}.\vec{s} = (5\hat{i} + 3\hat{j}).(2\hat{i} - \hat{j}) = 10 - 3 = 7\,J$

9. (a) $v = \frac{dx}{dt} = 3 - 8t + 3t^2$

$\therefore\ v_0 = 3\,m/s$ and $v_4 = 19\,m/s$

$W = \frac{1}{2}m(v_4^2 - v_0^2)$ (According to work energy theorem)

$= \frac{1}{2} \times 0.03 \times (19^2 - 3^2) = 5.28\,J$

10. (d) As the body moves in the direction of force therefore work done by gravitational force will be positive.

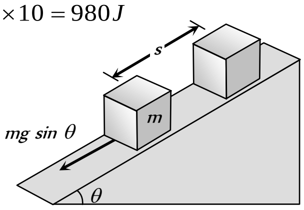
$W = Fs = mgh = 10 \times 9.8 \times 10 = 980\,J$

11. (d)

12. (b) $W = mg\sin\theta \times s$

$= 2 \times 10^3 \times \sin 15° \times 10$

$= 5.17\,kJ$



13. (d) $W = \vec{F}.\vec{s} = (5\hat{i} + 6\hat{j} - 4\hat{k}).(6\hat{i} + 5\hat{k}) = 30 - 20 = 10$ *units*

14. (b) $W = Fs = F \times \frac{1}{2}at^2$ $\left[\text{from } s = ut + \frac{1}{2}at^2\right]$

$$\Rightarrow W = F\left[\frac{1}{2}\left(\frac{F}{m}\right)t^2\right] = \frac{F^2t^2}{2m} = \frac{25 \times (1)^2}{2 \times 15} = \frac{25}{30} = \frac{5}{6}\ J$$

15. (b) Work done on the body = K.E. gained by the body

$$Fs\cos\theta = 1 \Rightarrow F\cos\theta = \frac{1}{s} = \frac{1}{0.4} = 2.5\,N$$

16. (b) Work done $= mgh = 10 \times 9.8 \times 1 = 98\,J$

17. (b)

18. (d) $s = \frac{t^2}{4}\ \therefore ds = \frac{t}{2}dt$

$$F = ma = \frac{md^2s}{dt^2} = \frac{6d^2}{dt^2}\left[\frac{t^2}{4}\right] = 3N$$

Now

$$W = \int_0^2 F\,ds = \int_0^2 3\frac{t}{2}dt = \frac{3}{2}\left[\frac{t^2}{2}\right]_0^2 = \frac{3}{4}\left[(2)^2 - (0)^2\right] = 3J$$

19. (d) Net force on body $= \sqrt{4^2 + 3^2} = 5N$

$\therefore\ a = F/m = 5/10 = 1/2\,m/s^2$

Kinetic energy $= \frac{1}{2}mv^2 = \frac{1}{2}m(at)^2 = 125\,J$

20. (d) $s = \frac{u^2}{2\mu g} = \frac{10 \times 10}{2 \times 0.5 \times 10} = 10\,m$

21. (d) $W = \vec{F}.\vec{s} = (3\hat{i} + 4\hat{j}).(3\hat{i} + 4\hat{j}) = 9 + 16 = 25\,J$

22. (d) Total mass = (50 + 20) = 70 *kg*

Total height = 20 × 0.25 = 5*m*

∴ Work done = *mgh* = 70 × 9.8 × 5 = 3430 *J*

23. (d) $W = \vec{F}.\vec{s} = (6\hat{i} + 2\hat{j} - 3\hat{k}).(2\hat{i} - 3\hat{j} + x\hat{k}) = 0$

$12 - 6 - 3x = 0 \Rightarrow x = 2$

24. (a) $W = \vec{F}.(\vec{r_2} - \vec{r_1}) = (4\hat{i} + \hat{j} + 3\hat{k})(11\hat{i} + 11\hat{j} + 15\hat{k})$

$W = 44 + 11 + 45 = 100\ Joule$

25. (c) $W = (3\hat{i} + c\hat{j} + 2\hat{k}).(-4\hat{i} + 2\hat{j} + 3\hat{k}) = 6\,J$

$W = -12 + 2c + 6 = 6 \Rightarrow c = 6$

26. (a) Both part will have numerically equal momentum and lighter part will have more velocity.

27. (d) *Watt* and *Horsepower* are the unit of power

28. (b) Work = Force × Displacement

If force and displacement both are doubled then work would be four times.

29. (d) $W = FS\cos\theta = 10 \times 4 \times \cos 60° = 20\ Joule$

30. (a) $W = \vec{F}.\vec{s} = (5\hat{i} + 4\hat{j}).(6\hat{i} - 5\hat{j} + 3\hat{k}) = 30 - 20 = 10\,J$

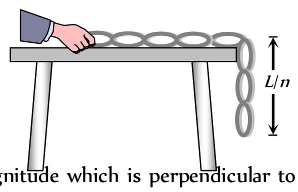
31. (b) Fraction of length of the chain hanging from the table

$= \frac{1}{n} = \frac{60cm}{200cm} = \frac{3}{10} \Rightarrow n = \frac{10}{3}$

Work done in pulling the chain on the table

$W = \frac{mgL}{2n^2}$

$= \frac{4 \times 10 \times 2}{2 \times (10/3)^2} = 3.6\,J$



32. (c) When a force of constant magnitude which is perpendicular to the velocity of particle acts on a particle, work done is zero and hence change in kinetic energy is zero.

33. (a) The ball rebounds with the same speed. So change in it's Kinetic energy will be zero *i.e.* work done by the ball on the wall is zero.

34. (b) $W = \vec{F}.\vec{r} = (5\hat{i} + 3\hat{j} + 2\hat{k}).(2\hat{i} - \hat{j}) = 10 - 3 = 7\,J$

35. (a) K.E. acquired by the body = work done on the body

$K.E. = \frac{1}{2}mv^2 = Fs$ *i.e.* it does not depend upon the mass of the body although velocity depends upon the mass

$v^2 \propto \frac{1}{m}$ [If F and s are constant]

36. (d) $W = \vec{F}.\vec{s} = (4\hat{i} + 5\hat{j} + 0\hat{k}).(3\hat{i} + 0\hat{j} + 6\hat{k}) = 4 \times 3$ *units*

37. (a) As surface is smooth so work done against friction is zero. Also the displacement and force of gravity are perpendicular so work done against gravity is zero.

38. (c) Opposing force in vertical pulling = mg

But opposing force on an inclined plane is $mg \sin\theta$, which is less than mg.

39. (c) Velocity of fall is independent of the mass of the falling body.

40. (a) Work done $= \vec{F}.\vec{s}$

$= (6\hat{i} + 2\hat{j}) \cdot (3\hat{i} - \hat{j}) = 6 \times 3 - 2 \times 1 = 18 - 2 = 16\ J$

41. (c) When the ball is released from the top of tower then ratio of distances covered by the ball in first, second and third second

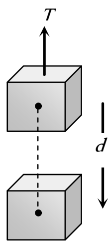
$h_I : h_{II} : h_{III} = 1 : 3 : 5 :$ [because $h_n \propto (2n-1)$]

$\therefore$ Ratio of work done $mgh_I : mgh_{II} : mgh_{III}$ = 1:3:5

Work Done by Variable Force

1. (b) $W \int_0^{x_1} F.dx = \int_0^{x_1} Cx\ dx = C\left[\frac{x^2}{2}\right]_0^{x_1} = \frac{1}{2}Cx_1^2$

2. (c) When the block moves vertically downward with acceleration $\frac{g}{4}$ then tension in the cord



$T = M\left(g - \frac{g}{4}\right) = \frac{3}{4}Mg$

Work done by the cord = $\vec{F}.\vec{s} = Fs\cos\theta$

= $Td\cos(180°) = -\left(\frac{3Mg}{4}\right) \times d = -3Mg\frac{d}{4}$

3. (c) $W = \frac{F^2}{2k}$

If both springs are stretched by same force then $W \propto \frac{1}{k}$

As $k_1 > k_2$ therefore $W_1 < W_2$

i.e. more work is done in case of second spring.

4. (a) $\Delta P.E. = \frac{1}{2}k(x_2^2 - x_1^2) = \frac{1}{2} \times 10[(0.25)^2 - (0.20)^2]$

$= 5 \times 0.45 \times 0.05 = 0.1\ J$

5. (a) $\frac{1}{2}kS^2 = 10\ J$ (given in the problem)

$\frac{1}{2}k\left[(2S)^2 - (S)^2\right] = 3 \times \frac{1}{2}kS^2$ = 3 × 10 = 30 J

6. (c) $U = \frac{F^2}{2k} \Rightarrow \frac{U_1}{U_2} = \frac{k_2}{k_1}$ (if force are same)

$\therefore \frac{U_1}{U_2} = \frac{3000}{1500} = \frac{2}{1}$

7. (d) Here $k = \frac{F}{x} = \frac{10}{1 \times 10^{-3}} = 10^4\ N/m$

$W = \frac{1}{2}kx^2 = \frac{1}{2} \times 10^4 \times (40 \times 10^{-3})^2 = 8\ J$

8. (d) $W = \int_0^5 F dx = \int_0^5 (7 - 2x + 3x^2)\,dx = [7x - x^2 + x^3]_0^5$

= 35 − 25 + 125 = 135 J

9. (d) $S = \frac{t^3}{3} \therefore dS = t^2 dt$

$a = \frac{d^2S}{dt^2} = \frac{d^2}{dt^2}\left[\frac{t^3}{3}\right] = 2t\ m/s^2$

Now work done by the force $W = \int_0^2 F.dS = \int_0^2 ma.dS$

$\int_0^2 3 \times 2t \times t^2 dt = \int_0^2 6t^3 dt = \frac{3}{2}\left[t^4\right]_0^2$ = 24 J

10. (b) $W = \frac{1}{2}kx^2$

If both wires are stretched through same distance then $W \propto k$. As $k_2 = 2k_1$ so $W_2 = 2W_1$

11. (b) $\frac{1}{2}mv^2 = \frac{1}{2}kx^2 \Rightarrow x = v\sqrt{\frac{m}{k}} = 10\sqrt{\frac{0.1}{1000}} = 0.1\,m$

12. (c) Force constant of a spring

$k = \frac{F}{x} = \frac{mg}{x} = \frac{1 \times 10}{2 \times 10^{-2}} \Rightarrow k = 500\ N/m$

Increment in the length = 60 − 50 = 10 cm

$U = \frac{1}{2}kx^2 = \frac{1}{2}500(10 \times 10^{-2})^2 = 2.5\ J$

13. (b) $W = \frac{1}{2}k(x_2^2 - x_1^2) = \frac{1}{2} \times 800 \times (15^2 - 5^2) \times 10^{-4} = 8\ J$

14. (c) $100 = \frac{1}{2}kx^2$ (given)

$W = \frac{1}{2}k(x_2^2 - x_1^2) = \frac{1}{2}k[(2x)^2 - x^2]$

$= 3 \times \left(\frac{1}{2}kx^2\right) = 3 \times 100 = 300\ J$

15. (d) $U = \frac{1}{2}kx^2$ if x becomes 5 times then energy will become 25 times *i.e.* $4 \times 25 = 100\ J$

16. (c) $W = \frac{1}{2}k(x_2^2 - x_1^2) = \frac{1}{2} \times 5 \times 10^3 (10^2 - 5^2) \times 10^{-4}$

$= 18.75\ J$

17. (a) The kinetic energy of mass is converted into potential energy of a spring

$\frac{1}{2}mv^2 = \frac{1}{2}kx^2 \Rightarrow x = \sqrt{\frac{mv^2}{k}} = \sqrt{\frac{0.5 \times (1.5)^2}{50}} = 0.15\,m$

18. (a) This condition is applicable for simple harmonic motion. As particle moves from mean position to extreme position its potential energy increases according to expression $U = \frac{1}{2}kx^2$ and accordingly kinetic energy decreases.

19. (c) Potential energy $U = \frac{1}{2}kx^2$

$\therefore U \propto x^2$ [if k = constant]

If elongation made 4 times then potential energy will become 16 times.

20. (b)

21. (d) $U \propto x^2 \Rightarrow \frac{U_2}{U_1} = \left(\frac{x_2}{x_1}\right)^2 = \left(\frac{0.1}{0.02}\right)^2 = 25 \therefore U_2 = 25U$

22. (a) If x is the extension produced in spring.

$$F = kx \Rightarrow x = \frac{F}{k} = \frac{mg}{k} = \frac{20 \times 9.8}{4000} = 4.9\,cm$$

23. (a) $U = \frac{F^2}{2k} = \frac{T^2}{2k}$

24. (b) $U = A - Bx^2 \Rightarrow F = -\frac{dU}{dx} = 2Bx \Rightarrow F \propto x$

25. (d) Condition for stable equilibrium $F = -\frac{dU}{dx} = 0$

$$\Rightarrow -\frac{d}{dx}\left[\frac{a}{x^{12}} - \frac{b}{x^6}\right] = 0 \Rightarrow -12ax^{-13} + 6bx^{-7} = 0$$

$$\Rightarrow \frac{12a}{x^{13}} = \frac{6b}{x^7} \Rightarrow \frac{2a}{b} = x^6 \Rightarrow x = \sqrt[6]{\frac{2a}{b}}$$

26. (d) Friction is a non-conservative force.

Conservation of Energy and Momentum

1. (c) $P = \sqrt{2mE}$ $\therefore P \propto \sqrt{m}$ (if E = const.) $\therefore \frac{P_1}{P_2} = \sqrt{\frac{m_1}{m_2}}$

2. (c) Work in raising a box

= (weight of the box) × (height by which it is raised)

3. (a) $E = \frac{P^2}{2m}$ if P = constant then $E \propto \frac{1}{m}$

4. (a) Body at rest may possess potential energy.

5. (b) Due to theory of relativity.

6. (d) $E = \frac{P^2}{2m}$ $\therefore$ $E \propto P^2$

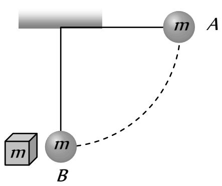
i.e. if P is increased n times then E will increase n times.

7. (c)

8. (c) P.E. of bob at point A = mgl

This amount of energy will be converted into kinetic energy

$\therefore$ K.E. of bob at point B = mgl



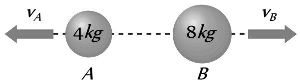
and as the collision between bob and block (of same mass) is elastic so after collision bob will come to rest and total Kinetic energy will be transferred to block. So kinetic energy of block = mgl

9. (b) According to conservation of momentum

Momentum of tank = Momentum of shell

$125000 \times v_{tank} = 25 \times 1000 \Rightarrow v_{tank} = 0.2$ *ft/sec.*

10. (d) As the initial momentum of bomb was zero, therefore after explosion two parts should possess numerically equal momentum



i.e. $m_A v_A = m_B v_B \Rightarrow 4 \times v_A = 8 \times 6 \Rightarrow v_A = 12\,m/s$

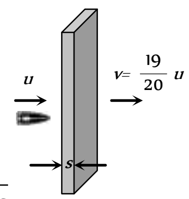
$\therefore$ Kinetic energy of other mass A, = $\frac{1}{2}m_A v_A^2$

= $\frac{1}{2} \times 4 \times (12)^2$ = 288 *J*.

11. (c) Let the thickness of one plank is s

if bullet enters with velocity u then it leaves with velocity

$$v = \left(u - \frac{u}{20}\right) = \frac{19}{20}u$$



from $v^2 = u^2 - 2as$

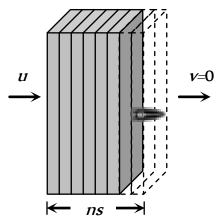
$$\Rightarrow \left(\frac{19}{20}u\right)^2 = u^2 - 2as \Rightarrow \frac{400}{39} = \frac{u^2}{2as}$$

Now if the n planks are arranged just to stop the bullet then again from $v^2 = u^2 - 2as$

$$0 = u^2 - 2ans$$

$$\Rightarrow n = \frac{u^2}{2as} = \frac{400}{39}$$

$$\Rightarrow n = 10.25$$



As the planks are more than 10 so we can consider n = 11

12. (b) Let h is that height at which the kinetic energy of the body becomes half its original value *i.e.* half of its kinetic energy will convert into potential energy

$\therefore$ $mgh = \frac{490}{2} \Rightarrow 2 \times 9.8 \times h = \frac{490}{2} \Rightarrow h = 12.5m.$

13. (c) $P = \sqrt{2mE}$. If E are same then $P \propto \sqrt{m}$

$$\Rightarrow \frac{P_1}{P_2} = \sqrt{\frac{m_1}{m_2}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

14. (a) Let initial kinetic energy, $E_1 = E$

Final kinetic energy, $E_2 = E + 300\%$ of E = $4E$

As $P \propto \sqrt{E} \Rightarrow \frac{P_2}{P_1} = \sqrt{\frac{E_2}{E_1}} = \sqrt{\frac{4E}{E}} = 2 \Rightarrow P_2 = 2P_1$

$\Rightarrow P_2 = P_1 + 100\%$ of P_1

i.e. Momentum will increase by 100%.

15. (b) $P = \sqrt{2mE}$ if E are equal then $P \propto \sqrt{m}$

i.e. heavier body will possess greater momentum.

16. (c) Let $P_1 = P$, $P_2 = P_1 + 50\%$ of $P_1 = P_1 + \frac{P_1}{2} = \frac{3P_1}{2}$

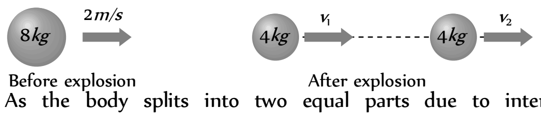
$$E \propto P^2 \Rightarrow \frac{E_2}{E_1} = \left(\frac{P_2}{P_1}\right)^2 = \left(\frac{3P_1/2}{P_1}\right)^2 = \frac{9}{4}$$

$$\Rightarrow E_2 = 2.25E = E_1 + 1.25E_1$$

$$\therefore E_2 = E_1 + 125\% \text{ of } E_1$$

i.e. kinetic energy will increase by 125%.

17. (b)



As the body splits into two equal parts due to internal explosion therefore momentum of system remains conserved *i.e.* $8 \times 2 = 4v_1 + 4v_2 \Rightarrow v_1 + v_2 = 4$ ...(i)

By the law of conservation of energy

Initial kinetic energy + Energy released due to explosion

= Final kinetic energy of the system

$$\Rightarrow \frac{1}{2}\times 8 \times (2)^2 + 16 = \frac{1}{2}4v_1^2 + \frac{1}{2}4v_2^2$$

$$\Rightarrow v_1^2 + v_2^2 = 16 \quad \text{...(ii)}$$

By solving eq. (i) and (ii) we get $v_1 = 4$ and $v_2 = 0$

i.e. one part comes to rest and other moves in the same direction as that of original body.

18. (d) $P = \sqrt{2mE}$ $\therefore P \propto \sqrt{E}$

i.e. if kinetic energy of a particle is doubled the its momentum will becomes $\sqrt{2}$ times.

19. (b) Potential energy = mgh

Potential energy is maximum when h is maximum

20. (c) If particle is projected vertically upward with velocity of $2m/s$ then it returns with the same velocity.

So its kinetic energy $= \frac{1}{2}mv^2 = \frac{1}{2}\times 2 \times (2)^2 = 4\ J$

21. (b)

22. (c) $E = \frac{P^2}{2m}$ if bodies possess equal linear momenta then

$E \propto \frac{1}{m}$ *i.e.* $\frac{E_1}{E_2} = \frac{m_2}{m_1}$

23. (d) $s \propto u^2$ *i.e.* if speed becomes double then stopping distance will become four times *i.e.* $8 \times 4 = 32m$

24. (c) $s \propto u^2$ *i.e.* if speed becomes three times then distance needed for stopping will be nine times.

25. (a) $P = \sqrt{2mE}$ $\therefore P \propto \sqrt{E}$

Percentage increase in $P = \frac{1}{2}$ (percentage increase in E)

$= \frac{1}{2}(0.1\%) = 0.05\%$

26. (c) Kinetic energy = $\frac{1}{2}mv^2$ $\therefore$ K.E. $\propto v$

If velocity is doubled then kinetic energy will become four times.

27. (d) $P = \sqrt{2mE}$ $\therefore \frac{P_1}{P_2} = \sqrt{\frac{m_1}{m_2}}$ (if E = constant)

$\therefore \frac{P_1}{P_2} = \sqrt{\frac{3}{1}}$

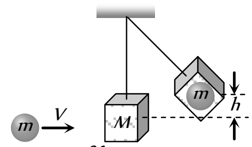
28. (d) In compression or extension of a spring work is done against restoring force.

In moving a body against gravity work is done against gravitational force of attraction.

It means in all three cases potential energy of the system increases.

But when the bubble rises in the direction of upthrust force then system works so the potential energy of the system decreases.

29. (a)



By the conservation of linear momentum

Initial momentum of sphere

= Final momentum of system

$$mV = (m + M)v_{sys.} \quad \text{...(i)}$$

If the system rises up to height h then by the conservation of energy

$$\frac{1}{2}(m + M)v_{sys.}^2 = (m + M)gh \quad \text{...(ii)}$$

$$\Rightarrow v_{sys.} = \sqrt{2gh}$$

Substituting this value in equation (i)

$$V = \left(\frac{m + M}{m}\right)\sqrt{2gh}$$

30. (b) $E = \frac{P^2}{2m}$. If momentum are same then $E \propto \frac{1}{m}$

$\therefore \frac{E_1}{E_2} = \frac{m_2}{m_1} = \frac{2m}{m} = \frac{2}{1}$

31. (d) $P = \sqrt{2mE}$. If kinetic energy are equal then $P \propto \sqrt{m}$

i.e., heavier body posses large momentum

As $M_1 < M_2$ therefore $M_1V_1 < M_2V_2$

32. (d) Condition for vertical looping $h = \frac{5}{2}r = 5cm$ $\therefore r = 2\ cm$

33. (a) Max. K.E. of the system = Max. P.E. of the system

$\frac{1}{2}kx^2 = = \frac{1}{2}\times(16)\times(5\times 10^{-2})^2 = 2\times 10^{-2}\ J$

34. (d) $E = \frac{p^2}{2m}$ $\therefore m \propto \frac{1}{E}$ (If momentum are constant)

$\frac{m_1}{m_2} = \frac{E_2}{E_1} = \frac{1}{4}$

35. (a) $P = \sqrt{2mE}$ $\therefore P \propto \sqrt{E}$ *i.e.* if kinetic energy becomes four time then new momentum will become twice.

36. (a) $E = \dfrac{P^2}{2m}$. If P = constant then $E \propto \dfrac{1}{m}$

i.e. kinetic energy of heavier body will be less. As the mass of gun is more than bullet therefore it possess less kinetic energy.

37. (b) Potential energy of water = kinetic energy at turbine

$$mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 19.6} = 19.6\, m/s$$

38. (c) $p = \sqrt{2mE}$ $\therefore$ $\dfrac{p_1}{p_2} = \sqrt{\dfrac{m_1}{m_2}\dfrac{E_1}{E_2}} = \sqrt{\dfrac{2}{1} \times \dfrac{8}{1}} = \dfrac{4}{1}$

39. (a) The bomb of mass 12*kg* divides into two masses

m_1 and m_2 then $m_1 + m_2 = 12$ ...(i)

and $\dfrac{m_1}{m_2} = \dfrac{1}{3}$ ...(ii)

by solving we get $m_1 = 3kg$ and $m_2 = 9kg$

Kinetic energy of smaller part = $\dfrac{1}{2}m_1 v_1^2 = 216J$

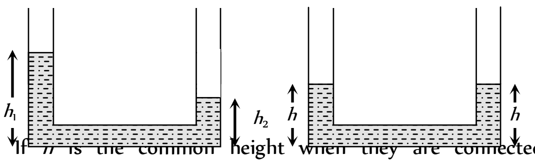
$\therefore$ $v_1^2 = \dfrac{216 \times 2}{3} \Rightarrow v_1 = 12m/s$

So its momentum = $m_1 v_1 = 3 \times 12 = 36\; kg\text{-}m/s$

As both parts possess same momentum therefore momentum of each part is $36\; kg\text{-}m/s$

40. (c) $P = \sqrt{2mE}$. If E are const. then $\dfrac{P_1}{P_2} = \sqrt{\dfrac{m_1}{m_2}} = \sqrt{\dfrac{4}{1}} = 2$

41. (d)



If h is the common height when they are connected, by conservation of mass

$$\rho A_1 h_1 + \rho A_2 h_2 = \rho h(A_1 + A_2)$$

$$h = (h_1 + h_2)/2 \qquad [\text{as } A_1 = A_2 = A \text{ given}]$$

As $(h_1/2)$ and $(h_2/2)$ are heights of initial centre of gravity of liquid in two vessels., the initial potential energy of the system

$$U_i = (h_1 A\rho)g\frac{h_1}{2} + (h_2 A\rho)\frac{h_2}{2} = \rho g A\frac{(h_1^2 + h_2^2)}{2} \quad ...(i)$$

When vessels are connected the height of centre of gravity of liquid in each vessel will be $h/2$,

i.e. $\left(\dfrac{(h_1 + h_2)}{4}\right.$ [as $h = (h_1 + h_2)/2$]

Final potential energy of the system

$$U_F = \left[\frac{(h_1 + h_2)}{2}A\rho\right]g\left(\frac{h_1 + h_2}{4}\right)$$

$$= A\rho g\left[\frac{(h_1 + h_2)^2}{4}\right] \quad ...(ii)$$

Work done by gravity

$$W = U_i - U_f = \frac{1}{4}\rho g A[2(h_1^2 + h_2^2) - (h_1 + h_2)^2]$$

$$= \frac{1}{4}\rho g A(h_1 \sim h_2)^2$$

42. (c) $P = \sqrt{2mE}$. If m is constant then

$$\frac{P_2}{P_1} = \sqrt{\frac{E_2}{E_1}} = \sqrt{\frac{1.22E}{E}} \Rightarrow \frac{P_2}{P_1} = \sqrt{1.22} = 1.1$$

$\Rightarrow P_2 = 1.1P_1 \Rightarrow P_2 = P_1 + 0.1P_1 = P_1 + 10\%$ of P_1

So the momentum will increase by 10%

43. (b) $\Delta U = mgh = 0.2 \times 10 \times 200$ = 400 J

$\therefore$ Gain in K.E. = decrease in P.E. = 400 J.

44. (a) $E = \dfrac{P^2}{2m}$. If m is constant then $E \propto P^2$

$$\Rightarrow \frac{E_2}{E_1} = \left(\frac{P_2}{P_1}\right)^2 = \left(\frac{1.2P}{P}\right)^2 = 1.44$$

$\Rightarrow$ $E_2 = 1.44E_1 = E_1 + 0.44E_1$

$E_2 = E_1 + 44\%$ of E_1

i.e. the kinetic energy will increase by 44%

45. (a) $E = \dfrac{P^2}{2m} = \dfrac{(2)^2}{2 \times 2} = 1J$

46. (b) $\Delta U = mgh = 20 \times 9.8 \times 0.5 = 98\; J$

47. (b) $E = \dfrac{P^2}{2m} = \dfrac{(10)^2}{2 \times 1} = 50\; J$

48. (b) Because 50% loss in kinetic energy will affect its potential energy and due to this ball will attain only half of the initial height.

49. (d) If there is no air drag then maximum height

$$H = \frac{u^2}{2g} = \frac{14 \times 14}{2 \times 9.8} = 10\, m$$

But due to air drag ball reaches up to height 8*m* only. So loss in energy

$= mg(10 - 8) = 0.5 \times 9.8 \times 2 = 9.8\; J$

50. (a) $1\,kcal = 10^3\; Calorie = 4200\; J = \dfrac{4200}{3.6 \times 10^6}kWh$

$$\therefore\; 700\,kcal = \frac{700 \times 4200}{3.6 \times 10^6}kWh = 0.81\,kWh$$

51. (b) $v = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 0.1} = \sqrt{1.96} = 1.4\; m/s$

52. (a)

53. (c) Let m = mass of boy, M = mass of man

v = velocity of boy, V = velocity of man

$\dfrac{1}{2}MV^2 = \dfrac{1}{2}\left[\dfrac{1}{2}mv^2\right]$ .....(i)

$\dfrac{1}{2}M(V+1)^2 = 1\left[\dfrac{1}{2}mv^2\right]$ .....(ii)

Putting $m = \dfrac{M}{2}$ and solving $V = \dfrac{1}{\sqrt{2} - 1}$

54. (d) $P = \sqrt{2mE}$ $\Rightarrow$ $\dfrac{P_1}{P_2} = \sqrt{\dfrac{m_1}{m_2}} = \sqrt{\dfrac{4}{9}} = \dfrac{2}{3}$

55. (d) $E = \dfrac{P^2}{2m} \Rightarrow E_2 = E_1\left(\dfrac{P_2}{P_1}\right)^2 = E_1\left(\dfrac{2P}{P}\right)^2$

$\Rightarrow E_2 = 4E = E + 3E = E + 300\%$ of E

56. (a) For first condition

Initial velocity = u, Final velocity = $u/2$, s = 3 cm

From $v^2 = u^2 - 2as \Rightarrow \left(\dfrac{u}{2}\right)^2 = u^2 - 2as \Rightarrow a = \dfrac{3u^2}{8s}$

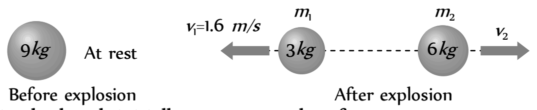
Second condition

Initial velocity = $u/2$, Final velocity = 0

From $v^2 = u^2 - 2ax \Rightarrow 0 = \dfrac{u^2}{4} - 2ax$

$\therefore\ x = \dfrac{u^2}{4 \times 2a} = \dfrac{u^2 \times 8s}{4 \times 2 \times 3u^2} = s/3 = 1\,cm$

57. (c)



As the bomb initially was at rest therefore

Initial momentum of bomb = 0

Final momentum of system = $m_1v_1 + m_2v_2$

As there is no external force

$\therefore\ m_1v_1 + m_2v_2 = 0 \Rightarrow 3 \times 1.6 + 6 \times v_2 = 0$

velocity of 6 kg mass $v_2 = 0.8\,m/s$ (numerically)

Its kinetic energy $= \dfrac{1}{2}m_2v_2^2 = \dfrac{1}{2} \times 6 \times (0.8)^2 = 1.92\,J$

58. (b) $P = \sqrt{2mE}$. $P \propto \sqrt{m}$ $\therefore$ $\dfrac{P_1}{P_2} = \sqrt{\dfrac{1}{16}} = \dfrac{1}{4}$

59. (c) Potential energy of a body = 75% of 12 J

$mgh = 9\,J \Rightarrow h = \dfrac{9}{1 \times 10} = 0.9m$

Now when this mass allow to fall then it acquire velocity

$v = \sqrt{2gh} = \sqrt{2 \times 10 \times 0.9} = \sqrt{18}$ m/s.

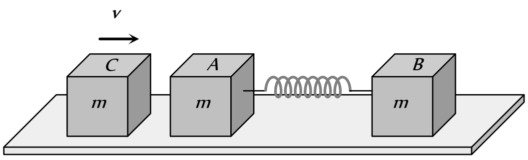
60. (a)

61. (b) Kinetic energy $E = \dfrac{P^2}{2m} = \dfrac{(Ft)^2}{2m} = \dfrac{F^2t^2}{2m}$ [As $P = F\,t$]

62. (b) Potential energy of spring = $\dfrac{1}{2}Kx^2$

$\therefore\ PE \propto x^2 \Rightarrow PE \propto a^2$

63. (a)



Initial momentum of the system (block C) = mv

After striking with A, the block C comes to rest and now both block A and B moves with velocity V, when compression in spring is maximum.

By the law of conservation of linear momentum

$mv = (m + m)\,V \Rightarrow V = \dfrac{v}{2}$

By the law of conservation of energy

K.E. of block C = K.E. of system + P.E. of system

$\dfrac{1}{2}mv^2 = \dfrac{1}{2}(2m)V^2 + \dfrac{1}{2}kx^2$

$\Rightarrow \dfrac{1}{2}mv^2 = \dfrac{1}{2}(2m)\left(\dfrac{v}{2}\right)^2 + \dfrac{1}{2}kx^2$

$\Rightarrow kx^2 = \dfrac{1}{2}mv^2$

$\Rightarrow x = v\sqrt{\dfrac{m}{2k}}$

64. (c) $P = \sqrt{2mE}$ $\therefore$ $P \propto \sqrt{m} \Rightarrow \dfrac{P_1}{P_2} = \sqrt{\dfrac{m_1}{m_2}} = \sqrt{\dfrac{m}{4m}} = \dfrac{1}{2}$

65. (d) $E = \dfrac{P^2}{2m} \Rightarrow E \propto \dfrac{1}{m} \Rightarrow \dfrac{E_1}{E_2} = \dfrac{m_2}{m_1}$

66. (b) $E = \dfrac{P^2}{2m} = \dfrac{4}{2 \times 3} = \dfrac{2}{3}\,J$

67. (d) Both fragment will possess the equal linear momentum

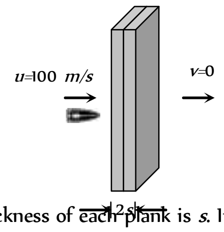
$m_1v_1 = m_2v_2 \Rightarrow 1 \times 80 = 2 \times v_2 \Rightarrow v_2 = 40\,m/s$

$\therefore$ Total energy of system $= \dfrac{1}{2}m_1v_1^2 + \dfrac{1}{2}m_2v_2^2$

$= \dfrac{1}{2} \times 1 \times (80)^2 + \dfrac{1}{2} \times 2 \times (40)^2$

= 4800 J = 4.8 kJ

68. (b)



Let the thickness of each plank is s. If the initial speed of bullet is 100 m/s then it stops by covering a distance $2s$

By applying $v^2 = u^2 - 2as \Rightarrow 0 = u^2 - 2as$

$s = \dfrac{u^2}{2a}$ $s \propto u^2$ [If retardation is constant]

If the speed of the bullet is double then bullet will cover four times distance before coming to rest

i.e. $s_2 = 4(s_1) = 4(2s) \Rightarrow s_2 = 8s$

So number of planks required = 8

69. (a) $E = \dfrac{P^2}{2m}$ if P = constant then $E \propto \dfrac{1}{m}$

According to problem $m_1 > m_2$ $\therefore$ $E_1 < E_2$

70. (c) Kinetic energy $= \frac{1}{2}mv^2$

As both balls are falling through same height therefore they possess same velocity.

but $KE \propto m$ (If v = constant)

$$\therefore \frac{(KE)_1}{(KE)_2} = \frac{m_1}{m_2} = \frac{2}{4} = \frac{1}{2}$$

71. (b) $E = \frac{P^2}{2m}$ $\therefore$ $E \propto \frac{1}{m}$ (If P = constant)

i.e. the lightest particle will possess maximum kinetic energy and in the given option mass of electron is minimum.

72. (a) $P = E \Rightarrow mv = \frac{1}{2}mv^2 \Rightarrow v = 2\,m/s$

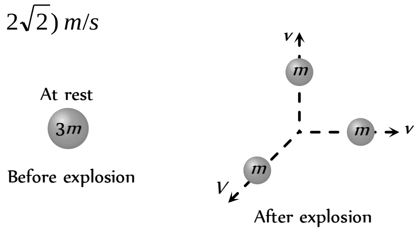
73. (c) Initial kinetic energy $E = \frac{1}{2}mv^2$ ...(i)

Final kinetic energy $2E = \frac{1}{2}m(v+2)^2$ ...(ii)

by solving equation (i) and (ii) we get

$v = (2 + 2\sqrt{2})\,m/s$

74. (c)



Initial momentum of $3m$ mass = 0 ...(i)

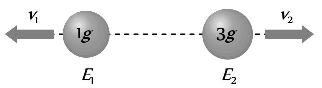
Due to explosion this mass splits into three fragments of equal masses.

Final momentum of system = $m\vec{V} + mv\hat{i} + mv\hat{j}$ ...(ii)

By the law of conservation of linear momentum

$m\vec{V} + mv\hat{i} + mv\hat{j} = 0 \Rightarrow \vec{V} = -v(\hat{i} + \hat{j})$

75. (c)



As the momentum of both fragments are equal therefore

$\frac{E_1}{E_2} = \frac{m_2}{m_1} = \frac{3}{1}$ *i.e.* $E_1 = 3E_2$ ...(i)

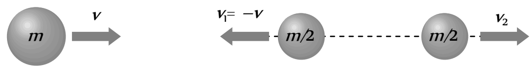
According to problem $E_1 + E_2 = 6.4 \times 10^4\,J$ ...(i)

By solving equation (i) and (ii) we get

$E_1 = 4.8 \times 10^4\,J$ and $E_2 = 1.6 \times 10^4\,J$

76. (a)

77. (b)



Before explosion After explosion

Let the initial mass of body = m

Initial linear momentum = mv ...(i)

When it breaks into equal masses then one of the fragment retrace back with same velocity

$\therefore$ Final linear momentum = $\frac{m}{2}(-v) + \frac{m}{2}(v_2)$ ...(ii)

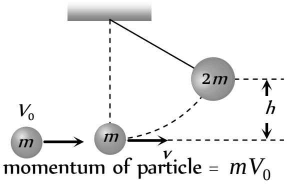
By the conservation of linear momentum

$\Rightarrow mv = \frac{-mv}{2} + \frac{mv_2}{2} \Rightarrow v_2 = 3v$

i.e. other fragment moves with velocity $3v$ in forward direction

78. (a)

79. (a)



Initial momentum of particle = mV_0

Final momentum of system (particle + pendulum) = $2mv$

By the law of conservation of momentum

$\Rightarrow mV_0 = 2mv \Rightarrow$ Initial velocity of system $v = \frac{V_0}{2}$

$\therefore$ Initial K.E. of the system = $\frac{1}{2}(2m)v^2 = \frac{1}{2}(2m)\left(\frac{V_0}{2}\right)^2$

If the system rises up to height h then P.E. = $2mgh$

By the law of conservation of energy

$$\frac{1}{2}(2m)\left(\frac{V_0}{2}\right)^2 = 2mgh \Rightarrow h = \frac{V_0^2}{8g}$$

80. (d) $\frac{P_1}{P_2} = \sqrt{\frac{m_1}{m_2}} = \sqrt{\frac{1}{9}} = \frac{1}{3}$

81. (d) Change in momentum = Force × time

$P_2 - P_1 = F \times t = 0.2 \times 10 = 2$

$\Rightarrow P_2 = 2 + P_1 = 2 + 10 = 12\,kg\text{-}m/s$

Increase in K.E. = $\frac{1}{2m}(P_2^2 - P_1^2) = \frac{1}{2 \times 5}\left[(12)^2 - (10)^2\right]$

$= \frac{44}{10} = 4.4\,J$

82. (b) $E \propto P^2$ (if m = constant)

Percentage increase in E = 2(Percentage increase in P)

= 2 × 0.01%=0.02%

83. (c) 1 $amu = 1.66 \times 10^{-27}\,kg$

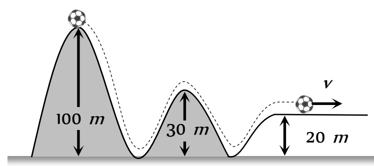
$E = mc^2 = 1.66 \times 10^{-27} \times (3 \times 10^8)^2 = 1.5 \times 10^{-10}\,J$

84. (b) Change in gravitational potential energy

= Elastic potential energy stored in compressed spring

$\Rightarrow mg(h + x) = \frac{1}{2}kx^2$

85. (c)



Ball starts from the top of a hill which is 100 m high and finally rolls down to a horizontal base which is 20 m above the ground so from the conservation of energy

$mg(h_1 - h_2) = \frac{1}{2}mv^2$

$\Rightarrow v = \sqrt{2g(h_1 - h_2)} = \sqrt{2 \times 10 \times (100 - 20)}$

$= \sqrt{1600} = 40\, m/s$.

86. (c) When block of mass M collides with the spring its kinetic energy gets converted into elastic potential energy of the spring.

From the law of conservation of energy

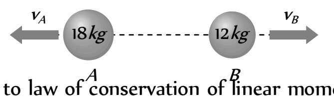
$\frac{1}{2}Mv^2 = \frac{1}{2}KL^2 \quad \therefore \quad v = \sqrt{\frac{K}{M}}L$

Where v is the velocity of block by which it collides with spring. So, its maximum momentum

$P = Mv = M\sqrt{\frac{K}{M}}\, L = \sqrt{MK}\, L$

After collision the block will rebound with same linear momentum.

87. (b)



According to law of conservation of linear momentum

$m_A v_A = m_B v_B = 18 \times 6 = 12 \times v_B \Rightarrow v_B = 9\ m/s$

K.E. of mass 12 kg, $E_B = \frac{1}{2}m_B v_B^2$

$= \frac{1}{2} \times 12 \times (9)^2 = 486 J$

88. (c) Force = Rate of change of momentum

Initial momentum $\vec{P}_1 = mv\sin\theta\,\hat{i} + mv\cos\theta\,\hat{j}$

Final momentum $\vec{P}_2 = -mv\sin\theta\,\hat{i} + mv\cos\theta\,\hat{j}$

$\therefore \vec{F} = \frac{\Delta\vec{P}}{\Delta t} = \frac{-2mv\sin\theta}{2 \times 10^{-3}}$

Substituting m = 0.1 kg, v = 5 m/s, θ = 60°

Force on the ball $\vec{F} = -250\sqrt{3}N$

Negative sign indicates direction of the force

Power

1. (a)

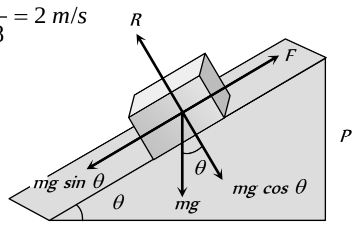
2. (d) $P = \vec{F}.\vec{v} = ma \times at = ma^2 t$ [as u = 0]

$= m\left(\frac{v_1}{t_1}\right)^2 t = \frac{mv_1^2 t}{t_1^2}$ $[\text{As}\, a = v_1/t_1]$

3. (d) $v = 7.2\frac{km}{h} = 7.2 \times \frac{5}{18} = 2\ m/s$

Slope is given 1 in 20

$\therefore \sin\theta = \frac{1}{20}$



When man and cycle moves up then component of weight opposes it motion *i.e.* $F = mg\sin\theta$

So power of the man $P = F \times v = mg\sin\theta \times v$

$= 100 \times 9.8 \times \left(\frac{1}{20}\right) \times 2 = 98\ Watt$

4. (b) If a motor of 12 HP works for 10 days at the rate of 8 hr/day then energy consumption = power × time

$= 12 \times 746 \frac{J}{\sec} \times (80 \times 60 \times 60)\sec$

$= 12 \times 746 \times 80 \times 60 \times 60\ J = 2.5 \times 10^9\ J$

Rate of energy = $50\frac{paisa}{kWh}$

i.e. $3.6 \times 10^6\ J$ energy cost 0.5 Rs

So $2.5 \times 10^9\ J$ energy cost = $\frac{2.5 \times 10^9}{2 \times 3.6 \times 10^6} = 358\ Rs$

5. (c) $P = Fv = 500 \times 3 = 1500\, W = 1.5\, kW$

6. (a) $P = Fv = F \times \frac{s}{t} = 40 \times \frac{30}{60} = 20W$

7. (b) $P = Fv = 4500 \times 2 = 9000\, W = 9\, kW$

8. (d) $P = \frac{\text{Workdone}}{\text{Time}} = \frac{mgh}{t} = \frac{300 \times 9.8 \times 2}{3} = 1960\, W$

9. (d) $P = \frac{mgh}{t} \Rightarrow m = \frac{p \times t}{gh} = \frac{2 \times 10^3 \times 60}{10 \times 10} = 1200\ kg$

As volume = $\frac{\text{mass}}{\text{density}} \Rightarrow v = \frac{1200kg}{10^3\, kg/m^3} = 1.2m^3$

Volume = $1.2m^3 = 1.2 \times 10^3\, litre = 1200\, litre$

10. (c) $P = \frac{mgh}{t} = 10 \times 10^3 \Rightarrow t = \frac{200 \times 40 \times 10}{10 \times 10^3} = 8\sec$

11. (c) Force required to move with constant velocity

$\therefore$ Power = FV

Force is required to oppose the resistive force R and also to accelerate the body of mass with acceleration a.

$\therefore$ Power = $(R + ma)V$

12. (d) $P = \frac{mgh}{t} = \frac{100 \times 9.8 \times 50}{50} = 980\, J/s$

13. (a) $P = \left(\frac{m}{t}\right)gh = 100 \times 10 \times 100 = 10^5\, W = 100\ kW$

14. (a) $p = \frac{mgh}{t} = \frac{200 \times 10 \times 200}{10} = 40\, kW$

15. (c) Volume of water to raise = 22380 l = $22380 \times 10^{-3}\, m^3$

$P = \frac{mgh}{t} = \frac{V\rho gh}{t} \Rightarrow t = \frac{V\rho gh}{P}$

$t = \frac{22380 \times 10^{-3} \times 10^3 \times 10 \times 10}{10 \times 746} = 15\ min$

16. (c) Force produced by the engine $F = \frac{P}{v} = \frac{30 \times 10^3}{30} = 10^3 N$

Acceleration $= \frac{\text{Forward force by engine} - \text{resistive force}}{\text{mass of car}}$

$$= \frac{1000-750}{1250} = \frac{250}{1250} = \frac{1}{5}\, m/s^2$$

17. (b) Power $= \dfrac{\text{Work done}}{\text{time}} = \dfrac{\frac{1}{2}m(v^2-u^2)}{t}$

$$P = \frac{1}{2}\times\frac{2.05\times10^6\times[(25)^2-(5^2)]}{5\times60}$$

$$P = 2.05\times10^6\,W = 2.05\,MW$$

18. (a) As truck is moving on an incline plane therefore only component of weight $(mg\sin\theta)$ will oppose the upward motion

Power = force × velocity = $mg\sin\theta\times v$

$$= 30000\times10\times\left(\frac{1}{100}\right)\times\frac{30\times5}{18} = 25\,kW$$

19. (c) $P = \dfrac{mgh}{t} \Rightarrow \dfrac{P_1}{P_2} = \dfrac{m_1}{m_2}\times\dfrac{t_2}{t_1}$ (As h = constant)

$$\therefore \frac{P_1}{P_2} = \frac{60}{50}\times\frac{11}{12} = \frac{11}{10}$$

20. (c) Power of a pump $= \dfrac{1}{2}\rho A v^3$

To get twice amount of water from same pipe v has to be made twice. So power is to be made 8 times.

21. (a) $p = \dfrac{mgh}{t} = \dfrac{80\times9.8\times6}{10}W = \dfrac{470}{746}HP = 0.63\,HP$

22. (b) Power $= \dfrac{\text{Work done}}{\text{time}} = \dfrac{\text{Increase in K.E.}}{\text{time}}$

$$P = \frac{\frac{1}{2}mv^2}{t} = \frac{\frac{1}{2}\times10^3\times(15)^2}{5} = 22500\,W$$

23. (a) Motor makes 600 revolution per minute

$$\therefore n = 600\frac{\text{revolution}}{\text{minute}} = 10\frac{rev}{sec}$$

$\therefore$ Time required for one revolution $= \dfrac{1}{10}\,sec$

Energy required for one revolution = power × time

$$= \frac{1}{4}\times746\times\frac{1}{10} = \frac{746}{40}\,J$$

But work done = 40% of input

$$= 40\%\times\frac{746}{40} = \frac{40}{100}\times\frac{746}{40} = 7.46\,J$$

24. (a) Work output of engine = $mgh = 100\times10\times10 = 10^4\,J$

Efficiency $(\eta) = \dfrac{\text{output}}{\text{input}}$ $\therefore$ Input energy $= \dfrac{\text{output}}{\eta}$

$$= \frac{10^4}{60}\times100 = \frac{10^5}{6}\,J$$

$$\therefore \text{Power} = \frac{\text{input energy}}{\text{time}} = \frac{10^5/6}{5} = \frac{10^5}{30} = 3.3\,kW$$

25. (a) $P = \dfrac{\vec F.\vec s}{t} = \dfrac{(2\hat i+3\hat j+4\hat k).(3\hat i+4\hat j+5\hat k)}{4} = \dfrac{38}{4} = 9.5\,W$

26. (a) $P = \dfrac{W}{t} = \dfrac{mgh}{t} = \dfrac{200\times10\times50}{10} = 10\times10^3\,W$

27. (a) Power of gun $= \dfrac{\text{Total K.E. of fired bullet}}{\text{time}}$

$$= \frac{n\times\frac{1}{2}mv^2}{t} = \frac{360}{60}\times\frac{1}{2}\times2\times10^{-2}\times(100)^2 = 600\,W$$

28. (a) Energy supplied to liquid per second by the pump

$$= \frac{1}{2}\frac{mv^2}{t} = \frac{1}{2}\frac{V\rho v^2}{t} = \frac{1}{2}A\times\left(\frac{l}{t}\right)\times\rho\times v^2 \quad \left[\frac{l}{t} = v\right]$$

$$= \frac{1}{2}A\times v\times\rho\times v^2 = \frac{1}{2}A\rho v^3$$

29. (a) Power $= \dfrac{\text{workdone}}{\text{time}} = \dfrac{\text{pressure}\times\text{change in volume}}{\text{time}}$

$$= \frac{20000\times1\times10^{-6}}{1} = 2\times10^{-2} = 0.02\,W$$

30. (c) Power $= \dfrac{W}{t}$. If W is constant then $P\propto\dfrac{1}{t}$

i.e. $\dfrac{P_1}{P_2} = \dfrac{t_2}{t_1} = \dfrac{20}{10} = \dfrac{2}{1}$

Elastic and Inelastic Collision

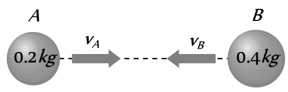
1. (a)

2. (a)

3. (c) According to law of conservation of linear momentum both pieces should possess equal momentum after explosion. As their masses are equal therefore they will possess equal speed in opposite direction.

4. (a)

5. (c)



Initial linear momentum of system = $m_A\vec v_A + m_B\vec v_B$

= 0.2 × 0.3 + 0.4 × v_B

Finally both balls come to rest

$\therefore$ final linear momentum = 0

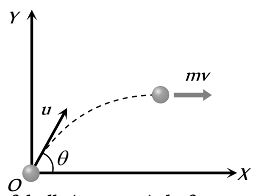
By the law of conservation of linear momenum

0.2 × 0.3 + 0.4 × v_B = 0

$$\therefore \quad v_B = -\frac{0.2\times0.3}{0.4} = -0.15\,m/s$$

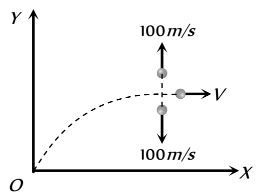
6. (c) For a collision between two identical perfectly elastic particles of equal mass, velocities after collision get interchanged.

7. (b)



Momentum of ball (mass m) before explosion at the highest point $= mv\hat i = mu\cos60°\hat i$

$$= m\times200\times\frac{1}{2}\hat i = 100\,m\hat i\;kgms^{-1}$$



Let the velocity of third part after explosion is V

After explosion momentum of system = $\vec{P}_1 + \vec{P}_2 + \vec{P}_3$

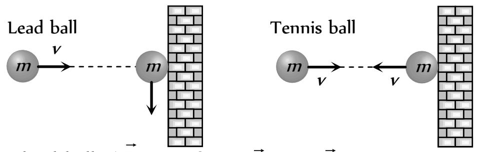
$= \frac{m}{3}\times 100\hat{j} - \frac{m}{3}\times 100\hat{j} + \frac{m}{3}\times V\hat{i}$

By comparing momentum of system before and after the explosion

$$\frac{m}{3}\times 100\hat{j} - \frac{m}{3}\times 100\hat{j} + \frac{m}{3}V\hat{i} = 100m\hat{i} \Rightarrow V = 300\,m/s$$

8. (c) Change in the momentum

= Final momentum – initial momentum



For lead ball $\Delta\vec{P}_{\text{lead}} = 0 - m\vec{v} = -m\vec{v}$

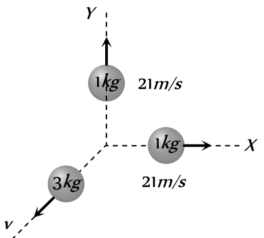
For tennis ball $\Delta\vec{P}_{\text{tennis}} = -m\vec{v} - m\vec{v} = -2m\vec{v}$

i.e. tennis ball suffers a greater change in momentum.

9. (c)

10. (d)

11. (d)



$P_x = m\times v_x = 1\times 21 = 21\;kg\;m/s$

$P_y = m\times v_y = 1\times 21 = 21\;kg\;m/s$

$\therefore$ Resultant = $\sqrt{P_x^2 + P_y^2} = 21\sqrt{2}\;kg\;m/s$

The momentum of heavier fragment should be numerically equal to resultant of $\vec{P}_x$ and $\vec{P}_y$.

$3\times v = \sqrt{P_x^2 + P_y^2} = 21\sqrt{2}\;\therefore\; v = 7\sqrt{2}$ = 9.89 m/s

12. (b) We know that when heavier body strikes elastically with a lighter body then after collision lighter body will move with double velocity that of heavier body.

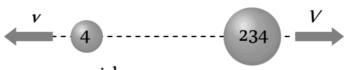
*i.e.*the ping pong ball move with speed of $2\times 2 = 4\;m/s$

13. (d) Change in momentum $= m\vec{v}_2 - m\vec{v}_1 = -mv - mv = -2mv$

14. (c) $m_G = \frac{m_B v_B}{v_G}$ $\frac{50\times 10^{-3}\times 30}{1} = 1.5\;kg$

15. (d)

16. (a) Initially ^{238}U nucleus was at rest and after decay its part moves in opposite direction.

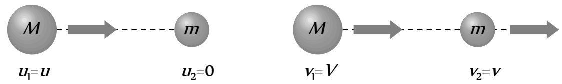


α particle Residual nucleus

According to conservation of momentum

$$4v + 234V = 238\times 0 \Rightarrow V = -\frac{4v}{234}$$

17. (c)



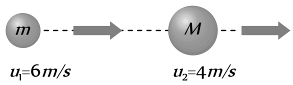
Before collision After collision

$$v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)u_2 + \frac{2m_1u_1}{m_1 + m_2} = \frac{2Mu}{M+m} = \frac{2u}{1+\frac{m}{M}}$$

18. (c) Velocity exchange takes place when the masses of bodies are equal

19. (d) In perfectly elastic head on collision of equal masses velocities gets interchanged

20. (a)



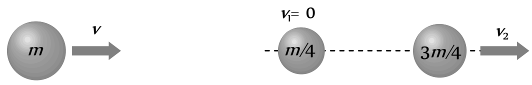
$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1 + \frac{2m_2u_2}{m_1 + m_2}$$

Substituting $m = 0$, $v_1 = -u_1 + 2u_2$

$\Rightarrow v_1 = -6 + 2(4) = 2m/s$

i.e. the lighter particle will move in original direction with the speed of 2 *m/s*.

21. (d)

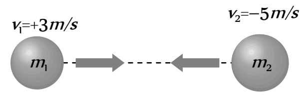


Before explosion After explosion

According to conservation of momentum

$$mv = \left(\frac{m}{4}\right)v_1 + \left(\frac{3m}{4}\right)v_2 \Rightarrow v_2 = \frac{4}{3}v$$

22. (d)

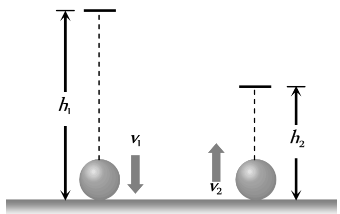


As $m_1 = m_2$ therefore after elastic collision velocities of masses get interchanged

i.e. velocity of mass $m_1 = -5\;m/s$

and velocity of mass $m_2 = +3\;m/s$

23. (b) If ball falls from height h_1 and bounces back up to height h_2 then $e = \sqrt{\frac{h_2}{h_1}}$



Similarly if the velocity of ball before and after collision are v_1 and v_2 respectively then $e = \frac{v_2}{v_1}$

So $\frac{v_2}{v_1} = \sqrt{\frac{h_2}{h_1}} = \sqrt{\frac{1.8}{5}} = \sqrt{\frac{9}{25}} = \frac{3}{5}$

i.e. fractional loss in velocity $= 1 - \frac{v_2}{v_1} = 1 - \frac{3}{5} = \frac{2}{5}$

24. (a) $h_n = he^{2n} = 32\left(\frac{1}{2}\right)^4 = \frac{32}{16} = 2m$ (here $n = 2$, $e = 1/2$)

25. (c) As the body at rest explodes into two equal parts, they acquire equal velocities in opposite directions according to conservation of momentum.

When the angle between the radius vectors connecting the point of explosion to the fragments is 90°, each radius vector makes an angle 45° with the vertical.

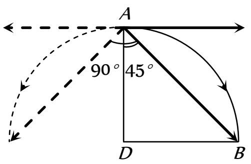
To satisfy this condition, the distance of free fall *AD* should be equal to the horizontal range in same interval of time.

$AD = DB$

$AD = 0 + \frac{1}{2} \times 10t^2 = 5t^2$

$DB = ut = 10t$

$\therefore\ 5t^2 = 10t \Rightarrow t = 2\sec$



26. (a) $v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1 + \left(\frac{2m_2}{m_1 + m_2}\right)u_2$ and

$v_2 = \left(\frac{2m_1}{m_1 + m_2}\right)u_1 + \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_2$

on putting the values $v_1 = 6\,m/s$ and $v_2 = 12\,m/s$

27. (b) $F = \frac{dp}{dt} = m\frac{dv}{dt} = \frac{m \times 2v}{1/50} = \frac{2 \times 2 \times 100}{1/50} = 2 \times 10^4\ N$

28. (d) $h_n = he^{2n} = 1 \times e^{2\times 1} = 1 \times (0.6)^2 = 0.36m$

29. (d) $h_n = he^{2n}$, if $n = 2$ then $h_n = he^4$

30. (b) Impulse = change in momentum

$mv_2 - mv_1 = 0.1 \times 40 - 0.1 \times (-30)$

31. (b) In elastic head on collision velocities gets interchanged.

32. (a) Impulse = change in momentum = 2 *mv*

$= 2 \times 0.06 \times 4 = 0.48\ kg\ m/s$

33. (b) When ball falls vertically downward from height h_1 its velocity $\vec{v}_1 = \sqrt{2gh_1}$

and its velocity after collision $\vec{v}_2 = \sqrt{2gh_2}$

Change in momentum

$\Delta\vec{P} = m(\vec{v}_2 - \vec{v}_1) = m(\sqrt{2gh_1} + \sqrt{2gh_2})$

(because $\vec{v}_1$ and $\vec{v}_2$ are opposite in direction)

34. (a) Velocity of 50 *kg*. mass after 5 *sec* of projection $v = u - gt = 100 - 9.8 \times 5 = 51\ m/s$

At this instant momentum of body is in upward direction

$P_{\text{initial}} = 50 \times 51 = 2550\ kg-m/s$

After breaking 20 *kg* piece travels upwards with 150 *m/s* let the speed of 30 *kg* mass is *V*

$P_{\text{final}} = 20 \times 150 + 30 \times V$

By the law of conservation of momentum

$P_{\text{initial}} = P_{\text{final}}$

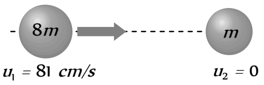
$\Rightarrow 2550 = 20 \times 150 + 30 \times V \Rightarrow V = -15\ m/s$

i.e. it moves in downward direction.

35. (c) Ratio in radius of steel balls = 1/2

So, ratio in their masses $= \frac{1}{8}$ [As $M \propto V \propto r^3$]

Let $m_1 = 8m$ and $m_2 = m$



$v_2 = \frac{2m_1u_1}{m_1 + m_2} = \frac{2 \times 8m \times 81}{8m + m} = 144\ cm/s$

36. (a) After explosion *m* mass comes at rest and let Rest $(M - m)$ mass moves with velocity *v*.

By the law of conservation of momentum $MV = (M - m)v$

$\Rightarrow v = \frac{MV}{(M - m)}$

37. (c) As the ball bounces back with same speed so change in momentum = 2 *mv*

and we know that force = rate of change of momentum

i.e. force will act on the ball so there is an acceleration.

38. (d) According to conservation of momentum

$m_Bv_B + m_Gv_G = 0 \Rightarrow v_G = -\frac{m_Bv_B}{m_G}$

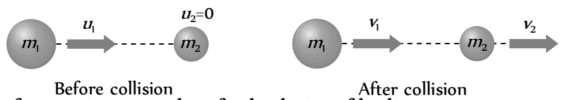
$v_G = \frac{-50 \times 10^{-3} \times 10^3}{5} = -10\ m/s$

39. (a) As 20% energy lost in collision therfore

$mgh_2 = 80\%\ \text{of}\ mgh_1 \Rightarrow \frac{h_2}{h_1} = 0.8$

but $e = \sqrt{\frac{h_2}{h_1}} = \sqrt{0.8} = 0.89$

40. (b)



If target is at rest then final velocity of bodies are

$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1$ ...(i) and $v_2 = \frac{2m_1u_1}{m_1 + m_2}$ ...(ii)

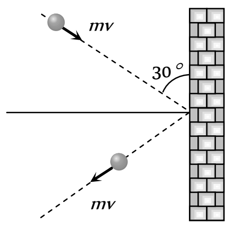
From (i) and (ii) $\frac{v_1}{v_2} = \frac{m_1 - m_2}{2m_1} = \frac{2}{5} \Rightarrow \frac{m_1}{m_2} = 5$

41. (b) *F* = Rate of change in momentum

$= \frac{2mv\sin\theta}{t}$

$= \frac{2 \times 10^{-1} \times 10\sin 30°}{0.1}$

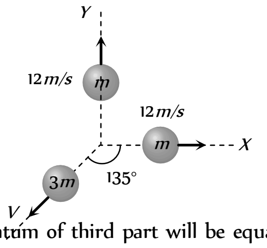
$\therefore F = 10\ N$



42. (d) By the conservation of momentum

$$40\times 10+(40)\times(-7)=80\times v \Rightarrow v=1.5\,m/s$$

43. (d)



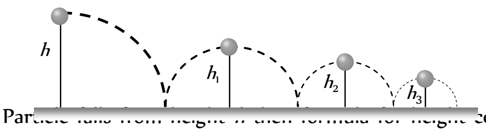
The momentum of third part will be equal and opposite to the resultant of momentum of rest two equal parts

let V is the velocity of third part.

By the conservation of linear momentum

$$3m\times V=m\times 12\sqrt{2} \Rightarrow V=4\sqrt{2}\ m/s$$

44. (a)



Particle falls from height h then formula for height covered by it in nth rebound is given by

$$h_n=he^{2n}$$

where e = coefficient of restitution, n = No. of rebound

Total distance travelled by particle before rebounding has stopped

$$H=h+2h_1+2h_2+2h_3+2h_n+........$$

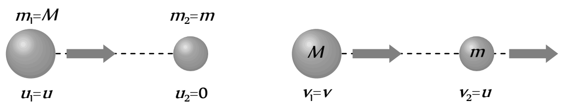
$$=h+2he^2+2he^4+2he^6+2he^8+.........$$

$$=h+2h(e^2+e^4+e^6+e^8+.......)$$

$$=h+2h\left[\frac{e^2}{1-e^2}\right]=h\left[1+\frac{2e^2}{1-e^2}\right]=h\left(\frac{1+e^2}{1-e^2}\right)$$

45. (d) Due to the same mass of A and B as well as due to elastic collision velocities of spheres get interchanged after the collision.

46. (a)



Before collision After collision

From the formulae $v_1=\left(\frac{m_1-m_2}{m_1+m_2}\right)u_1$

We get $v=\left(\frac{M-m}{M+m}\right)u$

47. (a) Momentum conservation

$$5\times 10+20\times 0=5\times 0+20\times v \Rightarrow v=2.5\,m/s$$

48. (d) Due to elastic collision of bodies having equal mass, their velocities get interchanged.

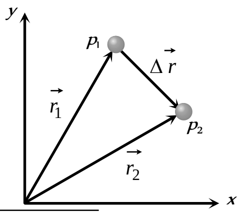
49. (c)

50. (b) $m_1=2\,kg$ and $v_1=\left(\frac{m_1-m_2}{m_1+m_2}\right)u_1=\frac{u_1}{4}$ (given)

By solving we get $m_2=1.2\,kg$

51. (c)

52. (d) It is clear from figure that the displacement vector $\Delta\vec{r}$ between particles p_1 and p_2 is $\Delta\vec{r}=\vec{r}_2-\vec{r}_1=-8\hat{i}-8\hat{j}$



$$|\Delta\vec{r}|=\sqrt{(-8)^2+(-8)^2}=8\sqrt{2} \quad .....(i)$$

Now, as the particles are moving in same direction ($\because \vec{v}_1$ and $\vec{v}_2$ are $+ve$), the relative velocity is given by

$$\vec{v}_{rel}=\vec{v}_2-\vec{v}_1=(\alpha-4)\hat{i}+4\hat{j}$$

$$\vec{v}_{rel}=\sqrt{(\alpha-4)^2+16} \quad .....(ii)$$

Now, we know $|\vec{v}_{rel}|=\frac{|\Delta\vec{r}|}{t}$

Substituting the values of $\vec{v}_{rel}$ and $|\Delta\vec{r}|$ from equation (i) and (ii) and $t=2s$, then on solving we get $\alpha=8$

53. (b) Fractional decrease in kinetic energy of neutron

$$=1-\left(\frac{m_1-m_2}{m_1+m_2}\right)^2 \quad [\text{As } m_1=1 \text{ and } m_2=2]$$

$$=1-\left(\frac{1-2}{1+2}\right)^2=1-\left(\frac{1}{3}\right)^2=1-\frac{1}{9}=\frac{8}{9}$$

54. (a)

55. (b) When target is very light and at rest then after head on elastic collision it moves with double speed of projectile *i.e.* the velocity of body of mass m will be $2v$.

56. (a) In head on elastic collision velocity get interchanged (if masses of particle are equal). *i.e.* the last ball will move with the velocity of first ball *i.e* 0.4 *m/s*

57. (a) By the principle of conservation of linear momentum,

$$Mv=mv_1+mv_2 \Rightarrow Mv=0+(M-m)v_2 \Rightarrow v_2=\frac{Mv}{M-m}$$

58. (a) Since bodies exchange their velocities, hence their masses are equal so that $\frac{m_A}{m_B}=1$

59. (d) $mgh=$ initial potential energy

$mgh'=$ final potential energy after rebound

As 40% energy lost during impact $\therefore mgh'=60\%$ of mgh

$$\Rightarrow h'=\frac{60}{100}\times h=\frac{60}{100}\times 10=6\,m$$

60. (c)

61. (a) Fractional loss $=\frac{\Delta U}{U}=\frac{mg(h-h')}{mgh}=\frac{2-1.5}{2}=\frac{1}{4}$

62. (c) $$\frac{\Delta K}{K}=\left[1-\left(\frac{m_1-m_2}{m_1+m_2}\right)^2\right]=\left[1-\left(\frac{m-2m}{m+2m}\right)^2\right]=\frac{8}{9}$$

$\Delta K = \frac{8}{9}K$ *i.e.* loss of kinetic energy of the colliding body is $\frac{8}{9}$ of its initial kinetic energy.

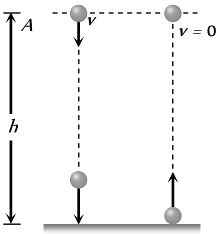
63. (d)

64. (a) $mgh = \frac{80}{100} \times mg \times 100 \Rightarrow h = 80\,m$

65. (a) Let ball is projected vertically downward with velocity v from height h

Total energy at point $A = \frac{1}{2}mv^2 + mgh$

During collision loss of energy is 50% and the ball rises up to same height. It means it possess only potential energy at same level.



$50\%\left(\frac{1}{2}mv^2 + mgh\right) = mgh$

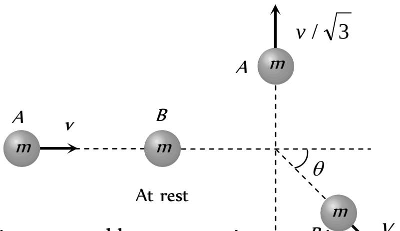
$\frac{1}{2}\left(\frac{1}{2}mv^2 + mgh\right) = mgh$

$v = \sqrt{2gh} = \sqrt{2 \times 10 \times 20}$

$\therefore v = 20\,m/s$

66. (a) $h_n = he^{2n}$ after third collision $h_3 = he^6$ [as $n = 3$]

67. (a) Let mass A moves with velocity v and collides inelastically with mass B, which is at rest.



According to problem mass A moves in a perpendicular direction and let the mass B moves at angle θ with the horizontal with velocity v.

Initial horizontal momentum of system (before collision) $= mv$ ....(i)

Final horizontal momentum of system (after collision) $= mV\cos\theta$ ....(ii)

From the conservation of horizontal linear momentum $mv = mV\cos\theta \Rightarrow v = V\cos\theta$ ...(iii)

Initial vertical momentum of system (before collision) is zero.

Final vertical momentum of system $\frac{mv}{\sqrt{3}} - mV\sin\theta$

From the conservation of vertical linear momentum

$\frac{mv}{\sqrt{3}} - mV\sin\theta = 0 \Rightarrow \frac{v}{\sqrt{3}} = V\sin\theta$ ...(iv)

By solving (iii) and (iv)

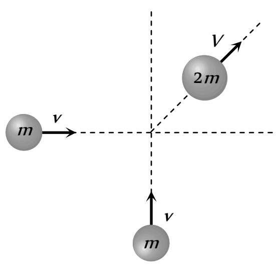
$$v^2 + \frac{v^2}{3} = V^2(\sin^2\theta + \cos^2\theta)$$

$$\Rightarrow \frac{4v^2}{3} = V^2 \Rightarrow V = \frac{2}{\sqrt{3}}v.$$

68. (d) Angle will be 90° if collision is perfectly elastic

Perfectly Inelastic Collision

1. (c)



Initial momentum of the system

$\vec{P}_i = mv\hat{i} + mv\hat{j}$

$|\vec{P}_i| = \sqrt{2}mv$

Final momentum of the system $= 2mV$

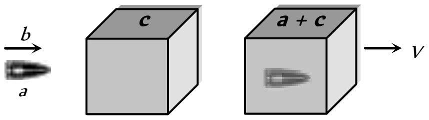
By the law of conservation of momentum

$\sqrt{2}mv = 2mV \Rightarrow V = \frac{v}{\sqrt{2}}$

2. (b)

3. (c)

4. (b)



Initially bullet moves with velocity b and after collision bullet get embedded in block and both move together with common velocity.

By the conservation of momentum

$\Rightarrow a \times b + 0 = (a + c)\,V \Rightarrow V = \frac{ab}{a+c}$

5. (d) Initially mass 10 gm moves with velocity 100 cm/s

$\therefore$ Initial momentum $= 10 \times 100 = 1000\,\frac{gm \times m}{\text{sec}}$

After collision system moves with velocity $v_{sys.}$ then

Final momentum $= (10 + 10) \times v_{sys.}$

By applying the conservation of momentum

$10000 = 20 \times v_{sys.} \Rightarrow v_{sys.} == 50\,cm/s$

If system rises upto height h then

$h = \frac{v_{sys.}^2}{2g} = \frac{50 \times 50}{2 \times 1000} = \frac{2.5}{2} = 1.25\,cm$

6. (b)

7. (c)

8. (c) $m_1v_1 - m_2v_2 = (m_1 + m_2)v$

$\Rightarrow 2 \times 3 - 1 \times 4 = (2 + 1)v \Rightarrow v = \frac{2}{3}\,m/s$

9. (c) Initial momentum of the system $= mv - mv = 0$

As body sticks together $\therefore$ final momentum $= 2mV$

By conservation of momentum $2mV = 0 \therefore V = 0$

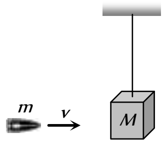
10. (a) If initially second body is at rest then

Initial momentum $= mv$

Final momentum $= 2mV$

By conservation of momentum $2mV = mv \Rightarrow V = \frac{v}{2}$

11. (d)



Initial momentum = mv

Final momentum = $(m+M)V$

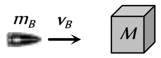
By conservation of momentum $mv = (m+M)V$

$\therefore$ Velocity of (bag + bullet) system $V = \dfrac{mv}{M+m}$

$\therefore$ Kinetic energy = $\dfrac{1}{2}(m+M)\,V^2$

$= \dfrac{1}{2}(m+M)\left(\dfrac{mv}{M+m}\right)^2 = \dfrac{1}{2}\dfrac{m^2v^2}{M+m}$

12. (b)



Initial K.E. of system = K.E. of the bullet = $\dfrac{1}{2}m_B v_B^2$

By the law of conservation of linear momentum

$m_B v_B + 0 = m_{sys.} \times v_{sys.}$

$\Rightarrow v_{sys.} = \dfrac{m_B v_B}{m_{sys.}} = \dfrac{50\times 10}{50+950} = 0.5\ m/s$

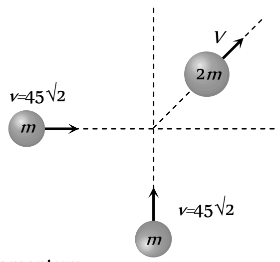
Fractional loss in K.E. = $\dfrac{\frac{1}{2}m_B v_B^2 - \frac{1}{2}m_{sys.}v_{sys.}^2}{\frac{1}{2}m_B v_B^2}$

By substituting $m_B = 50\times 10^{-3}\,kg, v_B = 10\ m/s$

$m_{sys.} = 1kg, v_s = 0.5\ m/s$ we get

Fractional loss = $\dfrac{95}{100}$ $\therefore$ Percentage loss = 95%

13. (b)



Initial momentum

$\vec{P} = m45\sqrt{2}\ \hat{i} + m45\sqrt{2}\ \hat{j} \Rightarrow |\vec{P}| = m\times 90$

Final momentum $2m\times V$

By conservation of momentum $2m\times V = m\times 90$

$\therefore V = 45\,m/s$

14. (c)

Before collision After collision

Initial momentum = mv

Final momentum = $3mV$

By the law of conservation of momentum $mv = 3mV$

$\therefore V = v/3$

15. (c)

Before collision After collision

Initial momentum = $m\times 3 + 2m\times 0 = 3m$

Final momentum = $3m\times V$

By the law of conservation of momentum

$3m = 3m\times V \Rightarrow V = 1\,km/h$

16. (d) Loss in K.E. = (initial K.E. – Final K.E.) of system

$\dfrac{1}{2}m_1u_1^2 + \dfrac{1}{2}m_2u_2^2 - \dfrac{1}{2}(m_1+m_2)V^2$

$= \dfrac{1}{2}3\times(32)^2 + \dfrac{1}{2}\times 4\times(5)^2 - \dfrac{1}{2}\times(3+4)\times(5)^2$

= 986.5 J

17. (a) Momentum of earth-ball system remains conserved.

18. (b) $v = 36\,km/h = 10\,m/s$

By law of conservation of momentum

$2\times 10 = (2+3)V \Rightarrow V = 4\,m/s$

Loss in K.E. $= \dfrac{1}{2}\times 2\times(10)^2 - \dfrac{1}{2}\times 5\times(4)^2 = 60\ J$

19. (d) Initial momentum = $\vec{P} = mv\hat{i} + mv\hat{j}$

$|\vec{P}| = \sqrt{2}mv$

Final momentum = $2m\times V$

By the law of conservation of momentum

$2m\times V = \sqrt{2}\ mv \Rightarrow V = \dfrac{v}{\sqrt{2}}$

In the problem $v = 10m/s$ (given) $\therefore V = \dfrac{10}{\sqrt{2}} = 5\sqrt{2}\ m/s$

20. (a) Because in perfectly inelastic collision the colliding bodies stick together and move with common velocity

21. (b) $m_1v_1 + m_2v_2 = (m_1+m_2)v_{sys.}$

$20\times 10 + 5\times 0 = (20+5)\,v_{sys.} \Rightarrow v_{sys.} = 8\,m/s$

K.E. of composite mass $= \dfrac{1}{2}(20+5)\times(8)^2 = 800\ J$

22. (c) According to law of conservation of momentum.

Momentum of neutron = Momentum of combination

$\Rightarrow 1.67\times 10^{-27}\times 10^{8} = (1.67\times 10^{-27} + 3.34\times 10^{-27})\,v$

$\therefore v = 3.33\times 10^{7}\ m/s$

23. (b)

24. (c) Loss in kinetic energy

$= \dfrac{1}{2}\dfrac{m_1m_2(u_1-u_2)^2}{m_1+m_2} = \dfrac{1}{2}\left(\dfrac{40\times 60}{40+60}\right)(4-2)^2 = 48\ J$

25. (b) By momentum conservation before and after collision.

$m_1V + m_2\times 0 = (m_1+m_2)v \Rightarrow v = \dfrac{m_1}{m_1+m_2}V$

i.e. Velocity of system is less than *V*.

26. (a) By conservation of momentum, $mv + M \times 0 = (m+M)V$

Velocity of composite block $V = \left(\frac{m}{m+M}\right)v$

K.E. of composite block $= \frac{1}{2}(M+m)V^2$

$= \frac{1}{2}(M+m)\left(\frac{m}{M+m}\right)^2 v^2 = \frac{1}{2}mv^2\left(\frac{m}{m+M}\right)$

27. (b)

28. (d) Velocity of combined mass, $v = \frac{m_1 v_1 - m_2 v_2}{m_1 + m_2}$

$= \frac{0.1 \times 1 - 0.4 \times 0.1}{0.5} = 0.12\ m/s$

$\therefore$ Distance travelled by combined mass

$= v \times t = 0.12 \times 10 = 1.2\ m.$

29. (c) Loss in K.E. $= \frac{m_1 m_2}{2(m_1 + m_2)}(u_1 - u_2)^2$

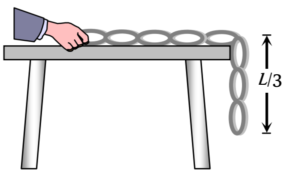
$= \frac{4 \times 6}{2 \times 10} \times (12-0)^2 = 172.8\ J$

30. (d) In case of perfectly inelastic collision, the bodies stick together after impact.

Critical Thinking Questions

1. (c) By the conservation of momentum in the absence of external force total momentum of the system (ball + earth) remains constant.

2. (d)



$W = \frac{MgL}{2n^2} = \frac{MgL}{2(3)^2} = \frac{MgL}{18}$ (n = 3 given)

3. (b) Gravitational force is a conservative force and work done against it is a point function *i.e.* does not depend on the path.

4. (b) Here $\frac{mv^2}{r} = \frac{K}{r^2}$ $\therefore$ K.E. $= \frac{1}{2}mv^2 = \frac{K}{2r}$

$U = -\int_\infty^r F.dr = -\int_\infty^r \left(-\frac{K}{r^2}\right)dr = -\frac{K}{r}$

Total energy $E = \text{K.E.} + \text{P.E.} = \frac{K}{2r} - \frac{K}{r} = -\frac{K}{2r}$

5. (c) $x = (t-3)^2 \Rightarrow v = \frac{dx}{dt} = 2(t-3)$

at $t = 0$; $v_1 = -6\ m/s$ and at $t = 6$ sec , $v_2 = 6\ m/s$

so, change in kinetic energy $= W = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = 0$

6. (c) While moving from (0,0) to (a,0)

Along positive x-axis, y = 0 $\therefore \vec{F} = -kx\hat{j}$

i.e. force is in negative y-direction while displacement is in positive x-direction.

$\therefore W_1 = 0$

Because force is perpendicular to displacement

Then particle moves from $(a,0)$ to (a,a) along a line parallel to y-axis $(x = +a)$ during this $\vec{F} = -k(y\hat{i} + a\hat{J})$

The first component of force, $-ky\hat{i}$ will not contribute any work because this component is along negative x-direction $(-\hat{i})$ while displacement is in positive y-direction (a,0) to (a,a). The second component of force *i.e.* $-ka\hat{j}$ will perform negative work

$\therefore W_2 = (-ka\hat{j})(a\hat{j}) = (-ka)(a) = -ka^2$

So net work done on the particle $W = W_1 + W_2$

$= 0 + (-ka^2) = -ka^2$

7. (a) Gain in potential energy $\Delta U = \frac{mgh}{1 + \frac{h}{R}}$

If $h = R$ then $\Delta U = \frac{mgR}{1 + \frac{R}{R}} = \frac{1}{2}mgR$

8. (c) $\text{Stopping distance} = \frac{\text{kinetic energy}}{\text{retarding force}} \Rightarrow s = \frac{1}{2}\frac{mu^2}{F}$

If lorry and car both possess same kinetic energy and retarding force is also equal then both come to rest in the same distance.

9. (d) Potential energy of the particle $U = k(1 - e^{-x^2})$

Force on particle $F = \frac{-dU}{dx} = -k[-e^{-x^2} \times (-2x)]$

$F = -2kxe^{-x^2} = -2kx\left[1 - x^2 + \frac{x^4}{2!} - ......\right]$

For small displacement $F = -2kx$

$\Rightarrow F \propto -x$ *i.e.* motion is simple harmonic motion.

10. (b) Kinetic energy acquired by the body

= Force applied on it × Distance covered by the body

K.E. = $F \times d$

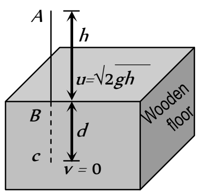
If F and d both are same then K.E. acquired by the body will be same

11. (c) Let the blade stops at depth d into the wood.

$v^2 = u^2 + 2aS$

$\Rightarrow 0 = (\sqrt{2gh})^2 + 2(g-a)d$

by solving $a = \left(1 + \frac{h}{d}\right)g$



So the resistance offered by the wood $= mg\left(1 + \frac{h}{d}\right)$

12. (d) Because linear momentum is vector quantity where as kinetic energy is a scalar quantity.

13. (c) $P = Fv = mav = m\left(\frac{dv}{dt}\right)v \Rightarrow \frac{P}{m}dt = v\,dv$

$$\Rightarrow \frac{P}{m}\times t = \frac{v^2}{2} \Rightarrow v = \left(\frac{2P}{m}\right)^{1/2}(t)^{1/2}$$

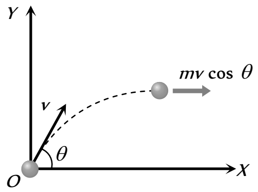
Now $s = \int v\,dt = \int \left(\frac{2P}{m}\right)^{1/2} t^{1/2}dt$

$$\therefore\ s = \left(\frac{2P}{m}\right)^{1/2}\left[\frac{2t^{3/2}}{3}\right] \Rightarrow s \propto t^{3/2}$$

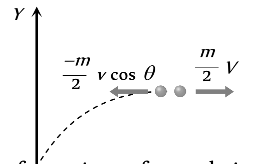
14. (a) Shell is fired with velocity v at an angle θ with the horizontal. So its velocity at the highest point

= horizontal component of velocity = $v\cos\theta$

So momentum of shell before explosion = $mv\cos\theta$



When it breaks into two equal pieces and one piece retrace its path to the canon, then other part move with velocity V.



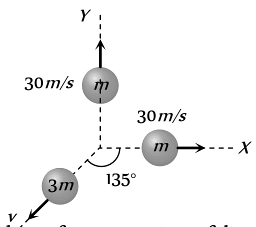
So momentum of two pieces after explosion

$$= \frac{m}{2}(-v\cos\theta) + \frac{m}{2}V$$

By the law of conservation of momentum

$$mv\cos\theta = \frac{-m}{2}v\cos\theta + \frac{m}{2}V \Rightarrow V = 3v\cos\theta$$

15. (a) Let two pieces are having equal mass m and third piece have a mass of $3m$.



According to law of conservation of linear momentum. Since the initial momentum of the system was zero, therefore final momentum of the system must be zero *i.e.* the resultant of momentum of two pieces must be equal to the momentum of third piece. We know that if two particle possesses same momentum and angle in between them is 90° then resultant will be given by $P\sqrt{2} = mv\sqrt{2} = m30\sqrt{2}$

Let the velocity of mass $3m$ is V. So $3mV = 30m\sqrt{2}$

$\therefore\ V = 10\sqrt{2}$ and angle 135° from either.

(as it is clear from the figure)

16. (c) The momentum of the two-particle system, at $t = 0$ is

$$\vec{P}_i = m_1\vec{v}_1 + m_2\vec{v}_2$$

Collision between the two does not affect the total momentum of the system.

A constant external force $(m_1 + m_2)g$ acts on the system.

The impulse given by this force, in time $t = 0$ to $t = 2t_0$ is $(m_1 + m_2)g \times 2t_0$

$\therefore$ |Change in momentum in this interval

$$= |m_1\vec{v}'_1 + m_2\vec{v}'_2 - (m_1\vec{v}_1 + m_2\vec{v}_2)| = 2(m_1 + m_2)gt_0$$

17. (b) If the masses are equal and target is at rest and after collision both masses moves in different direction. Then angle between direction of velocity will be 90°, if collision is elastic.

18. (d) K.E. of colliding body before collision $= \frac{1}{2}mv^2$

After collision its velocity becomes

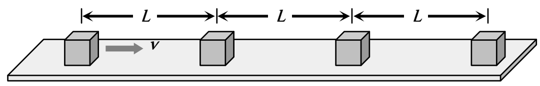
$$v' = \frac{(m_1 - m_2)}{(m_1 + m_2)}v = \frac{m}{3m}v = \frac{v}{3}$$

$\therefore$ K.E. after collision $\frac{1}{2}mv'^2 = \frac{1}{2}\frac{mv^2}{9}$

Ratio of kinetic energy = $\frac{K.E._{before}}{K.E._{after}} = \frac{\frac{1}{2}mv^2}{\frac{1}{2}\frac{mv^2}{9}} = 9:1$

19. (c)

20. (b,d)



Since collision is perfectly inelastic so all the blocks will stick together one by one and move in a form of combined mass.

Time required to cover a distance 'L' by first block $= \frac{L}{v}$

Now first and second block will stick together and move with $v/2$ velocity (by applying conservation of momentum) and combined system will take time $\frac{L}{v/2} = \frac{2L}{v}$ to reach up to block third.

Now these three blocks will move with velocity $v/3$ and combined system will take time $\frac{L}{v/3} = \frac{3L}{v}$ to reach upto the block fourth.

So, total time $= \frac{L}{v} + \frac{2L}{v} + \frac{3L}{v} + \ldots \frac{(n-1)L}{v} = \frac{n(n-1)L}{2v}$

and velocity of combined system having n blocks as $\frac{v}{n}$.

Graphical questions

1. (c) At time t_1 the velocity of ball will be maximum and it goes on decreasing with respect to time.

At the highest point of path its velocity becomes zero, then it increases but direction is reversed

This explanation match with graph (c).

2. (a) Work done = area between the graph and position axis

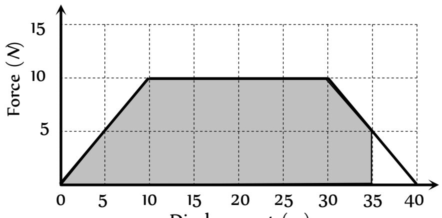
$W = 10\times1 + 20\times1 - 20\times1 + 10\times1 = 20\,erg$

3. (a) Spring constant $k = \frac{F}{x}$ = Slope of curve

$\therefore k = \frac{4-1}{30} = \frac{3}{30} = 0.1\,kg/cm$

4. (b) As the area above the time axis is numerically equal to area below the time axis therefore net momentum gained by body will be zero because momentum is a vector quantity.

5. (c)



Work done = (Shaded area under the graph between $x = 0$ to $x = 35\ m$) $= 287.5\ J$

6. (a) Work done = Area covered in between force displacement curve and displacement axis

= Mass × Area covered in between acceleration-displacement curve and displacement axis.

$= 10\times\frac{1}{2}(8\times10^{-2}\times20\times10^{-2})$

$= 8\times10^{-2}\ J$

7. (c) Work done = Gain in potential energy

Area under curve = mgh

$\Rightarrow \frac{1}{2}\times11\times100 = 5\times10\times h$

$\Rightarrow h = 11m$

8. (d) Initial K.E. of the body = $\frac{1}{2}mv^2 = \frac{1}{2}\times25\times4 = 50\ J$

Work done against resistive force

= Area between F-x graph

$= \frac{1}{2}\times4\times20 = 40J$

Final K.E. = Initial K.E. – Work done against resistive force

$= 50 - 40 = 10\ J$

9. (d) Area between curve and displacement axis

$= \frac{1}{2}\times(12+4)\times10 = 80\ J$

In this time body acquire kinetic energy = $\frac{1}{2}mv^2$

by the law of conservation of energy

$\frac{1}{2}mv^2 = 80J$

$\Rightarrow \frac{1}{2}\times0.1\times v^2 = 80$

$\Rightarrow v = 1600$

$\Rightarrow v = 40\ m/s$

10. (a) Work done = Area under curve and displacement axis

= Area of trapezium

$= \frac{1}{2}\times$(sum of two parallel lines) × distance between them

$= \frac{1}{2}(10+4)\times(2.5-0.5)$

$= \frac{1}{2}14\times2 = 14\ J$

As the area actually is not trapezium so work done will be more than $14\ J$ *i.e.* approximately $16\ J$

11. (a) As particle is projected with some velocity therefore its initial kinetic energy will not be zero.

As it moves downward under gravity then its velocity increases with time K.E. $\propto v \propto t$ (As $v \propto t$)

So the graph between kinetic energy and time will be parabolic in nature.

12. (a) From the graph it is clear that force is acting on the particle in the region AB and due to this force kinetic energy (velocity) of the particle increases. So the work done by the force is positive.

13. (d) $F = \frac{-dU}{dx} \Rightarrow dU = -F\ dx$

$\Rightarrow U = -\int_0^x(-Kx + ax^3)dx = \frac{kx^2}{2} - \frac{ax^4}{4}$

$\therefore$ We get $U = 0$ at $x = 0$ and $x = \sqrt{2k/a}$

and also U = negative for $x > \sqrt{2k/a}$.

So $F = 0$ at $x = 0$

i.e. slope of $U - x$ graph is zero at $x = 0$.

14. (b) Work done = Area enclosed by $F - x$ graph

$= \frac{1}{2}\times(3+6)\times3 = 13.5\ J$

15. (c) As slope of problem graph is positive and constant upto certain distance and then it becomes zero.

So from $F = \frac{-dU}{dx}$, up to distance a, F = constant (negative) and becomes zero suddenly.

16. (d) Work done = change in kinetic energy

$W = \frac{1}{2}mv^2 \; \therefore \; W \propto v^2$ graph will be parabolic in nature

17. (a) Potential energy increases and kinetic energy decreases when the height of the particle increases it is clear from the graph (*a*).

18. (c) $P = \sqrt{2mE}$ it is clear that $P \propto \sqrt{E}$

So the graph between P and $\sqrt{E}$ will be straight line.

but graph between $\frac{1}{P}$ and $\sqrt{E}$ will be hyperbola

19. (b) When particle moves away from the origin then at position $x = x_1$ force is zero and at $x > x_1$, force is positive (repulsive in nature) so particle moves further and does not return back to original position.

i.e. the equilibrium is not stable.

Similarly at position $x = x_2$ force is zero and at $x > x_2$, force is negative (attractive in nature)

So particle return back to original position *i.e.* the equilibrium is stable.

20. (c) $F = \frac{-dU}{dx}$ it is clear that slope of $U - x$ curve is zero at point B and C. $\therefore$ $F = 0$ for point B and C

21. (a) Work done = area under curve and displacement axis

$= 1 \times 10 - 1 \times 10 + 1 \times 10 = 10 \; J$

22. (a) When the length of spring is halved, its spring constant will becomes double. (because $k \propto \frac{1}{x} \propto \frac{1}{L} \therefore k \propto \frac{1}{L}$)

Slope of force displacement graph gives the spring constant (k) of spring.

If k becomes double then slope of the graph increases *i.e.* graph shifts towards force-axis.

23. (a) Kinetic energy $E = \frac{1}{2}mv^2 \Rightarrow E \propto v^2$

graph will be parabola symmetric to *E*-axis.

24. (c) Change in momentum = Impulse

= Area under force-time graph

$\therefore \; mv =$ Area of trapezium

$$\Rightarrow mv = \frac{1}{2}\left(T + \frac{T}{2}\right)F_0$$

$$\Rightarrow mv = \frac{3T}{4}F_0 \Rightarrow F_0 = \frac{4mu}{3T}$$

25. (c) When body moves under action of constant force then kinetic energy acquired by the body K.E. = $F \times S$

$\therefore$ KE $\propto$ S (If F = constant)

So the graph will be straight line.

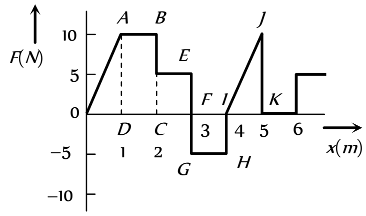
26. (a) When the distance between atoms is large then interatomic force is very weak. When they come closer, force of attraction increases and at a particular distance force becomes zero. When they are further brought closer force becomes repulsive in nature.

This can be explained by slope of $U - x$ curve shown in graph (a).

27. (b) Work done = area under *F-x* graph

= area of rectangle *ABCD* + area of rectangle *LCEF*

+ area of rectangle *GFIH* + area of triangle *IJK*



$$= (2-1) \times (10-0) + (3-2)(5-0) + (4-3)(-5-0)$$

$$+ \frac{1}{2}(5-4)(10-0) = 15 \; J$$

28. (a) $U = -\int F dx = -\int kx \; dx = -k\frac{x^2}{2}$

This is the equation of parabola symmetric to U axis in negative direction

Assertion and Reason

1. (a) The work done, $W = \vec{F}.\vec{s} = Fs\cos\theta$, when a person walk on a horizontal road with load on his head then $\theta = 90°$.

Hence $W = Fs\cos 90° = 0$

Thus no work is done by the person.

2. (d) In a round trip work done is zero only when the force is conservative in nature.

Force is always required to move a body in a conservative or non-conservative field

3. (e) When a body slides down on inclined plane,

work done by friction is negative because it opposes the motion (θ = 180° between force and displacement)

If $\theta < 90°$ then W =positive because $W = F.s.\cos\theta$

4. (a) Since the gaseous pressure and the displacement (of piston) are in the same direction. Therefore $\theta = 0°$

$\therefore$ Work done = $Fs\cos\theta = Fs =$ Positive

Thus during expansion work done by gas is positive.

5. (d) When two bodies have same momentum then lighter body possess more kinetic energy because $E = \frac{P^2}{2m}$

$\therefore E \propto \frac{1}{m}$ when P = constant

6. (b) $P = \vec{F}.\vec{v}$ and unit of power is *Watt*.

7. (c) Change in kinetic energy = work done by net force.

This relationship is valid for particle as well as system of particles.

8. (a) The work done on the spring against the restoring force is stored as potential energy in both conditions when it is compressed or stretched.

9. (c) The gravitational force on the comet due to the sun is a conservative force. Since the work done by a conservative force over a closed path is always zero (irrespective of the nature of path), the work done by the gravitational forces over every complete orbit of the comet is zero.

10. (e) Rate of change of momentum is proportional to external forces acting on the system. The total momentum of whole system remain constant when no external force is acted upon it.

Internal forces can change the kinetic energy of the system.

11. (a) When the water is at the top of the fall it has potential energy mgh (where m is the mass of the water and h is the height of the fall). On falling, this potential energy is converted into kinetic energy, which further converted into heat energy and so temperature of water increases.

12. (b) The power of the pump is the work done by it per sec.

$\therefore$ Power $= \dfrac{\text{work}}{\text{time}} = \dfrac{mgh}{t} = \dfrac{100 \times 10 \times 100}{10}$

$= 10^4 W = 10\, kW$

Also 1 *Horse power* (hp) =746 W.

13. (c) For conservative forces the sum of kinetic and potential energies at any point remains constant throughout the motion. This is known as law of conservation of mechanical energy. According to this law,

Kinetic energy + Potential energy = constant

or, $\Delta K + \Delta U = 0$ or, $\Delta K = -\Delta U$

14. (e) When the force retards the motion, the work done is negative.

Work done depends on the angle between force and displacement $W = Fs\cos\theta$

15. (d) In an elastic collision both the momentum and kinetic energy remains conserved. But this rule is not for individual bodies, but for the system of bodies before and after the collision. While collision in which there occurs some loss of kinetic energy is called inelastic collision. Collision in daily life are generally inelastic. The collision is said to be perfectly inelastic, if two bodies stick to each other.

16. (d) A body can have energy without having momentum if it possess potential energy but if body possess momentum then it must posses kinetic energy. Momentum and energy have different dimensions.

17. (e) Work done and power developed is zero in uniform circular motion only.

18. (a) $K = \dfrac{1}{2}mv^2 \;\therefore\; K \propto v^2$

If velocity is doubled then K.E. will be quadrupled.

19. (a) In a quick collision, time t is small. As $F \times t =$ constant, therefore, force involved is large, *i.e.* collision is more violent in comparison to slow collision.

20. (a) From, definition, work done in moving a body against a conservative force is independent of the path followed.

21. (c) When we supply current through the cell, chemical reactions takes place, so chemical energy of cell is converted into electrical energy. If a large amount of current is drawn from wire for a long time only then wire get heated.

22. (e) Potential energy $U = \dfrac{1}{2}kx^2$ *i.e.* $U \propto x^2$

This is a equation of parabola, so graph between U and x is a parabola, not straight line.

23. (c) When two bodies of same mass undergo an elastic collision, their velocities get interchanged after collision. Water and heavy water are hydrogenic materials containing protons having approximately the same mass as that of a neutron. When fast moving neutrons collide with protons, the neutrons come to rest and protons move with the velocity of that of neutrons.

24. (a) From Einstein equation $E = mc^2$

it can be observed that if mass is conserved then only energy is conserved and vice versa. Thus, both cannot be treated separately.

25. (b) If two protons are brought near one another, work has to be done against electrostatic force because same charge repel each other. This work done is stored as potential energy in the system.

26. (a) $E = \dfrac{P^2}{2m}$. In firing momentum is conserved $\therefore E \propto \dfrac{1}{m}$

So $\dfrac{E_{\text{gun}}}{E_{\text{bullet}}} = \dfrac{m_{\text{bullet}}}{m_{\text{gun}}}$

27. (a) K.E. of one bullet = k $\therefore$ K.E. of n bullet = nk

According to law of conservation of energy, the kinetic energy of bullets be equal to the work done by machine gun per sec.

28. (d) Work done in the motion of a body over a closed loop is zero only when the body is moving under the action of conservative forces (like gravitational or electrostatic forces). *i.e.* work done depends upon the nature of force.

29. (a) If roads of the mountain were to go straight up, the slope θ would have been large, the frictional force $\mu mg\cos\theta$ would be small. Due to small friction, wheels of vehicle would slip. Also for going up a large slope, a greater power shall be required.

30. (a) The rise in temperature of the soft steel is an example of transferring energy into a system by work and having it appear as an increase in the internal energy of the system. This works well for the soft steel because it is soft. This softness results in a deformation of the steel under blow of the hammer. Thus the point of application of the force is displaced by the hammer and positive work is done on the steel. With the hard steel, less deformation occur, thus, there is less displacement of point of application of the force and less work done on the steel. The soft steel is therefore better in absorbing energy from the hammer by means of work and its temperature rises more rapidly.

Work, Energy, Power and Collision

SET Self Evaluation Test - 6

1. How much work does a pulling force of 40 N do on the 20 kg box in pulling it 8 m across the floor at a constant speed. The pulling force is directed at 60° above the horizontal
   (a) 160 J (b) 277 J
   (c) 784 J (d) None of the above

2. A horizontal force of 5 N is required to maintain a velocity of 2 m/s for a block of 10 kg mass sliding over a rough surface. The work done by this force in one minute is
   (a) 600 J (b) 60 J
   (c) 6 J (d) 6000 J

3. Work done in time t on a body of mass m which is accelerated from rest to a speed v in time t_1 as a function of time t is given by
   (a) $\frac{1}{2}m\frac{v}{t_1}t^2$ (b) $m\frac{v}{t_1}t^2$
   (c) $\frac{1}{2}\left(\frac{mv}{t_1}\right)^2 t^2$ (d) $\frac{1}{2}m\frac{v^2}{t_1^2}t^2$

4. What is the shape of the graph between the speed and kinetic energy of a body
   (a) Straight line (b) Hyperbola
   (c) Parabola (d) Exponential

5. When a body moves with some friction on a surface
   (a) It loses kinetic energy but momentum is constant
   (b) It loses kinetic energy but gains potential energy
   (c) Kinetic energy and momentum both decrease
   (d) Mechanical energy is conserved

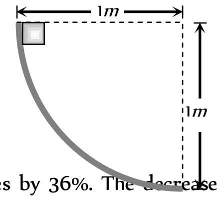
6. A bullet of mass m moving with velocity v strikes a suspended wooden block of mass M. If the block rises to a height h, the initial velocity of the block will be
   (a) $\sqrt{2gh}$ (b) $\frac{M+m}{m}\sqrt{2gh}$
   (c) $\frac{m}{M+m}2gh$ (d) $\frac{M+m}{M}\sqrt{2gh}$

7. There will be decrease in potential energy of the system, if work is done upon the system by
   (a) Any conservative or non-conservative force
   (b) A non-conservative force
   (c) A conservative force
   (d) None of the above

8. The slope of kinetic energy displacement curve of a particle in motion is
   (a) Equal to the acceleration of the particle
   (b) Inversely proportional to the acceleration
   (c) Directly proportional to the acceleration
   (d) None of the above

9. The energy required to accelerate a car from 10 m/s to 20 m/s is how many times the energy required to accelerate the car from rest to 10 m/s
   (a) Equal (b) 4 times
   (c) 2 times (d) 3 times

10. A body of mass 2 kg slides down a curved track which is quadrant of a circle of radius 1 $metre$. All the surfaces are frictionless. If the body starts from rest, its speed at the bottom of the track is



   (a) 4.43 m/sec
   (b) 2 m/sec
   (c) 0.5 m/sec
   (d) 19.6 m/sec

11. The kinetic energy of a body decreases by 36%. The decrease in its momentum is
   (a) 36% (b) 20%
   (c) 8% (d) 6%

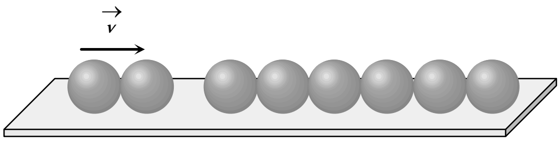
12. A bomb of mass $3m$ kg explodes into two pieces of mass m kg and $2m$ kg. If the velocity of m kg mass is 16 m/s, the total kinetic energy released in the explosion is
   (a) 192 mJ (b) 96 mJ
   (c) 384 mJ (d) 768 mJ

13. Which one of the following statement does not hold good when two balls of masses m_1 and m_2 undergo elastic collision
   (a) When $m_1 << m_2$ and m_2 at rest, there will be maximum transfer of momentum
   (b) When $m_1 >> m_2$ and m_2 at rest, after collision the ball of mass m_2 moves with four times the velocity of m_1
   (c) When $m_1 = m_2$ and m_2 at rest, there will be maximum transfer of K.E.
   (d) When collision is oblique and m_2 at rest with $m_1 = m_2$, after collision the balls move in opposite directions

14. A neutron travelling with a velocity v and K.E. E collides perfectly elastically head on with the nucleus of an atom of mass number A at rest. The fraction of total energy retained by neutron is
   (a) $\left(\frac{A-1}{A+1}\right)^2$ (b) $\left(\frac{A+1}{A-1}\right)^2$
   (c) $\left(\frac{A-1}{A}\right)^2$ (d) $\left(\frac{A+1}{A}\right)^2$

15. A body of mass m_1 moving with uniform velocity of 40 m/s collides with another mass m_2 at rest and then the two together begin to move with uniform velocity of 30 m/s. The ratio of their masses $\frac{m_1}{m_2}$ is
   (a) 0.75 (b) 1.33
   (c) 3.0 (d) 4.0

16. Six identical balls are lined in a straight groove made on a horizontal frictionless surface as shown. Two similar balls each moving with a velocity v collide elastically with the row of 6 balls from left. What will happen



(a) One ball from the right rolls out with a speed $2v$ and the remaining balls will remain at rest

(b) Two balls from the right roll out with speed v each and the remaining balls will remain stationary

(c) All the six balls in the row will roll out with speed $v/6$ each and the two colliding balls will come to rest

(d) The colliding balls will come to rest and no ball rolls out from right

17. A wooden block of mass M rests on a horizontal surface. A bullet of mass m moving in the horizontal direction strikes and gets embedded in it. The combined system covers a distance x on the surface. If the coefficient of friction between wood and the surface is μ, the speed of the bullet at the time of striking the block is (where m is mass of the bullet)

(a) $\sqrt{\dfrac{2Mg}{\mu m}}$ (b) $\sqrt{\dfrac{2\mu mg}{Mx}}$

(c) $\sqrt{2\mu gx}\left(\dfrac{M+m}{m}\right)$ (d) $\sqrt{\dfrac{2\mu mx}{M+m}}$

18. A ball moving with speed v hits another identical ball at rest. The two balls stick together after collision. If specific heat of the material of the balls is S, the temperature rise resulting from the collision is

(a) $\dfrac{v^2}{8S}$ (b) $\dfrac{v^2}{4S}$

(c) $\dfrac{v^2}{2S}$ (d) $\dfrac{v^2}{S}$

19. A bag of sand of mass M is suspended by a string. A bullet of mass m is fired at it with velocity v and gets embedded into it. The loss of kinetic energy in this process is

(a) $\dfrac{1}{2}mv^2$ (b) $\dfrac{1}{2}mv^2 \times \dfrac{1}{M+m}$

(c) $\dfrac{1}{2}mv^2 \times \dfrac{M}{m}$ (d) $\dfrac{1}{2}mv^2\left(\dfrac{M}{M+m}\right)$

AS Answers and Solutions

(SET -6)

1. (a) $W = \vec{F}.\vec{s} = 40 \times 8 \times \cos 60° = 160\ J$

2. (a) $W = F \times s = F \times v \times t = 5 \times 2 \times 60 = 600\ J$

3. (d) Work done $= F \times s = ma \times \dfrac{1}{2}at^2$ $\left[\text{from } s = ut + \dfrac{1}{2}at^2\right]$

$\therefore W = \dfrac{1}{2}ma^2t^2 = \dfrac{1}{2}m\left(\dfrac{v}{t_1}\right)^2 t^2$ $\left[\text{As } a = \dfrac{v}{t_1}\right]$